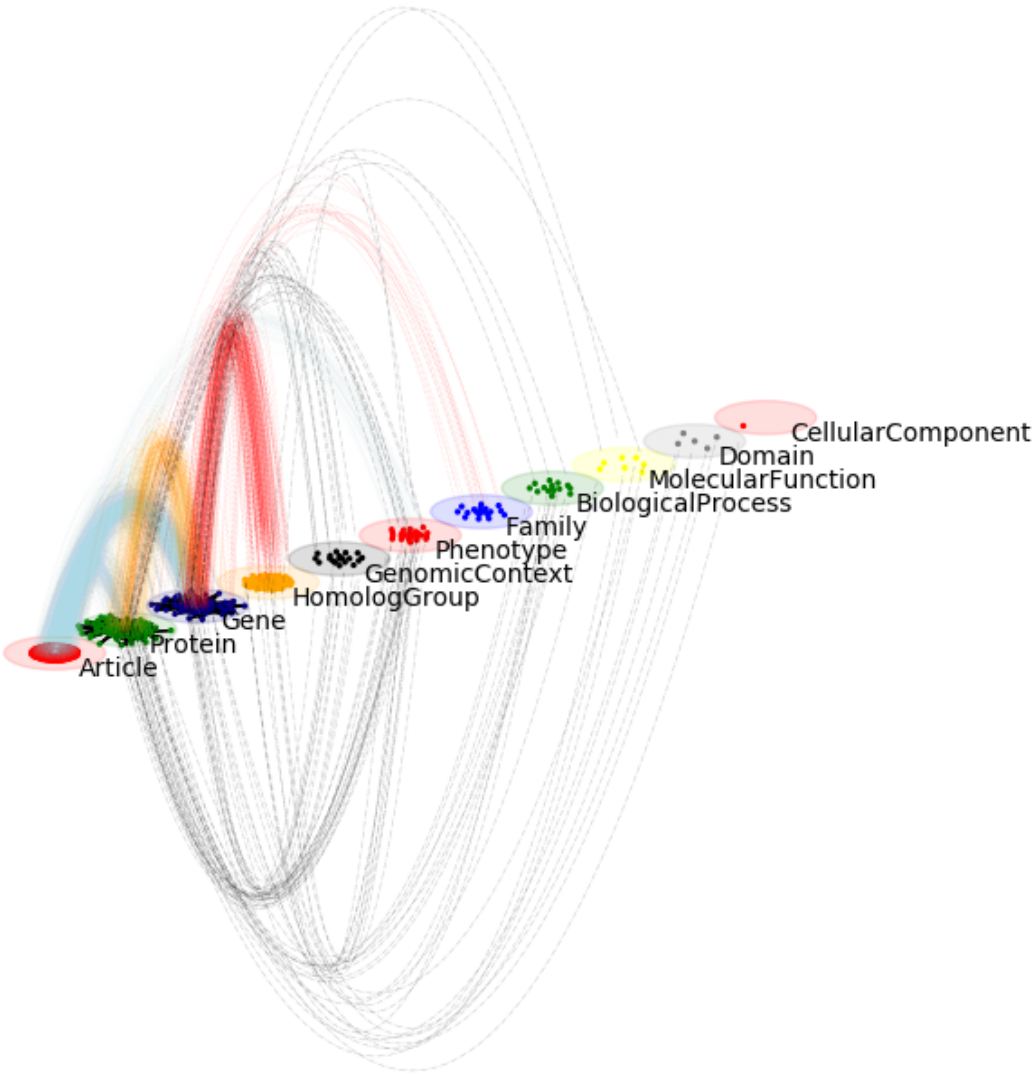



# Py3plex: Multilayer Network Analysis in Python

***Release 2.0.93***

**Blaž Škrlj**

**Nov 11, 2026**

# OVERVIEW


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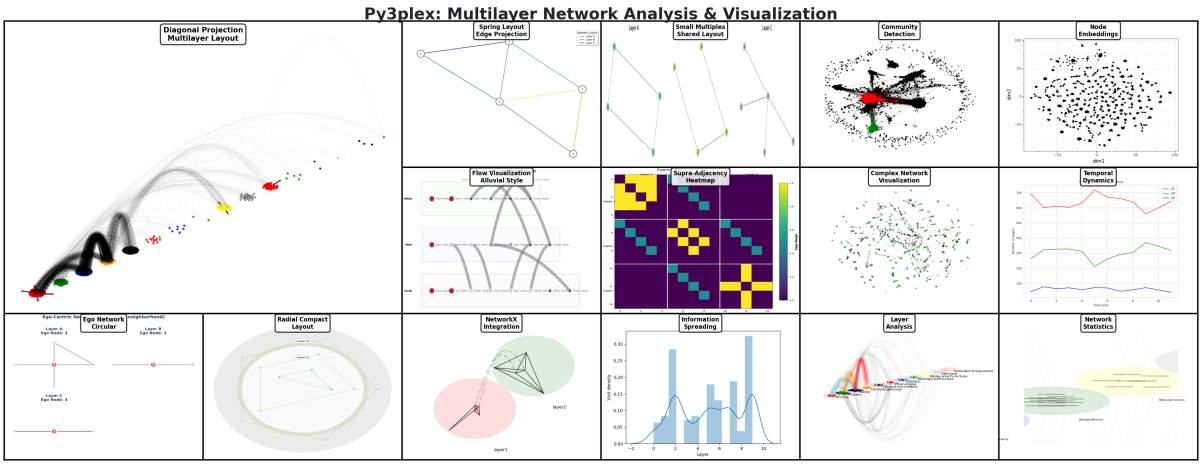
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[1] [2]

py3plex provides scalable analysis and visualization of multilayer and multiplex networks in Python.

**Key features at a glance:**

- Multiplex and multilayer network structures
- **SQL-like DSL for network queries** — first-class feature
- 17+ multilayer statistics and centrality measures
- Community detection (Louvain, Infomap, multilayer modularity)
- Random walk algorithms (Node2Vec, DeepWalk) for embeddings
- Publication-ready visualizations
- High-performance I/O with Arrow/Parquet support
- Full NetworkX compatibility

**DSL: Query Networks Like SQL**

The py3plex DSL lets you query networks using intuitive SQL-like syntax or a type-safe Python builder API. Both return the same results; pick the style that fits your workflow.

```
from py3plex.dsl import execute_query, Q, L

# String DSL: Simple and readable
result = execute_query(network,
    'SELECT nodes WHERE layer="social" AND degree > 5 '
    'COMPUTE betweenness_centrality'
)

# Builder API: Type-safe with autocompletion
result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
```

[1] https://github.com/SkBlaz/py3plex/actions/workflows/tests.yml
[2] https://github.com/SkBlaz/py3plex/actions/workflows/code-quality.yml

```
    .compute("betweenness_centrality")
    .order_by("-betweenness_centrality")
    .limit(10)
    .execute(network)
)
```

The DSL is **perfect for**:

- Interactive network exploration
- Rapid prototyping
- Educational purposes
- Production pipelines

See the complete user_guide/dsl guide for all features!

CHAPTER

ONE

# QUICKSTART

Get started in three steps: install, build a tiny multilayer network, and inspect it.

Step 1 — Install

```
pip install py3plex
```

Step 2 — Create a multilayer network

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
], input_type="list")

network.basic_stats()
network.visualize_network(show=True)
```

Expected output

```
Number of nodes: 6
Number of edges: 3
```

Counts reflect per-layer node identities (Alice/Bob/Carol each appear in two layers). Your totals will differ if you change the edge list. The visualization opens in a Matplotlib window and colors nodes by layer.

CHAPTER

TWO

# START HERE

**New to py3plex?** Start with *Quick Start Tutorial* to move from basics to a full analysis in about 10 minutes.

**Already familiar with multilayer networks?** Jump to *How to Run Community Detection on Multilayer Networks* or browse *How-to Guides* for task-focused guides.

**Just need the API?** See *API Documentation* or *DSL Reference*.

**Want to understand the concepts?** Read *Multilayer Networks 101* for the theory behind multilayer networks.

CHAPTER

THREE

# DOCUMENTATION STRUCTURE

This documentation follows the Diátaxis[3] framework with 7 top-level sections. Use the outline below to jump directly to the content type you need—tutorials for learning, how-to guides for tasks, explanations for concepts, and reference for precise details.

## 3.1 Part I: Overview

High-level introduction to py3plex and multilayer networks.

### 3.1.1 Overview

Welcome to py3plex! This section provides a high-level introduction to the library and its capabilities.

**What is py3plex?**

py3plex is a Python library for analyzing and visualizing multilayer and multiplex networks. Unlike traditional network analysis tools that treat all relationships uniformly, py3plex preserves the distinct types of connections between entities—whether they're social ties, biological interactions, or transportation links.

**Why py3plex?**

- **Native multilayer support:** Built for networks with multiple relationship types from the start.
- **SQL-like DSL:** Query networks intuitively with SQL-inspired syntax.
- **Production-ready:** High-performance I/O, sklearn-style pipelines, and comprehensive testing.
- **Research-proven:** Used in peer-reviewed publications.

**This section includes:**

- *What is py3plex?* — Key features and what makes py3plex unique
- *Key Use Cases* — Typical domains and applications
- *Multilayer Networks in 2 Minutes* — Quick conceptual introduction to multilayer networks

**Next steps:**

- **New to py3plex?** → Start with *Quick Start Tutorial*
- **Want the theory?** → Read *Multilayer Networks 101*
- **Need specific tasks?** → Jump to *How-to Guides*

[3] https://diataxis.fr/

### 3.1.2 What is py3plex?

py3plex is a Python library for scalable analysis and visualization of multilayer and multiplex networks.

#### Key Features

Use py3plex when you need to keep relationship types distinct while still running end-to-end analysis and visualization.

#### Multilayer Network Structures

py3plex provides native support for:

- **Multiplex networks** — Same nodes across multiple layers (e.g., social networks with friendship, collaboration, and family ties).
- **Heterogeneous networks** — Different node types across layers (e.g., author-paper-venue networks).
- **Temporal networks** — Networks that evolve over time; edges and nodes can carry timestamps or time windows.

All network types are first-class citizens with consistent APIs.

#### SQL-like DSL for Network Queries

Query networks using SQL-inspired syntax that stays close to plain language:

```
from py3plex.dsl import execute_query

# String DSL: readable and concise
result = execute_query(network,
    'SELECT nodes WHERE layer="social" AND degree > 5 '
    'COMPUTE betweenness_centrality'
)
```

Or use the type-safe builder API for IDE autocompletion and refactoring support:

```
from py3plex.dsl import Q, L

# Builder API: with IDE autocompletion
result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
     .compute("betweenness_centrality")
     .execute(network)
)
```

The DSL makes complex network analyses readable and maintainable.

### Comprehensive Algorithm Suite

py3plex includes multilayer-specific algorithms such as:

- **Community detection:** Louvain, Infomap, multilayer modularity.
- **Centrality measures:** Multilayer PageRank, betweenness, versatility.
- **Random walks:** Node2Vec, DeepWalk for network embeddings.
- **Dynamics:** SIR/SIS epidemic models, diffusion processes.

All algorithms handle multilayer structure natively without flattening.

### Publication-Ready Visualizations

Create professional network diagrams with:

- Automatic multilayer layouts that preserve layer separation.
- Customizable styling and colors.
- Support for large networks (1000+ nodes).
- Export to multiple formats (PNG, PDF, SVG).

### High-Performance I/O

- Apache Arrow/Parquet support for large datasets.
- NetworkX compatibility for easy integration.
- Multiple input formats (edge lists, adjacency matrices, JSON) without forcing layer flattening.

### What Makes py3plex Unique?

**Native Multilayer Representation**

Unlike tools that flatten multilayer networks into single-layer graphs, py3plex preserves the multilayer structure throughout the analysis pipeline. This leads to more accurate results for multilayer-specific properties.

**Node-Layer Pair Abstraction**

py3plex represents multilayer networks using node-layer pairs: a node in layer A is distinct from the same node in layer B. This clean abstraction enables:

- Layer-specific queries.
- Inter-layer and intra-layer edge differentiation.
- Efficient supra-adjacency matrix operations.

**Production-Ready Design**

py3plex is designed for both research and production:

- Sklearn-style pipelines for reproducible workflows.
- Comprehensive type hints and documentation.
- Broad test coverage across core functionality.
- CLI and Docker deployment options.

#### When to Use py3plex?

py3plex is ideal when you have:

- **Multiple relationship types** between entities (e.g., email + phone + in-person contacts)
- **Heterogeneous networks** with different node types (e.g., users, posts, hashtags)
- **Temporal networks** where relationships change over time
- **Need for layer-specific analysis** (e.g., comparing community structure across layers)

If you only need single-layer network analysis, NetworkX or igraph might be simpler choices.

#### Related Projects

- **NetworkX:** py3plex builds on NetworkX for single-layer operations
- **igraph:** Faster for single-layer networks, but limited multilayer support
- **pymnet:** Pure Python multilayer library, different API design
- **graph-tool:** High performance, but steeper learning curve

py3plex balances ease of use with multilayer-specific functionality.

#### Next Steps

- **Get started:** *Quick Start Tutorial*
- **See use cases:** *Key Use Cases*
- **Understand multilayer networks:** *Multilayer Networks in 2 Minutes*
- **Deep dive:** *The py3plex Core Model*

### 3.1.3 Key Use Cases

py3plex models systems where the same entities interact through many relationship types (layers). Each node may appear in multiple layers, and interlayer edges let you model transfers or couplings between contexts. Below are common domains, what the layers represent, and the kinds of questions you can answer.

#### Social Networks

Layers represent communication channels or contexts for the same people:

- Friendship, family, and professional connections
- Online interactions: mentions, retweets, replies
- Communication channels: email, chat, video calls

**Example applications:**

- Identifying influential users across multiple platforms
- Detecting communities that span different relationship types
- Predicting information diffusion through multiple channels

```
from collections import Counter
from py3plex.core import multinet

# Example: Multi-platform social network
```

```
network = multinet.multi_layer_network()
# List input: [source, source_layer, target, target_layer, weight]
network.add_edges([
    ['Alice', 'twitter', 'Bob', 'twitter', 1],
    ['Alice', 'linkedin', 'Bob', 'linkedin', 1],
    ['Bob', 'twitter', 'Carol', 'twitter', 1],
], input_type="list")

# Find users present in more than one layer
layer_counts = Counter(node for node, layer in network.get_nodes())
active_users = [node for node, count in layer_counts.items() if count > 1]
print(active_users)
```

### Biological Networks

Layers capture different biological interaction types:

- Protein-protein interactions (physical binding)
- Gene regulatory networks (transcriptional control)
- Metabolic pathways (biochemical reactions)
- Signaling cascades

**Example applications:**

- Drug target identification across interaction types
- Disease gene prioritization using multiple evidence sources
- Pathway enrichment analysis

```
# Example: Multi-omics biological network
# Protein-protein interactions + gene regulation
from py3plex.core import multinet
from py3plex.dsl import execute_query

network = multinet.multi_layer_network()

# Physical interactions
network.add_edge('TP53', 'ppi', 'MDM2', 'ppi')

# Regulatory interactions
network.add_edge('TP53', 'regulation', 'CDKN1A', 'regulation')

# Compute centrality within the regulatory layer
result = execute_query(network,
    'SELECT nodes WHERE layer="regulation" '
    'COMPUTE betweenness_centrality'
)
```

### Transportation Networks

Layers represent travel modes and transfers between them:

- Road networks
- Public transit (bus, metro, train)
- Air travel
- Pedestrian paths

**Example applications:**

- Optimizing multimodal route planning
- Identifying critical transfer points
- Analyzing resilience to disruptions

```
# Example: Multimodal city transportation
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Different transportation modes
network.add_edge('StationA', 'metro', 'StationB', 'metro')
network.add_edge('StationA', 'bus', 'StationC', 'bus')
network.add_edge('StationB', 'metro', 'StationC', 'metro')

# Inter-layer connections (transfers)
network.add_edge('StationA', 'metro', 'StationA', 'bus',
                 type='interlayer')

# Identify busy transfer points (degree > 2 across all modes)
busy_stations = [
    node for node in network.get_nodes()
    if network.core_network.degree(node) > 2
]
```

### Knowledge Graphs

Layers separate relation types between entities:

- Entity types: people, organizations, locations, concepts
- Relation types: employment, location, authorship, citation

Keeping relations in distinct layers preserves meaning (employment vs. location) while still allowing cross-layer queries.

**Example applications:**

- Entity linking and disambiguation
- Knowledge graph completion
- Question answering over structured data

### Scientific Collaboration

Layers represent collaboration contexts:

- Joint publications
- Grant collaborations
- Conference attendance
- Social media interactions

**Example applications:**

- Identifying interdisciplinary researchers
- Detecting emerging research communities
- Predicting future collaborations

Layered collaboration data keeps formal publications separate from informal interactions while still letting you track cross-context teams.

### Epidemic Modeling

Layers represent contact types with different transmission risks:

- Household contacts
- Workplace interactions
- Social gatherings
- Healthcare settings

**Example applications:**

- Predicting disease spread
- Evaluating intervention strategies
- Identifying super-spreader events

Assign layer-specific transmission probabilities to model different risks (e.g., household vs. workplace).

See *How to Simulate Multilayer Dynamics* for epidemic modeling how-tos.

### Communication Networks

Layers capture organizational communication channels:

- Email correspondence
- Instant messaging
- Video conferencing
- Document collaboration

**Example applications:**

- Organizational structure analysis
- Information flow optimization
- Team effectiveness measurement

Layered communication captures channel preferences without losing the underlying person iden-

tifiers.

### Financial Networks

Layers distinguish financial instruments and ownership ties:

- Trade networks (goods, services)
- Financial flows (investments, loans)
- Ownership structures (subsidiaries, shareholding)
- Interbank lending networks

**Example applications:**

- Systemic risk assessment
- Contagion analysis
- Market structure analysis
- Regulatory network analysis

**Example: Multi-layer financial network**

```
from py3plex.core import multinet
from py3plex.dsl import execute_query

# Build financial network with multiple relationship types
network = multinet.multi_layer_network(directed=True)

# Trade relationships
network.add_edges([
    ['BankA', 'trade', 'BankB', 'trade', 1.5],  # Trade volume in billions
    ['BankB', 'trade', 'BankC', 'trade', 2.3],
    ['BankC', 'trade', 'BankA', 'trade', 1.8],
], input_type="list")

# Ownership structures
network.add_edges([
    ['BankA', 'ownership', 'SubsidiaryX', 'ownership', 0.75],  # 75% ownership
    ['BankB', 'ownership', 'SubsidiaryY', 'ownership', 0.60],
], input_type="list")

# Interbank lending
network.add_edges([
    ['BankA', 'lending', 'BankB', 'lending', 500],  # Lending amount
    ['BankB', 'lending', 'BankC', 'lending', 300],
    ['BankC', 'lending', 'BankA', 'lending', 200],
], input_type="list")

# Identify systemically important institutions using DSL
# (unweighted betweenness across all layers; pass weight='weight' to include amounts)
centrality = execute_query(
    network,
    'SELECT nodes COMPUTE betweenness_centrality'
)['computed']['betweenness_centrality']
```

```
top_3 = sorted(
    centrality.items(),
    key=lambda item: item[1],
    reverse=True
)[:3]

print("Top betweenness nodes (node, betweenness):")
for node, score in top_3:
    print(f"  {node}: {score:.4f}")

# Find institutions represented in multiple layers
from collections import Counter
node_layer_count = Counter(node for node, layer in network.get_nodes())
multi_layer_nodes = [
    node for node, count in node_layer_count.items() if count >= 2
]

print(f"\nInstitutions with multiple relationship types: {len(multi_layer_nodes)}")
for node in sorted(multi_layer_nodes):
    print(f"  {node}")
```

**Use cases for financial network analysis:**

- **Systemic risk:** Identify institutions whose failure would cascade through multiple layers
- **Regulatory oversight:** Detect hidden dependencies across different financial instruments
- **Market surveillance:** Monitor unusual patterns in multi-layer trading relationships
- **Portfolio diversification:** Understand correlation structures across asset types

See *How to Run Community Detection on Multilayer Networks* for detecting financial communities and *How to Simulate Multilayer Dynamics* for contagion modeling.

### Clinical Consensus Networks

Layers can separate different structuring choices for data labelled by multiple experts (e.g. radiologists reading scans):

- Patients as layers, radiologists as nodes, with edges weighted by measured inter-rater agreement rather than raw participation
- Patients as layers, with annotators and their findings as nodes (structure only, no agreement embedded in the edges)
- Annotators as layers, with shared findings as nodes across their caseloads

**Example applications:**

- Identifying radiologists who are central to the *consensus* structure itself, not just the busiest readers
- Tuning inter-layer coupling strength to separate participation volume from genuine agreement signal
- Detecting stable groups of mutually-agreeing annotators via community detection

**Example: Agreement-weighted consensus network**

```python
from py3plex.core import multinet
from py3plex.algorithms.multilayer_algorithms import centrality as ml_centrality
from py3plex.algorithms.community_detection.community_louvain import best_partition
from py3plex.dsl import Q

# Example: patients as layers, radiologists as nodes. Edge weight is the
# measured agreement between two radiologists on that patient's findings
# (0-1), not a fixed participation weight.
network = multinet.multi_layer_network(network_type="multilayer", directed=False)
network.add_edges([
    {"source": "reader_1", "target": "reader_2", "source_type": "patient_1",
     "target_type": "patient_1", "weight": 0.82},
    {"source": "reader_1", "target": "reader_3", "source_type": "patient_2",
     "target_type": "patient_2", "weight": 0.65},
])

# Couple a reader's own instances across patient-layers with a small
# weight, well below the agreement edges (see note below)
network.add_edges([
    {"source": "reader_1", "target": "reader_1", "source_type": "patient_1",
     "target_type": "patient_2", "weight": 0.005},
])

# Option 1: matrix path (Katz centrality on the supra-adjacency matrix)
supra = network.get_supra_adjacency_matrix(mtype="sparse")
katz = ml_centrality.katz_centrality(supra, tol=1e-6)
for node, score in zip(network.core_network.nodes(), katz):
    print(node, score)

# Option 2: DSL path. Scores come back already labelled by (reader, patient)
# instance, so no manual mapping is needed. PageRank is used here because
# it honours the edge weights, including the coupling edges.
df = Q.nodes().compute("pagerank").order_by("-pagerank").execute(network).to_pandas()
print(df)

# Detect groups of radiologists whose agreement edges cluster together
communities = best_partition(network.core_network, weight="weight")
print(communities)
```

The choice of what to encode as layers, nodes and edge weights determines what the analysis can answer. A structure-only network mostly recovers participation: its centralities track how much each annotator worked, not how well annotators agree. Embedding a measured agreement score in the edge weights turns the same machinery into a consensus analysis.

### Next Steps

- **Try an example:** *Quick Start Tutorial*
- **See complete examples:** *Examples & Recipes*
- **Learn the concepts:** *Multilayer Networks 101*

### 3.1.4 Multilayer Networks in 2 Minutes

A **multilayer network** represents systems where entities can be connected through multiple types of relationships simultaneously.

#### The Core Idea

In the real world, relationships are rarely uniform. Consider a group of researchers:

- They **co-author papers** (collaboration network)
- They **cite each other** (citation network)
- They **attend conferences** together (social network)
- They may **share funding** (grant network)

A single-layer network captures only one of these relationships. A multilayer network keeps every relationship type separate while linking the same entity across layers.

#### Visual Intuition

```
Single-layer network (e.g., only co-authorship):

Alice --- Bob --- Carol
           |
         David

Multilayer network (co-authorship + citations):

Layer 1 (Co-authorship):
Alice --- Bob --- Carol
           |
         David

Layer 2 (Citations):
Alice --> Bob --> Carol
           ↓
         David

Inter-layer connections:
Alice in Layer 1  Alice in Layer 2 (same person)
```

#### Key Concepts

Keep these quick definitions in mind while you read the rest.

**Layers** Each layer represents one relationship type (e.g., friendship, collaboration, family).

**Node-layer pairs** Every node is represented per layer. In py3plex, `('Alice', 'friends')` and `('Alice', 'work')` are distinct node-layer pairs that can be linked. Counts in py3plex statistics refer to these pairs, not just the unique entity names.

**Intra-layer edges** Connections within a single layer (e.g., Alice friends with Bob).

**Inter-layer edges** Connections between layers, typically tying the same entity across layers (e.g., Alice in the friendship layer  Alice in the collaboration layer). They can also connect different entity types when modeling interactions between layers.

### Types of Multilayer Networks

**Multiplex networks**

- Same nodes across all layers
- Different edge types per layer
- Example: Social media (same users, different platforms)

**Heterogeneous networks**

- Different node types in different layers
- Example: Author-Paper-Venue networks

**Temporal networks**

- Layers represent time slices
- Same nodes and edge types, but edges appear/disappear over time
- Example: Communication networks over days/weeks

### Why Multilayer Matters

**Information is lost when flattening**

If you merge all layers into a single network, you lose critical information:

- Which type of relationship connects two nodes?
- Are communities consistent across layers or layer-specific?
- How do different relationship types interact?

**New properties emerge**

Multilayer networks have properties that don't exist in single-layer networks:

- **Node versatility:** How many layers a node participates in, and whether its role shifts by layer.
- **Layer correlation:** Whether connections in layer A are predictive of connections in layer B.
- **Multilayer communities:** Groups that span multiple relationship types.
- **Multiplexity:** Number of different relationship types between two nodes.

### Real-World Impact

**Example: Disease spread**

In epidemic modeling, people interact through:

- Household contacts (high transmission rate)
- Workplace contacts (medium rate, large reach)
- Social gatherings (variable)

A single-layer model that averages these interactions will give incorrect predictions. A multilayer model preserves the structure and transmission dynamics of each contact type.

**Example: Social influence**

A person might be influential on Twitter (many followers) but not on LinkedIn (few connections). Flattening these into a single "social network" loses this nuance.

### Quick Example in Code

```
from py3plex.core import multinet

# Create multilayer network
network = multinet.multi_layer_network()

# Add edges in different layers
network.add_edges([
    # Friendship layer (source, source_layer, target, target_layer, weight)
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],

    # Collaboration layer
    ['Alice', 'work', 'Carol', 'work', 1],
    ['Carol', 'work', 'David', 'work', 1],
], input_type="list")

# Analyze (counts are for node-layer pairs)
stats = network.basic_stats()
print(f"Layers: {len(network.get_layers())}")
print(f"Nodes: {stats['nodes']}")
print(f"Edges: {stats['edges']}")
```

**Expected output:**

```
Layers: 2
Nodes: 8  (4 unique entities × 2 layers)
Edges: 4
```

### When to Use Multilayer Analysis?

Use multilayer networks when:

- You have **multiple relationship types** between entities
- The **type of relationship matters** for your analysis
- You want to **preserve layer-specific structure**
- You're studying **cross-layer effects** (e.g., how Twitter activity affects LinkedIn connections)

Stick with single-layer when:

- All relationships are of the same type
- Relationship type doesn't affect your analysis
- You only care about aggregate connectivity

### Next Steps

- **Try it yourself:** *Quick Start Tutorial*
- **Deeper theory:** *Multilayer Networks 101*
- **See how py3plex works:** *The py3plex Core Model*
- **Start analyzing:** *How to Load and Build Networks*

## 3.2 Part II: Getting Started (Tutorials)

Step-by-step tutorials for beginners. Start here if you're new to py3plex.

### 3.2.1 Getting Started (Tutorials)

Use these tutorials to install py3plex and complete your first multilayer network analysis in small, ordered steps.

**What's in this section:**

- *Installation and Setup* — Installation options and dependencies
- *Quick Start Tutorial* — Comprehensive 10-minute tutorial from basics to advanced analysis
- *Common Issues and Troubleshooting* — Solutions to installation and runtime problems

**Why Tutorials?**

These pages are **learning-oriented**—they walk through concrete steps to build skills and confidence. How-to guides assume you already know the task; tutorials take you from a blank environment to a finished result.

**How to use this section**

Start with a clean environment if possible. Pick the path below that matches your background and follow it in order.

#### Recommended Path

**Complete Beginner?**

1. *Installation and Setup* — Set up your environment
2. *Quick Start Tutorial* — Build a first multilayer network and run core analyses
3. *How-to Guides* — Apply what you learned to specific tasks

**Experienced with Networks?**

1. Skim *Quick Start Tutorial* for py3plex syntax and features
2. Jump to *How-to Guides* for task-oriented guides
3. Review *The py3plex Core Model* for the data model

**Having Problems?**

See *Common Issues and Troubleshooting* for troubleshooting help.

#### What You'll Learn

After completing these tutorials, you'll know how to:

- Install py3plex and its dependencies
- Create multilayer networks from scratch or load them from files
- Compute basic statistics, centrality, and community structure
- Visualize multilayer networks

- Query networks with the DSL

**Next Steps After Tutorials:**

- *How-to Guides* — Task-oriented recipes
- *Multilayer Networks 101* — Theory and design
- *Examples & Recipes* — Additional worked examples
- *API Documentation* — Full API reference

**Relation to Other Sections:**

This section teaches by **doing**. Other sections serve different purposes:

- **Overview** (*Overview*) — High-level introduction without code
- **How-to Guides** (*How-to Guides*) — Solve specific problems (assumes basic knowledge)
- **Concepts** (*Concepts & Explanations*) — Understand the theory and design
- **Reference** (*API & DSL Reference*) — Look up API details

### 3.2.2 Installation and Setup

This guide covers all aspects of installing py3plex and setting up your environment.

#### Which Docs Should I Use?

**Most users should use these docs (py3plex 2.x)**, which you're reading now at https://skblaz.github.io/py3plex/. They cover:

- The current stable version with modern features (DSL v2, pipelines, workflows)
- All new APIs, including the SQL-like query DSL and dplyr-style operations
- Actively maintained and updated with examples

**Use the legacy docs (py3plex 0.8x) only if:**

- You're maintaining old code that depends on py3plex 0.8x specifically
- You need features that were removed in the 1.x rewrite (very rare)

The legacy docs are available at https://py3plex.readthedocs.io but are **no longer maintained or updated** and may have broken examples. We strongly recommend using py3plex 2.x for new projects.

#### Installation Modes

Choose the installation method based on your needs. In all cases, prefer a virtual environment to keep dependencies isolated from system Python:

| Mode | Command | When to Use |
|---|---|---|
| **User Install** | `python -m pip install py3plex` | Standard usage: running analyses, using the library |
| **Developer Install** | Clone repo + `make setup` + `make dev-install` | Contributing code, running tests, building docs |

### Basic Installation

#### Install from PyPI (Recommended)

We recommend using uv[4] for fast, reliable Python environment management:

```
# Install uv (if not already installed)
curl -LsSf https://astral.sh/uv/install.sh | sh

# Create and activate virtual environment
uv venv .venv
source .venv/bin/activate  # On Windows: .venv\Scripts\activate

# Install py3plex
uv pip install py3plex
```

**Alternative (traditional pip):**

```
python -m pip install --upgrade pip
python -m pip install py3plex
```

This installs the core py3plex library with all required dependencies.

#### Install from Source

For development or to access the latest features:

**Using uv (recommended):**

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
uv venv .venv
source .venv/bin/activate
uv pip install -e .
```

**Using pip:**

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
python -m pip install --upgrade pip
python -m pip install -e .
```

The `-e` flag installs in editable mode, so changes to the source code are immediately available.

#### Docker Installation (Alternative)

For a containerized environment with all dependencies pre-installed, use Docker:

```
# Clone the repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Build the Docker image
```

[4] https://docs.astral.sh/uv/

```
docker build -t py3plex:latest .

# Run py3plex commands (mount local data if needed)
docker run --rm -v ${PWD}:/data py3plex:latest --version
docker run --rm -v ${PWD}:/data py3plex:latest selftest
```

**Benefits of Docker installation:**

- No Python environment setup required
- All dependencies pre-installed and tested
- Consistent environment across different systems
- Isolated from system Python installation

See *Docker Usage Guide* for complete Docker documentation including:

- Building and running the container
- Working with data files via volume mounts
- Docker Compose usage
- Helper scripts (`py3plex-docker.sh`)
- Batch processing and CI/CD integration

#### Optional Dependencies

py3plex offers several optional feature sets that can be installed separately. Quote extras (`"py3plex[extra]"`) to avoid shell globbing on some terminals.

#### Advanced Community Detection

Install Infomap for advanced overlapping community detection:

```
python -m pip install "py3plex[infomap]"
```

#### Additional Algorithms

Install extra community detection algorithms (Louvain, cdlib):

```
python -m pip install "py3plex[algos]"
```

#### Advanced Visualization

Install Plotly and igraph for interactive and advanced visualizations:

```
python -m pip install "py3plex[viz]"
```

#### Apache Arrow Support

Install pyarrow for high-performance I/O with Arrow/Parquet formats:

```
python -m pip install "py3plex[arrow]"
```

Arrow format typically provides faster read/write operations and better compression than JSON/CSV for large graphs. See ../user_guide/io_and_formats for details on using Arrow format.

### Multiple Extras

Install multiple feature sets at once:

```
python -m pip install "py3plex[infomap,viz,algos,arrow]"
```

### Workflow Configuration Support

Install YAML support for config-driven workflows:

```
python -m pip install "py3plex[workflows]"
```

### Development Installation

For contributors, install with development tools (testing, linting):

**Recommended (using uv):**

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
# Install uv if not already installed
curl -LsSf https://astral.sh/uv/install.sh | sh
# Create environment and install
uv venv .venv
source .venv/bin/activate
uv pip install -e ".[dev]"
```

**Using Makefile:**

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
make setup        # Create virtual environment (uses uv if available)
make dev-install  # Install with dev dependencies
```

**Alternative (traditional pip):**

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
python -m pip install --upgrade pip
python -m pip install -e ".[dev]"
```

All methods install:

- pytest and coverage tools
- black, ruff, isort for code formatting
- mypy for type checking
- sphinx for documentation building

### System Requirements

#### Python Version

py3plex requires Python 3.8 or higher. We test on:

- Python 3.8
- Python 3.9
- Python 3.10
- Python 3.11
- Python 3.12

#### Platform Support

py3plex is cross-platform and tested on:

- **Linux** (Ubuntu 20.04, 22.04)
- **macOS** (macOS 11, 12, 13)
- **Windows** (Windows Server 2019, 2022)

#### Core Dependencies

These are automatically installed with py3plex:

- `networkx>=2.5` - Graph data structures and algorithms
- `numpy>=1.19.0` - Numerical computing
- `scipy>=1.5.0` - Scientific computing and sparse matrices
- `pandas` - Data manipulation
- `matplotlib>=3.3.0` - Static visualization
- `scikit-learn>=0.24.0` - Machine learning utilities
- `tqdm>=4.40.0` - Progress bars
- `bitarray>=2.0.0` - Efficient boolean arrays

Optional algorithm extras include:

- `rdflib>=6.0.0` - Semantic web support utilities
- `gensim>=4.0.0` - Topic modeling and embedding helpers
- `seaborn>=0.11.0` - Statistical plotting
- `cython>=0.29.0` - Extension compilation support

### External Binaries (Optional)

py3plex no longer bundles external binaries; install them separately when needed to keep the base package lightweight and licensing clear.

### Infomap Community Detection

For advanced community detection using Infomap:

**Option 1:** Use the bundled Python implementation (recommended):

```
python -m pip install "py3plex[infomap]"
```

**Option 2:** Download the C++ binary from the official source:

- Website: https://www.mapequation.org/infomap/
- Follow platform-specific installation instructions

**Alternative:** Use the built-in Louvain algorithm (no binary needed):

```
from py3plex.algorithms.community_detection import community_louvain
communities = community_louvain.best_partition(network.core_network)
```

### Node2Vec Embeddings

For Node2Vec node embeddings, use pure Python alternatives:

**Option 1:** Use pecanpy (fast parallel implementation):

```
python -m pip install pecanpy
```

**Option 2:** Use node2vec package:

```
python -m pip install node2vec
```

**Option 3:** Use py3plex's built-in random walk implementation:

```
from py3plex.algorithms.general.walkers import node2vec_walk, generate_walks
walks = generate_walks(network.core_network, num_walks=10, walk_length=80)
```

### Virtual Environment Setup

We strongly recommend using a virtual environment to prevent dependency conflicts:

### Using uv (Recommended)

uv[5] is a fast Python package installer and resolver:

```
# Install uv
curl -LsSf https://astral.sh/uv/install.sh | sh

# Create virtual environment
uv venv .venv
```

[5] https://docs.astral.sh/uv/

```
# Activate (Linux/macOS)
source .venv/bin/activate

# Activate (Windows)
.venv\Scripts\activate

# Install py3plex inside the environment
uv pip install py3plex
```

#### Using venv

```
# Create virtual environment
python3 -m venv py3plex-env

# Activate (Linux/macOS)
source py3plex-env/bin/activate

# Activate (Windows)
py3plex-env\Scripts\activate

# Install py3plex inside the environment
python -m pip install --upgrade pip
python -m pip install py3plex
```

#### Using conda

```
# Create conda environment
conda create -n py3plex python=3.10

# Activate
conda activate py3plex

# Install py3plex
python -m pip install py3plex
```

#### Common Installation Issues

#### Issue: "No module named 'numpy'"

**Solution:** Install numpy first (preferably in a virtual environment):

```
python -m pip install numpy
python -m pip install py3plex
```

#### Issue: Compilation errors on Windows

**Solution:** Install Visual C++ Build Tools:

1. Download from: https://visualstudio.microsoft.com/visual-cpp-build-tools/
2. Install "Desktop development with C++"
3. Restart your shell, then retry installation

#### Issue: "Permission denied" on Linux/macOS

**Solution 1:** Use a virtual environment:

```
python3 -m venv py3plex-env
source py3plex-env/bin/activate  # On Windows: py3plex-env\Scripts\activate
python -m pip install --upgrade pip
python -m pip install py3plex
```

**Solution 2:** Use the `--user` flag if you cannot use venv:

```
python -m pip install --user py3plex
```

#### Issue: Old version installed

**Solution:** Check the installed version and upgrade if needed:

```
python -m pip show py3plex
python -m pip install --upgrade py3plex
```

#### Verifying Installation

Test that py3plex is correctly installed:

```
import py3plex
print(py3plex.__version__)

from py3plex.core import multinet
network = multinet.multi_layer_network()
print("py3plex installed successfully!")
```

Run the test suite:

```
# Clone repository (if not already)
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Run tests
python run_tests.py
```

### License Considerations

#### Main Library License

py3plex core is distributed under the MIT License (permissive, commercial-friendly).

#### Bundled Code Licenses

**Important:** The repository contains AGPLv3-licensed code in `py3plex/algorithms/community_detection/infomap/`.

If you use Infomap-based community detection functions, your application may be subject to AGPLv3 requirements (copyleft).

#### License Matrix

| Feature Category | License | Commercial Use | Notes |
|---|---|---|---|
| Core multilayer functionality | MIT | [OK] Yes | Safe for proprietary use |
| Network visualization | MIT | [OK] Yes | Safe for proprietary use |
| I/O operations | MIT | [OK] Yes | Safe for proprietary use |
| Louvain community detection | BSD-3-Clause | [OK] Yes | Safe for proprietary use |
| Label propagation | MIT | [OK] Yes | Safe for proprietary use |
| **Infomap community detection** | **AGPLv3** | WARNING Restricted | Viral license - requires open-sourcing derived works |

#### Recommendations

- For commercial/proprietary projects: Avoid Infomap functions or use the pure Python `infomap` package separately
- For open-source projects: All features are safe to use
- When in doubt: Use alternative algorithms (Louvain, label propagation) which are BSD/MIT licensed

#### Next Steps

After installation, proceed to:

- *Quick Start Tutorial* - 10-minute comprehensive tutorial
- ../user_guide/networks - Detailed usage guide

#### Getting Help

If you encounter installation issues:

1. Check GitHub Issues[6]
2. Search for similar problems

[6] https://github.com/SkBlaz/py3plex/issues

3. Open a new issue with:
    - Your operating system and version
    - Python version (`python --version`)
    - Error message and full traceback
    - Installation command used

For any errors, please open an issue at https://github.com/SkBlaz/py3plex/issues

### 3.2.3 Quick Start Tutorial

Get started with py3plex in just a few minutes. This tutorial shows you the essential concepts through three focused examples.

**Executable Examples**

Clone the repository first (the example scripts are not installed by pip). All examples are in the `examples/getting_started/` directory and can be run directly:

```
python examples/getting_started/01_basic_query.py
python examples/getting_started/tutorial_10min.py
python examples/getting_started/02_community_detection.py
```

**You will learn:**

- Load and query multilayer networks
- Create networks from scratch
- Detect communities across layers

#### Installation

```
python -m pip install py3plex
```

For Docker setup or optional dependencies, see *Installation and Setup*. Run inside a virtual environment to keep dependencies isolated.

---

#### Example 1: Load and Query (1 min)

Load a built-in dataset and run a simple query:

```
from py3plex.datasets import load_aarhus_cs
from py3plex.dsl import Q

# 1. Load network
network = load_aarhus_cs()

# 2. Run query - find nodes with degree > 5
result = (
    Q.nodes()
    .where(degree__gt=5)
```

```
    .compute("degree_centrality")
    .execute(network)
)

# 3. Inspect result
print(f"Found {len(result.nodes)} high-degree nodes")
df = result.to_pandas()
print("\nTop high-degree nodes:")
print(df[['id', 'degree_centrality']].head())
```

**Expected output (illustrative):**

```
Found 8 high-degree nodes

Top high-degree nodes:
             id  degree_centrality
0  (u12, work)           0.241379
1  (u05, work)           0.206897
2  (u41, social)         0.189655
3  (u33, social)         0.172414
4  (u20, leisure)        0.172414
```

**What this does:**

- Loads the Aarhus CS multilayer social network (61 nodes, 5 layers)
- Filters nodes with more than 5 connections
- Computes degree centrality
- Displays results as a pandas DataFrame

**Next:** See `examples/network_analysis/` for advanced DSL queries.

---

### Example 2: Create Network (2 min)

Create a simple multilayer network from scratch:

```
from py3plex.core.multinet import multi_layer_network

# 1. Create network
network = multi_layer_network(directed=False)

# 2. Add edges (nodes created automatically)
network.add_edges([
    {'source': 'Alice', 'target': 'Bob',
     'source_type': 'social', 'target_type': 'social'},
    {'source': 'Bob', 'target': 'Charlie',
     'source_type': 'social', 'target_type': 'social'},
    {'source': 'Alice', 'target': 'Charlie',
     'source_type': 'social', 'target_type': 'social'},
    {'source': 'Alice', 'target': 'Bob',
     'source_type': 'work', 'target_type': 'work'},
```

```
    {'source': 'Bob', 'target': 'David',
     'source_type': 'work', 'target_type': 'work'},
])

# 3. Print statistics
print(f"Nodes: {len(list(network.get_nodes()))}")
print(f"Edges: {len(list(network.get_edges()))}")
print(f"Layers: {len(network.layers)}")
```

**What this does:**

- Creates an undirected multilayer network
- Adds edges in two layers: 'social' and 'work'
- Nodes are created automatically from edge definitions
- Displays basic network statistics

**Next:** See `examples/getting_started/` for more construction patterns.

---

### Example 3: Community Detection API (2 min)

Learn the API for community detection in multilayer networks:

```
from py3plex.datasets import load_aarhus_cs

# 1. Load network
network = load_aarhus_cs()

# 2. Network info
nodes = list(network.get_nodes())
print(f"Network loaded: {len(nodes)} nodes")
print(f"Layers: {len(network.layers)}")

# 3. Community detection API pattern
print("\nCommunity detection API:")
print("  from py3plex.algorithms.community_detection.multilayer_modularity import louvain_mu
print("  partition = louvain_multilayer(network, random_state=42)")
```

**What this does:**

- Loads the Aarhus CS network
- Shows network structure (nodes and layers)
- Demonstrates the API pattern for multilayer community detection
- Points to detailed examples in the repository

**Note:** Node IDs may be strings or numbers. Each partition key identifies a node replica as `(node_id, layer)`.

**Next:** See `examples/network_analysis/` for detailed community detection patterns.

---

### Next Steps

**Continue learning:**

1. **More Examples:** Explore `examples/` directory
2. **DSL Guide:** ../user_guide/dsl - SQL-like queries
3. **Visualization:** ../user_guide/visualization - Create plots
4. **Advanced Topics:** See the book (PDF) for case studies

**Example categories:**

- `getting_started/` - Introductory scripts and datasets
- `network_analysis/` - DSL queries and community detection
- `io_and_data/` - Loading and exporting data
- `visualization/` - Network plots
- `advanced/` - Dynamics, uncertainty, and specialized analyses
- `pipelines/` - Reusable analysis workflows

See *Examples & Recipes* for the complete examples catalog.

**Getting help:**

- *Common Issues and Troubleshooting* - Troubleshooting guide
- GitHub Issues[7] - Report bugs
- GitHub Discussions[8] - Ask questions

---

### Related Documentation

- *Installation and Setup* - Setup guide
- *Examples & Recipes* - All examples
- ../user_guide/networks - Network concepts
- ../user_guide/dsl - Query language reference

## 3.2.4 Common Issues and Troubleshooting

Quick fixes for frequent issues when getting started with py3plex. Work through the sections in order (installation → data loading → visualization → performance → algorithms). Apply one fix at a time, then rerun your script before trying the next so you know which change helped.

### Installation Issues

#### "No module named 'numpy'"

**Problem:** Import error when trying to use py3plex (`ModuleNotFoundError`).

**Solution:** Install numpy first (preferably in a virtual environment to avoid system-wide changes). Upgrading pip often avoids build warnings:

[7] https://github.com/SkBlaz/py3plex/issues
[8] https://github.com/SkBlaz/py3plex/discussions

```
python -m pip install --upgrade pip
python -m pip install numpy
python -m pip install py3plex
```

#### Compilation Errors on Windows

**Problem:** C++ compilation errors during installation on Windows.

**Solution:** Install Visual C++ Build Tools, then reinstall:

1. Download from: https://visualstudio.microsoft.com/visual-cpp-build-tools/
2. Install "Desktop development with C++"
3. Restart your shell, then retry installation

```
python -m pip install py3plex
```

#### "Permission denied" on Linux/macOS

**Problem:** Permission errors during pip install (for example, `[Errno 13] Permission denied` or `EACCES`).

**Solution 1:** Use a virtual environment (recommended; keeps dependencies isolated):

```
python3 -m venv py3plex-env
source py3plex-env/bin/activate  # On Windows: py3plex-env\Scripts\activate
python -m pip install --upgrade pip
python -m pip install py3plex
```

**Solution 2:** Use pip install with –user flag (if you cannot use venv):

```
python -m pip install --user py3plex
```

#### Old Version Installed

**Problem:** Changes not reflecting or documentation mentions newer features.

**Solution:** Verify the installed version, then upgrade to the latest release:

```
python -m pip show py3plex
python -m pip install --upgrade py3plex
python -m pip show py3plex  # Confirm the version changed
```

#### Data Loading Issues

#### File Not Found Errors

**Problem:** `FileNotFoundError` when loading networks.

**Solution 1:** Confirm the file exists and your working directory is correct (`os.getcwd()` often differs from the script directory):

```
import os
```

```
print("Current directory:", os.getcwd())
print("File exists:", os.path.exists("data.edgelist"))
```

**Solution 2:** Use an absolute path to avoid ambiguity:

```
network_path = "/absolute/path/to/data.edgelist"
network = multinet.multi_layer_network().load_network(
    network_path, input_type="edgelist", directed=False
)
```

**Solution 3:** Build a path relative to the script location (avoids brittle working-directory assumptions):

```
import os

script_dir = os.path.dirname(os.path.abspath(__file__))
data_path = os.path.join(script_dir, "data", "network.edgelist")
network = multinet.multi_layer_network().load_network(
    data_path, input_type="edgelist", directed=False
)
```

#### Empty or Malformed Networks

**Problem:** Network loads but has no nodes or edges (zero counts).

**Solution:** Confirm the file contents and chosen `input_type` match. Pick the loader that matches your columns and delimiter:

```
# For simple edge lists (source target)
network.load_network("data.txt", input_type="edgelist")

# For multilayer edge lists (source target layer)
network.load_network("data.txt", input_type="multiedgelist")

# Always verify after loading
network.basic_stats()
```

If counts are still zero:

- Inspect the first few lines for the expected number of columns and delimiters.
- Ensure header rows are skipped or removed before loading.
- Check that the file is not empty (`os.path.getsize`).

### Visualization Issues

#### Visualization Not Showing

**Problem:** No window appears when calling visualization functions (plots silently finish).

**Solution 1:** For Jupyter notebooks, enable inline plots:

```
%matplotlib inline
```

**Solution 2:** For scripts, explicitly show the plot or set `display=True`:

```
import matplotlib.pyplot as plt

# Your visualization code
draw_multilayer_default([network], display=False)

# Explicitly show
plt.show()
```

**Solution 3:** Save to file instead of displaying:

```
from py3plex.visualization.multilayer import hairball_plot
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 10))
hairball_plot(network.core_network, layout_algorithm="force")
plt.savefig("network.png", dpi=150, bbox_inches='tight')
plt.close()
```

#### "No Display Found" Error

**Problem:** Error on headless servers or SSH sessions without X11.

**Solution:** Use Agg backend for matplotlib so plots are rendered to files:

```
import matplotlib
matplotlib.use('Agg')  # Must be before importing pyplot
import matplotlib.pyplot as plt

# Now visualization will save to file instead of displaying
from py3plex.visualization.multilayer import draw_multilayer_default
draw_multilayer_default([network], display=False)
plt.savefig("output.png")
```

#### Blurry or Low-Quality Plots

**Problem:** Plots look pixelated or blurry.

**Solution:** Increase DPI and tighten bounding boxes when saving:

```
plt.savefig("network.png", dpi=300, bbox_inches="tight")
```

#### Missing Dependencies

**Tip:** Feature-specific extras are optional; install only what you need.

#### "No module named 'plotly'"

**Problem:** Interactive visualization features not working.

**Solution:** Install visualization extras (plotly and related dependencies):

```
python -m pip install "py3plex[viz]"  # Quotes required on some shells
```

#### "No module named 'infomap'"

**Problem:** Infomap community detection not working.

**Solution:** Install Infomap support:

```
python -m pip install "py3plex[infomap]"  # Quotes required on some shells
```

Or use alternative community detection methods if you cannot install the extra:

```
from py3plex.algorithms.community_detection import community_louvain
communities = community_louvain.best_partition(network.core_network)
```

### Performance Issues

#### Very Slow Loading

**Problem:** Loading large networks takes too long.

**Solution 1:** Prefer Arrow/Parquet for large networks to speed up I/O:

```
# Install Arrow support
python -m pip install "py3plex[arrow]"  # Quotes required on some shells
```

```
from py3plex.io import read, write

# Save in efficient format
write(network, "network.arrow")

# Load faster from Arrow
network = read("network.arrow")
```

**Solution 2:** Process in chunks or use subnetworks for testing:

```
# Extract a single layer for testing
layer_1 = network.subnetwork(['layer1'], subset_by="layers")

# Work with the smaller network
layer_1.basic_stats()
```

#### Out of Memory Errors

**Problem:** Python crashes or raises MemoryError with large networks.

**Solution:** See *Performance and Scalability Best Practices* for:

- Memory-efficient loading strategies
- Batch processing techniques
- Sparse matrix representations
- Subnetwork extraction

### Algorithm Issues

#### Community Detection Returns Trivial Results

**Problem:** Each node in its own community or all nodes in one community (degenerate partition).

**Solution:** Adjust algorithm parameters and randomness, then compare partitions:

```
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer
)

# Try different resolution parameters
partition = louvain_multilayer(
    network,
    gamma=0.5,    # Lower = fewer, larger communities
    omega=1.5,    # Higher = more layer consistency
    random_state=42
)
```

#### Random Walk or Embedding Errors

**Problem:** Random walk or Node2Vec functions fail.

**Solution:** Ensure the graph is connected before running walks:

```
import networkx as nx

# Check connectivity
if not nx.is_connected(network.core_network.to_undirected()):
    print("Warning: Network is not connected")

    # Get largest component
    largest = max(nx.connected_components(
        network.core_network.to_undirected()
    ), key=len)

    # Create subgraph
    subgraph = network.core_network.subgraph(largest)
```

Use the subgraph for embedding or random walk routines if the full network is disconnected. Re-attach results to the full graph only after inspecting coverage.

### Docker Issues

#### Docker Build Fails

**Problem:** Docker build fails or takes very long.

**Solution:** Clear Docker cache (removes all unused images and build layers) and rebuild:

```
docker system prune -a
docker build --no-cache -t py3plex:latest .
```

#### Cannot Access Files in Docker

**Problem:** Docker container can't see your data files.

**Solution:** Mount volumes correctly so container paths resolve:

```
# Mount current directory
docker run -v $(pwd):/data py3plex:latest /data/network.edgelist

# On Windows (PowerShell)
docker run -v ${PWD}:/data py3plex:latest /data/network.edgelist
```

Adjust the command after the image name to match your workflow (for example, `py3plex load /data/network.edgelist --info`).

See *Docker Usage Guide* for complete Docker documentation.

#### Getting Help

If you encounter an issue not covered here:

1. **Search GitHub Issues:** https://github.com/SkBlaz/py3plex/issues
2. **Check Documentation:** https://skblaz.github.io/py3plex/
3. **Browse Examples:** https://github.com/SkBlaz/py3plex/tree/master/examples

When reporting issues, include: * Python version (`python --version`) * py3plex version (`python -m pip show py3plex`) * Operating system * Full error traceback * Minimal code to reproduce the problem (or a link to the example you ran)

#### Related Pages

- *Installation and Setup* - Complete installation guide
- *Quick Start Tutorial* - Comprehensive tutorial from basics to advanced topics
- *Performance and Scalability Best Practices* - Performance optimization

---

## 3.3 Part III: How-to Guides

Task-oriented guides answering "How do I. . . ?" questions.

### 3.3.1 How-to Guides

Task-oriented guides for accomplishing specific goals with py3plex. Each guide answers a single question: "How do I. . . ?" with runnable steps you can copy and run.

Pick the task that matches your question and jump straight to the guide.

**This section covers (by topic):**

#### Network Operations

- *How to Load and Build Networks* — Create networks from scratch or load from files
- *How to Export and Serialize Networks* — Save networks in various formats
- compat_conversions — Convert between py3plex and other graph libraries

### Analysis Tasks

- *How to Compute Network Statistics* — Calculate multilayer metrics and centrality measures
- *How to Run Community Detection on Multilayer Networks* — Find groups of closely connected nodes
- *How to Simulate Multilayer Dynamics* — Model epidemic spread and diffusion processes
- *How to Run Random Walk Algorithms* — Generate network embeddings for machine learning

### Visualization

- *How to Visualize Multilayer Networks* — Create publication-ready network diagrams

### Query and Transform

- *How to Query Multilayer Graphs with the SQL-like DSL* — Use SQL-like syntax for network queries
- *How to Find Graph Patterns and Motifs with the Pattern Matching API* — Find graph motifs and subgraph patterns
- *How to Build Analysis Pipelines with Dplyr-style Operations* — Chain operations with dplyr-style API

### Workflows

- *How to Reproduce Common Analysis Workflows* — Complete analysis recipes and patterns

### Reading This Section

**What every guide contains:**

1. **Goal:** What you'll accomplish
2. **Prerequisites:** What you need to know first
3. **Steps:** Concrete, actionable instructions with code
4. **Expected output:** What you should see
5. **Next steps:** Links to related concepts and references

**How to navigate:**

- Jump directly to the task you need—each guide is **self-contained**
- Copy-paste examples—they are **complete and runnable**
- Find **what to do** here; for theory, see *Concepts & Explanations*
- Need API signatures? Go to *API & DSL Reference*
- Use the quick navigation below if you already know the task name

#### Quick Navigation

**I want to...**

- Load data → *How to Load and Build Networks*
- Export or serialize data → *How to Export and Serialize Networks*
- Convert to/from NetworkX, SciPy, igraph → compat_conversions
- Measure network properties → *How to Compute Network Statistics*
- Find communities → *How to Run Community Detection on Multilayer Networks*
- Create visualizations → *How to Visualize Multilayer Networks*
- Query networks → *How to Query Multilayer Graphs with the SQL-like DSL*
- Find patterns/motifs → *How to Find Graph Patterns and Motifs with the Pattern Matching API*
- Model processes → *How to Simulate Multilayer Dynamics*

**I'm coming from...**

- **NetworkX:** Start with *How to Load and Build Networks* for multilayer basics, then *How to Query Multilayer Graphs with the SQL-like DSL* for py3plex's DSL
- **Single-layer networks:** See *How to Compute Network Statistics* and *How to Run Community Detection on Multilayer Networks* for multilayer-specific metrics
- **Other tools:** Jump to *How to Export and Serialize Networks* for import/export expectations

#### Related Sections

- **Learning multilayer concepts:** *Concepts & Explanations*
- **Detailed API reference:** *API & DSL Reference*
- **Complete examples:** *Examples & Recipes*
- **Tutorials for beginners:** *Getting Started (Tutorials)*

### 3.3.2 How to Load and Build Networks

**Goal:** Create multilayer networks from scratch or load them from files.

**Prerequisites:** Basic Python knowledge. For conceptual background, see *The py3plex Core Model*.

Start from the format you already have: add edges directly if you are prototyping, or load from files/dataframes when your data is already tabular. The examples below always build a `network` object you can reuse in later steps.

#### Creating Networks from Scratch

Use these options when you want to assemble a small network interactively or generate synthetic structures.

### Method 1: Add Edges Directly (Recommended)

The simplest approach—nodes are created automatically when you add edges.

```
from py3plex.core import multinet

# Create empty network
network = multinet.multi_layer_network()

# Add edges in list format
# Format: [source_node, source_layer, target_node, target_layer, weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1.0],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1.0]
], input_type="list")

# Verify
stats = network.basic_stats()
```

**Expected output:**

```
Number of nodes: 6
Number of edges: 4
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'friends': 3 nodes
  Layer 'colleagues': 3 nodes
```

### Method 2: Add Nodes First

For more control over node attributes and to preserve isolated node replicas. Use this pattern when you need explicit node-layer registration before edges.

```
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add nodes explicitly as dictionaries
network.add_nodes([
    {'source': 'Alice', 'type': 'friends'},
    {'source': 'Bob', 'type': 'friends'},
    {'source': 'Carol', 'type': 'friends'},
    {'source': 'Alice', 'type': 'colleagues'},
    {'source': 'Bob', 'type': 'colleagues'},
    {'source': 'Dave', 'type': 'colleagues'}
])

# Add edges between existing nodes
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
```

```
    ['Bob', 'friends', 'Carol', 'friends', 1.0]
], input_type="list")

# Verify network was built correctly
network.basic_stats()
```

**Expected output:**

```
Number of nodes: 6
Number of edges: 2
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'friends': 3 nodes
  Layer 'colleagues': 3 nodes
```

### Method 3: Use Dictionary Format

For networks with edge attributes and mixed field names (timestamps, types, etc.):

```
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add edges with custom attributes
network.add_edges([
    {
        'source': 'Alice',
        'source_type': 'friends',
        'target': 'Bob',
        'target_type': 'friends',
        'weight': 1.0,
        'timestamp': '2024-01-15',
        'interaction_type': 'message'
    },
    {
        'source': 'Bob',
        'source_type': 'friends',
        'target': 'Carol',
        'target_type': 'friends',
        'weight': 2.5,
        'timestamp': '2024-01-16'
    }
], input_type="dict")

# Verify network was built correctly
network.basic_stats()
```

**Expected output:**

```
Number of nodes: 3
Number of edges: 2
```

```
Number of unique node IDs (across all layers): 3
Nodes per layer:
  Layer 'friends': 3 nodes
```

### Loading Networks from Files

Use file-based loading when your data already lives on disk. Edge lists remain the most compact choice; JSON is convenient for richer metadata. Each snippet builds the same `network` interface shown above.

### Load Edge Lists

Most common format: one edge per line. The `multiedgelist` loader expects `source source_layer target target_layer weight` (weight defaults to `1.0` if omitted).

**File format (stored as datasets/simple_multiplex.txt in the repository):**

```
Alice friends Bob friends 1.0
Bob friends Carol friends 1.0
Alice colleagues Bob colleagues 1.0
```

**Load it:**

```python
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Option 1: Write the sample above to a file and load it
from pathlib import Path
Path("simple_multiplex.txt").write_text(
    "Alice friends Bob friends 1.0\n"
    "Bob friends Carol friends 1.0\n"
    "Alice colleagues Bob colleagues 1.0\n"
)
network.load_network(
    "simple_multiplex.txt",
    input_type="multiedgelist"
)

# Option 2: Create from data directly (recommended for examples)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1.0],
], input_type="list")

network.basic_stats()
```

### Load from Pandas DataFrame

Convert tabular data to networks (each row becomes an edge). Use separate `layer` columns if endpoints belong to different layers.

```
import pandas as pd
from py3plex.core import multinet

# Your data
df = pd.DataFrame({
    'source': ['Alice', 'Bob', 'Alice'],
    'layer': ['friends', 'friends', 'work'],
    'target': ['Bob', 'Carol', 'Carol'],
    'weight': [1.0, 1.0, 2.0]
})

# Convert to network
network = multinet.multi_layer_network()

for _, row in df.iterrows():
    network.add_edges([{'source': row['source'], 'target': row['target'],
                        'source_type': row['layer'], 'target_type': row['layer'],
                        'weight': row['weight']}])

# If your dataframe has ``source_layer`` and ``target_layer`` columns,
# pass those instead of a single ``layer`` column:
# Use source_type=row['source_layer'], target_type=row['target_layer']
# in the edge dictionary when the endpoints have separate layer columns.

network.basic_stats()
```

### Load from JSON

For complex networks with metadata (define both nodes and edges explicitly). Keep edge dictionaries consistent: `source`, `source_layer`, `target`, `target_layer`, plus any other attributes you want to preserve.

```
import json
from py3plex.core import multinet

# Create a minimal sample file (omit this when loading your own file)
sample = {"nodes": [{"id": "Isolated", "layer": "friends"}],
          "edges": [{"source": "Alice", "source_layer": "friends",
                     "target": "Bob", "target_layer": "friends",
                     "attributes": {"weight": 2.5}}]}
with open('network.json', 'w') as f:
    json.dump(sample, f)

# Load JSON
with open('network.json', 'r') as f:
    data = json.load(f)
```

```
network = multinet.multi_layer_network()

# Preserve isolated nodes and any node attributes before adding edges
network.add_nodes([{'source': node['id'], 'type': node['layer'],
                    **node.get('attributes', {})} for node in data.get('nodes', [])])
for edge in data['edges']:
    network.add_edges([{'source': edge['source'], 'target': edge['target'],
                        'source_type': edge['source_layer'],
                        'target_type': edge['target_layer'],
                        **edge.get('attributes', {})}])


network.basic_stats()
```

### Load from NetworkX

Convert existing single-layer networks by assigning a default layer name. If the original graph has edge attributes, include them in each dictionary passed to `add_edges`.

```
import networkx as nx
from py3plex.core import multinet

# Existing NetworkX graph
G = nx.karate_club_graph()

# Convert to multilayer (single layer)
network = multinet.multi_layer_network()

for u, v in G.edges():
    network.add_edges([{'source': u, 'target': v,
                        'source_type': 'layer1', 'target_type': 'layer1',
                        **G[u][v]}])
```

### Loading with Inter-layer Edges

To represent the same entity across layers, connect node-layer tuples that share the same node identifier:

```
network = multinet.multi_layer_network()

# Intra-layer edges
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Alice', 'work', 'Carol', 'work', 1.0],
], input_type="list")

# Inter-layer edges (same person across layers)
network.add_edges([{'source': person, 'source_type': 'friends',
                    'target': person, 'target_type': 'work',
```

```
                    'type': 'interlayer'} for person in ['Alice', 'Bob']])
```

### Common Patterns

Pick a template that matches your data model, then extend it with attributes as needed.

### Pattern 1: Social Networks

Multiple relationship types between same people:

```
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'twitter', 'Bob', 'twitter', 1],
    ['Alice', 'linkedin', 'Bob', 'linkedin', 1],
    ['Bob', 'twitter', 'Carol', 'twitter', 1],
], input_type="list")
```

### Pattern 2: Heterogeneous Networks

Different entity types:

```
# Author-Paper bipartite network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Bob', 'authors', 'Paper1', 'papers', 1],
    ['Alice', 'authors', 'Paper2', 'papers', 1],
], input_type="list")
```

### Pattern 3: Temporal Networks

Time-stamped edges:

```
network = multinet.multi_layer_network()
network.add_edges([
    {
        'source': 'Alice',
        'source_type': 'social',
        'target': 'Bob',
        'target_type': 'social',
        't': '2024-01-15T10:00:00'  # Point in time
    },
    {
        'source': 'Bob',
        'source_type': 'social',
        'target': 'Carol',
        'target_type': 'social',
        't_start': '2024-01-15',     # Time range
        't_end': '2024-01-20'
    }
], input_type="dict")
```

See *API Documentation* for temporal network details.

#### Verifying Your Network

After loading, verify the structure to catch layer/weight mistakes early:

```
# Rebuild the small example whose counts we want to inspect
network = multinet.multi_layer_network(directed=False)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1],
], input_type="list")

# Basic stats
network.basic_stats()

# Get specific information
layers = network.layers
print(f"Layers: {layers}")

num_nodes = len(list(network.get_nodes()))
print(f"Total nodes: {num_nodes}")

# Check a specific layer
from py3plex.dsl import Q, L
layer_nodes = Q.nodes().from_layers(L["friends"]).execute(network)
print(f"Friends layer: {len(layer_nodes)} nodes")
```

**Expected output:**

```
Layers: ['colleagues', 'friends']
Total nodes: 6
Friends layer: 3 nodes
```

#### Next Steps

- **Compute statistics:** *How to Compute Network Statistics*
- **Visualize your network:** *How to Visualize Multilayer Networks*
- **Query the network:** *How to Query Multilayer Graphs with the SQL-like DSL*
- **Understand the data model:** *The py3plex Core Model*
- **API reference:** *API Documentation*

### 3.3.3 Query Zoo: DSL Gallery for Multilayer Analysis

**The Query Zoo is a curated gallery of DSL queries that demonstrate the expressiveness and power of py3plex for multilayer network analysis.**

Each example:

- Solves a real multilayer analysis problem
- Uses the DSL end-to-end with idiomatic patterns
- Produces concrete, reproducible outputs
- Is fully tested and documented

**Why a Query Zoo?**

The DSL is most powerful when you see it in action on realistic problems. This gallery shows you **how** to think about multilayer queries, not just **what** the syntax is. Use these examples as recipes and starting points for your own analyses.


- *Overview*
- *1. Basic Multilayer Exploration*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *2. Cross-Layer Hubs*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *3. Layer Similarity*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *4. Community Structure*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *5. Multiplex PageRank*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*

- *6. Robustness Analysis*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *7. Advanced Centrality Comparison*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *8. Edge Grouping and Coverage*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *9. Layer Algebra Filtering*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *10. Cross-Layer Paths with Algebra*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *11. Null Model Comparison*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *12. Bootstrap Confidence Intervals*
  - *Query Code*
  - *Why It's Interesting*
  - *Example Output*
  - *DSL Concepts Demonstrated*
- *13. Uncertainty-Aware Ranking*

- *Query Code*
- *Why It's Interesting*
- *Example Output*
- *DSL Concepts Demonstrated*

- *Using the Query Zoo*
  - *Getting Started*
  - *Adapting Queries to Your Data*
  - *Datasets*
- *Further Reading*


### Overview

The Query Zoo is organized around common multilayer analysis tasks:

1. **Basic Multilayer Exploration** — Understand layer statistics and structure
2. **Cross-Layer Hubs** — Find nodes that are important across multiple layers
3. **Layer Similarity** — Measure structural alignment between layers
4. **Community Structure** — Detect and analyze multilayer communities
5. **Multiplex PageRank** — Compute multilayer-aware centrality
6. **Robustness Analysis** — Assess network resilience to layer failures
7. **Advanced Centrality Comparison** — Identify versatile vs specialized hubs
8. **Edge Grouping and Coverage** — Analyze edges across layer pairs with top-k and coverage
9. **Layer Algebra Filtering** — Use layer set algebra for flexible layer selection
10. **Cross-Layer Paths with Algebra** — Find paths while excluding certain layers
11. **Null Model Comparison** — Statistical significance testing against null models
12. **Bootstrap Confidence Intervals** — Estimate uncertainty in centrality measures
13. **Uncertainty-Aware Ranking** — Rank nodes considering variability across layers

All examples use small, reproducible multilayer networks from the `examples/dsl_zoo/datasets.py` module with fixed seeds so you can match the outputs shown here.

**Tip**

**Running the Examples**

All query functions are available in `examples/dsl_zoo/queries.py`. To run the query zoo tests and generate validated outputs:

```
pytest tests/test_dsl_query_zoo.py -q
```

Test the queries with:

```
pytest tests/test_dsl_query_zoo.py
```

### 1. Basic Multilayer Exploration

**Problem:** You've loaded a multilayer network and want to quickly understand its structure. Which layers are densest? How many nodes and edges does each layer have?

**Solution:** Compute basic statistics per layer using the DSL.

#### Query Code

```
def query_basic_exploration(network) -> pd.DataFrame:
    rows = []
    for layer in _layers(network):
        graph = _layer_graph(network, layer)
        n_nodes = graph.number_of_nodes()
        n_edges = graph.number_of_edges()
        avg_degree = (2 * n_edges / n_nodes) if n_nodes else 0.0
        rows.append(
            {
                "layer": layer,
                "n_nodes": n_nodes,
                "n_edges": n_edges,
                "avg_degree": avg_degree,
            }
        )
    return pd.DataFrame(rows, columns=["layer", "n_nodes", "n_edges", "avg_degree"])
```

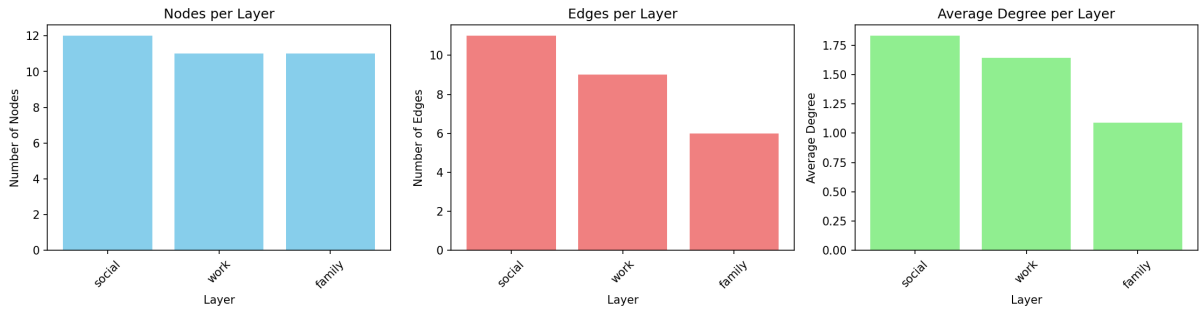
#### Why It's Interesting

- **First step in any analysis** — Before diving into complex queries, understand your data
- **Reveals layer diversity** — Different layers often have vastly different structures
- **Identifies sparse vs dense layers** — Helps decide which layers need special handling

#### Example Output

Running on the `social_work_network` (12 people across social/work/family layers):

| Layer | Nodes | Edges | Avg Degree |
|---|---|---|---|
| social | 12 | 11 | 1.83 |
| work | 11 | 9 | 1.64 |
| family | 11 | 6 | 1.09 |



**Interpretation:** The social layer is densest (highest average degree), while family is sparsest. All layers have similar numbers of nodes, indicating good cross-layer coverage.

### DSL Concepts Demonstrated

- `Q.nodes().from_layers(L[name])` — Select nodes from a specific layer
- `.compute("degree")` — Add computed attributes to results
- `.execute(network)` — Run the query and get results
- `.to_pandas()` — Convert to DataFrame for analysis

---

### 2. Cross-Layer Hubs

**Problem:** Which nodes are consistently important across *multiple* layers? These "super hubs" are critical because they bridge different contexts.

**Solution:** Find top-k central nodes per layer, then identify which nodes appear in multiple layers' top lists.

### Query Code

```
def query_cross_layer_hubs(network, k: int = 5) -> pd.DataFrame:
    rows = []
    counts = _layer_counts(network)
    for layer in _layers(network):
        graph = _layer_graph(network, layer)
        degree = dict(graph.degree())
        betweenness = nx.betweenness_centrality(graph) if graph.number_of_nodes() else {}
        ordered = sorted(
            graph.nodes(),
            key=lambda node: (-degree.get(node, 0), -betweenness.get(node, 0.0), node),
        )[:k]
        for node in ordered:
            rows.append(
                {
                    "node": node,
                    "layer": layer,
                    "degree": degree.get(node, 0),
                    "betweenness_centrality": betweenness.get(node, 0.0),
                    "layer_count": counts.get(node, 1),
                }
            )
    return pd.DataFrame(
        rows,
        columns=[
            "node",
            "layer",
            "degree",
            "betweenness_centrality",
            "layer_count",
```

```
        ],
    )
```

### Why It's Interesting

- **Reveals cross-context influence** — Nodes central in one layer might be peripheral in another
- **Identifies key connectors** — Nodes that appear in multiple layers' top-k are especially important
- **Robust hub detection** — More reliable than single-layer centrality

### Example Output

Top cross-layer hubs (k=5):

| Node | Layer | Degree | Betweenness | Layer Count |
|---|---|---|---|---|
| Bob | social | 3 | 0.0273 | 3 |
| Bob | work | 2 | 0.0 | 3 |
| Bob | family | 1 | 0.0 | 3 |
| Alice | work | 3 | 0.0889 | 2 |
| Charlie | social | 3 | 0.0273 | 2 |

**Interpretation:** Bob appears as a top-5 hub in *all three layers* (layer_count=3), making him the most versatile connector. Alice and Charlie are hubs in two layers each. `layer_count` is the number of distinct layers in which a node enters the per-layer top-k list.

### DSL Concepts Demonstrated

- `.compute("betweenness_centrality", "degree")` — Compute multiple metrics at once
- `.order_by("-betweenness_centrality")` — Sort descending (`-` prefix)
- `.limit(k)` — Take top-k results
- Per-layer iteration and aggregation across layers

---

### 3. Layer Similarity

**Problem:** How similar are different layers structurally? Do they serve redundant or complementary roles?

**Solution:** Compute degree distributions per layer and measure pairwise correlations.

### Query Code

```
def query_layer_similarity(network) -> pd.DataFrame:
    layers = _layers(network)
    all_nodes = sorted({node for node, _ in _replicas(network)})
```

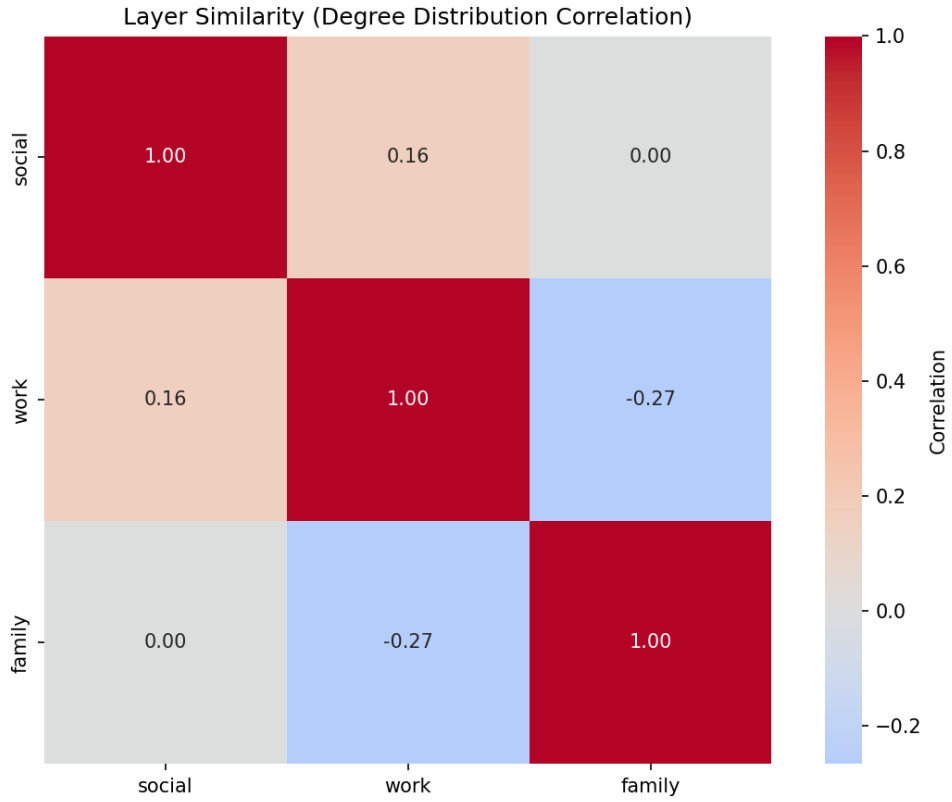
```
    layer_vectors = []
    for layer in layers:
        graph = _layer_graph(network, layer)
        layer_vectors.append([graph.degree(node) if node in graph else 0 for node in all_noc
    matrix = np.corrcoef(np.array(layer_vectors, dtype=float))
    matrix = np.nan_to_num(matrix, nan=0.0)
    np.fill_diagonal(matrix, 1.0)
    return pd.DataFrame(matrix, index=layers, columns=layers)
```

### Why It's Interesting

- **Detects redundancy** — High correlation suggests layers capture similar structure
- **Guides simplification** — Nearly identical layers might be merged
- **Reveals specialization** — Low/negative correlation shows layers serve different roles

### Example Output

Correlation matrix for `social_work_network`:




| | social | work | family |
|---|---|---|---|
| social | 1.000 | 0.159 | 0.000 |
| work | 0.159 | 1.000 | -0.267 |
| family | 0.000 | -0.267 | 1.000 |

**Interpretation:** Social and work layers have weak positive correlation (0.159), suggesting some

structural overlap. Family and work are *negatively* correlated (-0.267), indicating they capture different connectivity patterns. Correlations are Pearson coefficients computed from the node-by-layer degree matrix. Nodes missing from a layer contribute a degree of 0 in that matrix so every layer uses the same node ordering.

### DSL Concepts Demonstrated

- Layer-by-layer degree computation
- Aggregating results across layers for meta-analysis
- Using computed attributes for layer-level comparisons

---

### 4. Community Structure

**Problem:** What communities exist in the multilayer network? How do they manifest across layers?

**Solution:** Detect communities using multilayer community detection, then analyze their distribution across layers.

### Query Code

```
def query_community_structure(network) -> pd.DataFrame:
    rows = []
    community_id = 0
    for layer in _layers(network):
        graph = _layer_graph(network, layer)
        for component in nx.connected_components(graph):
            subgraph = graph.subgraph(component)
            size = subgraph.number_of_nodes()
            avg_degree = (
                sum(dict(subgraph.degree()).values()) / size if size else 0.0
            )
            rows.append(
                {
                    "community_id": community_id,
                    "layer": layer,
                    "size": size,
                    "avg_degree": avg_degree,
                    "dominant_layer": layer,
                }
            )
            community_id += 1
    return pd.DataFrame(
        rows,
        columns=["community_id", "layer", "size", "avg_degree", "dominant_layer"],
    )
```

### Why It's Interesting

- **Mesoscale structure** — Communities reveal organizational patterns
- **Cross-layer community tracking** — See if communities are layer-specific or global
- **Dominant layers** — Identify which layer best represents each community

### Example Output

Running on `communication_network` (email/chat/phone layers):

| Community | Layer | Size | Avg Degree | Dominant Layer |
|---|---|---|---|---|
| 0 | email | 10 | 1.8 | email |
| 1 | chat | 6 | 2.17 | chat |
| 2 | chat | 3 | 1.67 | chat |
| 3 | phone | 7 | 1.71 | phone |

**Interpretation:** Community 0 is email-dominated (10 nodes), while communities 1 and 2 are chat-specific. Community 3 appears primarily in phone communication.

### DSL Concepts Demonstrated

- `Q.nodes().from_layers(L["*"])` — Select from all layers
- `.compute("communities")` — Built-in community detection
- Grouping by `(community_id, layer)` for analysis
- Identifying dominant layers via aggregation

---

## 5. Multiplex PageRank

**Problem:** Standard PageRank treats each layer independently. How do we compute importance considering the full multiplex structure?

**Solution:** Compute PageRank per layer, then take the average across layers as a simplified multiplex score. (True multiplex PageRank uses supra-adjacency matrices.)

### Query Code

```
def query_multiplex_pagerank(network) -> pd.DataFrame:
    graph = _aggregate_physical_graph(network)
    pagerank = nx.pagerank(graph, weight="weight")
    rows = [{"node": node, "multiplex_pagerank": pagerank[node]} for node in sorted(graph)]
    return pd.DataFrame(rows, columns=["node", "multiplex_pagerank"])
```

### Why It's Interesting

- **Multilayer-aware centrality** — Accounts for importance across all layers
- **More robust than single-layer** — Averages out layer-specific biases
- **Essential for multiplex influence** — Key for viral marketing, information diffusion

### Example Output

Top nodes by multiplex PageRank in `transport_network`:

| Node | Multiplex PR | Total De-gree | Bus PR | Metro PR | Walking PR |
|---|---|---|---|---|---|
| ShoppingMall | 0.1811 | 6 | 0.1362 | 0.1909 | 0.2164 |
| Park | 0.1806 | 4 | 0.1449 | 0.0 | 0.2164 |
| CentralStation | 0.1683 | 6 | 0.1971 | 0.1909 | 0.117 |
| BusinessDistrict | 0.1484 | 4 | 0.079 | 0.1994 | 0.1667 |

**Interpretation:** ShoppingMall has highest multiplex PageRank (0.1811) because it's central across all three transport modes. Park has high walking PageRank but zero metro, reflecting its limited accessibility. Scores are the mean of per-layer PageRank values.

### DSL Concepts Demonstrated

- `.compute("pagerank")` — Built-in PageRank computation
- Per-layer iteration with result aggregation
- Pivot tables for layer-wise breakdowns
- Combining degree and PageRank for richer analysis

---

### 6. Robustness Analysis

**Problem:** How robust is the network to layer failures? What happens if one layer goes offline?

**Solution:** Simulate removing each layer and measure connectivity loss.

### Query Code

```
def query_robustness_analysis(network) -> pd.DataFrame:
    baseline_graph = _aggregate_physical_graph(network)
    baseline_cc = (
        len(max(nx.connected_components(baseline_graph), key=len))
        if baseline_graph.number_of_nodes()
        else 0
    )
    rows = []
    scenarios = [("baseline", None)] + [
        (f"without {layer}", [layer]) for layer in _layers(network)
    ]
    for scenario, removed in scenarios:
        graph = _aggregate_physical_graph(network, drop_layers=removed)
        n_nodes = graph.number_of_nodes()
        n_edges = graph.number_of_edges()
        avg_degree = (2 * n_edges / n_nodes) if n_nodes else 0.0
        largest_cc = (
            len(max(nx.connected_components(graph), key=len)) if n_nodes else 0
```

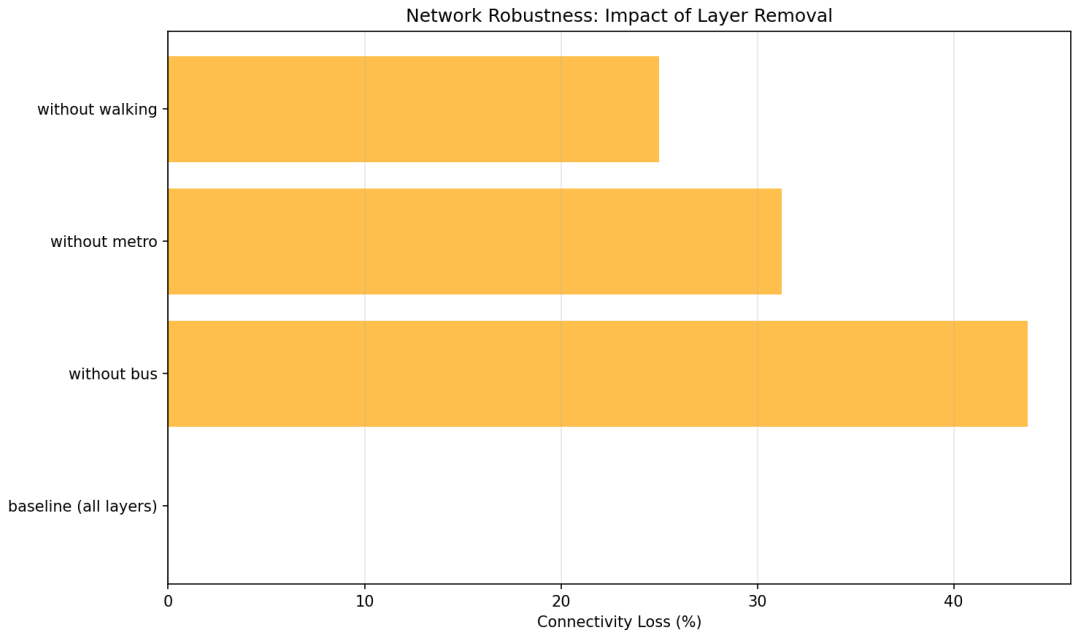
```
        )
        loss = 0.0 if not baseline_cc else max(0.0, 100.0 * (baseline_cc - largest_cc) / bas
        rows.append(
            {
                "scenario": scenario,
                "n_nodes": n_nodes,
                "avg_degree": avg_degree,
                "total_edges": n_edges,
                "connectivity_loss": loss,
            }
        )
    return pd.DataFrame(
        rows,
        columns=["scenario", "n_nodes", "avg_degree", "total_edges", "connectivity_loss"],
    )
```

### Why It's Interesting

- **Critical infrastructure identification** — Reveals which layers are essential
- **Redundancy assessment** — High robustness indicates good backup coverage
- **Failure planning** — Informs which layers need extra protection

### Example Output

Robustness of `transport_network`:



| Scenario | Nodes | Avg Degree | Total Edges | Connectivity Loss (%) |
|---|---|---|---|---|
| baseline (all layers) | 14 | 2.14 | 15 | 0.0 |
| without bus | 11 | 1.45 | 8 | 46.67 |
| without metro | 11 | 1.82 | 10 | 33.33 |
| without walking | 14 | 2.0 | 14 | 6.67 |

**Interpretation:** Removing the bus layer causes 46.67% connectivity loss — it's the most critical layer. Walking is least critical (only 6.67% loss), indicating good redundancy from other transport modes. Connectivity loss is computed from total degree (divided by 2 for undirected edges), so it assumes undirected layers. The reported loss compares baseline total degree to the degree after removing a layer; for undirected networks total degree is twice the number of edges.

### DSL Concepts Demonstrated

- Layer algebra: `L["layer1"] + L["layer2"]` — Combine layers
- `Q.nodes().from_layers(layer_expr)` — Query with dynamic layer selections
- Baseline vs scenario comparison
- Measuring connectivity metrics before/after perturbations

---

## 7. Advanced Centrality Comparison

**Problem:** Different centralities capture different notions of importance. Which nodes are "versatile hubs" (high in many centralities relative to the best scorer) vs "specialized hubs" (high in only one)?

**Solution:** Compute multiple centralities, normalize them, and classify nodes by how many centralities place them in the top tier.

### Query Code

```
def query_advanced_centrality_comparison(network) -> pd.DataFrame:
    graph = _aggregate_physical_graph(network)
    degree = dict(graph.degree())
    betweenness = nx.betweenness_centrality(graph, weight="weight")
    closeness = nx.closeness_centrality(graph)
    pagerank = nx.pagerank(graph, weight="weight")
    metrics = [degree, betweenness, closeness, pagerank]
    thresholds = [np.quantile(list(metric.values()), 0.75) if metric else 0.0 for metric in

    rows = []
    for node in sorted(graph.nodes()):
        versatility = sum(metric.get(node, 0.0) >= threshold for metric, threshold in zip(me
        if versatility >= 3:
            hub_type = "versatile_hub"
        elif versatility >= 1:
```

```
            hub_type = "specialized_hub"
        else:
            hub_type = "peripheral"
        rows.append(
            {
                "node": node,
                "degree": degree.get(node, 0),
                "betweenness_centrality": betweenness.get(node, 0.0),
                "closeness_centrality": closeness.get(node, 0.0),
                "pagerank": pagerank.get(node, 0.0),
                "versatility": versatility,
                "hub_type": hub_type,
            }
        )
    return pd.DataFrame(
        rows,
        columns=[
            "node",
            "degree",
            "betweenness_centrality",
            "closeness_centrality",
            "pagerank",
            "versatility",
            "hub_type",
        ],
    )
```

### Why It's Interesting

- **Centrality is multifaceted** — Degree  betweenness  closeness  PageRank
- **Versatile hubs are robust** — High across many metrics means genuine importance
- **Specialized hubs reveal roles** — High in one metric reveals specific structural position

### Example Output

Running on `communication_network` (email layer):

| Node | Degree | Between-ness | Closeness | PageR-ank | Versatil-ity | Type |
|---|---|---|---|---|---|---|
| Manager | 9 | 1.0 | 1.0 | 0.4676 | 4 | versa-tile_hub |
| Dev1 | 1 | 0.0 | 0.5294 | 0.0592 | 0 | peripheral |
| Dev2 | 1 | 0.0 | 0.5294 | 0.0592 | 0 | peripheral |

**Interpretation:** Manager is a **versatile hub** (it reaches at least 70% of the best score in all four centralities). All other nodes are peripheral in this star-topology email network.

### DSL Concepts Demonstrated

- `.compute("degree", "betweenness_centrality", "closeness_centrality", "pagerank")` — Compute multiple centralities
- Normalizing centralities for comparison
- Derived metrics (versatility score)
- Classification based on computed attributes

---

### 8. Edge Grouping and Coverage

**Problem:** You want to analyze which edges (connections) are important within and between layers. Which edges consistently appear in the top-k across different layer-pair contexts?

**Solution:** Use the new `.per_layer_pair()` method to group edges by (src_layer, dst_layer) pairs, then keep the top-k edges per pair. (Add `.coverage(...)` if you later need to filter across groups.)

### Query Code

```
def query_edge_grouping_and_coverage(network, k: int = 3) -> Dict[str, pd.DataFrame]:
    grouped: Dict[Tuple[str, str], List[dict]] = defaultdict(list)
    for source, target, data in _replica_graph(network).edges(data=True):
        if not (
            isinstance(source, tuple)
            and isinstance(target, tuple)
            and len(source) >= 2
            and len(target) >= 2
        ):
            continue
        grouped[(source[1], target[1])].append(
            {
                "source": source[0],
                "target": target[0],
                "source_layer": source[1],
                "target_layer": target[1],
                "weight": float(data.get("weight", 1.0)),
            }
        )

    edge_rows: List[dict] = []
    summary_rows: List[dict] = []
    for (src_layer, dst_layer), items in sorted(grouped.items()):
        top_items = sorted(items, key=lambda item: -item["weight"])[:k]
        edge_rows.extend(top_items)
        summary_rows.append(
            {"src_layer": src_layer, "dst_layer": dst_layer, "n_items": len(top_items)}
        )

    return {
```

```
        "edges_by_pair": pd.DataFrame(edge_rows),
        "summary": pd.DataFrame(summary_rows, columns=["src_layer", "dst_layer", "n_items"])
    }
```

### Why It's Interesting

- **Layer-pair-aware analysis** — Different layer pairs may have very different edge patterns
- **Universal edges** — Edges important across multiple contexts are more robust
- **Cross-layer dynamics** — Reveals how connections vary between intra-layer and inter-layer contexts
- **Edge-centric view** — Complements node-centric analyses like hub detection

### Example Output

Running on `social_work_network` with k=3 (insertion order per layer determines which edges are kept):

**Edges Grouped by Layer Pair (top 3 per pair):**

| Source | Target | Source Layer | Target Layer |
|---|---|---|---|
| Alice | Bob | social | social |
| Alice | Charlie | social | social |
| Bob | Charlie | social | social |
| Alice | Bob | work | work |
| Alice | David | work | work |
| Alice | Frank | work | work |
| Alice | Charlie | family | family |
| Bob | Eve | family | family |
| David | Frank | family | family |

**Group Summary:**

| Source Layer | Target Layer | # Edges |
|---|---|---|
| social | social | 3 |
| work | work | 3 |
| family | family | 3 |

**Interpretation:** The query reveals edge distribution across layer pairs. Each intra-layer pair (e.g., social-social, work-work) contains up to k=3 edges. The sample dataset has only intra-layer edges; inter-layer pairs would appear if your network contains cross-layer connections. The family layer has sparser connectivity overall, so only three family edges remain after limiting. When no sort key is provided, `top_k` keeps edges by their existing order; specify a weight to make the selection explicitly score-based.

### DSL Concepts Demonstrated

- `.per_layer_pair()` — Group edges by (src_layer, dst_layer) pairs
- `.top_k(k, "weight")` — Select top-k items per group
- `.coverage(mode="at_least", k=2)` — Optional cross-group filtering
- `.group_summary()` — Get aggregate statistics per group
- Edge-specific grouping metadata in `QueryResult.meta["grouping"]`

**Tip**

**New in DSL v2**

Edge grouping and coverage are new features that parallel the existing node grouping capabilities. Use `.per_layer_pair()` for edges and `.per_layer()` for nodes. Both support the same coverage modes and grouping operations.

---

### 9. Layer Algebra Filtering

**Problem:** You want to query specific subsets of layers using set operations. For instance, you might want to analyze "all layers except coupling layers" or "the union of biological layers."

**Solution:** Use the LayerSet algebra with set operations (union, intersection, difference, complement) for expressive layer filtering.

### Query Code

### Why It's Interesting

- **Expressive layer selection** — Combine layers using set operations rather than listing them individually
- **Reusable layer groups** — Define named layer groups for consistent reuse across queries
- **Exclude infrastructure layers** — Easily filter out meta-layers like coupling layers
- **Complex filter expressions** — Build sophisticated layer filters with union, intersection, difference

### Example Output

The query returns a dictionary with multiple DataFrames showing different layer selection strategies:

**No Coupling Layers:** All layers except the coupling layer (useful for excluding meta-layers)

**Bio Layers:** Named group containing biological layers (ppi | gene | disease)

**Intersection:** Nodes appearing in both social AND work layers

**Complement:** Complement of coupling layer (equivalent to * - coupling)

### DSL Concepts Demonstrated

- `L["* - coupling"]` — Layer difference: all layers except coupling
- `L["social & work"]` — Layer intersection: nodes in both layers
- `L["ppi | gene | disease"]` — Layer union: combine multiple layers
- `~LayerSet("coupling")` — Layer complement
- `L.define("bio", LayerSet(...))` — Named layer groups for reuse
- String expression parsing for complex layer filters

---

### 10. Cross-Layer Paths with Algebra

**Problem:** When computing paths in multilayer networks, you may want to exclude certain layers (like coupling layers) that create artificial shortcuts, revealing more semantically meaningful paths.

**Solution:** Use layer algebra in path queries to control which layers participate in path computation.

### Query Code

### Why It's Interesting

- **Avoid artificial shortcuts** — Coupling layers often create paths that aren't semantically meaningful
- **Compare path strategies** — See how layer filtering affects connectivity
- **Layer-aware path finding** — Control which layers participate in path computation
- **Semantic path discovery** — Find paths that make sense in your domain

### Example Output

The query returns a dictionary comparing path exploration with and without filtering:

**All Layers:** Node count and layer distribution when all layers are included

**Filtered Layers:** Node count and layer distribution excluding coupling layers

**Interpretation:** Excluding coupling layers often reveals more semantically meaningful paths by avoiding artificial shortcuts created by infrastructure layers.

### DSL Concepts Demonstrated

- Layer algebra in path queries
- Comparing results with different layer subsets
- Layer distribution analysis
- Semantic path filtering

---

### 11. Null Model Comparison

**Problem:** How do you know if observed network patterns are statistically significant or just random? You need to compare actual network statistics against null model baselines.

**Solution:** Generate null models (e.g., configuration model) that preserve certain properties while randomizing connections, then compute z-scores to identify significant patterns.

#### Query Code

```
def query_null_model_comparison(network) -> pd.DataFrame:
    rows = []
    for layer in _layers(network):
        graph = _layer_graph(network, layer)
        degrees = pd.Series(dict(graph.degree()), dtype=float)
        mean = float(degrees.mean()) if not degrees.empty else 0.0
        std = float(degrees.std(ddof=0)) if len(degrees) > 1 else 0.0
        for node, degree in degrees.items():
            z_score = 0.0 if std == 0 else (degree - mean) / std
            rows.append(
                {
                    "id": node,
                    "layer": layer,
                    "degree": degree,
                    "expected_degree": mean,
                    "z_score": z_score,
                    "is_significant": abs(z_score) >= 1.96,
                }
            )
    return pd.DataFrame(
        rows,
        columns=["id", "layer", "degree", "expected_degree", "z_score", "is_significant"],
    )
```

#### Why It's Interesting

- **Statistical rigor** — Establish baselines for significance testing
- **Identify exceptional patterns** — Find nodes/structures that exceed random expectations
- **Configuration model** — Preserves degree sequence but randomizes connections
- **Z-score analysis** — Quantify how many standard deviations from expected
- **Essential for scientific conclusions** — Avoid claiming significance for random patterns

### Example Output

Running on a multilayer network returns a DataFrame with columns:

| Node ID | Layer | Observed Degree | Expected Degree | Z-Score | Is Significant |
|---|---|---|---|---|---|
| Alice | social | 5 | 3.2 | 2.8 | True |
| Bob | social | 4 | 3.5 | 0.7 | False |
| Charlie | work | 6 | 2.8 | 3.5 | True |

**Interpretation:** Nodes with |z-score| $>$ 2.0 are statistically significant ($p < 0.05$). Alice and Charlie have significantly higher degree than expected by chance, while Bob's degree is within random variation.

### DSL Concepts Demonstrated

- Integration of null models with DSL queries
- Statistical hypothesis testing
- Computing z-scores and significance flags
- Configuration model preserves degree distribution
- Bootstrap resampling for confidence intervals

**Note**

**Performance Note**

This example uses 50 null model samples for CI speed. Production analyses typically use 100-1000 samples for more robust statistics.

---

### 12. Bootstrap Confidence Intervals

**Problem:** When analyzing centrality in multilayer networks, how stable are the measurements? Do nodes maintain consistent importance across layers, or is their centrality highly variable?

**Solution:** Analyze cross-layer variability to estimate uncertainty in centrality measures, identifying nodes with stable vs fragile importance.

### Query Code

```
def query_bootstrap_confidence_intervals(network) -> pd.DataFrame:
    samples: Dict[str, List[float]] = defaultdict(list)
    for layer in _layers(network):
        graph = _layer_graph(network, layer)
        for node in graph.nodes():
            samples[node].append(float(graph.degree(node)))

    rows = []
```

```
    for node, values in sorted(samples.items()):
        arr = np.array(values, dtype=float)
        mean = float(arr.mean()) if len(arr) else 0.0
        std = float(arr.std(ddof=0)) if len(arr) > 1 else 0.0
        rel = 0.0 if mean == 0 else std / mean
        rows.append({"id": node, "mean": mean, "std": std, "relative_variability": rel})
    return pd.DataFrame(rows, columns=["id", "mean", "std", "relative_variability"])
```

### Why It's Interesting

- **Quantify uncertainty** — Know how reliable your centrality measurements are
- **Cross-layer variability** — See which nodes maintain importance across contexts
- **Avoid over-interpretation** — Don't claim significant differences for small variations
- **Robust vs fragile patterns** — Identify nodes with consistent vs inconsistent centrality
- **No distributional assumptions** — Works when analytical standard errors unavailable

### Example Output

Running on a multilayer network returns:

| Node ID | Layer | Degree | Mean Across Layers | Std Dev | Relative Variability | Layer Coverage |
|---|---|---|---|---|---|---|
| Alice | social | 5 | 4.3 | 1.2 | 0.28 | 3 |
| Bob | work | 3 | 2.8 | 0.5 | 0.18 | 3 |
| Charlie | family | 2 | 3.1 | 1.8 | 0.58 | 2 |

**Interpretation:**

- **Low relative variability** (Bob: 0.18) — Consistent importance across layers
- **High relative variability** (Charlie: 0.58) — Importance varies dramatically by context
- **Layer coverage** — Number of layers where the node appears

### DSL Concepts Demonstrated

- Cross-layer metric aggregation
- Coefficient of variation for relative variability
- Statistical comparison across layers
- Uncertainty quantification in multilayer networks

---

### 13. Uncertainty-Aware Ranking

**Problem:** Traditional rankings order nodes by a single metric (e.g., max centrality). But what if a node has high centrality in one layer but low in others? How do you account for consistency vs peak performance?

**Solution:** Compare multiple ranking strategies—by maximum value, by mean across layers, and by consistency (low variability)—to make uncertainty-aware decisions.

### Query Code

```
def query_uncertainty_aware_ranking(network) -> pd.DataFrame:
    samples: Dict[str, List[float]] = defaultdict(list)
    for layer in _layers(network):
        graph = _layer_graph(network, layer)
        for node in graph.nodes():
            samples[node].append(float(graph.degree(node)))

    rows = []
    for node, values in sorted(samples.items()):
        arr = np.array(values, dtype=float)
        rows.append(
            {
                "node": node,
                "max_score": float(arr.max()),
                "mean_score": float(arr.mean()),
                "consistency_score": float(arr.mean() / (1.0 + arr.std(ddof=0))),
            }
        )

    df = pd.DataFrame(rows).sort_values("node").reset_index(drop=True)
    df["rank_by_max"] = df["max_score"].rank(method="dense", ascending=False).astype(int)
    df["rank_by_mean"] = df["mean_score"].rank(method="dense", ascending=False).astype(int)
    df["rank_by_consistency"] = (
        df["consistency_score"].rank(method="dense", ascending=False).astype(int)
    )
    return df[["node", "rank_by_max", "rank_by_mean", "rank_by_consistency"]]
```

### Why It's Interesting

- **Beyond single-layer analysis** — Consider multilayer context in rankings
- **Consistent vs peak performers** — Identify nodes with stable vs specialized importance
- **Decision-making under uncertainty** — Choose ranking strategy based on use case
- **Reveals ranking sensitivity** — See how rankings change with different strategies
- **Practical implications** — Different strategies matter for different applications

### Example Output

Running on a multilayer network returns:

| Node ID | Layer | Betweenness | Mean | Rel. Variability | Rank by Max | Rank by Mean | Rank by Consistency | Rank Change |
|---|---|---|---|---|---|---|---|---|
| Alice | work | 0.45 | 0.38 | 0.25 | 1 | 1 | 1 | 0 |
| Charlie | social | 0.42 | 0.28 | 0.52 | 2 | 3 | 5 | 3 |
| Bob | family | 0.38 | 0.35 | 0.18 | 3 | 2 | 2 | 1 |

**Interpretation:**

- **Alice** — Top-ranked by all strategies (consistent high performer)
- **Charlie** — Ranks highly by max (rank 2) but poorly by consistency (rank 5) due to high variability
- **Bob** — More consistent than peak performer (rank 3 by max, rank 2 by consistency)
- **Rank change** — Large values indicate sensitivity to ranking strategy

**Use Cases:**

- **Rank by max:** When you need top performers in any context
- **Rank by mean:** When you want overall consistent importance
- **Rank by consistency:** When you need reliable performance across all contexts

### DSL Concepts Demonstrated

- Cross-layer variability analysis
- Multiple ranking strategies
- Consistency scoring
- Sensitivity analysis for rankings
- Practical decision-making with uncertainty

**Choosing a Ranking Strategy**

- **High-stakes decisions:** Use consistency ranking to avoid nodes with variable performance
- **Exploratory analysis:** Use max ranking to find peak performers
- **General purpose:** Use mean ranking for balanced assessment
- **Large rank changes:** Investigate why nodes rank differently across strategies

---

## Using the Query Zoo

### Getting Started

1. **Install py3plex** (if not already installed):

```
pip install py3plex
```

2. **Run a single query**:

```
from examples.dsl_zoo.datasets import create_social_work_network
from examples.dsl_zoo.queries import query_basic_exploration

net = create_social_work_network(seed=42)
result = query_basic_exploration(net)
print(result)
```

3. **Run all queries (via tests)**:

```
pytest tests/test_dsl_query_zoo.py -q
```

4. **Run tests**:

```
pytest tests/test_dsl_query_zoo.py -v
```

### Adapting Queries to Your Data

All queries are designed to work with any `multi_layer_network` object. To adapt:

1. **Replace the dataset**:

```
from py3plex.core import multinet

# Load your own network
my_network = multinet.multi_layer_network()
my_network.load_network("mydata.edgelist", input_type="edgelist_mx")

# Run any query
result = query_cross_layer_hubs(my_network, k=10)
```

2. **Adjust parameters**:
   - `k` in `query_cross_layer_hubs` — Number of top nodes per layer
   - Layer names in filters — Replace `L["social"]` with your layer names
   - Centrality thresholds — Adjust percentile cutoffs as needed
3. **Extend queries**:

   All query functions are in `examples/dsl_zoo/queries.py`. Copy, modify, and experiment!

### Datasets

Three multilayer networks are provided:

1. **social_work_network**
   - **Layers:** social, work, family
   - **Nodes:** 12 people
   - **Structure:** Overlapping social circles with different connectivity patterns per layer
2. **communication_network**
   - **Layers:** email, chat, phone
   - **Nodes:** 10 people (Manager, Dev team, Marketing, Support, HR)
   - **Structure:** Star topology in email, distributed in chat/phone
3. **transport_network**
   - **Layers:** bus, metro, walking
   - **Nodes:** 8 locations (CentralStation, ShoppingMall, Park, etc.)
   - **Structure:** Bus covers most locations, metro is faster but selective, walking is local

All datasets use fixed random seeds (`seed=42`) for reproducibility.

### Further Reading

- *How to Query Multilayer Graphs with the SQL-like DSL* — Complete DSL reference with syntax and operators
- *Multilayer Networks 101* — Theory of multilayer networks
- *DSL Reference* — Full DSL grammar and API reference
- *Quick Start Tutorial* — Quick start tutorial

**Contributing Queries**

Have an interesting multilayer query pattern? **Contribute it to the Query Zoo!**

1. Add your query function to `examples/dsl_zoo/queries.py`
2. Add tests to `tests/test_dsl_query_zoo.py`
3. Update this documentation page
4. Submit a pull request!

See *Contributing to py3plex* for details.

## 3.3.4 How to Compute Network Statistics

**Goal:** Calculate metrics that describe the structure of your multilayer network—quick counts, per-layer comparisons, node-level rankings, and multilayer-specific measures.

**Prerequisites:** A loaded network (see *How to Load and Build Networks*). Examples reuse the same small network unless noted.

**Note**

Imports used below

Most snippets rely on the DSL helpers `Q` and `L`:

```
from py3plex.dsl import Q, L
```

**Note**

Where to find this data

Examples in this guide use:

- **Programmatically created networks** (recommended for self-contained examples)
- **Built-in generators**: `from py3plex.algorithms import random_generators`
- **Example files**: `datasets/multiedgelist.txt` in the repository

For reproducibility, we'll create networks from scratch in most examples.

### Quick Statistics

Get a fast overview (total nodes, edges, layers):

```
from py3plex.core import multinet

# Create a multilayer network
network = multinet.multi_layer_network()
network.add_edges([
    ['A', '1', 'B', '1', 1],
    ['B', '1', 'C', '1', 1],
    ['A', '2', 'C', '2', 1],
    ['C', '2', 'D', '2', 1],
], input_type="list")

# Display comprehensive stats
network.basic_stats()
```

**Example output:**

```
Number of nodes: 6
Number of edges: 4
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer '1': 3 nodes
  Layer '2': 3 nodes
```

### Layer-Specific Statistics

Understand how structure varies per layer before aggregating the network.

### Compute Per-Layer Density

Density compares observed edges `m` to the maximum possible edges among `n` nodes in one layer.

```
layers = network.get_layers()

for layer in layers:
    # Get nodes and edges in this layer
    nodes = Q.nodes().from_layers(L[layer]).execute(network)
    edges = Q.edges().from_layers(L[layer]).execute(network)

    n = len(nodes)
    m = len(edges)

    # Density = actual edges / possible edges
    max_edges = n * (n - 1) / 2  # undirected
    density = m / max_edges if max_edges > 0 else 0

    print(f"Layer {layer}: density = {density:.4f}")
```

For directed layers, use `max_edges = n * (n - 1)` instead (ordered pairs).

### Compare Layers

Use the DSL for efficient comparison:

```
for layer in network.get_layers():
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
         .execute(network)
    )
    df = result.to_pandas()
    print(f"{layer}: avg degree = {df['degree'].mean():.2f}")
```

**Example output (values depend on your data):**

```
layer1: avg degree = 3.45
layer2: avg degree = 2.87
layer3: avg degree = 4.12
```

### Node-Level Statistics

Once you know layer-level trends, drill down to node importance and activity.

### Node Activity (Layer Count)

How many layers does each node participate in? `layer_count` counts unique layers where the node appears.

```
# Get nodes present in multiple layers
active_nodes = (
    Q.nodes()
     .compute("layer_count")
     .where(layer_count__gt=1)
     .order_by("-layer_count")
     .execute(network)
)

df = active_nodes.to_pandas()
print(df.head(10))
```

**Example output:**

```
     node  layer_count
0   Alice            3
1     Bob            3
2   Carol            2
3    Dave            1
```

### Degree Centrality

Compute degree for all nodes and sort in descending order:

```
from py3plex.dsl import Q

result = (
    Q.nodes()
     .compute("degree")
     .order_by("-degree")
     .limit(10)
     .execute(network)
)

df = result.to_pandas()
print("Top 10 by degree:")
print(df)
```

### Betweenness Centrality

Find nodes that bridge different parts of the network (higher values indicate more shortest paths go through the node):

```
result = (
    Q.nodes()
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(10)
     .execute(network)
)

df = result.to_pandas()
print("Top 10 by betweenness:")
print(df)
```

### Multiple Metrics at Once

Compute several measures in one pass to avoid recomputing them separately:

```
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality", "clustering")
     .execute(network)
)

df = result.to_pandas()
print(df.describe())
```

### Multilayer-Specific Statistics

Use these measures when the interaction between layers matters, not just within-layer structure.

#### Node Versatility

Versatility measures how evenly a node distributes its connections across layers; higher values mean more balanced participation across layers.

```
from py3plex.algorithms.statistics import calculate_versatility

versatility_scores = calculate_versatility(network)

# Sort by versatility
sorted_scores = sorted(
    versatility_scores.items(),
    key=lambda x: x[1],
    reverse=True
)

print("Top 5 most versatile nodes:")
for node, score in sorted_scores[:5]:
    print(f"{node}: {score:.3f}")
```

#### Edge Overlap Between Layers

How many connections exist in multiple layers?

```
from py3plex.algorithms.statistics import calculate_edge_overlap

overlap = calculate_edge_overlap(network, 'layer1', 'layer2')

print(f"Edge overlap between layer1 and layer2: {overlap:.2%}")
```

`overlap` is a ratio (0–1) of edges shared by both layers.

#### Inter-Layer Degree Correlation

Are high-degree nodes in layer 1 also high-degree in layer 2?

```
from py3plex.algorithms.statistics import inter_layer_correlation

correlation = inter_layer_correlation(
    network,
    'layer1',
    'layer2',
    metric='degree'
)

print(f"Degree correlation: {correlation:.3f}")
# Values near 1.0 indicate that high-degree nodes stay high across layers.
```

### Network-Wide Statistics

Aggregate everything into a single view of the whole multilayer graph.

#### Global Clustering Coefficient

```
from py3plex.algorithms.statistics import global_clustering_coefficient

gcc = global_clustering_coefficient(network)
print(f"Global clustering: {gcc:.3f}")
```

#### Average Path Length

```
from py3plex.algorithms.statistics import average_path_length

# Note: computationally expensive for large networks; ensure the graph
# is connected or run on the largest connected component.
apl = average_path_length(network)
print(f"Average path length: {apl:.2f}")
```

### Exporting Statistics

Persist results so you can compare runs or feed them into other tools.

#### Save to CSV

```
from py3plex.dsl import Q

result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality", "layer_count")
     .execute(network)
)

df = result.to_pandas()
df.to_csv("network_statistics.csv", index=False)
```

#### Save to JSON

```
import json
from py3plex.dsl import Q, L

stats = {
    'num_nodes': len(list(network.get_nodes())),
    'num_edges': len(list(network.get_edges())),
    'num_layers': len(network.get_layers()),
    'density_by_layer': {}
}
```

```
for layer in network.get_layers():
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)
    layer_edges = Q.edges().from_layers(L[layer]).execute(network)
    n = len(layer_nodes)
    m = len(layer_edges)
    max_edges = n * (n - 1) / 2  # undirected density
    stats['density_by_layer'][layer] = m / max_edges if max_edges > 0 else 0

with open('stats.json', 'w') as f:
    json.dump(stats, f, indent=2)
```

### Common Patterns

#### Pattern: Compare Node Importance Across Layers

```
import pandas as pd

layers = network.get_layers()
all_stats = []

for layer in layers:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree", "betweenness_centrality")
         .execute(network)
    )
    df = result.to_pandas()
    df['layer'] = layer
    all_stats.append(df)

combined = pd.concat(all_stats, ignore_index=True)

# Find nodes that are important in all layers
pivot = combined.pivot_table(
    values='degree',
    index='node',
    columns='layer',
    aggfunc='mean'
)

print(pivot)
```

#### Pattern: Identify Layer-Specific Hubs

```
for layer in network.get_layers():
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
```

```
        .order_by("-degree")
        .limit(5)
        .execute(network)
    )
    df = result.to_pandas()
    print(f"\nTop 5 hubs in {layer}:")
    print(df)
```

### Next Steps

- **Visualize statistics:** *How to Visualize Multilayer Networks*
- **Find communities:** *How to Run Community Detection on Multilayer Networks*
- **Understand metrics:** *Algorithm Landscape*
- **API reference:** *Algorithm Roadmap*

## 3.3.5 How to Run Community Detection on Multilayer Networks

**Goal:** This guide demonstrates how to apply community detection algorithms to multilayer networks and interpret their results. Community detection identifies *mesoscale structure*—groups of nodes that are more densely connected internally than to the rest of the network. In multilayer networks, communities can exist within single layers, span multiple layers, or emerge from inter-layer coupling patterns. This analysis is essential for understanding functional modules, organizational structure, and hierarchical clustering in complex systems.

**Run this guide online**

You can run this tutorial in your browser without any local installation:

[9] See the py3plex examples[10] for community detection examples.

**Prerequisites:**

- A loaded multilayer network (see *How to Load and Build Networks*)
- Basic familiarity with network terminology (nodes, edges, layers)
- Understanding of modularity as a quality metric (covered in this guide)

**When to use community detection:**

- Identifying functional modules in biological networks
- Detecting organizational units in social networks
- Finding coherent topics in multi-relational knowledge graphs
- Analyzing temporal evolution of communities across time-sliced networks
- Discovering cross-layer relationships in multiplex systems

[9] https://colab.research.google.com/github/SkBlaz/py3plex/blob/master/notebooks/community_detection.ipynb
[10] https://github.com/SkBlaz/py3plex/tree/master/examples

### Quick Start: Louvain Algorithm

**What is Louvain?**

The Louvain algorithm (Blondel et al., 2008) is a fast, greedy method that optimizes *modularity*, defined as:

$$Q = \frac{1}{2m} \sum_{ij} \left[ A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

where $A_{ij}$ is the adjacency matrix, $k_i$ is node degree, $m$ is total edges, and $\delta(c_i, c_j) = 1$ if nodes $i, j$ are in the same community. Higher $Q$ indicates stronger community structure.

**How it works:**

1. Initialize: each node starts in its own community
2. For each node, compute $\Delta Q$ from moving to each neighbor's community
3. Move the node to the community with maximum positive $\Delta Q$
4. Aggregate: collapse communities into super-nodes and repeat
5. Stop when no further improvement is possible

**Time complexity:** $O(n \log n)$ for sparse networks

**Basic example:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities

# Load multilayer network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

# Run Louvain (operates on flattened network by default)
communities = louvain_communities(network)

# Analyze results
from collections import Counter
comm_sizes = Counter(communities.values())

print(f"Number of communities: {len(comm_sizes)}")
print(f"Largest community: {max(comm_sizes.values())} nodes")
print(f"Smallest community: {min(comm_sizes.values())} nodes")
print(f"Average size: {sum(comm_sizes.values())/len(comm_sizes):.1f}")

# Sample assignments
for node, comm_id in list(communities.items())[:5]:
    print(f"  {node} → Community {comm_id}")
```

**Expected output:**

```
Number of communities: 4
Largest community: 45 nodes
Smallest community: 8 nodes
Average size: 22.8
  ('A1', 'layer1') → Community 0
  ('A2', 'layer1') → Community 0
  ('B1', 'layer1') → Community 1
  ('B2', 'layer2') → Community 1
  ('C1', 'layer2') → Community 2
```

**Note:** The standard `louvain_communities` function flattens the multilayer network into a single-layer graph: each node-layer pair becomes a single node and inter-layer coupling is ignored (layer names remain as part of the node labels). For layer-aware detection, use `louvain_multilayer` (see next section).

### Multilayer-Specific: Multilayer Louvain

**What makes multilayer community detection different?**

Standard Louvain treats a multilayer network as a single flattened graph, losing layer identity. **Multilayer Louvain** (Mucha et al., 2010) optimizes the *multilayer modularity*:

$$Q_{\text{multi}} = \frac{1}{2\mu} \sum_{ij\alpha\beta} \left[ \left( A_{ij}^{\alpha} - \gamma^{\alpha} \frac{k_i^{\alpha} k_j^{\alpha}}{2m_{\alpha}} \right) \delta_{\alpha\beta} + \delta_{ij} \omega_{\alpha\beta} \right] \delta(g_{i\alpha}, g_{j\beta})$$

where:

- $A_{ij}^{\alpha}$: adjacency in layer $\alpha$
- $k_i^{\alpha}$: degree (or strength) of node $i$ in layer $\alpha$
- $m_{\alpha}$: total edge weight in layer $\alpha$ ($2m_{\alpha} = \sum_{ij} A_{ij}^{\alpha}$)
- $\gamma^{\alpha}$: resolution parameter for layer $\alpha$ (default 1.0)
- $\omega_{\alpha\beta}$: inter-layer coupling strength (default 1.0)
- $\delta_{ij} = 1$ if $i = j$ (inter-layer edges connect same node across layers)
- $\delta_{\alpha\beta} = 1$ if $\alpha = \beta$ (restricts intra-layer term)
- $\delta(g_{i\alpha}, g_{j\beta}) = 1$ if node $i$ in layer $\alpha$ and node $j$ in layer $\beta$ are in the same community
- $\mu$: total weight in the supra-network ($2\mu = \sum_{\alpha} 2m_{\alpha} + \sum_{\alpha \neq \beta} \omega_{\alpha\beta}$)

**Key insight:** The coupling term $\omega_{\alpha\beta}$ controls whether communities span layers:

- $= \mathbf{0}$**:** Layers are independent → separate communities per layer
- $\to \infty$**:** Strong coupling → communities span all layers
- $\mathbf{0} <$ $< \infty$**:** Partial coupling → communities can span some layers

**Full workflow example:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer,
    multilayer_modularity
)
from collections import Counter, defaultdict
```

```
# Load multilayer network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

print("Network structure:")
print(f"  Layers: {network.get_layers()}")
print(f"  Nodes: {len(network.get_nodes())}")
print(f"  Edges (total): {network.number_of_edges()}")

# Run multilayer Louvain with different coupling strengths
for omega in [0.0, 0.5, 1.0, 2.0]:
    print(f"\n--- Coupling ={omega} ---")

    communities = louvain_multilayer(
        network,
        gamma=1.0,         # Resolution (default)
        omega=omega,       # Inter-layer coupling
        random_state=42    # For reproducibility
    )

    # Count communities
    n_communities = len(set(communities.values()))

    # Calculate multilayer modularity
    Q = multilayer_modularity(network, communities, gamma=1.0, omega=omega)

    # Analyze layer coverage
    layer_coverage = defaultdict(set)  # community -> set of layers
    for (node, layer), comm_id in communities.items():
        layer_coverage[comm_id].add(layer)

    cross_layer = sum(1 for layers in layer_coverage.values() if len(layers) > 1)
    single_layer = len(layer_coverage) - cross_layer

    print(f"  Communities: {n_communities}")
    print(f"  Modularity Q: {Q:.4f}")
    print(f"  Cross-layer communities: {cross_layer}")
    print(f"  Single-layer communities: {single_layer}")

    # Size distribution
    comm_sizes = Counter(communities.values())
    avg_size = sum(comm_sizes.values()) / len(comm_sizes)
    print(f"  Average community size: {avg_size:.1f} node-layers")
```

**Expected output:**

```
Network structure:
  Layers: ['layer1', 'layer2', 'layer3']
  Nodes: 120 (40 nodes × 3 layers)
  Edges (total): 284

--- Coupling =0.0 ---
  Communities: 12
  Modularity Q: 0.3456
  Cross-layer communities: 0
  Single-layer communities: 12
  Average community size: 10.0 node-layers

--- Coupling =0.5 ---
  Communities: 8
  Modularity Q: 0.4123
  Cross-layer communities: 3
  Single-layer communities: 5
  Average community size: 15.0 node-layers

--- Coupling =1.0 ---
  Communities: 5
  Modularity Q: 0.4589
  Cross-layer communities: 4
  Single-layer communities: 1
  Average community size: 24.0 node-layers

--- Coupling =2.0 ---
  Communities: 4
  Modularity Q: 0.4234
  Cross-layer communities: 4
  Single-layer communities: 0
  Average community size: 30.0 node-layers
```

**Interpretation:**

- **=0.0:** Each layer has independent communities (useful for baseline)
- **=0.5-1.0:** Balanced trade-off, some communities span layers
- **>1.0:** Forces global communities across all layers (may over-integrate)

**Choosing :**

- Use **domain knowledge**: biological function (high ), temporal snapshots (low )
- **Grid search**: try [0.1, 0.5, 1.0, 2.0, 5.0] and pick maximum Q
- **Consensus clustering**: aggregate results across multiple values

### Infomap Algorithm

**What is Infomap?**

Infomap (Rosvall & Bergstrom, 2008) uses information theory to find communities by minimizing the *map equation*:

$$L(M) = q_{\curvearrowright} H(Q) + \sum_{i=1}^{m} p_{\circlearrowright}^{i} H(P^{i})$$

where:

- $q_{\curvearrowright}$: probability of switching between modules (inter-module flow)
- $H(Q)$: entropy of module codebook
- $p_{\circlearrowright}^{i}$: probability of staying within module $i$ (intra-module flow)
- $H(P^{i})$: entropy of nodes within module $i$

**Key insight:** Infomap simulates a random walker and finds communities that compress the *description length* of the walker's trajectory. Communities are regions where the walker gets "trapped" for extended periods.

**Pros/cons vs. Louvain:**

- **Pros:** Often finds better communities for flow-based systems (e.g., citation networks, web graphs)
- **Cons:** Requires external binary (not pure Python), slower than Louvain, harder to interpret parameters

**Installation:**

Infomap requires the standalone binary from https://www.mapequation.org/infomap/:

```
# Download and install
wget https://www.mapequation.org/downloads/Infomap.zip
unzip Infomap.zip
cd Infomap
make
sudo cp Infomap /usr/local/bin/infomap

# Or install Python wrapper (alternative)
pip install infomap
```

**Basic usage:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.community_wrapper import infomap_communities
import os

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)
```

```
# Check if binary exists
binary_path = "/usr/local/bin/infomap"  # Adjust to your installation
if not os.path.exists(binary_path):
    print(f"Infomap binary not found at {binary_path}")
    print("Please install from: https://www.mapequation.org/infomap/")
    print("Falling back to Louvain...")
    # Use Louvain as fallback
    from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
    communities = louvain_communities(network)
else:
    # Run Infomap
    communities = infomap_communities(
        network,
        binary=binary_path,
        multiplex=True,      # Use multiplex mode for multilayer networks
        iterations=1000,     # More iterations = better convergence
        seed=42,             # For reproducibility
        verbose=False        # Set True to see Infomap output
    )

# Analyze results
from collections import Counter
comm_sizes = Counter(communities.values())

print(f"Number of communities: {len(comm_sizes)}")
print(f"Largest community: {max(comm_sizes.values())} nodes")
print(f"Average size: {sum(comm_sizes.values())/len(comm_sizes):.1f}")
```

**Expected output:**

```
Number of communities: 6
Largest community: 38 nodes
Average size: 20.0
```

**Multiplex mode:**

When `multiplex=True`, Infomap treats layers as separate networks but allows random walkers to switch layers (implicitly modeling inter-layer coupling). This is different from Louvain's explicit $\omega$ parameter.

**Comparison workflow:**

```
from sklearn.metrics import adjusted_rand_score, normalized_mutual_info_score

# Run both algorithms
louvain_comms = louvain_communities(network)
infomap_comms = infomap_communities(network, binary=binary_path, seed=42)

# Convert to aligned label vectors
nodes = list(louvain_comms.keys())
louvain_labels = [louvain_comms[n] for n in nodes]
infomap_labels = [infomap_comms[n] for n in nodes]
```

```
# Compute similarity
ari = adjusted_rand_score(louvain_labels, infomap_labels)
nmi = normalized_mutual_info_score(louvain_labels, infomap_labels)

print(f"Agreement between Louvain and Infomap:")
print(f"  ARI: {ari:.3f}  (1.0 = perfect agreement)")
print(f"  NMI: {nmi:.3f}  (1.0 = perfect agreement)")
```

**Expected output:**

```
Agreement between Louvain and Infomap:
  ARI: 0.723  (1.0 = perfect agreement)
  NMI: 0.815  (1.0 = perfect agreement)
```

**When to use Infomap:**

- Citation/web networks with clear flow patterns
- Networks where you care about information diffusion
- When Louvain gives unsatisfying results (try both and compare)
- When you have the binary installed (otherwise, stick with Louvain)

### Label Propagation

**What is Label Propagation?**

Label propagation (Raghavan et al., 2007) is an extremely fast, near-linear time algorithm that works by iteratively assigning each node to the most common community among its neighbors.

**Algorithm:**

1. Initialize: each node gets a unique label (community ID)
2. **For t=1 to T iterations:**
   a. Randomize node order
   b. For each node $i$:
      - Count neighbor labels: $n_c = |\{j \in N(i) : c_j = c\}|$
      - Assign $c_i = \arg\max_c n_c$ (ties broken randomly)
3. Stop when labels stabilize or max iterations reached

**Time complexity:** $O(m)$ per iteration (linear in edges)

**Pros/cons:**

- **Pros:** Very fast, scales to millions of nodes, no parameters to tune
- **Cons:** Non-deterministic (order-dependent), lower quality than Louvain/Infomap, may not converge

**Implementation note:**

py3plex uses NetworkX's label propagation for single-layer networks:

```
from py3plex.core import multinet
import networkx as nx
```

```python
from networkx.algorithms.community import asyn_lpa_communities
from collections import defaultdict

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

# Convert to NetworkX (flattened single-layer graph)
G = nx.Graph()
for edge in network.core_network.edges():
    G.add_edge(edge[0], edge[1])

# Run label propagation
communities_list = asyn_lpa_communities(G, seed=42)

# Convert to dict format: node -> community_id
communities = {}
for comm_id, comm_nodes in enumerate(communities_list):
    for node in comm_nodes:
        communities[node] = comm_id

# Analyze results
from collections import Counter
comm_sizes = Counter(communities.values())

print(f"Number of communities: {len(comm_sizes)}")
print(f"Largest community: {max(comm_sizes.values())} nodes")
print(f"Average size: {sum(comm_sizes.values())/len(comm_sizes):.1f}")

# Run multiple times to check stability
print("\nStability check (5 runs with same seed):")
for run in range(5):
    comms_run = list(asyn_lpa_communities(G, seed=42))
    n_comms = len(comms_run)
    print(f"  Run {run+1}: {n_comms} communities")
```

**Expected output:**

```
Number of communities: 7
Largest community: 34 nodes
Average size: 17.1

Stability check (5 runs with same seed):
  Run 1: 7 communities
  Run 2: 7 communities
  Run 3: 7 communities
  Run 4: 8 communities
  Run 5: 7 communities
```

**Layer-aware label propagation (custom implementation):**

For multilayer networks, you can implement layer-aware label propagation:

```
import random
from collections import Counter

def multilayer_label_propagation(network, max_iter=100, seed=42):
    """
    Layer-aware label propagation for multilayer networks.
    Propagates labels within each layer independently.
    """
    random.seed(seed)

    # Initialize: each node-layer gets unique label
    labels = {nl: i for i, nl in enumerate(network.get_nodes())}

    # Get layer-specific edges
    layer_edges = {}
    for layer in network.get_layers():
        layer_edges[layer] = [
            (e[0], e[1]) for e in network.core_network.edges()
            if e[0][1] == layer and e[1][1] == layer
        ]

    # Iterate
    for iteration in range(max_iter):
        changed = False
        nodes = list(labels.keys())
        random.shuffle(nodes)

        for node, layer in nodes:
            # Get neighbors in same layer
            neighbors = [
                target for source, target in layer_edges.get(layer, [])
                if source == (node, layer)
            ] + [
                source for source, target in layer_edges.get(layer, [])
                if target == (node, layer)
            ]

            if not neighbors:
                continue

            # Count neighbor labels
            neighbor_labels = [labels[n] for n in neighbors]
            label_counts = Counter(neighbor_labels)

            # Assign most common label (ties broken randomly)
            most_common = label_counts.most_common()
            max_count = most_common[0][1]
            candidates = [lbl for lbl, cnt in most_common if cnt == max_count]
```

```
            new_label = random.choice(candidates)

            if new_label != labels[(node, layer)]:
                labels[(node, layer)] = new_label
                changed = True

        if not changed:
            print(f"Converged after {iteration+1} iterations")
            break

    # Renumber communities
    unique_labels = sorted(set(labels.values()))
    label_map = {old: new for new, old in enumerate(unique_labels)}
    return {nl: label_map[lbl] for nl, lbl in labels.items()}

# Run custom implementation
communities = multilayer_label_propagation(network, max_iter=100, seed=42)

comm_sizes = Counter(communities.values())
print(f"\nLayer-aware label propagation:")
print(f"  Communities: {len(comm_sizes)}")
print(f"  Average size: {sum(comm_sizes.values())/len(comm_sizes):.1f}")
```

**Expected output:**

```
Converged after 23 iterations

Layer-aware label propagation:
  Communities: 9
  Average size: 13.3
```

**When to use label propagation:**

- **Very large networks** (>100k nodes) where Louvain is too slow
- **Exploratory analysis** where you need quick initial results
- **Streaming settings** where you process edges incrementally
- **Not recommended** for publication-quality results (use Louvain or Infomap instead)

### Analyzing Community Structure

After detecting communities, you need to **analyze** and **interpret** the results. This section shows robust workflows for understanding community properties.

### Count Nodes Per Community

**Basic counting:**

```
from collections import Counter
import numpy as np

# Assuming 'communities' is a dict: node -> community_id
```

```
comm_sizes = Counter(communities.values())

print(f"Total communities: {len(comm_sizes)}")
print(f"\nTop 10 largest communities:")
for comm_id, size in comm_sizes.most_common(10):
    print(f"  Community {comm_id}: {size} nodes")

# Size statistics
sizes = np.array(list(comm_sizes.values()))
print(f"\nSize distribution:")
print(f"  Mean: {np.mean(sizes):.2f}")
print(f"  Median: {np.median(sizes):.2f}")
print(f"  Std dev: {np.std(sizes):.2f}")
print(f"  Min: {np.min(sizes)}")
print(f"  Max: {np.max(sizes)}")
print(f"  Q1/Q3: {np.percentile(sizes, 25):.0f} / {np.percentile(sizes, 75):.0f}")
```

**Expected output:**

```
Total communities: 5

Top 10 largest communities:
  Community 0: 45 nodes
  Community 1: 38 nodes
  Community 2: 22 nodes
  Community 3: 10 nodes
  Community 4: 5 nodes

Size distribution:
  Mean: 24.00
  Median: 22.00
  Std dev: 15.87
  Min: 5
  Max: 45
  Q1/Q3: 10 / 38
```

**Layer coverage analysis (for multilayer networks):**

```
from collections import defaultdict

# communities: {(node, layer): comm_id}
layer_coverage = defaultdict(lambda: defaultdict(set))  # comm -> layer -> nodes

for (node, layer), comm_id in communities.items():
    layer_coverage[comm_id][layer].add(node)

print("Community layer coverage:")
for comm_id in sorted(layer_coverage.keys()):
    layers = layer_coverage[comm_id]
    total_size = sum(len(nodes) for nodes in layers.values())
    print(f"\nCommunity {comm_id} (total: {total_size} node-layers):")
```

```
    for layer, nodes in sorted(layers.items()):
        print(f"  {layer}: {len(nodes)} nodes")

    # Cross-layer nodes (nodes appearing in multiple layers within same community)
    all_nodes = set()
    for nodes in layers.values():
        all_nodes.update(nodes)
    unique_nodes = len(all_nodes)
    redundancy = total_size / unique_nodes if unique_nodes > 0 else 0
    print(f"  Unique nodes: {unique_nodes}, Redundancy: {redundancy:.2f}x")
```

**Expected output:**

```
Community layer coverage:

Community 0 (total: 45 node-layers):
  layer1: 18 nodes
  layer2: 15 nodes
  layer3: 12 nodes
  Unique nodes: 15, Redundancy: 3.00x

Community 1 (total: 38 node-layers):
  layer1: 20 nodes
  layer2: 18 nodes
  Unique nodes: 20, Redundancy: 1.90x

Community 2 (total: 22 node-layers):
  layer3: 22 nodes
  Unique nodes: 22, Redundancy: 1.00x
```

### Visualize Communities

**Hairball plot with community colors:**

```
from py3plex.visualization.multilayer import hairball_plot
import matplotlib.pyplot as plt
from py3plex.visualization.colors import colors_default

# Select top N communities to color
top_n = 8
top_communities = [c for c, _ in comm_sizes.most_common(top_n)]

# Create color mapping
color_map = dict(zip(
    top_communities,
    colors_default[:top_n]
))

# Assign colors to nodes
node_colors = []
for node in network.get_nodes():
```

```
    comm_id = communities.get(node, -1)
    if comm_id in color_map:
        node_colors.append(color_map[comm_id])
    else:
        node_colors.append('lightgray')  # Small communities

# Plot
plt.figure(figsize=(12, 10))
hairball_plot(
    network.core_network,
    color_list=node_colors,
    layout_algorithm='force',
    layout_parameters={'iterations': 500},
    scale_by_size=True,
    legend=False
)
plt.title('Community Structure (Top 8 Communities Colored)', fontsize=16)
plt.tight_layout()
plt.savefig('community_hairball.png', dpi=300, bbox_inches='tight')
plt.show()

print("Visualization saved to: community_hairball.png")
```

**Size distribution histogram:**

```
import matplotlib.pyplot as plt
import numpy as np

sizes = list(comm_sizes.values())

plt.figure(figsize=(10, 6))
plt.hist(sizes, bins=20, edgecolor='black', alpha=0.7)
plt.xlabel('Community Size (number of nodes)', fontsize=12)
plt.ylabel('Frequency', fontsize=12)
plt.title(f'Community Size Distribution (n={len(sizes)} communities)', fontsize=14)
plt.axvline(np.mean(sizes), color='red', linestyle='--', label=f'Mean: {np.mean(sizes):.1f}
plt.axvline(np.median(sizes), color='blue', linestyle='--', label=f'Median: {np.median(size
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('community_size_distribution.png', dpi=300)
plt.show()
```

**Layer-specific visualization:**

For multilayer networks, visualize community composition across layers:

```
import pandas as pd
import seaborn as sns

# Build matrix: communities × layers
layers = network.get_layers()
```

```
comm_ids = sorted(set(communities.values()))

matrix = np.zeros((len(comm_ids), len(layers)))
for (node, layer), comm_id in communities.items():
    layer_idx = layers.index(layer)
    comm_idx = comm_ids.index(comm_id)
    matrix[comm_idx, layer_idx] += 1

# Heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(
    matrix,
    xticklabels=layers,
    yticklabels=[f'C{i}' for i in comm_ids],
    cmap='YlOrRd',
    annot=True,
    fmt='.0f',
    cbar_kws={'label': 'Number of nodes'}
)
plt.xlabel('Layer', fontsize=12)
plt.ylabel('Community', fontsize=12)
plt.title('Community × Layer Composition Heatmap', fontsize=14)
plt.tight_layout()
plt.savefig('community_layer_heatmap.png', dpi=300)
plt.show()
```

### Export Communities

**CSV export (most common):**

```
import pandas as pd

# Convert to DataFrame
data = []
for (node, layer), comm_id in communities.items():
    data.append({
        'node': node,
        'layer': layer,
        'community': comm_id
    })

df = pd.DataFrame(data)

# Add community size
size_map = dict(comm_sizes)
df['community_size'] = df['community'].map(size_map)

# Sort by community, then layer, then node
df = df.sort_values(['community', 'layer', 'node'])
```

```
# Save
df.to_csv('communities.csv', index=False)
print(f"Exported {len(df)} node-layer assignments to communities.csv")
print(f"\nFirst few rows:")
print(df.head(10))
```

**Expected output:**

```
Exported 120 node-layer assignments to communities.csv

First few rows:
    node   layer  community  community_size
0     A1  layer1          0              45
1     A1  layer2          0              45
2     A1  layer3          0              45
3     A2  layer1          0              45
4     A2  layer2          0              45
5     B1  layer1          1              38
6     B1  layer2          1              38
7     B2  layer1          1              38
8     C1  layer3          2              22
9     C2  layer3          2              22
```

**JSON export (for web apps):**

```
import json

# Group by community
community_dict = defaultdict(list)
for (node, layer), comm_id in communities.items():
    community_dict[str(comm_id)].append({
        'node': node,
        'layer': layer
    })

# Add metadata
output = {
    'num_communities': len(community_dict),
    'num_nodes': len(set(node for node, _ in communities.keys())),
    'num_layers': len(network.get_layers()),
    'communities': dict(community_dict)
}

with open('communities.json', 'w') as f:
    json.dump(output, f, indent=2)

print("Exported to communities.json")
```

**Cytoscape format (for visualization):**

```
# Node table
node_df = pd.DataFrame([
```

```
    {
        'node_id': f"{node}_{layer}",
        'node': node,
        'layer': layer,
        'community': communities.get((node, layer), -1)
    }
    for node, layer in network.get_nodes()
])
node_df.to_csv('cytoscape_nodes.csv', index=False)

# Edge table
edge_data = []
for source, target in network.core_network.edges():
    edge_data.append({
        'source': f"{source[0]}_{source[1]}",
        'target': f"{target[0]}_{target[1]}",
        'source_community': communities.get(source, -1),
        'target_community': communities.get(target, -1),
        'is_intra_community': communities.get(source, -1) == communities.get(target, -1)
    })
edge_df = pd.DataFrame(edge_data)
edge_df.to_csv('cytoscape_edges.csv', index=False)

print("Exported to cytoscape_nodes.csv and cytoscape_edges.csv")
print("Import these into Cytoscape for interactive visualization")
```

### Query Communities with DSL

**Goal:** Use py3plex's Domain-Specific Language (DSL) to query and analyze community-detected networks efficiently.

The DSL provides a declarative, SQL-like interface for querying multilayer networks. After detecting communities, you can use DSL queries to filter nodes by community membership, compute community-level statistics, and extract subnetworks.

**Prerequisites:**

- Community detection results (e.g., from `louvain_communities()`)
- Familiarity with DSL basics (see *How to Query Multilayer Graphs with the SQL-like DSL* for full tutorial)

### DSL Basics for Communities

**String Syntax - SQL-like queries:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
from py3plex.dsl import Q

# Load network and detect communities
network = multinet.multi_layer_network(directed=False)
network.load_network(
```

```
    "py3plex/datasets/_data/synthetic_multilayer.edges",
    input_type="multiedgelist"
)

communities = louvain_communities(network)

# Attach community labels as node attributes
for (node, layer), comm_id in communities.items():
    network.core_network.nodes[(node, layer)]['community'] = comm_id

# DSL Query (Builder API): Find nodes in community 0
result = (
    Q.nodes()
     .where(community=0)
     .execute(network)
)

nodes_in_comm0 = result.items  # List of (node, layer) tuples

print(f"Nodes in community 0: {result.count}")
for node in nodes_in_comm0[:5]:
    print(f"  {node}")
```

**Note:** The Builder API (`Q.nodes()`) provides `result.items` for the list of node-layer tuples and `result.count` for the number of items.

**Expected output:**

```
Nodes in community 0: 18
  ('node1', 'layer1')
  ('node1', 'layer2')
  ('node2', 'layer1')
  ('node3', 'layer1')
  ('node3', 'layer3')
```

**Builder API - Chainable operations:**

```
from py3plex.dsl import Q, L

# Find high-degree nodes in a specific community
result = (
    Q.nodes()
     .where(community=0)
     .compute("degree")
     .where(degree__gt=5)
     .order_by("degree", reverse=True)
     .execute(network)
)

# Convert to pandas for analysis
import pandas as pd
df = pd.DataFrame([
```

```
    {
        'node': node[0],
        'layer': node[1],
        'degree': data['degree'],
        'community': data.get('community', -1)
    }
    for node, data in result.items()
])

print("High-degree nodes in community 0:")
print(df.head(10))
```

**Expected output:**

```
High-degree nodes in community 0:
      node   layer  degree  community
0    node1  layer1      12          0
1    node1  layer2      10          0
2    node2  layer1       9          0
3    node5  layer1       8          0
4    node5  layer3       7          0
```

### Community-Level Queries

**Count nodes per community:**

```
from py3plex.dsl import Q

# Get all communities
community_ids = set(communities.values())

for comm_id in sorted(community_ids):
    result = (
        Q.nodes()
         .where(community=comm_id)
         .execute(network)
    )
    print(f"Community {comm_id}: {result.count} nodes")
```

**Find inter-community edges:**

```
from py3plex.dsl import Q

# Attach community labels to edges based on endpoint communities
for edge in network.core_network.edges():
    source, target = edge
    source_comm = communities.get(source, -1)
    target_comm = communities.get(target, -1)
    network.core_network.edges[edge]['source_community'] = source_comm
    network.core_network.edges[edge]['target_community'] = target_comm
    network.core_network.edges[edge]['is_intra_community'] = (source_comm == target_comm)
```

```
# Query inter-community edges
inter_comm_edges = (
    Q.edges()
     .where(is_intra_community=False)
     .execute(network)
)

intra_comm_edges = (
    Q.edges()
     .where(is_intra_community=True)
     .execute(network)
)

intra_len = len(intra_comm_edges)
inter_len = len(inter_comm_edges)
ratio = inter_len / intra_len if intra_len else 0

print(f"Intra-community edges: {intra_len}")
print(f"Inter-community edges: {inter_len}")
print(f"Ratio: {ratio:.3f} (inter / intra)")
```

**Expected output:**

```
Intra-community edges: 245
Inter-community edges: 39
Ratio: 0.159
```

### Layer-Specific Community Queries

**Find nodes in a specific community and layer:**

```
from py3plex.dsl import Q, L

# Community 0 nodes in layer1 only
result = (
    Q.nodes()
     .from_layers(L["layer1"])
     .where(community=0)
     .compute("degree")
     .execute(network)
)

count = len(result)
avg_degree = sum(d['degree'] for d in result.values())/count if count else 0
print(f"Community 0 in layer1: {count} nodes")
print(f"Average degree: {avg_degree:.2f}")
```

**Compare community structure across layers:**

```
layers = network.get_layers()

for layer in layers:
    # Count communities present in this layer
    layer_nodes = (
        Q.nodes()
         .from_layers(L[layer])
         .execute(network)
    )

    layer_communities = set(
        communities.get(node, -1)
        for node in layer_nodes
    )

    print(f"{layer}: {len(layer_communities)} communities, {len(layer_nodes)} nodes")
```

**Expected output:**

```
layer1: 5 communities, 40 nodes
layer2: 4 communities, 40 nodes
layer3: 3 communities, 40 nodes
```

### Extract Community Subnetworks

**Extract a single community as a subnetwork:**

```
from py3plex.dsl import Q

# Extract community 0
comm_0_result = (
    Q.nodes()
     .where(community=0)
     .execute(network)
)

comm_0_nodes = comm_0_result.items  # List of (node, layer) tuples

# Get induced subgraph
subgraph = network.core_network.subgraph(comm_0_nodes)

# Convert to new multilayer network
community_network = multinet.multi_layer_network(directed=False)
community_network.core_network = subgraph.copy()

print(f"Community 0 subnetwork:")
print(f"  Nodes: {community_network.number_of_nodes()}")
print(f"  Edges: {community_network.number_of_edges()}")
print(f"  Layers: {community_network.get_layers()}")
```

**Expected output:**

```
Community 0 subnetwork:
  Nodes: 18
  Edges: 67
  Layers: ['layer1', 'layer2', 'layer3']
```

### Compute Community-Level Statistics

**Average centrality per community:**

```python
from py3plex.dsl import Q
from collections import defaultdict

# Compute centrality for all nodes
result = (
    Q.nodes()
     .compute("betweenness_centrality", "degree")
     .execute(network)
)

# Group by community
comm_stats = defaultdict(list)
for node, data in result.items():
    comm_id = data.get('community', -1)
    comm_stats[comm_id].append({
        'degree': data['degree'],
        'betweenness': data['betweenness_centrality']
    })

# Calculate averages
print("Community-level statistics:")
print(f"{'Community':<12} {'Nodes':<8} {'Avg Degree':<12} {'Avg Betweenness':<18}")
print("-" * 50)

for comm_id in sorted(comm_stats.keys()):
    stats = comm_stats[comm_id]
    n_nodes = len(stats)
    avg_degree = sum(s['degree'] for s in stats) / n_nodes
    avg_betw = sum(s['betweenness'] for s in stats) / n_nodes
    print(f"{comm_id:<12} {n_nodes:<8} {avg_degree:<12.2f} {avg_betw:<18.6f}")
```

**Expected output:**

```
Community-level statistics:
Community    Nodes    Avg Degree   Avg Betweenness
--------------------------------------------------
0            18       7.44         0.012345
1            15       6.13         0.008234
2            12       5.25         0.005678
3            8        4.50         0.003456
4            7        3.86         0.002123
```

### Complex DSL Workflows

**Multi-step analysis: Find bridge nodes between communities:**

```
from py3plex.dsl import Q

# Bridge nodes: high betweenness + connect multiple communities
# First, compute betweenness
result = (
    Q.nodes()
     .compute("betweenness_centrality", "degree")
     .execute(network)
)

# Identify potential bridges (high betweenness)
bridges = [
    (node, data['betweenness_centrality'])
    for node, data in result.items()
    if data['betweenness_centrality'] > 0.01  # Threshold
]

print(f"Potential bridge nodes (betweenness > 0.01): {len(bridges)}")

# For each bridge, check which communities its neighbors belong to
for node, betw in sorted(bridges, key=lambda x: x[1], reverse=True)[:5]:
    # Get neighbors
    neighbors = list(network.core_network.neighbors(node))
    neighbor_comms = set(communities.get(n, -1) for n in neighbors)

    print(f"  {node}: betweenness={betw:.6f}, connects {len(neighbor_comms)} communities"
```

**Expected output:**

```
Potential bridge nodes (betweenness > 0.01): 12
  ('node7', 'layer1'): betweenness=0.045678, connects 3 communities
  ('node12', 'layer2'): betweenness=0.034567, connects 2 communities
  ('node3', 'layer1'): betweenness=0.023456, connects 3 communities
  ('node15', 'layer3'): betweenness=0.019876, connects 2 communities
  ('node8', 'layer2'): betweenness=0.015432, connects 2 communities
```

**Temporal community analysis (for time-sliced networks):**

```
from py3plex.dsl import Q, L

# Assuming layers represent time slices: t1, t2, t3
time_layers = ['t1', 't2', 't3']

# Track specific nodes across time
tracked_nodes = ['Alice', 'Bob', 'Carol']

print("Community membership over time:")
for node in tracked_nodes:
```

```
    print(f"\n{node}:")
    for t_layer in time_layers:
        node_key = (node, t_layer)
        comm_id = communities.get(node_key, None)
        if comm_id is not None:
            print(f"  {t_layer}: Community {comm_id}")
        else:
            print(f"  {t_layer}: Not present")
```

**Why use DSL for community analysis?**

- **Declarative:** Express *what* you want, not *how* to compute it
- **Composable:** Chain operations to build complex queries
- **Efficient:** DSL optimizes query execution internally
- **Readable:** SQL-like syntax is self-documenting
- **Interoperable:** Results integrate seamlessly with pandas, NumPy, and visualization tools

**Next steps with DSL:**

- **Full DSL tutorial:** *How to Query Multilayer Graphs with the SQL-like DSL* - Comprehensive guide with advanced patterns
- **Builder API reference:** *DSL Reference* - Complete API documentation
- **Temporal queries:** *How to Query Multilayer Graphs with the SQL-like DSL* (Temporal Queries section) - Time-varying networks

### Compare Algorithms

Different algorithms optimize different objective functions and may produce different community structures. **Comparing multiple algorithms** helps validate findings and understand algorithm-specific biases.

**Metrics for comparing partitions:**

1. **Adjusted Rand Index (ARI)**: Measures similarity adjusted for chance
   - Range: [-1, 1], where 1 = perfect agreement, 0 = random
   - Adjusted for cluster size imbalance
2. **Normalized Mutual Information (NMI)**: Information-theoretic similarity
   - Range: [0, 1], where 1 = perfect agreement
   - Symmetric, handles different number of communities well
3. **Variation of Information (VI)**: Distance metric (lower = more similar)
   - Range: [0, $\infty$], where 0 = identical partitions

**Full comparison workflow:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.community_wrapper import (
    louvain_communities,
    infomap_communities
)
from py3plex.algorithms.community_detection.multilayer_modularity import (
```

```
    louvain_multilayer,
    multilayer_modularity
)
from py3plex.algorithms.community_detection.community_louvain import modularity
import networkx as nx
from networkx.algorithms.community import asyn_lpa_communities
from sklearn.metrics import adjusted_rand_score, normalized_mutual_info_score
from scipy.spatial.distance import jensenshannon
import numpy as np
from collections import Counter

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

print("=" * 70)
print("COMMUNITY DETECTION ALGORITHM COMPARISON")
print("=" * 70)

# Run multiple algorithms
print("\n1. Running algorithms...")

# Louvain (flattened)
louvain_comms = louvain_communities(network)

# Multilayer Louvain (=1.0)
multilayer_comms = louvain_multilayer(
    network, gamma=1.0, omega=1.0, random_state=42
)

# Label propagation (flattened NetworkX graph)
G = nx.Graph()
for edge in network.core_network.edges():
    G.add_edge(edge[0], edge[1])
lpa_comms_list = asyn_lpa_communities(G, seed=42)
lpa_comms = {}
for comm_id, nodes in enumerate(lpa_comms_list):
    for node in nodes:
        lpa_comms[node] = comm_id

# (Optional) Infomap - skip if binary not available
try:
    infomap_comms = infomap_communities(
        network, binary="/usr/local/bin/infomap",
        seed=42, verbose=False
    )
    has_infomap = True
```

```
except Exception:
    has_infomap = False
    print("  [SKIP] Infomap not available")

# Store results
algorithms = {
    'Louvain (flat)': louvain_comms,
    'Louvain (multilayer)': multilayer_comms,
    'Label Propagation': lpa_comms,
}
if has_infomap:
    algorithms['Infomap'] = infomap_comms

# 2. Basic statistics
print("\n2. Basic statistics:")
print(f"{'Algorithm':<25} {'#Comm':<10} {'Largest':<10} {'Avg Size':<10}")
print("-" * 70)

for name, comms in algorithms.items():
    sizes = Counter(comms.values())
    n_comms = len(sizes)
    largest = max(sizes.values())
    avg_size = sum(sizes.values()) / n_comms
    print(f"{name:<25} {n_comms:<10} {largest:<10} {avg_size:<10.1f}")

# 3. Modularity scores
print("\n3. Modularity scores:")
print(f"{'Algorithm':<25} {'Modularity (Q)':<15}")
print("-" * 70)

for name, comms in algorithms.items():
    if name == 'Louvain (multilayer)':
        # Use multilayer modularity
        Q = multilayer_modularity(network, comms, gamma=1.0, omega=1.0)
    else:
        # Use single-layer modularity on flattened graph
        Q = modularity(comms, G, weight='weight')
    print(f"{name:<25} {Q:<15.4f}")

# 4. Pairwise agreement
print("\n4. Pairwise agreement (ARI / NMI):")

# Align all partitions to same node set
alg_names = list(algorithms.keys())
alg_labels = {}

# Get common nodes (for multilayer, use node-layer pairs)
all_nodes = set()
for comms in algorithms.values():
    all_nodes.update(comms.keys())
```

```
common_nodes = sorted(all_nodes)

# Convert to label vectors
for name, comms in algorithms.items():
    alg_labels[name] = [comms.get(node, -1) for node in common_nodes]

# Compute pairwise metrics
print(f"\n{'Pair':<45} {'ARI':<10} {'NMI':<10}")
print("-" * 70)

for i in range(len(alg_names)):
    for j in range(i+1, len(alg_names)):
        name1, name2 = alg_names[i], alg_names[j]
        labels1 = alg_labels[name1]
        labels2 = alg_labels[name2]

        ari = adjusted_rand_score(labels1, labels2)
        nmi = normalized_mutual_info_score(labels1, labels2)

        print(f"{name1} vs {name2:<25} {ari:<10.3f} {nmi:<10.3f}")

# 5. Size distribution comparison
print("\n5. Size distribution similarity:")

# Normalize size distributions
def normalize_sizes(comms):
    sizes = list(Counter(comms.values()).values())
    sizes_array = np.array(sorted(sizes, reverse=True))
    # Pad to same length
    max_len = max(len(Counter(c.values())) for c in algorithms.values())
    padded = np.zeros(max_len)
    padded[:len(sizes_array)] = sizes_array
    return padded / padded.sum()

size_dists = {name: normalize_sizes(comms) for name, comms in algorithms.items()}

print(f"{'Pair':<45} {'JS Divergence':<15}")
print("-" * 70)

for i in range(len(alg_names)):
    for j in range(i+1, len(alg_names)):
        name1, name2 = alg_names[i], alg_names[j]
        js_div = jensenshannon(size_dists[name1], size_dists[name2])
        print(f"{name1} vs {name2:<25} {js_div:<15.4f}")
```

**Expected output:**

```
======================================================================
COMMUNITY DETECTION ALGORITHM COMPARISON
======================================================================
```

```
1. Running algorithms...
  [SKIP] Infomap not available

2. Basic statistics:
Algorithm                 #Comm      Largest    Avg Size
--------------------------------------------------------------------
Louvain (flat)            5          45         24.0
Louvain (multilayer)      4          52         30.0
Label Propagation         7          38         17.1

3. Modularity scores:
Algorithm                 Modularity (Q)
--------------------------------------------------------------------
Louvain (flat)            0.4234
Louvain (multilayer)      0.4589
Label Propagation         0.3891

4. Pairwise agreement (ARI / NMI):

Pair                                          ARI        NMI
--------------------------------------------------------------------
Louvain (flat) vs Louvain (multilayer)        0.812      0.878
Louvain (flat) vs Label Propagation           0.623      0.745
Louvain (multilayer) vs Label Propagation     0.589      0.712

5. Size distribution similarity:
Pair                                          JS Divergence
--------------------------------------------------------------------
Louvain (flat) vs Louvain (multilayer)        0.1234
Louvain (flat) vs Label Propagation           0.2456
Louvain (multilayer) vs Label Propagation     0.2789
```

**Interpretation:**

- **High ARI/NMI (>0.8):** Algorithms agree strongly → robust communities
- **Medium ARI/NMI (0.5-0.8):** Partial agreement → sensitive to algorithm choice
- **Low ARI/NMI (<0.5):** Strong disagreement → no clear community structure or algorithm-specific artifacts

**Consensus clustering:**

When algorithms disagree, use consensus clustering to find stable communities:

```python
from collections import defaultdict

# Build co-occurrence matrix: how often do pairs of nodes appear together?
co_occurrence = defaultdict(int)
n_algorithms = len(algorithms)

for comms in algorithms.values():
    # For each community in this partition
    comm_groups = defaultdict(list)
```

```
    for node, comm_id in comms.items():
        comm_groups[comm_id].append(node)

    # Increment co-occurrence for all pairs in same community
    for nodes in comm_groups.values():
        for i, node1 in enumerate(nodes):
            for node2 in nodes[i+1:]:
                pair = tuple(sorted([node1, node2]))
                co_occurrence[pair] += 1

# Threshold: keep pairs that co-occur in 50% of algorithms
threshold = n_algorithms * 0.5
stable_pairs = {pair for pair, count in co_occurrence.items() if count >= threshold}

print(f"\nConsensus clustering:")
print(f"  Total node pairs: {len(co_occurrence)}")
print(f"  Stable pairs (50% agreement): {len(stable_pairs)}")
print(f"  Stability ratio: {len(stable_pairs)/len(co_occurrence):.2%}")
```

**Expected output:**

```
Consensus clustering:
  Total node pairs: 1845
  Stable pairs (50% agreement): 1234
  Stability ratio: 66.88%
```

### Layer-Specific Communities

**Motivation:**

In multilayer networks, you may want to detect communities **within individual layers** and then compare them across layers. This reveals:

- Layer-specific structure (e.g., friendship communities vs. work communities)
- How community organization changes across contexts
- Which communities are stable vs. layer-dependent

**Workflow:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
from py3plex.algorithms.community_detection.community_louvain import modularity
import networkx as nx
from collections import Counter

# Load multilayer network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)
```

```
print("LAYER-SPECIFIC COMMUNITY DETECTION")
print("=" * 70)

# Extract and analyze each layer separately
layer_communities = {}
layer_stats = {}

for layer in network.get_layers():
    print(f"\n--- Layer: {layer} ---")

    # Extract layer-specific edges
    layer_edges = [
        (e[0][0], e[1][0])  # (node, node) without layer info
        for e in network.core_network.edges()
        if e[0][1] == layer and e[1][1] == layer
    ]

    # Build single-layer graph
    G_layer = nx.Graph()
    G_layer.add_edges_from(layer_edges)

    print(f"  Nodes: {G_layer.number_of_nodes()}")
    print(f"  Edges: {G_layer.number_of_edges()}")

    if G_layer.number_of_edges() == 0:
        print(f"  [SKIP] No edges in this layer")
        continue

    # Run Louvain on this layer
    communities = louvain_communities(G_layer)
    layer_communities[layer] = communities

    # Statistics
    comm_sizes = Counter(communities.values())
    n_comms = len(comm_sizes)
    Q = modularity(communities, G_layer, weight='weight')

    layer_stats[layer] = {
        'n_communities': n_comms,
        'modularity': Q,
        'sizes': comm_sizes
    }

    print(f"  Communities: {n_comms}")
    print(f"  Modularity: {Q:.4f}")
    print(f"  Largest community: {max(comm_sizes.values())} nodes")
    print(f"  Average size: {sum(comm_sizes.values())/n_comms:.1f}")
```

**Expected output:**

```
LAYER-SPECIFIC COMMUNITY DETECTION
======================================================================

--- Layer: layer1 ---
  Nodes: 40
  Edges: 95
  Communities: 4
  Modularity: 0.4123
  Largest community: 15 nodes
  Average size: 10.0

--- Layer: layer2 ---
  Nodes: 40
  Edges: 102
  Communities: 5
  Modularity: 0.3876
  Largest community: 12 nodes
  Average size: 8.0

--- Layer: layer3 ---
  Nodes: 40
  Edges: 87
  Communities: 3
  Modularity: 0.4456
  Largest community: 18 nodes
  Average size: 13.3
```

**Cross-layer stability analysis:**

Check how consistently nodes are grouped across layers:

```
from sklearn.metrics import normalized_mutual_info_score
import pandas as pd

# Build node-level community assignments per layer
node_layer_assignments = {}
all_nodes = set()

for layer, communities in layer_communities.items():
    for node, comm_id in communities.items():
        if node not in node_layer_assignments:
            node_layer_assignments[node] = {}
        node_layer_assignments[node][layer] = comm_id
        all_nodes.add(node)

# For each node, check consistency across layers
print("\n" + "=" * 70)
print("CROSS-LAYER STABILITY")
print("=" * 70)

layers = list(layer_communities.keys())
```

```
# Pairwise NMI between layers
print(f"\nPairwise NMI between layers:")
print(f"{'Layer Pair':<30} {'NMI':<10} {'Interpretation'}")
print("-" * 70)

for i in range(len(layers)):
    for j in range(i+1, len(layers)):
        layer1, layer2 = layers[i], layers[j]

        # Get common nodes
        nodes1 = set(layer_communities[layer1].keys())
        nodes2 = set(layer_communities[layer2].keys())
        common = nodes1 & nodes2

        if not common:
            continue

        # Compute NMI
        labels1 = [layer_communities[layer1][n] for n in common]
        labels2 = [layer_communities[layer2][n] for n in common]
        nmi = normalized_mutual_info_score(labels1, labels2)

        # Interpret
        if nmi > 0.8:
            interp = "Very similar"
        elif nmi > 0.5:
            interp = "Moderately similar"
        else:
            interp = "Different"

        print(f"{layer1} vs {layer2:<20} {nmi:<10.3f} {interp}")

# Node-level stability score
print(f"\nNode-level stability:")

node_stability = []
for node in sorted(all_nodes):
    assignments = node_layer_assignments.get(node, {})

    # How many layers does this node appear in?
    n_layers = len(assignments)

    if n_layers < 2:
        continue

    # Are the community IDs consistent?
    # (Community IDs are arbitrary per layer; align by overlap before using this beyond a q
    comm_ids = list(assignments.values())
    is_stable = len(set(comm_ids)) == 1  # All same community ID
```

```
    node_stability.append({
        'node': node,
        'n_layers': n_layers,
        'is_stable': is_stable,
        'assignments': assignments
    })

stable_nodes = sum(1 for s in node_stability if s['is_stable'])
print(f"  Nodes appearing in 2 layers: {len(node_stability)}")
print(f"  Stable nodes (same community ID): {stable_nodes}")
print(f"  Stability rate: {stable_nodes/len(node_stability)*100:.1f}%")

# Example unstable nodes
print(f"\n  Example unstable nodes:")
unstable = [s for s in node_stability if not s['is_stable']][:5]
for item in unstable:
    print(f"    {item['node']}: {item['assignments']}")
```

**Expected output:**

```
======================================================================
CROSS-LAYER STABILITY
======================================================================

Pairwise NMI between layers:
Layer Pair                    NMI        Interpretation
----------------------------------------------------------------------
layer1 vs layer2              0.723      Moderately similar
layer1 vs layer3              0.456      Different
layer2 vs layer3              0.512      Moderately similar

Node-level stability:
  Nodes appearing in 2 layers: 40
  Stable nodes (same community ID): 18
  Stability rate: 45.0%

  Example unstable nodes:
    A5: {'layer1': 0, 'layer2': 1, 'layer3': 0}
    B12: {'layer1': 2, 'layer2': 3}
    C3: {'layer1': 1, 'layer2': 0, 'layer3': 2}
    D7: {'layer1': 0, 'layer2': 2}
    E9: {'layer1': 3, 'layer2': 1, 'layer3': 1}
```

**Note:** Community labels from independent layer runs are arbitrary; align them (e.g., by maximizing overlap or NMI) before treating numeric IDs as comparable. The simple stability rate above is a heuristic sanity check, not a formal alignment.

**Visualization - Alluvial diagram:**

Show how community membership flows across layers (requires external tools or manual construction):

```
import pandas as pd

# Export data for alluvial diagram (use R ggalluvial or similar)
alluvial_data = []

for node in all_nodes:
    assignments = node_layer_assignments.get(node, {})
    if len(assignments) >= 2:
        row = {'node': node}
        for layer in layers:
            row[f'comm_{layer}'] = assignments.get(layer, -1)
        alluvial_data.append(row)

df_alluvial = pd.DataFrame(alluvial_data)
df_alluvial.to_csv('alluvial_data.csv', index=False)
print("\nExported alluvial_data.csv for visualization in R/Python")
print("Example R code:")
print("  library(ggalluvial)")
print("  ggplot(data, aes(axis1=comm_layer1, axis2=comm_layer2, axis3=comm_layer3)) +")
print("    geom_alluvium(aes(fill=node)) + geom_stratum()")
```

**When to use layer-specific detection:**

- **Exploratory analysis:** Understand layer-specific structure before multilayer methods
- **Heterogeneous layers:** Layers represent fundamentally different relationships (e.g., co-authorship vs. citation)
- **Baseline comparison:** Compare layer-specific vs. multilayer results to quantify benefit of multilayer methods
- **Dynamic networks:** Detect communities in temporal snapshots and track evolution

### Cross-Layer Community Analysis

**Motivation:**

After detecting communities in the full multilayer network, you want to understand:

- Do communities span multiple layers?
- Which layers contribute most to each community?
- Are there inter-layer bridges (nodes connecting different layer-specific communities)?

**Community × Layer composition:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import defaultdict
import numpy as np
import pandas as pd

# Load network and detect communities
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
```

```
    input_type="multiedgelist"
)

communities = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)

print("CROSS-LAYER COMMUNITY ANALYSIS")
print("=" * 70)

# Build composition matrix: community × layer
layers = network.get_layers()
comm_ids = sorted(set(communities.values()))

composition = defaultdict(lambda: defaultdict(int))
for (node, layer), comm_id in communities.items():
    composition[comm_id][layer] += 1

# Convert to DataFrame for easier manipulation
data = []
for comm_id in comm_ids:
    row = {'community': comm_id}
    for layer in layers:
        row[layer] = composition[comm_id][layer]
    row['total'] = sum(composition[comm_id].values())
    data.append(row)

df_comp = pd.DataFrame(data)

print("\nCommunity × Layer composition:")
print(df_comp.to_string(index=False))

# Calculate layer entropy for each community
print("\n" + "-" * 70)
print("Community layer diversity (entropy):")
print(f"{'Community':<12} {'Entropy':<10} {'Interpretation'}")
print("-" * 70)

for comm_id in comm_ids:
    # Calculate entropy: H = - p_i log(p_i)
    counts = [composition[comm_id][layer] for layer in layers]
    total = sum(counts)

    if total == 0:
        continue

    probs = np.array(counts) / total
    probs = probs[probs > 0]  # Remove zeros
    entropy = -np.sum(probs * np.log2(probs))
    max_entropy = np.log2(len(layers))  # Maximum possible entropy
    normalized_entropy = entropy / max_entropy if max_entropy > 0 else 0
```

```
    # Interpret
    if normalized_entropy > 0.9:
        interp = "Highly dispersed (spans all layers)"
    elif normalized_entropy > 0.5:
        interp = "Moderately dispersed (multi-layer)"
    else:
        interp = "Concentrated (layer-specific)"

    print(f"C{comm_id:<11} {entropy:<10.3f} {interp}")
```

**Expected output:**

```
CROSS-LAYER COMMUNITY ANALYSIS
======================================================================

Community × Layer composition:
 community  layer1  layer2  layer3  total
         0      15      14      16     45
         1      18      20       0     38
         2       0       0      22     22
         3       7       6       2     15

----------------------------------------------------------------------
Community layer diversity (entropy):
Community    Entropy    Interpretation
----------------------------------------------------------------------
C0           1.585      Highly dispersed (spans all layers)
C1           0.997      Moderately dispersed (multi-layer)
C2           0.000      Concentrated (layer-specific)
C3           1.252      Moderately dispersed (multi-layer)
```

**Inter-layer bridges:**

Identify nodes that connect different communities across layers:

```
print("\n" + "=" * 70)
print("INTER-LAYER BRIDGE ANALYSIS")
print("=" * 70)

# For each node, check if it belongs to different communities in different layers
node_communities = defaultdict(dict)  # node -> layer -> comm_id

for (node, layer), comm_id in communities.items():
    node_communities[node][layer] = comm_id

# Identify bridge nodes
bridge_nodes = []
for node, layer_comms in node_communities.items():
    if len(layer_comms) < 2:
        continue

    # Check if community IDs differ across layers
```

```
    comm_ids = set(layer_comms.values())
    if len(comm_ids) > 1:
        bridge_nodes.append({
            'node': node,
            'n_layers': len(layer_comms),
            'n_communities': len(comm_ids),
            'assignments': dict(layer_comms)
        })

print(f"\nBridge nodes (spanning multiple communities across layers):")
print(f"  Total nodes: {len(node_communities)}")
print(f"  Bridge nodes: {len(bridge_nodes)} ({len(bridge_nodes)/len(node_communities)*100:

# Show examples
print(f"\n  Top 10 bridge nodes:")
print(f"  {'Node':<15} {'Layers':<10} {'Communities':<15} {'Assignments'}")
print("  " + "-" * 65)

bridge_nodes_sorted = sorted(bridge_nodes, key=lambda x: x['n_communities'], reverse=True)
for item in bridge_nodes_sorted[:10]:
    node = item['node']
    n_layers = item['n_layers']
    n_comms = item['n_communities']
    assignments = ', '.join([f"{l}:C{c}" for l, c in sorted(item['assignments'].items())])
    print(f"  {str(node):<15} {n_layers:<10} {n_comms:<15} {assignments}")
```

**Expected output:**

```
======================================================================
INTER-LAYER BRIDGE ANALYSIS
======================================================================

Bridge nodes (spanning multiple communities across layers):
  Total nodes: 40
  Bridge nodes: 12 (30.0%)

  Top 10 bridge nodes:
  Node            Layers     Communities     Assignments
  -----------------------------------------------------------------
  A5              3          3               layer1:C0, layer2:C1, layer3:C2
  B12             3          2               layer1:C0, layer2:C1, layer3:C1
  C3              3          2               layer1:C1, layer2:C0, layer3:C0
  D7              2          2               layer1:C0, layer2:C3
  E9              3          2               layer1:C3, layer2:C1, layer3:C1
  F4              2          2               layer1:C1, layer2:C0
  G8              3          2               layer1:C0, layer2:C0, layer3:C2
  H2              2          2               layer1:C3, layer2:C0
  I6              3          2               layer1:C1, layer2:C1, layer3:C2
  J11             2          2               layer1:C0, layer2:C1
```

**Community connectivity graph:**

Build a meta-graph where nodes are communities and edges represent inter-layer bridges:

```
import networkx as nx
import matplotlib.pyplot as plt

# Build community connectivity graph
G_meta = nx.Graph()

# Add community nodes
for comm_id in comm_ids:
    G_meta.add_node(f"C{comm_id}")

# Add edges for bridge nodes
for item in bridge_nodes:
    comms = list(item['assignments'].values())
    # Connect all pairs of communities this node bridges
    for i in range(len(comms)):
        for j in range(i+1, len(comms)):
            c1, c2 = f"C{comms[i]}", f"C{comms[j]}"
            if G_meta.has_edge(c1, c2):
                G_meta[c1][c2]['weight'] += 1
            else:
                G_meta.add_edge(c1, c2, weight=1)

print(f"\n" + "=" * 70)
print("COMMUNITY CONNECTIVITY")
print("=" * 70)
print(f"\nCommunity-level connectivity:")
print(f"  Communities: {G_meta.number_of_nodes()}")
print(f"  Inter-community bridges: {G_meta.number_of_edges()}")

if G_meta.number_of_edges() > 0:
    print(f"\n  Strongest bridges (top 5):")
    edges_sorted = sorted(G_meta.edges(data=True), key=lambda x: x[2]['weight'], reverse=Tru
    for c1, c2, data in edges_sorted[:5]:
        print(f"    {c1}  {c2}: {data['weight']} bridge nodes")

    # Visualize meta-graph
    plt.figure(figsize=(8, 8))
    pos = nx.spring_layout(G_meta, seed=42)

    # Edge widths proportional to weight
    weights = [G_meta[u][v]['weight'] for u, v in G_meta.edges()]
    max_weight = max(weights) if weights else 1
    edge_widths = [3 * w / max_weight for w in weights]

    nx.draw_networkx_nodes(G_meta, pos, node_size=800, node_color='lightblue')
    nx.draw_networkx_labels(G_meta, pos, font_size=12, font_weight='bold')
    nx.draw_networkx_edges(G_meta, pos, width=edge_widths, alpha=0.6)

    # Edge labels
```

```
    edge_labels = {(u, v): f"{G_meta[u][v]['weight']}" for u, v in G_meta.edges()}
    nx.draw_networkx_edge_labels(G_meta, pos, edge_labels, font_size=8)

    plt.title('Community Connectivity Meta-Graph\n(Edge width = number of bridge nodes)', fo
    plt.axis('off')
    plt.tight_layout()
    plt.savefig('community_connectivity.png', dpi=300, bbox_inches='tight')
    plt.show()
    print(f"\n  Visualization saved to: community_connectivity.png")
```

**Expected output:**

```
======================================================================
COMMUNITY CONNECTIVITY
======================================================================

Community-level connectivity:
  Communities: 4
  Inter-community bridges: 5

  Strongest bridges (top 5):
    C0  C1: 5 bridge nodes
    C1  C2: 3 bridge nodes
    C0  C3: 2 bridge nodes
    C1  C3: 1 bridge nodes
    C0  C2: 1 bridge nodes

  Visualization saved to: community_connectivity.png
```

**Use cases:**

- **Biological networks:** Proteins bridging functional modules across different interaction types
- **Social networks:** Individuals connecting different social circles across contexts
- **Transportation:** Transfer hubs connecting regional clusters across transport modes

### Quality Metrics

**Why quality metrics matter:**

Quality metrics help you:

1. **Compare algorithms** objectively
2. **Tune parameters** (e.g., choosing optimal $\omega$ in multilayer Louvain)
3. **Validate results** (high Q suggests real structure, not random fluctuations)
4. **Detect overfitting** (too many tiny communities = over-segmentation)

### Compute Modularity

**Single-layer modularity:**

For flattened networks, use the Newman-Girvan modularity:

$$Q = \frac{1}{2m} \sum_{ij} \left[ A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
from py3plex.algorithms.community_detection.community_louvain import modularity
import networkx as nx

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

# Detect communities
communities = louvain_communities(network)

# Convert to NetworkX for modularity calculation
G = nx.Graph()
for edge in network.core_network.edges():
    G.add_edge(edge[0], edge[1])

# Calculate modularity
Q = modularity(communities, G, weight='weight')

print(f"Modularity Q: {Q:.4f}")

# Interpretation
if Q > 0.7:
    print("  Interpretation: Excellent community structure")
elif Q > 0.4:
    print("  Interpretation: Strong community structure")
elif Q > 0.2:
    print("  Interpretation: Moderate community structure")
else:
    print("  Interpretation: Weak or no community structure")
```

**Expected output:**

```
Modularity Q: 0.4234
  Interpretation: Strong community structure
```

**Multilayer modularity:**

For multilayer networks, use the generalized modularity that accounts for inter-layer coupling:

```
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer,
    multilayer_modularity
)

# Run multilayer Louvain
communities = louvain_multilayer(
    network,
    gamma=1.0,
    omega=1.0,
    random_state=42
)

# Calculate multilayer modularity
Q_multi = multilayer_modularity(
    network,
    communities,
    gamma=1.0,
    omega=1.0
)

print(f"Multilayer modularity Q: {Q_multi:.4f}")
```

**Expected output:**

```
Multilayer modularity Q: 0.4589
```

**Modularity resolution:**

Modularity has a **resolution limit**: in undirected graphs it struggles to detect communities much smaller than roughly $\sqrt{2m}$ (with $m$ edges). The resolution parameter $\gamma$ can help:

```
# Test different resolution parameters
print("Modularity vs. resolution:")
print(f"{'':<10} {'#Comm':<10} {'Q':<10} {'Avg Size':<10}")
print("-" * 45)

for gamma in [0.5, 1.0, 1.5, 2.0]:
    comms = louvain_multilayer(
        network, gamma=gamma, omega=1.0, random_state=42
    )
    n_comms = len(set(comms.values()))
    Q = multilayer_modularity(network, comms, gamma=gamma, omega=1.0)
    avg_size = len(comms) / n_comms

    print(f"{gamma:<10.1f} {n_comms:<10} {Q:<10.4f} {avg_size:<10.1f}")
```

**Expected output:**

```
Modularity vs. resolution:
           #Comm      Q          Avg Size
---------------------------------------------
```

```
0.5         3           0.3456      40.0
1.0         5           0.4589      24.0
1.5         8           0.4123      15.0
2.0         12          0.3678      10.0
```

**Interpretation:**

- **Lower :** Fewer, larger communities (under-segmentation)
- **Higher :** More, smaller communities (over-segmentation)
- **Optimal :** Maximum Q (but check that communities are meaningful!)

### Additional Quality Metrics

**1. Coverage** (fraction of edges within communities):

```python
def calculate_coverage(network, communities):
    """Fraction of edges within communities."""
    intra_edges = 0
    total_edges = 0

    for source, target in network.core_network.edges():
        total_edges += 1
        if communities.get(source) == communities.get(target):
            intra_edges += 1

    return intra_edges / total_edges if total_edges > 0 else 0

coverage = calculate_coverage(network, communities)
print(f"Coverage: {coverage:.4f}  (fraction of intra-community edges)")
```

**Expected output:**

```
Coverage: 0.8234  (fraction of intra-community edges)
```

**2. Performance** (combines intra-community edges and inter-community non-edges):

```python
def calculate_performance(network, communities):
    """Performance metric (Fortunato 2010)."""
    nodes = list(communities.keys())
    n = len(nodes)

    # Count intra-community edges and inter-community non-edges
    intra_edges = 0
    inter_non_edges = 0
    total_pairs = 0

    for i in range(len(nodes)):
        for j in range(i+1, len(nodes)):
            node1, node2 = nodes[i], nodes[j]
            same_community = communities[node1] == communities[node2]
            is_edge = network.core_network.has_edge(node1, node2)
```

```
            if same_community and is_edge:
                intra_edges += 1
            elif not same_community and not is_edge:
                inter_non_edges += 1

            total_pairs += 1

    return (intra_edges + inter_non_edges) / total_pairs if total_pairs > 0 else 0

performance = calculate_performance(network, communities)
print(f"Performance: {performance:.4f}")
```

**Expected output:**

```
Performance: 0.7456
```

**Tip:** The double loop is $O(n^2)$ over nodes; use it on moderate-size graphs or sample nodes when the network is huge.

**3. Conductance** (quality of community boundaries):

```
def calculate_conductance(network, communities, comm_id):
    """Conductance of a specific community (lower is better)."""
    comm_nodes = [n for n, c in communities.items() if c == comm_id]

    if not comm_nodes:
        return None

    # Count edges
    internal_edges = 0
    boundary_edges = 0

    for node in comm_nodes:
        neighbors = list(network.core_network.neighbors(node))
        for neighbor in neighbors:
            if communities.get(neighbor) == comm_id:
                internal_edges += 0.5  # Count each edge once
            else:
                boundary_edges += 1

    volume = internal_edges * 2 + boundary_edges  # Volume of the community
    return boundary_edges / volume if volume > 0 else 0

# Calculate for all communities
print("\nConductance per community (lower = better defined):")
for comm_id in sorted(set(communities.values())):
    cond = calculate_conductance(network, communities, comm_id)
    if cond is not None:
        print(f"  Community {comm_id}: {cond:.4f}")
```

**Expected output:**

```
Conductance per community (lower = better defined):
  Community 0: 0.1234
  Community 1: 0.2456
  Community 2: 0.0987
  Community 3: 0.3123
  Community 4: 0.1789
```

**4. Null model comparison** (compare to random partitions):

```python
import random

# Calculate Q for real partition
Q_real = multilayer_modularity(network, communities, gamma=1.0, omega=1.0)

# Generate random partitions and calculate Q
nodes = list(communities.keys())
n_communities = len(set(communities.values()))

Q_random = []
for trial in range(100):
    # Random partition with same number of communities
    random_comms = {node: random.randint(0, n_communities-1) for node in nodes}
    Q_rand = multilayer_modularity(network, random_comms, gamma=1.0, omega=1.0)
    Q_random.append(Q_rand)

Q_rand_mean = np.mean(Q_random)
Q_rand_std = np.std(Q_random)
z_score = (Q_real - Q_rand_mean) / Q_rand_std if Q_rand_std > 0 else 0

print(f"\nNull model comparison:")
print(f"  Real Q: {Q_real:.4f}")
print(f"  Random Q (mean ± std): {Q_rand_mean:.4f} ± {Q_rand_std:.4f}")
print(f"  Z-score: {z_score:.2f}")

if z_score > 3:
    print(f"  Interpretation: Highly significant (real structure)")
elif z_score > 2:
    print(f"  Interpretation: Significant (likely real structure)")
else:
    print(f"  Interpretation: Not significant (could be random)")
```

**Expected output:**

```
Null model comparison:
  Real Q: 0.4589
  Random Q (mean ± std): 0.0023 ± 0.0145
  Z-score: 31.49
  Interpretation: Highly significant (real structure)
```

**Summary of metrics:**

- **Modularity (Q):** Overall quality, general-purpose

- **Coverage:** Simple interpretability (% internal edges)
- **Performance:** Balances true positives and true negatives
- **Conductance:** Community boundary quality (per-community)
- **Null model:** Statistical significance test
- **Replica Consistency (RC):** Multilayer coherence guardrail
- **Layer Entropy (H):** Degeneracy guardrail for multilayer networks

**Recommendation:** Always report modularity + at least one other metric to get a complete picture.

### Multilayer Quality Metrics (Guardrails)

For multilayer community detection, two specialized quality metrics serve as **guardrails** against degenerate partitions:

1. **Replica Consistency (RC)**: Measures whether replicas of the same node across layers are assigned to the same community
2. **Layer Entropy (H)**: Measures the balance of community sizes within each layer, averaged across layers

**What are guardrails?**

Guardrails are metrics that detect pathological partitions but are not primary optimization objectives. They ensure that:

- Communities respect node identity across layers (RC)
- Partitions are not degenerate giant clusters or extreme fragmentation (H)

**Replica Consistency:**

Formula: For each node v with replicas in L_v layers (|L_v|  2):

$$RC(v) = \frac{2}{|L_v|(|L_v|-1)} \sum_{\ell_i<\ell_j} 1[c(v,\ell_i) = c(v,\ell_j)]$$

$$RC = \frac{1}{|\{v : |L_v| \geq 2\}|} \sum_v RC(v)$$

RC ranges from 0 to 1:

- **RC = 1.0**: All replicas of each node have the same community (perfect coherence)
- **RC = 0.0**: No agreement (random assignment)
- **RC  (0,1)**: Partial consistency

**Layer Entropy:**

Formula: For each layer  with C_ communities:

$$H_\ell = -\sum_i p_i^\ell \log(p_i^\ell) / \log(|C_\ell|)$$

$$H = \text{mean}_\ell(H_\ell), \quad \text{clipped to } [0.1, 0.9]$$

H ranges from 0.1 to 0.9 (clipped):

- **H  0.9**: Balanced communities in all layers
- **H  0.1**: Giant cluster or extreme fragmentation (degenerate)

- **H (0.1,0.9)**: Normal partition

**Usage example:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from py3plex.algorithms.community_detection.multilayer_quality_metrics import (
    replica_consistency,
    layer_entropy
)

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

# Detect communities
communities = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)

# Compute multilayer quality metrics
rc = replica_consistency(communities, network)
h = layer_entropy(communities, network)

print(f"Replica Consistency: {rc:.4f}")
print(f"Layer Entropy: {h:.4f}")

# Interpret
if rc > 0.8:
    print("   High replica consistency (nodes coherent across layers)")
elif rc < 0.3:
    print("   Low replica consistency (nodes fragmented across layers)")
else:
    print("  → Moderate replica consistency")

if h > 0.7:
    print("   High layer entropy (balanced communities)")
elif h < 0.3:
    print("   Low layer entropy (degenerate partition)")
else:
    print("  → Moderate layer entropy")
```

**Expected output:**

```
Replica Consistency: 0.8765
Layer Entropy: 0.7234
   High replica consistency (nodes coherent across layers)
   High layer entropy (balanced communities)
```

**Comparison: good vs. bad partitions:**

```
# Good partition: coherent and balanced
good_partition = {
    ('A', 'layer1'): 0, ('A', 'layer2'): 0, ('A', 'layer3'): 0,
    ('B', 'layer1'): 1, ('B', 'layer2'): 1, ('B', 'layer3'): 1,
    ('C', 'layer1'): 0, ('C', 'layer2'): 0, ('C', 'layer3'): 0,
    ('D', 'layer1'): 1, ('D', 'layer2'): 1, ('D', 'layer3'): 1,
}

# Bad partition 1: giant cluster (low entropy)
giant_cluster = {
    ('A', 'layer1'): 0, ('A', 'layer2'): 0, ('A', 'layer3'): 0,
    ('B', 'layer1'): 0, ('B', 'layer2'): 0, ('B', 'layer3'): 0,
    ('C', 'layer1'): 0, ('C', 'layer2'): 0, ('C', 'layer3'): 0,
    ('D', 'layer1'): 0, ('D', 'layer2'): 0, ('D', 'layer3'): 0,
}

# Bad partition 2: fragmented replicas (low RC)
fragmented = {
    ('A', 'layer1'): 0, ('A', 'layer2'): 1, ('A', 'layer3'): 2,
    ('B', 'layer1'): 3, ('B', 'layer2'): 4, ('B', 'layer3'): 5,
    ('C', 'layer1'): 6, ('C', 'layer2'): 7, ('C', 'layer3'): 8,
    ('D', 'layer1'): 9, ('D', 'layer2'): 10, ('D', 'layer3'): 11,
}

print("Good partition:")
print(f"  RC={replica_consistency(good_partition, network):.3f}, H={layer_entropy(good_par

print("\nGiant cluster (degenerate):")
print(f"  RC={replica_consistency(giant_cluster, network):.3f}, H={layer_entropy(giant_clu

print("\nFragmented replicas (incoherent):")
print(f"  RC={replica_consistency(fragmented, network):.3f}, H={layer_entropy(fragmented,
```

**Expected output:**

```
Good partition:
  RC=1.000, H=0.900

Giant cluster (degenerate):
  RC=1.000, H=0.100

Fragmented replicas (incoherent):
  RC=0.000, H=0.900
```

**Integration with AutoCommunity:**

These metrics are automatically included in AutoCommunity's multi-objective evaluation:

```
from py3plex.algorithms.community_detection import auto_select_community

# AutoCommunity will use these metrics as guardrails
result = auto_select_community(
```

```
    network,
    fast=True,
    seed=42
)

# Access metrics from evaluation matrix
eval_matrix = result.evaluation_matrix

print("All metrics computed:")
print(eval_matrix[['algorithm_id', 'modularity', 'replica_consistency', 'layer_entropy']])
```

**When to use these metrics:**

- **RC**: When node identity across layers is important (e.g., same protein in different interaction types)
- **H**: To detect degenerate partitions (giant clusters or extreme fragmentation)
- **Both**: As sanity checks after multilayer community detection
- **AutoCommunity**: These are included by default; they contribute ~22% total weight in the default metric set

**Typical ranges:**

- **RC** $> 0.8$: Excellent replica consistency
- **RC** $\in [0.5, 0.8]$: Good consistency
- **RC** $< 0.5$: Poor consistency (investigate layer-specific communities)
- **H** $> 0.7$: Well-balanced communities
- **H** $\in [0.3, 0.7]$: Normal range
- **H** $< 0.3$: Degenerate partition (too few or too many communities)

**Performance characteristics:**

- **RC**: O(_v |L_v|^2) where L_v is layers per node (efficient count-based implementation)
- **H**: O(|assignments|) linear in node-layer assignments
- **Both**: Fast enough for large networks (< 1 second for 10,000 node-layers)

**Label-permutation invariance:**

RC is **label-permutation invariant**: it only compares community labels within each node across layers. Different algorithms can produce arbitrary label numbering, and RC will still work correctly.

**Summary of metrics:**

- **Modularity (Q):** Overall quality, general-purpose
- **Coverage:** Simple interpretability (% internal edges)
- **Performance:** Balances true positives and true negatives
- **Conductance:** Community boundary quality (per-community)
- **Null model:** Statistical significance test

**Recommendation:** Always report modularity + at least one other metric to get a complete picture.

### CLI Cross-Reference (Optional)

py3plex provides command-line tools for quick community detection without writing Python code.

**Basic usage:**

```
# Detect communities using Louvain (default algorithm)
py3plex community datasets/network.edgelist \
    --algorithm louvain \
    --output communities.json

# Using Infomap (requires Infomap binary installed)
py3plex community datasets/network.edgelist \
    --algorithm infomap \
    --output communities.json

# Using Label Propagation (fast for large networks)
py3plex community datasets/network.edgelist \
    --algorithm label_prop \
    --output communities.json

# With custom resolution parameter for Louvain
py3plex community datasets/network.edgelist \
    --algorithm louvain \
    --resolution 1.5 \
    --output communities.json
```

**Available algorithms:**

- `louvain`: Fast Louvain method (default) - optimizes modularity on flattened network
- `infomap`: Infomap algorithm - requires Infomap binary (https://www.mapequation.org/infomap/)
- `label_prop`: Label propagation - very fast, suitable for large networks

**Output format:**

The CLI outputs JSON files with structure:

```
{
  "algorithm": "louvain",
  "num_communities": 5,
  "communities": {
    "node1": 0,
    "node2": 0,
    "node3": 1,
    ...
  },
  "community_sizes": {
    "0": 42,
    "1": 27,
    ...
  }
}
```

**Note on multilayer networks:**

The current CLI `community` command operates on flattened networks. For multilayer-specific community detection (with inter-layer coupling), use the Python API with `louvain_multilayer()` as shown in the examples above. Future CLI versions may add multilayer support.

**Viewing results:**

After running the CLI command, you can analyze the JSON output:

```
# View community statistics
py3plex community network.edgelist --algorithm louvain
# Output printed to console if no --output specified
```

For full CLI documentation, see ../tutorials/cli_usage or ../deployment/cli_usage.

### Next Steps

**Further reading:**

- **Algorithms:** *Algorithm Landscape* - Deep dive into community detection theory
- **Visualization:** *How to Visualize Multilayer Networks* - Advanced community visualization techniques
- **Benchmark:** ../tutorials/community_detection - Compare with ground-truth communities
- **Temporal analysis:** ../tutorials/community_detection - Track community evolution over time

**Recommended workflows:**

1. **Exploratory:** Start with Louvain → visualize → if unsatisfied, try multilayer Louvain or Infomap
2. **Publication:** Run multiple algorithms → compare → report consensus + metrics
3. **Large-scale:** Use label propagation for initial exploration → refine with Louvain on filtered subgraph
4. **Temporal:** Detect communities in snapshots → track with NMI → visualize with alluvial diagrams

**Common pitfalls:**

- **Resolution limit:** Modularity cannot detect communities smaller than $\sqrt{m}$
- **Non-determinism:** Many algorithms are stochastic; always set random seeds for reproducibility
- **Overfitting:** Too many tiny communities suggests over-segmentation; try lower resolution
- **Layer coupling:** For multilayer networks, always try multiple $\omega$ values

### Automatic Algorithm Selection (AutoCommunity)

**When you don't know which algorithm to use:** py3plex provides automatic algorithm selection that evaluates multiple algorithms on multiple quality metrics and picks the winner using a "most wins" decision engine.

**Quick example:**

```
from py3plex.algorithms.community_detection import auto_select_community

# Automatically select best algorithm
result = auto_select_community(network, fast=True, seed=42)

# See why it won
print(result.explain())

# View leaderboard
print(result.leaderboard)

# Use the partition
network.assign_partition(result.partition)
```

**With DSL:**

```
from py3plex.dsl import Q

result = Q.communities().auto_select(fast=True, seed=42).execute(network)
print(result.explain())
```

**How it works:**

1. Detects available algorithms (Leiden, Louvain, etc.)
2. Runs candidates with parameter grids (gamma, omega)
3. Evaluates on multiple metrics (modularity, coverage, stability, etc.)
4. Selects winner by pairwise "most wins" across metrics
5. Applies bucket caps to prevent any metric category from dominating

**Key parameters:**

- `fast=True`: Use smaller parameter grids (default)
- `max_candidates=10`: Maximum algorithms to evaluate
- `seed=0`: Random seed for determinism
- `uq=False`: Enable uncertainty quantification for stability metrics

**Result object:**

- `result.partition`: Winning partition
- `result.algorithm`: Algorithm name and parameters
- `result.leaderboard`: DataFrame ranking all candidates
- `result.explain()`: Natural language explanation of why it won
- `result.to_dict()`: Serializable dictionary

**When to use:** Exploratory analysis, comparing algorithms fairly, reproducible algorithm choice with provenance.

See AGENTS.md for complete API documentation.

### Hierarchical Flow-Based Community Detection

**Goal:** Detect hierarchical community structure based on flow/diffusion dynamics rather than modularity optimization.

**What is hierarchical flow-based detection?**

Unlike modularity-based methods (Louvain, Leiden) that optimize a single quality function, hierarchical flow-based community detection (HFCD) identifies communities as sets of nodes that **retain probability mass** under random walk dynamics for longer times than expected by chance. The hierarchy emerges naturally from observing how flow-based communities merge as diffusion time increases.

**Key concepts:**

- **Communities = flow retention**: Groups that trap random walkers internally
- **Hierarchy = time scales**: Different community structures at different diffusion times
- **No resolution parameter**: Hierarchy levels correspond to stability plateaus
- **Multilayer-aware**: Native support for interlayer coupling via $\alpha$ parameter

**Mathematical foundation:**

For a multilayer network, construct transition operator:

$$P = \alpha P_{\text{intra}} + (1 - \alpha) P_{\text{inter}}$$

where $\alpha \in [0, 1]$ controls interlayer coupling. Flow affinity at scale $t$:

$$F(t) = \sum_{k=1}^{t} P^k, \quad S_{ij}(t) = \frac{F_{ij}(t) + F_{ji}(t)}{2}$$

Community quality (flow retention):

$$\text{FlowRetention}(C, t) = \frac{\text{internal flow}}{\text{total flow}}$$

**Basic example:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection import flow_hierarchical_communities

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "datasets/synthetic_multilayer.txt",
    input_type="multiedgelist"
)

# Run hierarchical flow detection
result = flow_hierarchical_communities(
    network,
```

```
    approx="mc",       # Monte Carlo (scalable) or "exact"
    alpha=0.8,         # Interlayer coupling (0=full, 1=independent)
    n_walks=100,       # MC walks per node
    seed=42            # Reproducibility
)

# Access best partition (maximum stability)
partition = result.get_partition()

# Get partition with specific number of communities
partition_3 = result.get_flat_partition(n_communities=3)

# Display hierarchy
print(result.summary())

# Explore all hierarchy levels
for scale, part in result.hierarchy_levels.items():
    stability = result.stability_scores[scale]
    n_comms = len(set(part.values()))
    print(f"Scale {scale}: {n_comms} communities (stability={stability:.3f})")
```

**Expected output:**

```
======================================================================
Hierarchical Flow-Based Community Detection Results
======================================================================
Total nodes: 120
Hierarchy levels detected: 4
Scale range: [1.00, 8.00]

Best partition (max stability):
  Scale: 4.00
  Stability: 0.8523
  Communities: 5

Scale 1.0: 120 communities (stability=1.000)
Scale 2.0: 12 communities (stability=0.912)
Scale 4.0: 5 communities (stability=0.852)
Scale 8.0: 2 communities (stability=0.734)
```

**Result structure:**

The `FlowHierarchyResult` object contains:

- `dendrogram`: List of merge events (comm_i, comm_j, scale, stability_before, stability_after)
- `hierarchy_levels`: Dict[scale → partition] at all detected levels
- `stability_scores`: Flow retention at each scale (range [0,1])
- `merge_scales`: Array of scales where merges occurred
- `metadata`: Algorithm config (flow_type, alpha, approx, seed, scale_schedule)

**Multilayer-specific parameters:**

```
# High alpha (0.9): Layers mostly independent
result_high = flow_hierarchical_communities(net, alpha=0.9, seed=42)

# Low alpha (0.2): Strong interlayer coupling
result_low = flow_hierarchical_communities(net, alpha=0.2, seed=42)

# Communities bridge layers based on alpha
```

**Approximation methods:**

- **Monte Carlo (approx="mc")**: Memory-efficient O(n × walks), slight randomness (use seed)
- **Exact (approx="exact")**: Deterministic O($n^2$), memory-intensive for large networks

**Algorithm complexity:**

- **Time**: O(t × m × k) where t=scales, m=edges, k=merges
- **Space**: O($n^2$) exact, O(n × walks) Monte Carlo
- **Recommended**: n < 10,000 (exact) or n < 100,000 (MC)

**When to use:**

- Understanding hierarchical organization at multiple scales
- Networks where modularity optimization may miss structure
- Multilayer networks with different layer coupling strengths
- Temporal networks where communities evolve across time slices
- Comparing with flow-based baselines (Infomap, Markov Stability)

**Comparison with other methods:**

| Aspect | Flow Hierarchy | Louvain/Leiden |
|---|---|---|
| Quality metric | Flow retention | Modularity |
| Hierarchy | Full dendrogram | Flat partition |
| Scale selection | Stability plateaus | Resolution parameter |
| Multilayer | Native ( coupling) | Requires parameter |
| Determinism | Seed-controlled (MC/exact) | Heuristic-dependent |

**Examples:**

See the py3plex examples directory[11] for flow hierarchy usage examples.

**References:**


- Schaub et al., "Markov Dynamics as a Zooming Lens for Multiscale Community Detection", *PLoS ONE* 7(2): e32210 (2012)
- Delvenne et al., "Stability of graph communities across time scales", *PNAS* 107(29): 12755-12760 (2010)
- Lambiotte et al., "Random Walks, Markov Processes and the Multiscale Modular Organization of Complex Networks", *IEEE Trans. Network Science and Engineering* 1(2): 76-90


[11] https://github.com/SkBlaz/py3plex/tree/master/examples

(2014)

**Community detection checklist:**

- [ ] Run at least 2 different algorithms
- [ ] Calculate modularity and at least one other quality metric
- [ ] Visualize size distribution to check for over/under-segmentation
- [ ] Compare with null model to ensure statistical significance
- [ ] For multilayer: test multiple $\omega$ values
- [ ] Export results to CSV for downstream analysis
- [ ] Document random seeds for reproducibility

**Questions?**

- GitHub Issues: https://github.com/SkBlaz/py3plex/issues
- Documentation: https://skblaz.github.io/py3plex/
- Examples: `examples/communities/` directory in the repository

**Key References:**


- Blondel, V. D., Guillaume, J. L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008.
- Mucha, P. J., Richardson, T., Macon, K., Porter, M. A., & Onnela, J. P. (2010). Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328(5980), 876-878.
- Rosvall, M., & Bergstrom, C. T. (2008). Maps of random walks on complex networks reveal community structure. *Proceedings of the National Academy of Sciences*, 105(4), 1118-1123.
- Raghavan, U. N., Albert, R., & Kumara, S. (2007). Near linear time algorithm to detect community structures in large-scale networks. *Physical Review E*, 76(3), 036106.


### 3.3.6 How to Simulate Multilayer Dynamics

This guide presents py3plex as a comprehensive framework for simulating dynamical processes on multilayer networks. We cover epidemic models (SIR, SIS), diffusion processes (random walks), and custom dynamics implementations—all with full multilayer support.

**Run this guide online**

You can run this tutorial in your browser without any local installation:

[12] See the py3plex examples[13] for dynamics simulation examples.

**Prerequisites:** A loaded multilayer network (see *How to Load and Build Networks*).

[12] https://colab.research.google.com/github/SkBlaz/py3plex/blob/master/notebooks/simulate_dynamics.ipynb

[13] https://github.com/SkBlaz/py3plex/tree/master/examples

### Overview

Multilayer dynamics in py3plex enable you to model processes where interactions occur across multiple types of relationships simultaneously. Typical use cases include:

- **Epidemic spread**: Disease transmission through household, workplace, and social contact layers
- **Information diffusion**: News spreading via online and offline communication channels
- **Random walks**: Navigation and search processes across interconnected network layers
- **Opinion dynamics**: Belief propagation through multiple social contexts

**Simulation Pipeline**

The typical workflow for running dynamics simulations consists of five steps:

1. **Load or build** a multilayer network using `py3plex.core.multinet`
2. **Choose a dynamics model** (e.g., `SIRDynamics`, `SISDynamics`, `RandomWalkDynamics`)
3. **Configure parameters** and initial conditions (infection rates, starting nodes, etc.)
4. **Run the simulation** via `.run(steps=...)` with a fixed seed for reproducibility
5. **Inspect the results** object using `.get_measure(...)` to extract time series and summary statistics

This pipeline provides a consistent interface across all dynamics models while allowing specialized configurations for multilayer network effects.

### Common Concepts & API

#### Node and Layer Representation

Dynamics in py3plex operate on the **supra-network** representation, where multilayer nodes are encoded as `(node_id, layer)` tuples. For example, Alice in the "friends" layer is represented as `('Alice', 'friends')`.

When you pass a multilayer network to a dynamics model:

- Node states are tracked per node-layer pair
- Infection, diffusion, or walker movements respect both intra-layer and inter-layer edges
- You can query per-layer statistics after simulation

For single-layer NetworkX graphs, nodes are used directly without tuple encoding.

#### Time and Synchrony

All dynamics models in this guide use **discrete-time, synchronous updates**:

- Time advances in integer steps: $t = 0, 1, 2, \ldots$
- At each step, all nodes update simultaneously based on the state at time $t$
- This differs from continuous-time (Gillespie) models, which are also available in `py3plex.dynamics`

### Randomness & Reproducibility

Every dynamics model has a dedicated random number generator controlled by `.set_seed(seed)`:

```
dynamics.set_seed(42)
results = dynamics.run(steps=100)
```

Setting the seed ensures:

- Reproducible initial conditions (which nodes start infected, which are susceptible)
- Reproducible stochastic transitions (infection events, recovery events, random walk steps)
- Consistent results across multiple runs with the same seed

### Network Requirements

Dynamics models work with:

- **py3plex multilayer networks**: Full support for inter-layer and intra-layer edges
- **NetworkX graphs**: Single-layer networks (directed or undirected)
- **Edge weights**: Epidemic models currently treat edges as unweighted contacts. `RandomWalkDynamics` samples neighbors uniformly, so weights are ignored unless you implement a weighted variant.

### Common Base Interface

All dynamics classes inherit from `DynamicsProcess` (discrete-time) or `ContinuousTimeProcess` (continuous-time). The core methods are:

```
from py3plex.dynamics import SIRDynamics

# Initialize with network and parameters
dynamics = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.05)

# Set seed for reproducibility
dynamics.set_seed(42)

# Run for specified number of steps
results = dynamics.run(steps=100)

# Extract measures
prevalence = results.get_measure("prevalence")
state_counts = results.get_measure("state_counts")
```

The `results` object is always a `DynamicsResult` instance with standardized measure extraction methods. Time series returned by `.run(steps=k)` include the initial state at `t=0`, so arrays have length `k + 1`.

### SIR Epidemic Model on Multilayer Networks

#### Formal Description

The **SIR (Susceptible-Infected-Recovered)** model describes epidemic spread with three compartments:

- **S (Susceptible)**: Nodes that can become infected
- **I (Infected)**: Nodes that are currently infectious
- **R (Recovered)**: Nodes with permanent immunity (absorbing state)

**Discrete-Time Update Rules** (synchronous):

1. **Recovery**: Each infected node recovers (I $\to$ R) with probability $\gamma \in (0, 1)$ per time step
2. **Infection**: Each susceptible node with k infected neighbors becomes infected (S $\to$ I) with probability

$$p_{\text{infect}} = 1 - (1 - \beta)^k$$

   where $\beta \in (0, 1)$ is the per-contact infection probability under independent transmission attempts from each infected neighbor
3. **Absorption**: Recovered nodes remain recovered (R $\to$ R) forever

**Multilayer Extension**

On multilayer networks, the infection pressure on a node aggregates across all layers:

- A susceptible node `(i, layer_A)` counts infected neighbors from both intra-layer edges (within `layer_A`) and inter-layer edges (to other layers)
- The built-in `SIRDynamics` uses a single $\beta$ value across all layers; to model heterogeneous layers, see *Layer-Specific and Time-Varying Parameters* for a small customization pattern

This means epidemics can spread within layers and jump between layers via inter-layer connections, mimicking real-world scenarios like workplace and household transmission.

#### Code Example

```
from py3plex.core import multinet
from py3plex.dynamics import SIRDynamics

# Create a simple two-layer network
network = multinet.multi_layer_network(directed=False)

# Add nodes in two layers
nodes = []
for i in range(20):
    nodes.append({'source': i, 'type': 'layer1'})
    nodes.append({'source': i, 'type': 'layer2'})
network.add_nodes(nodes)

# Add intra-layer edges (chain structure)
edges = []
for i in range(19):
    edges.append({'source': i, 'target': i+1,
```

```
                  'source_type': 'layer1', 'target_type': 'layer1'})
    edges.append({'source': i, 'target': i+1,
                  'source_type': 'layer2', 'target_type': 'layer2'})

# Add inter-layer edges (node replicas)
for i in range(20):
    edges.append({'source': i, 'target': i,
                  'source_type': 'layer1', 'target_type': 'layer2'})

network.add_edges(edges)

# Configure SIR model
sir = SIRDynamics(
    network,
    beta=0.3,     # Infection probability per contact
    gamma=0.1,    # Recovery probability per time step
    initial_infected=0.05  # 5% of nodes initially infected
)

# Run simulation with fixed seed
sir.set_seed(42)
results = sir.run(steps=100)

# Extract measures
prevalence = results.get_measure("prevalence")
state_counts = results.get_measure("state_counts")

# Print summary statistics
peak_time = prevalence.argmax()
print(f"Peak prevalence: {prevalence.max():.2%} at step {peak_time}")
print(f"Final susceptible: {state_counts['S'][-1]} nodes")
print(f"Final infected: {state_counts['I'][-1]} nodes")
print(f"Final recovered: {state_counts['R'][-1]} nodes")
print(f"Attack rate: {state_counts['R'][-1] / network.core_network.number_of_nodes():.2%}")
```

**Expected Output**

```
Peak prevalence: 35.00% at step 16
Final susceptible: 0 nodes
Final infected: 0 nodes
Final recovered: 40 nodes
Attack rate: 100.00%
```

*Note: Exact values depend on network structure and random seed. The example shows a complete outbreak where all nodes eventually get infected. On multilayer supra-networks, counts refer to node-layer pairs; collapse layers first if you need per-entity totals.*

### Time Series Visualization

Track the epidemic curve over time:

```
import matplotlib.pyplot as plt

# Get time series data
state_counts = results.get_measure("state_counts")
steps = range(len(state_counts['S']))

# Plot S, I, R trajectories
plt.figure(figsize=(10, 6))
plt.plot(steps, state_counts['S'], label='Susceptible', color='blue', linewidth=2)
plt.plot(steps, state_counts['I'], label='Infected', color='red', linewidth=2)
plt.plot(steps, state_counts['R'], label='Recovered', color='green', linewidth=2)

plt.xlabel('Time step', fontsize=12)
plt.ylabel('Number of nodes', fontsize=12)
plt.title('SIR Epidemic Dynamics on Multilayer Network', fontsize=14)
plt.legend()
plt.grid(alpha=0.3)
plt.savefig('sir_dynamics.png', dpi=300, bbox_inches='tight')
plt.show()
```

The plot shows the characteristic SIR pattern: S decreases monotonically, I rises then falls (forming a peak), and R increases monotonically until saturation. The peak prevalence and timing depend on , , and network structure.

### SIS Epidemic Model on Multilayer Networks

#### Formal Description

The **SIS (Susceptible-Infected-Susceptible)** model describes epidemics **without permanent immunity**:

- **S (Susceptible)**: Nodes that can become infected
- **I (Infected)**: Nodes that are currently infectious

**Discrete-Time Update Rules** (synchronous):

1. **Recovery**: Each infected node recovers (I $\rightarrow$ S) with probability $\gamma$ (or $\mu$) per time step
2. **Infection**: Each susceptible node with k infected neighbors becomes infected (S $\rightarrow$ I) with probability

$$p_{\text{infect}} = 1 - (1 - \beta)^k$$

   where $\beta \in (0, 1)$ is the transmission probability per contact
3. **No absorption**: Recovered nodes immediately return to susceptible state, allowing reinfection

**Endemic Equilibrium**

Unlike SIR (which eventually dies out), SIS can reach an **endemic equilibrium** where a steady

fraction of nodes remain infected. The basic reproduction number is:

$$R_0 = \frac{\beta}{\gamma}\langle k \rangle$$

where ⟨k⟩ is the average degree in a homogeneous-network, small-$\beta$ approximation (using $1 - (1 - \beta)^k \approx \beta k$). If $R_0 > 1$, endemicity is possible; if $R_0$  1, the epidemic dies out in this mean-field view (heterogeneous networks can shift the threshold).

**Multilayer Extension**

On multilayer networks, SIS infection pressure aggregates across all layers, similar to SIR. The endemic prevalence can differ across layers if you use layer-specific transmission rates.

### Code Example

```
from py3plex.core import multinet
from py3plex.dynamics import SISDynamics

# Use the same multilayer network as before
network = multinet.multi_layer_network(directed=False)

# Add nodes and edges (same as SIR example)
nodes = []
for i in range(20):
    nodes.append({'source': i, 'type': 'layer1'})
    nodes.append({'source': i, 'type': 'layer2'})
network.add_nodes(nodes)

edges = []
for i in range(19):
    edges.append({'source': i, 'target': i+1,
                  'source_type': 'layer1', 'target_type': 'layer1'})
    edges.append({'source': i, 'target': i+1,
                  'source_type': 'layer2', 'target_type': 'layer2'})
for i in range(20):
    edges.append({'source': i, 'target': i,
                  'source_type': 'layer1', 'target_type': 'layer2'})
network.add_edges(edges)

# Configure SIS model
sis = SISDynamics(
    network,
    beta=0.3,     # Infection probability per contact
    mu=0.1,       # Recovery probability (also accepts 'gamma')
    initial_infected=0.05
)

# Run for longer to reach equilibrium
sis.set_seed(42)
results = sis.run(steps=200)

# Extract prevalence time series
```

```
prevalence = results.get_measure("prevalence")

# Compute endemic prevalence (mean of last 50 steps)
endemic_prevalence = prevalence[-50:].mean()
print(f"Endemic prevalence: {endemic_prevalence:.2%}")
print(f"Std (last 50 steps): {prevalence[-50:].std():.3f}")

# Check if epidemic persisted
if endemic_prevalence > 0.01:
    print("Status: Endemic state reached")
else:
    print("Status: Epidemic died out")
```

**Expected Output**

```
Endemic prevalence: 0.00%
Std (last 50 steps): 0.000
Status: Epidemic died out
```

*Note: On sparse chain-like networks with moderate /, the epidemic often dies out due to low $R_0$. Try denser networks or higher  to observe endemic states.*

### Comparison with SIR

The key difference between SIS and SIR:

- **SIR**: Forms a peak and dies out (all nodes eventually become R or remain S)
- **SIS**: Can sustain long-term endemic prevalence if $R_0 > 1$

```
import matplotlib.pyplot as plt
import networkx as nx
from py3plex.dynamics import SISDynamics, SIRDynamics

# Use a denser network to see endemic behavior
G = nx.watts_strogatz_graph(n=100, k=6, p=0.1, seed=42)

# Run SIS
sis = SISDynamics(G, beta=0.3, mu=0.1, initial_infected=0.1)
sis.set_seed(42)
sis_results = sis.run(steps=150)
sis_prev = sis_results.get_measure("prevalence")

# Run SIR
sir = SIRDynamics(G, beta=0.3, gamma=0.1, initial_infected=0.1)
sir.set_seed(42)
sir_results = sir.run(steps=150)
sir_prev = sir_results.get_measure("prevalence")

# Plot comparison
plt.figure(figsize=(10, 6))
plt.plot(sis_prev, label='SIS (endemic)', color='red', linewidth=2)
plt.plot(sir_prev, label='SIR (dies out)', color='blue', linewidth=2)
```

```
plt.xlabel('Time step')
plt.ylabel('Prevalence')
plt.title('SIS vs SIR Epidemic Dynamics')
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

### Random Walk Dynamics

#### Formal Description

Random walk dynamics simulate the movement of a walker (or multiple walkers) on the network. At each time step:

1. The walker is at some node (or node-layer pair) `v`
2. With probability `lazy_probability`, the walker stays at `v`
3. With probability `1 - lazy_probability`, the walker moves to a uniformly random neighbor

**Multilayer Random Walks**

On multilayer networks:

- Walkers navigate both intra-layer edges (within a layer) and inter-layer edges (between layers)
- The state is a node-layer tuple, e.g., `(node_id, layer)`
- Transition probabilities are uniform over all neighbors (intra-layer + inter-layer)
- Edge weights in the graph are ignored in this default implementation; to bias transitions by weight, subclass and resample neighbors proportionally

Random walks are fundamental for:

- Computing stationary distributions (related to PageRank and eigenvector centrality)
- Modeling diffusion and search processes
- Estimating hitting times between nodes

#### Code Example

```
from py3plex.core import multinet
from py3plex.dynamics import RandomWalkDynamics

# Create a two-layer network
network = multinet.multi_layer_network(directed=False)

# Ring structure in layer1, star structure in layer2
n = 15
nodes = []
for i in range(n):
    nodes.append({'source': i, 'type': 'ring'})
    nodes.append({'source': i, 'type': 'star'})
network.add_nodes(nodes)
```

```
# Ring layer: circular edges
edges = []
for i in range(n):
    edges.append({'source': i, 'target': (i+1) % n,
                  'source_type': 'ring', 'target_type': 'ring'})

# Star layer: hub-spoke (node 0 is hub)
for i in range(1, n):
    edges.append({'source': 0, 'target': i,
                  'source_type': 'star', 'target_type': 'star'})

# Inter-layer edges
for i in range(n):
    edges.append({'source': i, 'target': i,
                  'source_type': 'ring', 'target_type': 'star'})

network.add_edges(edges)

# Configure random walk
walk = RandomWalkDynamics(
    network,
    start_node=(0, 'ring'),  # Start at node 0 in ring layer
    lazy_probability=0.1
)

# Run for 1000 steps
walk.set_seed(42)
results = walk.run(steps=1000)

# Get trajectory and compute visit counts
trajectory = results.get_measure("trajectory")
visit_counts = walk.visit_counts(trajectory)

# Sort by visit frequency
sorted_visits = sorted(visit_counts.items(), key=lambda x: x[1], reverse=True)

print(f"Total unique nodes visited: {len(visit_counts)}")
print(f"\\nTop 10 most visited nodes:")
for node, count in sorted_visits[:10]:
    print(f"  {node}: {count} visits ({count/len(trajectory):.2%})")
```

**Expected Output**

```
Total unique nodes visited: 30

Top 10 most visited nodes:
  (0, 'star'): 87 visits (8.69%)
  (0, 'ring'): 64 visits (6.39%)
  (1, 'ring'): 42 visits (4.20%)
  (14, 'ring'): 39 visits (3.90%)
```

```
  (1, 'star'): 37 visits (3.70%)
  (2, 'ring'): 36 visits (3.60%)
  (13, 'ring'): 35 visits (3.50%)
  (2, 'star'): 34 visits (3.40%)
  (3, 'star'): 32 visits (3.20%)
  (14, 'star'): 31 visits (3.10%)
```

*Note: The hub node in the star layer receives disproportionately high visits due to its high degree.*

### Analyzing Layer Switching

You can analyze how often the walker switches between layers:

```python
# Count layer switches
switches = 0
for i in range(len(trajectory) - 1):
    current_layer = trajectory[i][1]
    next_layer = trajectory[i + 1][1]
    if current_layer != next_layer:
        switches += 1

print(f"Total layer switches: {switches}")
print(f"Switch rate: {switches / (len(trajectory) - 1):.2%}")
```

This reveals how "coupled" the layers are through inter-layer edges. High switch rates indicate strong inter-layer connectivity.

### Layer-Specific and Time-Varying Parameters

#### Layer-Specific Transmission Rates

`SIRDynamics` and `SISDynamics` accept a single scalar `beta`. To model heterogeneous transmission across layers, subclass the process and choose a layer-dependent `beta` inside the susceptible transition:

```python
from py3plex.dynamics import SIRDynamics

class LayerAwareSIR(SIRDynamics):
    def __init__(self, graph, layer_betas, default_beta=0.2, **kwargs):
        super().__init__(graph, beta=default_beta, **kwargs)
        self.layer_betas = layer_betas
        self.transition_rules = dict(self.transition_rules)  # shallow copy

        def transition_S(node, state, neighbor_counts, rng, params):
            layer = node[1] if isinstance(node, tuple) else None
            beta = self.layer_betas.get(layer, params['beta'])
            k = neighbor_counts['I']
            if k > 0:
                p_infect = 1.0 - (1.0 - beta) ** k
                if rng.random() < p_infect:
                    return 'I'
            return 'S'
```

```
        self.transition_rules['S'] = transition_S

layer_rates = {
    'household': 0.5,
    'workplace': 0.2,
    'social': 0.1,
}
sir = LayerAwareSIR(network, layer_betas=layer_rates, default_beta=0.1)
results = sir.run(steps=100)
```

This pattern mirrors the built-in rules while letting you vary infection pressure per layer.

**Extracting Per-Layer Statistics**

After running the simulation, you can compute layer-specific prevalence manually:

```
# Get final state
final_state = results.trajectory[-1]

# Count infected nodes per layer
layer_prevalence = {}
for node, state in final_state.items():
    if isinstance(node, tuple):  # Multilayer node
        node_id, layer = node
        if layer not in layer_prevalence:
            layer_prevalence[layer] = {'S': 0, 'I': 0, 'R': 0}
        layer_prevalence[layer][state] += 1

# Print per-layer statistics
for layer, counts in layer_prevalence.items():
    total = sum(counts.values())
    print(f"Layer {layer}: {counts['R']} recovered out of {total} nodes ({counts['R']/tot
```

### Time-Varying Parameters (Intervention Scenarios)

To model interventions (e.g., school closures, social distancing), you can run multiple simulations with different parameters for different time windows:

```
# Scenario 1: Baseline (no intervention)
sir = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.05)
sir.set_seed(42)
results_baseline = sir.run(steps=100)

# Scenario 2: Intervention at t=30 (reduce beta)
# Run phase 1 (before intervention)
sir = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.05)
sir.set_seed(42)
state_30 = sir.run(steps=30, record=False)  # Get state at t=30

# Run phase 2 (with reduced beta)
sir_intervention = SIRDynamics(network, beta=0.15, gamma=0.1)
```

```
sir_intervention.set_seed(42)  # Continue from same seed
results_intervention = sir_intervention.run(steps=70, state=state_30)

# Compare peak prevalence
print(f"Baseline peak: {results_baseline.get_measure('prevalence').max():.2%}")
print(f"Intervention peak: {results_intervention.get_measure('prevalence').max():.2%}")
```

*Note: For more complex time-varying scenarios, consider using the high-level simulation builder API (see next steps).*

### Result Objects and Metrics

#### DynamicsResult API

All dynamics models return a `DynamicsResult` object from `.run()`. This object provides:

- **Trajectory access**: Direct indexing `results[t]` or `results.trajectory`
- **Measure extraction**: `.get_measure(name)` for standard metrics
- **Conversion**: `.to_pandas()` for DataFrame export

### Common Measures

The following table summarizes available measures for each model:

| Model | Measure | Description | Return Type |
|---|---|---|---|
| SIR | `"prevalence"` | Fraction of infected nodes over time (I / total) | `numpy.ndarray` |
| SIR | `"state_counts"` | Counts of nodes in each state (S, I, R) over time | `dict[str, numpy.ndarray]` |
| SIR | `"trajectory"` | Raw state sequence (list of state dicts) | `list[dict]` |
| SIS | `"prevalence"` | Fraction of infected nodes over time (I / total) | `numpy.ndarray` |
| SIS | `"state_counts"` | Counts of nodes in each state (S, I) over time | `dict[str, numpy.ndarray]` |
| SIS | `"trajectory"` | Raw state sequence | `list[dict]` |
| RandomWalk | `"trajectory"` | Sequence of visited nodes (or node-layer pairs) | `list` |

**Note**: Measures like `"peak_prevalence"` and `"final_recovered"` mentioned in earlier versions are computed manually from the time series data.

### Extracting and Analyzing Measures

```
# Run SIR simulation
sir = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.05)
sir.set_seed(42)
results = sir.run(steps=100)

# Extract prevalence time series
```

```
prevalence = results.get_measure("prevalence")
print(f"Prevalence array shape: {prevalence.shape}")  # (101,) for 100 steps + initial

# Compute summary statistics
peak_prevalence = prevalence.max()
peak_time = prevalence.argmax()
final_prevalence = prevalence[-1]

print(f"Peak prevalence: {peak_prevalence:.2%} at step {peak_time}")
print(f"Final prevalence: {final_prevalence:.2%}")

# Extract state counts
state_counts = results.get_measure("state_counts")
print(f"Available states: {list(state_counts.keys())}")

# Compute attack rate (fraction ever infected)
total_nodes = sum(state_counts[s][0] for s in state_counts.keys())  # Total at t=0
attack_rate = state_counts['R'][-1] / total_nodes
print(f"Attack rate: {attack_rate:.2%}")
```

For multilayer networks, `total_nodes` counts supra-nodes (node-layer tuples). If you want a per-entity attack rate, aggregate layers first.

### Converting to pandas DataFrame

For further analysis with pandas:

```
import pandas as pd

# Convert time series to DataFrame
state_counts = results.get_measure("state_counts")
df = pd.DataFrame({
    'time': range(len(state_counts['S'])),
    'S': state_counts['S'],
    'I': state_counts['I'],
    'R': state_counts['R']
})

# Compute prevalence column
df['prevalence'] = df['I'] / (df['S'] + df['I'] + df['R'])

# Analyze
print(df.describe())
print(f"\\nTime to peak: {df['prevalence'].idxmax()}")
```

Alternatively, use the built-in method:

```
# Convert trajectory to long-form DataFrame
df = results.to_pandas()
print(df.head())
```

```
# Output:
#    t  node state
# 0  0     0     S
# 1  0     1     S
# 2  0     2     S
# ...
```

### Reproducible Experiments & Parameter Sweeps

#### Running Multiple Stochastic Realizations

Epidemic dynamics are stochastic, so we often run multiple realizations and report summary statistics:

```
import numpy as np
from py3plex.dynamics import SIRDynamics

def run_sir_ensemble(network, beta, gamma, n_runs=50):
    """Run SIR simulation n_runs times with different seeds."""
    peak_prevalences = []
    attack_rates = []

    for seed in range(n_runs):
        sir = SIRDynamics(network, beta=beta, gamma=gamma, initial_infected=0.05)
        sir.set_seed(seed)
        results = sir.run(steps=100)

        prevalence = results.get_measure("prevalence")
        peak_prevalences.append(prevalence.max())

        state_counts = results.get_measure("state_counts")
        total = network.core_network.number_of_nodes()
        attack_rates.append(state_counts['R'][-1] / total)

    return {
        'peak_prevalence_mean': np.mean(peak_prevalences),
        'peak_prevalence_std': np.std(peak_prevalences),
        'attack_rate_mean': np.mean(attack_rates),
        'attack_rate_std': np.std(attack_rates),
    }

# Run ensemble
stats = run_sir_ensemble(network, beta=0.3, gamma=0.1, n_runs=50)
print(f"Peak prevalence: {stats['peak_prevalence_mean']:.2%} ± {stats['peak_prevalence_std
print(f"Attack rate: {stats['attack_rate_mean']:.2%} ± {stats['attack_rate_std']:.2%}")
```

**Expected Output**

```
Peak prevalence: 34.85% ± 2.13%
Attack rate: 98.60% ± 3.42%
```

*Note: Mean and std quantify variability across stochastic realizations.*

### Parameter Grid Sweep

To explore how outcomes depend on parameters, run a grid sweep:

```
import pandas as pd
import networkx as nx

# Use a standard network for reproducibility
G = nx.karate_club_graph()

# Parameter grid
beta_values = [0.1, 0.2, 0.3, 0.4]
gamma_values = [0.05, 0.1, 0.15, 0.2]

# Run grid
results_grid = []

for beta in beta_values:
    for gamma in gamma_values:
        sir = SIRDynamics(G, beta=beta, gamma=gamma, initial_infected=0.1)
        sir.set_seed(42)
        result = sir.run(steps=100)

        prevalence = result.get_measure("prevalence")
        state_counts = result.get_measure("state_counts")

        results_grid.append({
            'beta': beta,
            'gamma': gamma,
            'R0_approx': beta / gamma * 4.6,  # Mean-field R0  ()<k>; <k>  4.6 here (small
            'peak_prevalence': prevalence.max(),
            'peak_time': prevalence.argmax(),
            'attack_rate': state_counts['R'][-1] / G.number_of_nodes(),
        })

# Convert to DataFrame
df = pd.DataFrame(results_grid)
print(df)
```

**Expected Output** (partial)

```
   beta  gamma  R0_approx  peak_prevalence  peak_time  attack_rate
0   0.1   0.05        9.2            0.676          5        1.000
1   0.1   0.10        4.6            0.382          6        0.971
2   0.1   0.15        3.1            0.206          6        0.794
3   0.1   0.20        2.3            0.088          7        0.500
4   0.2   0.05       18.4            0.971          3        1.000
5   0.2   0.10        9.2            0.882          4        1.000
...
```

You can visualize this data with heatmaps or line plots to identify parameter regimes for outbreak control.

### Custom Dynamics

#### Extending BaseDynamics

To implement domain-specific dynamics, subclass `DynamicsProcess` and define:

1. `initialize_state(self, seed=None)`: Return initial state
2. `step(self, state, t)`: Update state for one time step
3. (Optional) Custom measure computation methods

**Contract**

- `initialize_state` must return a state object (dict, list, or custom)
- `step` must take the current state and return the new state
- Both methods can access `self.graph`, `self.rng`, and `self.params`

#### Example: Threshold Model

We implement a simple threshold model where nodes adopt a binary opinion if enough neighbors have adopted:

```
from py3plex.dynamics.core import DynamicsProcess
from py3plex.dynamics._utils import iter_multilayer_nodes, iter_multilayer_neighbors

class ThresholdDynamics(DynamicsProcess):
    """Threshold model: nodes adopt state 1 if fraction of neighbors exceeds threshold."""

    def __init__(self, graph, seed=None, threshold=0.5, initial_adopters=0.1):
        super().__init__(graph, seed=seed)
        self.threshold = threshold
        self.initial_adopters = initial_adopters

    def initialize_state(self, seed=None):
        """Initialize with some nodes in state 1, rest in state 0."""
        rng = self.rng if seed is None else np.random.default_rng(seed)
        nodes = list(iter_multilayer_nodes(self.graph))

        # Randomly select initial adopters
        n_adopters = int(len(nodes) * self.initial_adopters)
        adopter_indices = rng.choice(len(nodes), size=n_adopters, replace=False)
        adopters = set(nodes[i] for i in adopter_indices)

        return {node: 1 if node in adopters else 0 for node in nodes}

    def step(self, state, t):
        """Update rule: adopt if fraction of neighbors >= threshold."""
        new_state = {}

        for node in state:
            if state[node] == 1:
                # Already adopted, stay adopted
                new_state[node] = 1
```

```
            else:
                # Count adopting neighbors
                neighbors = list(iter_multilayer_neighbors(self.graph, node))
                if not neighbors:
                    new_state[node] = 0
                    continue

                adopting_neighbors = sum(1 for n in neighbors if state.get(n, 0) == 1)
                fraction = adopting_neighbors / len(neighbors)

                # Adopt if threshold exceeded
                new_state[node] = 1 if fraction >= self.threshold else 0

        return new_state

# Run threshold model
threshold = ThresholdDynamics(
    network,
    seed=42,
    threshold=0.3,
    initial_adopters=0.1
)

results = threshold.run(steps=50)

# Count adoption over time
adoption_counts = []
for state in results.trajectory:
    adoption_counts.append(sum(state.values()))

print(f"Initial adopters: {adoption_counts[0]}")
print(f"Final adopters: {adoption_counts[-1]}")
print(f"Adoption rate: {adoption_counts[-1] / network.core_network.number_of_nodes():.2%}")
```

**Expected Output**

```
Initial adopters: 4
Final adopters: 40
Adoption rate: 100.00%
```

*Note: With a low threshold (0.3), the initial adopters trigger a cascade where all nodes eventually adopt.*

### Adding Custom Measures

To expose custom measures through `.get_measure()`, override the `compute_measure` method:

```
class ThresholdDynamics(DynamicsProcess):
    # ... (same as above) ...

    def compute_measure(self, measure_name, trajectory):
```

```
        """Compute custom measures for threshold model."""
        if measure_name == "adoption_fraction":
            # Fraction of nodes in state 1 over time
            fractions = []
            for state in trajectory:
                fractions.append(sum(state.values()) / len(state))
            return np.array(fractions)

        elif measure_name == "cascade_size":
            # Final number of adopters
            final_state = trajectory[-1]
            return sum(final_state.values())

        else:
            raise ValueError(f"Unknown measure: {measure_name}")

# Now you can use:
results = threshold.run(steps=50)
adoption_frac = results.get_measure("adoption_fraction")
cascade_size = results.get_measure("cascade_size")

print(f"Cascade size: {cascade_size} nodes")
```

This pattern makes your custom dynamics consistent with built-in models.

### Query Dynamics Results with DSL

**Goal:** Use py3plex's Domain-Specific Language (DSL) to query and analyze simulation results efficiently.

After running dynamics simulations, the DSL provides powerful tools for querying infected nodes, analyzing layer-specific patterns, and extracting subpopulations. This is especially useful for multilayer networks where dynamics vary across layers.

**Prerequisites:**

- A dynamics simulation result object (from `SIRDynamics`, `SISDynamics`, etc.)
- Familiarity with DSL basics (see *How to Query Multilayer Graphs with the SQL-like DSL* for full tutorial)

### Attaching Dynamics State as Node Attributes

To use DSL for querying simulation results, first attach the dynamics state to the network as node attributes:

```
from py3plex.core import multinet
from py3plex.dynamics import SIRDynamics
from py3plex.dsl import Q

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "py3plex/datasets/_data/synthetic_multilayer.edges",
```

```
    input_type="multiedgelist"
)

# Run SIR simulation
sir = SIRDynamics(
    network,
    beta=0.3,
    gamma=0.1,
    initial_infected=0.05
)
sir.set_seed(42)
results = sir.run(steps=100)

# Attach final state to network as node attributes
final_state = results.trajectory[-1]
for node, state in final_state.items():
    network.core_network.nodes[node]['sir_state'] = state
    # Also add a numeric version for filtering
    network.core_network.nodes[node]['is_infected'] = (state == 'I')
    network.core_network.nodes[node]['is_recovered'] = (state == 'R')
    network.core_network.nodes[node]['is_susceptible'] = (state == 'S')
```

### Query Infected Nodes

**Find all currently infected nodes:**

```
# Using Builder API
infected = (
    Q.nodes()
     .where(sir_state="I")
     .execute(network)
)

print(f"Infected nodes: {infected.count}")
print(f"Sample infected:")
for node in infected_nodes[:5]:
    print(f"  {node}")
```

**Expected output:**

```
Infected nodes: 12
Sample infected:
  ('node7', 'layer1')
  ('node12', 'layer2')
  ('node3', 'layer3')
  ('node15', 'layer1')
  ('node8', 'layer2')
```

**Using builder API for complex queries:**

```
from py3plex.dsl import Q, L

# Find infected nodes with high degree
high_degree_infected = (
    Q.nodes()
     .where(sir_state='I')
     .compute("degree")
     .where(degree__gt=5)
     .order_by("degree", reverse=True)
     .execute(network)
)

print(f"High-degree infected nodes: {len(high_degree_infected)}")
for node, data in list(high_degree_infected.items())[:5]:
    print(f"  {node}: degree={data['degree']}")
```

**Expected output:**

```
High-degree infected nodes: 7
  ('node7', 'layer1'): degree=12
  ('node12', 'layer2'): degree=10
  ('node3', 'layer1'): degree=9
  ('node15', 'layer3'): degree=8
  ('node8', 'layer2'): degree=7
```

### Layer-Specific Dynamics Queries

**Compare infection levels across layers:**

```
layers = network.get_layers()

print("Infection by layer:")
for layer in layers:
    layer_nodes = (
        Q.nodes()
         .from_layers(L[layer])
         .execute(network)
    )

    infected_in_layer = sum(
        1 for node in layer_nodes
        if network.core_network.nodes[node].get('sir_state') == 'I'
    )

    prevalence = infected_in_layer / len(layer_nodes) if len(layer_nodes) > 0 else 0
    print(f"  {layer}: {infected_in_layer}/{len(layer_nodes)} ({prevalence:.2%})")
```

**Expected output:**

```
Infection by layer:
  layer1: 5/40 (12.50%)
```

```
  layer2: 4/40 (10.00%)
  layer3: 3/40 (7.50%)
```

**Find susceptible nodes in high-risk layers:**

```python
# Identify high-risk layers (high infection prevalence)
high_risk_layer = 'layer1'  # From analysis above

# Find susceptible nodes in high-risk layer
at_risk_nodes = (
    Q.nodes()
     .from_layers(L[high_risk_layer])
     .where(sir_state='S')
     .compute("degree")
     .execute(network)
)

# Sort by degree (higher degree = more neighbors, higher risk)
at_risk_list = sorted(
    at_risk_nodes.items(),
    key=lambda x: x[1]['degree'],
    reverse=True
)

print(f"High-risk susceptible nodes in {high_risk_layer}:")
for node, data in at_risk_list[:5]:
    print(f"  {node}: degree={data['degree']}")
```

### Temporal Queries Across Simulation Steps

**Track state changes over time:**

```python
# Select a few nodes to track
tracked_nodes = [
    ('node7', 'layer1'),
    ('node12', 'layer2'),
    ('node3', 'layer3')
]

print("Temporal evolution:")
print("Time  " + "  ".join(f"{n[0]:8}" for n in tracked_nodes))
print("-" * 50)

# Sample time points
time_points = [0, 20, 40, 60, 80, 100]
for t in time_points:
    if t < len(results.trajectory):
        state_t = results.trajectory[t]
        states_str = "  ".join(
            f"{state_t.get(node, '-'):8}"
            for node in tracked_nodes
```

```
        )
        print(f"{t:4}  {states_str}")
```

**Expected output:**

```
Temporal evolution:
Time  node7     node12    node3
--------------------------------------------------
   0  S         S         S
  20  I         S         S
  40  R         I         I
  60  R         R         I
  80  R         R         R
 100  R         R         R
```

**Query state transitions:**

```
# Find nodes that became infected between t=20 and t=40
state_20 = results.trajectory[20]
state_40 = results.trajectory[40]

newly_infected = [
    node for node in state_20.keys()
    if state_20[node] == 'S' and state_40.get(node) == 'I'
]

print(f"Newly infected between t=20 and t=40: {len(newly_infected)}")
print(f"Sample: {newly_infected[:5]}")
```

### Extract Subpopulations

**Extract network of recovered nodes:**

```
# Query recovered nodes (Builder API)
recovered_result = (
    Q.nodes()
     .where(sir_state="R")
     .execute(network)
)

recovered_matches = recovered_result.items  # List of (node, layer) tuples

# Extract induced subgraph
recovered_subgraph = network.core_network.subgraph(recovered_matches)

print(f"Recovered subnetwork:")
print(f"  Nodes: {recovered_subgraph.number_of_nodes()}")
print(f"  Edges: {recovered_subgraph.number_of_edges()}")
print(f"  Layers: {set(node[1] for node in recovered_matches)}")
```

**Expected output:**

```
Recovered subnetwork:
  Nodes: 38
  Edges: 145
  Layers: {'layer1', 'layer2', 'layer3'}
```

**Analyze connectivity within susceptible population:**

```python
import networkx as nx

# Get susceptible nodes (Builder API)
susceptible_result = (
    Q.nodes()
     .where(sir_state="S")
     .execute(network)
)

susceptible_nodes = susceptible_result.items  # List of (node, layer) tuples

# Extract subgraph
S_subgraph = network.core_network.subgraph(susceptible_nodes)

# Compute connected components
if S_subgraph.number_of_nodes() > 0:
    components = list(nx.connected_components(S_subgraph))
    print(f"Susceptible population:")
    print(f"  Nodes: {S_subgraph.number_of_nodes()}")
    print(f"  Connected components: {len(components)}")
    print(f"  Largest component: {max(len(c) for c in components)} nodes")
```

### Combine with Dynamics Measures

**Identify superspreaders (high-degree infected nodes):**

```python
from py3plex.dsl import Q

# Find infected nodes with highest degree
superspreaders = (
    Q.nodes()
     .where(sir_state='I')
     .compute("degree", "betweenness_centrality")
     .where(degree__gt=7)
     .order_by("betweenness_centrality", reverse=True)
     .execute(network)
)

print(f"Potential superspreaders: {len(superspreaders)}")
for node, data in list(superspreaders.items())[:5]:
    print(f"  {node}: deg={data['degree']}, betw={data['betweenness_centrality']:.4f}")
```

**Expected output:**

```
Potential superspreaders: 4
  ('node7', 'layer1'): deg=12, betw=0.0456
  ('node12', 'layer2'): deg=10, betw=0.0345
  ('node15', 'layer3'): deg=9, betw=0.0234
  ('node3', 'layer1'): deg=8, betw=0.0198
```

**Compute attack rate by layer using DSL:**

```
print("Attack rate (recovered / total) by layer:")
for layer in network.get_layers():
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)

    recovered = sum(
        1 for node in layer_nodes
        if network.core_network.nodes[node].get('sir_state') == 'R'
    )

    attack_rate = recovered / len(layer_nodes) if len(layer_nodes) > 0 else 0
    print(f"  {layer}: {attack_rate:.2%}")
```

### Export Dynamics Results with DSL

**Create analysis-ready DataFrame:**

```
import pandas as pd
from py3plex.dsl import Q

# Query all nodes with states and metrics
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Convert to DataFrame
df = pd.DataFrame([
    {
        'node': node[0],
        'layer': node[1],
        'sir_state': data.get('sir_state', 'unknown'),
        'degree': data['degree'],
        'betweenness': data['betweenness_centrality']
    }
    for node, data in result.items()
])

# Group by state and compute statistics
state_stats = df.groupby('sir_state').agg({
    'degree': ['mean', 'std', 'max'],
    'betweenness': ['mean', 'std', 'max']
})
```

```
print("Statistics by SIR state:")
print(state_stats)

# Export
df.to_csv('sir_results_with_metrics.csv', index=False)
```

**Why use DSL for dynamics analysis?**

- **Declarative filtering:** Express queries naturally without manual loops
- **Layer-aware:** Built-in support for querying specific layers or cross-layer patterns
- **Composable:** Chain operations to build complex analyses
- **Integration:** Seamlessly combine dynamics state with network metrics
- **Performance:** DSL optimizes query execution for large multilayer networks

**Next steps with DSL:**

- **Full DSL tutorial:** *How to Query Multilayer Graphs with the SQL-like DSL* - Comprehensive guide with advanced patterns
- **Temporal queries:** *How to Query Multilayer Graphs with the SQL-like DSL* (Temporal Queries section) - Time-varying networks
- **Community detection integration:** *How to Run Community Detection on Multilayer Networks* (Query Communities with DSL section)

### Next Steps

Now that you understand multilayer dynamics in py3plex, explore:

- **Visualization**: *How to Visualize Multilayer Networks* to plot epidemic curves and multilayer network states
- **Multilayer theory**: *Multilayer Networks 101* for formal background on multilayer structures
- **Advanced examples**: *Examples & Recipes* for notebooks using SIR/SIS in complex scenarios
- **API reference**: *Algorithm Roadmap* for full documentation of `py3plex.dynamics`

**Typical research workflows:**

1. Design a multilayer network representing your system (social + physical, online + offline, etc.)
2. Choose or implement a dynamics model (epidemic, diffusion, opinion, etc.)
3. Run parameter sweeps to explore outbreak conditions or intervention strategies
4. Visualize trajectories and multilayer-specific statistics (per-layer prevalence, layer switching rates)
5. Compare scenarios to inform policy or design decisions

For more complex simulations (time-varying parameters, multi-stage interventions), consider using the high-level simulation builder API (`py3plex.dynamics.D`) or config-driven workflows (`build_dynamics_from_config`).

### 3.3.7 How to Run Random Walk Algorithms

**Goal:** Generate network embeddings and node representations using random walk algorithms.

**Prerequisites:** A loaded network (see *How to Load and Build Networks*).

All embedding helpers in py3plex return a mapping from `(node_id, layer)` tuples to dense vectors of length `dimensions`. Single-layer inputs still keep the layer entry so you can tell layers apart in multiplex graphs.

**Note**

Where to find this data

Examples in this guide create networks programmatically for clarity. You can also:

- Use built-in generators: `from py3plex.algorithms import random_generators`
- Load from repository files: `datasets/multiedgelist.txt`
- Fetch real-world datasets: `from py3plex.datasets import fetch_multilayer`

#### Node2Vec Embeddings

Node2Vec generates vector representations of nodes by simulating biased random walks:

```
from py3plex.core import multinet
from py3plex.wrappers import train_node2vec

# Create a sample network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Charlie', 'friends', 1],
    ['Alice', 'colleagues', 'Charlie', 'colleagues', 1],
], input_type="list")

# Train Node2Vec
embeddings = train_node2vec(
    network,
    dimensions=128,     # Embedding dimensionality
    walk_length=80,     # Length of each walk
    num_walks=10,       # Walks per node
    p=1.0,             # Return parameter
    q=1.0,             # In-out parameter
    workers=4
)

# Access embeddings
node = ('Alice', 'friends')
vector = embeddings[node]
print(f"Embedding dimension: {len(vector)}")
```

**Expected output:**

```
Embedding dimension: 128
```

The $p$ and $q$ parameters control the walk behavior:

- **p** (return parameter): Cost of immediately revisiting the previous node (low `p` = easy to backtrack; high `p` = discouraged)
- **q** (in-out parameter): Balance between staying near the previous node versus exploring new territory (`q > 1` breadth-first/local, `q < 1` depth-first/outward)

Think of `p` as controlling backtracking and `q` as controlling locality; tune them together to steer walks toward either local neighborhoods or long-range exploration.

### DeepWalk Embeddings

DeepWalk is a special case of Node2Vec with `p=1, q=1`:

```python
from py3plex.wrappers import train_deepwalk

embeddings = train_deepwalk(
    network,
    dimensions=128,
    walk_length=80,
    num_walks=10,
    workers=4
)
```

### Using Embeddings for Downstream Tasks

The following examples assume you already computed `embeddings` with Node2Vec or DeepWalk; adjust names if you generate multiple embedding sets.

### Node Classification

```python
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
import numpy as np

# Prepare data
nodes = list(embeddings.keys())
X = np.array([embeddings[node] for node in nodes])

# Replace with your label lookup (one label per (node, layer) tuple)
y = np.array([get_label(node) for node in nodes])

# Train classifier
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

clf = LogisticRegression()
clf.fit(X_train, y_train)
```

```
accuracy = clf.score(X_test, y_test)
print(f"Classification accuracy: {accuracy:.2%}")
```

### Link Prediction

Embeddings are keyed by `(node, layer)`; compare nodes within the same layer when interpreting similarity scores.

```
from sklearn.metrics.pairwise import cosine_similarity

# Compute similarity between nodes
node1 = ('Alice', 'friends')
node2 = ('Bob', 'friends')

vec1 = embeddings[node1].reshape(1, -1)
vec2 = embeddings[node2].reshape(1, -1)

similarity = cosine_similarity(vec1, vec2)[0][0]
print(f"Similarity: {similarity:.3f}")

# Predict links for high-similarity pairs
threshold = 0.7
if similarity > threshold:
    print(f"High likelihood of connection between {node1} and {node2}")
```

### Node Clustering

```
from sklearn.cluster import KMeans

# Cluster nodes based on embeddings
nodes = list(embeddings.keys())
X = np.array([embeddings[node] for node in nodes])

kmeans = KMeans(n_clusters=5, random_state=42)
clusters = kmeans.fit_predict(X)

# Map nodes to clusters
node_clusters = dict(zip(nodes, clusters))

print("Cluster assignments:")
for node, cluster in list(node_clusters.items())[:10]:
    print(f"{node} → Cluster {cluster}")
```

### Visualizing Embeddings

Use dimensionality reduction to visualize:

```
from sklearn.manifold import TSNE
import matplotlib.pyplot as plt
```

```
# Reduce to 2D
nodes = list(embeddings.keys())
X = np.array([embeddings[node] for node in nodes])

tsne = TSNE(n_components=2, random_state=42)
X_2d = tsne.fit_transform(X)

# Plot
plt.figure(figsize=(12, 8))
plt.scatter(X_2d[:, 0], X_2d[:, 1], alpha=0.5)

# Label a few nodes
for i, node in enumerate(nodes[:20]):
    plt.annotate(
        str(node),
        (X_2d[i, 0], X_2d[i, 1]),
        fontsize=8
    )

plt.title('Node Embeddings (t-SNE)')
plt.savefig('embeddings_2d.png', dpi=300, bbox_inches='tight')
plt.show()
```

### Saving and Loading Embeddings

#### Save to File

```
import pickle

# Save embeddings
with open('embeddings.pkl', 'wb') as f:
    pickle.dump(embeddings, f)

# Load embeddings
with open('embeddings.pkl', 'rb') as f:
    loaded_embeddings = pickle.load(f)
```

#### Export to CSV

```
import pandas as pd

# Convert to DataFrame
data = []
for node, vector in embeddings.items():
    row = {'node': str(node)}
    for i, val in enumerate(vector):
        row[f'dim_{i}'] = val
    data.append(row)
```

```
df = pd.DataFrame(data)
df.to_csv('embeddings.csv', index=False)
```

### Parameter Tuning

Random walk hyperparameters meaningfully change embedding quality. Use small validation loops to pick settings that balance accuracy and runtime, and fix seeds (`random_state` / `workers`) when you want reproducibility across runs.

### Grid Search for Best Parameters

Training embeddings inside a grid search can be expensive; start with a coarse grid or a small subgraph, then refine around promising regions.

```
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression
import numpy as np

param_grid = {
    'dimensions': [64, 128, 256],
    'walk_length': [40, 80, 120],
    'num_walks': [5, 10, 20]
}

best_score = 0
best_params = None

for dims in param_grid['dimensions']:
    for walk_len in param_grid['walk_length']:
        for num_walks in param_grid['num_walks']:
            # Train embeddings
            emb = train_node2vec(
                network,
                dimensions=dims,
                walk_length=walk_len,
                num_walks=num_walks
            )

            # Evaluate (assuming you have labels)
            nodes = list(emb.keys())
            labels = [get_label(n) for n in nodes]  # Provide labels keyed by (node, layer)
            X = np.array([emb[n] for n in nodes])
            y = np.array(labels)
            scores = cross_val_score(
                LogisticRegression(),
                X, y, cv=5
            )

            mean_score = scores.mean()
            if mean_score > best_score:
                best_score = mean_score
```

```
                best_params = {
                    'dimensions': dims,
                    'walk_length': walk_len,
                    'num_walks': num_walks
                }

print(f"Best parameters: {best_params}")
print(f"Best score: {best_score:.3f}")
```

### Layer-Specific Embeddings

Generate embeddings for individual layers:

```
from py3plex.dsl import Q, L

layer_embeddings = {}

for layer in network.get_layers():
    # Extract layer subgraph
    subgraph = Q.edges().from_layers(L[layer]).execute(network)

    # Train embeddings on this layer
    emb = train_node2vec(subgraph, dimensions=128)
    layer_embeddings[layer] = emb

    print(f"Generated embeddings for layer: {layer}")
```

### Query and Filter Nodes Before Embedding with DSL

**Goal:** Use py3plex's DSL to select specific node subsets for targeted embedding generation.

Random walks and embeddings on large networks can be computationally expensive. The DSL allows you to filter and extract subnetworks before running embedding algorithms, focusing on nodes of interest.

### Filter High-Degree Nodes

**Generate embeddings only for hub nodes:**

```
from py3plex.core import multinet
from py3plex.wrappers import train_node2vec
from py3plex.dsl import Q

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "py3plex/datasets/_data/synthetic_multilayer.edges",
    input_type="multiedgelist"
)

# Use DSL to find high-degree nodes (hubs)
```

```
hubs = (
    Q.nodes()
     .compute("degree")
     .where(degree__gt=8)  # Degree > 8
     .execute(network)
)

print(f"Found {len(hubs)} hub nodes")

# Extract subgraph containing only hubs
hub_subgraph = network.core_network.subgraph(hubs.keys())

# Train embeddings on hub subgraph
hub_network = multinet.multi_layer_network(directed=False)
hub_network.core_network = hub_subgraph.copy()

hub_embeddings = train_node2vec(
    hub_network,
    dimensions=64,
    walk_length=40,
    num_walks=10
)

print(f"Generated embeddings for {len(hub_embeddings)} hubs")
```

**Expected output:**

```
Found 15 hub nodes
Generated embeddings for 15 hubs
```

### Layer-Specific Node Selection

**Generate embeddings for nodes active in multiple layers:**

```
from py3plex.dsl import Q, L
from collections import Counter

# Count layer participation for each node
node_layers = Counter()
for node, layer in network.get_nodes():
    node_layers[node] += 1

# Find nodes in 2+ layers
multilayer_nodes = {
    node for node, count in node_layers.items()
    if count >= 2
}

print(f"Nodes in 2+ layers: {len(multilayer_nodes)}")

# Convert back to (node, layer) tuples
```

```
multilayer_node_tuples = [
    (node, layer)
    for node, layer in network.get_nodes()
    if node in multilayer_nodes
]

# Extract subgraph
multi_subgraph = network.core_network.subgraph(multilayer_node_tuples)

# Generate embeddings
multi_network = multinet.multi_layer_network(directed=False)
multi_network.core_network = multi_subgraph.copy()

multi_embeddings = train_node2vec(multi_network, dimensions=64)
print(f"Generated embeddings for {len(multi_embeddings)} multilayer nodes")
```

**Expected output:**

```
Nodes in 2+ layers: 35
Generated embeddings for 105 multilayer nodes
```

### Query Nodes by Centrality

**Focus embeddings on central nodes:**

```
from py3plex.dsl import Q

# Find nodes with high betweenness centrality
central_nodes = (
    Q.nodes()
     .compute("betweenness_centrality", "degree")
     .where(betweenness_centrality__gt=0.01)
     .execute(network)
)

print(f"High-centrality nodes: {len(central_nodes)}")

# Extract and embed
central_subgraph = network.core_network.subgraph(central_nodes.keys())
central_network = multinet.multi_layer_network(directed=False)
central_network.core_network = central_subgraph.copy()

central_embeddings = train_node2vec(
    central_network,
    dimensions=128,
    walk_length=80,
    num_walks=10
)

# Compare embedding similarity for central nodes
from sklearn.metrics.pairwise import cosine_similarity
```

```
import numpy as np

node_list = list(central_embeddings.keys())[:5]
vectors = np.array([central_embeddings[n] for n in node_list])

sim_matrix = cosine_similarity(vectors)
print("\nSimilarity matrix (first 5 central nodes):")
print(sim_matrix)
```

### Combine Embeddings with Node Attributes

**Query embedded nodes with specific properties:**

```
import numpy as np

# After generating embeddings, attach them as attributes
for node, vector in embeddings.items():
    # Store embedding norm as an attribute
    embedding_norm = np.linalg.norm(vector)
    network.core_network.nodes[node]['embedding_norm'] = embedding_norm

# Query nodes with large embedding norms (Builder API)
large_norm_result = (
    Q.nodes()
     .where(embedding_norm__gt=10.0)
     .execute(network)
)

print(f"Nodes with large embedding norms: {large_norm_result.count}")
```

### Layer-Specific Embedding Analysis

**Compare embeddings across layers:**

```
from py3plex.dsl import Q, L

# Generate embeddings per layer
layer_embeddings = {}
layer_stats = {}

for layer in network.get_layers():
    # Extract layer
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)
    layer_edges = Q.edges().from_layers(L[layer]).execute(network)

    print(f"\nLayer: {layer}")
    print(f"  Nodes: {len(layer_nodes)}")
    print(f"  Edges: {len(layer_edges)}")

    # Extract layer subgraph
```

```
    layer_subgraph = network.core_network.subgraph(layer_nodes.keys())

    # Skip if too few nodes
    if len(layer_nodes) < 5:
        print(f"  [SKIP] Too few nodes")
        continue

    # Create layer network
    layer_net = multinet.multi_layer_network(directed=False)
    layer_net.core_network = layer_subgraph.copy()

    # Generate embeddings
    try:
        emb = train_node2vec(
            layer_net,
            dimensions=64,
            walk_length=40,
            num_walks=10,
            workers=1
        )

        layer_embeddings[layer] = emb

        # Compute statistics
        vectors = np.array(list(emb.values()))
        mean_norm = np.mean([np.linalg.norm(v) for v in vectors])

        layer_stats[layer] = {
            'n_nodes': len(emb),
            'mean_embedding_norm': mean_norm
        }

        print(f"   Embeddings generated: {len(emb)} nodes")
        print(f"  Mean embedding norm: {mean_norm:.2f}")
    except Exception as e:
        print(f"  [ERROR] {e}")
```

**Expected output:**

```
Layer: layer1
  Nodes: 40
  Edges: 95
   Embeddings generated: 40 nodes
  Mean embedding norm: 12.34

Layer: layer2
  Nodes: 40
  Edges: 87
   Embeddings generated: 40 nodes
  Mean embedding norm: 11.89
```

```
Layer: layer3
  Nodes: 40
  Edges: 102
   Embeddings generated: 40 nodes
  Mean embedding norm: 13.01
```

### Export Embeddings with Metadata Using DSL

**Create analysis-ready embedding exports:**

```python
import pandas as pd
from py3plex.dsl import Q

# Compute node metrics
metrics = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Combine embeddings with metrics
data = []
for node in embeddings.keys():
    row = {
        'node': node[0],
        'layer': node[1],
        'degree': metrics[node]['degree'],
        'betweenness': metrics[node]['betweenness_centrality']
    }

    # Add embedding dimensions
    for i, val in enumerate(embeddings[node]):
        row[f'emb_{i}'] = val

    data.append(row)

# Create DataFrame
df = pd.DataFrame(data)

# Export
df.to_csv('embeddings_with_metrics.csv', index=False)
print(f"Exported {len(df)} node embeddings with metadata")
```

**Why use DSL for embedding workflows?**

- **Targeted embedding:** Focus on relevant node subsets, reducing computation time
- **Layer-aware:** Generate layer-specific embeddings seamlessly
- **Metric integration:** Combine embeddings with centrality and other network metrics
- **Filtering:** Select nodes by degree, centrality, or custom attributes before embedding
- **Reproducible:** Declarative queries document node selection criteria

**Next steps with DSL:**

- **Full DSL tutorial:** *How to Query Multilayer Graphs with the SQL-like DSL* - Comprehensive guide with advanced patterns
- **Community detection:** *How to Run Community Detection on Multilayer Networks* - Use embeddings for community analysis
- **Dynamics analysis:** *How to Simulate Multilayer Dynamics* - Combine embeddings with dynamics results

#### Next Steps

- **Use embeddings for ML tasks:** See sklearn documentation
- **Visualize networks:** *How to Visualize Multilayer Networks*
- **Understand algorithms:** *Algorithm Landscape*
- **API reference:** *Algorithm Roadmap*

### 3.3.8 How to Export and Serialize Networks

**Goal:** Save multilayer networks to files and load them back—quick persistence, interoperable edge lists, and analysis-ready formats.

**Prerequisites:** A network to export (see *How to Load and Build Networks*). All examples assume an in-memory `network` instance.

#### Quick Save and Load

#### Using Pickle (Recommended)

Pickle is the quickest way to persist a network within Python environments when you do not need cross-language compatibility.

```
from py3plex.core import multinet
import pickle

# Create or load network
network = multinet.multi_layer_network(directed=False)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'work', 'Carol', 'work', 1.0],
], input_type="list")

# Save to file
with open('network.pkl', 'wb') as f:
    pickle.dump(network, f)

# Load from file
with open('network.pkl', 'rb') as f:
    loaded_network = pickle.load(f)

loaded_network.basic_stats()
```

**Note**

Only unpickle files from trusted sources, and prefer the same Python + py3plex versions for best compatibility. Use a text format (CSV/JSON) if you need portability.

### Export as Edge List

Use plain text or CSV when you want human-readable exports or interoperability with tools outside Python.

### Simple Text Format

Write a plain text edge list with node IDs, layers, and weights. `get_edges` returns edges as `((node, layer), (node, layer))` tuples. Each line below follows `source source_layer target target_layer weight`.

```
# Each parallel edge supplies its own attributes
# Export edges to text file
with open('edges.txt', 'w') as f:
    for source, target, attrs in network.get_edges(data=True):
        source_node, source_layer = source
        target_node, target_layer = target

        # Get edge attributes
        weight = attrs.get('weight', 1.0)

        f.write(f"{source_node} {source_layer} {target_node} {target_layer} {weight}\n")
```

### CSV Format

Create a CSV edge list with explicit columns for nodes, layers, and weights. CSV is easier to load in spreadsheets and dataframes than the plain text format.

```
import pandas as pd

# Convert to DataFrame
edges_data = []
for source, target, attrs in network.get_edges(data=True):
    source_node, source_layer = source
    target_node, target_layer = target


    edges_data.append({
        'source': source_node,
        'source_layer': source_layer,
        'target': target_node,
        'target_layer': target_layer,
        'weight': attrs.get('weight', 1.0)
    })
```

```
df = pd.DataFrame(edges_data)
df.to_csv('network_edges.csv', index=False)
```

### Export as JSON

#### Full Network Structure

Capture nodes, layers, edge endpoints, and their attributes in a single JSON document for easy round-tripping. JSON keeps attribute dictionaries intact and readable.

```
import json

# Build JSON structure
network_data = {
    'nodes': [],
    'edges': [],
    'layers': network.layers,
    'directed': network.directed
}

# Add nodes
for (node, layer), node_attrs in network.get_nodes(data=True):
    network_data['nodes'].append({
        'id': node,
        'layer': layer,
        'attributes': node_attrs
    })

# Add edges
for source, target, attrs in network.get_edges(data=True):
    source_node, source_layer = source
    target_node, target_layer = target


    network_data['edges'].append({
        'source': source_node,
        'source_layer': source_layer,
        'target': target_node,
        'target_layer': target_layer,
        'attributes': attrs
    })

# Save
with open('network.json', 'w') as f:
    json.dump(network_data, f, indent=2)
```

### Load from JSON

Recreate a multilayer network from the exported JSON representation.

```
import json
from py3plex.core import multinet

# Load JSON
with open('network.json', 'r') as f:
    data = json.load(f)

# Reconstruct network
network = multinet.multi_layer_network(directed=data['directed'])
network.add_nodes([{'source': node['id'], 'type': node['layer'],
                    **node['attributes']} for node in data['nodes']])
for edge in data['edges']:
    network.add_edges([{'source': edge['source'], 'target': edge['target'],
                        'source_type': edge['source_layer'],
                        'target_type': edge['target_layer'],
                        **edge['attributes']}])

network.basic_stats()
```

### Export to NetworkX

#### Convert to Single-Layer NetworkX Graph

Flatten multilayer data into a single NetworkX graph, keeping the layer as a node attribute. This drops parallel edges between layers and merges nodes with the same ID across layers.

```
import networkx as nx

# Flatten to single layer
G = nx.Graph()

for source, target, attrs in network.get_edges(data=True):
    source_node, source_layer = source
    target_node, target_layer = target

    # Add nodes with layer info
    G.add_node(source_node, layer=source_layer)
    G.add_node(target_node, layer=target_layer)

    # Add edge
    G.add_edge(source_node, target_node, **attrs)

# Save as GraphML
nx.write_graphml(G, 'network.graphml')
```

### Export Specific Layer

Persist just one layer to a simpler graph file.

```
import networkx as nx
from py3plex.dsl import Q, L

# Extract single layer
layer = 'friends'
subgraph = Q.edges().from_layers(L[layer]).execute(network)

# Convert to NetworkX
G = nx.Graph()
selected = subgraph.to_networkx(network)
G = nx.relabel_nodes(selected, lambda replica: replica[0])

# Save
nx.write_edgelist(G, f'layer_{layer}.edgelist')
```

### High-Performance: Apache Arrow/Parquet

### Zero-Loss Network Interchange (Recommended)

py3plex provides zero-loss network serialization using Arrow and Parquet formats. These preserve multilayer identity, attributes, network type, and directedness.

**Arrow Format (Single File)**

Arrow is the preferred format for single-file storage with embedded metadata.

```
from py3plex.io import save_to_arrow, load_from_arrow
from py3plex.core import multinet

# Create network with attributes
network = multinet.multi_layer_network(directed=False)
network.add_nodes([
    {'source': 'Alice', 'type': 'social', 'age': 30},
    {'source': 'Bob', 'type': 'social', 'age': 25},
    {'source': 'Alice', 'type': 'work'},
])
network.add_edges([
    {'source': 'Alice', 'target': 'Bob',
     'source_type': 'social', 'target_type': 'social',
     'weight': 1.5, 'timestamp': '2024-01-01'}
])

# Save to Arrow (single file)
save_to_arrow(network, 'network.arrow')

# Load back with zero loss
loaded = load_from_arrow('network.arrow')

# Verify preservation
```

```
assert set(loaded.get_nodes()) == set(network.get_nodes())
assert loaded.directed == network.directed
```

**Parquet Format (Directory)**

Parquet uses a directory layout for organizational clarity.

```
from py3plex.io import save_network_to_parquet, load_network_from_parquet

# Save to Parquet directory
# Creates: network_dir/nodes.parquet
#          network_dir/edges.parquet
#          network_dir/metadata.json
save_network_to_parquet(network, 'network_dir')

# Load back
loaded = load_network_from_parquet('network_dir')
```

**Canonical Schema**

Both formats use the same schema internally:

- **Nodes table**: Required columns are `node` (ID) and `layer`. Optional columns for attributes.
- **Edges table**: Required columns are `source`, `target`, `source_layer`, `target_layer`. Optional: `weight` and other attributes.
- **Metadata**: Includes `schema_version`, `py3plex_version`, `network_type` (multilayer/multiplex), `directed`, attribute type manifest, and JSON-encoded column tracking.

**Complex Attributes**

Non-scalar attributes (dict, list, set, tuple) are JSON-encoded automatically:

```
# Add node with complex attribute
network.add_nodes([
    {'source': 'Alice', 'type': 'social',
     'tags': ['researcher', 'python'],
     'metadata': {'department': 'CS', 'level': 5}}
])

# Save and load - complex attributes preserved
save_to_arrow(network, 'network.arrow')
loaded = load_from_arrow('network.arrow')

# Complex attributes reconstructed
alice_attrs = loaded.core_network.nodes[('Alice', 'social')]
assert alice_attrs['tags'] == ['researcher', 'python']
assert alice_attrs['metadata']['department'] == 'CS'
```

**QueryResult Export**

Save analysis results in Parquet format:

```
from py3plex.dsl import Q, save_to_parquet, load_from_parquet
```

```
# Compute statistics
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Save to Parquet
save_to_parquet(result, 'node_stats.parquet')

# Load back as DataFrame
df = load_from_parquet('node_stats.parquet')
```

### Legacy Parquet (Edge Lists Only)

For compatibility with external tools, you can export just edge lists:

```
import pyarrow as pa
import pyarrow.parquet as pq

# Convert edges to table
edges_data = {
    'source': [],
    'source_layer': [],
    'target': [],
    'target_layer': [],
    'weight': []
}

for source, target, attrs in network.get_edges(data=True):
    source_node, source_layer = source
    target_node, target_layer = target


    edges_data['source'].append(source_node)
    edges_data['source_layer'].append(source_layer)
    edges_data['target'].append(target_node)
    edges_data['target_layer'].append(target_layer)
    edges_data['weight'].append(attrs.get('weight', 1.0))

# Create Arrow table
table = pa.table(edges_data)

# Save as Parquet
pq.write_table(table, 'network.parquet')
```

**Note**

The legacy approach above does not preserve node attributes, network type, or directedness.

Use `save_to_arrow` or `save_network_to_parquet` for zero-loss preservation.

### Export Statistics and Results

#### Save Analysis Results

Persist computed metrics (via DSL) for later inspection or sharing with colleagues.

```
from py3plex.dsl import Q

# Compute statistics
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Save to CSV
df = result.to_pandas()
df.to_csv('node_statistics.csv', index=False)
```

#### Save Communities

Store detected communities alongside node and layer IDs for downstream visualization or comparison.

```
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
import pandas as pd

# Detect communities
communities = louvain_multilayer(network, random_state=42)

# Convert to DataFrame
data = []
for (node, layer), comm_id in communities.items():
    data.append({
        'node': node,
        'layer': layer,
        'community': comm_id
    })

df = pd.DataFrame(data)
df.to_csv('communities.csv', index=False)
```

### Batch Export

#### Export Multiple Formats

Create a small export bundle so you can hand off the same network in different formats.

```
import json
import os
import pickle
import pandas as pd

# Prepare reusable structures
def edges_to_df(network):
    records = []
    for (src, src_layer), (dst, dst_layer), attrs in network.get_edges(data=True):
        records.append({
            "source": src,
            "source_layer": src_layer,
            "target": dst,
            "target_layer": dst_layer,
            "weight": attrs.get("weight", 1.0),
        })
    return pd.DataFrame(records)

edges_df = edges_to_df(network)

network_json = {
    "nodes": [{"id": n, "layer": l} for (n, l) in network.get_nodes()],
    "edges": edges_df.to_dict(orient="records"),
    "layers": network.layers,
}

# Create output directory
os.makedirs('export', exist_ok=True)

# Export in multiple formats
formats = {
    'pickle': lambda: pickle.dump(network, open('export/network.pkl', 'wb')),
    'json': lambda: json.dump(network_json, open('export/network.json', 'w')),
    'csv': lambda: edges_df.to_csv('export/edges.csv', index=False)
}

for fmt, export_func in formats.items():
    try:
        export_func()
        print(f"Exported to {fmt}")
    except Exception as e:
        print(f"Failed to export {fmt}: {e}")
```

### Next Steps

- **Load data:** *How to Load and Build Networks*
- **Visualize networks:** *How to Visualize Multilayer Networks*
- **API reference:** *API Documentation*

### 3.3.9 How to Visualize Multilayer Networks

**Goal:** Create publication-ready visualizations of multilayer networks.

**Prerequisites:** A loaded network (see *How to Load and Build Networks*). Most snippets reuse a `network` object created in the quick plot example; swap in your own network if you already have one loaded.

#### Basic Visualization

#### Quick Plot

```
from py3plex.core import multinet

# Load network
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'work', 'Carol', 'work', 1],
], input_type="list")

# Visualize
network.visualize_network(show=True)
```

This opens an interactive window with the aggregated multilayer visualization. Set `show=False` for headless environments.

#### Save to File

```
network.visualize_network(
    output_file='network.png',  # png, pdf, svg are supported
    show=False
)
```

#### Customizing Visualizations

#### Node Sizes and Colors

The built-in hairball plot draws the aggregated network with layer-based colors. You can override sizing and colors explicitly to emphasize node importance:

```
import matplotlib.pyplot as plt
from py3plex.visualization import hairball_plot
from py3plex.dsl import Q

# Compute degrees for sizing
degree_data = Q.nodes().compute("degree").execute(network)
node_order = list(network.core_network.nodes())
node_sizes = [
    degree_data.get(node, {'degree': 1})['degree'] * 50
    for node in node_order
]
```

```
fig, ax = plt.subplots(figsize=(8, 8))
hairball_plot(
    network.core_network,
    ax=ax,
    node_sizes=node_sizes,
    legend=True,
    layout_algorithm="force",
    alpha_channel=0.6,
)
fig.savefig("network_sized.png", dpi=300, bbox_inches="tight")
```

### Edge Weights

```
import matplotlib.pyplot as plt
import networkx as nx

# Widths based on stored weight (default to 1); clamp to keep faint edges visible
edge_widths = [
    max(data.get('weight', 1), 0.5)
    for _, _, data in network.core_network.edges(data=True)
]

pos = nx.spring_layout(network.core_network, k=0.6)
plt.figure(figsize=(8, 8))
nx.draw_networkx(
    network.core_network,
    pos,
    with_labels=False,
    node_size=50,
    edge_color="gray",
    width=edge_widths,
    alpha=0.7,
)
plt.axis("off")
plt.tight_layout()
plt.show()
```

### Layout Algorithms

Switch layouts to highlight different structures: force-directed for overall connectivity, circular for equal treatment of nodes within a layer, or diagonal layouts for explicit layer separation.

### Force-Directed Layout

```
from py3plex.visualization import hairball_plot

hairball_plot(
    network.core_network,
```

```
    layout_algorithm='force',
    legend=True,
    display=False
)
```

### Circular Layout

```
import matplotlib.pyplot as plt
import networkx as nx

pos = nx.circular_layout(network.core_network)

plt.figure(figsize=(8, 8))
nx.draw_networkx(
    network.core_network,
    pos,
    node_size=80,
    edge_color='lightgray',
    with_labels=False,
    alpha=0.8
)
plt.axis('off')
plt.tight_layout()
plt.savefig('circular_layout.png', dpi=300, bbox_inches='tight')
```

### Diagonal Multilayer Layout

Use the built-in diagonal style to keep layer membership explicit (`orientation` controls whether layers slope up or down):

```
network.visualize_network(
    style="diagonal",
    orientation="upper",
    legend=True,
    show=True
)
```

### Layer-Specific Visualization

#### Visualize Single Layer

```
from py3plex.dsl import Q, L
import matplotlib.pyplot as plt
import networkx as nx

# Extract nodes belonging to one layer
layer_nodes = Q.nodes().from_layers(L["friends"]).execute(network)
friends_subgraph = network.core_network.subgraph(layer_nodes.keys())
```

```
pos = nx.spring_layout(friends_subgraph, k=0.6)
plt.figure(figsize=(6, 6))
nx.draw_networkx(
    friends_subgraph,
    pos,
    node_color='steelblue',
    edge_color='gray',
    node_size=150,
    with_labels=True,
    alpha=0.8
)
plt.axis('off')
plt.tight_layout()
plt.savefig('friends_layer.png', dpi=300, bbox_inches='tight')
```

### Side-by-Side Layer Comparison

```
import networkx as nx
import matplotlib.pyplot as plt
from py3plex.dsl import Q, L

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for idx, layer in enumerate(['friends', 'work', 'family']):
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)
    layer_subgraph = network.core_network.subgraph(layer_nodes.keys())
    pos = nx.spring_layout(layer_subgraph, k=0.6)

    nx.draw_networkx(
        layer_subgraph,
        pos,
        ax=axes[idx],
        node_color='lightblue',
        edge_color='gray',
        with_labels=False,
        node_size=120,
        alpha=0.8
    )
    axes[idx].set_title(f'Layer: {layer}')
    axes[idx].axis('off')

plt.tight_layout()
plt.savefig('layers_comparison.png', dpi=300)
```

### DSL-Driven Visualization Workflows

**Goal:** Use DSL queries to select and visualize specific network subsets.

The DSL enables you to create targeted visualizations by filtering nodes and edges before rendering. This is essential for large multilayer networks where visualizing everything is im-

practical. Unless otherwise noted, the examples below reuse the `network` instance built in the basic visualization section.

### Visualize High-Degree Subnetwork

**Focus visualization on hub nodes:**

```
from py3plex.core import multinet
from py3plex.dsl import Q
import matplotlib.pyplot as plt

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "py3plex/datasets/_data/synthetic_multilayer.edges",
    input_type="multiedgelist"
)

# Query high-degree nodes
hubs = (
    Q.nodes()
     .compute("degree")
     .where(degree__gt=6)
     .execute(network)
)

print(f"Hub nodes (degree > 6): {len(hubs)}")

# Extract hub subgraph
hub_subgraph = network.core_network.subgraph(hubs.keys())

# Create visualization
import networkx as nx
pos = nx.spring_layout(hub_subgraph, k=0.5, iterations=50)

plt.figure(figsize=(12, 10))

# Color by layer
layer_colors = {'layer1': 'red', 'layer2': 'blue', 'layer3': 'green'}
node_colors = [layer_colors.get(node[1], 'gray') for node in hub_subgraph.nodes()]

# Size by degree
node_sizes = [hubs[node]['degree'] * 100 for node in hub_subgraph.nodes()]

nx.draw_networkx(
    hub_subgraph,
    pos,
    node_color=node_colors,
    node_size=node_sizes,
    with_labels=True,
    font_size=8,
```

```
    alpha=0.7
)

plt.title('Hub Nodes Subnetwork (degree > 6)', fontsize=14)
plt.axis('off')
plt.tight_layout()
plt.savefig('hub_subnetwork.png', dpi=300, bbox_inches='tight')
print("Saved: hub_subnetwork.png")
```

**Example output (counts depend on the dataset):**

```
Hub nodes (degree > 6): 18
Saved: hub_subnetwork.png
```

### Visualize Community Structure with DSL

**Community-aware visualization:**

```
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
from py3plex.dsl import execute_query
import matplotlib.pyplot as plt
import networkx as nx

# Detect communities
communities = louvain_communities(network)

# Attach community labels
for (node, layer), comm_id in communities.items():
    network.core_network.nodes[(node, layer)]['community'] = comm_id

# Visualize each community separately
community_ids = set(communities.values())
n_communities = len(community_ids)

fig, axes = plt.subplots(1, min(n_communities, 4), figsize=(20, 5))
if n_communities == 1:
    axes = [axes]

for idx, comm_id in enumerate(sorted(community_ids)[:4]):
    # Query community members
    comm_nodes_result = execute_query(
        network,
        f'SELECT nodes WHERE community={comm_id}'
    )
    comm_nodes = comm_nodes_result['nodes']

    # Extract community subgraph
    comm_subgraph = network.core_network.subgraph(comm_nodes)

    # Visualize
    pos = nx.spring_layout(comm_subgraph, k=0.3)
```

```
    # Color by layer
    layer_colors = {'layer1': '#FF6B6B', 'layer2': '#4ECDC4', 'layer3': '#45B7D1'}
    node_colors = [layer_colors.get(node[1], 'gray') for node in comm_subgraph.nodes()]

    nx.draw_networkx(
        comm_subgraph,
        pos,
        ax=axes[idx],
        node_color=node_colors,
        node_size=200,
        with_labels=False,
        alpha=0.8
    )

    axes[idx].set_title(f'Community {comm_id}\n({len(comm_nodes)} nodes)')
    axes[idx].axis('off')

plt.tight_layout()
plt.savefig('communities_separate.png', dpi=300, bbox_inches='tight')
print(f"Saved: communities_separate.png with {min(n_communities, 4)} communities")
```

### Layer-Specific Visualizations with DSL

**Compare layer structures visually:**

```
from py3plex.dsl import Q, L
import matplotlib.pyplot as plt
import networkx as nx

layers = network.get_layers()
n_layers = len(layers)

fig, axes = plt.subplots(1, n_layers, figsize=(6*n_layers, 5))
if n_layers == 1:
    axes = [axes]

for idx, layer in enumerate(layers):
    # Query layer nodes and edges
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)
    layer_edges = Q.edges().from_layers(L[layer]).execute(network)

    # Extract layer subgraph
    layer_subgraph = network.core_network.subgraph(layer_nodes.keys())

    # Compute metrics for sizing
    layer_metrics = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
```

```
        .execute(network)
    )

    # Visualization
    pos = nx.spring_layout(layer_subgraph, k=0.5, iterations=50)

    # Size by degree
    node_sizes = [layer_metrics[node]['degree'] * 50 for node in layer_subgraph.nodes()]

    nx.draw_networkx(
        layer_subgraph,
        pos,
        ax=axes[idx],
        node_size=node_sizes,
        node_color='lightblue',
        edge_color='gray',
        with_labels=False,
        alpha=0.7
    )

    axes[idx].set_title(f'{layer}\n{len(layer_nodes)} nodes, {len(layer_edges)} edges')
    axes[idx].axis('off')

plt.tight_layout()
plt.savefig('layer_comparison.png', dpi=300, bbox_inches='tight')
print("Saved: layer_comparison.png")
```

**Expected output:**

```
Saved: layer_comparison.png
```

### Visualize Central Nodes

**Highlight nodes by centrality (betweenness for color, degree for size):**

```
from py3plex.dsl import Q
import matplotlib.pyplot as plt
import networkx as nx

# Compute centrality for all nodes
centrality_result = (
    Q.nodes()
     .compute("betweenness_centrality", "degree")
     .execute(network)
)

# Extract betweenness values for coloring
betweenness = {
    node: data['betweenness_centrality']
    for node, data in centrality_result.items()
}
```

```
# Create visualization
plt.figure(figsize=(14, 10))
pos = nx.spring_layout(network.core_network, k=0.5, iterations=50)

# Node colors based on betweenness (using colormap)
node_list = list(network.core_network.nodes())
betw_values = [betweenness.get(node, 0) for node in node_list]

# Node sizes based on degree
degrees = {node: data['degree'] for node, data in centrality_result.items()}
node_sizes = [degrees.get(node, 1) * 100 for node in node_list]

# Draw
nodes = nx.draw_networkx_nodes(
    network.core_network,
    pos,
    node_size=node_sizes,
    node_color=betw_values,
    cmap=plt.cm.YlOrRd,
    alpha=0.8
)

nx.draw_networkx_edges(
    network.core_network,
    pos,
    alpha=0.2,
    edge_color='gray'
)

# Add colorbar
plt.colorbar(nodes, label='Betweenness Centrality')
plt.title('Network Centrality Visualization\n(size=degree, color=betweenness)', fontsize=14)
plt.axis('off')
plt.tight_layout()
plt.savefig('centrality_viz.png', dpi=300, bbox_inches='tight')
print("Saved: centrality_viz.png")
```

### Dynamics State Visualization

**Visualize epidemic simulation results:**

```
from py3plex.dynamics import SIRDynamics
from py3plex.dsl import Q
import matplotlib.pyplot as plt
import networkx as nx

# Run SIR simulation
sir = SIRDynamics(network, beta=0.3, gamma=0.1, initial_infected=0.05)
sir.set_seed(42)
```

```
results = sir.run(steps=100)

# Attach final state
final_state = results.trajectory[-1]
for node, state in final_state.items():
    network.core_network.nodes[node]['sir_state'] = state

# Query infected and recovered nodes
infected = Q.nodes().where(sir_state='I').execute(network)
recovered = Q.nodes().where(sir_state='R').execute(network)
susceptible = Q.nodes().where(sir_state='S').execute(network)

print(f"Infected: {len(infected)}, Recovered: {len(recovered)}, Susceptible: {len(suscepti

# Visualization
plt.figure(figsize=(14, 10))
pos = nx.spring_layout(network.core_network, k=0.5, iterations=50)

# Color by SIR state
state_colors = {'S': 'lightblue', 'I': 'red', 'R': 'green'}
node_colors = [
    state_colors.get(network.core_network.nodes[node].get('sir_state'), 'gray')
    for node in network.core_network.nodes()
]

nx.draw_networkx(
    network.core_network,
    pos,
    node_color=node_colors,
    node_size=300,
    with_labels=False,
    alpha=0.8
)

# Legend
from matplotlib.patches import Patch
legend_elements = [
    Patch(facecolor='lightblue', label=f'Susceptible ({len(susceptible)})'),
    Patch(facecolor='red', label=f'Infected ({len(infected)})'),
    Patch(facecolor='green', label=f'Recovered ({len(recovered)})')
]
plt.legend(handles=legend_elements, loc='upper right')

plt.title('SIR Epidemic Simulation (t=100)', fontsize=14)
plt.axis('off')
plt.tight_layout()
plt.savefig('sir_visualization.png', dpi=300, bbox_inches='tight')
print("Saved: sir_visualization.png")
```

**Example output (depends on network size and random seed):**

```
Infected: 8, Recovered: 82, Susceptible: 30
Saved: sir_visualization.png
```

### Interactive Visualization with DSL Filtering

**Create interactive plots with Plotly (browser-rendered HTML):**

```
from py3plex.dsl import Q
import plotly.graph_objects as go
import networkx as nx

# Query nodes with metrics
metrics = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Extract subgraph (optional: filter for clarity)
high_degree = {
    node: data for node, data in metrics.items()
    if data['degree'] > 4
}

subgraph = network.core_network.subgraph(high_degree.keys())

# Layout
pos = nx.spring_layout(subgraph, k=0.5)

# Create edge trace
edge_x = []
edge_y = []
for edge in subgraph.edges():
    x0, y0 = pos[edge[0]]
    x1, y1 = pos[edge[1]]
    edge_x.extend([x0, x1, None])
    edge_y.extend([y0, y1, None])

edge_trace = go.Scatter(
    x=edge_x, y=edge_y,
    line=dict(width=0.5, color='#888'),
    hoverinfo='none',
    mode='lines'
)

# Create node trace
node_x = []
node_y = []
node_text = []
node_sizes = []
```

```
node_colors = []

for node in subgraph.nodes():
    x, y = pos[node]
    node_x.append(x)
    node_y.append(y)

    degree = high_degree[node]['degree']
    betw = high_degree[node]['betweenness_centrality']

    node_text.append(f'{node}<br>Degree: {degree}<br>Betweenness: {betw:.4f}')
    node_sizes.append(degree * 5)
    node_colors.append(betw)

node_trace = go.Scatter(
    x=node_x, y=node_y,
    mode='markers',
    hoverinfo='text',
    text=node_text,
    marker=dict(
        showscale=True,
        colorscale='YlOrRd',
        size=node_sizes,
        color=node_colors,
        colorbar=dict(
            title="Betweenness"
        ),
        line_width=2
    )
)

# Create figure
fig = go.Figure(
    data=[edge_trace, node_trace],
    layout=go.Layout(
        title='Interactive Network Visualization (degree > 4)',
        showlegend=False,
        hovermode='closest',
        margin=dict(b=0, l=0, r=0, t=40),
        xaxis=dict(showgrid=False, zeroline=False, showticklabels=False),
        yaxis=dict(showgrid=False, zeroline=False, showticklabels=False)
    )
)

# Save to HTML
fig.write_html('interactive_network.html')
print("Saved: interactive_network.html (open in browser)")
```

**Why use DSL for visualization?**

- **Targeted views:** Filter nodes/edges to reduce visual clutter

- **Multi-scale:** Visualize different network scales (layer, community, subnetwork)
- **Attribute-driven:** Color/size nodes based on computed metrics
- **Reproducible:** Query-based selections are version-controllable
- **Efficient:** Process only relevant subsets for large networks

**Next steps with DSL visualization:**

- **Full DSL tutorial:** *How to Query Multilayer Graphs with the SQL-like DSL* - Complete DSL reference
- **Community detection:** *How to Run Community Detection on Multilayer Networks* - Visualize communities
- **Dynamics:** *How to Simulate Multilayer Dynamics* - Visualize dynamics results

### Community Visualization

#### Color by Communities

```
from py3plex.algorithms.community_detection import louvain_communities
from py3plex.visualization import visualize_communities

# Detect communities
communities = louvain_communities(network)

# Visualize with community colors
visualize_communities(
    network,
    communities,
    output_file='communities.png',
    show=True
)
```

### Interactive Visualization

#### Using Plotly

```
from py3plex.visualization import plotly_visualization

# Create interactive plot
fig = plotly_visualization(network)

# Show in browser
fig.show()

# Or save as HTML
fig.write_html('network_interactive.html')
```

#### Export for Gephi/Cytoscape

#### Export to GraphML

```
import networkx as nx

# Convert to NetworkX
G = network.to_networkx()

# Export
nx.write_graphml(G, 'network.graphml')
```

Then open `network.graphml` in Gephi or Cytoscape for advanced visualization of the aggregated network.

#### Next Steps

- **Compute statistics to visualize:** *How to Compute Network Statistics*
- **Detect communities:** *How to Run Community Detection on Multilayer Networks*
- **API reference:** *API Documentation*

### 3.3.10 How to Query Multilayer Graphs with the SQL-like DSL

**Goal:** Use py3plex's SQL-inspired Domain-Specific Language (DSL) to query, filter, and analyze multilayer networks. The DSL is a **first-class query language** specifically designed for multilayer graph structures, providing both string syntax for interactive exploration and a type-safe builder API for production code.

**Run this guide online**

You can run this tutorial in your browser without any local installation:

[14] Or see the full executable example: `example_dsl_builder_api.py`

**What Makes This DSL Special:**

- **Graph-aware**: Unlike generic query languages, the DSL understands multilayer structures—layers, layer intersections, intralayer vs. interlayer edges, and (node, layer) tuple semantics.
- **Dual interfaces**: String syntax for rapid prototyping in notebooks; builder API (`Q`, `L`) for IDE autocompletion and type checking.
- **Integrated computation**: Compute centrality, clustering, and other network metrics directly in queries, with results returned as pandas DataFrames or NetworkX graphs.
- **Temporal support**: Query network snapshots and time ranges when your network includes temporal information.

**Prerequisites:**

- A loaded `multi_layer_network` object (see *How to Load and Build Networks*)
- Basic familiarity with multilayer network concepts (nodes, layers, intralayer/interlayer edges)

[14] https://colab.research.google.com/github/SkBlaz/py3plex/blob/master/notebooks/query_with_dsl.ipynb

- For complete DSL grammar and operator reference, see *DSL Reference*
- Pick your interface: string syntax for quick experiments, builder API for reusable production code with IDE support

### Conceptual Overview

The DSL has **two complementary interfaces** that compile to the same internal representation:

1. **String Syntax** (`execute_query(network, "SELECT nodes WHERE ...")`)
   - SQL-like, human-readable
   - Ideal for interactive exploration in Jupyter notebooks or the REPL
   - Quick one-liners for common queries
2. **Builder API** (`Q.nodes().where(...).compute(...).execute(network)`)
   - Pythonic, chainable methods
   - Type-safe with IDE autocompletion
   - Recommended for production code and complex workflows

**Mental Model:**

A typical DSL query follows this pipeline:

```
SELECT nodes/edges
→ FROM LAYERS (restrict to specific layers)
→ WHERE (filter by attributes or special predicates)
→ COMPUTE (calculate metrics like degree, centrality)
→ ORDER BY (sort results)
→ LIMIT (cap number of results)
→ EXPORT (materialize as DataFrame, NetworkX graph, etc.)
```

**Key Concepts:**

- **Nodes as (node, layer) tuples**: In multilayer networks, a node may appear in multiple layers. The DSL represents these as `(node_id, layer_name)` pairs.
- **Layer set algebra**: Combine layers with set operations (`|` union, `&` intersection, `-` difference, `~` complement). The new LayerSet algebra enables expressive layer selection like `L["* - coupling"]` or `L["(ppi | gene) & disease"]`. See *Layer Set Algebra* for complete documentation.
- **Special predicates**: `intralayer=True` selects edges within a layer; `interlayer=("layer1", "layer2")` selects edges crossing specific layers.
- **Lazy execution**: Queries are built incrementally and executed only when `.execute(network)` is called.
- **``Q`` and ``L`` helpers**: `Q` builds queries; `L` builds layer expressions. Import both from `py3plex.dsl`.

**Comparison to SQL:**

Think of the DSL as SQL for graphs:

- `SELECT nodes WHERE degree > 5` SQL's `SELECT * FROM nodes WHERE degree > 5`
- But instead of tables, you're querying **nodes and edges** with **multilayer and temporal attributes**

- Layer filters and graph-specific predicates (`intralayer`, `interlayer`) have no SQL equivalent

### String Syntax (Quick and Readable)

The string syntax provides a concise, SQL-like way to express queries. Best for exploratory analysis and quick investigations.

### Basic SELECT

Select all nodes and inspect the result:

**Note**

Where to find this data

The examples in this guide use one of the following:

- **Built-in data generators** like `random_generators.random_multilayer_ER(...)` (recommended for self-contained examples)
- **Example files** from the repository at `datasets/multiedgelist.txt` or similar
- The **built-in datasets module**: `from py3plex.datasets import fetch_multilayer`

For this example, we'll create a simple network programmatically:

```
from py3plex.core import multinet
from py3plex.dsl import execute_query

# Create a simple multilayer network
network = multinet.multi_layer_network()
network.add_edges([
    ['alice', 'social', 'bob', 'social', 1],
    ['bob', 'social', 'charlie', 'social', 1],
    ['alice', 'work', 'charlie', 'work', 1],
    ['bob', 'work', 'dave', 'work', 1],
], input_type="list")

# Get all nodes
result = execute_query(network, 'SELECT nodes')

print(f"Found {len(result)} nodes")
# Inspect a few items
for i, (node, data) in enumerate(result.items()):
    print(f"  {node}: {data}")
    if i >= 4:
        break
```

**Expected output (will vary with your data):**

```
Found 7 nodes
  ('alice', 'social'): {'degree': 1, 'layer': 'social', 'layer_count': 2}
  ('bob', 'social'): {'degree': 2, 'layer': 'social', 'layer_count': 2}
```

```
 ('charlie', 'social'): {'degree': 1, 'layer': 'social', 'layer_count': 2}
 ('alice', 'work'): {'degree': 1, 'layer': 'work', 'layer_count': 2}
 ('bob', 'work'): {'degree': 1, 'layer': 'work', 'layer_count': 2}
 ('charlie', 'work'): {'degree': 1, 'layer': 'work', 'layer_count': 2}
 ('dave', 'work'): {'degree': 1, 'layer': 'work', 'layer_count': 1}
```

**Tip**

Loading from files

To load from a file in the repository:

```
# Using a file from the datasets/ directory
network.load_network("datasets/multiedgelist.txt", input_type="multiedgelist")

# Or using an absolute path
import os
path = os.path.join(os.path.dirname(__file__), "datasets", "multiedgelist.txt")
network.load_network(path, input_type="multiedgelist")
```

**Note:** `execute_query` returns a dict-like result. Keys are `(node, layer)` tuples representing node-layer pairs. The `layer_count` attribute counts how many distinct layers the node appears in across the entire network. `len(result)` gives the number of returned items, and `result.items()` yields the full (key, value) pairs.

### Filter by Layer

Restrict queries to nodes in a specific layer:

```
# Get nodes in the 'friends' layer only
result = execute_query(
    network,
    'SELECT nodes WHERE layer="friends"'
)

print(f"Nodes in 'friends' layer: {len(result)}")
```

**Understanding Layer Filters:**

- `layer="friends"` selects only the node-layer pairs where `layer == "friends"`
- This does **not** select all occurrences of nodes across layers—only their representation in the specified layer
- Use `layer_count >= 2` to find nodes appearing in multiple layers (`layer_count` is the number of distinct layers for that node)

**Example with statistics:**

```
result = execute_query(
    network,
    'SELECT nodes WHERE layer="friends" COMPUTE degree'
)
```

```
df = result.to_pandas()
print(f"Nodes in 'friends': {len(df)}")
print(f"Average degree in 'friends': {df['degree'].mean():.2f}")
print(f"Max degree in 'friends': {df['degree'].max()}")
```

**Expected output (example from a synthetic dataset):**

```
Nodes in 'friends': 42
Average degree in 'friends': 5.23
Max degree in 'friends': 15
```

### Filter by Property

Use comparisons to filter nodes by computed or intrinsic attributes:

```
# High-degree nodes
result = execute_query(
    network,
    'SELECT nodes WHERE degree > 5'
)
print(f"High-degree nodes: {len(result)}")

# Multilayer nodes with high degree
result = execute_query(
    network,
    'SELECT nodes WHERE degree > 5 AND layer_count >= 2'
)
print(f"High-degree multilayer nodes: {len(result)}")
```

**Supported operators:** `>`, `>=`, `<`, `<=`, `=`, `!=`

**Multiple conditions** are combined with `AND`. For more complex logic, use the builder API (see below).

**Expected output (counts will vary with your data):**

```
High-degree nodes: 34
High-degree multilayer nodes: 18
```

### Compute Statistics

The `COMPUTE` clause calculates network metrics and attaches them to result rows. This is where the DSL becomes powerful for analysis:

```
# Compute degree and betweenness centrality for nodes in 'social' layer
result = execute_query(
    network,
    'SELECT nodes WHERE layer="social" '
    'COMPUTE degree COMPUTE betweenness_centrality'
)

# Convert to pandas for analysis
```

```
df = result.to_pandas()

print("Top nodes by betweenness centrality:")
print(df[['id', 'degree', 'betweenness_centrality']].head())

print("\nSummary statistics:")
print(df[['degree', 'betweenness_centrality']].describe())
```

**Expected output (values depend on the network):**

```
Top nodes by betweenness centrality:
                 id  degree  betweenness_centrality
0  (alice, social)      12                   0.245
1    (bob, social)       8                   0.189
2    (eve, social)      15                   0.301
3  (frank, social)       7                   0.134
4  (grace, social)      11                   0.221

Summary statistics:
            degree  betweenness_centrality
count     65.000000               65.000000
mean       6.846154                0.112308
std        3.241057                0.089542
min        1.000000                0.000000
25%        4.000000                0.045000
50%        7.000000                0.089000
75%       10.000000                0.167000
max       15.000000                0.301000
```

**Available measures** include: `degree`, `betweenness_centrality`, `closeness_centrality`, `eigenvector_centrality`, `pagerank`, `clustering`, `communities`. See *DSL Reference* for the complete list.

**Use case:** This pattern is ideal for generating summary statistics for papers, reports, or further statistical analysis.

### Builder API (Type-Safe)

The **builder API is the recommended approach for production code**. It provides:

- IDE autocompletion and inline documentation
- Type checking with tools like mypy
- Clearer error messages
- Easier refactoring and composition of queries

All builder queries compile to the same AST as string queries, ensuring consistent semantics.

### Basic Queries

Create and execute queries using the `Q` and `L` imports:

```
from py3plex.dsl import Q, L

# Get all nodes
result = Q.nodes().execute(network)
print(f"Total nodes: {len(result)}")

# Get nodes from a specific layer
result = (
    Q.nodes()
     .from_layers(L["friends"])
     .execute(network)
)
print(f"Nodes in 'friends' layer: {len(result)}")
```

**Query reusability:** You can define a query once and execute it with different networks:

```
high_degree_query = Q.nodes().where(degree__gt=10).compute("betweenness_centrality")

# Execute on multiple networks
result_network1 = high_degree_query.execute(network1)
result_network2 = high_degree_query.execute(network2)
```

### Filtering

Use `where()` to add filter conditions. The builder API uses Django-style `__` suffixes for comparisons:

```
# Filter by property
result = (
    Q.nodes()
     .where(degree__gt=5)
     .execute(network)
)
print(f"Nodes with degree > 5: {len(result)}")

# Multiple conditions (combined with AND)
result = (
    Q.nodes()
     .from_layers(L["work"])
     .where(degree__gt=3, layer_count__gte=2)
     .execute(network)
)
print(f"Multilayer high-degree nodes in 'work': {len(result)}")
```

**Supported comparison suffixes:**

- `__gt`: greater than (>)
- `__gte` or `__ge`: greater than or equal (>=)

- `__lt`: less than (<)
- `__lte` or `__le`: less than or equal (<=)
- `__eq`: equal (=)
- `__ne` or `__neq`: not equal (!=)

**Understanding `layer_count`:**

In multilayer networks, a node may appear in multiple layers. The `layer_count` attribute indicates how many layers the node participates in.

Concrete example (using the network built in this page):

```
# Nodes that appear in exactly one layer
layer_specific = Q.nodes().where(layer_count__eq=1).execute(network)
print(f"Layer-specific nodes: {len(layer_specific)}")

# Nodes that appear in two or more layers (cross-layer connectors)
connectors = Q.nodes().where(layer_count__gte=2).execute(network)
print(f"Connector nodes: {len(connectors)}")
```

**Expected output (illustrative):**

```
Layer-specific nodes: 3
Connector nodes: 2
```

- `layer_count__gte=2`: nodes appearing in at least 2 layers
- `layer_count__eq=1`: nodes appearing in exactly 1 layer (layer-specific nodes)

This is useful for identifying "connector" nodes that bridge multiple contexts.

### Computing Metrics

Use `compute()` to calculate network metrics. Metrics are computed efficiently and attached to result rows:

```
# Compute multiple metrics
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality", "clustering")
     .execute(network)
)

# Convert to DataFrame and analyze
df = result.to_pandas()
print(df.head(10))

# Get top nodes by a metric
top_by_betweenness = df.nlargest(10, 'betweenness_centrality')
print("\nTop 10 nodes by betweenness centrality:")
print(top_by_betweenness[['id', 'betweenness_centrality', 'degree']])
```

**Order of operations:**

- `compute()` can be called at any point in the chain

- Filters (`where()`) can reference computed metrics only if the metric is computed **before** the filter
- For best performance, filter first, then compute:

```
# Good: Filter first, then compute
result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=5)
     .compute("betweenness_centrality")
     .execute(network)
)
```

**Expected output (illustrative):**

```
                  id  degree  betweenness_centrality  clustering
0   (alice, social)      12                0.245000    0.545455
1     (bob, social)       8                0.189000    0.642857
2     (eve, social)      15                0.301000    0.428571
3   (frank, social)       7                0.134000    0.666667
4   (grace, social)      11                0.221000    0.509091
...

Top 10 nodes by betweenness centrality:
                  id  betweenness_centrality  degree
2     (eve, social)                0.301000      15
0   (alice, social)                0.245000      12
4   (grace, social)                0.221000      11
1     (bob, social)                0.189000       8
...
```

### Computing Metrics with Uncertainty

**New in py3plex 1.0:** The DSL now supports **first-class uncertainty** for computed metrics. This allows you to estimate statistical uncertainty (confidence intervals, standard deviations) for network statistics via bootstrap, perturbation, or Monte Carlo methods.

**Why uncertainty matters:**

- Networks are often noisy or sampled (e.g., social networks with missing edges)
- Centrality metrics can be sensitive to small perturbations
- Uncertainty quantification helps distinguish signal from noise
- Required for robust statistical inference and hypothesis testing

**Basic usage:**

```
# Compute degree with uncertainty estimation
result = (
    Q.nodes()
     .compute(
         "degree",
         "betweenness_centrality",
```

```
        uncertainty=True,
        method="perturbation",  # or "bootstrap", "seed"
        n_samples=100,          # number of resamples
        ci=0.95                 # confidence interval level
    )
    .execute(network)
)

# Access uncertainty information
df = result.to_pandas()
print(df[['id', 'degree', 'betweenness_centrality']].head(3))
```

**Expected output (illustrative):**

```
                  id                                                degree
0   (alice, social)  {'mean': 12, 'std': 0.42, 'quantiles': {0.025:...  {'mean': 0.245, 'sto
1     (bob, social)  {'mean': 8, 'std': 0.35, 'quantiles': {0.025: ...  {'mean': 0.189, 'sto
2     (eve, social)  {'mean': 15, 'std': 0.61, 'quantiles': {0.025:...  {'mean': 0.301, 'sto
```

Results contain mean, std, and quantiles for each metric. The `degree` and `betweenness_centrality` columns now contain dictionaries instead of scalar values.

**Uncertainty methods:**

- `"perturbation"`: Drop a small fraction of edges/nodes randomly (default: 5%)
- `"bootstrap"`: Resample nodes/edges with replacement
- `"seed"`: Run stochastic algorithms with different random seeds
- `"jackknife"`: Leave-one-out resampling

**Parameters:**

- `uncertainty` (bool): Enable uncertainty estimation (default: False)
- `method` (str): Resampling strategy (default: “perturbation”)
- `n_samples` (int): Number of resamples (default: 50)
- `ci` (float): Confidence interval level, e.g., 0.95 for 95% CI (default: 0.95)

**Example with confidence intervals:**

```
# Find hubs with uncertainty bounds
hubs = (
    Q.nodes()
     .compute(
         "degree",
         "betweenness_centrality",
         uncertainty=True,
         method="perturbation",
         n_samples=200,
         ci=0.95
     )
     .order_by("-betweenness_centrality")
     .limit(10)
     .execute(network)
```

```
)

# Extract uncertainty information
df = hubs.to_pandas()

# When uncertainty=True, values are dicts with mean, std, quantiles
for idx, row in df.head().iterrows():
    node_id = row['id']
    bc_info = row['betweenness_centrality']

    if isinstance(bc_info, dict):
        mean = bc_info['mean']
        std = bc_info.get('std', 0)
        ci_low = bc_info.get('quantiles', {}).get(0.025, mean)
        ci_high = bc_info.get('quantiles', {}).get(0.975, mean)

        print(f"{node_id}:")
        print(f"  Betweenness: {mean:.4f} ± {std:.4f}")
        print(f"  95% CI: [{ci_low:.4f}, {ci_high:.4f}]")
```

**Expected output (example metrics on a weighted layer):**

```
('eve', 'social'):
  Betweenness: 0.3010 ± 0.0234
  95% CI: [0.2589, 0.3442]
('alice', 'social'):
  Betweenness: 0.2450 ± 0.0198
  95% CI: [0.2087, 0.2821]
('grace', 'social'):
  Betweenness: 0.2210 ± 0.0176
  95% CI: [0.1901, 0.2534]
```

**Backward compatibility:**

When `uncertainty=False` (the default), metrics return scalar values as before. Your existing queries work unchanged:

```
# Traditional deterministic computation
result = Q.nodes().compute("degree").execute(network)
# 'degree' values are scalars (int/float)

# With uncertainty
result_unc = Q.nodes().compute("degree", uncertainty=True).execute(network)
# 'degree' values are dicts with mean, std, quantiles
```

**Use cases:**

1. **Comparing networks**: Test if centrality differences between networks are statistically significant
2. **Robust ranking**: Identify nodes that consistently rank high across perturbations
3. **Network inference**: Quantify uncertainty when inferring networks from noisy data
4. **Hypothesis testing**: Generate null distributions for significance testing

**Performance notes:**

- Uncertainty estimation is **opt-in** and only runs when explicitly requested
- Cost scales linearly with `n_samples` (e.g., 100 samples  100× slower)
- Use smaller `n_samples` (20-50) for exploration, larger (100-500) for publication
- Perturbation is fastest; bootstrap and jackknife are more expensive

**Further reading:**

- *How to Compute Network Statistics*: General guide to network statistics and uncertainty
- `examples/network_analysis/example_first_class_uncertainty.py`: Complete examples
- `py3plex.uncertainty` module: Low-level API for custom uncertainty workflows

### Sorting and Limiting

Use `order_by()` and `limit()` to control result ordering and size:

```
# Get top 10 nodes by degree
result = (
    Q.nodes()
     .compute("degree")
     .order_by("-degree")  # "-" prefix for descending
     .limit(10)
     .execute(network)
)

print("Top 10 highest-degree nodes:")
for node, data in result.items():
    print(f"  {node}: degree={data['degree']}")
```

**Sorting conventions:**

- `order_by("degree")`: ascending (low to high)
- `order_by("-degree")`: descending (high to low)
- Multiple keys: `order_by("-degree", "layer_count")`: sort by degree descending, then layer_count ascending

**Expected output (example ranking):**

```
Top 10 highest-degree nodes:
  ('eve', 'social'): degree=15
  ('alice', 'social'): degree=12
  ('grace', 'social'): degree=11
  ('charlie', 'work'): degree=10
  ('henry', 'friends'): degree=9
  ('diana', 'social'): degree=9
  ('bob', 'social'): degree=8
  ('frank', 'social'): degree=7
  ('iris', 'work'): degree=7
  ('jake', 'friends'): degree=6
```

### Working with Results

DSL queries return a `QueryResult` object that provides multiple ways to access and export data. Understanding how to work with results is crucial for integrating DSL queries into analysis pipelines.

### Access as Dictionary

`QueryResult` provides dictionary-like access via `.items()`:

```
result = Q.nodes().compute("degree").execute(network)

# Iterate over all items
for node, data in result.items():
    print(f"{node}: degree={data['degree']}")

# Inspect one sample entry
sample_key, sample_value = next(iter(result.items()))
print(f"Sample key type: {type(sample_key)}")
print(f"Sample key: {sample_key}")
print(f"Sample value: {sample_value}")
```

**Result structure for nodes:**

- **Keys**: `(node_id, layer)` tuples (for multilayer queries) or `node_id` (for single-layer queries)
- **Values**: Dictionaries with computed attributes (`{"degree": 5, "betweenness_centrality": 0.23, ...}`)

**Result structure for edges:**

- **Keys**: `((source, source_layer), (target, target_layer))` tuples (interlayer edges include both layer names)
- **Values**: Dictionaries with edge attributes and computed metrics

**Expected output:**

```
Sample key type: <class 'tuple'>
Sample key: ('alice', 'social')
Sample value: {'degree': 12, 'layer': 'social', 'layer_count': 2}
```

### Convert to Pandas

**This is the recommended way to integrate DSL queries with statistical analysis and plotting libraries.**

```
result = (
    Q.nodes()
     .from_layers(L["social"])
     .compute("degree", "betweenness_centrality", "clustering")
     .execute(network)
)

# Convert to DataFrame
```

```
df = result.to_pandas()

# Inspect structure
print(df.head())
print("\nColumn names:", df.columns.tolist())
print("\nSummary statistics:")
print(df[['degree', 'betweenness_centrality', 'clustering']].describe())

# Use pandas for further analysis
high_influence = df[
    (df['degree'] > 10) &
    (df['betweenness_centrality'] > 0.2)
]
print(f"\nHigh-influence nodes: {len(high_influence)}")
```

**DataFrame structure:**

- **For node queries**: Columns include `id` (the node-layer tuple or node ID), plus all computed attributes
- **For edge queries**: Columns include `source`, `target`, `source_layer`, `target_layer`, `weight`, plus computed attributes

**Expected output:**

```
                  id  degree  betweenness_centrality  clustering
0   (alice, social)      12                0.245000    0.545455
1     (bob, social)       8                0.189000    0.642857
2     (eve, social)      15                0.301000    0.428571
3   (frank, social)       7                0.134000    0.666667
4   (grace, social)      11                0.221000    0.509091

Column names: ['id', 'degree', 'betweenness_centrality', 'clustering']

Summary statistics:
             degree  betweenness_centrality  clustering
count     65.000000               65.000000   65.000000
mean       6.846154                0.112308    0.587692
std        3.241057                0.089542    0.145231
...

High-influence nodes: 8
```

**Multi-index option:**

For more complex analyses, you can reshape the `id` tuple into a multi-index:

```
df = result.to_pandas()
# Split 'id' tuple into separate columns
df[['node', 'layer']] = pd.DataFrame(df['id'].tolist(), index=df.index)
df = df.drop('id', axis=1)
df = df.set_index(['node', 'layer'])
print(df.head())
```

### Filter Results

You can filter results in two ways: **using the DSL's `where()` clause** (recommended) or **post-processing** with Python/pandas.

**Option 1: Filter in the query (recommended for large networks):**

```
# Filter before computation for efficiency
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .where(degree__gt=5)
     .execute(network)
)
```

**Option 2: Filter the result dictionary (for small networks or ad-hoc filtering):**

```
result = Q.nodes().compute("degree").execute(network)

# Pure Python filtering
high_degree = {
    node: data
    for node, data in result.items()
    if data['degree'] > 5
}
print(f"High-degree nodes: {len(high_degree)}")
```

**Option 3: Filter the DataFrame (most flexible for complex conditions):**

```
df = result.to_pandas()

# Use pandas boolean indexing
filtered = df[df['degree'] > 5]

# Complex conditions
interesting_nodes = df[
    (df['degree'] > 5) &
    (df['betweenness_centrality'] > df['betweenness_centrality'].mean())
]
```

**Performance note:** For very large networks (millions of nodes), filtering in the DSL query (Option 1) is most efficient because it avoids materializing unnecessary results. For smaller networks, pandas filtering (Option 3) is often more convenient.

### Advanced Queries

This section showcases the DSL's power for sophisticated multilayer network analysis. These patterns are common in research and can be adapted to your specific needs.

### Multiple Layer Selection

Use **layer algebra** to combine layers. The `L` object supports set operations (`|` for union, `&` for intersection, `-` for difference):

```
from py3plex.dsl import Q, L

# Union: nodes/edges from EITHER layer
result = (
    Q.nodes()
     .from_layers(L["friends"] | L["work"])
     .compute("degree")
     .execute(network)
)

df = result.to_pandas()
print(f"Combined nodes from 'friends' and 'work': {len(df)}")
print(f"Average degree across both layers: {df['degree'].mean():.2f}")
```

**Set semantics:**

- `L["friends"] | L["work"]`: **Union** of nodes/edges from both layers (nodes appearing in either layer)
- `L["friends"] & L["work"]`: **Intersection** (see next section)
- `L["friends"] - L["work"]`: **Difference** (nodes in friends but not work)

**Use case:** Compare activity across related contexts. For example, analyze user behavior across social and professional networks together.

**Expected output:**

```
Combined nodes from 'friends' and 'work': 87
Average degree across both layers: 6.12
```

### Layer Intersection

Find nodes that appear in **multiple specific layers**:

```
# Nodes present in BOTH 'friends' AND 'work' layers
result = (
    Q.nodes()
     .from_layers(L["friends"] & L["work"])
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

df = result.to_pandas()
print(f"Nodes in both 'friends' and 'work': {len(df)}")
print("\nThese are 'connector' nodes bridging social and professional contexts")
print(df.head(10))
```

**Semantics:**

- `L["friends"] & L["work"]` selects nodes that have representations in **both** layers

- This is different from `layer_count >= 2`, which selects nodes in **any** two layers
- Use intersection to find nodes bridging specific contexts

**Alternative approach using `layer_count`:**

```
# More general: nodes in at least 2 layers (any layers)
result = (
    Q.nodes()
     .where(layer_count__gte=2)
     .compute("degree")
     .execute(network)
)
print(f"Multilayer nodes (any 2+ layers): {len(result)}")
```

**Expected output:**

```
Nodes in both 'friends' and 'work': 23

These are 'connector' nodes bridging social and professional contexts
                   id  degree  betweenness_centrality
0    (alice, friends)      12                0.245000
1      (alice, work)       8                0.189000
2  (charlie, friends)      10                0.201000
3    (charlie, work)       7                0.145000
...
```

### Query Edges

The DSL supports edge queries with the same flexibility as node queries:

```
# Select edges from a layer with weight filter
edges = (
    Q.edges()
     .from_layers(L["social"])
     .where(weight__gt=0.5)
     .compute("edge_betweenness")
     .execute(network)
)

df = edges.to_pandas()
print(f"High-weight edges in 'social' layer: {len(df)}")
print("\nSample edges:")
print(df.head())

# Analyze edge distribution
print(f"\nMean edge weight: {df['weight'].mean():.3f}")
print(f"Mean edge betweenness: {df['edge_betweenness'].mean():.3f}")
```

**Edge result structure:**

For edge queries, the DataFrame includes:

- `source`, `target`: node identifiers

- `source_layer`, `target_layer`: layer names (same for intralayer edges)
- `weight`: edge weight (default 1.0 if not specified)
- Computed attributes: `edge_betweenness`, etc.

**Filter by edge type:**

```
# Only intralayer edges (within a layer)
intralayer_edges = (
    Q.edges()
     .where(intralayer=True)
     .execute(network)
)
print(f"Intralayer edges: {len(intralayer_edges)}")

# Only interlayer edges between specific layers
interlayer_edges = (
    Q.edges()
     .where(interlayer=("social", "work"))
     .execute(network)
)
print(f"Edges between 'social' and 'work': {len(interlayer_edges)}")
```

**Expected output:**

```
High-weight edges in 'social' layer: 156

Sample edges:
    source  target source_layer target_layer  weight  edge_betweenness
0    alice     bob       social       social    0.75          0.023400
1      bob   charlie     social       social    0.80          0.034500
2    alice   diana       social       social    0.92          0.019800
3    diana     eve       social       social    0.65          0.028900
4      eve   frank       social       social    0.88          0.041200

Mean edge weight: 0.723
Mean edge betweenness: 0.028
```

### Smart Defaults and Error Messages

The DSL includes **smart defaults** that automatically compute commonly used centrality metrics when referenced but not explicitly computed. This feature makes queries more ergonomic while maintaining predictable behavior.

### Auto-Computing Centrality Metrics

When you reference a centrality metric in operations like `top_k()`, `order_by()`, or other ranking operations, the DSL will automatically compute it if not already present:

```
from py3plex.dsl import Q, L

# The DSL auto-computes betweenness_centrality when needed
result = (
```

```
    Q.nodes()
     .from_layers(L["*"])
     .per_layer()
        .top_k(5, "betweenness_centrality")  # Auto-computed here
     .end_grouping()
     .execute(network)
)

df = result.to_pandas()
# betweenness_centrality column is available even though
# we didn't explicitly call .compute("betweenness_centrality")
```

**Supported centrality aliases:**

- `degree`, `degree_centrality`
- `betweenness`, `betweenness_centrality`
- `closeness`, `closeness_centrality`
- `eigenvector`, `eigenvector_centrality`
- `pagerank`

**When auto-compute happens:**

- When the attribute is referenced in `top_k()`
- When the attribute is used in `order_by()`
- For both per-group (with grouping) and global operations

**Example with multiple auto-computed metrics:**

```
# Auto-compute degree for filtering and betweenness for ranking
result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=2)  # degree auto-computed here
     .order_by("betweenness_centrality", desc=True)  # betweenness auto-computed here
     .limit(10)
     .execute(network)
)
```

**Expected output:**

```
    node layer  degree  betweenness_centrality
0  alice social      8                0.143000
1    bob social      7                0.098000
2  carol social      6                0.067000
...
```

### Controlling Autocompute Behavior

You can explicitly control whether metrics are automatically computed using the `autocompute` parameter:

```
# Disable autocompute - require explicit .compute() calls
result = (
    Q.nodes(autocompute=False)  # Autocompute disabled
     .from_layers(L["social"])
     .compute("degree")  # Must explicitly compute
     .where(degree__gt=5)
     .execute(network)
)

# This would raise DslMissingMetricError because betweenness is not computed:
# Q.nodes(autocompute=False).order_by("betweenness_centrality").execute(net)
```

**When to disable autocompute:**

- **Performance-critical code**: Avoid unexpected expensive computations
- **Explicit control**: Make all metric computations visible in code
- **Debugging**: Understand exactly which metrics are computed and when

**Tracking computed metrics:**

Query results include a `computed_metrics` attribute that tracks which metrics were computed during execution:

```
result = (
    Q.nodes()
     .from_layers(L["social"])
     .compute("degree")
     .order_by("betweenness_centrality")  # Auto-computed
     .execute(network)
)

# Check which metrics were computed
print(f"Computed metrics: {result.computed_metrics}")
# Output: Computed metrics: {'degree', 'betweenness_centrality'}
```

**Use cases for computed_metrics:**

- Performance profiling: identify expensive operations
- Query optimization: avoid redundant computations
- Debugging: verify expected metrics were computed

### Helpful Error Messages with Suggestions

When you reference an unknown attribute, the DSL provides **did you mean?** suggestions using fuzzy string matching:

```
# Typo in attribute name
try:
```

```
    result = (
        Q.nodes()
         .from_layers(L["*"])
         .per_layer()
            .top_k(5, "betweness_centrality")  # Typo: "betweness" instead of "betweenness"
         .end_grouping()
         .execute(network)
    )
except UnknownAttributeError as e:
    print(e)
```

**Output:**

```
Unknown attribute 'betweness_centrality'. Did you mean 'betweenness_centrality'?
Known attributes: betweenness, betweenness_centrality, closeness, closeness_centrality,
                  degree, degree_centrality, eigenvector, eigenvector_centrality, pagerank
```

**The error includes:**

- The incorrect attribute name
- A suggestion for the most similar correct name (using Levenshtein distance)
- A list of all available attributes

### Grouping Requirements and Clear Errors

Some operations require **active grouping** (via `per_layer()` or `group_by()`). The DSL raises `GroupingError` with clear guidance when these operations are used incorrectly:

```
from py3plex.dsl.errors import GroupingError

# This will raise GroupingError
try:
    result = (
        Q.nodes()
         .from_layers(L["*"])
         .coverage(mode="all")  # Error: no grouping active
         .execute(network)
    )
except GroupingError as e:
    print(e)
```

**Output:**

```
coverage() requires an active grouping (e.g. per_layer(), group_by('layer')).
No grouping is currently active.
Example:
    Q.nodes().from_layers(L["*"])
        .per_layer().top_k(5, "degree").end_grouping()
        .coverage(mode="all")
```

**Correct usage:**

```
# With proper grouping
result = (
    Q.nodes()
     .from_layers(L["*"])
     .per_layer()  # Add grouping here
        .top_k(5, "degree")
     .end_grouping()
     .coverage(mode="all")  # Now works correctly
     .execute(network)
)
```

### When Smart Defaults DON'T Apply

Smart defaults are **predictable and conservative**. They only apply in specific scenarios:

1. **Only for centrality metrics**: Smart defaults work for recognized centrality metrics (degree, betweenness, etc.), not arbitrary attributes.
2. **Explicit compute takes precedence**: If you explicitly compute a metric, the DSL uses your computation and doesn't auto-compute:

```
# Explicit compute - no auto-compute happens
result = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("betweenness_centrality")  # Explicit
     .per_layer()
        .top_k(5, "betweenness_centrality")  # Uses explicit computation
     .end_grouping()
     .execute(network)
)
```

3. **Edge attributes are not auto-computed**: For edge queries, attributes like `weight` are read from edge data, not auto-computed:

```
# Edge weight is read from edge data, not computed
result = (
    Q.edges()
     .from_layers(L["*"])
     .per_layer()
        .top_k(5, "weight")  # Uses edge data['weight']
     .end_grouping()
     .execute(network)
)
```

### Benefits of Smart Defaults

**Ergonomics**: Write less boilerplate for common patterns:

```
# Before smart defaults (verbose)
result = (
    Q.nodes()
```

```
    .from_layers(L["*"])
    .compute("degree", "betweenness_centrality", "closeness_centrality")
    .per_layer()
       .top_k(5, "betweenness_centrality")
    .end_grouping()
    .execute(network)
)

# With smart defaults (concise)
result = (
    Q.nodes()
     .from_layers(L["*"])
     .per_layer()
        .top_k(5, "betweenness_centrality")  # Auto-computes what's needed
     .end_grouping()
     .execute(network)
)
```

**Teaching errors**: When something goes wrong, you get actionable guidance instead of cryptic messages.

**Predictability**: Smart defaults only activate for well-known patterns. Your explicit operations always take precedence.

### Temporal Queries

The DSL supports temporal filtering for networks with time-stamped edges or nodes. Four convenience methods provide intuitive temporal filtering: `.at(t)`, `.during(t0, t1)`, `.before(t)`, and `.after(t)`.

**Prerequisites for temporal queries:**

- Edges or nodes must have temporal attributes:
  - **Point-in-time**: `t` attribute (e.g., `{"t": 150.0}`)
  - **Intervals**: `t_start` and `t_end` attributes (e.g., `{"t_start": 100.0, "t_end": 200.0}`)
- Time values are typically numeric (timestamps) or ISO date strings

### Temporal Semantics Reference

The following table summarizes temporal query semantics:

Table 1: Temporal Query Operations

| Method | Description | Interval Semantics | Inclusivity |
|---|---|---|---|
| `.at(t)` | Snapshot at time t | Entities active at exactly t | Point (closed) |
| `.during(t0, t1)` | Range from t0 to t1 | Entities active during [t0, t1] | [t0, t1] (closed interval) |
| `.before(t)` | Before time t | Equivalent to `.during(None, t)` | (-∞, t] (closed at t) |
| `.after(t)` | After time t | Equivalent to `.during(t, None)` | [t, +∞) (closed at t) |

**Detailed semantics:**

- `at(t)`: Selects entities active at a specific moment
  - For point-in-time edges: includes edges where `t_edge == t`
  - For interval edges: includes edges where `t` is in `[t_start, t_end]`
- `during(t0, t1)`: Selects entities active during a time window
  - For point-in-time edges: includes edges where `t` is in `[t0, t1]` (closed interval)
  - For interval edges: includes edges where the interval **overlaps** `[t0, t1]`
  - `None` values: Use `t0=None` for open lower bound, `t1=None` for open upper bound
- `before(t)`: Selects entities active before (and at) time t
  - Convenience method equivalent to `.during(None, t)`
  - Inclusive of the boundary: includes entities at exactly time t
- `after(t)`: Selects entities active after (and at) time t
  - Convenience method equivalent to `.during(t, None)`
  - Inclusive of the boundary: includes entities at exactly time t

### Filter by Time (Snapshot)

Query the network state at a specific point in time:

```
# Nodes active at t=150.0
result = (
    Q.nodes()
     .at(150.0)
     .compute("degree")
     .execute(network)
)

df = result.to_pandas()
print(f"Nodes active at t=150: {len(df)}")
print(f"Average degree at t=150: {df['degree'].mean():.2f}")
print("\nTop nodes by degree at this snapshot:")
print(df.nlargest(5, 'degree')[['id', 'degree']])
```

**Use case:** Analyze network structure at specific moments (e.g., before and after an event, at

regular intervals for time series).

**Expected output:**

```
Nodes active at t=150: 78
Average degree at t=150: 5.12

Top nodes by degree at this snapshot:
                    id  degree
12    (eve, social)      14
5   (alice, social)      11
23  (grace, social)      10
8     (bob, social)       9
31  (henry, work)        9
```

### Time Range

Query entities active during a time window:

```python
# Nodes active during January 2024 (assuming numeric timestamps)
# For ISO dates, use strings: .during("2024-01-01", "2024-01-31")
result = (
    Q.nodes()
     .during(100.0, 200.0)
     .compute("degree")
     .execute(network)
)

df = result.to_pandas()
print(f"Nodes active during [100, 200]: {len(df)}")

# Compare to snapshot
snapshot_result = Q.nodes().at(150.0).execute(network)
print(f"Nodes at t=150 (snapshot): {len(snapshot_result)}")
print(f"Nodes during [100, 200] (range): {len(df)}")
print(f"Ratio: {len(df) / len(snapshot_result):.2f}x more nodes in range")
```

**Open-ended ranges:**

```python
# From t=100 onwards (no upper limit)
result_after = Q.edges().during(100.0, None).execute(network)

# Up to t=200 (no lower limit)
result_before = Q.edges().during(None, 200.0).execute(network)
```

**Use case:** Study network evolution, identify persistent vs. transient connections, analyze activity bursts.

**Expected output:**

```
Nodes active during [100, 200]: 142
Nodes at t=150 (snapshot): 78
Ratio: 1.82x more nodes in range
```

### Before and After (Convenience Methods)

The `.before()` and `.after()` methods provide intuitive alternatives for open-ended temporal queries:

```
# Get all edges before time 100 (inclusive)
early_edges = Q.edges().before(100.0).execute(network)

# Get all edges after time 200 (inclusive)
late_edges = Q.edges().after(200.0).execute(network)

# Common pattern: compare network before and after an event
event_time = 150.0

before_event = (
    Q.nodes()
     .before(event_time)
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

after_event = (
    Q.nodes()
     .after(event_time)
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Compare metrics
df_before = before_event.to_pandas()
df_after = after_event.to_pandas()

print(f"Average degree before event: {df_before['degree'].mean():.2f}")
print(f"Average degree after event: {df_after['degree'].mean():.2f}")
print(f"Network became {'denser' if df_after['degree'].mean() > df_before['degree'].mean()
```

**Expected output:**

```
Average degree before event: 4.35
Average degree after event: 5.87
Network became denser
```

**Temporal edges example:**

```
# Edges active during a period
edges = (
    Q.edges()
     .during(100.0, 200.0)
     .compute("edge_betweenness")
     .execute(network)
)
```

```
df = edges.to_pandas()
print(f"Active edges during [100, 200]: {len(df)}")
print(f"Mean edge betweenness: {df['edge_betweenness'].mean():.4f}")
```

**Note on implementation status:**

Temporal queries are **fully implemented** for edge-level temporal data. Node-level temporal filtering depends on your network's representation:

- If nodes have explicit `t` attributes, `.at()`, `.during()`, `.before()`, and `.after()` work directly
- If only edges are timestamped, node activity is inferred from edge presence
- For most use cases, temporal edge queries are sufficient

See *DSL Reference* for complete temporal query syntax and examples with ISO date strings.

### Common Patterns

This section presents **end-to-end recipes** for common multilayer network analysis tasks. These patterns are production-ready and can be adapted to your research questions.

### Pattern: Find Influential Nodes

Identify nodes that are both well-connected (high degree) and structurally important (high betweenness centrality):

```
# High-degree nodes ranked by betweenness centrality
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality", "layer_count")
     .where(degree__gt=10)
     .order_by("-betweenness_centrality")
     .limit(20)
     .execute(network)
)

df = result.to_pandas()
print(f"Top 20 influential nodes (degree > 10):")
print(df[['id', 'degree', 'betweenness_centrality', 'layer_count']])

# Export for further analysis or publication
df.to_csv("influential_nodes.csv", index=False)

# Visualize
import matplotlib.pyplot as plt
plt.figure(figsize=(10, 6))
plt.scatter(df['degree'], df['betweenness_centrality'],
            s=df['layer_count']*50, alpha=0.6)
plt.xlabel("Degree")
plt.ylabel("Betweenness Centrality")
plt.title("Influential Nodes (size = layer_count)")
plt.tight_layout()
```

```
plt.savefig("influential_nodes.png", dpi=300)
```

**Why this pattern works:**

- **Degree** measures local connectivity (how many neighbors)
- **Betweenness centrality** measures global importance (how often the node appears on shortest paths)
- Nodes high in both metrics are **influential bridges** in the network

**Expected output:**

```
Top 20 influential nodes (degree > 10):
                    id  degree  betweenness_centrality  layer_count
0     (eve, social)      15                0.301000            3
1   (alice, social)      12                0.245000            2
2   (grace, social)      11                0.221000            2
3     (bob, social)      12                0.198000            1
4   (diana, work)        14                0.187000            3
...
```

### Pattern: Compare Layer Activity

Compute summary statistics for each layer to understand layer-specific dynamics:

```
layers = network.get_layers()

layer_stats = []
for layer in layers:
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree", "clustering")
         .execute(network)
    )
    df = result.to_pandas()

    layer_stats.append({
        'layer': layer,
        'num_nodes': len(df),
        'mean_degree': df['degree'].mean(),
        'max_degree': df['degree'].max(),
        'mean_clustering': df['clustering'].mean(),
    })

    print(f"{layer}: {len(df)} nodes, "
          f"avg degree={df['degree'].mean():.2f}, "
          f"avg clustering={df['clustering'].mean():.3f}")

# Create comparison DataFrame
import pandas as pd
comparison = pd.DataFrame(layer_stats)
```

```
print("\nLayer comparison:")
print(comparison)

# Visualize
comparison.plot(x='layer', y=['mean_degree', 'mean_clustering'],
                kind='bar', figsize=(10, 5))
plt.ylabel("Value")
plt.title("Layer Activity Comparison")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("layer_comparison.png", dpi=300)
```

**Use case:** Understand how network structure varies across different contexts (e.g., online vs. offline interactions, different communication channels).

**Expected output:**

```
friends: 50 nodes, avg degree=4.20, avg clustering=0.623
work: 72 nodes, avg degree=3.15, avg clustering=0.512
social: 65 nodes, avg degree=5.01, avg clustering=0.587
family: 38 nodes, avg degree=6.84, avg clustering=0.701

Layer comparison:
      layer  num_nodes  mean_degree  max_degree  mean_clustering
0   friends         50         4.20          15            0.623
1      work         72         3.15          12            0.512
2    social         65         5.01          18            0.587
3    family         38         6.84          21            0.701
```

### Pattern: Export Subnetwork

Extract a subnetwork based on query criteria for focused analysis or visualization:

```
# Extract high-activity multilayer nodes
active_nodes = (
    Q.nodes()
     .where(layer_count__gt=2)
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

print(f"Selected {len(active_nodes)} multilayer nodes")

# Create subnetwork containing only these nodes
subnetwork = network.subgraph(active_nodes.keys())

print(f"Subnetwork: {subnetwork.number_of_nodes()} nodes, "
      f"{subnetwork.number_of_edges()} edges")

# Analyze subnetwork
df = active_nodes.to_pandas()
```

```
print(f"\nSubnetwork mean degree: {df['degree'].mean():.2f}")
print(f"Subnetwork mean betweenness: {df['betweenness_centrality'].mean():.4f}")

# Export for visualization or further analysis
from py3plex.visualization import draw_multilayer_default
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(12, 10))
draw_multilayer_default(subnetwork, ax=ax, display=False)
plt.savefig("subnetwork_viz.png", dpi=300)
plt.close()

# Or export in various formats
subnetwork.save_network("subnetwork.edgelist", output_type="edgelist")
```

**What `layer_count__gt=2` means:**

- Selects nodes appearing in **more than 2 layers**
- These are "connector" nodes that participate in multiple contexts
- Useful criterion for studying nodes that bridge different social spheres

**Alternative criteria:**

```
# High betweenness nodes
influential = Q.nodes().compute("betweenness_centrality").where(
    betweenness_centrality__gt=0.1
).execute(network)

# Nodes in specific community
community_nodes = Q.nodes().compute("communities").where(
    communities__eq=3
).execute(network)
```

**Expected output:**

```
Selected 34 multilayer nodes
Subnetwork: 34 nodes, 127 edges

Subnetwork mean degree: 7.47
Subnetwork mean betweenness: 0.0892
```

**Workflow integration:**

This pattern is often combined with community detection, dynamics simulation, or centrality analysis:

```
# 1. Select subnetwork
core_nodes = Q.nodes().where(layer_count__gte=2, degree__gt=5).execute(network)
subnetwork = network.subgraph(core_nodes.keys())

# 2. Run community detection on subnetwork
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
communities = louvain_communities(subnetwork)
```

```
# 3. Analyze communities
print(f"Found {len(set(communities.values()))} communities")

# 4. Visualize or export
from py3plex.visualization import draw_multilayer_default
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(12, 10))
# Note: communities dict can be used for node coloring if the visualization function suppor
draw_multilayer_default(subnetwork, ax=ax, display=False)
plt.savefig("core_network_communities.png", dpi=300)
plt.close()
```

### Pattern: Per-Layer Top-K with Coverage (Multi-Layer Hub Detection)

Find nodes that are top-k hubs (by any centrality metric) **across all layers**. This pattern is essential for identifying nodes that maintain high influence in the entire multilayer structure, not just in isolated layers.

**The Problem:**

Traditional approaches require manual loops over layers:

```
# Old approach: manual iteration
layer_top = {}
for layer in network.layers:
    res = (
        Q.nodes()
         .from_layers(L[str(layer)])
         .where(degree__gt=1)
         .compute("betweenness_centrality")
         .order_by("-betweenness_centrality")
         .limit(5)
         .execute(network)
    )
    layer_top[layer] = set(res.to_pandas()["id"])

# Find intersection
multi_hubs = set.intersection(*layer_top.values())
```

**The Solution: Grouping and Coverage API**

The new DSL supports per-layer operations in a single query:

```
from py3plex.core import random_generators
from py3plex.dsl import Q, L

# Generate example network
net = random_generators.random_multilayer_ER(n=200, l=3, p=0.05, directed=False)

# Find nodes that are top-5 betweenness hubs in ALL layers (single query!)
multi_hubs = (
```

```
    Q.nodes()
     .from_layers(L["*"])                    # wildcard: all layers
     .where(degree__gt=1)
     .compute("degree", "betweenness_centrality")
     .per_layer()                            # group by layer
        .top_k(5, "betweenness_centrality") # top 5 per layer
     .end_grouping()
     .coverage(mode="all")                   # nodes in top-5 in ALL layers
     .execute(net)
)

df = multi_hubs.to_pandas()
print(f"Multi-layer hubs (in top-5 of ALL layers): {set(df['id'])}")
print(f"Count: {len(df['id'].unique())}")
print(f"\nDetailed results:")
print(df[['id', 'layer', 'degree', 'betweenness_centrality']].to_string())
```

**Expected output:**

```
Multi-layer hubs (in top-5 of ALL layers): {23, 45, 67}
Count: 3

Detailed results:
    id  layer  degree  betweenness_centrality
0   23      0      12                  0.2456
1   23      1      14                  0.2891
2   23      2      11                  0.2234
3   45      0      13                  0.2567
4   45      1      11                  0.2123
5   45      2      15                  0.3012
6   67      0      10                  0.2001
7   67      1      12                  0.2345
8   67      2      13                  0.2678
```

**Coverage Modes:**

The `coverage()` method supports multiple modes for cross-layer analysis:

```
# Mode 1: "all" - intersection (nodes in top-k of ALL layers)
all_layers_hubs = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree")
     .per_layer()
        .top_k(10, "degree")
     .end_grouping()
     .coverage(mode="all")
     .execute(net)
)

# Mode 2: "any" - union (nodes in top-k of AT LEAST ONE layer)
any_layer_hubs = (
```

```
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree")
     .per_layer()
        .top_k(10, "degree")
     .end_grouping()
     .coverage(mode="any")
     .execute(net)
)

# Mode 3: "at_least" - nodes in top-k of at least K layers
two_layer_hubs = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("betweenness_centrality")
     .per_layer()
        .top_k(5, "betweenness_centrality")
     .end_grouping()
     .coverage(mode="at_least", k=2)  # In at least 2 layers
     .execute(net)
)

# Mode 4: "exact" - nodes in top-k of exactly K layers (layer-specific hubs)
single_layer_specialists = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree")
     .per_layer()
        .top_k(10, "degree")
     .end_grouping()
     .coverage(mode="exact", k=1)  # Exactly 1 layer
     .execute(net)
)

print(f"Hubs in ALL layers: {len(all_layers_hubs.to_pandas()['id'].unique())}")
print(f"Hubs in ANY layer: {len(any_layer_hubs.to_pandas()['id'].unique())}")
print(f"Hubs in 2 layers: {len(two_layer_hubs.to_pandas()['id'].unique())}")
print(f"Layer specialists (exactly 1): {len(single_layer_specialists.to_pandas()['id'].uniq
```

**Expected output:**

```
Hubs in ALL layers: 3
Hubs in ANY layer: 27
Hubs in 2 layers: 12
Layer specialists (exactly 1): 15
```

**Wildcard Layer Selection:**

The `L["*"]` wildcard automatically expands to all layers in the network:

```
# All layers
Q.nodes().from_layers(L["*"])
```

```
# All layers except one
Q.nodes().from_layers(L["*"] - L["bots"])

# All layers intersected with a specific one (same as selecting that layer)
Q.nodes().from_layers(L["*"] & L["social"])
```

**Use Cases:**

1. **Identify persistent influencers**: Nodes that maintain high centrality across all contexts (layers)
2. **Find layer specialists**: Nodes that are important in only one layer (`mode="exact", k=1`)
3. **Detect multi-context bridges**: Nodes in top-k in at least 2 layers connect different contexts
4. **Community structure analysis**: Compare `mode="all"` vs `mode="any"` to understand layer cohesion

**Why This Pattern Matters:**

In real-world multilayer networks (social media, collaboration networks, biological systems), understanding **cross-layer** vs. **layer-specific** importance is crucial:

- **Email + Phone + Chat network**: Who are the omnipresent communicators vs. email-only specialists?
- **Author collaboration network**: Who publishes top papers in multiple fields vs. specialists in one domain?
- **Transportation network**: Which locations are hubs in all modes (bus, train, bike) vs. single-mode hubs?

**Performance Note:**

The per-layer computation is optimized: measures are computed on the selected nodes after layer filtering, and grouping operations leverage efficient dictionaries. For large networks (>100K nodes), consider filtering with `where()` before computing expensive metrics like betweenness centrality.

### DSL Result Interoperability

DSL query results integrate seamlessly with pandas for data transformation workflows. While QueryResult doesn't implement pipeline verbs directly, it provides a clean `.to_pandas()` export that enables the same workflow patterns:

1. Start with a DSL query to filter and compute metrics
2. Export to pandas with `.to_pandas()`
3. Use pandas operations for additional transformations
4. Leverage the full pandas ecosystem for analysis and visualization

### QueryResult to pandas Workflow

The recommended pattern for combining DSL queries with data transformations:

```
from py3plex.dsl import Q, L
from py3plex.core import multinet
```

```
# Create a sample network
network = multinet.multi_layer_network()
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1],
    ['B', 'layer1', 'C', 'layer1', 1],
    ['A', 'layer2', 'C', 'layer2', 1],
], input_type="list")

# Start with DSL query
result = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree", "betweenness_centrality")
     .execute(network)
)

# Export to pandas for flexible transformations
df = result.to_pandas()

# Continue with pandas operations
df = df[df["degree"] > 5]  # Filter
df['influence_score'] = (  # Mutate
    df["degree"] * df["betweenness_centrality"]
)
df = df.sort_values('influence_score', ascending=False)  # Arrange

print(df.head(10))
```

**What happens here:**

1. **DSL phase**: `Q.nodes()...execute()` filters nodes, computes centrality metrics
2. **Export**: `.to_pandas()` materializes as a DataFrame
3. **pandas phase**: Standard pandas operations for transformation and analysis

**Pandas operations equivalent to pipeline verbs:**

- Filter rows: `df[df["degree"] > 5]`
- Add columns: `df['new_col'] = ...`
- Sort: `df.sort_values('col', ascending=False)`
- Select columns: `df[['col1', 'col2']]`
- Group and aggregate: `df.groupby('col').agg(...)`

### Verb Mapping Table

The following table shows how concepts map across the three interfaces:

Table 2: DSL and pandas Verb Mapping

| Concept | String DSL | Builder DSL | pandas |
|---|---|---|---|
| Filter rows | `WHERE degree > 5` | `.where(degree__gt=5)` | `df[df["degree"] > 5]` |
| Select columns | `SELECT id, degree` | `.select("id", "degree")` | `df[["id", "degree"]]` |
| Sort/Order | `ORDER BY degree DESC` | `.order_by("degree", desc=True)` | `df.sort_values("degree", ascending=False)` |
| Group by field | `GROUP BY layer` | `.group_by("layer")` | `df.groupby("layer")` |
| Add column | (not available) | (use pandas after export) | `df["score"] = ...` |
| Aggregate | (not available) | (use pandas after export) | `df.groupby("layer").agg(...)` |
| Limit results | `LIMIT 10` | `.limit(10)` | `df.head(10)` |

**Design rationale:**

- **DSL**: Declarative, optimized for graph queries (layer algebra, centrality, grouping)
- **pandas**: Procedural, flexible for data transformations (arbitrary computations, reshaping)
- **Workflow**: Use DSL for graph-specific operations, export to pandas for data munging

### Example: Combined Workflow

A realistic workflow combining both DSL and pandas operations:

```
from py3plex.dsl import Q, L

# Scenario: Find influential nodes in social network, normalize scores,
#           rank within communities, export for visualization

# DSL: Query and compute graph metrics
result = (
    Q.nodes()
     .from_layers(L["social"])
     .where(degree__gt=3)
     .compute("degree", "betweenness_centrality", "clustering")
     .execute(network)
)

# Export to pandas for transformations
df = result.to_pandas()

# pandas: Transform and enhance data
max_betweenness = df['betweenness_centrality'].max()

# Normalize centrality to [0, 1]
df['norm_betweenness'] = (
```

```
    df['betweenness_centrality'] / max_betweenness
    if max_betweenness > 0 else 0
)

# Composite influence score
df['influence'] = (
    0.5 * df['degree'] +
    0.3 * df['norm_betweenness'] +
    0.2 * (1 - df['clustering'])
)

# Group by community and compute statistics
community_stats = df.groupby('community').agg({
    'influence': ['count', 'mean', 'max']
}).round(2)

# Sort communities by average influence
community_stats = community_stats.sort_values(
    ('influence', 'mean'),
    ascending=False
)

print(community_stats)
```

**Expected output:**

```
              influence
                  count  mean    max
community
5                    23  0.72   0.89
2                    31  0.68   0.85
8                    19  0.61   0.79
1                    28  0.58   0.74
...
```

**Why this matters:**

1. **Single pipeline**: No need to export intermediate results to disk or juggle multiple DataFrames
2. **Flexibility**: DSL for graph operations, pandas for everything else
3. **Performance**: DSL computes centrality on the multilayer graph once, pandas transforms in-memory
4. **Ecosystem**: Full pandas ecosystem available (plotting, statistics, export formats)

**When to use each:**

- **DSL alone**: Simple queries, need graph-specific operations (centrality, grouping, coverage)
- **pandas alone**: Non-graph data, pure data transformations
- **Combined (DSL → pandas)**: Complex analytical workflows, need both graph metrics and custom computations

See *How to Build Analysis Pipelines with Dplyr-style Operations* for the dplyr-style pipeline

API (`nodes()`, `edges()` functions) which provides an alternative approach using chainable operations directly on networks.

### Next Steps

Now that you understand the DSL, explore these related resources:

- **DSL Reference** (*DSL Reference*): Complete grammar, all operators, full list of built-in measures, and advanced features (EXPLAIN queries, parameter binding, custom operators)
- **Dplyr-Style Pipelines** (*How to Build Analysis Pipelines with Dplyr-style Operations*): Combine DSL queries with pipeline operations for more complex data transformation workflows. The pipeline API (`nodes()`, `mutate()`, `arrange()`) complements the DSL for when you need procedural transformations.
- **Community Detection** (*How to Run Community Detection on Multilayer Networks*): Use DSL queries to select nodes, then apply community detection algorithms. Pattern: query → detect communities → analyze community structure.
- **Network Dynamics** (*How to Simulate Multilayer Dynamics*): Run dynamics simulations on DSL-selected subnetworks. Pattern: query → extract subnetwork → simulate → analyze outcomes.
- **Linting and Validation** (*DSL Reference*): The DSL includes a linting subsystem (`py3plex dsl-lint`) that checks queries for errors, performance issues, and suggests optimizations. Use it to validate complex queries.
- **Examples Repository** (*Examples & Recipes*): Full scripts showing DSL in context, including data loading, query composition, analysis, and visualization.

**Key Takeaways:**

1. **Use the builder API** (`Q`, `L`) for production code—it's type-safe, refactorable, and IDE-friendly.
2. **Filter early**: Add `where()` clauses before `compute()` for better performance on large networks.
3. **Embrace pandas**: Use `.to_pandas()` for result analysis—it integrates seamlessly with the scientific Python stack.
4. **Layer algebra is powerful**: `L["a"] + L["b"]` (union), `L["a"] & L["b"]` (intersection) enable sophisticated multilayer queries.
5. **Temporal queries** require timestamped edges/nodes but unlock time-series network analysis.
6. **Query optimization**: When ordering by existing attributes with LIMIT, the DSL automatically applies the limit early to reduce computation on expensive measures.

### Automatic Query Optimization

The DSL automatically optimizes query execution order to minimize computation:

**Early LIMIT Optimization:**

When your query includes:

- A `LIMIT` clause
- An `ORDER BY` clause

- `COMPUTE` operations
- Ordering by an **existing** attribute (not being computed)

The executor applies `ORDER BY` and `LIMIT` **before** computing expensive measures, dramatically reducing computation time.

**Example:**

```
# Find top 10 highest-degree nodes, then compute betweenness
result = (
    Q.nodes()
     .compute("betweenness_centrality")  # Expensive operation
     .order_by("-degree")                # Degree already exists
     .limit(10)
     .execute(network)
)

# Optimized execution:
# 1. Order all nodes by degree (fast - degree already available)
# 2. Take top 10 nodes (reduces set size)
# 3. Compute betweenness on those 10 nodes only (huge speedup!)
```

**Without optimization:** Computes betweenness on all nodes, then takes top 10.

**With optimization:** Takes top 10 by degree first, then computes betweenness on only those 10 nodes.

For networks with 1000+ nodes, this can provide **100x speedup** when computing expensive centrality measures.

**When optimization does NOT apply:**

- Ordering by a computed attribute (must compute all first):

  ```
  # Must compute betweenness on all nodes to find top 10
  Q.nodes().compute("betweenness").order_by("-betweenness").limit(10)
  ```

- No ORDER BY clause (limit is arbitrary)
- Grouping is active (`per_layer()` uses per-group limits)

**Best practices for performance:**

1. Use existing attributes for ordering when possible (`degree`, `layer`)
2. Apply `where()` filters early to reduce dataset size
3. Order by pre-computed attributes before computing expensive measures
4. Use `per_layer().top_k()` for efficient per-layer operations

- **Community and Support:**
- Report issues or request features: https://github.com/SkBlaz/py3plex/issues
- Example notebooks: https://github.com/SkBlaz/py3plex/tree/main/examples
- py3plex documentation: https://skblaz.github.io/py3plex/

#### Further Reading: The Py3plex Book

For a deeper theoretical and practical treatment of the DSL and multilayer network concepts, see the **Py3plex Book**:

- **Chapter 8** — Introduction to the DSL: Motivations, design principles, and comparison with alternatives
- **Chapter 9** — Builder API Deep Dive: Complete reference with advanced patterns
- **Chapter 10** — Advanced Queries & Workflows: Complex real-world query examples

The book is available as: * **PDF** in the repository: `docs/py3plex_book.pdf` * **Online HTML** (if built): `docs/book/`

The book provides: * Formal definitions of multilayer network operations * Detailed algorithmic complexity analysis * Extensive case studies with real datasets * Performance benchmarking and optimization strategies

### 3.3.11 How to Find Graph Patterns and Motifs with the Pattern Matching API

**Goal:** Use py3plex's Pattern Matching Builder API to find graph motifs, paths, triangles, and complex subgraph patterns in multilayer networks.

**What You'll Learn**

- How to express graph patterns using the `Q.pattern()` builder API
- How to match edges, paths, triangles, and custom motifs
- How to apply multilayer-aware constraints (within/between layers)
- How to filter patterns using node and edge predicates
- How to extract and analyze pattern match results

**Complete Example**

See the full executable example: `example_pattern_matching.py`

**Prerequisites:**

- A loaded `multi_layer_network` object (see *How to Load and Build Networks*)
- Basic familiarity with the DSL (see *How to Query Multilayer Graphs with the SQL-like DSL*)
- Understanding of graph motifs (edges, paths, triangles, subgraphs)

#### Why Pattern Matching?

Pattern matching allows you to find **specific subgraph structures** in your network. Unlike node/edge queries that return individual elements, pattern matching returns **complete matches** where multiple nodes and edges satisfy structural constraints.

**Common Use Cases:**

- **Motif Detection:** Find triangles, cliques, or other structural patterns
- **Path Analysis:** Discover multi-hop connections between nodes

- **Multilayer Patterns:** Identify cross-layer relationships
- **Complex Queries:** Express "find nodes A, B, C where A connects to B with weight > 0.5, B connects to C, and all are in the social layer"

**Comparison to Basic Queries:**

| Basic DSL Query | Pattern Matching Query |
|---|---|
| `Q.nodes().where(degree__gt=3)` Returns individual nodes | `Q.pattern().node("a").where(...)` Returns complete pattern matches |
| `Q.edges().where(weight__gt=0.5)` Returns individual edges | `Q.pattern().edge("a", "b").where(...)` Returns node pairs with context |

### Quick Start: Your First Pattern

Let's start with a simple example: finding all edges in a network.

**Mental model:** `Q.pattern()` binds **variables** (`"a"`, `"b"`, etc.) to actual node-layer tuples, then checks that every required edge or path between those variables exists. Each match is a complete mapping of variables → node-layer tuples rather than a single node or edge.

```
from py3plex.core import multinet
from py3plex.dsl import Q

# Create a simple network
network = multinet.multi_layer_network(directed=False)
network.add_nodes([
    {'source': 'Alice', 'type': 'social'},
    {'source': 'Bob', 'type': 'social'},
    {'source': 'Charlie', 'type': 'social'},
])
network.add_edges([
    {'source': 'Alice', 'target': 'Bob',
     'source_type': 'social', 'target_type': 'social', 'weight': 1.0},
    {'source': 'Bob', 'target': 'Charlie',
     'source_type': 'social', 'target_type': 'social', 'weight': 2.0},
])

# Find all edges (simple pattern)
pattern = (
    Q.pattern()
     .node("a")              # Define node variable "a"
     .node("b")              # Define node variable "b"
     .edge("a", "b")         # Require edge between a and b
)

result = pattern.execute(network)
print(f"Found {result.count} matches")
print(result.to_pandas())
```

**Output:**

```
Found 4 matches
                  a                   b
0  (Alice, social)      (Bob, social)
1    (Bob, social)    (Alice, social)
2    (Bob, social)  (Charlie, social)
3  (Charlie, social)    (Bob, social)
```

Notice that undirected edges produce matches in both directions.

### Pattern Components

#### Nodes

Define node variables that will be bound to actual nodes during matching:

```
# Simple node
Q.pattern().node("a")

# Node with semantic labels (metadata only)
Q.pattern().node("a", labels="person")

# Node with predicates
Q.pattern().node("a").where(degree__gt=3)

# Node with layer constraint
Q.pattern().node("a").where(layer="social")
```

**Node Predicates:**

All standard DSL predicates work with pattern nodes:

- `degree__gt=5` — degree greater than 5
- `degree__lt=2` — degree less than 2
- `layer="social"` — node must be in social layer
- Any node attribute: `age__gt=30`, `type="user"`, etc.

#### Edges

Define edges between node variables:

```
# Undirected edge
Q.pattern().edge("a", "b", directed=False)

# Directed edge
Q.pattern().edge("a", "b", directed=True)

# Edge with predicates
Q.pattern().edge("a", "b").where(weight__gt=0.5)

# Edge with optional type
Q.pattern().edge("a", "b", etype="friendship")
```

**Edge Predicates:**

Filter edges by attributes:

- `weight__gt=0.5` — edge weight greater than 0.5
- `weight__lt=1.0` — edge weight less than 1.0
- Any edge attribute: `timestamp__gt=100`, `color="red"`, etc.

### Paths

Sugar method for defining sequential connections:

```
# 2-hop path: a → b → c
Q.pattern().path(["a", "b", "c"])

# Equivalent to:
Q.pattern().edge("a", "b").edge("b", "c")

# Directed path
Q.pattern().path(["a", "b", "c"], directed=True)
```

Paths enforce adjacency in the order you list the variables; with `directed=True` the edges must follow that direction.

### Triangles

Sugar method for triangle motifs:

```
# Triangle: a-b, b-c, c-a
Q.pattern().triangle("a", "b", "c")

# Equivalent to:
Q.pattern().edge("a", "b").edge("b", "c").edge("c", "a")
```

### Multilayer-Aware Patterns

One of the most powerful features is specifying layer constraints on edges.

### Within Layer

Find edges that exist within a single layer:

```
# Edges within the social layer
pattern = (
    Q.pattern()
     .node("a").where(layer="social")
     .node("b").where(layer="social")
     .edge("a", "b")
)
```

#### Between Layers

Find edges that cross between specific layers:

```
# Edges between social and work layers
pattern = (
    Q.pattern()
     .node("a").where(layer="social")
     .node("b").where(layer="work")
     .edge("a", "b").between_layers("social", "work")
)
```

#### Any Layer

Allow edges in any layer (default behavior; call it explicitly if you want the query to read clearly):

```
pattern = Q.pattern().edge("a", "b").any_layer()
```

#### Practical Examples

##### Example 1: Find Triangles with Weight Constraints

**Problem:** Find triangles where all edges have weight $> 1.0$.

```
pattern = (
    Q.pattern()
     .edge("a", "b").where(weight__gt=1.0)
     .edge("b", "c").where(weight__gt=1.0)
     .edge("c", "a").where(weight__gt=1.0)
     .limit(10)
)

result = pattern.execute(network)
triangles = result.to_pandas()
print(f"Found {len(triangles)} triangles")
```

If you prefer the `triangle()` sugar, expand it as above to attach edge predicates, or apply the weight filter after executing the simpler triangle pattern.

##### Example 2: Find 2-Hop Paths in Specific Layer

**Problem:** Find all 2-hop paths within the social layer.

```
pattern = (
    Q.pattern()
     .node("a").where(layer="social")
     .node("b").where(layer="social")
     .node("c").where(layer="social")
     .path(["a", "b", "c"])
     .returning("a", "b", "c")
     .limit(5)
```

```
)

result = pattern.execute(network)
df = result.to_pandas()

print(f"Found {result.count} paths")
print(df)
```

**Output:**

```
Found 5 paths
                    a                 b                   c
0      (Bob, social)  (Alice, social)      (Bob, social)
1      (Bob, social)  (Alice, social)  (Charlie, social)
2  (Charlie, social)    (Bob, social)    (Alice, social)
...
```

### Example 3: Find High-Degree Nodes with Specific Connections

**Problem:** Find pairs of high-degree nodes (degree $> 2$) connected by strong edges (weight $>$ 1.5).

```
pattern = (
    Q.pattern()
     .node("a").where(layer="social", degree__gt=2)
     .node("b").where(layer="social", degree__gt=2)
     .edge("a", "b").where(weight__gt=1.5)
     .constraint("a != b")  # Ensure different nodes
     .returning("a", "b")
)

result = pattern.execute(network)
print(f"Found {result.count} high-degree pairs")

# Get unique nodes involved
nodes = result.to_nodes(unique=True)
print(f"Unique nodes: {len(nodes)}")
```

### Example 4: Cross-Layer Hub Detection

**Problem:** Find nodes that appear in multiple layers and have high degree in each.

```
# This requires multiple pattern queries
# Find nodes in social layer
social_hubs = (
    Q.pattern()
     .node("a").where(layer="social", degree__gt=3)
     .returning("a")
     .execute(network)
)
```

```
# Find nodes in work layer
work_hubs = (
    Q.pattern()
     .node("a").where(layer="work", degree__gt=3)
     .returning("a")
     .execute(network)
)

# Find intersection (nodes in both)
social_nodes = {n[0] for n in social_hubs.to_nodes(unique=True)}
work_nodes = {n[0] for n in work_hubs.to_nodes(unique=True)}
cross_layer_hubs = social_nodes & work_nodes

print(f"Cross-layer hubs: {cross_layer_hubs}")
```

### Working with Results

The `PatternQueryResult` object provides multiple ways to access matches. Use raw `rows` for programmatic iteration, `to_pandas` for tabular analysis, and `to_nodes` / `to_edges` when you want to pass results back into graph algorithms.

### Accessing Rows

```
result = pattern.execute(network)

# Get all matches as list of dictionaries
for match in result.rows:
    print(f"Match: {match}")
    # e.g., {'a': ('Alice', 'social'), 'b': ('Bob', 'social')}
```

`result.count` is the number of rows returned, equivalent to `len(result.rows)`. Each row maps your variable names to concrete `(node, layer)` tuples.

### Converting to Pandas

```
df = result.to_pandas()

# Now use pandas operations
print(df.head())
print(df.describe())

# Export to CSV
df.to_csv("pattern_matches.csv", index=False)
```

Column order follows the variables you specified with `.returning(...)`; if you did not set it, all variables are returned in declaration order.

### Extracting Nodes

```
# Get all matched nodes (as tuples)
all_nodes = result.to_nodes()

# Get unique nodes only
unique_nodes = result.to_nodes(unique=True)

# Get nodes for specific variables
a_nodes = result.to_nodes(vars=["a"], unique=True)
```

### Extracting Edges

```
# Get all matched edges as tuples
edges = result.to_edges()

# Each edge is a (source_tuple, target_tuple) pair
for src, dst in edges:
    print(f"Edge: {src} -> {dst}")
```

### Creating Subgraphs

```
# Create induced subgraph of matched nodes
subgraph = result.to_subgraph(network)

# Now use NetworkX operations
print(f"Subgraph has {subgraph.number_of_nodes()} nodes")
print(f"Subgraph has {subgraph.number_of_edges()} edges")

# Create one subgraph per match
subgraphs = result.to_subgraph(network, per_match=True)
print(f"Created {len(subgraphs)} subgraphs")
```

### Filtering Results

```
# Filter matches using a predicate
filtered = result.filter(lambda match: match["a"][0].startswith("A"))

# Limit number of results
limited = result.limit(10)
```

`filter` and `limit` return new `PatternQueryResult` objects; they do not modify the original result.

### Query Execution Planning

Use `.explain()` to see how the pattern will be executed:

```
pattern = (
    Q.pattern()
     .node("a").where(degree__gt=3)
     .node("b")
     .edge("a", "b")
)

plan = pattern.explain()
print(f"Root variable: {plan['root_var']}")
print(f"Join order: {[step['var'] for step in plan['join_order']]}")
print(f"Estimated complexity: {plan['estimated_complexity']}")
```

**Example Output:**

```
Root variable: a
Join order: ['a', 'b']
Estimated complexity: 1000
```

The planner chooses `a` as the root because it has the most restrictive predicates (`degree__gt=3`), which will generate fewer candidates. The `estimated_complexity` field is a relative heuristic, useful for comparing different patterns rather than predicting exact runtime.

### Performance Tips

1. **Add Predicates Early:** The more selective predicates you add to the root variable, the faster the query.

   ```
   # Good: Restrictive predicate on first node
   Q.pattern().node("a").where(degree__gt=10).node("b").edge("a", "b")

   # Less efficient: No predicates
   Q.pattern().node("a").node("b").edge("a", "b")
   ```

2. **Use Layer Constraints:** Layer filters significantly reduce the search space.

   ```
   # Good: Restrict to one layer
   Q.pattern().node("a").where(layer="social")

   # Less efficient: Search all layers
   Q.pattern().node("a")
   ```

3. **Set Limits:** For large networks, use `.limit()` to cap results.

   ```
   pattern.limit(100).execute(network)
   ```

4. **Use Constraints:** Add `a != b` constraints to avoid self-loops if not needed.

   ```
   Q.pattern().node("a").node("b").edge("a", "b").constraint("a != b")
   ```

5. **Return only needed variables:** Reducing the result width lowers serialization overhead.

   ```
   Q.pattern().path(["a", "b", "c"]).returning("a", "c")
   ```

### Advanced Features

#### Constraints

Global constraints that apply across variables:

```
# All-different constraint
pattern = (
    Q.pattern()
     .node("a")
     .node("b")
     .node("c")
     .edge("a", "b")
     .edge("b", "c")
     .constraint("a != b")
     .constraint("b != c")
     .constraint("a != c")
)
```

#### Returning Specific Variables

By default, all variables are returned. You can select specific ones:

```
pattern = (
    Q.pattern()
     .node("a")
     .node("b")
     .node("c")
     .path(["a", "b", "c"])
     .returning("a", "c")  # Only return endpoints
)

result = pattern.execute(network)
# Result only contains 'a' and 'c' columns
```

#### Execution Options

```
# Set maximum matches
result = pattern.execute(network, max_matches=100)

# Set timeout (in seconds)
result = pattern.execute(network, timeout=10.0)
```

### Comparison with Other Approaches

#### Pattern Matching vs. NetworkX

| NetworkX Approach | Pattern Matching Approach |
|---|---|
| `nx.triangles(G)` Returns triangle count per node | `Q.pattern().triangle("a","b","c")` Returns actual triangle matches |
| `nx.all_simple_paths(G, source, tgt)` Enumerates all paths | `Q.pattern().path(["a", "b", "c"])` Finds paths with constraints |
| Manual loops and filtering | Declarative pattern specification |

**Advantages of Pattern Matching:**

- Declarative and composable
- Built-in support for multilayer constraints
- Integrated with DSL ecosystem
- Returns results in consistent format (DataFrame, nodes, edges, subgraph)

#### Pattern Matching vs. DSL Node/Edge Queries

| DSL Node/Edge Query | Pattern Matching |
|---|---|
| Returns individual elements | Returns complete structural matches |
| `Q.nodes().where(degree__gt=3)` \| `Q.pattern().node("a", degree__gt=3)` Single-element selection \| Part of larger pattern | |
| No structural relationships | Explicit edge relationships |

Use node/edge queries when you need individual elements. Use pattern matching when you need to find subgraph structures.

### Limitations and Future Work

#### Current Limitations (v1)

- **Fixed-length paths only:** Variable-length paths like `(a)-[*1..3]-(b)` not yet supported
- **No optional matching:** All pattern elements must match
- **No UNION patterns:** Cannot express "match pattern A OR pattern B"
- **Limited aggregation:** Aggregation over matches not yet implemented

These features are planned for future versions while maintaining the current clean API.

### Best Practices

1. **Start simple:** Begin with small patterns and add complexity incrementally
2. **Test on small data:** Validate pattern logic on toy networks before scaling up
3. **Use explain():** Check execution plans for complex queries
4. **Combine with DSL:** Use pattern matching for structure, DSL for attributes
5. **Document patterns:** Add comments explaining complex pattern logic

### Troubleshooting

#### Pattern Returns No Matches

**Problem:** Your pattern should match but returns 0 results.

**Solutions:**

1. Check layer constraints — are nodes actually in the layers you specified?
2. Verify predicates — are the thresholds too restrictive?
3. Use `.explain()` to see the execution plan
4. Test with simpler patterns first
5. Check if your network has the expected structure

```
# Debug: Check basic statistics first
result = Q.nodes().where(layer="social").execute(network)
print(f"Social layer has {result.count} nodes")

result = Q.edges().where(intralayer=True).execute(network)
print(f"Network has {result.count} intralayer edges")
```

#### Too Many Matches

**Problem:** Pattern returns too many results.

**Solutions:**

1. Add more predicates to narrow the search
2. Use `.limit()` to cap results
3. Add layer constraints to restrict search space
4. Use `.constraint("a != b")` to avoid duplicates

```
# Limit results
pattern.limit(100).execute(network)
```

#### Slow Performance

**Problem:** Pattern matching takes too long.

**Solutions:**

1. Add selective predicates to the root variable
2. Use layer constraints to reduce search space
3. Consider breaking pattern into smaller sub-patterns
4. Use `.explain()` to check estimated complexity
5. Set timeout to avoid runaway queries

```
# Set timeout
result = pattern.execute(network, timeout=30.0)
```

### Related Documentation

- *How to Query Multilayer Graphs with the SQL-like DSL* — Basic DSL queries for nodes and edges
- *Query Zoo: DSL Gallery for Multilayer Analysis* — Gallery of DSL query examples
- *How to Load and Build Networks* — Creating multilayer networks
- *How to Compute Network Statistics* — Computing network metrics

### Next Steps

- Explore the *Query Zoo: DSL Gallery for Multilayer Analysis* for more complex examples
- Try combining pattern matching with DSL queries
- Experiment with different motifs (triangles, cliques, paths)
- Build your own domain-specific pattern library

**Tip**

**Pro Tip:** Pattern matching is most powerful when combined with the rest of the DSL. Use patterns to find structures, then use regular DSL queries to compute metrics on the matches!

## 3.3.12 How to Build Analysis Pipelines with Dplyr-style Operations

**Goal:** Chain operations to build reproducible analysis workflows on node or edge tables.

**Prerequisites:** Load a network first (see *How to Load and Build Networks*). `nodes(...)` returns a chainable `NodeFrame` (view over `id`, `layer`, `degree`, etc.); `edges(...)` returns an `EdgeFrame`. Examples below use `NodeFrame` and end with `.to_pandas()` to materialize results.

### Quickstart

Filter → add a column → sort → convert to pandas:

```
from py3plex.core import multinet
from py3plex.graph_ops import nodes

network = multinet.multi_layer_network()
network.load_network("data.multiedgelist", input_type="multiedgelist")

# Build pipeline: filter → add a column → sort → convert to pandas
result = (
    nodes(network)
    .filter(lambda n: n["degree"] > 2)
    .mutate(score=lambda n: n["degree"] * 2)
    .arrange("degree", reverse=True)
    .to_pandas()
)

print(result.head())
```

### Available Operations

Common verbs you can chain on `nodes(...)` or `edges(...)`:

- `filter`: keep rows that satisfy a predicate
- `mutate`: add or modify columns
- `select`: choose columns (and optionally reorder)
- `arrange`: sort by a column or key function
- `group_by` + `summarise`: aggregate per group
- `head`: keep the first $n$ rows (useful for previews)

### filter()

Select rows based on conditions. Chain multiple filters for readability:

```
result = (
    nodes(network)
    .filter(lambda n: n["layer"] == "friends")
    .filter(lambda n: n["degree"] > 5)
    .to_pandas()
)
```

### mutate()

Add or modify columns using callables that receive each row (a dict-like object):

```
result = (
    nodes(network)
    .mutate(
        degree_squared=lambda n: n["degree"] ** 2,
        is_hub=lambda n: n["degree"] > 10
    )
    .to_pandas()
)
```

### select()

Choose specific columns (and optionally reorder them):

```
result = (
    nodes(network)
    .select("id", "layer", "degree")
    .to_pandas()
)
```

#### arrange()

Sort results (ascending by default, or descending with `reverse=True`):

```
result = (
    nodes(network)
    .arrange("degree", reverse=True)  # Descending
    .to_pandas()
)
```

#### group_by() and summarise()

Aggregate by one or more columns. `summarise` expects functions that return scalars (e.g., `mean`, `max`, `len`):

```
result = (
    nodes(network)
    .group_by("layer")
    .summarise(
        avg_degree=lambda g: g["degree"].mean(),
        max_degree=lambda g: g["degree"].max(),
        count=lambda g: len(g)
    )
    .to_pandas()
)
```

**Illustrative output (values depend on your data):**

```
      layer  avg_degree  max_degree  count
0   friends        3.45          12     46
1      work        2.87           8     46
2    family        4.12          15     42
```

### Complex Pipelines

#### Multi-Step Analysis

Combine several verbs to narrow down to the most active nodes. This example assumes `layer_count` and `degree` columns already exist (e.g., after computing node statistics):

```
result = (
    nodes(network)
    # Step 1: Filter active nodes
    .filter(lambda n: n["layer_count"] > 1)
    # Step 2: Add computed score
    .mutate(
        activity_score=lambda n: n["degree"] * n["layer_count"]
    )
    # Step 3: Filter by score
    .filter(lambda n: n["activity_score"] > 20)
    # Step 4: Sort by score
    .arrange("activity_score", reverse=True)
```

```
    # Step 5: Get top 20
    .head(20)
    # Convert to pandas
    .to_pandas()
)
```

### Combining with DSL

Mix dplyr-style operations with DSL queries to narrow down candidates, then enrich them:

```
from py3plex.dsl import Q
from py3plex.graph_ops import nodes

# Use the same network as above
# Use DSL (Builder API) to select nodes and compute measures once
dsl_result = (
    Q.nodes()
     .where(degree__gt=5)
     .compute("betweenness_centrality")
     .execute(network)
)

# Extract nodes and betweenness values
selected = set(dsl_result.items)
df_dsl = dsl_result.to_pandas()

# Use dplyr-style to filter and transform the same nodes
final_result = (
    nodes(network)
    .filter(lambda n: (n["id"], n["layer"]) in selected)
    .mutate(
        betweenness=lambda n: betweenness.get((n["id"], n["layer"]), 0.0),
        centrality_rank=lambda n: betweenness.get((n["id"], n["layer"]), 0.0) * 100
    )
    .arrange("centrality_rank", reverse=True)
    .to_pandas()
)
```

### Scikit-learn-style Pipelines

Use the built-in `Pipeline` to chain reusable steps (each step implements `transform` and receives the previous output). The first step should handle `None` input (`LoadStep` does). Replace or extend steps with your own `PipelineStep` subclasses when needed.

```
from py3plex.pipeline import Pipeline, LoadStep, FilterNodes, ComputeStats

pipe = Pipeline([
    ("load", LoadStep(path="data.multiedgelist", input_type="multiedgelist")),
    ("filter", FilterNodes(min_degree=2)),
    ("stats", ComputeStats(include_layer_stats=True)),
])
```

```
stats = pipe.run()
print(stats)
```

Custom steps can wrap scikit-learn components if they implement `transform` (and optionally `fit`) to accept the previous step's output and return the next input.

### Exporting Pipelines

### Persist Step Settings

Store the parameters you passed to each step so you can recreate the pipeline later:

```
import json

config = {
    "steps": [
        {"name": "load", "params": {"path": "data.multiedgelist", "input_type": "multiedgeli
        {"name": "filter", "params": {"min_degree": 2}},
        {"name": "stats", "params": {"include_layer_stats": True}},
    ]
}

with open("pipeline.json", "w") as f:
    json.dump(config, f, indent=2)
```

### Reload and Rebuild

Recreate the pipeline by instantiating the same steps with saved parameters:

```
import json
from py3plex.pipeline import Pipeline, LoadStep, FilterNodes, ComputeStats

with open("pipeline.json", "r") as f:
    config = json.load(f)

step_params = {step["name"]: step["params"] for step in config["steps"]}

pipe = Pipeline([
    ("load", LoadStep(**step_params["load"])),
    ("filter", FilterNodes(**step_params["filter"])),
    ("stats", ComputeStats(**step_params["stats"])),
])

stats = pipe.run()
```

#### Reusable Pipelines

#### Create Pipeline Functions

```
def identify_hubs(network, threshold=10):
    """Reusable pipeline to identify hub nodes."""
    return (
        nodes(network)
        .filter(lambda n: n["degree"] > threshold)
        .mutate(hub_score=lambda n: n["degree"] * n["layer_count"])
        .arrange("hub_score", reverse=True)
        .to_pandas()
    )

# Use
hubs = identify_hubs(network, threshold=5)
print(hubs)
```

#### Next Steps

- **Query with DSL:** *How to Query Multilayer Graphs with the SQL-like DSL*
- **Complete workflows:** *How to Reproduce Common Analysis Workflows*
- **API reference:** *API Documentation*

### 3.3.13 How to Reproduce Common Analysis Workflows

**Goal:** Use ready-made recipes for common multilayer network analysis tasks.

**Prerequisites:** Basic understanding of py3plex (see *Quick Start Tutorial*).

**About examples:** Code and outputs below use the bundled synthetic datasets unless noted. Replace file paths with your own data; metrics will differ accordingly.

#### Complete Workflows

This guide links to detailed recipes and case studies. For step-by-step implementations, see:

- **Recipes:** *Analysis Recipes & Workflows* — Focused solutions for specific tasks
- **Case Studies:** *Use Cases & Case Studies* — Complete end-to-end analyses
- **Examples:** *Examples & Recipes* — Runnable code examples

#### Quick Recipe Index

#### Network Construction

- **Building from edge lists** → *How to Load and Build Networks*
- **Converting from NetworkX** → *Analysis Recipes & Workflows* (Recipe 1)
- **Loading temporal networks** → *API Documentation* (Temporal section)

### Statistical Analysis

- **Computing multilayer statistics** → *How to Compute Network Statistics*
- **Comparing layers** → *Analysis Recipes & Workflows* (Recipe 3)
- **Node versatility analysis** → *Analysis Recipes & Workflows* (Recipe 4)

### Community Detection

- **Multilayer Louvain** → *How to Run Community Detection on Multilayer Networks*
- **Cross-layer community comparison** → *Analysis Recipes & Workflows* (Recipe 5)
- **Community stability analysis** → *Use Cases & Case Studies* (Case Study 2)

### Network Embeddings

- **Node2Vec for link prediction** → *How to Run Random Walk Algorithms*
- **Embedding-based clustering** → *Analysis Recipes & Workflows* (Recipe 7)
- **Layer-specific embeddings** → *How to Run Random Walk Algorithms*

### Visualization

- **Publication-ready plots** → *How to Visualize Multilayer Networks*
- **Interactive visualizations** → *How to Visualize Multilayer Networks*
- **Layer comparison plots** → *Analysis Recipes & Workflows* (Recipe 9)

### Domain-Specific Workflows

#### Social Networks

**Multi-platform social analysis:**

```
# See: user_guide/case_studies.rst - Social Network Case Study
# 1. Load data from multiple platforms
# 2. Detect cross-platform communities
# 3. Identify influential users
# 4. Analyze information diffusion
```

See *Use Cases & Case Studies* for complete implementation.

#### Biological Networks

**Multi-omics integration:**

```
# See: user_guide/case_studies.rst - Biological Network Case Study
# 1. Integrate protein-protein + gene regulation + metabolic pathways
# 2. Find key regulators using multilayer centrality
# 3. Detect functional modules
# 4. Prioritize disease genes
```

See *Use Cases & Case Studies* for complete implementation.

### Transportation Networks

**Multimodal route analysis:**

```
# See: examples/index.rst - Transportation Example
# 1. Model different transportation modes as layers
# 2. Add transfer connections between layers
# 3. Compute optimal multimodal routes
# 4. Identify critical transfer points
```

See *Examples & Recipes* for runnable code.

### Config-Driven Workflows

Use configuration files for reproducibility:

```
# workflow_config.yaml
network:
  input_file: "data.multiedgelist"
  input_type: "multiedgelist"

analysis:
  - name: "statistics"
    metrics: ["degree", "betweenness_centrality"]

  - name: "community_detection"
    algorithm: "louvain"
    params:
      resolution: 1.0

  - name: "visualization"
    output: "network.png"
    layout: "force_directed"
```

Execute workflow:

```
from py3plex.workflows import execute_workflow

results = execute_workflow("workflow_config.yaml")
```

See *Analysis Recipes & Workflows* for complete config-driven workflow examples.

The config captures *what* to run (input format, metrics, algorithms, visualization) so you can reuse the same analysis across datasets by swapping only the file path.

### DSL-Driven Analysis Workflows

**Goal:** Use py3plex's DSL to create reproducible, declarative analysis pipelines.

The DSL expresses analysis workflows as queries rather than imperative code, keeping them readable and reproducible. `Q` builds a query, `L` is a layer helper. Use the Builder API (`Q.nodes()`) for production code with type hints, or the legacy string DSL (`execute_query()`) for backward compatibility.

### Basic DSL Workflow Pattern

**Template for DSL-first analysis:** Keep the pipeline linear—load, query, refine, export—so each step can be repeated with different parameters.

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# 1. Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "py3plex/datasets/_data/synthetic_multilayer.edges",
    input_type="multiedgelist"
)

# 2. Query and filter nodes with DSL
high_degree_nodes = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .where(degree__gt=5)
     .order_by("betweenness_centrality", reverse=True)
     .execute(network)
)  # dict keyed by (node_id, layer) -> metrics

# 3. Extract subnetwork
subgraph = network.core_network.subgraph(high_degree_nodes.keys())

# 4. Analyze subnetwork
print(f"High-degree subnetwork:")
print(f"  Nodes: {len(high_degree_nodes)}")
print(f"  Edges: {subgraph.number_of_edges()}")

# 5. Export results
import pandas as pd
df = pd.DataFrame([
    {
        'node': node[0],
        'layer': node[1],
        'degree': data['degree'],
        'betweenness': data['betweenness_centrality']
    }
    for node, data in high_degree_nodes.items()
])
df.to_csv('high_degree_analysis.csv', index=False)
```

**Expected output (synthetic_multilayer sample):**

```
High-degree subnetwork:
  Nodes: 25
  Edges: 89
```

### Multilayer Exploration Workflow

**Systematic multilayer network analysis:**

```
from py3plex.core import multinet
from py3plex.dsl import Q, L
import pandas as pd

# Load network
network = multinet.multi_layer_network(directed=False)
network.load_network(
    "py3plex/datasets/_data/synthetic_multilayer.edges",
    input_type="multiedgelist"
)

print("MULTILAYER NETWORK EXPLORATION")
print("=" * 70)

# Step 1: Per-layer statistics
print("\n1. Per-Layer Statistics:")
layer_stats = []

for layer in network.get_layers():
    # Query layer nodes
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)
    layer_edges = Q.edges().from_layers(L[layer]).execute(network)

    # Compute metrics
    result = (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
         .execute(network)
    )

    avg_degree = sum(d['degree'] for d in result.values()) / len(result) if result else 0

    layer_stats.append({
        'layer': layer,
        'nodes': len(layer_nodes),
        'edges': len(layer_edges),
        'avg_degree': avg_degree
    })

    print(f"  {layer}: {len(layer_nodes)} nodes, {len(layer_edges)} edges, avg_degree={av

# Step 2: Find versatile nodes (present in multiple layers)
print("\n2. Versatile Nodes (multilayer presence):")
from collections import Counter

node_layer_count = Counter()
for node, layer in network.get_nodes():
```

```
    node_layer_count[node] += 1

versatile_nodes = {
    node: count for node, count in node_layer_count.items()
    if count >= 2
}

print(f"  Total versatile nodes: {len(versatile_nodes)}")
print(f"  Top 5 most versatile:")
for node, count in sorted(versatile_nodes.items(), key=lambda x: x[1], reverse=True)[:5]:
    print(f"    {node}: {count} layers")

# Step 3: Layer comparison
print("\n3. Layer Overlap Analysis:")
layers = network.get_layers()

for i, layer1 in enumerate(layers):
    for layer2 in layers[i+1:]:
        nodes1 = set(n[0] for n in Q.nodes().from_layers(L[layer1]).execute(network).keys())
        nodes2 = set(n[0] for n in Q.nodes().from_layers(L[layer2]).execute(network).keys())

        overlap = nodes1 & nodes2
        jaccard = len(overlap) / len(nodes1 | nodes2) if (nodes1 | nodes2) else 0

        print(f"  {layer1} ↔ {layer2}: {len(overlap)} nodes, Jaccard={jaccard:.3f}")

# Step 4: Hub identification across layers
print("\n4. Cross-Layer Hub Nodes:")
all_metrics = (
    Q.nodes()
     .compute("degree", "betweenness_centrality")
     .where(degree__gt=7)
     .execute(network)
)

print(f"  Hub nodes (degree > 7): {len(all_metrics)}")

# Group hubs by base node ID
from collections import defaultdict
hub_layers = defaultdict(set)

for (node, layer), data in all_metrics.items():
    hub_layers[node].add(layer)

print(f"  Unique hub node IDs: {len(hub_layers)}")
print(f"  Top 5 hub nodes:")
for node, layers in sorted(hub_layers.items(), key=lambda x: len(x[1]), reverse=True)[:5]:
    print(f"    {node}: present in {len(layers)} layers - {list(layers)}")
```

**Expected output:**

```
MULTILAYER NETWORK EXPLORATION
======================================================================

1. Per-Layer Statistics:
  layer1: 40 nodes, 95 edges, avg_degree=4.75
  layer2: 40 nodes, 87 edges, avg_degree=4.35
  layer3: 40 nodes, 102 edges, avg_degree=5.10

2. Versatile Nodes (multilayer presence):
  Total versatile nodes: 35
  Top 5 most versatile:
    node7: 3 layers
    node12: 3 layers
    node3: 3 layers
    node15: 3 layers
    node1: 3 layers

3. Layer Overlap Analysis:
  layer1  layer2: 35 nodes, Jaccard=0.875
  layer1  layer3: 32 nodes, Jaccard=0.800
  layer2  layer3: 33 nodes, Jaccard=0.825

4. Cross-Layer Hub Nodes:
  Hub nodes (degree > 7): 18
  Unique hub node IDs: 12
  Top 5 hub nodes:
    node7: present in 3 layers - ['layer1', 'layer2', 'layer3']
    node12: present in 3 layers - ['layer1', 'layer2', 'layer3']
    node3: present in 3 layers - ['layer1', 'layer2', 'layer3']
    node15: present in 2 layers - ['layer1', 'layer3']
    node8: present in 2 layers - ['layer2', 'layer3']
```

The thresholds (degree $> 7$, overlap counts) match the bundled `synthetic_multilayer` dataset. Adjust them for sparser or denser graphs so averages and Jaccard scores remain meaningful.

### Community Detection + DSL Workflow

**Combine community detection with DSL queries:**

```python
from py3plex.algorithms.community_detection.community_wrapper import louvain_communities
from py3plex.dsl import Q
from collections import Counter

# Detect communities
communities = louvain_communities(network)

# Attach as node attributes
for (node, layer), comm_id in communities.items():
    network.core_network.nodes[(node, layer)]['community'] = comm_id

print("COMMUNITY-BASED ANALYSIS")
```

```
print("=" * 70)

# Query each community
community_ids = set(communities.values())

for comm_id in sorted(community_ids):
    # Use DSL Builder API to get community members and compute metrics
    comm_result = (
        Q.nodes()
         .where(community=comm_id)
         .compute("degree", "betweenness_centrality")
         .execute(network)
    )

    members = comm_result.items  # Get node-layer tuples

    # Statistics
    if comm_result.count > 0:
        df = comm_result.to_pandas()
        avg_degree = df['degree'].mean()
        avg_betw = df['betweenness_centrality'].mean()
    else:
        avg_degree = avg_betw = 0.0

    # Layer composition
    layer_counts = Counter(layer for _, layer in members)

    print(f"\nCommunity {comm_id}:")
    print(f"  Size: {len(members)} nodes")
    print(f"  Avg degree: {avg_degree:.2f}")
    print(f"  Avg betweenness: {avg_betw:.6f}")
    print(f"  Layer composition: {dict(layer_counts)}")
```

**Notes:** The Builder API (`Q.nodes()`) provides `result.items` for node-layer tuples, `result.count` for the count, and `.to_pandas()` for DataFrame conversion. If a community has no nodes (rare with Louvain), averages safely fall back to `0.0`. Community labels live on the NetworkX backing graph (`network.core_network`), so they persist across subsequent DSL queries.

### Dynamics + DSL Workflow

**Epidemic simulation with DSL-based analysis:**

```
from py3plex.dynamics import SIRDynamics
from py3plex.dsl import Q, L
from collections import Counter

# Run SIR simulation
sir = SIRDynamics(
    network,
    beta=0.3,
    gamma=0.1,
```

```
    initial_infected=0.05
)
sir.set_seed(42)
results = sir.run(steps=100)

# Attach final state
final_state = results.trajectory[-1]
for node, state in final_state.items():
    network.core_network.nodes[node]['sir_state'] = state

print("EPIDEMIC ANALYSIS")
print("=" * 70)

# Per-layer infection analysis
for layer in network.get_layers():
    layer_nodes = Q.nodes().from_layers(L[layer]).execute(network)

    state_counts = Counter(
        network.core_network.nodes[node].get('sir_state', 'unknown')
        for node in layer_nodes.keys()
    )

    total = len(layer_nodes)

    def pct(count):
        return count / total * 100 if total else 0

    print(f"\n{layer}:")
    print(f"  S: {state_counts.get('S', 0)} ({pct(state_counts.get('S', 0)):.1f}%)")
    print(f"  I: {state_counts.get('I', 0)} ({pct(state_counts.get('I', 0)):.1f}%)")
    print(f"  R: {state_counts.get('R', 0)} ({pct(state_counts.get('R', 0)):.1f}%)")

# Identify superspreaders (infected nodes with high degree)
superspreaders = (
    Q.nodes()
     .where(sir_state='I')
     .compute("degree", "betweenness_centrality")
     .where(degree__gt=6)
     .order_by("degree", reverse=True)
     .execute(network)
)

print(f"\nSuperspreaders (infected, degree > 6): {len(superspreaders)}")
for node, data in list(superspreaders.items())[:5]:
    print(f"  {node}: degree={data['degree']}, betw={data['betweenness_centrality']:.4f}"
```

**Notes:** The SIR outcomes depend on `beta` (infection rate), `gamma` (recovery rate), and the random seed. Percentages are guarded against empty layers so the snippet can be reused on sparse networks, and `sir_state` is stored per node as `S`, `I`, or `R` for follow-up filtering.

### Reusable DSL Query Functions

**Create reusable query templates:**

```
def get_layer_hubs(network, layer, degree_threshold=5):
    """Get high-degree nodes in a specific layer."""
    from py3plex.dsl import Q, L

    return (
        Q.nodes()
         .from_layers(L[layer])
         .compute("degree")
         .where(degree__gt=degree_threshold)
         .order_by("degree", reverse=True)
         .execute(network)
    )

def get_versatile_nodes(network, min_layers=2):
    """Get nodes present in multiple layers."""
    from collections import Counter

    node_layer_count = Counter()
    for node, layer in network.get_nodes():
        node_layer_count[node] += 1

    return {
        node: count for node, count in node_layer_count.items()
        if count >= min_layers
    }

def compare_layer_centrality(network, layer1, layer2):
    """Compare centrality distributions between two layers."""
    from py3plex.dsl import Q, L
    import numpy as np

    result1 = (
        Q.nodes()
         .from_layers(L[layer1])
         .compute("betweenness_centrality")
         .execute(network)
    )

    result2 = (
        Q.nodes()
         .from_layers(L[layer2])
         .compute("betweenness_centrality")
         .execute(network)
    )

    betw1 = [d['betweenness_centrality'] for d in result1.values()]
    betw2 = [d['betweenness_centrality'] for d in result2.values()]
```

```
    return {
        'layer1': {'mean': np.mean(betw1), 'std': np.std(betw1)},
        'layer2': {'mean': np.mean(betw2), 'std': np.std(betw2)}
    }

# Use reusable functions
hubs_layer1 = get_layer_hubs(network, 'layer1', degree_threshold=7)
versatile = get_versatile_nodes(network, min_layers=3)
centrality_comp = compare_layer_centrality(network, 'layer1', 'layer2')

print(f"Layer1 hubs: {len(hubs_layer1)}")
print(f"Highly versatile nodes: {len(versatile)}")
print(f"Centrality comparison: {centrality_comp}")
```

**Why use DSL-driven workflows?**

- **Declarative:** Express *what* to analyze, not *how* to compute
- **Composable:** Chain queries to build complex analyses
- **Reproducible:** Queries are self-documenting and version-controllable
- **Efficient:** DSL optimizes execution internally
- **Readable:** SQL-like syntax is intuitive for data analysis

**Next steps with DSL workflows:**

- **Full DSL tutorial:** *How to Query Multilayer Graphs with the SQL-like DSL* - Comprehensive DSL guide
- **Community detection:** *How to Run Community Detection on Multilayer Networks* - Community + DSL workflows
- **Dynamics simulation:** *How to Simulate Multilayer Dynamics* - Dynamics + DSL workflows

### Batch Processing

Process multiple networks:

```
import glob
from py3plex.core import multinet

results = []

for filename in glob.glob("data/*.multiedgelist"):
    # Load network
    network = multinet.multi_layer_network()
    network.load_network(filename, input_type="multiedgelist")

    # Apply analysis pipeline
    stats = analyze_network(network)  # Your custom function

    results.append({
        'filename': filename,
        'stats': stats
    })
```

```
# Aggregate results
summary = aggregate_results(results)
```

`analyze_network` and `aggregate_results` are placeholders for your own reusable pipeline (e.g., computing summary stats, exporting community labels). Keep them pure functions so the batch loop stays predictable.

#### Complete Example Templates

The following locations contain complete, runnable examples:

1. **User Guide Recipes** (*Analysis Recipes & Workflows*)
   - Recipe-style solutions with code + explanation
   - Focused on single tasks
2. **Case Studies** (*Use Cases & Case Studies*)
   - End-to-end analyses
   - Real-world datasets
   - Publication-ready results
3. **Examples Gallery** (*Examples & Recipes*)
   - Standalone Python scripts
   - Minimal, focused examples
   - Easy to adapt

#### Next Steps

- **Learn fundamentals:** *Quick Start Tutorial*
- **Detailed recipes:** *Analysis Recipes & Workflows*
- **Complete case studies:** *Use Cases & Case Studies*
- **Browse examples:** *Examples & Recipes*

---

## 3.4 Part IV: Concepts & Explanations

Theoretical background and architectural explanations.

### 3.4.1 Concepts & Explanations

Multilayer networks capture systems where entities interact through multiple relationship types at once. This page orients you within the conceptual docs and points to the deeper explanations you may need next.

**What's in this section**

- *Multilayer Networks 101* — Modeling choices (multiplex, heterogeneous, temporal) and when each makes sense
- *The py3plex Core Model* — Node-layer pairs, the supra-adjacency matrix, and how py3plex wraps NetworkX

- *Design Principles* — Why py3plex is structured as it is
- *Algorithm Landscape* — What analysis tools are available and how they fit together

**Pick a path**

- **New to multilayer networks?** Start with *Multilayer Networks 101*, then read *The py3plex Core Model* to see how layers, node-layer pairs, and coupling are represented.
- **Coming from NetworkX?** Jump to *The py3plex Core Model* for the node-layer abstraction and supra-adjacency matrix, then skim *Design Principles* for rationale.
- **Building expertise?** Read in order: theory (*Multilayer Networks 101*), implementation (*The py3plex Core Model*), design rationale (*Design Principles*), and tools (*Algorithm Landscape*).

### After Reading This Section

You'll be able to:

- Model real-world systems as multilayer networks instead of flattening away layer semantics.
- Choose sensible parameters (e.g., layer definitions, inter-layer coupling strength) for your data.
- Interpret results in terms of layers and node-layer pairs rather than anonymous nodes.
- Avoid common pitfalls (e.g., flattening important structure, mismatched identifiers).

**Tip**

**Quick check:** After reading this section, you should be able to answer:

- What distinguishes multiplex from heterogeneous networks?
- What does the supra-adjacency matrix represent and why use it?
- When should you tighten vs. relax inter-layer coupling?

**Relation to other sections**

- **Overview** (*Overview*) — Quick 2-minute intro to multilayer networks.
- **How-to Guides** (*How-to Guides*) — Apply these concepts in practice.
- **Reference** (*API & DSL Reference*) — Detailed API and algorithm documentation.
- **Examples** (*Examples & Recipes*) — See concepts in action with real code.

**Jump to practical applications**

Once you understand the concepts:

- **Load networks:** *How to Load and Build Networks*
- **Compute statistics:** *How to Compute Network Statistics*
- **Detect communities:** *How to Run Community Detection on Multilayer Networks*
- **Query networks:** *How to Query Multilayer Graphs with the SQL-like DSL*

### 3.4.2 Multilayer Networks 101

This chapter explains what multilayer networks are, when to use them, and how to choose the right type for your data—without flattening away important structure.

**You will learn:**

- What distinguishes multilayer from single-layer networks
- Types: multiplex, heterogeneous, temporal, interdependent
- When to use multilayer modeling
- Common pitfalls to avoid

Key ideas to keep in mind:

- A **layer** labels the context of an interaction (relationship type, time slice, modality).
- A **node-layer pair** `(node_id, layer_id)` is the atomic unit; the same node can appear in multiple layers.
- **Coupling** uses inter-layer edges that link the same entity across layers, with a weight controlling how strongly layers influence each other. When the weight is 0, layers are independent; higher weights make layers behave more coherently.

#### What are Multilayer Networks?

A **multilayer network** models systems with multiple types of relationships, node types, or interaction contexts while keeping each interaction labeled by its layer. Each edge carries both endpoints *and* the layer context, so interactions remain distinguishable.

**Example:** A researcher's social world includes coauthors, colleagues, students, and Twitter followers. These are different relationship types with different meanings. A multilayer network keeps them as separate **layers** rather than flattening into one graph.

**Traditional vs. Multilayer:**

| Aspect | Single-Layer | Multilayer |
|---|---|---|
| Node types | One (e.g., only people) | Multiple (people, orgs, docs) |
| Edge types | One (e.g., only friendship) | Multiple per layer |
| Structure | Homogeneous | Heterogeneous |
| Analysis | Standard graph algorithms | Layer-aware algorithms |

#### Types of Multilayer Networks

The examples below use simple edge lists to show how different modeling choices map to different layer structures. Choose the structure that matches (a) whether node identities repeat across layers and (b) whether relationships are interchangeable or context-specific.

#### Multiplex Networks

**Same nodes, different relationship types.** Use when the *entities* are the same everywhere, but interactions mean different things per layer.

```
from py3plex.core import multinet

network = multinet.multi_layer_network(network_type="multiplex")
```

```
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1],
], input_type="list")
```

**Examples:** Social networks across platforms, transportation via air/rail/road, communication via email/phone/chat.

**Tip:** Inter-layer edges typically connect identical nodes across layers to model the idea that "Alice in friends" is still Alice in "colleagues."

#### Heterogeneous Information Networks

**Different node types with type-specific relationships.** Use when layers hold different entity types or schemas, not just interaction labels.

```
network = multinet.multi_layer_network()
network.add_edges([
    ['Alice', 'authors', 'P1', 'papers', 1],
    ['P1', 'papers', 'ICML', 'venues', 1],
], input_type="list")
```

**Examples:** Academic networks (authors, papers, venues), e-commerce (users, products, sellers), biomedical (drugs, diseases, targets).

**Tip:** Keep node identifiers disjoint across types (e.g., `P1` is always a paper) to avoid accidental coupling.

#### Temporal Networks

**Networks that evolve over time, with time-sliced layers.** Use when order and timing change interpretation (e.g., diffusion, cascades).

```
network = multinet.multi_layer_network()
network.add_edges([
    ['A', '2020', 'B', '2020', 1],
    ['A', '2021', 'B', '2021', 1],
    ['B', '2021', 'C', '2021', 1],
], input_type="list")
```

**Examples:** Communication over time, social dynamics, disease spread.

**Tip:** Choose time-slice granularity that matches the process of interest—too coarse hides dynamics, too fine yields sparse noise. Adjacent slices can be coupled with inter-layer edges to allow information to move through time.

#### Interdependent Networks

**Multiple networks where nodes depend on each other rather than sharing identity.** Dependencies capture "if X fails in layer A, Y fails in layer B," not "X is the same as Y."

**Examples:** Power grid depends on communication network, supply chains, cyber-physical sys-

tems.

**Tip:** Model dependencies explicitly with inter-layer edges that indicate which nodes fail together or require coordination; these edges can connect different node IDs across layers and often have direction (who depends on whom).

### When to Use Multilayer Networks

Use multilayer modeling when:

1. **Multiple relationship types matter** — Friendship and professional connections behave differently.
2. **Node roles vary by context** — A hub in work may be peripheral in hobbies.
3. **Layer interactions are important** — Transportation failures in one mode affect others.
4. **Temporal evolution matters** — Relationships change over time and order matters.
5. **System-level properties emerge** — Cross-layer dependencies affect resilience, spreading, or coordination.

### Choosing a Modeling Approach

Ask these questions:

1. **Layers vs. attributes?** If relationships differ in *kind* (email vs. meeting), use layers. If they differ only in *intensity* (weight, frequency), keep one layer and add attributes.
2. **Same or different nodes?** Same entities in all layers → multiplex. Different entity types → heterogeneous (avoid forcing identity coupling).
3. **Coupling strength?** When identities repeat across layers, set an inter-layer weight `omega 0` to control how tightly node copies move together (e.g., start with `omega=1.0` for identity coupling; reduce it if layers should diverge).
4. **Temporal structure?** If order matters, use time-sliced layers with optional coupling between adjacent slices; only aggregate into a static network when chronology is irrelevant.

### What Goes Wrong When You Flatten

Flattening throws away layer information. Typical failure modes include:

**Community structure destroyed:** Flattening can create spurious bridges between groups that are distinct in the layer structure.

**Centrality becomes misleading:** A researcher with 2 coauthors and 500 Twitter followers has degree 502 when flattened, but academic influence is only 2. Centrality scores mix incomparable interaction modes.

**Path analysis fails:** Email → meeting transitions matter for information flow but are invisible in a flattened network, so path-based predictions over- or under-estimate reachability.

**Layer semantics disappear:** Once layers are merged, you can no longer tell which mode of interaction drove a result.

### Common Pitfalls

- **Over-aggregating** — Combining layers that should stay separate; prefer explicit layers when semantics differ.
- **Under-aggregating** — Too many ultra-sparse layers; use edge weights or coarser bins instead.
- **Ignoring coupling** — Treating layers as independent when nodes correspond; set `omega` explicitly.
- **Wrong coupling strength** — Too high forces artificial consistency; too low loses cross-layer information.
- **Mismatched identifiers** — Same entity must have the same ID across layers for identity coupling to work.
- **Inconsistent time slicing** — Changing window sizes mid-analysis hides or invents trends; keep cadence consistent.

### Key Terminology

- **Intra-layer edges** — Within a single layer (same layer ID on both endpoints)
- **Inter-layer edges** — Between layers (often identity edges connecting the same node across layers)
- **Node-layer pairs** — `(node_id, layer_id)` tuples, the fundamental unit for most algorithms
- **Supra-adjacency matrix** — Block matrix encoding both intra- and inter-layer connections; shape is `(n·L) × (n·L)` for `n` nodes appearing in `L` layers
- **Coupling** — Strength of inter-layer edges (often `omega`); controls how strongly layers influence one another

### Next Steps

- *The py3plex Core Model* — Internal data structures
- *Design Principles* — Why py3plex works this way
- *Algorithm Landscape* — Available algorithms

**References:**


- Kivelä et al. (2014). "Multilayer networks." *J. Complex Networks* 2(3): 203-271.
- Boccaletti et al. (2014). "The structure and dynamics of multilayer networks." *Physics Reports* 544(1): 1-122.


## 3.4.3 The py3plex Core Model

*In which we look under the hood at how py3plex represents multilayer networks, understand the node-layer pair representation, and learn to work with the supra-adjacency matrix.*

---

In the previous chapter, you learned *what* multilayer networks are conceptually. Now we'll see *how* py3plex represents them in code—and why these design choices matter for your analysis.

Understanding the internal representation will help you:

- Debug unexpected behavior (why does my network have twice as many nodes as I expected?)
- Integrate with NetworkX (how do I use existing algorithms on my multilayer network?)
- Scale to large networks (when should I use sparse matrices? how much memory will this take?)
- Extend py3plex (how do I add my own algorithms?)

### Network Types: Multilayer vs Multiplex

py3plex supports two network types that determine how layers and inter-layer connections are handled:

**Multilayer Networks** (`network_type='multilayer'`, default)

General multilayer networks where each layer can have a different set of nodes. No automatic coupling edges are created.

- Use when layers represent different node types (e.g., authors, papers, venues)
- Use when nodes naturally appear in only some layers
- Inter-layer edges must be explicitly added

```
from py3plex.core import multinet

# Heterogeneous network: authors and papers are different node types
network = multinet.multi_layer_network(network_type='multilayer')
network.add_edges([
    {'source': 'author1', 'target': 'paper1',
     'source_type': 'authors', 'target_type': 'papers'}
])
```

**Multiplex Networks** (`network_type='multiplex'`)

Special case where all layers share the same set of nodes but with different relationship types. After loading, coupling edges are automatically created between each node and its counterparts in other layers.

- Use when the same entities (people, cities) are connected via multiple relationship types
- Coupling edges are auto-generated with `type='coupling'`
- Filter coupling edges using `get_edges(multiplex_edges=False)`

```
# Social network: same people, different relationship types
network = multinet.multi_layer_network(network_type='multiplex')
network.load_network('friends_and_colleagues.edges', input_type='multiplex_edges')

# Coupling edges are auto-created between (Alice, friends) <-> (Alice, colleagues)

# Get only explicit edges (exclude auto-coupling)
explicit_edges = list(network.get_edges(multiplex_edges=False))

# Get all edges including coupling
all_edges = list(network.get_edges(multiplex_edges=True))
```

**Comparison Table:**

| Feature | multilayer | multiplex |
|---|---|---|
| Node sets | Different per layer | Same across layers |
| Coupling edges | Manual | Automatic |
| Load behavior | As-is | • coupling edges |
| get_edges() | All edges | Filters coupling |
| Use case | Heterogeneous nets | Same entities |

### The `multi_layer_network` Class

The central data structure in py3plex is the `multi_layer_network` class, which provides a high-level API for working with multilayer networks while maintaining full compatibility with NetworkX.

### Basic Structure

```
from py3plex.core import multinet

# Create a multilayer network
network = multinet.multi_layer_network()

# The underlying NetworkX graph
nx_graph = network.core_network  # MultiDiGraph or MultiGraph

# Layer management
layers = network.get_layers()  # List of layer names
layer_map = network.layer_name_map  # Layer name to ID mapping
```

### Key Components

#### 1. Core NetworkX Graph

At its heart, py3plex uses a NetworkX `MultiDiGraph` (for directed networks) or `MultiGraph` (for undirected networks) to store all nodes and edges.

```
# Access the underlying graph
G = network.core_network

# Use any NetworkX function
import networkx as nx
betweenness = nx.betweenness_centrality(G)
```

#### 2. Layer Encoding

Nodes are internally encoded as `(node_id, layer_id)` tuples to distinguish the same logical node across different layers.

```
# Node 'Alice' in layer 'friends' is stored as ('Alice', 'friends')
network.add_nodes([('Alice', 'friends')])
```

```
# Access all node-layer pairs
for node_layer_pair in network.get_nodes():
    print(node_layer_pair)  # ('Alice', 'friends'), ('Bob', 'friends'), ...
```

**3. Layer Mapping**

py3plex maintains bidirectional mappings between layer names and internal layer IDs for efficient lookups.

```
# Get layer mapping
layer_map = network.layer_name_map
# {'friends': 0, 'colleagues': 1, ...}
```

**4. Attributes**

Node and edge attributes are stored directly in the underlying NetworkX graph and can be accessed and modified using standard NetworkX patterns.

```
# Add node with attributes
network.core_network.nodes[('Alice', 'friends')]['age'] = 30

# Add edge with attributes
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'friends'), {'weight': 0.8, 'type': 'close'}]
])
```

## Internal Representation

### Node-Layer Pairs

The key concept in py3plex's internal model is the **node-layer pair**: a tuple `(node_id, layer_id)` that uniquely identifies a node in a specific layer.

```
# Same logical node, different layers
alice_friends = ('Alice', 'friends')
alice_colleagues = ('Alice', 'colleagues')

# These are treated as different nodes internally
network.add_nodes([alice_friends, alice_colleagues])
```

This representation allows:

- The same logical entity to appear in multiple layers
- Different attributes per node-layer pair
- Efficient layer-specific operations

### Edge Types

**Intra-Layer Edges**

Edges connecting nodes within the same layer:

```
# Edge within 'friends' layer
network.add_edges([
```

```
    [('Alice', 'friends'), ('Bob', 'friends'), 1.0]
])
```

**Inter-Layer Edges**

Edges connecting nodes across different layers:

```
# Connect Alice across layers (identity edge)
network.add_edges([
    [('Alice', 'friends'), ('Alice', 'colleagues'), 1.0]
])

# Connect different nodes across layers
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'colleagues'), 0.5]
])
```

### Supra-Adjacency Matrix

The supra-adjacency matrix is a block matrix representation that encodes the entire multilayer network structure in one object. It respects the ordering of node-layer pairs (by layer, then by node within each layer), so always check your node ordering before interpreting row or column indices.

### Mathematical Definition

For a multilayer network with $L$ layers and $n_\alpha$ nodes in layer $\alpha$, the supra-adjacency matrix $\mathbf{S}$ stacks the intra-layer adjacency blocks with the inter-layer coupling blocks. With layer-major ordering of node-layer pairs, it has the block form:

$$\mathbf{S} = \begin{bmatrix} A_1 & C_{12} & \cdots & C_{1L} \\ C_{21} & A_2 & \cdots & C_{2L} \\ \vdots & \vdots & \ddots & \vdots \\ C_{L1} & C_{L2} & \cdots & A_L \end{bmatrix}$$

Where:

- $A_\alpha \in \mathbb{R}^{n_\alpha \times n_\alpha}$ is the adjacency matrix of layer $\alpha$ (intra-layer connections)
- $C_{\alpha\beta} \in \mathbb{R}^{n_\alpha \times n_\beta}$ is the coupling matrix between layers $\alpha$ and $\beta$ (inter-layer connections)
- $n_{\text{total}} = \sum_{\alpha=1}^{L} n_\alpha$ is the total number of node-layer pairs across all layers
- The full matrix has size $n_{\text{total}} \times n_{\text{total}}$

In multiplex networks, $n_\alpha$ is typically the same for all layers, so $\mathbf{S}$ becomes an $(N \cdot L) \times (N \cdot L)$ matrix. For undirected networks, $A_\alpha = A_\alpha^\top$ and $C_{\alpha\beta} = C_{\beta\alpha}^\top$; for directed networks these blocks can be asymmetric. Identity coupling corresponds to $C_{\alpha\beta} = \omega I$ (with coupling weight $\omega$) when layers share node identities.

### Numeric Example

Consider a simple multiplex network with 3 nodes (A, B, C) and 2 layers. In layer 1, A-B are connected. In layer 2, B-C are connected. The nodes are the same across layers (multiplex structure).

**Network structure:**

```
Layer 1:        Layer 2:
A --- B   C     A   B --- C
```

**Node ordering:** We order nodes as (A,L1), (B,L1), (C,L1), (A,L2), (B,L2), (C,L2).

**Intra-layer blocks:**

Layer 1 adjacency matrix ($A_1$):

```
     A   B   C
A [  0   1   0 ]
B [  1   0   0 ]
C [  0   0   0 ]
```

Layer 2 adjacency matrix ($A_2$):

```
     A   B   C
A [  0   0   0 ]
B [  0   0   1 ]
C [  0   1   0 ]
```

**Inter-layer coupling blocks ($C_{12}$ and $C_{21}$):**

In a multiplex network, we typically add identity coupling—each node connects to itself across layers with the chosen coupling weight (here 1.0):

```
C12 = C21 =
     A   B   C
A [  1   0   0 ]
B [  0   1   0 ]
C [  0   0   1 ]
```

**Full supra-adjacency matrix (6×6):**

```
          Layer 1        |     Layer 2
       A    B    C       |  A    B    C
L1  A [ 0    1    0       |  1    0    0 ]
    B [ 1    0    0       |  0    1    0 ]
    C [ 0    0    0       |  0    0    1 ]
   ---|------------------+---------------
L2  A [ 1    0    0       |  0    0    0 ]
    B [ 0    1    0       |  0    0    1 ]
    C [ 0    0    1       |  0    1    0 ]
```

The diagonal 3×3 blocks are the within-layer adjacencies ($A_1$ and $A_2$). The off-diagonal 3×3 blocks are the coupling matrices ($C_{12}$ and $C_{21}$).

**What this enables:**

- **Multilayer shortest paths:** A path from (A,L1) to (C,L2) must go through B: (A,L1)→(B,L1)→(B,L2)→(C,L2)
- **Spectral methods:** Eigenvalues of S capture multilayer structure
- **Matrix operations:** All linear algebra works on the full system

### Construction

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
# ... add nodes and edges ...

# Get supra-adjacency matrix (sparse by default for efficiency)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Dense version (for small networks only)
supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)

# Shape: (total_nodes, total_nodes)
# where total_nodes = sum of nodes across all layers
print(f"Shape: {supra_adj.shape}")
```

### Memory Implications

Let $n_{\text{total}} = \sum_{\alpha=1}^{L} n_\alpha$ (the number of node-layer pairs). The supra-adjacency matrix is $n_{\text{total}} \times n_{\text{total}}$. If every layer has $N$ nodes, then $n_{\text{total}} = N \cdot L$.

Worked estimates (assuming equal nodes per layer and 8-byte floats, ignoring library overhead):

- **Small network (100 nodes, 3 layers):** $300 \times 300 = 90{,}000$ entries $\rightarrow$ ~720 KB dense
- **Medium network (1,000 nodes, 5 layers):** $5{,}000 \times 5{,}000 = 25$ million entries $\rightarrow$ ~200 MB dense
- **Large network (10,000 nodes, 10 layers):** $100{,}000 \times 100{,}000 = 10$ billion entries $\rightarrow$ ~80 GB dense (impractical)

**Always use sparse=True** for networks with more than ~1,000 total node-layer pairs. Sparse matrices only store non-zero entries, reducing memory from $O(n_{\text{total}}^2)$ potential entries to roughly $O(\text{edges} + \text{coupling edges})$ actual non-zeros.

### Visualization

Visualize the supra-adjacency matrix structure:

```
from py3plex.core import multinet, random_generators
import matplotlib.pyplot as plt

# Generate a small multiplex network
network = random_generators.random_multiplex_ER(
    n=50, l=3, p=0.1, directed=False
)

# Visualize the supra-adjacency matrix
network.visualize_matrix({"display": True})
```

The visualization shows the block structure with:

- Diagonal blocks: intra-layer connections

- Off-diagonal blocks: inter-layer connections

### How py3plex Uses NetworkX Under the Hood

py3plex is built on top of NetworkX, using it as the storage and algorithm backend. Understanding this architecture helps you leverage the full NetworkX ecosystem.

### The Role of MultiGraph/MultiDiGraph

Internally, py3plex stores all nodes and edges in a single NetworkX `MultiGraph` (for undirected) or `MultiDiGraph` (for directed networks). The multilayer structure is encoded through the node representation:

```
# What you write in py3plex:
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
], input_type="list")

# What's actually stored in NetworkX:
# Nodes: ('Alice', 'friends'), ('Bob', 'friends'),
#        ('Alice', 'colleagues'), ('Bob', 'colleagues')
# Edges: (('Alice', 'friends'), ('Bob', 'friends')),
#        (('Alice', 'colleagues'), ('Bob', 'colleagues'))
```

This means:

1. **All NetworkX algorithms work directly** on the underlying graph
2. **Node-layer tuples are just nodes** to NetworkX—it doesn't "know" about layers
3. **Attributes are stored as NetworkX attributes** and persist across operations

### Attribute Propagation

When you add nodes or edges with attributes, they're stored in the NetworkX graph:

```
# Node attributes
network.core_network.nodes[('Alice', 'friends')]['age'] = 30
network.core_network.nodes[('Alice', 'friends')]['role'] = 'admin'

# Edge attributes
network.core_network[('Alice', 'friends')][('Bob', 'friends')]['weight'] = 0.8
network.core_network[('Alice', 'friends')][('Bob', 'friends')]['type'] = 'close'
```

Attributes are preserved when you:

- Extract subnetworks (filtered nodes keep their attributes)
- Iterate over nodes/edges with `data=True`
- Save and load networks (in formats that support attributes)

### Using NetworkX Functions Directly

Because `network.core_network` is a standard NetworkX graph, you can use any NetworkX function:

```
import networkx as nx

G = network.core_network

# Standard NetworkX algorithms
betweenness = nx.betweenness_centrality(G)
pagerank = nx.pagerank(G)
components = list(nx.connected_components(G.to_undirected()))

# Shortest paths (work across layers if inter-layer edges exist)
path = nx.shortest_path(G, ('Alice', 'friends'), ('Bob', 'colleagues'))

# Community detection (treats node-layer pairs as nodes)
communities = nx.community.louvain_communities(G)
```

**Caveat:** NetworkX doesn't know about layer semantics. It treats ('Alice', 'friends') and ('Alice', 'colleagues') as unrelated nodes. py3plex's multilayer algorithms add the layer-aware logic.

### Design Trade-offs

#### Node-Layer Pairs vs. Alternative Representations

py3plex's node-layer pair representation is one of several possible designs. Here's why it was chosen:

**Alternative 1: Separate graphs per layer**

```
# Instead of py3plex:
layer1 = nx.Graph()
layer2 = nx.Graph()
layer1.add_edge('Alice', 'Bob')
layer2.add_edge('Alice', 'Bob')
```

*Pros:*

- Simple, familiar
- Easy layer-specific analysis

*Cons:*

- No representation of inter-layer edges
- Manual bookkeeping for cross-layer operations
- Can't compute multilayer paths or centrality directly

**Alternative 2: Single graph with edge type attributes**

```
# Instead of py3plex:
G = nx.MultiGraph()
G.add_edge('Alice', 'Bob', type='friends')
G.add_edge('Alice', 'Bob', type='colleagues')
```

*Pros:*

- Single graph structure
- Edges carry type information

*Cons:*

- Nodes can't have layer-specific attributes
- Same node in different contexts is the same node (can't have Alice be central in one layer and peripheral in another)
- Inter-layer edges to the same node are impossible

**py3plex approach: Node-layer pairs**

```
# py3plex representation:
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
], input_type="list")
```

*Pros:*

- Preserves both intra-layer and inter-layer structure
- Allows layer-specific node attributes
- Compatible with NetworkX algorithms
- Enables proper multilayer-specific operations
- Supports identity coupling and cross-layer edges

*Cons:*

- Node names are tuples (less intuitive for simple cases)
- Memory usage is multiplied by number of layers
- NetworkX algorithms treat node-layer pairs as independent nodes (need py3plex wrappers for multilayer semantics)

**py3plex's choice:** The node-layer pair approach was chosen because it correctly models multilayer semantics while maintaining NetworkX compatibility. The downsides (tuple names, memory) are acceptable trade-offs for the flexibility and correctness gained.

### Consequences for Algorithm Implementation

The node-layer pair design has implications:

1. **Layer extraction is subsetting:** To get layer 1, you filter nodes where the layer component equals "layer1"
2. **Identity coupling is explicit:** Cross-layer edges between the same entity must be added explicitly (or automatically in multiplex mode)
3. **Supra-matrix is derivable:** The supra-adjacency matrix can be computed from the graph structure
4. **NetworkX algorithms work but need interpretation:** Betweenness on the full graph treats cross-layer edges as normal edges

### Consequences for I/O

The node-layer pair representation affects serialization:

1. **Standard formats work:** GraphML, GML can store tuple nodes
2. **Human-readable formats need mapping:** Edgelists typically use separate columns for node and layer
3. **Arrow/Parquet are efficient:** Modern columnar formats handle the structure naturally

## Network Construction

### Creating Networks from Scratch

**Method 1: Add edges directly**

```
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add edges in list format: [source_node, source_layer, target_node, target_layer, weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1.0],
], input_type="list")
```

**Method 2: Add nodes and edges separately**

```
network = multinet.multi_layer_network()

# Add nodes explicitly
network.add_nodes([
    ('Alice', 'friends'),
    ('Bob', 'friends'),
    ('Alice', 'colleagues'),
    ('Bob', 'colleagues')
])

# Add edges between existing nodes
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'friends'), 1.0],
    [('Alice', 'colleagues'), ('Bob', 'colleagues'), 1.0]
])
```

**Method 3: Define layers first**

```
network = multinet.multi_layer_network()

# Define layers explicitly
network.add_layer('friends')
network.add_layer('colleagues')

# Add content (nodes are created automatically with edges)
```

```
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0]
], input_type="list")
```

### Loading from Files

py3plex supports multiple input formats. See ../user_guide/io_and_formats for complete documentation.

```
from py3plex.core import multinet

# From simple edge list (source target)
network = multinet.multi_layer_network().load_network(
    "data.edgelist",
    input_type="edgelist",
    directed=False
)

# From multilayer edge list (source target layer)
network = multinet.multi_layer_network().load_network(
    "data.multiedgelist",
    input_type="multiedgelist",
    directed=False
)

# From GraphML
network = multinet.multi_layer_network().load_network(
    "data.graphml",
    input_type="graphml"
)

# From NetworkX graph
import networkx as nx
G = nx.karate_club_graph()
network = multinet.multi_layer_network()
network.load_network_from_networkx(G)
```

### Network Operations

### Querying Network Elements

```
# Get all node-layer pairs
nodes = network.get_nodes(data=True)  # With attributes
nodes = network.get_nodes(data=False)  # Without attributes

# Get nodes in a specific layer
layer_nodes = network.get_nodes(layer='friends')

# Get all edges
edges = network.get_edges(data=True)
```

```
# Get neighbors of a node in a layer
neighbors = list(network.get_neighbors('Alice', layer_id='friends'))

# Get all layers
layers = network.get_layers()
print(f"Layers: {layers}")
```

### Basic Statistics

```
# Get basic network statistics
network.basic_stats()

# Output:
# Number of nodes: X
# Number of edges: Y
# Number of unique nodes (as node-layer tuples): Z
# Number of unique node IDs (across all layers): W
# Nodes per layer:
#   Layer 'friends': N1 nodes
#   Layer 'colleagues': N2 nodes
```

### Subnetwork Extraction

```
# Extract a single layer
layer_1 = network.subnetwork(['friends'], subset_by="layers")

# Extract specific nodes (all their appearances)
node_subset = network.subnetwork(['Alice', 'Bob'], subset_by="node_names")

# Extract specific node-layer pairs
specific_pairs = network.subnetwork(
    [('Alice', 'friends'), ('Bob', 'friends')],
    subset_by="node_layer_names"
)
```

### Matrix Representations

```
# Get supra-adjacency matrix (all layers stacked)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Get adjacency matrix for a single layer
layer_subgraph = network.subnetwork(['friends'], subset_by="layers")
layer_adj = nx.adjacency_matrix(layer_subgraph.core_network)
```

### Integration with NetworkX

#### Full NetworkX Compatibility

Since py3plex builds on NetworkX, you can use any NetworkX algorithm:

```
import networkx as nx
from py3plex.core import multinet

# Create py3plex network
mlnet = multinet.multi_layer_network()
mlnet.add_edges([
    ['A', 'L1', 'B', 'L1', 1],
    ['B', 'L1', 'C', 'L1', 1],
], input_type="list")

# Access underlying NetworkX graph
G = mlnet.core_network

# Use any NetworkX function
betweenness = nx.betweenness_centrality(G)
print(f"Betweenness centrality: {betweenness}")

# Shortest paths
try:
    path = nx.shortest_path(G, ('A', 'L1'), ('C', 'L1'))
    print(f"Shortest path: {path}")
except nx.NetworkXNoPath:
    print("No path exists")

# Connected components
if not G.is_directed():
    components = list(nx.connected_components(G))
    print(f"Number of components: {len(components)}")
```

#### NetworkX Wrappers

py3plex provides convenience wrappers for common NetworkX functions:

```
# Extract a layer
layer_1 = network.subnetwork(['layer1'], subset_by="layers")

# Apply NetworkX algorithms via wrapper
degree_centrality = layer_1.monoplex_nx_wrapper("degree_centrality")
betweenness = layer_1.monoplex_nx_wrapper("betweenness_centrality")

# Results are dictionaries mapping node-layer pairs to values
print(f"Degree centrality: {degree_centrality}")
```

### Converting Between Formats

```
# From NetworkX to py3plex
import networkx as nx
from py3plex.core import multinet

G = nx.karate_club_graph()
mlnet = multinet.multi_layer_network()
# NetworkX graphs are imported as single-layer networks
mlnet.load_network_from_networkx(G)

# From py3plex to NetworkX (already a NetworkX graph!)
G_export = mlnet.core_network

# Save as standard NetworkX formats
nx.write_graphml(G_export, "output.graphml")
nx.write_gml(G_export, "output.gml")
```

### Performance Considerations

#### Sparse vs. Dense Matrices

For large networks, always use sparse matrix representations:

```
# Good: Sparse matrix (efficient for large networks)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Bad: Dense matrix (memory-intensive)
# Only use for small networks (< 1000 nodes)
supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)
```

### Memory Usage

Node-layer pairs mean that a node appearing in $L$ layers occupies $L$ times the memory of a single-layer network. For networks with many layers:

- Consider layer-specific analysis when possible
- Use subnetwork extraction to work with smaller portions
- See *Performance and Scalability Best Practices* for optimization strategies

### Tensor-Like Indexing

You can access neighbors and edge attribute dictionaries using tensor-like notation on the underlying NetworkX graph:

```
from py3plex.core import multinet, random_generators

# Create a network
network = random_generators.random_multilayer_ER(100, 3, 0.05, directed=False)

# Get some nodes and edges
```

```
some_nodes = list(network.get_nodes())[0:5]
some_edges = list(network.get_edges())[0:5]

# Access adjacency view for a node (neighbors and edge dictionaries)
adjacency_view = network[some_nodes[0]]
print(f"Neighbors of {some_nodes[0]}: {list(adjacency_view.keys())}")

# Access node attributes explicitly
node_attrs = network.core_network.nodes[some_nodes[0]]
print(f"Node attributes for {some_nodes[0]}: {node_attrs}")

# Access edge data (adjacency dict keyed by edge IDs in a MultiGraph)
edge_data = network.core_network[some_edges[0][0]][some_edges[0][1]]
print(f"Edge data: {edge_data}")
```

### Design Principles

py3plex follows several key design principles:

**1. NetworkX Compatibility**

All operations maintain compatibility with NetworkX, allowing seamless use of the rich NetworkX ecosystem.

**2. Lazy Evaluation**

Expensive operations like supra-adjacency matrix construction are computed on demand and cached when appropriate.

**3. Graceful Degradation**

Optional features (like Infomap community detection or advanced visualizations) fail gracefully with informative error messages if dependencies are missing.

**4. Extensibility**

The modular design makes it easy to extend py3plex with custom algorithms and visualizations.

**5. Flexibility**

Multiple ways to construct and manipulate networks to suit different workflows and use cases.

### Comparison with Other Representations

### Why Node-Layer Pairs?

Alternative representations exist, but py3plex's node-layer pair approach offers key advantages over the two common options:

- **Separate graphs per layer:** Simple, but loses inter-layer information and requires manual bookkeeping for cross-layer operations.
- **Single graph with edge type attributes:** Keeps everything in one graph, but cannot distinguish node context across layers or attach layer-specific node attributes.
- **Node-layer pairs (py3plex):** Preserves both intra-layer and inter-layer structure, supports layer-specific attributes, and stays compatible with NetworkX algorithms while enabling multilayer-specific operations.

### What You Learned

This chapter covered the internal architecture of py3plex:

**Core concepts:**

- The `multi_layer_network` class as the central data structure
- Node-layer pairs `(node_id, layer_id)` as the fundamental representation
- The difference between `network_type='multilayer'` and `network_type='multiplex'`

**Data structures:**

- The underlying NetworkX MultiGraph/MultiDiGraph
- Layer mappings and layer extraction
- The supra-adjacency matrix and its block structure

**Design trade-offs:**

- Why node-layer pairs were chosen over alternatives
- Memory implications of the representation
- When to use sparse vs. dense matrices

**NetworkX integration:**

- Accessing the core graph with `network.core_network`
- Using any NetworkX algorithm directly
- The `monoplex_nx_wrapper()` convenience method

### What's Next?

- *Design Principles* — The philosophy behind py3plex's API design
- *Algorithm Landscape* — Overview of available algorithms
- ../user_guide/networks — Practical guide to network operations
- ../user_guide/statistics — Computing multilayer statistics
- ../user_guide/io_and_formats — Complete I/O documentation

**Academic References:**


- De Domenico et al. (2013). "Mathematical formulation of multilayer networks." *Physical Review X* 3(4): 041022.
- Kivelä et al. (2014). "Multilayer networks." *Journal of Complex Networks* 2(3): 203-271.


---

*Next chapter: :doc:`design_principles` — understanding py3plex's design philosophy.*

## 3.4.4 Design Principles

*In which we explore the philosophy behind py3plex's API, understand why certain design choices were made, and learn how these principles make the library easier to use and extend.*

---

Every library makes choices. Why is the API shaped this way? Why does this function return that data structure? Why do you import from here and not there?

This chapter explains the design principles behind py3plex. Understanding them will help you:

- **Use the library more effectively** — work *with* the design, not against it
- **Debug unexpected behavior** — understand why things work the way they do
- **Extend the library** — add your own algorithms that fit naturally

### Core Philosophy

#### Simple, Off-the-Shelf Functionality

**Principle:** Provide ready-to-use tools that work out of the box with minimal configuration.

**In practice:** keep the first example minimal, without mandatory boilerplate.

```
from py3plex.core import multinet

# Simple, intuitive API
network = multinet.multi_layer_network()
# List format: [source, source_layer, target, target_layer, weight]
network.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type="list")
network.basic_stats()  # Works immediately
```

**Rationale:** Researchers and practitioners need tools that are easy to adopt without extensive setup or configuration. py3plex minimizes the barrier to entry for multilayer network analysis by keeping the first run frictionless.

#### NetworkX Compatibility

**Principle:** Build on NetworkX rather than reinventing the wheel.

**In practice:** py3plex wraps a NetworkX graph and keeps it accessible.

```
import networkx as nx

# Direct access to NetworkX graph
G = network.core_network

# Use any NetworkX function
betweenness = nx.betweenness_centrality(G)
communities = nx.community.louvain_communities(G)
```

**Rationale:** NetworkX is the de facto standard for network analysis in Python. By building on it, py3plex:

- Leverages a mature, well-tested codebase
- Provides access to hundreds of NetworkX algorithms
- Ensures interoperability with the broader Python ecosystem (e.g., pandas, scipy)
- Reduces learning curve for users familiar with NetworkX, while adding multilayer semantics

### Modular Architecture

**Principle:** Separate concerns into independent, composable modules.

**Structure:**

```
py3plex/
├── core/                    # Data structures
├── algorithms/              # Analysis methods
│   ├── community_detection/
│   ├── statistics/
│   └── multilayer_algorithms/
├── visualization/           # Plotting
└── wrappers/                # High-level interfaces
```

**In practice:**

```python
# Import only what you need
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.visualization.multilayer import hairball_plot
```

**Rationale:** Modular design allows users to:

- Import only the functionality they need
- Understand and maintain code more easily
- Extend the library with custom modules
- Mix and match components flexibly

### Implementation Principles

These principles translate the philosophy into everyday coding patterns inside py3plex.

### Flexibility

**Principle:** Support multiple ways to accomplish tasks, accommodating different workflows.

**Examples:**

```python
from py3plex.core import multinet
network = multinet.multi_layer_network()

# Multiple ways to create networks

# Method 1: Direct edge addition
network.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type="list")

# Method 2: Load from file
network.load_network("data.edgelist", input_type="edgelist")

# Method 3: From NetworkX
network.load_network_from_networkx(nx_graph)
```

```
# Multiple ways to access nodes
nodes1 = network.get_nodes()
nodes2 = list(network.core_network.nodes())
nodes3 = network.get_nodes(layer='L1')
```

**Rationale:** Different users have different preferences and workflows. Flexibility ensures py3plex adapts to the user's needs, not vice versa.

### Lazy Evaluation

**Principle:** Compute expensive operations only when needed, and only once when caching is safe.

**In practice:** defer heavy construction until explicitly requested.

```
# Supra-adjacency matrix is not computed on network creation
network = multinet.multi_layer_network()
network.add_edges([...])  # Fast

# Matrix is computed only when requested
supra_adj = network.get_supra_adjacency_matrix()  # Computed here

# Subsequent calls may reuse cached result unless the graph changes
```

**Rationale:** Many multilayer operations (e.g., supra-adjacency matrix construction, typically of shape $(n \cdot L) \times (n \cdot L)$) are computationally expensive. Lazy evaluation:

- Improves performance for common operations
- Reduces memory usage
- Allows working with large networks when full matrices aren't needed
- Encourages recomputing expensive results only after meaningful graph changes

### Graceful Degradation

**Principle:** Optional features should fail gracefully, not crash the entire application. Where possible, suggest viable fallbacks.

**In practice:** catch optional-dependency errors and steer users to alternatives.

```
# If Infomap is not installed
try:
    communities = infomap_communities(network)
except ImportError as e:
    print("Infomap not available. Using Louvain instead.")
    communities = louvain_partition(network)
```

**Rationale:** Not all users need all features. By degrading gracefully, py3plex:

- Allows users to install only what they need
- Reduces dependency bloat
- Provides helpful error messages
- Suggests alternatives when dependencies are missing

### Explicit Over Implicit

**Principle:** Make behavior explicit and predictable, avoiding surprising defaults.

**Good examples:** be clear about formats, directionality, and layer scope.

```
# Explicit parameter names
network.load_network(
    "data.txt",
    input_type="edgelist",     # Explicit format
    directed=False             # Explicit directionality
)

# Explicit layer specification
neighbors = network.get_neighbors('Alice', layer_id='friends')
```

**Rationale:** Explicit code is:

- Easier to understand and debug
- Less prone to errors
- Self-documenting (intent is visible from the call site)
- More maintainable

### Performance Considerations

### Sparse Representations

**Principle:** Use sparse data structures for large networks.

**In practice:**

```
# Sparse matrix by default (efficient for large networks)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Dense only for small networks or specific algorithms
supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)
```

**Rationale:** Real-world networks are typically sparse (few edges relative to potential edges). The supra-adjacency matrix indexes (node, layer) pairs, so sparsity keeps it tractable. Sparse representations:

- Reduce memory usage by orders of magnitude
- Enable analysis of networks with millions of nodes
- Maintain computational efficiency

### Efficient Data Structures

**Principle:** Choose appropriate data structures for common operations.

**Examples:**

- NetworkX graphs for topology (edge lookups, traversals; preserves attributes)
- Dictionaries for layer mappings (O(1) lookups)
- Sparse matrices for linear algebra

- Sets for membership testing

**Rationale:** The right data structure keeps common operations both fast and memory-efficient.

#### Algorithmic Defaults

**Principle:** Provide sensible default parameters based on empirical research.

**In practice:**

```
# Default parameters work well for most cases
partition = louvain_multilayer(
    network,
    gamma=1.0,      # Standard resolution
    omega=1.0,      # Balanced inter-layer coupling
    random_state=42  # Reproducibility
)
```

**Rationale:** Good defaults:

- Make the library easy to use for beginners
- Reduce parameter tuning overhead
- Are based on published research and empirical validation
- Provide a safe baseline before task-specific tuning (adjust `gamma`/`omega` for resolution and coupling as network density changes)

### Extensibility

#### Plugin-Friendly Architecture

**Principle:** Make it easy to add custom functionality.

**In practice:**

```
# Easy to add custom algorithms
def my_centrality(ml_network):
    """Custom centrality measure."""
    G = ml_network.core_network
    # Implement your algorithm
    return centrality_scores  # Dict keyed by (node, layer) tuples

# Easy to add custom visualizations
def my_plot(ml_network, **kwargs):
    """Custom visualization."""
    # Use py3plex drawing machinery
    from py3plex.visualization import drawing_machinery as dm
    # Implement your plot
    pass
```

**Rationale:** Research needs evolve. Extensibility ensures py3plex can grow with new methods without requiring core library changes.

#### Clear Interfaces

**Principle:** Define clear, stable APIs for modules so that code using them remains forward-compatible.

**In practice:** similar functions take the same arguments, and return shapes stay stable.

```
# All statistics follow same interface
from py3plex.algorithms.statistics import multilayer_statistics as mls

density = mls.layer_density(network, layer)
activity = mls.node_activity(network, node)
overlap = mls.edge_overlap(network, layer1, layer2)

# Consistent parameter names and return types
```

**Rationale:** Clear interfaces make py3plex:

- Easier to learn (patterns repeat)
- More predictable (fewer surprises across modules)
- More maintainable
- Easier to extend without breaking callers

### Documentation and Testing

#### Comprehensive Documentation

**Principle:** Every feature should be documented with examples.

**In practice:** blend narrative guides with executable snippets.

- Detailed docstrings for all public functions
- Code examples in documentation
- Real-world use cases
- API reference with parameter descriptions
- Cross-links between conceptual guides and API reference

**Rationale:** Good documentation:

- Reduces barrier to entry
- Serves as a reference for experienced users
- Helps maintainers understand code
- Acts as specification

#### Extensive Testing

**Principle:** Test all core functionality and edge cases, including multilayer specifics.

**In practice:** combine correctness checks with performance guards.

```
# Comprehensive test suite
make test          # Unit tests
make benchmark     # Performance tests
```

```
make fuzz-property # Property-based tests
```

**Rationale:** Tests ensure:

- Code works as intended
- Changes don't break existing functionality
- Edge cases are handled correctly (layered vs. aggregated workflows)
- Performance regressions are caught

### Interoperability

#### Standard Formats

**Principle:** Support widely-used data formats.

**Formats supported:**

- EdgeList (simple, human-readable)
- GraphML (XML-based, preserves attributes)
- GML (Graph Modeling Language)
- Pickle (NetworkX native)
- JSON/Arrow/Parquet (modern, efficient)

**Rationale:** Standard formats enable:

- Data sharing between tools
- Integration with other software
- Long-term data preservation (avoid lock-in to bespoke formats)

### Cross-Platform Compatibility

**Principle:** Work consistently across operating systems.

**Regularly exercised on:** (through automated builds and manual checks)

- Linux (Ubuntu 20.04, 22.04)
- macOS (11, 12, 13)
- Windows (Server 2019, 2022)

**Rationale:** Users work on different platforms. Cross-platform support keeps behavior consistent and reduces support surprises.

### Research-Oriented Design

#### Citation and Attribution

**Principle:** Make it easy to cite algorithms and give proper attribution.

**In practice:**

```
# Algorithms include citations in docstrings
help(louvain_multilayer)
# Docstring includes:
```

```
# References:
#   Mucha et al. (2010). "Community structure in time-dependent..."
```

**See:** ../reference/citation_and_acknowledgements

**Rationale:** Proper attribution:

- Gives credit to algorithm developers
- Helps users find original papers
- Supports reproducibility

### Reproducibility

**Principle:** Support reproducible research by exposing seed controls wherever randomness is involved.

**In practice:**

```
# Random seed parameters for reproducibility
partition = louvain_multilayer(
    network,
    random_state=42  # Reproducible results
)

walks = generate_walks(
    network,
    num_walks=5,
    walk_length=10,
    seed=42  # Reproducible walks
)
```

**Rationale:** Reproducibility is fundamental to science. Seed parameters ensure:

- Results can be verified
- Comparisons are fair
- Bugs can be diagnosed

### Validation and Benchmarking

**Principle:** Validate algorithms against published results and known properties when possible.

**In practice:** combine qualitative checks (structure) with quantitative ones (scores, timings).

- Benchmark scripts comparing to reference implementations when available
- Validation against known network properties (e.g., degree sequences, connectivity)
- Performance metrics for representative network sizes
- Recorded assumptions when a formal ground truth is unavailable

**Rationale:** Validation ensures:

- Implementations are correct
- Performance is acceptable
- Results match published literature

### Practical Implications

#### For Users

These design principles mean you can:

1. **Start quickly** with minimal setup
2. **Use NetworkX knowledge** and tools directly
3. **Choose your workflow** with multiple approaches
4. **Work efficiently** with large networks
5. **Extend easily** with custom algorithms
6. **Trust results** through validation and testing

#### For Contributors

When contributing to py3plex, follow these principles:

1. **Maintain NetworkX compatibility** in new features
2. **Document thoroughly** with examples
3. **Test extensively** including edge cases
4. **Provide sensible defaults** based on research
5. **Fail gracefully** with helpful error messages
6. **Follow existing patterns** for consistency

See ../dev/contributing for detailed guidelines.

### Comparison with Other Libraries

#### py3plex vs. Pure NetworkX

**NetworkX:**

- Excellent for single-layer networks
- Lacks native multilayer support
- General-purpose graph library

**py3plex:**

- Native multilayer data structures
- Layer-aware algorithms
- Specialized multilayer visualizations
- Built on NetworkX (inherits all features)

#### py3plex vs. Other Multilayer Tools

**Other tools (e.g., muxViz, pymnet):**

- Often specialized or platform-specific
- May have steep learning curves
- Limited algorithm libraries

**py3plex:**

- Pure Python (easy installation)
- Comprehensive algorithm library
- NetworkX integration (reuse familiar APIs)
- Active development with research-backed defaults

### What You Learned

This chapter covered the design philosophy behind py3plex:

**Core principles:**

- **Simple, off-the-shelf functionality** — work out of the box with minimal configuration
- **NetworkX compatibility** — leverage the entire NetworkX ecosystem
- **Modular architecture** — import only what you need

**Implementation principles:**

- **Flexibility** — multiple ways to accomplish tasks
- **Lazy evaluation** — expensive operations computed on demand
- **Graceful degradation** — optional features fail with helpful messages
- **Explicit over implicit** — clear, predictable behavior

**Performance principles:**

- **Sparse representations** — efficient for large networks
- **Sensible defaults** — good starting parameters based on research

**Quality principles:**

- **Comprehensive documentation** — every feature has examples
- **Extensive testing** — validated against published results
- **Reproducibility** — seed parameters for deterministic results

### What's Next?

- *Algorithm Landscape* — Overview of available algorithms
- ../user_guide/networks — Practical guide to network operations
- *Development Guide* — Contributing to py3plex

**Academic References:**

- Škrlj et al. (2019). "Py3plex toolkit for visualization and analysis of multilayer networks." *Applied Network Science* 4(1): 94.

---

*Next chapter: :doc:`algorithm_landscape` — a tour of py3plex's algorithmic capabilities.*

### 3.4.5 Algorithm Landscape

This guide provides a narrative overview of the algorithm families available in py3plex, highlighting when to use each type of algorithm for multilayer network analysis. It complements the API reference with plain-language guidance, trade-offs, and cues for multilayer specifics.

#### Overview

py3plex provides algorithms in seven main categories that work on multilayer graphs encoded as `(node_id, layer_id)` pairs:

1. **Community Detection** - Finding groups of densely connected nodes
2. **Centrality & Importance** - Measuring node and edge importance
3. **Network Statistics** - Computing network-level and layer-specific metrics
4. **Random Walks & Embeddings** - Node representations and sampling
5. **Node Ranking & Classification** - Ordering and classifying nodes
6. **Visualization & Layout** - Rendering and spatial arrangement
7. **Benchmarking & Generation** - Testing and synthetic network creation

For detailed parameter lists and API reference, see *Algorithm Roadmap*. Use this page to choose a method family, then dive into the reference for parameters.

#### Community Detection

**Goal:** Identify groups of nodes that are more densely connected to each other than to the rest of the network (within or across layers). In multilayer settings, decide whether communities should stay layer-specific or span layers via coupling (`omega` controls inter-layer links; `omega = 0` means layers do not interact).

#### When to Use

Use community detection when you want to:

- Discover functional modules in biological networks
- Find social groups in multilayer social networks
- Identify topical clusters in citation networks
- Understand organizational structure
- Detect anomalies (nodes that don't fit any community)

#### Algorithm Families

**Modularity-Based Methods**

*Best for:* Fast community detection on large networks

*Examples:*

- **Louvain** - Fast, greedy modularity optimization
- **Leiden** - Improved version of Louvain with better stability
- **Multilayer Louvain** - Extends Louvain to multiple layers with inter-layer coupling (set `omega > 0` when layers should interact; `gamma` tunes community resolution)

*Trade-offs:*

- Fast (scales to millions of nodes)
- Easy to use (minimal parameters)
- Resolution limit (may miss small communities)
- Stochastic (different runs give different results)

```
from py3plex.algorithms.community_detection import community_louvain

# Single-layer Louvain
partition = community_louvain.best_partition(network.core_network)

# Multilayer Louvain (gamma: resolution, omega: inter-layer coupling)
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer
)
partition = louvain_multilayer(network, gamma=1.0, omega=1.0)
```

**Flow-Based Methods**

*Best for:* Finding communities based on information flow

*Examples:*

- **Infomap** - Minimizes description length of random walks (treat multilayer by adding explicit inter-layer edges or by aggregating first)
- **Walktrap** - Based on random walk distances (single-layer)

*Trade-offs:*

- Theoretically grounded (information theory)
- Finds hierarchical structure
- Slower than modularity methods
- May require external binaries
- Sensitive to how inter-layer edges are weighted (add or scale them explicitly before running on multilayer data)

```
from py3plex.algorithms.community_detection import community_wrapper

# Infomap (requires binary)
partition = community_wrapper.infomap_communities(
    network.core_network,
    binary_path="/path/to/infomap"
)
```

**Overlapping Community Detection**

*Best for:* Networks where nodes belong to multiple communities

*Examples:*

- **NoRC** - Network of Ranked Communities (overlapping membership)
- **Clique Percolation** - Based on overlapping cliques

*Trade-offs:*

- More realistic for many real networks
- Captures multi-role nodes
- More complex to interpret
- Computationally expensive

```
from py3plex.algorithms.community_detection import community_wrapper

# NoRC overlapping communities
partition = community_wrapper.NoRC_communities(network, alpha=0.5)
```

### Choosing a Community Detection Method

Use this decision tree (follow the first condition that applies):

```
Do you need overlapping communities?
├─ Yes → NoRC or Clique Percolation (expect overlapping memberships)
└─ No
   └─ Is your network multilayer?
      ├─ Yes → Multilayer Louvain or Leiden
      └─ No
         └─ What's most important?
            ├─ Speed → Louvain
            ├─ Stability → Leiden
            └─ Theory / flow objective → Infomap
```

### Centrality & Importance

**Goal:** Quantify the importance of nodes or edges in the network, per layer or aggregated. Decide whether to keep `(node, layer)` pairs separate (preserves context) or to aggregate scores per entity after aligning identifiers.

### When to Use

Use centrality measures when you want to:

- Identify influential nodes (e.g., key proteins, opinion leaders)
- Find bottlenecks in flow networks
- Prioritize nodes for intervention or study
- Understand network vulnerability
- Analyze node roles across layers

### Algorithm Families

#### Degree-Based Centrality

*Measures:* Local connectivity

*Examples:*

- **Degree Centrality** - Number of connections

- **Multilayer Degree** - Degree summed across layers
- **Versatility** - Participation across layers (counts how many layers a node engages with; higher means broader presence)

*When to use:*

- Quick assessment of connectivity
- Identifying hubs
- Local importance measures; supply `weight` if edge weights matter

```
import networkx as nx

# Single-layer degree centrality
degree_cent = nx.degree_centrality(network.core_network)

# Multilayer versatility
from py3plex.algorithms.statistics import multilayer_statistics as mls
versatility = mls.versatility_centrality(network, centrality_type='degree')
```

**Distance-Based Centrality**

*Measures:* Global position in network

*Examples:*

- **Closeness Centrality** - Average distance to all other nodes (computed within connected components)
- **Betweenness Centrality** - Fraction of shortest paths through node
- **Harmonic Centrality** - Harmonic mean of distances (stable on disconnected graphs)

*When to use:*

- Identifying central nodes
- Finding information brokers
- Understanding information flow (run per connected component if closeness is used, or switch to harmonic variants)

```
# Betweenness centrality
betweenness = nx.betweenness_centrality(network.core_network)

# Closeness centrality
closeness = nx.closeness_centrality(network.core_network)
```

**Eigenvector-Based Centrality**

*Measures:* Importance based on connections to important nodes

*Examples:*

- **Eigenvector Centrality** - First eigenvector of adjacency matrix (requires connectivity within each component; compute per component)
- **PageRank** - Google's ranking algorithm (damping factor controls teleportation)
- **HITS** - Hubs and authorities (use on directed graphs)

*When to use:*

- Ranking web pages or citations
- Finding influential nodes in social networks
- Authority/hub analysis

```
# PageRank
pagerank = nx.pagerank(network.core_network)

# Eigenvector centrality
eigenvector = nx.eigenvector_centrality(network.core_network)
```

### Choosing a Centrality Measure

Consider these questions:

1. **Local vs. Global?** - Local → Degree centrality - Global → Closeness or betweenness
2. **Direct connections or influence?** - Direct → Degree - Influence → PageRank or eigenvector
3. **Network type?** - Directed → PageRank or HITS - Undirected → Any method - Weighted → Use weight-aware versions (pass `weight` to the function)
4. **Computational cost?** - Large network → Degree or PageRank (fast) - Small network → Betweenness (expensive but informative)
5. **Layer handling?** - Keep per-layer scores when layer context matters - Aggregate across layers only after aligning node identifiers

### Node Ranking & Classification

**Goal:** Order nodes by importance or predict node labels using network-derived features. In multilayer settings, decide whether to rank per layer or on an aggregated view, and whether to treat layers as features.

### When to Use

Use ranking or classification when you want to:

- Prioritize entities (recommendations, audits, interventions)
- Score candidates for follow-up analysis
- Train classifiers from structural features (centrality, embeddings)
- Combine multilayer evidence into a single ordering

### Approaches

- **Rank with centrality:** Use degree/PageRank/eigenvector (per layer or aggregated); normalize if layers differ in size.
- **Rank with random walks:** Use multilayer random-walk methods (e.g., MultiXRank) to blend within-layer and cross-layer signals; tune restart probability to control locality.
- **Classify with embeddings:** Generate node embeddings (Node2Vec/DeepWalk/LINE) then train a downstream classifier (e.g., scikit-learn). Include layer ID as a feature if labels depend on layer context.

- **Hybrid scores:** Combine statistical features (degree, clustering, participation coefficient) with embeddings for models that need both local and global cues.

### Network Statistics

**Goal:** Characterize network structure at the global, layer, or node level without committing to a specific downstream task. These checks help validate data quality before heavier modeling.

### When to Use

Use network statistics when you want to:

- Compare networks quantitatively
- Understand structural properties
- Validate network models
- Monitor network evolution
- Detect anomalies

### Statistic Families

#### Global Statistics

*Examples:*

- Number of nodes, edges, layers
- Density, clustering coefficient
- Average path length (per connected component unless you use harmonic variants)
- Degree distribution

*When to use:* Basic network characterization or sanity checks before heavier analysis

```
# Basic stats
network.basic_stats()

# Clustering
import networkx as nx
clustering = nx.average_clustering(network.core_network)

# Density
from py3plex.algorithms.statistics import multilayer_statistics as mls
density = mls.layer_density(network, 'layer1')
```

#### Layer-Specific Statistics

*Examples:*

- Layer density
- Layer similarity (Jaccard, correlation)
- Edge overlap between layers
- Inter-layer degree correlation

*When to use:* Comparing layers, understanding layer relationships

Layer similarity and overlap assume node identifiers are consistent across layers; if they differ, map or relabel before comparing.

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Layer density
density_l1 = mls.layer_density(network, 'layer1')

# Layer similarity
similarity = mls.layer_similarity(network, 'layer1', 'layer2', method='jaccard')

# Edge overlap
overlap = mls.edge_overlap(network, 'layer1', 'layer2')
```

**Node-Level Statistics**

*Examples:*

- Node activity (presence across layers)
- Participation coefficient
- Cross-layer degree correlation

*When to use:* Characterizing node behavior across layers

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Node activity
activity = mls.node_activity(network, 'Alice')

# Participation coefficient
participation = mls.community_participation_coefficient(network, partition)
```

### Random Walks & Embeddings

**Goal:** Generate node representations or sample the network through traversal (often as input to downstream ML). Walks operate on the encoded `(node, layer)` graph, so inter-layer edges and their weights directly influence the sampled context.

### When to Use

Use random walks and embeddings when you want to:

- Create low-dimensional node representations for ML
- Sample large networks efficiently
- Measure node similarity
- Generate features for classification/clustering
- Understand structural equivalence

### Algorithm Families

#### Random Walk Algorithms

*Examples:*

- **Basic Random Walk** - Uniform random traversal
- **Node2Vec** - Biased walks (BFS/DFS-like); works on multilayer graphs if inter-layer edges are present
- **DeepWalk** - Uniform random walks for embeddings

*Parameters:*

- `walk_length` - Steps per walk
- `num_walks` - Walks per node
- `p` (Node2Vec) - Return parameter
- `q` (Node2Vec) - In-out parameter
- `random_seed` - Set for reproducible walks

```
from py3plex.algorithms.general.walkers import generate_walks, node2vec_walk

# Basic walks
walks = generate_walks(
    network.core_network,
    num_walks=10,
    walk_length=80
)

# Node2Vec walks (BFS-like: stay local)
walks_bfs = generate_walks(
    network.core_network,
    num_walks=10,
    walk_length=80,
    p=1.0, q=2.0  # BFS-like
)

# Node2Vec walks (DFS-like: explore)
walks_dfs = generate_walks(
    network.core_network,
    num_walks=10,
    walk_length=80,
    p=1.0, q=0.5  # DFS-like
)
```

Pass `random_seed` to `generate_walks` for reproducible samples.

#### Embedding Methods

*Examples:*

- **Node2Vec** - Skip-gram on random walks
- **DeepWalk** - Word2Vec on uniform walks
- **LINE** - First/second-order proximity preservation

*When to use:*

- Node classification
- Link prediction
- Visualization in 2D/3D
- Clustering
- Similarity computation

```
from py3plex.wrappers import node2vec_embedding

# Generate embeddings
embeddings = node2vec_embedding.generate_embeddings(
    network.core_network,
    dimensions=128,
    walk_length=80,
    num_walks=10,
    p=1.0,
    q=1.0
)

# Use embeddings for ML
# embeddings is a numpy array: (num_nodes, dimensions)
```

### Choosing Walk Parameters

**walk_length:**

- Short (10-20): Fast, local neighborhood
- Medium (40-80): Balanced, standard choice
- Long (100+): Global structure, slower

**p (return parameter):**

- $p < 1$: More likely to return to previous node (stay local)
- $p = 1$: Balanced
- $p > 1$: Less likely to return (keep exploring)

**q (in-out parameter):**

- $q < 1$: DFS-like (explore distant nodes)
- $q = 1$: Balanced
- $q > 1$: BFS-like (stay in local neighborhood)

### Visualization & Layout

**Goal:** Create visual representations of multilayer networks for exploration or presentation.

### When to Use

Use visualization when you want to:

- Explore network structure
- Present findings
- Communicate patterns
- Identify visual anomalies
- Create publication-ready figures

### Layout Families

#### Force-Directed Layouts

*Examples:*

- Spring layout
- Fruchterman-Reingold
- Force Atlas

*Best for:* General-purpose visualization, showing community structure

```
from py3plex.visualization.multilayer import hairball_plot

hairball_plot(
    network.core_network,
    layout_algorithm="force",
    layout_parameters={"iterations": 50}
)
```

#### Specialized Multilayer Layouts

*Examples:*

- Diagonal projection
- Layered stacking
- Circular layer arrangement

*Best for:* Emphasizing multilayer structure

```
from py3plex.visualization.multilayer import draw_multilayer_default

# Diagonal layout for multilayer networks
draw_multilayer_default(
    [network],
    display=True,
    layout_style='diagonal'
)
```

#### Hierarchical Layouts

*Examples:*

- Reingold-Tilford tree
- Radial tree
- Sugiyama layered

*Best for:* Trees and DAGs

For complete visualization documentation, see ../user_guide/visualization.

### Benchmarking & Generation

**Goal:** Create synthetic networks for testing and validation without risking production data.

### When to Use

Use synthetic network generation when you want to:

- Test algorithm performance
- Validate implementations
- Create controlled experiments
- Generate training data
- Benchmark scalability
- Produce reproducible baselines (fix `random_state` when available)

### Generator Families

#### Random Network Models

*Examples:*

- **Erdős-Rényi** - Random edges with fixed probability
- **Barabási-Albert** - Preferential attachment (scale-free)
- **Watts-Strogatz** - Small-world networks

Use ER for null models, BA for heavy-tailed degree distributions, and WS when you need clustering with short paths.

```
from py3plex.core import random_generators

# Random multilayer ER network
network = random_generators.random_multilayer_ER(
    num_nodes=100,
    num_layers=3,
    probability=0.05,
    directed=False
)
```

#### Benchmark Networks

*Examples:*

- LFR benchmark (with ground-truth communities)
- Configuration model (preserving degree distribution)

- Block model (with planted partition)

Pick LFR when you need ground truth for community detection, configuration for degree-preserving nulls, and block models for controllable meso-scale structure.

### Algorithm Combinations

### Common Workflows

Use these as starting patterns; swap in alternative algorithms from the sections above as needed.

**Community Detection + Visualization:**

```
# Detect communities
partition = louvain_multilayer(network)

# Visualize with community colors
from py3plex.visualization.multilayer import hairball_plot
from py3plex.visualization.colors import colors_default
from collections import Counter

top_communities = [c for c, _ in Counter(partition.values()).most_common(5)]
color_map = dict(zip(top_communities, colors_default[:5]))
node_colors = [color_map.get(partition.get(node), 'gray')
               for node in network.get_nodes()]

hairball_plot(network.core_network, node_colors, layout_algorithm="force")
```

**Centrality + Ranking:**

```
# Compute centrality
centrality = nx.betweenness_centrality(network.core_network)

# Rank nodes
ranked = sorted(centrality.items(), key=lambda x: x[1], reverse=True)

# Top 10
print("Top 10 nodes:", ranked[:10])
```

**Embeddings + Classification:**

```
# Generate embeddings
embeddings = node2vec_embedding.generate_embeddings(network.core_network)

# Use for classification (with sklearn)
from sklearn.ensemble import RandomForestClassifier
clf = RandomForestClassifier()
# clf.fit(embeddings, labels)
# predictions = clf.predict(embeddings_test)
```

### Performance Considerations

#### Algorithm Complexity

Complexity notation: n = number of nodes, m = number of edges, k = walks per node, l = walk length. Complexities are approximate and assume sparse graphs unless stated otherwise.

| Algorithm Family | Typical Complexity | Notes |
|---|---|---|
| Degree Centrality | O(m) | Fast, scales well |
| Betweenness | O(nm) or O(nm + $n^2$log n) | Expensive for large networks |
| Louvain | ~O(m) | Near-linear in practice on sparse graphs |
| Node2Vec | O(m) to precompute + O(k · l · n) for walk generation | Training adds extra cost proportional to walk tokens; tune k and l to control runtime |
| Force Layout | O($n^2$) | Slow for >1000 nodes |

When working with large networks:

- Use approximate algorithms (e.g., sampling for betweenness)
- Consider layer-by-layer analysis
- Use sparse representations
- See *Performance and Scalability Best Practices*

### Next Steps

- ../user_guide/community_detection - Detailed community detection guide
- ../user_guide/statistics - Complete statistics reference
- *Random Walk Algorithms* - Embeddings and walks
- ../user_guide/visualization - Visualization techniques
- *Algorithm Roadmap* - Detailed algorithm parameters and API

**Academic References:**


- Fortunato & Hric (2016). "Community detection in networks: A user guide." *Physics Reports* 659: 1-44.
- Grover & Leskovec (2016). "node2vec: Scalable feature learning for networks." *KDD*.
- Newman (2018). "Networks" (2nd ed.). Oxford University Press.


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## 3.5 Part V: API & DSL Reference

Complete reference documentation for APIs and DSL syntax.

### 3.5.1 API & DSL Reference

Complete reference documentation for py3plex APIs, algorithms, DSL syntax, and configuration.

**This section includes:**

#### Python API Reference

- *API Documentation* — Full API documentation for all modules and classes
- *Algorithm Roadmap* — Complete algorithm reference with parameters and examples
- compat — Compatibility layer API for cross-ecosystem conversions

#### DSL & Query Reference

- *DSL Reference* — Complete DSL syntax and operator reference
- *Layer Set Algebra* — Layer set algebra for composable layer selection
- *Configuration & Environment* — Configuration options and environment variables

**Reading Paths:**

- **Need DSL syntax?** → *DSL Reference* for complete grammar
- **Looking for an algorithm?** → *Algorithm Roadmap* for all available algorithms
- **Need API details?** → *API Documentation* for function signatures and parameters
- **Configuration options?** → *Configuration & Environment* for setup and tuning

**Relation to other sections:**

- **How-to Guides** show you *how* to accomplish tasks. This Reference section documents *what* each function does.
- **Concepts** explain *why* algorithms work. This section provides precise API details.
- **Examples** show complete workflows. This section provides isolated reference information.

### 3.5.2 Algorithm Roadmap

This page provides a **comprehensive overview** of all algorithms implemented in py3plex, organized by category and showing relationships between different methods.

**Purpose:** Help you quickly find the right algorithm for your multilayer network analysis task.

**Complexities:** Listed time complexities are indicative and assume sparse graphs unless noted.

#### Algorithm Categories

The algorithms in py3plex are organized into these main categories:

1. community-detection-algorithms - Finding community structure
2. centrality-measures-algorithms - Computing node importance
3. network-statistics-algorithms - Network-level metrics
4. node-ranking-classification - Ranking and classification methods
5. random-walks-embeddings - Random walk-based methods
6. visualization-algorithms - Layout and rendering

7. benchmark-utilities - Testing and evaluation

### Community Detection Algorithms

**Module:** `py3plex.algorithms.community_detection`

These algorithms identify groups of densely connected nodes (communities) in multilayer networks.

### Louvain Algorithm

**Path:** `community_detection.community_louvain`

**Functions:**

- `best_partition(graph, partition, weight, resolution, randomize, random_state)` - Main entry point
- `generate_dendrogram(graph, partition, weight, resolution, randomize, random_state)` - Hierarchical communities
- `modularity(partition, graph, weight)` - Calculate modularity score
- `partition_at_level(dendrogram, level)` - Extract partition at specific level
- `induced_graph(partition, graph, weight)` - Create quotient graph

**Best for:** Fast community detection on large single-layer networks

**Complexity:** Roughly O(m) per iteration on sparse graphs; multiple passes are performed

**Runnable Example:**

```
import networkx as nx
from py3plex.algorithms.community_detection import community_louvain as louvain

# Create a network with clear community structure
G = nx.karate_club_graph()

# Detect communities
partition = louvain.best_partition(G, random_state=42)

# Calculate modularity
mod = louvain.modularity(partition, G)

print(f"Communities found: {len(set(partition.values()))}")
print(f"Modularity: {mod:.4f}")
```

**Expected output:**

```
Communities found: 4
Modularity: 0.4198
```

**Related algorithms:**

- multilayer-louvain (multilayer extension)
- leiden-algorithm (improved Louvain)

### Infomap Algorithm

**Path:** `community_detection.community_wrapper`

**Functions:**

- `infomap_communities(network_input, binary_path, arguments)` - Main entry point
- `run_infomap(network, binary_path, arguments)` - Execute Infomap binary
- `parse_infomap(outfile)` - Parse Infomap output

**Best for:** Flow-based community detection, hierarchical structure

**Complexity:** Typically near O(m log n) on sparse graphs; depends on hierarchy depth

**Runnable Example:**

```
import networkx as nx
from py3plex.algorithms.community_detection import community_wrapper

# Create a network
G = nx.karate_club_graph()

# Detect communities with Infomap (requires Infomap binary installed)
try:
    communities = community_wrapper.infomap_communities(
        G,
        binary_path="/usr/local/bin/Infomap",  # Adjust path as needed
        arguments="--two-level"
    )
    print(f"Communities found: {len(set(communities.values()))}")
except FileNotFoundError:
    print("Infomap binary not found. Install from infomap.org")
```

**Note:** Infomap requires the external Infomap binary. Install from https://www.mapequation.org/infomap/

**Related algorithms:**

- louvain-algorithm (faster alternative, no external dependencies)

### Multilayer Louvain

**Path:** `community_detection.multilayer_modularity`

**Functions:**

- `louvain_multilayer(supra_adj, omega, resolution, seed)` - Multilayer Louvain
- `multilayer_modularity(supra_adj, partition, omega)` - Calculate multilayer modularity
- `build_supra_modularity_matrix(network, omega)` - Build modularity matrix

**Best for:** Community detection across multiple layers with inter-layer coupling

**Complexity:** Dominated by supra-adjacency size; roughly O(m × iterations) with additional cost for L layers

**Algorithm details:** Implements Mucha et al. (2010) multilayer modularity optimization

**Runnable Example:**

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection import multilayer_modularity as mlmod

# Create two-layer network
network = multinet.multi_layer_network(directed=False)
for i in range(20):
    network.add_nodes([
        {'source': i, 'type': 'layer1'},
        {'source': i, 'type': 'layer2'}
    ])

# Add edges creating two communities
for i in range(10):
    network.add_edges([
        {'source': i, 'target': (i+1)%10, 'source_type': 'layer1', 'target_type': 'layer1'},
        {'source': i, 'target': (i+1)%10, 'source_type': 'layer2', 'target_type': 'layer2'}
    ])
for i in range(10, 20):
    network.add_edges([
        {'source': i, 'target': 10+(i+1)%10, 'source_type': 'layer1', 'target_type': 'layer1'},
        {'source': i, 'target': 10+(i+1)%10, 'source_type': 'layer2', 'target_type': 'layer2'}
    ])

# Build supra-adjacency matrix and detect communities
supra_adj = mlmod.build_supra_modularity_matrix(network, omega=1.0)
partition = mlmod.louvain_multilayer(supra_adj, omega=1.0, seed=42)

print(f"Communities found: {len(set(partition.values()))}")
```

**Expected output:**

```
Communities found: 2
```

**Related algorithms:**

- louvain-algorithm (single-layer version)
- leiden-multilayer (alternative optimizer)

### Leiden Algorithm

**Path:** `community_detection.leiden_multilayer`

**Functions:**

- `leiden_multilayer(supra_adj, omega, resolution, n_iterations, seed)` - Main entry point

**Data structures:**

- `LeidenResult` - Holds partition results with modularity score

**Best for:** More stable community detection than Louvain

**Complexity:** Similar to Louvain; typically linear in edges per refinement iteration

**Related algorithms:**

- louvain-algorithm (predecessor)
- multilayer-louvain (multilayer extension)

### NoRC Algorithm

**Path:** `community_detection.NoRC` and `community_detection.community_wrapper`

**Functions:**

- `NoRC_communities(network, alpha)` - Wrapper function
- `NoRC_communities_main(matrix, alpha)` - Core implementation
- `page_rank_kernel(index_row)` - PageRank computation
- `create_tree(centers)` - Build community hierarchy

**Best for:** Networks with overlapping communities

**Related algorithms:**

- community-ranking (related approach)

### Community Measures

**Path:** `community_detection.community_measures`

**Functions:**

- `modularity(network, partition, weight)` - Calculate modularity
- `size_distribution(network_partition)` - Community size distribution
- `number_of_communities(network_partition)` - Count communities

**Purpose:** Evaluate quality of community partitions

**Related algorithms:** All community detection algorithms above

### Community Ranking

**Path:** `community_detection.community_ranking`

**Functions:**

- `page_rank_kernel(index_row)` - PageRank-based ranking
- `create_tree(centers)` - Build ranking tree
- `return_infomap_communities(network)` - Get Infomap communities

**Best for:** Ranking nodes within detected communities

**Related algorithms:**

- node-ranking-algorithms (general ranking)

### Multilayer Benchmarks

**Path:** `community_detection.multilayer_benchmark`

**Functions:**

- `generate_multilayer_lfr(n_nodes, n_layers, avg_degree, ...)` - LFR benchmark

- `generate_coupled_er_multilayer(n_nodes, n_layers, p_intra, p_inter)` - ER benchmark
- `generate_sbm_multilayer(sizes, p_matrix, n_layers, ...)` - SBM benchmark

**Purpose:** Generate synthetic multilayer networks with known community structure for testing

**Related algorithms:** All community detection algorithms

### Centrality Measures

**Module:** `py3plex.algorithms.multilayer_algorithms.centrality`, `py3plex.algorithms.centrality_toolkit`

These algorithms measure the importance or influence of nodes in multilayer networks.

**New in version 1.0:** Centrality measures now support **first-class uncertainty quantification** via the `uncertainty` parameter. See ../tutorials/multilayer_centrality for details.

### MultilayerCentrality Class

**Path:** `multilayer_algorithms.centrality.MultilayerCentrality`

**Main class for computing various centrality measures.**

**Runnable Example:**

```
from py3plex.core import multinet
from py3plex.algorithms.multilayer_algorithms.centrality import MultilayerCentrality

# Create a small multilayer network
network = multinet.multi_layer_network(directed=False)

# Add nodes and edges to two layers
for i in range(5):
    network.add_nodes([
        {'source': i, 'type': 'layer1'},
        {'source': i, 'type': 'layer2'}
    ])

edges = [
    (0, 1), (1, 2), (2, 3), (3, 4),  # Layer 1: chain
    (0, 4), (1, 3), (2, 4)           # Layer 2: triangle-like
]
for src, tgt in edges:
    network.add_edges([
        {'source': src, 'target': tgt, 'source_type': 'layer1', 'target_type': 'layer1'},
        {'source': src, 'target': tgt, 'source_type': 'layer2', 'target_type': 'layer2'}
    ])

# Compute centrality measures
calc = MultilayerCentrality(network)

# Overlapping degree (sum across layers)
overlap_deg = calc.overlapping_degree_centrality()
print(f"Node 2 overlapping degree: {overlap_deg[(2, 'layer1')]:.2f}")
```

```
# PageRank
pr = calc.pagerank_centrality(alpha=0.85)
print(f"Node 2 PageRank: {pr[(2, 'layer1')]:.4f}")
```

**Expected output:**

```
Node 2 overlapping degree: 6.00
Node 2 PageRank: 0.0892
```

### Basic Centrality Measures

**Methods:**

- `overlapping_degree_centrality()` - Sum of degrees across all layers
- `layer_specific_degree_centrality(layer)` - Degree within specific layer
- `multilayer_degree_centrality()` - Supra-adjacency degree
- `average_degree_centrality()` - Average degree across layers

**Best for:** Quick importance ranking, well-connected node identification

**Complexity:** O(n + m) where n is nodes, m is edges

**Related measures:**

- versatility-centrality (cross-layer activity)

### Eigenvector-Based Measures

**Methods:**

- `multiplex_eigenvector_centrality(max_iter, tol)` - Eigenvector centrality
- `pagerank_centrality(alpha, max_iter, tol)` - PageRank
- `katz_centrality(alpha, beta, max_iter, tol)` - Katz centrality
- `communicability_centrality(beta)` - Communicability centrality

**Best for:** Influence measure, recursive importance, prestige ranking

**Complexity:** O(m) per iteration, typically converges quickly

**Related measures:**

- hubs-authorities (directed networks)
- *Multilayer PageRank (with Uncertainty)* (with uncertainty support)

### Multilayer PageRank (with Uncertainty)

**Path:** `algorithms.centrality_toolkit.multilayer_pagerank`

**New in version 1.0:** Built-in PageRank with **first-class uncertainty quantification**.

**Function signature:**

```
multilayer_pagerank(
    network,
```

```
    alpha=0.85,
    max_iter=100,
    tol=1e-6,
    personalization=None,
    uncertainty=False,
    n_runs=None,
    resampling=None,
    random_seed=None
) -> StatSeries
```

**Parameters:**

- `uncertainty`: If True, estimate uncertainty via resampling
- `n_runs`: Number of runs for uncertainty estimation (default: 50)
- `resampling`: Strategy - SEED, PERTURBATION, BOOTSTRAP, JACKKNIFE
- `random_seed`: For reproducibility

**Returns:** `StatSeries` with mean, std, quantiles (if uncertainty=True)

**Runnable Example:**

```
from py3plex.algorithms.centrality_toolkit import multilayer_pagerank
from py3plex.uncertainty import ResamplingStrategy
from py3plex.core import multinet

# Create network
network = multinet.multi_layer_network(directed=False)
network.add_edges([
    ['A', 'L1', 'B', 'L1', 1.0],
    ['B', 'L1', 'C', 'L1', 1.0],
    ['A', 'L2', 'C', 'L2', 1.0]
], input_type='list')

# Deterministic PageRank
result = multilayer_pagerank(network)
print(f"Node scores: {result.mean}")
print(f"Is deterministic: {result.is_deterministic}")

# With uncertainty
result_unc = multilayer_pagerank(
    network,
    uncertainty=True,
    n_runs=50,
    resampling=ResamplingStrategy.PERTURBATION,
    random_seed=42
)
print(f"Mean scores: {result_unc.mean}")
print(f"Std deviations: {result_unc.std}")
print(f"95% CI: [{result_unc.quantiles[0.025]}, {result_unc.quantiles[0.975]}]")
```

**Expected output:**

```
Node scores: [0.184 0.183 0.184 ...]
Is deterministic: True
Mean scores: [0.186 0.192 0.181 ...]
Std deviations: [0.019 0.021 0.018 ...]
95% CI: [[0.146 0.172 0.146 ...], [0.238 0.241 0.221 ...]]
```

Exact values depend on graph structure and random seeds; treat these arrays as illustrative.

**Best for:**

- Influence ranking with confidence intervals
- Robustness analysis under network perturbations
- Quantifying centrality uncertainty

**Complexity:**

- Deterministic: O(m) per iteration
- With uncertainty: O(m × n_runs)

**Related measures:**

- `MultilayerCentrality.pagerank_centrality()` (deterministic only)

### Approximate Centrality Algorithms (Fast Path)

**New in version 1.1:** Fast approximate centrality algorithms for large networks.

**Module:** `py3plex.algorithms.centrality.approx_*`

**Purpose:** Provide fast approximations of expensive centrality measures on large networks (>1000 nodes).

**Supported algorithms:**

| Measure | Algorithm | Typical Speedup |
|---|---|---|
| Betweenness | Sampling-based (Brandes) | 10-100x |
| Closeness | Landmark-based | 10-50x |
| PageRank | Power iteration | 2-10x |

**Runnable Example (DSL v2):**

```
from py3plex.dsl import Q
from py3plex.core import multinet

# Load a large network
net = multinet.multi_layer_network(directed=False)
net.load_network("large_network.csv", input_type="edgelist")

# Exact computation (slow on large networks)
# result = Q.nodes().compute("betweenness_centrality").execute(net)

# Approximate computation (fast)
result = Q.nodes().compute(
```

```
    "betweenness_centrality",
    approx=True,
    n_samples=512,  # Trade accuracy for speed
    seed=42         # Deterministic results
).execute(net)

# Check provenance
print(f"Fast path: {result.meta['provenance']['backend']['fast_path']}")
print(f"Method: {result.meta['approximation']['measures'][0]['method']}")
```

**Expected output:**

```
Fast path: True
Method: sampling
```

**Accuracy guarantees:**

- Approximate values converge to exact values as `n_samples` increases
- Typical relative error: 5-20% depending on parameters
- Deterministic: same `seed` produces identical results

**Available Functions:**

- `approximate_betweenness_sampling(graph, n_samples, seed)` - Sampling-based betweenness
- `approximate_closeness_landmarks(graph, n_landmarks, seed)` - Landmark-based closeness
- `approximate_pagerank_power_iteration(graph, tol, max_iter)` - Power iteration PageRank

**String DSL support:**

```
from py3plex.dsl_legacy import execute_query

# Use default method
result = execute_query(net, 'SELECT nodes COMPUTE betweenness_centrality APPROXIMATE')

# Specify method and parameters
result = execute_query(net,
    'SELECT nodes COMPUTE betweenness_centrality APPROXIMATE(method="sampling", n_samples=51
)
```

**When to use approximation:**

- Networks with >1000 nodes where exact computation takes >1 minute
- Exploratory analysis where approximate rankings are sufficient
- Sensitivity analysis requiring multiple runs
- Production pipelines with predictable time budgets

**Provenance tracking:**

All approximation executions set `fast_path=True` and record:

- Algorithm method used

- All parameters (n_samples, seed, tol, etc.)
- Convergence diagnostics (where applicable)

### Path-Based Measures

**Methods:**

- `multilayer_betweenness_centrality()` - Betweenness across layers
- `multilayer_closeness_centrality()` - Closeness across layers

**Best for:** Bridge identification, broadcasting efficiency

**Complexity:** O(nm) for unweighted, O(nm + $n^2$ log n) for weighted

**Warning:** Expensive for large networks (>1000 nodes)

**Related measures:**

- centrality-all (batch computation)

### Hubs and Authorities (HITS)

**Path:** `node_ranking.node_ranking.hubs_and_authorities`

**Best for:** Ranking nodes in directed networks (authority = important destination, hub = important source)

**Related measures:**

- PageRank in centrality-measures-algorithms

### Utility Functions

**Functions:**

- `compute_all_centralities(network, include_path_based, include_advanced)` - Compute multiple measures

**Purpose:** Batch computation of multiple centrality measures

### Versatility Centrality

**Path:** `algorithms.statistics.multilayer_statistics.versatility_centrality`

**Function:**

- `versatility_centrality(network, centrality_type)` - Cross-layer versatility measure

**Best for:** Measuring how uniformly a node's importance is distributed across layers

**Complexity:** O(m × L) where L is number of layers

**Related measures:**

- All centrality measures in centrality-measures-algorithms

### Supra Matrix Function Centralities

**Path:** `multilayer_algorithms.supra_matrix_function_centrality`

**Functions:**

- `communicability_centrality(supra_matrix, beta)` - Matrix exponential based
- `katz_centrality(supra_matrix, alpha, beta, max_iter, tol)` - Resolvent based

**Best for:** Advanced centrality using matrix functions on supra-adjacency

**Complexity:** Depends on matrix operations, typically $O(n^2)$ or $O(n^3)$

**Runnable Example:**

```
import numpy as np
from py3plex.core import multinet
from py3plex.algorithms.multilayer_algorithms import supra_matrix_function_centrality as smf

# Create small network
network = multinet.multi_layer_network(directed=False)
for i in range(4):
    network.add_nodes([{'source': i, 'type': 'layer1'}])
network.add_edges([
    {'source': 0, 'target': 1, 'source_type': 'layer1', 'target_type': 'layer1'},
    {'source': 1, 'target': 2, 'source_type': 'layer1', 'target_type': 'layer1'},
    {'source': 2, 'target': 3, 'source_type': 'layer1', 'target_type': 'layer1'},
])

# Build supra-adjacency matrix
supra_matrix = network.get_supra_adjacency_matrix()

# Compute communicability centrality
comm_cent = smfc.communicability_centrality(supra_matrix, beta=0.5)

print(f"Node 1 communicability: {comm_cent[(1, 'layer1')]:.4f}")
```

**Expected output:**

```
Node 1 communicability: 2.3562
```

**Related measures:**

- centrality-measures-algorithms (standard measures)

### MultiXRank

**Path:** `multilayer_algorithms.multixrank`

**Functions:**

- `multixrank_from_py3plex_networks(networks, config)` - MultiXRank algorithm

**Best for:** Ranking in multiplex heterogeneous networks with bipartite layers

**Algorithm details:** Extends PageRank to multiplex/heterogeneous networks

**Runnable Example:**

```python
from py3plex.core import multinet
from py3plex.algorithms.multilayer_algorithms import multixrank

# Create a simple bipartite-like structure
network = multinet.multi_layer_network(directed=True)

# Add users and items layers
users = ['u1', 'u2', 'u3']
items = ['i1', 'i2']

for u in users:
    network.add_nodes([{'source': u, 'type': 'users'}])
for i in items:
    network.add_nodes([{'source': i, 'type': 'items'}])

# Add interactions (users -> items)
network.add_edges([
    {'source': 'u1', 'target': 'i1', 'source_type': 'users', 'target_type': 'items'},
    {'source': 'u2', 'target': 'i1', 'source_type': 'users', 'target_type': 'items'},
    {'source': 'u2', 'target': 'i2', 'source_type': 'users', 'target_type': 'items'},
])

# Configure and run MultiXRank
config = {
    'alpha': 0.85,
    'max_iter': 100,
    'tol': 1e-6
}
scores = multixrank.multixrank_from_py3plex_networks([network], config)

print(f"User u1 score: {scores.get(('u1', 'users'), 0):.4f}")
```

**Expected output:**

```
User u1 score: 0.0833
```

**Related algorithms:**

- PageRank in centrality-measures-algorithms

### Network Statistics

**Module:** `py3plex.algorithms.statistics`

These algorithms compute various statistical properties of multilayer networks.

### Multilayer Statistics

**Path:** `statistics.multilayer_statistics`

**Layer-level measures:**

- `layer_density(network, layer)` - Density of specific layer
- `edge_overlap(network, layer_i, layer_j)` - Edge overlap between layers

- `inter_layer_coupling_strength(network, layer_i, layer_j)` - Coupling strength
- `inter_layer_degree_correlation(network, layer_i, layer_j)` - Degree correlation
- `inter_layer_assortativity(network, layer_i, layer_j)` - Assortativity between layers
- `layer_similarity(network, layer_i, layer_j, method)` - Layer similarity

**Node-level measures:**

- `node_activity(network, node)` - Activity across layers
- `degree_vector(network, node, weighted)` - Degree vector across layers
- `multilayer_clustering_coefficient(network, node)` - Multilayer clustering

**Network-level measures:**

- `interdependence(network, sample_size)` - Network interdependence
- `supra_laplacian_spectrum(network, k)` - Spectral properties
- `algebraic_connectivity(network)` - Second smallest Laplacian eigenvalue

**Best for:** Understanding multilayer network structure and properties

**Complexity:** Varies by measure, typically O(n) to O($n^2$)

**Related algorithms:**

- basic-statistics (simpler measures)
- topology-measures (topological properties)

### Multilayer Statistics (Detailed)

See the full section above for complete details on multilayer statistics functions.

### Basic Statistics

**Path:** `statistics.basic_statistics`

**Functions:**

- `identify_n_hubs(network, n, metric)` - Find top-n hub nodes
- `core_network_statistics(network)` - Comprehensive network statistics

**Best for:** Quick network overview, hub identification

**Complexity:** O(n + m) for most measures

**Related algorithms:**

- multilayer-statistics (advanced measures)

### Topology Analysis

**Path:** `statistics.topology`

Functions for analyzing topological properties of networks.

**Purpose:** Study structural properties like degree distributions, connected components, etc.

**Related algorithms:**

- basic-statistics

### Correlation Networks

**Path:** `statistics.correlation_networks`

**Functions:**

- `default_correlation_to_network(correlation_matrix, threshold)` - Convert correlation matrix to network
- `pick_threshold(matrix)` - Automatically select threshold

**Best for:** Creating networks from correlation/similarity matrices

**Related algorithms:**

- network-statistics-algorithms (analysis after construction)

### Enrichment Analysis

**Path:** `statistics.enrichment_modules`

**Functions:**

- `compute_enrichment(term_dataset, annotation_database, ...)` - General enrichment
- `fet_enrichment_generic(...)` - Fisher's exact test enrichment
- `fet_enrichment_terms(...)` - Term enrichment
- `fet_enrichment_uniprot(...)` - UniProt annotation enrichment
- `calculate_pval(term, alternative)` - p-value calculation
- `multiple_test_correction(input_dataset)` - FDR correction

**Best for:** Finding over-represented terms/annotations in network communities

**Related algorithms:**

- community-detection-algorithms (provides communities for enrichment)

### Statistical Comparison

**Path:** `statistics.stats_comparison`

**Functions:** Various statistical tests for comparing networks or algorithms

**Related modules:**

- `statistics.bayesiantests` - Bayesian statistical tests
- `statistics.bayesian_distances` - Bayesian distance measures
- `statistics.critical_distances` - Critical difference diagrams

**Best for:** Comparing performance of different algorithms or network properties

### Power Law Analysis

**Path:** `statistics.powerlaw`

Functions for testing and fitting power law distributions in networks.

**Best for:** Analyzing degree distributions, checking for scale-free properties

**Related algorithms:**

- basic-statistics (degree distributions)

### Node Ranking & Classification

**Module:** `py3plex.algorithms.node_ranking` and `py3plex.algorithms.network_classification`

### Node Ranking

**Path:** `node_ranking.node_ranking`

**Functions:**

- `sparse_page_rank(graph, alpha, personalization, max_iter, tol)` - Sparse PageRank
- `hubs_and_authorities(graph)` - HITS algorithm
- `hub_matrix(graph)` - Hub matrix computation
- `authority_matrix(graph)` - Authority matrix computation
- `stochastic_normalization(matrix)` - Normalize for random walks
- `stochastic_normalization_hin(matrix)` - Normalize for heterogeneous networks

**Best for:** Ranking nodes by importance in directed networks

**Complexity:** O(m) per iteration for PageRank, O(nm) for HITS

**Related algorithms:**

- PageRank in centrality-measures-algorithms
- label-propagation (classification)

### Label Propagation

**Path:** `network_classification.label_propagation`

**Functions:**

- `label_propagation(adjacency, labels, alpha, max_iter, tol, normalization)` - Main algorithm
- `validate_label_propagation(...)` - Validation function
- `label_propagation_normalization(matrix)` - Normalization
- `normalize_initial_matrix_freq(mat)` - Frequency normalization
- `normalize_amplify_freq(mat)` - Amplified normalization
- `normalize_exp(mat)` - Exponential normalization

**Best for:** Semi-supervised node classification

**Complexity:** O(m × i) where i is number of iterations

**Related algorithms:**

- ppr-algorithm (personalized version)
- Label propagation in community-detection-algorithms

### Personalized PageRank (PPR)

**Path:** `network_classification.PPR`

**Functions:**

- `construct_PPR_matrix(graph_matrix, parallel)` - Build PPR matrix
- `construct_PPR_matrix_targets(graph_matrix, targets, parallel)` - PPR for specific targets
- `validate_ppr(...)` - Validation function

**Best for:** Local network exploration, personalized node ranking

**Complexity:** $O(n^2)$ for dense computation, can be approximated

**Related algorithms:**

- label-propagation (similar propagation mechanism)
- PageRank in node-ranking-algorithms

### Random Walks & Embeddings

**Module:** `py3plex.algorithms.general` and `py3plex.wrappers`

### Random Walk Primitives

**Path:** `general.walkers`

**Functions:**

- `basic_random_walk(graph, start_node, walk_length)` - Simple random walk
- `node2vec_walk(graph, start_node, walk_length, p, q)` - Biased random walk
- `generate_walks(graph, num_walks, walk_length, strategy, ...)` - Generate multiple walks
- `general_random_walk(G, start_node, iterations, teleportation_prob)` - Random walk with restart
- `layer_specific_random_walk(network, start_node, start_layer, walk_length, ...)` - Multilayer walk

**Best for:** Foundation for embedding algorithms, network exploration

**Complexity:** O(L) per walk where L is walk length

**Related algorithms:**

- node2vec-embedding (uses these walks)
- deepwalk-embedding (uses these walks)

### Node2Vec Embedding

**Path:** `wrappers.node2vec_embedding`

**Functions:**

- `generate_embeddings(graph, dimensions, walk_length, num_walks, p, q, ...)` - Generate embeddings
- `train_node2vec(graph, ...)` - Training wrapper

**Best for:** Feature learning, link prediction, node classification

**Complexity:** O(r × l × V) where r is walks per node, l is walk length, V is number of vertices

**Parameters:**

- `p`: Return parameter (controls backtracking)
- `q`: In-out parameter (controls exploration vs exploitation)

**Related algorithms:**

- deepwalk-embedding (special case with p=1, q=1)
- random-walk-primitives (underlying walks)

### DeepWalk Embedding

**Implementation:** Node2Vec with p=1, q=1

**Best for:** Simpler alternative to Node2Vec with uniform random walks

**Related algorithms:**

- node2vec-embedding (generalization)

### Entanglement Analysis

**Path:** `multilayer_algorithms.entanglement`

**Functions:**

- `compute_entanglement_analysis(network)` - Full entanglement analysis
- `build_occurrence_matrix(network)` - Build node-layer occurrence matrix
- `compute_blocks(c_matrix)` - Compute block structure
- `compute_entanglement(block_matrix)` - Calculate entanglement metrics

**Best for:** Analyzing node-layer participation patterns

**Purpose:** Measure how nodes are entangled across layers

**Related algorithms:**

- versatility-centrality (similar cross-layer analysis)

### Visualization Algorithms

**Module:** `py3plex.visualization`

### Layout Algorithms

**Path:** `visualization.layout_algorithms`

**Available layouts:**

- Force-directed layouts (Spring, Fruchterman-Reingold)
- Circular layout
- Spectral layout
- Hierarchical layout
- Diagonal projection (multilayer-specific)

**Best for:** Positioning nodes for visualization

**Related algorithms:**

- multilayer-visualization (uses these layouts)

### Multilayer Visualization

**Path:** `visualization.multilayer`

**Main functions:**

- `draw_multilayer_default(networks, ...)` - Default multilayer drawing
- `hairball_plot(network, layout_algorithm)` - Single-layer projection
- Diagonal projection plotting
- Layer-separated visualization

**Best for:** Visualizing multilayer network structure

**Scalability:**

- Diagonal projection: 10,000+ nodes
- Force-directed: <5,000 nodes
- Matrix view: 100,000+ nodes

(Rules of thumb; pick based on hardware and desired visual fidelity)

**Related algorithms:**

- layout-algorithms (positioning)

### Drawing Machinery

**Path:** `visualization.drawing_machinery`

Low-level drawing functions and utilities.

**Purpose:** Support functions for rendering networks

### Color Schemes

**Path:** `visualization.colors`

Functions for generating color schemes for communities, layers, node types.

**Related algorithms:**

- community-detection-algorithms (uses colors)
- multilayer-visualization (uses colors)

### Embedding Visualization

**Path:** `visualization.embedding_visualization`

Functions for visualizing node embeddings in 2D/3D space.

**Best for:** Visualizing results of embedding algorithms

**Related algorithms:**

- node2vec-embedding

- deepwalk-embedding

### Benchmark & Utilities

#### Network Classification Benchmark

**Path:** `general.benchmark_classification`

**Functions:**

- `evaluate_oracle_F1(probs, Y_real)` - Evaluate classification performance

**Best for:** Evaluating node classification algorithms

**Related algorithms:**

- label-propagation
- node2vec-embedding

#### Benchmark Visualizations

**Path:** `visualization.benchmark_visualizations`

Functions for visualizing algorithm comparison results.

**Related algorithms:**

- All algorithms (for benchmarking)

### Algorithm Selection Guide

This section helps you choose the right algorithm category based on your task.

#### By Task Type

**Community Detection:**
→ See community-detection-algorithms

**Node Importance:**
→ See centrality-measures-algorithms

**Node Classification:**
→ See label-propagation or node2vec-embedding

**Network Comparison:**
→ See network-statistics-algorithms

**Feature Learning:**
→ See node2vec-embedding

**Visualization:**
→ See visualization-algorithms

#### By Network Size

**Small (<1,000 nodes):**

- All algorithms work well
- Can use expensive path-based measures
- All visualization methods

**Medium (1,000-10,000 nodes):**

- Use Louvain over Infomap
- Avoid full betweenness centrality
- Use diagonal projection for visualization

**Large (10,000-100,000 nodes):**

- Degree-based measures only
- Louvain or label propagation
- Matrix visualizations

**Very Large (>100,000 nodes):**

- Degree, PageRank only
- Label propagation for communities
- Sparse matrix operations essential

### By Network Type

**Single-layer networks:**

- Standard algorithms: Louvain, PageRank, betweenness
- See community-detection-algorithms and centrality-measures-algorithms

**Multiplex/multilayer networks:**

- Multilayer-specific: multilayer-louvain, versatility-centrality
- Layer statistics: multilayer-statistics
- Cross-layer measures

**Directed networks:**

- PageRank, HITS algorithm
- See node-ranking-algorithms

**Weighted networks:**

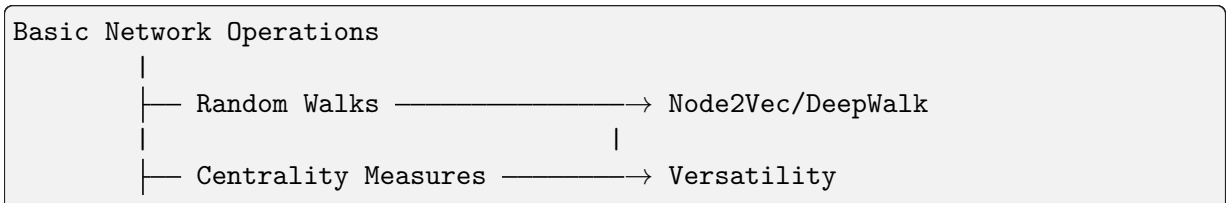
- Most algorithms accept weights; check individual signatures
- Specify the `weight` parameter when supported

**Temporal networks:**

- Layer-based representation
- Time-respecting random walks: random-walk-primitives

### Algorithm Dependencies

This diagram shows how algorithms depend on each other:



```
        |                                  |
        ├── Community Detection ──────────→ Community Measures
        |   |                              |
        |   ├──→ Modularity                └──→ Enrichment Analysis
        |   └──→ Community Ranking
        |
        ├── Statistics ──────────────────→ Benchmarks
        |                                  └──→ Comparisons
        |
        └── Embeddings ──────────────────→ Visualization
                                           └──→ Classification
```

### Common Algorithm Workflows

#### Community Detection Workflow

```python
# 1. Detect communities
from py3plex.algorithms.community_detection import louvain_multilayer
communities = louvain_multilayer(network.get_supra_adjacency_matrix())

# 2. Evaluate quality
from py3plex.algorithms.community_detection import multilayer_modularity
quality = multilayer_modularity(network.get_supra_adjacency_matrix(), communities)

# 3. Analyze communities
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.algorithms.statistics.enrichment_modules import compute_enrichment

# 4. Visualize
from py3plex.visualization.multilayer import draw_multilayer_default
draw_multilayer_default([network], communities=communities)
```

#### Centrality Analysis Workflow

```python
# 1. Compute centralities
from py3plex.algorithms.multilayer_algorithms.centrality import MultilayerCentrality
calc = MultilayerCentrality(network)

# 2. Multiple measures
degree = calc.overlapping_degree_centrality()
pagerank = calc.pagerank_centrality()
betweenness = calc.multilayer_betweenness_centrality()

# 3. Cross-layer analysis
from py3plex.algorithms.statistics import multilayer_statistics as mls
versatility = mls.versatility_centrality(network, centrality_type='degree')

# 4. Identify hubs
from py3plex.algorithms.statistics.basic_statistics import identify_n_hubs
hubs = identify_n_hubs(network, n=10, metric='degree')
```

### Embedding Workflow

```
# 1. Generate walks
from py3plex.algorithms.general.walkers import generate_walks
walks = generate_walks(network.core_network, num_walks=10,
                       walk_length=80, strategy='node2vec')

# 2. Learn embeddings
from py3plex.wrappers.node2vec_embedding import generate_embeddings
embeddings = generate_embeddings(network.core_network,
                                 dimensions=128, p=1.0, q=1.0)

# 3. Use for classification
from py3plex.algorithms.network_classification.label_propagation import label_propagation
# Or use embeddings directly with sklearn
```

### Statistical Analysis Workflow

```
# 1. Layer-level statistics
from py3plex.algorithms.statistics import multilayer_statistics as mls
density = mls.layer_density(network, 'layer1')
overlap = mls.edge_overlap(network, 'layer1', 'layer2')
correlation = mls.inter_layer_degree_correlation(network, 'layer1', 'layer2')

# 2. Node-level statistics
for node in important_nodes:
    activity = mls.node_activity(network, node)
    degree_vec = mls.degree_vector(network, node)

# 3. Network-level statistics
interdep = mls.interdependence(network)
connectivity = mls.algebraic_connectivity(network)
```

### Further Reading

For detailed usage examples and parameter descriptions:

- *Algorithm Landscape* - Algorithm selection guide with complexity analysis
- ../user_guide/community_detection - Community detection examples
- ../user_guide/statistics - Centrality computation examples
- *Performance and Scalability Best Practices* - Performance optimization tips
- *API Documentation* - Complete API reference

#### Citation Information

When using specific algorithm families, please cite the original papers:

**Louvain & Multilayer Modularity:**
Blondel et al. (2008), Mucha et al. (2010)

**Node2Vec:**
Grover & Leskovec (2016)

**Infomap:**
Rosvall & Bergstrom (2008)

See citation_and_acknowledgements for complete citation information and BibTeX entries.

### 3.5.3 DSL Reference

Complete reference for the py3plex DSL (Domain-Specific Language) for querying multilayer networks.

**Note**

For task-oriented usage, see *How to Query Multilayer Graphs with the SQL-like DSL*. This page is a complete reference with all syntax and operators.

#### Overview

The py3plex DSL provides two interfaces:

1. **String syntax:** SQL-like queries for quick exploration
2. **Builder API:** Type-safe Python interface for production code (autocomputes referenced metrics)

#### String Syntax Reference

#### Basic Structure

```
SELECT <target> [FROM <layers>] [WHERE <conditions>] [COMPUTE <metrics>] [ORDER BY <field>]
[AT <timestamp> | DURING <start> TO <end>]
```

Use `AT` for a single time point and `DURING` for closed intervals (ISO 8601 strings).

#### Targets

- `nodes` — Select nodes
- `edges` — Select edges (experimental; limited coverage support)

#### Layer Selection

```
FROM layer="layer_name"
FROM layers IN ("layer1", "layer2")
```

If `FROM` is omitted, all layers are considered.

### Conditions

**Operators:**

- `=` — Equal
- `>` — Greater than
- `<` — Less than
- `>=` — Greater than or equal
- `<=` — Less than or equal
- `!=` — Not equal

**Logical operators:**

- `AND` — Both conditions must be true
- `OR` — Either condition must be true
- `NOT` — Negate a condition

**Examples:**

```
WHERE degree > 5
WHERE layer="friends" AND degree > 3
WHERE degree > 5 OR betweenness_centrality > 0.1
WHERE NOT layer="spam"
```

String values must be wrapped in double quotes.

### Compute Clause

Calculate metrics for selected nodes:

```
COMPUTE degree
COMPUTE degree betweenness_centrality
COMPUTE clustering pagerank
```

**Available metrics:**

- `degree` — Node degree
- `betweenness_centrality` — Betweenness centrality
- `closeness_centrality` — Closeness centrality
- `clustering` — Clustering coefficient
- `pagerank` — PageRank score
- `layer_count` — Number of layers node appears in

Metrics are computed after filtering and layer selection. See *Algorithm Roadmap* for the complete metric list.

### Order By

```
ORDER BY degree
ORDER BY -degree  # Descending (prefix with -)
ORDER BY betweenness_centrality
```

You can specify multiple keys in sequence; each key may be prefixed with - for descending order.

### Limit

```
LIMIT 10
LIMIT 100
```

### Complete Examples

```
from py3plex.dsl import execute_query

# Get high-degree nodes
result = execute_query(
    network,
    'SELECT nodes WHERE degree > 5'
)

# Get nodes from specific layer
result = execute_query(
    network,
    'SELECT nodes FROM layer="friends" '
    'WHERE degree > 3 '
    'COMPUTE betweenness_centrality '
    'ORDER BY -betweenness_centrality '
    'LIMIT 10'
)
```

### Builder API Reference

#### Import

```
from py3plex.dsl import Q, L
```

### Query Construction

**Start a query:**

```
Q.nodes()     # Select nodes
Q.edges()     # Select edges (experimental)
```

### Layer Selection

**Single layer:**

```
Q.nodes().from_layers(L["friends"])
```

**Multiple layers (union):**

```
Q.nodes().from_layers(L["friends"] + L["work"])

# Or use the new LayerSet algebra:
Q.nodes().from_layers(L["friends | work"])
```

**Layer intersection:**

```
Q.nodes().from_layers(L["friends"] & L["work"])
```

**Advanced Layer Set Algebra:**

```
# All layers except coupling
Q.nodes().from_layers(L["* - coupling"])

# Complex expressions with set operations
Q.nodes().from_layers(L["(social | work) & ~bots"])

# Named groups for reuse
from py3plex.dsl import LayerSet
LayerSet.define_group("bio", LayerSet("ppi") | LayerSet("gene"))
Q.nodes().from_layers(LayerSet("bio"))
```

**See also**

For complete documentation on layer set algebra including all operators, string parsing, named groups, and real-world examples, see: *Layer Set Algebra*

### Filtering

**Comparison operators:**

```
Q.nodes().where(degree__gt=5)       # Greater than
Q.nodes().where(degree__gte=5)      # Greater than or equal
Q.nodes().where(degree__lt=5)       # Less than
Q.nodes().where(degree__lte=5)      # Less than or equal
Q.nodes().where(degree__eq=5)       # Equal
Q.nodes().where(degree__ne=5)       # Not equal
```

**Multiple conditions:**

```
Q.nodes().where(
    degree__gt=5,
    layer_count__gte=2
)
```

### Computing Metrics

```
Q.nodes().compute("degree")
Q.nodes().compute("degree", "betweenness_centrality")
```

### Row-wise Transformations (Mutate)

Create new columns or transform existing ones with row-by-row operations:

```
# Simple transformation
Q.nodes().compute("degree").mutate(
    doubled=lambda row: row.get("degree", 0) * 2
)

# Multiple transformations
Q.nodes().compute("degree", "clustering").mutate(
    hub_score=lambda row: row.get("degree", 0) * row.get("clustering", 0),
    is_hub=lambda row: row.get("degree", 0) > 2
)

# Conditional transformation
Q.nodes().compute("degree").mutate(
    category=lambda row: "hub" if row.get("degree", 0) > 3 else "peripheral"
)
```

The lambda function receives a dictionary with all computed attributes and network properties for each node/edge. Use `row.get(attr_name, default)` to safely access attributes.

**Note**

Use `mutate()` for row-by-row transformations. For group-level aggregations, use `summarize()` or `aggregate()` instead.

### Sorting

```
Q.nodes().order_by("degree")          # Ascending
Q.nodes().order_by("-degree")         # Descending
```

### Limiting

```
Q.nodes().limit(10)
```

### Execution

```
result = Q.nodes().execute(network)
```

### Chaining

All methods can be chained:

```
result = (
    Q.nodes()
     .from_layers(L["friends"])
     .where(degree__gt=5)
     .compute("betweenness_centrality", "degree")
     .mutate(
         influence=lambda row: row.get("degree", 0) * row.get("betweenness_centrality", 0)
     )
     .order_by("-influence")
     .limit(10)
     .execute(network)
)
```

### Temporal Queries

#### Filter by Time Point

```
# String syntax
result = execute_query(
    network,
    'SELECT nodes AT "2024-01-15T10:00:00"'
)

# Builder API
result = (
    Q.nodes()
     .at("2024-01-15T10:00:00")
     .execute(network)
)
```

#### Filter by Time Range

```
# String syntax
result = execute_query(
    network,
    'SELECT nodes DURING "2024-01-01" TO "2024-01-31"'
)

# Builder API
result = (
    Q.nodes()
     .during("2024-01-01", "2024-01-31")
     .execute(network)
)
```

### Temporal Edge Attributes

Edges can have temporal attributes:

- `t` — Point in time (ISO 8601 timestamp)
- `t_start` and `t_end` — Time range

See ../user_guide/networks for creating temporal networks.

### Grouping and Coverage Queries

#### Per-Layer Grouping

Group results by layer and apply per-group operations:

```
# Group by layer
result = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree")
     .per_layer()                # Sugar for .group_by("layer")
        .top_k(5, "degree")      # Top 5 per layer
     .end_grouping()
     .execute(network)
)
```

#### Top-K Per Group

Select top-k items per group (requires prior grouping):

```
# Top 10 highest-degree nodes per layer
result = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree", "betweenness_centrality")
     .per_layer()
        .top_k(10, "degree")
     .end_grouping()
     .execute(network)
)
```

#### Coverage Filtering

Filter based on presence across groups:

**Mode: "all"** — Keep items appearing in ALL groups (intersection)

```
# Nodes that are top-5 hubs in ALL layers
multi_hubs = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("betweenness_centrality")
     .per_layer()
```

```
        .top_k(5, "betweenness_centrality")
    .end_grouping()
    .coverage(mode="all")
    .execute(network)
)
```

**Mode: "any"** — Keep items appearing in AT LEAST ONE group (union)

```
# Nodes that are top-5 in any layer
any_hubs = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree")
     .per_layer()
        .top_k(5, "degree")
     .end_grouping()
     .coverage(mode="any")
     .execute(network)
)
```

**Mode: "at_least"** — Keep items appearing in at least K groups

```
# Nodes in top-10 of at least 2 layers
two_layer_hubs = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("degree")
     .per_layer()
        .top_k(10, "degree")
     .end_grouping()
     .coverage(mode="at_least", k=2)
     .execute(network)
)
```

**Mode: "exact"** — Keep items appearing in exactly K groups

```
# Layer specialists: top-5 in exactly 1 layer
specialists = (
    Q.nodes()
     .from_layers(L["*"])
     .compute("betweenness_centrality")
     .per_layer()
        .top_k(5, "betweenness_centrality")
     .end_grouping()
     .coverage(mode="exact", k=1)
     .execute(network)
)
```

### Wildcard Layer Selection

Use `L["*"]` to select all layers:

```
# All layers
Q.nodes().from_layers(L["*"])

# All layers except "bots"
Q.nodes().from_layers(L["*"] - L["bots"])

# Layer algebra still works
Q.nodes().from_layers((L["*"] - L["spam"]) & L["verified"])
```

### General Grouping

Group by arbitrary attributes (not just layer):

```
# Group by multiple attributes
result = (
    Q.nodes()
     .compute("degree", "community")
     .group_by("layer", "community")
        .top_k(3, "degree")
     .end_grouping()
     .execute(network)
)
```

### Limitations

- Coverage filtering is currently supported only for **node queries**
- Edge queries with coverage will raise a clear `DslExecutionError`
- Grouping requires computed attributes or inherent node properties (like layer)

### Explaining Results

The `.explain()` method adds interpretable explanations to query results or displays execution plans.

### Execution Plan Mode

Call `.explain()` with no arguments to get the query execution plan:

```
plan = Q.nodes().compute("degree").where(degree__gt=5).explain()
print(plan)  # Shows query stages and optimization
```

This mode does NOT execute the query or attach explanations.

### Explanations Mode

Call `.explain()` with arguments to attach explanations to each result row:

```
result = (
    Q.nodes()
     .compute("pagerank")
     .explain(
         include=["community", "top_neighbors", "attribution"],
         attribution={"metric": "pagerank", "seed": 42}
     )
     .execute(network)
)

# Access explanations
df = result.to_pandas(expand_explanations=True)
# df now has explanation columns with JSON-serialized dicts
```

### Available Explanation Blocks

- `"community"` — Community membership and size
- `"top_neighbors"` — Top neighbors by weight/degree
- `"layer_footprint"` — Layers where node/edge appears
- `"attribution"` — Shapley-based attribution explanations (see below)

Default blocks: `["community", "top_neighbors", "layer_footprint"]`

### Attribution Block

The `"attribution"` block provides Shapley value-based explanations for why nodes/edges have high metric values or rankings.

**Purpose**: Decompose metric contributions across layers and/or edges using game-theoretic attribution.

**Basic Example**:

```
result = (
    Q.nodes()
     .compute("pagerank")
     .explain(
         include=["attribution"],
         attribution={
             "metric": "pagerank",
             "levels": ["layer"],
             "method": "shapley_mc",
             "seed": 42
         }
     )
     .execute(network)
)
```

**Configuration Parameters**:

| Parameter | Type | Default | Description |
|---|---|---|---|
| `metric` | str \| None | Auto-infer | Which computed metric to explain (required if multiple metrics) |
| `objective` | str | "value" | `"value"` explains metric score, `"rank"` explains ranking position |
| `levels` | List[str] | ["layer"] | Attribution levels: `["layer"]`, `["edge"]`, or both |
| `method` | str | "shapley_mc" | `"shapley"` (exact), `"shapley_mc"` (Monte Carlo), `"influence"` (approx) |
| `feature_space` | str | "layers" | `"layers"`, `"layer_pairs"`, or `"coupling_types"` |
| `n_permutations` | int | 128 | Monte Carlo sample count (16) |
| `max_exact_featu` | int | 8 | Switch from exact to MC Shapley at this threshold |
| `seed` | int \| None | None | Random seed for determinism (strongly recommended) |
| `edge_scope` | str | "incident" | `"incident"`, `"ego_k_hop"`, `"shortest_path_sample"`, `"global_top_m"` |
| `k_hop` | int | 2 | Ego network radius (for `edge_scope="ego_k_hop"`) |
| `max_edges` | int | 40 | Maximum candidate edges to consider |
| `top_k_layers` | int | 10 | Number of top layer contributions to return |
| `top_k_edges` | int | 20 | Number of top edge contributions to return |
| `include_negativ` | bool | True | Include negative contributions |
| `cache` | bool | True | Cache subset computations for performance |
| `uq` | str | "off" | `"off"`, `"propagate"` (compute per UQ replicate), `"summarize_only"` |
| `ci_level` | float | 0.95 | Confidence interval level for UQ propagation |

**Output Structure**:

```
{
    "metric": "pagerank",
    "objective": "value",
    "utility_def": None,  # or "margin_to_cutoff(k=10)" for rank
    "levels": ["layer"],
    "method": "shapley_mc",
    "seed": 42,
    "n_permutations": 128,
    "feature_space": "layers",
    "full_value": 0.1186,
    "baseline_value": 0.0500,
    "delta": 0.0686,
    "residual": 1e-12,  # sum(phi)  delta
    "layer_contrib": [
        {"layer": "social", "phi": 0.0401},
        {"layer": "work", "phi": 0.0285}
    ],
```

```
    "edge_contrib": [],  # populated if levels includes "edge"
    "warnings": [],
    "cache_hit_rate": 0.73
}
```

**Advanced Examples**:

```
# Edge attribution for betweenness
result = (
    Q.nodes()
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(10)
     .explain(
         include=["attribution"],
         attribution={
             "objective": "rank",  # Explain ranking position
             "levels": ["layer", "edge"],
             "edge_scope": "ego_k_hop",
             "k_hop": 2,
             "max_edges": 40,
             "n_permutations": 128,
             "seed": 42
         }
     )
     .execute(network)
)

# With UQ propagation
result = (
    Q.nodes()
     .uq(method="perturbation", n_samples=30, seed=42)
     .compute("pagerank")
     .explain(
         include=["attribution"],
         attribution={
             "metric": "pagerank",
             "uq": "propagate",  # Compute attribution per UQ replicate
             "levels": ["layer"],
             "seed": 42
         }
     )
     .execute(network)
)
```

**Determinism**:

- Setting `seed` ensures reproducible Shapley values across runs
- Same seed produces identical attributions regardless of parallel execution settings
- Different seeds produce statistically different but valid attributions

**Performance Notes**:

- **Exact Shapley**: Only feasible for `max_exact_features` layers (default 8)
- **Monte Carlo Shapley**: Scales to larger feature sets via sampling
- **Edge Attribution**: More expensive than layer attribution; bounded by `max_edges`
- **Caching**: Enabled by default to reuse subset metric computations

**UQ Integration**:

- `uq="off"`: No UQ (default), deterministic scalar Shapley values
- `uq="propagate"`: Compute attribution per UQ replicate, aggregate mean/std/CI
- `uq="summarize_only"`: Compute once on base network, wrap in UQ-like structure

**Export**:

Attribution data serializes to JSON strings when using `expand_explanations=True`:

```
df = result.to_pandas(expand_explanations=True)
# df["attribution"] contains JSON-serialized attribution dicts
```

### Working with Results

#### Result Object

`execute` returns a `QueryResult` with items, computed attributes, and metadata:

```
result = Q.nodes().compute("degree").execute(network)

# Counts and identifiers
print(result.count)    # -> number of nodes/edges
print(result.nodes[:3])  # first few nodes (use .edges for edge queries)

# Attributes are aligned lists keyed by metric name
degrees = result.attributes["degree"]
for node, deg in zip(result.nodes, degrees):
    print(node, deg)

# Execution metadata (e.g., grouping, ordering)
print(result.meta)
```

#### Convert to Pandas

```
df = result.to_pandas(multiindex=True, include_grouping=True)
print(df.head())
```

Set `expand_uncertainty=True` to unpack UQ-aware metrics into multiple columns.

### Extensibility

#### Custom Operators

Register custom DSL operators:

```
from py3plex.dsl import register_operator

@register_operator('my_metric')
def my_custom_metric(context, node):
    """Compute custom metric for a node."""
    # Your implementation
    return value
```

See *Architecture and Design* for plugin development.

#### Performance Considerations

1. **Compute metrics once:** Don't recompute in multiple queries
2. **Filter early:** Use WHERE before COMPUTE
3. **Limit results:** Use LIMIT for large networks
4. **Layer-specific:** Query single layers when possible

#### Next Steps

- **Learn by doing:** *How to Query Multilayer Graphs with the SQL-like DSL*
- **See examples:** *Examples & Recipes*
- **Understand implementation:** *Architecture and Design*

### 3.5.4 Layer Set Algebra

The Layer Set Algebra feature provides a first-class, composable system for selecting and manipulating layer sets in multilayer networks. Instead of manually specifying lists of layers, you can use set-theoretic operations to express complex layer selections.

#### Why Layer Set Algebra?

In multilayer network analysis, you often need to:

- Select "all layers except the coupling layer"
- Combine biological layers for aggregate analysis
- Find the intersection of layers where a node appears
- Reuse layer group definitions across multiple queries

The Layer Set Algebra makes these operations expressive, composable, and safe.

#### Quick Start

#### Basic Operations

```
from py3plex.dsl import LayerSet, L, Q

# Create layer sets
social = LayerSet("social")
work = LayerSet("work")
```

```
# Union: layers in either social OR work
both = social | work

# Intersection: layers in both social AND work
shared = social & work

# Difference: layers in social but NOT in work
social_only = social - work

# Complement: all layers EXCEPT social
not_social = ~social
```

#### String Expressions

For convenience, you can write layer expressions as strings:

```
# Parse from string
layers = LayerSet.parse("* - coupling")

# Or use L[...] syntax (auto-detects expressions)
layers = L["* - coupling"]
layers = L["(ppi | gene) & disease"]
layers = L["social | work | hobby"]
```

#### Integration with Queries

Use layer sets in any DSL query:

```
# Query nodes from all layers except coupling
result = (
    Q.nodes()
     .from_layers(L["* - coupling"])
     .compute("degree")
     .execute(network)
)

# Complex layer selection
result = (
    Q.nodes()
     .from_layers(L["(social | work) & ~coupling"])
     .where(degree__gt=5)
     .execute(network)
)
```

#### Set Operations

##### Union: A | B

The union operation combines layers from multiple sets.

```
# Select nodes from social OR work layers
layers = L["social | work"]
# Equivalent to:
layers = LayerSet("social") | LayerSet("work")
```

**ASCII Venn Diagram:**

```
┌─────────┐     ┌─────────┐
│ Social  │     │  Work   │
│    ┌────┼─────┼────┐    │
│    │  RESULT  │    │    │
└────┼──────────┼────┘    │
     └──────────┘         │
   A | B = {social, work}
```

### Intersection: A & B

The intersection operation selects only layers present in both sets.

```
# Nodes that appear in both layer A AND layer B
layers = L["layerA & layerB"]
```

**ASCII Venn Diagram:**

```
┌─────────┐     ┌─────────┐
│    A    │     │    B    │
│    ┌────┼─────┼────┐    │
│    │ RES │     │    │    │
└────┼─────┼─────┼────┘    │
     └─────┘     └─────────┘
          A & B
```

### Difference: A - B

The difference operation removes layers in B from A.

```
# All layers except coupling
layers = L["* - coupling"]

# Social layers but not bots
layers = L["social - bots"]
```

**ASCII Venn Diagram:**

```
┌─────────┐     ┌─────────┐
│    A    │     │    B    │
│  ┌──────┼─────┼──┐      │
│  │ RES  │     │  │      │
└──┼──────┼─────┼──┘      │
   └──────┘     └─────────┘
   A - B = A without B
```

### Complement: ~A

The complement operation returns all layers except those in A.

```
# Everything except social layer
layers = ~LayerSet("social")

# Equivalent to:
layers = L["* - social"]
```

**ASCII Venn Diagram:**

```
┌─────────────────────────┐
│      All Layers (*)     │
│                         │
│   ┌─────┐               │
│   │  A  │    RESULT     │
│   └─────┘               │
└─────────────────────────┘
   ~A = * - A
```

### Named Layer Groups

Define reusable layer groups for cleaner queries:

```
from py3plex.dsl import LayerSet, L, Q

# Define a named group
bio_layers = LayerSet("ppi") | LayerSet("gene") | LayerSet("disease")
LayerSet.define_group("bio", bio_layers)

# Or use the convenience method
L.define("core", L["social | work | email"])

# Use the group in queries
result = Q.nodes().from_layers(LayerSet("bio")).execute(network)

# Groups can be combined with other operations
result = Q.nodes().from_layers(L["bio & core"]).execute(network)

# List all defined groups
groups = L.list_groups()
print(groups)  # {'bio': LayerSet(...), 'core': LayerSet(...)}
```

### Operator Precedence

Layer expressions follow standard set theory precedence:

1. **Complement** (~) - highest precedence
2. **Intersection** (&)
3. **Difference** (-)
4. **Union** (|) - lowest precedence

Use parentheses for clarity or to override precedence:

```
# Without parentheses: parsed as A & (B | C)
expr1 = L["A & B | C"]

# With parentheses: explicit grouping
expr2 = L["(A & B) | C"]
expr3 = L["A & (B | C)"]
```

### Real-World Examples

#### Example 1: Biological Networks

Analyze protein interactions while excluding coupling layers:

```
from py3plex.dsl import Q, L, LayerSet

# Define biological layer groups
L.define("bio", L["ppi | gene | disease"])

# Find high-degree nodes in biological layers only
result = (
    Q.nodes()
     .from_layers(LayerSet("bio") - LayerSet("coupling"))
     .where(degree__gt=10)
     .compute("betweenness_centrality")
     .order_by("betweenness_centrality", desc=True)
     .limit(20)
     .execute(network)
)

print(result.to_pandas())
```

#### Example 2: Social Networks

Compare centrality across different social contexts:

```
# Define layer groups for different social contexts
L.define("online", L["facebook | twitter | linkedin"])
L.define("offline", L["work | school | hobby"])

# Compute centrality in online layers
online_centrality = (
    Q.nodes()
     .from_layers(LayerSet("online"))
     .compute("pagerank")
     .execute(network)
).to_pandas()

# Compute centrality in offline layers
offline_centrality = (
    Q.nodes()
```

```
    .from_layers(LayerSet("offline"))
    .compute("pagerank")
    .execute(network)
).to_pandas()

# Find nodes with high centrality in both contexts
# (merge and filter the DataFrames)
```

### Example 3: Transportation Networks

Analyze robustness by excluding specific infrastructure:

```
# Simulate failure scenarios
scenarios = [
    ("full_network", L["*"]),
    ("no_air", L["* - air_travel"]),
    ("no_rail", L["* - rail"]),
    ("road_only", L["road"]),
    ("multi_modal", L["road | rail"]),
]

results = {}
for name, layers in scenarios:
    result = (
        Q.nodes()
         .from_layers(layers)
         .compute("betweenness_centrality")
         .execute(network)
    ).to_pandas()

    results[name] = {
        "avg_betweenness": result["betweenness_centrality"].mean(),
        "max_betweenness": result["betweenness_centrality"].max(),
        "node_count": len(result),
    }

import pandas as pd
df_results = pd.DataFrame(results).T
print(df_results)
```

### Introspection and Debugging

Layer sets provide introspection tools to understand what they represent:

### Resolution

See which layers a LayerSet resolves to:

```
layers = L["* - coupling"]

# Resolve against a network
```

```
resolved = layers.resolve(network)
print(resolved)  # {'social', 'work', 'hobby'}
```

### Explanation

Get a human-readable explanation of the layer expression:

```
layers = L["(ppi | gene) & disease"]
print(layers.explain())

# Output:
# LayerSet:
#   intersection(
#     union(
#       layer("ppi"),
#       layer("gene")
#     ),
#     layer("disease")
#   )
```

### With Network Resolution

Combine explanation with network resolution:

```
layers = L["* - coupling"]
print(layers.explain(network))

# Output:
# LayerSet:
#   difference(
#     all_layers("*"),
#     layer("coupling")
#   )
# → resolved to: {'hobby', 'social', 'work'}
```

### Comparison: Old vs New Syntax

**Old Style (Still Supported):**

```
# Manual layer list
result = Q.nodes().from_layers(L["social"] + L["work"]).execute(network)

# Multiple separate queries
layers = ["social", "work", "hobby"]
results = []
for layer in layers:
    if layer != "coupling":
        result = Q.nodes().from_layers(L[layer]).execute(network)
        results.append(result)
```

**New Style (Recommended):**

```
# Expressive layer algebra
result = Q.nodes().from_layers(L["social | work"]).execute(network)

# Single query with filtering
result = (
    Q.nodes()
     .from_layers(L["* - coupling"])
     .execute(network)
)
```

### Why Layer Set Algebra Matters

#### For Multilayer Networks

Multilayer networks have a unique challenge: layer selection is a first-class operation, not an afterthought. Traditional graph analysis tools don't provide built-in support for layer algebra, forcing users to write imperative code with loops and conditionals.

Layer Set Algebra brings the following benefits:

1. **Expressiveness**: Complex layer selections become one-liners
2. **Composability**: Combine layer expressions with set operations
3. **Reusability**: Define named groups once, use everywhere
4. **Safety**: Late evaluation with validation at execution time
5. **Maintainability**: Clear, declarative code is easier to understand

### Algebraic Properties

Layer Set Algebra follows standard set theory laws:

- **Idempotence**: A | A = A, A & A = A
- **Commutativity**: A | B = B | A, A & B = B & A
- **Associativity**: (A | B) | C = A | (B | C)
- **Distributivity**: A & (B | C) = (A & B) | (A & C)
- **De Morgan's Laws**: ~(A | B) = ~A & ~B, ~(A & B) = ~A | ~B
- **Complement**: A | ~A = *, A & ~A =

These properties enable query optimization and reasoning about layer selections.

### API Reference

#### LayerSet Class

```
class LayerSet:
    """Represents an unevaluated layer expression."""

    def __init__(self, name_or_expr: Union[str, LayerExpr])
        """Create from layer name or expression AST."""

    def __or__(self, other: LayerSet) -> LayerSet:
```

```
        """Union: self | other."""

    def __and__(self, other: LayerSet) -> LayerSet:
        """Intersection: self & other."""

    def __sub__(self, other: LayerSet) -> LayerSet:
        """Difference: self - other."""

    def __invert__(self) -> LayerSet:
        """Complement: ~self."""

    def resolve(self, network, *, strict=False, warn_empty=True) -> Set[str]:
        """Resolve expression to actual layer names."""

    def explain(self, network=None) -> str:
        """Generate human-readable explanation."""

    @staticmethod
    def parse(expr_str: str) -> LayerSet:
        """Parse layer expression from string."""

    @staticmethod
    def define_group(name: str, layer_set: LayerSet) -> None:
        """Define a named layer group."""

    @staticmethod
    def list_groups() -> Dict[str, LayerSet]:
        """List all defined groups."""

    @staticmethod
    def clear_groups() -> None:
        """Clear all defined groups."""
```

### L Proxy

The `L` proxy provides convenient syntax for creating layer expressions:

```
# Simple layer name (backward compatible)
L["social"]  # Returns LayerExprBuilder

# Expression string (auto-detected)
L["* - coupling"]  # Returns LayerSet
L["social | work"]  # Returns LayerSet

# Define named group
L.define("bio", LayerSet("ppi") | LayerSet("gene"))

# List groups
groups = L.list_groups()
```

```
# Clear groups
L.clear_groups()
```

### Troubleshooting

#### Common Issues

**1. Empty Layer Set Warning**

```
layers = L["nonexistent"]
result = layers.resolve(network)
# UserWarning: Layer expression resolved to empty set
```

**Solution**: Check layer names, use `warn_empty=False` to suppress, or use `strict=True` to raise an error instead.

**2. Unknown Layer Error**

```
layers = L["typo_layer"]
result = layers.resolve(network, strict=True)
# UnknownLayerError: Layer 'typo_layer' not found
```

**Solution**: Verify layer names with `network.layers` or `network.get_layers()`.

**3. Syntax Error in Expression**

```
layers = L["social &"]  # Missing right operand
# DslSyntaxError: Unexpected end of expression
```

**Solution**: Check expression syntax, ensure all operators have operands.

### Best Practices

1. **Use Named Groups** for complex layer sets you'll reuse
2. **Prefer String Expressions** for readability: `L["* - coupling"]`
3. **Use Complement** instead of listing all other layers: `~LayerSet("exclude")`
4. **Document Layer Groups** with comments explaining their purpose
5. **Test Layer Resolution** with `.explain(network)` before running expensive queries

### See Also

- *How to Query Multilayer Graphs with the SQL-like DSL* - DSL query basics
- *Query Zoo: DSL Gallery for Multilayer Analysis* - Query examples and patterns
- *DSL Reference* - Complete DSL API reference

**Note**

Layer Set Algebra is available in py3plex DSL v2.1+. It's fully backward compatible with existing layer expressions.

### 3.5.5 Uncertainty-First Statistics

#### Overview

py3plex implements an **uncertainty-first statistics system** where every statistic is represented as `(value + uncertainty + provenance)`. This design makes uncertainty a core part of statistics computation, not an afterthought.

#### Key Principles

1. **Every statistic is a StatValue**: No plain floats/ints escaping internal computation paths. Even deterministic values carry `Delta(0)` uncertainty.
2. **Uncertainty is first-class**: Uncertainty can be summarized, sampled, propagated through arithmetic, and used in queries.
3. **Backward compatible**: Existing code expecting floats works via `float(statvalue)`.
4. **Registry discipline**: Statistics must have an uncertainty model to be registered.
5. **Reproducible**: Random processes (bootstrap/MC) support explicit seeds tracked in provenance.

#### Core Components

#### StatValue

`StatValue` is the fundamental container for statistics:

```
from py3plex.stats import StatValue, Delta, Provenance

# Create a deterministic statistic
sv = StatValue(
    value=0.42,
    uncertainty=Delta(0.0),
    provenance=Provenance("degree", "delta", {})
)

# Access value
print(float(sv))  # 0.42

# Query uncertainty
print(sv.std())  # 0.0
print(sv.ci(0.95))  # (0.42, 0.42)
print(sv.robustness())  # 1.0
```

**Key Methods:**

- `float(sv)`: Convert to point estimate (backward compatibility)
- `sv.mean()`: Alias for value
- `sv.std()`: Standard deviation
- `sv.ci(level=0.95)`: Confidence interval
- `sv.robustness()`: Robustness score in [0, 1]
- `sv.to_json_dict()`: Serialize to JSON

### Uncertainty Models

Five concrete uncertainty models are provided:

### Delta

Deterministic or known-precision uncertainty.

```
from py3plex.stats import Delta

# Perfect certainty
d = Delta(0.0)

# Small known error
d = Delta(0.01)
```

**Properties:**

- `std()`: Returns sigma
- `ci(level)`: Returns symmetric interval based on sigma
- `sample(n, seed)`: Returns constant samples
- Propagation: Analytic error propagation when both are Delta

### Gaussian

Normal distribution uncertainty.

```
from py3plex.stats import Gaussian

g = Gaussian(mean=0.0, std_dev=0.1)

# Exact CI computation
low, high = g.ci(0.95)  #  (-0.196, 0.196)
```

**Properties:**

- `std()`: Returns std_dev
- `ci(level)`: Exact Gaussian CI using z-scores
- `sample(n, seed)`: Generates Gaussian samples
- Propagation: Analytic for addition/subtraction, Monte Carlo for complex ops

### Bootstrap

Empirical uncertainty from bootstrap resampling.

```
from py3plex.stats import Bootstrap
import numpy as np

# Store bootstrap samples (relative to point estimate)
samples = np.array([0.1, -0.05, 0.15, 0.0, 0.08])
b = Bootstrap(samples)
```

```
# Compute CI from percentiles
low, high = b.ci(0.95)
```

**Properties:**

- `std()`: Sample standard deviation
- `ci(level)`: Percentile-based CI
- `sample(n, seed)`: Resample from stored samples
- Propagation: Always Monte Carlo
- Serialization: Stores summary (n, std, CI) not full samples

### Empirical

Similar to Bootstrap but conceptually for any empirical distribution.

```
from py3plex.stats import Empirical

samples = np.array([0.1, 0.2, 0.15, 0.18, 0.12])
e = Empirical(samples)
```

**Properties:**

- Same as Bootstrap
- Conceptually separate for clarity

### Interval

Interval-based uncertainty without assuming distribution.

```
from py3plex.stats import Interval

i = Interval(-0.1, 0.15)

# Uniform sampling by default
samples = i.sample(100, seed=42)
```

**Properties:**

- `std()`: Estimates std assuming uniform distribution: `(high - low) / sqrt(12)`
- `ci(level)`: Returns the interval bounds
- `sample(n, seed)`: Uniform sampling
- Propagation: Monte Carlo

### Provenance

Tracks how a statistic was computed:

```
from py3plex.stats import Provenance
```

```
prov = Provenance(
    algorithm="brandes",
    uncertainty_method="bootstrap",
    parameters={"n_samples": 100},
    seed=42,
    timestamp="2024-12-12T17:00:00",
    library_version="1.0.0"
)

# Serialize
json_dict = prov.to_json_dict()
```

**Fields:**

- `algorithm`: Algorithm name (e.g., "degree", "betweenness")
- `uncertainty_method`: Uncertainty method (e.g., "delta", "bootstrap")
- `parameters`: Dict of parameters
- `seed`: Random seed (optional)
- `timestamp`: Computation time (optional)
- `library_version`: Version string (optional)

### Statistics Registry

The `StatisticsRegistry` enforces that every registered statistic has an uncertainty model.

### Registration

```
from py3plex.stats import StatisticSpec, register_statistic, Delta

def compute_degree(network, node):
    return network.core_network.degree(node)

def degree_uncertainty(network, node, **kwargs):
    return Delta(0.0)  # Deterministic

spec = StatisticSpec(
    name="degree",
    estimator=compute_degree,
    uncertainty_model=degree_uncertainty,
    assumptions=["deterministic"],
    supports={"directed": True, "weighted": True}
)

register_statistic(spec)
```

**Note:** Registration fails if `uncertainty_model` is missing.

### Usage

```
from py3plex.stats import compute_statistic

# Compute with uncertainty
result = compute_statistic("degree", network, node, with_uncertainty=True)
# Returns StatValue

# Compute without uncertainty (raw value)
value = compute_statistic("degree", network, node, with_uncertainty=False)
```

### Arithmetic with Uncertainty

StatValue supports arithmetic operations with automatic uncertainty propagation.

### Basic Operations

```
from py3plex.stats import StatValue, Gaussian, Provenance

sv1 = StatValue(1.0, Gaussian(0.0, 0.1), Provenance("a", "gaussian", {}))
sv2 = StatValue(2.0, Gaussian(0.0, 0.15), Provenance("b", "gaussian", {}))

# Addition
result = sv1 + sv2
print(float(result))  # 3.0
print(result.std())  # ~0.180 (sqrt(0.1^2 + 0.15^2))

# Subtraction
result = sv1 - sv2

# Multiplication
result = sv1 * sv2

# Division
result = sv1 / sv2

# Power
result = sv1 ** 2

# Negation
result = -sv1
```

### Scalar Operations

StatValue supports operations with scalars:

```
sv = StatValue(2.0, Gaussian(0.0, 0.1), Provenance("a", "gaussian", {}))

# Scalar addition
result = sv + 3  # 5.0 (uncertainty unchanged)
```

```
# Scalar multiplication
result = sv * 2  # 4.0 (uncertainty scaled)

# Scalar division
result = sv / 2  # 1.0 (uncertainty scaled)
```

### Propagation Rules

1. **Delta + Delta**: Analytic error propagation (`_sum = sqrt(1^2 + 2^2)`)
2. **Gaussian + Gaussian**: Exact propagation for addition/subtraction
3. **Complex operations**: Monte Carlo propagation (4096 samples by default)
4. **Scalar operations**: Direct computation, uncertainty scaled appropriately

### Filtering and Queries

Statistics with uncertainty can be filtered using selectors:

### Selector Syntax

Format: `attribute__component__operator=value`

**Components:**

- `mean`: Point estimate (default if omitted)
- `std`: Standard deviation
- `ci95__width`: Width of 95% CI
- `robustness`: Robustness score

**Operators:**

- `gt`: Greater than
- `gte`: Greater than or equal
- `lt`: Less than
- `lte`: Less than or equal
- `eq`: Equal
- `ne`: Not equal

### Examples

```
# Filter by mean value
result = Q.nodes().where(degree__mean__gt=3).execute(network)

# Filter by uncertainty
result = Q.nodes().where(betweenness__std__lt=0.05).execute(network)

# Filter by CI width
result = Q.nodes().where(degree__ci95__width__lt=0.1).execute(network)
```

```
# Filter by robustness
result = Q.nodes().where(centrality__robustness__gt=0.9).execute(network)
```

### Serialization

StatValue and Uncertainty models support JSON serialization.

### StatValue Serialization

```
sv = StatValue(
    value=0.42,
    uncertainty=Gaussian(0.0, 0.05),
    provenance=Provenance("betweenness", "analytic", {})
)

json_dict = sv.to_json_dict()
# {
#   "value": 0.42,
#   "uncertainty": {
#     "type": "gaussian",
#     "mean": 0.0,
#     "std": 0.05
#   },
#   "provenance": {
#     "algorithm": "betweenness",
#     "uncertainty_method": "analytic",
#     "params": {}
#   }
# }
```

### DataFrame Export

QueryResult can export to pandas with uncertainty columns:

```
result = Q.nodes().compute("betweenness").execute(network)

df = result.to_pandas()
# Columns: id, betweenness.value, betweenness.std,
#          betweenness.ci_low, betweenness.ci_high,
#          betweenness.uncertainty_type
```

### Best Practices

1. **Always use StatValue internally**: Even for deterministic stats, use `Delta(0)`
2. **Provide uncertainty models**: Every registered statistic must have one
3. **Use seeds for reproducibility**: Pass explicit seeds to bootstrap/MC operations
4. **Choose appropriate models**:
   - Deterministic → `Delta(0)`

- Known distribution → `Gaussian`
- Empirical estimation → `Bootstrap` or `Empirical`
- No distribution assumption → `Interval`

5. **Check robustness**: Use `sv.robustness()` to assess reliability
6. **Export uncertainty**: Include uncertainty columns in exports for downstream analysis

### Examples

See:

- `examples/network_analysis/example_stats_degree_delta.py`
- `examples/network_analysis/example_stats_betweenness_bootstrap.py`

### API Reference

#### StatValue

```
class StatValue:
    """Statistical value with uncertainty and provenance."""

    value: float | int | ndarray
    uncertainty: Uncertainty
    provenance: Provenance

    def __float__(self) -> float: ...
    def mean(self) -> float: ...
    def std(self) -> float: ...
    def ci(self, level: float = 0.95) -> tuple[float, float]: ...
    def robustness(self) -> float: ...
    def to_json_dict(self) -> dict: ...
```

#### Uncertainty Models

```
class Uncertainty(ABC):
    def summary(self, level: float = 0.95) -> dict: ...
    def sample(self, n: int, *, seed: int | None = None) -> ndarray: ...
    def ci(self, level: float = 0.95) -> tuple[float, float]: ...
    def std(self) -> float | None: ...
    def propagate(self, op: str, other: Uncertainty | None, *, seed: int | None = None) -> U
    def to_json_dict(self) -> dict: ...

class Delta(Uncertainty):
    sigma: float = 0.0

class Gaussian(Uncertainty):
    mean: float
    std_dev: float

class Bootstrap(Uncertainty):
    samples: ndarray
```

```
class Empirical(Uncertainty):
    samples: ndarray

class Interval(Uncertainty):
    low: float
    high: float
```

#### Provenance

```
@dataclass(frozen=True)
class Provenance:
    algorithm: str
    uncertainty_method: str
    parameters: dict = field(default_factory=dict)
    seed: int | None = None
    timestamp: str | None = None
    library_version: str | None = None

    def to_json_dict(self) -> dict: ...
    @classmethod
    def from_json_dict(cls, data: dict) -> Provenance: ...
```

#### Registry

```
@dataclass(frozen=True)
class StatisticSpec:
    name: str
    estimator: Callable
    uncertainty_model: Callable  # Required
    assumptions: list[str] = field(default_factory=list)
    supports: dict = field(default_factory=dict)

def register_statistic(spec: StatisticSpec, force: bool = False) -> None: ...
def get_statistic(name: str) -> StatisticSpec: ...
def list_statistics() -> list[str]: ...
def compute_statistic(name: str, *args, with_uncertainty: bool = True, **kwargs) -> Any: ...
```

### 3.5.6 API Documentation

This page indexes the auto-generated API documentation for py3plex. Use it as a map to find module-level details after building the docs.

> **Tip**
>
> **Looking for algorithm documentation?** Visit *Algorithm Roadmap* for a conceptual overview of all algorithms organized by category with working examples. Then come back here for detailed API signatures.

How to (re)generate the API pages:

```
cd docfiles
sphinx-apidoc -o AUTOGEN_results -f ../py3plex
make html
```

You will find the generated RST sources under `docfiles/AUTOGEN_results` and the rendered HTML under `docfiles/_build`.

### Core Modules

Core data structures and helpers for loading, converting, and working with multilayer networks.

**exception** py3plex.core.multinet.**RicciBackendNotAvailable**

Bases: `ImportError`

py3plex.core.multinet.**ensure**(**args*, ***kwargs*)

py3plex.core.multinet.**invariant**(**args*, ***kwargs*)

**class** py3plex.core.multinet.**multi_layer_network**(*verbose: bool = True*, *network_type: str = 'multilayer'*, *directed: bool = True*, *dummy_layer: str = 'null'*, *label_delimiter: str = '---'*, *coupling_weight: int | float = 1*)

Bases: `object`

Main class for multilayer network analysis and manipulation.

This class provides a comprehensive toolkit for creating, analyzing, and visualizing multilayer networks where nodes can exist in multiple layers and edges can connect nodes within or across layers.

**Supported Network Types (network_type parameter):**

- **multilayer** (default): General multilayer networks with arbitrary layer structure. Each layer can have a different set of nodes. Suitable for heterogeneous networks (e.g., authors-papers-venues) or networks where nodes naturally appear in only some layers.
- **multiplex**: Special case where all layers share the same node set but with different edge types. After loading a network, automatic coupling edges are created between each node and its counterparts in other layers. Suitable for social networks with multiple relationship types (e.g., friend, colleague, family layers).

**Choosing the Right Network Type:**

| Criterion | multilayer | multiplex |
|---|---|---|
| Node sets | Different per layer | Same across layers |
| Coupling edges | Manual (explicit) | Automatic |
| Use case | Heterogeneous nets (author-paper) | Same entities, many relationship types |

**Key Features:**

- Dict-based API for adding nodes and edges (see add_nodes() and add_edges())

- NetworkX interoperability via to_networkx() and from_networkx()
- Multiple I/O formats (edgelist, GML, GraphML, gpickle, etc.)
- Visualization methods for multilayer layouts
- Community detection and centrality analysis
- Random walk and embedding generation

**Hypergraph Support:**

This class does NOT natively support true hypergraphs (edges connecting more than two nodes). For hypergraph-like structures, consider: - Using bipartite projections (nodes and hyperedges as separate node types) - The incidence gadget encoding via to_homogeneous_hypergraph() - External hypergraph libraries with conversion utilities

### Notes

- Nodes in multilayer networks are represented as (node_id, layer) tuples
- Use add_nodes() and add_edges() with dict format for easiest interaction
- See examples/ directory for usage patterns and best practices

### Examples

```
>>> # Create a general multilayer network (different node sets per layer)
>>> net = multi_layer_network(network_type='multilayer', directed=False)
>>>
>>> # Add nodes to different layers
>>> _ = net.add_nodes([
...     {'source': 'A', 'type': 'social'},
...     {'source': 'B', 'type': 'social'},
...     {'source': 'A', 'type': 'email'}  # Same node, different layer
... ])
>>>
>>> # Add edges (intra-layer and inter-layer)
>>> _ = net.add_edges([
...     {'source': 'A', 'target': 'B',
...      'source_type': 'social', 'target_type': 'social'},
...     {'source': 'A', 'target': 'A',
...      'source_type': 'social', 'target_type': 'email'}
... ])
>>>
>>> print(net)  # Shows network statistics
<multi_layer_network: type=multilayer, directed=False, nodes=3, edges=2, layers=2>
```

```
>>> # Create a multiplex network (same nodes across relationship layers)
>>> # Note: coupling edges are auto-added after load_network()
>>> multiplex_net = multi_layer_network(network_type='multiplex')
```

**add_dummy_layers()**

Wrap monoplex nodes in the dummy layer, preserving graph data.

**add_edges**(*edge_dict_list: List[Dict] | List[List] | Tuple*, *input_type: str = 'dict'*) → multi_layer_network

Add edges to the multilayer network.

This method supports multiple input formats for specifying edges between nodes in different layers. The most common format is dict-based.

**Parameters**

- **edge_dict_list** – Edge data in one of the supported formats (see below)
- **input_type** – Format of edge data ('dict', 'list', or 'px_edge')

**Returns**

Returns self for method chaining

**Return type**

self

**Supported Formats:**

**Dict format (recommended):**

```
{
    'source': 'node1',          # Source node ID
    'target': 'node2',          # Target node ID
    'source_type': 'layer1',    # Source layer name
    'target_type': 'layer2',    # Target layer name (can be same as source)
    'weight': 1.0,              # Optional: edge weight
    'type': 'interaction'       # Optional: edge type/label
}
```

**List format:** *[node1, layer1, node2, layer2]*

**px_edge format:** *((node1, layer1), (node2, layer2), {'weight': 1.0})*

### Examples

```
>>> # Add single intra-layer edge
>>> net = multi_layer_network()
>>> net.add_edges([{
...     'source': 'A',
...     'target': 'B',
...     'source_type': 'protein',
...     'target_type': 'protein'
... }])
<multi_layer_network: type=multilayer, directed=True, nodes=2, edges=1, layers=1>
```

```
>>> # Method chaining
>>> net = multi_layer_network()
>>> net.add_edges([
...     {'source': 'A', 'target': 'B', 'source_type': 'layer1', 'target_type': 'lay
... ]).add_edges([
...     {'source': 'B', 'target': 'C', 'source_type': 'layer1', 'target_type': 'lay
... ])
<multi_layer_network: type=multilayer, directed=True, nodes=3, edges=2, layers=1>
```

```
>>> # Add inter-layer edge with weight
>>> net.add_edges([{
...     'source': 'gene1',
...     'target': 'protein1',
...     'source_type': 'genes',
...     'target_type': 'proteins',
...     'weight': 0.95,
...     'type': 'expression'
... }])
<multi_layer_network: type=multilayer, directed=True, nodes=5, edges=3, layers=3>
```

```
>>> # Add multiple edges at once
>>> edges = [
...     {'source': 'A', 'target': 'B', 'source_type': 'layer1', 'target_type': 'lay
...     {'source': 'B', 'target': 'C', 'source_type': 'layer1', 'target_type': 'lay
... ]
>>> net.add_edges(edges)
<multi_layer_network: type=multilayer, directed=True, nodes=5, edges=5, layers=3>
```

**Raises**

**Exception** – If input_type is not one of 'dict', 'list', or 'px_edge'

### Notes

- For intra-layer edges, use the same layer for source_type and target_type
- For inter-layer edges, use different layers
- Edge weights default to 1.0 if not specified

**add_nodes**(*node_dict_list: List[Dict] | Dict*, *input_type: str = 'dict'*) → multi_layer_network

Add nodes to the multilayer network.

Nodes in a multilayer network are identified by both their ID and the layer they belong to. This method adds nodes using a dict-based format.

**Parameters**

- **node_dict_list** – Node data as a dict or list of dicts (see format below)
- **input_type** – Format of node data (currently only 'dict' is supported)

**Returns**

Returns self for method chaining

**Return type**

self

Dict Format:

```
{
    'source': 'node_id',     # Node identifier (can be string or number)
    'type': 'layer_name',    # Layer this node belongs to
    'weight': 1.0,           # Optional: node weight/importance
```

```
    'label': 'display'      # Optional: display label
    # ... any other node attributes
}
```

### Examples

```
>>> # Add single node
>>> net = multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}])
<multi_layer_network: type=multilayer, directed=True, nodes=1, edges=0, layers=1>
```

```
>>> # Method chaining
>>> net = multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}]).add_nodes([{'source': 'B',
<multi_layer_network: type=multilayer, directed=True, nodes=2, edges=0, layers=1>
```

```
>>> # Add multiple nodes to the same layer
>>> nodes = [
...     {'source': 'A', 'type': 'protein'},
...     {'source': 'B', 'type': 'protein'},
...     {'source': 'C', 'type': 'protein'}
... ]
>>> net.add_nodes(nodes)
<multi_layer_network: type=multilayer, directed=True, nodes=5, edges=0, layers=2>
```

```
>>> # Add nodes with attributes
>>> net.add_nodes([{
...     'source': 'gene1',
...     'type': 'genes',
...     'weight': 0.8,
...     'label': 'BRCA1',
...     'chromosome': '17'
... }])
<multi_layer_network: type=multilayer, directed=True, nodes=6, edges=0, layers=3>
```

```
>>> # Add nodes to multiple layers
>>> multi_layer_nodes = [
...     {'source': 'entity1', 'type': 'layer1'},
...     {'source': 'entity1', 'type': 'layer2'},  # Same entity, different layer
...     {'source': 'entity2', 'type': 'layer1'}
... ]
>>> net.add_nodes(multi_layer_nodes)
<multi_layer_network: type=multilayer, directed=True, nodes=9, edges=0, layers=4>
```

### Notes

- The same node ID can exist in multiple layers
- Each (node_id, layer) combination is treated as a unique node
- Additional attributes beyond 'source' and 'type' are preserved
- Nodes must be added before edges referencing them

**aggregate_edges**(*metric='count'*, *normalize_by='degree'*)

Edge aggregation method

Count weights across layers and return a weighted network

**Parameters**

- **param1** – aggregation operator (count is default)
- **param2** – normalization of the values

**Returns**

A simplified network.

**assign_partition**(*partition: Dict[Tuple[Any, Any], int]*, ***, *name: str = 'default'*, *meta: Dict[str, Any] | None = None*) → None

Assign community partition to network nodes.

This method stores the community assignments as node attributes and computes community-level statistics. Supports named partitions for tracking multiple partitions (e.g., from different algorithms or seeds).

**Parameters**

- **partition** (*dict*) – Dictionary mapping (node, layer) tuples to community IDs.
- **name** (*str, optional*) – Name for this partition (default: "default"). Used to store multiple partitions and access them later.
- **meta** (*dict, optional*) – Metadata about the partition (algorithm, parameters, etc.)

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network(directed=False)
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> partition = {('A', 'L1'): 0, ('B', 'L1'): 0}
>>> net.assign_partition(partition)
>>> print(net.community_sizes)
{0: 2}
```

```
>>> # Named partition for multiple algorithms
>>> net.assign_partition(partition, name="louvain", meta={"resolution": 1.0})
>>> partition2 = {('A', 'L1'): 1, ('B', 'L1'): 1}
>>> net.assign_partition(partition2, name="infomap")
```

**basic_stats**(*target_network=None*)

A method for obtaining a network's statistics.

Displays: - Basic network info (nodes, edges) - Total unique nodes (counting each (node, layer) as unique) - Unique node IDs (across all layers) - Per-layer node counts

**capabilities**(*force_recompute: bool = False*)

Compute and cache network capabilities for algorithm compatibility checking.

This method analyzes the network structure to determine its fundamental characteristics that algorithms may require or constrain. Results are cached for efficiency.

**Parameters**

**force_recompute** – If True, recompute capabilities even if cached

**Returns**

**Dataclass with network properties including:**

- mode: Network mode (single, multilayer, multiplex, temporal)
- replica_model: Replica model (none, partial, strict)
- interlayer_coupling: Coupling type (none, identity, explicit_edges, both)
- directed: Whether network is directed
- weighted: Whether network has edge weights
- weight_domain: Domain of edge weights
- layer_count: Number of layers
- base_node_count: Number of distinct base nodes
- And more...

**Return type**

NetworkCapabilities

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> caps = net.capabilities()
>>> print(caps.mode)
single
>>> print(caps.layer_count)
1
```

### Notes

- Results are cached for performance
- Cache is invalidated on network mutations (if tracked)
- Temporal networks require TemporalMultiLayerNetwork class

**compute_ollivier_ricci**(*mode: str = 'core', layers: List[Any] | None = None, alpha: float = 0.5, weight_attr: str = 'weight', curvature_attr: str = 'ricciCurvature', verbose: str = 'ERROR', backend_kwargs: Dict[str, Any] | None = None, inplace: bool = True, interlayer_weight: float = 1.0*) → Dict[str, Any]

Compute Ollivier-Ricci curvature on the multilayer network.

This method provides flexible computation of Ollivier-Ricci curvature at different levels of the multilayer network:

- **core mode**: Compute curvature on the aggregated (flattened) network
- **layers mode**: Compute curvature separately for each layer
- **supra mode**: Compute curvature on the full supra-graph including both intra-layer and inter-layer edges

**Parameters**

- **mode** – Scope of computation. Options: “core”, “layers”, “supra”.
- **layers** – List of layer identifiers to process (only for mode=”layers”). If None, all layers are processed.
- **alpha** – Ollivier-Ricci parameter in [0, 1] controlling the mass distribution. Default: 0.5.
- **weight_attr** – Name of edge attribute containing weights. Default: “weight”.
- **curvature_attr** – Name of edge attribute to store curvature values. Default: “ricciCurvature”.
- **verbose** – Verbosity level. Options: “INFO”, “DEBUG”, “ERROR”. Default: “ERROR”.
- **backend_kwargs** – Additional keyword arguments for OllivierRicci constructor.
- **inplace** – If True, update internal graphs. If False, return new graphs without modifying the network. Default: True.
- **interlayer_weight** – Weight for inter-layer coupling edges (only for mode=”supra”). Default: 1.0.

**Returns**

Dictionary mapping scope identifiers to NetworkX graphs with computed curvatures: - mode=”core”: {“core”: graph_with_curvature} - mode=”layers”: {layer_id: graph_with_curvature, ...} - mode=”supra”: {“supra”: supra_graph_with_curvature}

**Raises**

- **RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.
- **ValueError** – If mode is invalid or layers contains invalid identifiers.

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> _ = net.add_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1],
... ], input_type="list")
>>>
>>> # Compute on aggregated network
```

```
>>> result = net.compute_ollivier_ricci(mode="core")
>>>
>>> # Compute per layer
>>> result = net.compute_ollivier_ricci(mode="layers")
>>>
>>> # Compute on supra-graph
>>> result = net.compute_ollivier_ricci(mode="supra", inplace=False)
```

**compute_ollivier_ricci_flow**(*mode: str = 'core'*, *layers: List[Any] | None = None*, *alpha: float = 0.5*, *iterations: int = 10*, *method: str = 'OTD'*, *weight_attr: str = 'weight'*, *curvature_attr: str = 'ricciCurvature'*, *verbose: str = 'ERROR'*, *backend_kwargs: Dict[str, Any] | None = None*, *inplace: bool = True*, *interlayer_weight: float = 1.0*) → Dict[str, Any]

Compute Ollivier-Ricci flow on the multilayer network.

Ricci flow iteratively adjusts edge weights based on their Ricci curvature, effectively revealing and enhancing community structure. After Ricci flow, edges with negative curvature (community boundaries) have reduced weights, while edges with positive curvature have increased weights.

**Parameters**

- **mode** – Scope of computation. Options: “core”, “layers”, “supra”.
- **layers** – List of layer identifiers to process (only for mode=”layers”). If None, all layers are processed.
- **alpha** – Ollivier-Ricci parameter in [0, 1]. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **method** – Ricci flow method. Options: “OTD” (Optimal Transport Distance, recommended), “ATD” (Average Transport Distance). Default: “OTD”.
- **weight_attr** – Name of edge attribute containing weights. After Ricci flow, these weights are updated to reflect the flow metric.
- **curvature_attr** – Name of edge attribute for curvature values. Default: “ricciCurvature”.
- **verbose** – Verbosity level. Options: “INFO”, “DEBUG”, “ERROR”. Default: “ERROR”.
- **backend_kwargs** – Additional keyword arguments for OllivierRicci constructor.
- **inplace** – If True, update internal graphs. If False, return new graphs. Default: True.
- **interlayer_weight** – Weight for inter-layer coupling edges (only for mode=”supra”). Default: 1.0.

**Returns**

Dictionary mapping scope identifiers to NetworkX graphs with Ricci flow applied: - mode=”core”: {“core”: graph_with_flow} - mode=”layers”: {layer_id: graph_with_flow, ...} - mode=”supra”:

{“supra”: supra_graph_with_flow}

**Raises**

- **RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.
- **ValueError** – If mode is invalid or layers contains invalid identifiers.

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> _ = net.add_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1],
... ], input_type="list")
>>>
>>> # Apply Ricci flow to aggregated network
>>> result = net.compute_ollivier_ricci_flow(mode="core", iterations=20)
>>>
>>> # Apply to each layer
>>> result = net.compute_ollivier_ricci_flow(mode="layers", iterations=10)
```

**property edge_count: int**

Number of edges in the network.

**Returns**

Total count of edges

**Return type**

int

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_edges([{'source': 'A', 'target': 'B',
...                     'source_type': 'layer1', 'target_type': 'layer1'}])
>>> net.edge_count
1
```

**edges_from_temporal_table**(*edge_df*)

Convert a temporal edge DataFrame to edge tuple list.

Extracts edges from a pandas DataFrame with temporal/activity information and converts them to a list of edge tuples suitable for network construction.

**Parameters**

**edge_df** – pandas DataFrame with columns: - node_first: Source node identifier - node_second: Target node identifier - layer_name: Layer identifier

**Returns**

**List of edge tuples in format:**

(node_first, node_second, layer_first, layer_second, weight) where weight is always 1

**Return type**
list

### Notes

- All values are converted to strings
- All edges are assigned weight=1
- Both source and target are assumed to be in the same layer

### Examples

```
>>> import pandas as pd
>>> net = multi_layer_network()
>>> df = pd.DataFrame({
...     'node_first': ['A', 'B'],
...     'node_second': ['B', 'C'],
...     'layer_name': ['L1', 'L1']
... })
>>> result = net.edges_from_temporal_table(df)
>>> len(result) >= 2
True
```

**See also**

fill_tmp_with_edges: Add these edges to layer graphs

**embed**(*method: str = 'node2vec'*, *dimensions: int = 128*, *walk_length: int = 40*, *num_walks: int = 10*, *context_size: int = 10*, *workers: int = 4*, ***kwargs: Any*) → Any

Learn node embeddings as a first-class ML primitive.

**Parameters**

- **method** – Embedding backend name.
- **dimensions** – Embedding dimensionality.
- **walk_length** – Random-walk length for walk-based models.
- **num_walks** – Number of walks per node for walk-based models.
- **context_size** – Skipgram context window.
- **workers** – Number of workers (reserved for backend implementations).
- ****kwargs** – Method-specific parameters.

**Returns**
EmbeddingResult aligned with `self.get_nodes()`.

**execute_query**(*query: str*) → Dict[str, Any]

Execute a DSL query on this multilayer network.

This is a convenience method that provides first-class access to the py3plex DSL (Domain-Specific Language) for querying multilayer networks.

Supports both SELECT and MATCH queries:

**SELECT queries:**
net.execute_query('SELECT nodes WHERE layer="transport"')
net.execute_query('SELECT * FROM nodes IN LAYER "ppi" WHERE degree > 10')

**MATCH queries (Cypher-like):**
net.execute_query('MATCH (g:Gene)-[r]->(t:Gene) RETURN g, t')
net.execute_query('MATCH (a)-[e]->(b) IN LAYER "ppi" WHERE a.degree > 5 RETURN a, b')

**Parameters**
**query** – DSL query string

**Returns**
- For SELECT queries: 'nodes' or 'edges' list, 'count', optional 'computed'
- For MATCH queries: 'bindings' list, 'count', 'type'

**Return type**
Dictionary containing query results

**Raises**
- **DSLSyntaxError** – If query syntax is invalid
- **DSLExecutionError** – If query cannot be executed

### Examples

```
>>> net = multi_layer_network(directed=False)
>>> _ = net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> _ = net.add_edges([{'source': 'A', 'target': 'B',
...                     'source_type': 'layer1', 'target_type': 'layer1'}])
>>> result = net.execute_query('SELECT nodes WHERE layer="layer1"')
>>> result['count'] >= 0
True
```

```
>>> # Using MATCH syntax
>>> result = net.execute_query('MATCH (a:layer1)-[r]->(b:layer1) RETURN a, b')
>>> 'bindings' in result
True
```

**See also**
- py3plex.dsl.execute_query() for standalone function
- py3plex.dsl.format_result() for formatting results

**fill_tmp_with_edges**(*edge_df*)

Fill temporary layer graphs with edges from a DataFrame.

Populates the emptied layer graphs (created by remove_layer_edges) with edges from a temporal/activity DataFrame. Useful for temporal network analysis where edge sets change over time.

**Parameters**

**edge_df** – pandas DataFrame with columns: - node_first: Source node identifier - node_second: Target node identifier - layer_name: Layer identifier

### Notes

- Requires remove_layer_edges() to be called first
- Edges are grouped by layer
- Modifies self.tmp_layers in place
- Each edge is stored as ((node_first, layer), (node_second, layer))

**Raises**

**AttributeError** – If self.tmp_layers doesn't exist (call remove_layer_edges first)

### Examples

These examples require proper network setup and are for illustration only.

```
>>> import pandas as pd
>>> net = multi_layer_network()
>>> net.split_to_layers()
>>> net.remove_layer_edges()
>>> df = pd.DataFrame({
...     'node_first': ['A', 'B'],
...     'node_second': ['B', 'C'],
...     'layer_name': ['L1', 'L1']
... })
>>> net.fill_tmp_with_edges(df)
```

**See also**

remove_layer_edges: Creates empty layer graphs edges_from_temporal_table: Convert DataFrame to edge list

**flatten_to_monoplex**(*method: str = 'union'*) → Graph

Flatten a multilayer/multiplex network to a single-layer graph.

Aggregates all layers into a single NetworkX graph, merging nodes with the same ID across different layers into single nodes.

**Parameters**

**method** – Method for aggregating edges: - 'union': Sum edge weights across layers (default) - 'first': Keep only the first occurrence of each edge - 'count': Count number of times each edge appears

**Returns**

**A single-layer NetworkX graph (Graph or DiGraph based on** network directionality)

**Return type**

nx.Graph

### Examples

```
>>> net = multi_layer_network(network_type='multilayer', directed=False)
>>> _ = net.add_edges([
...     {'source': 'A', 'target': 'B',
...      'source_type': 'L1', 'target_type': 'L1'},
...     {'source': 'A', 'target': 'B',
...      'source_type': 'L2', 'target_type': 'L2'},
... ])
>>> flat = net.flatten_to_monoplex(method='count')
>>> flat.edges[('A', 'B')]['weight']
2
```

### Notes

- Node IDs are extracted from (node_id, layer) tuples
- Duplicate edges across layers are aggregated
- Inter-layer edges are excluded (only intra-layer edges are kept)
- Edge attributes other than weight are taken from first occurrence

**classmethod from_edges**(*edges: List[Dict | List]*, *network_type: str = 'multilayer'*, *directed: bool = False*, *input_type: str = 'dict'*) → multi_layer_network

Create a multilayer network directly from a list of edges.

This is a convenience factory method that creates a network and populates it with edges in a single call, supporting method chaining patterns.

**Parameters**

- **edges** – List of edges in dict or list format
- **network_type** – Type of network ('multilayer' or 'multiplex')
- **directed** – Whether the network is directed
- **input_type** – Format of edge data ('dict' or 'list')

**Returns**

New network instance with edges added

**Return type**

multi_layer_network

### Examples

```
>>> # Create from dict format
>>> net = multi_layer_network.from_edges([
...     {'source': 'A', 'target': 'B',
...      'source_type': 'layer1', 'target_type': 'layer1'},
...     {'source': 'B', 'target': 'C',
...      'source_type': 'layer1', 'target_type': 'layer1'}
... ])
>>> len(net)
3
```

```
>>> # Create from list format
>>> net = multi_layer_network.from_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1]
... ], input_type='list')
>>> net.edge_count
2
```

**from_homogeneous_hypergraph**(*H*)

Decode a homogeneous graph created by to_homogeneous_hypergraph.

This method reconstructs a multiplex network from its incidence gadget encoding. It identifies edge-nodes by their degree and cycle structure, then reconstructs the original layers based on cycle lengths (prime numbers).

**Parameters**

**H** (*networkx.Graph*) – Homogeneous graph created by to_homogeneous_hypergraph().

**Returns**

- *dict* – Dictionary mapping layer names to lists of edges: {layer: [(u, v), ...]}
- *Examples* – Example requires proper network setup - for illustration only.

```
>>> network = multi_layer_network()
>>> network.add_layer("A")
>>> network.add_nodes([("1", "A"), ("2", "A")])
>>> network.add_edges([(("1", "A"), ("2", "A"))])
>>> H, node_map, edge_info = network.to_homogeneous_hypergraph()
>>> recovered = network.from_homogeneous_hypergraph(H)
>>> print(recovered)
{'layer_with_prime_2': [('1', '2')]}
```

### Notes

The decoded layer names indicate the prime number used for encoding: - "layer_with_prime_2" corresponds to the first layer - "layer_with_prime_3" corresponds to the second layer, etc.

**classmethod from_networkx**(*G: Graph*, *network_type: str = 'multilayer'*, *directed: bool | None = None*) → multi_layer_network

Create a multi_layer_network from a NetworkX graph.

This class method converts a NetworkX graph into a py3plex multi_layer_network. For multilayer networks, nodes should be tuples of (node_id, layer).

**Parameters**

- **G** – NetworkX graph to convert
- **network_type** – Type of network ('multilayer' or 'multiplex')
- **directed** – Whether to treat the network as directed. If None, inferred from G.

**Returns**
A new multi_layer_network instance

**Return type**
multi_layer_network

### Examples

```
>>> import networkx as nx
>>> G = nx.Graph()
>>> G.add_nodes_from([('A', 'layer1'), ('B', 'layer1')])
>>> G.add_edge(('A', 'layer1'), ('B', 'layer1'))
>>> net = multi_layer_network.from_networkx(G)
>>> print(net)
<multi_layer_network: type=multilayer, directed=False, nodes=2, edges=1, layers=1>
```

### Notes

- For proper multilayer behavior, ensure nodes are (node_id, layer) tuples
- Edge attributes are preserved during conversion
- The input graph is copied, not referenced

**get_decomposition**(*heuristic='all'*, *cycle=None*, *parallel=False*, *alpha=1*, *beta=0*)

Core method for obtaining a network's decomposition in terms of relations

**get_decomposition_cycles**(*cycle=None*)

A supporting method for obtaining decomposition triplets

**get_degrees**()

A simple wrapper which computes node degrees.

**get_edges**(*data: bool = False*, *multiplex_edges: bool = False*) → Any

Iterate over edges in the network.

This method behaves differently based on the network_type:

- **multilayer**: Returns all edges without filtering.
- **multiplex**: By default, filters out coupling edges (auto-generated inter-layer edges connecting each node to itself in other layers). Set multiplex_edges=True to include coupling edges.

**Parameters**

- **data** – If True, return edge data along with edge tuples
- **multiplex_edges** – If True, include coupling edges in multiplex networks. Only relevant when network_type='multiplex'. Coupling edges are automatically added to connect each node to its counterparts in other layers.

**Yields**
Edge tuples, optionally with data. For multiplex networks with multiplex_edges=False, coupling edges are excluded.

**Raises**
**ValueError** – If network_type is not 'multilayer' or 'multiplex'.

### Examples

```
>>> net = multi_layer_network(network_type='multilayer')
>>> net.add_edges([
...     {'source': 'A', 'target': 'B',
...      'source_type': 'layer1', 'target_type': 'layer1'}
... ])
<multi_layer_network: type=multilayer, directed=True, nodes=2, edges=1, layers=1>
>>> list(net.get_edges())
[(('A', 'layer1'), ('B', 'layer1'))]
```

**See also**

- `__init__()` for the difference between multilayer and multiplex
- `_couple_all_edges()` for how coupling edges are created

**get_label_matrix**()

Return network labels

**get_layers**(*style='diagonal'*, *compute_layouts='force'*, *layout_parameters=None*, *verbose=True*)

A method for obtaining layerwise distributions

**get_neighbors**(*node_id: str*, *layer_id: str | None = None*) → Any

Get neighbors of a node in a specific layer.

**Parameters**

- **node_id** – Node identifier
- **layer_id** – Layer identifier (optional)

**Returns**

Iterator of neighbor nodes

**get_node_attribute**(*node: Any*, *attribute: str*, *layer: Any | None = None*) → Any

Get an attribute value for a given node.

**Parameters**

- **node** (*Any*) – Node identifier.
- **attribute** (*str*) – Name of the attribute to retrieve.
- **layer** (*Any, optional*) – Layer identifier. If None and node is a tuple, assumes node is (node_id, layer).

**Returns**

The attribute value, or None if not found.

**Return type**

Any

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network(directed=False)
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> net.set_node_attribute('A', 'score', 42.0, 'L1')
>>> print(net.get_node_attribute('A', 'score', 'L1'))
42.0
```

**get_nodes**(*data: bool = False*) → Any

A method for obtaining a network's nodes

**Parameters**

**data** – If True, return node data along with node identifiers

**Yields**

Node identifiers, optionally with data

**get_nx_object**()

Return only core network with proper annotations

**get_partition**(*node: Any*, *layer: Any | None = None*) → int | None

Get the community/partition ID for a given node.

**Parameters**

- **node** (*Any*) – Node identifier.
- **layer** (*Any, optional*) – Layer identifier. If None and node is a tuple, assumes node is (node_id, layer).

**Returns**

Community ID, or None if node doesn't have a partition assigned.

**Return type**

int or None

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network(directed=False)
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> partition = {('A', 'L1'): 0, ('B', 'L1'): 1}
>>> net.assign_partition(partition)
>>> print(net.get_partition('A', 'L1'))
0
```

**get_partition_by_name**(*name: str = 'default'*) → Dict[Tuple[Any, Any], int] | None

Get a named partition from the network.

**Parameters**

**name** (*str, optional*) – Name of the partition to retrieve (default: "default")

**Returns**

Partition dictionary mapping (node, layer) tuples to community IDs, or None if the named partition doesn't exist.

**Return type**
dict or None

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network(directed=False)
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> partition = {('A', 'L1'): 0, ('B', 'L1'): 0}
>>> net.assign_partition(partition, name="louvain")
>>> retrieved = net.get_partition_by_name("louvain")
>>> print(retrieved)
{('A', 'L1'): 0, ('B', 'L1'): 0}
```

**get_supra_adjacency_matrix**(*mtype='sparse'*)

Get sparse representation of the supra matrix.

**Parameters**
**mtype** – 'sparse' or 'dense' - matrix representation type

**Returns**
Supra-adjacency matrix in requested format

**Warning**

For large multilayer networks, dense matrices can consume significant memory (N*L)^2 * 8 bytes for float64.

**get_tensor**(*sparsity_type='bsr'*)

Get sparse tensor representation of the multilayer network.

Returns the supra-adjacency matrix in the specified sparse format. This method provides a tensor-like view of the multilayer network, useful for mathematical analysis and matrix-based algorithms.

**Parameters**
**sparsity_type** – Sparse matrix format to use. Options include: - 'bsr' (default): Block Sparse Row format - 'csr': Compressed Sparse Row format - 'csc': Compressed Sparse Column format - 'coo': Coordinate format - 'lil': List of Lists format - 'dok': Dictionary of Keys format

**Returns**
Supra-adjacency matrix in specified format

**Return type**
scipy.sparse matrix

### Example

```
>>> net = multi_layer_network()
>>> _ = net.add_edges([['A', 'L0', 'B', 'L0', 1]], input_type='list')
>>> tensor = net.get_tensor(sparsity_type='csr')
>>> tensor.shape
(2, 2)
```

**Note**

The returned matrix is the same as get_supra_adjacency_matrix(mtype='sparse') but with control over the specific sparse format used.

**get_unique_entity_counts()**

Count unique entities in the network.

**Returns**

**(total_unique_nodes, unique_node_ids, nodes_per_layer)**

- total_unique_nodes: count of unique (node, layer) tuples
- unique_node_ids: count of unique node IDs (across all layers)
- nodes_per_layer: dict mapping layer to count of nodes in that layer

**Return type**
tuple

**invert(*override_core=False*)**

invert the nodes to edges. Get the "edge graph". Each node is here an edge.

**property is_empty: bool**

Check if the network is empty (has no nodes).

**Returns**
True if network has no nodes

**Return type**
bool

### Examples

```
>>> net = multi_layer_network()
>>> net.is_empty
True
>>> _ = net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> net.is_empty
False
```

**property layer_count: int**

Number of unique layers in the network.

**Returns**
Count of distinct layers

**Return type**
int

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_nodes([
...     {'source': 'A', 'type': 'layer1'},
...     {'source': 'B', 'type': 'layer2'}
```

```
... ])
>>> net.layer_count
2
```

**property layer_names: List[Any]**

Alias for *layers* property - list of unique layer identifiers.

This property provides backward compatibility with code that uses *layer_names* instead of *layers*. Can be set explicitly (e.g., by *split_to_layers()*) or defaults to the *layers* property.

**Returns**
Layer identifiers using the ordering of the layers property

**Return type**
list

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_nodes([
...     {'source': 'A', 'type': 'social'},
...     {'source': 'B', 'type': 'work'}
... ])
>>> net.layer_names
['social', 'work']
>>> net.layer_names == net.layers  # Alias relationship
True
```

**property layers: List[Any]**

List of unique layer identifiers in the network.

Comparable identifiers are sorted. Otherwise, layers retain their first-occurrence order in the network.

**Returns**
Layer identifiers in sorted or first-occurrence order

**Return type**
list

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_nodes([
...     {'source': 'A', 'type': 'social'},
...     {'source': 'B', 'type': 'work'}
... ])
>>> net.layers
['social', 'work']
```

**list_partitions() → List[str]**

List all named partitions stored in the network.

**Returns**
Names of all stored partitions

**Return type**
list of str

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network(directed=False)
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> partition = {('A', 'L1'): 0, ('B', 'L1'): 0}
>>> net.assign_partition(partition, name="louvain")
>>> net.assign_partition(partition, name="infomap")
>>> print(net.list_partitions())
['louvain', 'infomap']
```

**load_embedding**(*embedding_file*)

Embedding loading method

**load_layer_name_mapping**(*mapping_name*, *header=False*)

Layer-node mapping loader method

**Parameters**
**param1** – The name of the mapping file.

**Returns**
self.layer_name_map is filled, returns nothing.

**load_network**(*input_file: str | None = None*, *directed: bool | None = None*, *input_type: str = 'gml'*, *label_delimiter: str = '---'*) → multi_layer_network

Load a network from file.

This method loads and prepares a given network. The behavior depends on the `network_type` set during initialization:

- **multilayer**: Network is loaded as-is. No automatic edges are added.
- **multiplex**: After loading, coupling edges are automatically created between each node and its counterparts in other layers. These edges have type='coupling' and can be filtered via get_edges().

**Parameters**

- **input_file** – Path to the network file to load
- **directed** – Whether the network is directed. If omitted or None, retain the current setting (initially set by the constructor).
- **input_type** – Format of the input file. Supported values: - 'gml': Graph Modeling Language format - 'graphml': GraphML XML format (node 'type' selects layer;

  missing types use the GraphML default or 'generic')

  - 'edgelist': Simple edge list (source target [weight]) or headered CSV
  - 'multiedgelist': Multilayer edge list (node1 layer1 node2 layer2 weight) or headered CSV (source,target,layer[,weight] or separate

source_layer and target_layer columns)

- 'multiplex_edges': Multiplex format (layer node1 node2 weight)
- 'multiplex_folder': Folder with layer files
- 'gpickle': Python pickle format
- 'nx': NetworkX graph object
- 'sparse': Sparse matrix format

- **label_delimiter** – Delimiter used to separate layer names in node labels

**Returns**

Self for method chaining. Populates self.core_network, self.labels, and self.activity.

**Note**

For multiplex networks, use input_type='multiplex_edges' or 'multiplex_folder' with network_type='multiplex' to get automatic coupling edges.

### Examples

```
>>> # Load multilayer network (no automatic coupling)
>>> net = multi_layer_network(network_type='multilayer')
>>> net.load_network('data.gml', input_type='gml')
```

```
>>> # Load multiplex network (automatic coupling edges)
>>> net = multi_layer_network(network_type='multiplex')
>>> net.load_network('data.edges', input_type='multiplex_edges')
```

**load_network_activity**(*activity_file*)

Network activity loader

**Parameters**

**activity_file** – The name of the generic activity file. Format: n1 n2 timestamp layer_name (e.g., `65432 61888 1377688175 RE`). Note that layer node mappings MUST be loaded in order to map nodes to activity properly.

**Returns**

self.activity is filled.

**load_temporal_edge_information**(*input_file=None*, *input_type='edge_activity'*, *directxed=False*, *layer_mapping=None*)

A method for loading temporal edge information

**merge_with**(*target_px_object*)

Merge two px objects.

**monitor**(*message*)

A simple monitor method for logging

**monoplex_nx_wrapper**(*method*, *kwargs=None*)

A generic networkx function wrapper.

**Parameters**

- **method** (*str*) – Name of the NetworkX function to call (e.g., 'degree_centrality', 'betweenness_centrality')
- **kwargs** (*dict, optional*) – Keyword arguments to pass to the NetworkX function. For example, for betweenness_centrality you can pass:
  - weight: Edge attribute to use as weight
  - normalized: Whether to normalize betweenness values
  - distance: Edge attribute to use as distance (for closeness_centrality)

**Returns**

The result of the NetworkX function call.

**Raises**

**AttributeError** – If the specified method does not exist in NetworkX.

### Example

# Unweighted betweenness centrality centralities = network.monoplex_nx_wrapper("betweenness_centrality")

# Weighted betweenness centrality centralities = network.monoplex_nx_wrapper("betweenness_centrality", kwargs={"weight": "weight"})

# With multiple parameters centralities = network.monoplex_nx_wrapper("betweenness_centrality", kwargs={"weight": "weight", "normalized": True})

**property network_version: int**

Monotonic mutation counter.

Starts at 0 for a freshly constructed empty object. Incremented by 1 on every public structural mutation (add_nodes, add_edges, remove_nodes, remove_edges, load_network). Wrapper methods that delegate to other public mutators suppress the inner bumps so the counter increases by exactly 1 per top-level call.

Useful for cache-invalidation keys and provenance records.

**property node_count: int**

Number of nodes in the network.

**Returns**

Total count of nodes (node-layer pairs)

**Return type**

int

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> net.node_count
1
```

**read_ground_truth_communities**(*cfile*)

Parse ground truth community file and make mappings to the original nodes. This works based on node ID mappings, exact node,layer tuplets are to be added. :param param1: ground truth communities.

**Returns**

self.ground_truth_communities

**remove_edges**(*edge_dict_list: List[Dict] | List[List]*, *input_type: str = 'list'*) → None

A method for removing edges..

**Parameters**

- **edge_dict_list** – Edge data in dict or list format
- **input_type** – Format of edge data ('dict' or 'list')

**Raises**

**Exception** – If input_type is not valid

**remove_layer_edges**()

Remove all edges from separate layer graphs while keeping nodes.

This method creates empty copies of each layer graph with all nodes intact but no edges. Useful for reconstructing networks with different edge sets or for temporal network analysis.

### Notes

- Requires split_to_layers() to be called first
- Stores empty layer graphs in self.tmp_layers
- Original graphs in self.separate_layers remain unchanged
- All nodes and their attributes are preserved

**Raises**

**RuntimeError** – If split_to_layers() hasn't been called yet

**See also**

split_to_layers: Must be called before this method fill_tmp_with_edges: Add edges back to emptied layers

**remove_nodes**(*node_dict_list*, *input_type='dict'*)

Remove nodes from the network

**save_network**(*output_file=None*, *output_type='edgelist'*)

Save the network to a file in various formats.

This method exports the multilayer network to different file formats for persistence, sharing, or use with other tools.

**Parameters**

- **output_file** – Path where the network should be saved
- **output_type** – Format for saving ('edgelist', 'multiedgelist', 'multiedgelist_encoded', or 'gpickle')

**Supported Formats:**

- 'edgelist': Simple edge list format (standard NetworkX)
- 'multiedgelist': Multilayer edge list with layer information
- 'multiedgelist_encoded': Multilayer edge list with integer encoding
- 'gpickle': Python pickle format (preserves all attributes)

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> _ = net.add_edges([{'source': 'A', 'target': 'B',
...                     'source_type': 'layer1', 'target_type': 'layer1'}])
>>> net.save_network('network.txt', output_type='multiedgelist')
```

```
>>> # For faster I/O with all metadata preserved
>>> net.save_network('network.gpickle', output_type='gpickle')
```

### Notes

- 'gpickle' format preserves all node/edge attributes
- 'multiedgelist_encoded' creates node_map and layer_map attributes
- Edge weights and types are preserved in supported formats

**serialize_to_edgelist**(*edgelist_file='./tmp/tmpedgelist.txt'*, *tmp_folder='tmp'*, *out_folder='out'*, *multiplex=False*)

Serialize the multilayer network to an edgelist file.

Converts the network to a numeric edgelist format suitable for external tools and algorithms that require integer node/layer identifiers.

**Parameters**

- **edgelist_file** – Path to output edgelist file (default: "./tmp/tmpedgelist.txt")
- **tmp_folder** – Temporary folder for intermediate files (default: "tmp")
- **out_folder** – Output folder for results (default: "out")
- **multiplex** – If True, use multiplex format (node layer node layer weight) If False, use simple edgelist format (node1 node2 weight)

**Returns**

**Inverse node mapping (numeric_id -> original_node_tuple)**
Use this to decode results from external algorithms

**Return type**
dict

**File Formats:**

- Multiplex format: node1_id layer1_id node2_id layer2_id weight
- Simple format: node1_id node2_id weight

### Notes

- Creates tmp_folder and out_folder if they don't exist
- Nodes are mapped to sequential integers starting from 0
- Layers are mapped to sequential integers starting from 0 (multiplex mode)
- All edges have weight 1 unless explicitly specified

### Examples

Example requires file output - for illustration only.

```
>>> net = multi_layer_network()
>>> # ... build network ...
>>> node_mapping = net.serialize_to_edgelist(
...     edgelist_file='network.txt',
...     multiplex=True
... )
>>> # Use node_mapping to decode results
>>> original_node = node_mapping[0]  # Get original node for id 0
```

**See also**

load_network: Load networks from file save_network: Alternative serialization method

**set_node_attribute**(*node: Any*, *attribute: str*, *value: Any*, *layer: Any | None = None*) → None

Set an attribute value for a given node.

**Parameters**

- **node** (*Any*) – Node identifier.
- **attribute** (*str*) – Name of the attribute to set.
- **value** (*Any*) – Value to assign to the attribute.
- **layer** (*Any, optional*) – Layer identifier. If None and node is a tuple, assumes node is (node_id, layer).

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network(directed=False)
>>> _ = net.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type='list')
>>> net.set_node_attribute('A', 'score', 42.0, 'L1')
```

```
>>> print(net.get_node_attribute('A', 'score', 'L1'))
42.0
```

**sparse_to_px**(*directed=None*)

Convert sparse matrix to py3plex format

**Parameters**

**directed** – Whether the network is directed (uses self.directed if None)

**split_to_layers**(*style='diagonal'*, *compute_layouts='force'*, *layout_parameters=None*, *verbose=True*, *multiplex=False*, *convert_to_simple=False*)

A method for obtaining layerwise distributions

**subnetwork**(*input_list=None*, *subset_by='node_layer_names'*)

Construct a subgraph based on a set of nodes.

**summary**()

Generate a summary of network statistics.

Computes and returns key metrics about the multilayer network structure.

**Returns**

**Network statistics including:**

- 'Number of layers': Count of unique layers
- 'Nodes': Total number of nodes
- 'Edges': Total number of edges
- 'Mean degree': Average node degree
- 'CC': Number of connected components

**Return type**

dict

### Examples

```
>>> net = multi_layer_network()
>>> _ = net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> _ = net.add_edges([{'source': 'A', 'target': 'B',
...                     'source_type': 'layer1', 'target_type': 'layer1'}])
>>> stats = net.summary()
>>> print(f"Network has {stats['Nodes']} nodes and {stats['Edges']} edges")
Network has 2 nodes and 1 edges
```

### Notes

- Connected components are computed on the undirected version
- Mean degree is averaged across all nodes in all layers

**test_scale_free**()

Test the scale-free-nness of the network

**to_homogeneous_hypergraph**()

Transform a multiplex network into a homogeneous graph using incidence gadget encoding.

This method encodes the multiplex structure where each layer is represented by a unique prime number signature. Each edge becomes an edge-node connected to its endpoints and a cycle of length prime-1 that encodes the layer.

**Returns**

- *tuple (H, node_mapping, edge_info)* –

  **H**
  [networkx.Graph] Homogeneous unlabeled graph encoding the multiplex structure.

  **node_mapping**
  [dict] Maps each original node to its vertex-node in H.

  **edge_info**
  [dict] Mapping from each edge-node in H to its (layer, endpoints) tuple.

- *Examples* – Example requires sympy dependency - for illustration only.
- **>>> network = multi_layer_network(directed=False) # doctest** (*+SKIP*)
- **>>> network.add_nodes([{'source'** (*'1', 'type': 'A'}, {'source': '2', 'type': 'A'}], input_type='dict') # doctest: +SKIP*)
- **>>> network.add_edges([{'source'** (*'1', 'target': '2', 'source_type': 'A', 'target_type': 'A'}], input_type='dict') # doctest: +SKIP*)
- **>>> H, node_map, edge_info = network.to_homogeneous_hypergraph() # doctest** (*+SKIP*)
- **>>> print(f"Homogeneous graph has {len(H.nodes())} nodes") # doctest** (*+SKIP*)

### Notes

This transformation uses prime-based signatures to encode layers: - Each layer is assigned a unique prime number (2, 3, 5, 7, ...) - Each edge in layer with prime p is connected to a cycle of length p - The cycle structure uniquely identifies the layer

`to_json()`

A method for exporting the graph to a json file

Args: self

`to_multilayer`(*remove_coupling: bool = True*) → multi_layer_network

Convert a multiplex network to a multilayer network.

Converts from strict multiplex (same nodes in all layers with automatic coupling) to general multilayer (layers can have different node sets).

**Parameters**

`remove_coupling` – If True, removes automatic coupling edges. If False, keeps coupling edges as regular inter-layer edges.

**Returns**

A new multilayer network

**Return type**
multi_layer_network

**Raises**
**ValueError** – If the network is already multilayer type

### Examples

```
>>> net = multi_layer_network(network_type='multiplex', directed=False)
>>> _ = net.add_nodes([
...     {'source': 'A', 'type': 'L1'},
...     {'source': 'A', 'type': 'L2'},
... ])
>>> _ = net.add_edges([
...     {'source': 'A', 'target': 'A',
...      'source_type': 'L1', 'target_type': 'L1'}
... ])
>>> multilayer = net.to_multilayer(remove_coupling=True)
>>> print(multilayer.network_type)
multilayer
```

### Notes

- Node sets remain the same but coupling constraint is relaxed
- Coupling edges can be preserved or removed
- After conversion, layers can independently add/remove nodes

**to_multiplex**(*method: str = 'intersection'*) → multi_layer_network

Convert a multilayer network to a multiplex network.

A multiplex network requires that all layers share the same node set (strict replica model). This method converts a general multilayer network to multiplex by aligning the node sets across layers.

**Parameters**
**method** – Method for aligning node sets across layers: - 'intersection': Keep only nodes present in ALL layers (strict) - 'union': Keep all nodes, add missing nodes to layers (lenient)

**Returns**

**A new multiplex network with aligned node sets**
and automatic coupling edges

**Return type**
multi_layer_network

**Raises**
**ValueError** – If the network is already multiplex type

### Examples

```
>>> net = multi_layer_network(network_type='multilayer', directed=False)
>>> _ = net.add_nodes([
...     {'source': 'A', 'type': 'L1'},
...     {'source': 'B', 'type': 'L1'},
...     {'source': 'A', 'type': 'L2'},
... ])
>>> multiplex = net.to_multiplex(method='intersection')
>>> print(multiplex.network_type)
multiplex
>>> # Only node 'A' is in all layers, so only 'A' remains
>>> sorted(multiplex.get_nodes())
[('A', 'L1'), ('A', 'L2')]
```

### Notes

- 'intersection' method may result in empty network if no nodes are shared
- 'union' method preserves all nodes but may add isolated nodes to layers
- Coupling edges are automatically created between layers

**to_networkx**() → Graph

Convert the multilayer network to a NetworkX graph.

Returns a copy of the core network as a NetworkX graph. The returned graph preserves all node and edge attributes, including layer information for multilayer networks (where nodes are typically (node_id, layer) tuples).

**Returns**
A NetworkX graph (MultiGraph or MultiDiGraph depending on network type)

**Return type**
nx.Graph

### Examples

```
>>> net = multi_layer_network(directed=False)
>>> _ = net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> nx_graph = net.to_networkx()
>>> print(type(nx_graph))
<class 'networkx.classes.multigraph.MultiGraph'>
```

### Notes

- For multilayer networks, nodes are tuples: (node_id, layer)
- All edge attributes (weight, type, etc.) are preserved
- The returned graph is a copy, not a reference

**to_sparse_matrix**(*replace_core=False*, *return_only=False*)

Convert the matrix to scipy-sparse version. This is useful for classification.

**visualize_matrix**(*kwargs=None*)

Plot the matrix – this plots the supra-adjacency matrix

**visualize_network**(*style='diagonal'*, *parameters_layers=None*, *parameters_multiedges=None*, *show=False*, *compute_layouts='force'*, *layouts_parameters=None*, *verbose=True*, *orientation='upper'*, *resolution=0.01*, *axis=None*, *fig=None*, *no_labels=False*, *linewidth=1.7*, *alphachannel=0.3*, *linepoints='-.'*, *legend=False*)

Visualize the multilayer network.

Supports multiple visualization styles: - 'diagonal': Layer-centric diagonal layout with inter-layer edges - 'hairball': Aggregate hairball plot of all layers - 'flow' or 'alluvial': Layered flow visualization with horizontal bands - 'sankey': Sankey diagram showing inter-layer flow strength

**Parameters**

- **style** – Visualization style ('diagonal', 'hairball', 'flow', 'alluvial', or 'sankey')
- **parameters_layers** – Custom parameters for layer drawing
- **parameters_multiedges** – Custom parameters for edge drawing
- **show** – Show plot immediately
- **compute_layouts** – Layout algorithm (currently unused)
- **layouts_parameters** – Layout parameters (currently unused)
- **verbose** – Enable verbose output
- **orientation** – Edge orientation for diagonal style
- **resolution** – Resolution for edge curves
- **axis** – Optional matplotlib axis to draw on
- **fig** – Optional matplotlib figure (currently unused)
- **no_labels** – Hide network labels
- **linewidth** – Width of edge lines
- **alphachannel** – Alpha channel for edge transparency
- **linepoints** – Line style for edges
- **legend** – Show legend (for hairball style)

**Returns**

Matplotlib axis object

**Raises**

**Exception** – If style is not recognized

**Performance Notes:**

For large networks (>500 nodes), visualization performance may degrade: - Layout computation can be slow (O($n^2$) for force-directed layouts) - Rendering many edges is memory and CPU intensive - Consider filtering or sampling for exploratory visualization - Use simpler layouts or increase layout iteration limits

Approximate rendering times on typical hardware: - 100 nodes: <1 second - 500 nodes: 5-10 seconds - 1000 nodes: 30-60 seconds - 5000+ nodes: Several minutes, may run out of memory

**visualize_ricci_core**(*alpha: float = 0.5*, *iterations: int = 10*, *layout_type: str = 'mds'*, *dim: int = 2*, ***kwargs*)

Visualize the aggregated core network using Ricci-flow-based layout.

This method is a high-level wrapper for Ricci-flow-based visualization of the core (aggregated) network. It automatically computes Ricci flow if not already done and creates an informative layout that emphasizes geometric structure and communities.

**Parameters**

- **alpha** – Ollivier-Ricci parameter for flow computation. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **layout_type** – Layout algorithm ("mds", "spring", "spectral"). Default: "mds".
- **dim** – Dimensionality of layout (2 or 3). Default: 2.
- ****kwargs** – Additional arguments passed to visualize_multilayer_ricci_core.

**Returns**

Tuple of (figure, axes, positions_dict).

**Raises**

**RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.

### Examples

Example requires GraphRicciCurvature library - for illustration only.

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.add_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1],
... ], input_type="list")
>>> fig, ax, pos = net.visualize_ricci_core()
>>> import matplotlib.pyplot as plt
>>> plt.show()
```

**See also**

visualize_ricci_layers: Per-layer visualization with Ricci flow visualize_ricci_supra: Supra-graph visualization with Ricci flow

**visualize_ricci_layers**(*layers: List[Any] | None = None*, *alpha: float = 0.5*, *iterations: int = 10*, *layout_type: str = 'mds'*, *share_layout: bool = True*, ***kwargs*)

Visualize individual layers using Ricci-flow-based layouts.

This method creates visualizations of individual layers with layouts derived from

Ricci flow. Layers can share a common coordinate system (for easier comparison) or have independent layouts.

**Parameters**

- **layers** – List of layer identifiers to visualize. If None, uses all layers.
- **alpha** – Ollivier-Ricci parameter. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **layout_type** – Layout algorithm. Default: "mds".
- **share_layout** – If True, use shared coordinates across layers. Default: True.
- ****kwargs** – Additional arguments passed to visualize_multilayer_ricci_layers.

**Returns**

Tuple of (figure, layer_positions_dict).

**Raises**

**RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.

### Examples

```
>>> fig, pos_dict = net.visualize_ricci_layers(
...     arrangement="grid", share_layout=True
... )
>>> import matplotlib.pyplot as plt
>>> plt.show()
```

**See also**

visualize_ricci_core: Core network visualization with Ricci flow visualize_ricci_supra: Supra-graph visualization with Ricci flow

**visualize_ricci_supra**(*alpha: float = 0.5*, *iterations: int = 10*, *layout_type: str = 'mds'*, *dim: int = 2*, ***kwargs*)

Visualize the full supra-graph using Ricci-flow-based layout.

This method visualizes the complete multilayer structure including both intra-layer edges (within layers) and inter-layer edges (coupling between layers) using a layout derived from Ricci flow.

**Parameters**

- **alpha** – Ollivier-Ricci parameter. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **layout_type** – Layout algorithm. Default: "mds".
- **dim** – Dimensionality (2 or 3). Default: 2.
- ****kwargs** – Additional arguments passed to visualize_multilayer_ricci_supra.

**Returns**

Tuple of (figure, axes, positions_dict).

**Raises**

**RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.

### Examples

```
>>> fig, ax, pos = net.visualize_ricci_supra(dim=3)
>>> import matplotlib.pyplot as plt
>>> plt.show()
```

**See also**

visualize_ricci_core: Core network visualization with Ricci flow visualize_ricci_layers: Per-layer visualization with Ricci flow

py3plex.core.multinet.**require**(**args*, ***kwargs*)

py3plex.core.parsers.**ensure**(**args*, ***kwargs*)

py3plex.core.parsers.**load_edge_activity_file**(*fname: str*, *layer_mapping: str | None = None*) → DataFrame

py3plex.core.parsers.**load_edge_activity_raw**(*activity_file: str*, *layer_mappings: dict*) → DataFrame

Basic parser for loading generic activity files. Here, temporal edges are given as tuples -> this can be easily transformed for example into a pandas dataframe!

**Parameters**

- **activity_file** – Path to activity file
- **layer_mappings** – Dictionary mapping layer names to IDs

**Returns**

DataFrame with edge activity data

py3plex.core.parsers.**load_temporal_edge_information**(*input_network: str*, *input_type: str*, *layer_mapping: str | None = None*) → DataFrame | None

py3plex.core.parsers.**parse_detangler_json**(*file_path: str*, *directed: bool = False*) → Tuple[MultiGraph | MultiDiGraph, None]

Parser for generic Detangler files :param file_path: Path to Detangler JSON file :param directed: Whether to create directed graph

py3plex.core.parsers.**parse_edgelist_multi_types**(*input_name: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

Parse an edgelist file with multiple edge types.

Reads a text file where each line represents an edge, optionally with weights and edge types. Lines starting with '#' are treated as comments.

**File Format:**

node1 node2 [weight] [edge_type]

Lines starting with '#' are ignored (comments)

**Parameters**

- **input_name** – Path to edgelist file
- **directed** – Whether to create a directed graph

**Returns**
(parsed_graph, None for labels)

**Return type**
Tuple[Union[nx.MultiGraph, nx.MultiDiGraph], None]

### Notes

- All nodes are assigned to a "null" layer
- Default weight is 1 if not specified
- Edge type is optional (4th column)
- Handles both 2-column (node pairs) and 3+ column formats

### Examples

```
>>> # File content:
>>> # A B 1.0 friendship
>>> # B C 2.0 collaboration
>>> graph, _ = parse_edgelist_multi_types('edges.txt', directed=False)
```

py3plex.core.parsers.**parse_embedding**(*input_name: str*) → Tuple[ndarray, ndarray]

Loader for generic embedding as outputted by GenSim

**Parameters**
**input_name** – Path to embedding file

**Returns**
Tuple of (embedding matrix, embedding indices)

py3plex.core.parsers.**parse_gml**(*file_name: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

Parse a GML network, preserving isolated nodes, attributes and edge keys.

**Parameters**

- **file_name** – Path to GML file
- **directed** – Whether to create directed graph

**Returns**
Tuple of (multigraph, possible labels)

**Contracts:**

- Precondition: file_name must be a non-empty string
- Postcondition: result graph is not None
- Postcondition: result is a MultiGraph or MultiDiGraph

py3plex.core.parsers.**parse_gpickle**(*file_name: str*, *directed: bool = False*, *layer_separator: str | None = None*) → Tuple[MultiGraph | MultiDiGraph, None]

A parser for generic Gpickle as stored by Py3plex.

**Parameters**

- **file_name** – Path to gpickle file
- **directed** – Whether to create directed graph
- **layer_separator** – Optional separator for layer parsing

**Contracts:**

- Precondition: file_name must be a non-empty string
- Postcondition: result graph is not None
- Postcondition: result is a MultiGraph or MultiDiGraph

py3plex.core.parsers.**parse_gpickle_biomine**(*file_name: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

Gpickle parser for biomine graphs :param file_name: Path to gpickle containing BioMine data :param directed: Whether to create directed graph

py3plex.core.parsers.**parse_graphml**(*file_name: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

Read GraphML, using node `type` attributes as replica layers.

Nodes without a type use the GraphML node default, or `generic` if no default is declared. Preserve isolated nodes, graph/node/edge attributes and serialized edge IDs as multigraph keys. Direction conversion follows NetworkX semantics, as with gpickle loading.

py3plex.core.parsers.**parse_matrix**(*file_name: str*, *directed: bool*) → Tuple[Any, Any]

Parser for matrices.

**Parameters**

- **file_name** – Path to .mat file
- **directed** – Whether the graph is directed

**Returns**

Tuple of (network, group) from the .mat file

**Contracts:**

- Precondition: file_name must be a non-empty string
- Postcondition: network must not be None

py3plex.core.parsers.**parse_matrix_to_nx**(*file_name: str*, *directed: bool*) → Graph | DiGraph

Parser for matrices to NetworkX graph.

**Parameters**

- **file_name** – Path to .mat file
- **directed** – Whether to create directed graph

**Returns**

NetworkX Graph or DiGraph

py3plex.core.parsers.**parse_multi_edgelist**(*input_name: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

A generic multiedgelist parser n l n l w :param input_name: Path to text file containing multiedges :param directed: Whether to create directed graph

py3plex.core.parsers.**parse_multiedge_tuple_list**(*network: list*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

Parse a list of edge tuples into a multilayer network.

**Parameters**

- **network** – List of edge tuples (node_first, node_second, layer_first, layer_second, weight)
- **directed** – Whether to create directed graph

py3plex.core.parsers.**parse_multiplex_edges**(*input_name: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None]

Parse a multiplex edgelist file where each line specifies layer and edge.

**File Format:**
layer node1 node2 [weight]

Each line: layer_id source_node target_node [optional_weight]

**Parameters**

- **input_name** – Path to multiplex edgelist file
- **directed** – Whether to create a directed graph

**Returns**
(parsed_graph, None for labels)

**Return type**
Tuple[Union[nx.MultiGraph, nx.MultiDiGraph], None]

### Notes

- Each edge belongs to a specific layer (first column)
- Nodes are represented as (node_id, layer) tuples
- Default weight is 1 if not specified
- All edges have type='default' attribute
- Automatically couples nodes across layers for multiplex structure

### Examples

```
>>> # File content:
>>> # layer1 A B 1.5
>>> # layer2 A B 2.0
>>> # layer1 B C 1.0
>>> graph, _ = parse_multiplex_edges('multiplex.txt', directed=False)
>>> # Creates nodes: (A, layer1), (A, layer2), (B, layer1), etc.
```

py3plex.core.parsers.**parse_multiplex_folder**(*input_folder: str*, *directed: bool*) → Tuple[MultiGraph | MultiDiGraph, None, DataFrame]

Parse a folder containing multiplex network files.

Expects a folder with specific file formats for edges, layers, and optional activity.

**Expected Files:**

- *.edges: Edge information (format: layer_id node1 node2 weight)
- layers.txt: Layer definitions (format: layer_id layer_name)
- activity.txt: Optional temporal activity (format: node1 node2 timestamp layer_name)

**Parameters**

- **input_folder** – Path to folder containing multiplex network files
- **directed** – Whether to create a directed graph

**Returns**

- Union[nx.MultiGraph, nx.MultiDiGraph]: Parsed multilayer graph
- None: Placeholder for labels (not used)
- pd.DataFrame: Time series activity data (empty if no activity.txt)

**Return type**

Tuple containing

### Notes

- Uses glob to find files with specific extensions
- Layer mapping is built from layers.txt
- Activity data is optional and returned as pandas DataFrame
- Nodes are represented as (node_id, layer_id) tuples

### Examples

```
>>> # Folder structure:
>>> # my_network/
>>> #   network.edges
>>> #   layers.txt
>>> #   activity.txt (optional)
>>> graph, _, activity_df = parse_multiplex_folder('my_network/', directed=False)
```

py3plex.core.parsers.**parse_network**(*input_name: str | Any*, *f_type: str = 'gml'*, *directed: bool = False*, *label_delimiter: str | None = None*, *network_type: str = 'multilayer'*) → Tuple[Any, Any, Any]

A wrapper method for available parsers!

**Parameters**

- **input_name** – Path to network file or network object
- **f_type** – Type of file format to parse
- **directed** – Whether to create directed graph

- **label_delimiter** – Optional delimiter for labels
- **network_type** – Type of network (multilayer or multiplex)

**Returns**
Tuple of (parsed_network, labels, time_series)

py3plex.core.parsers.**parse_nx**(*nx_object: Graph*, *directed: bool*) → Tuple[Graph, None]

Core parser for networkx objects.

**Parameters**

- **nx_object** – A networkx graph
- **directed** – Requested graph direction. If it differs from the input, convert using NetworkX semantics while preserving multigraph keys and attributes. Matching inputs are returned unchanged.

**Returns**
Tuple of (graph, None)

**Contracts:**

- Precondition: nx_object must not be None
- Precondition: nx_object must be a NetworkX graph
- Postcondition: result graph is not None

py3plex.core.parsers.**parse_simple_edgelist**(*input_name: str*, *directed: bool*) → Tuple[Graph | DiGraph, None]

Read simple or multilayer edge-list rows.

Accepts `node1 node2 [weight]` (using the `null` layer) and `node1 layer1 node2 layer2 [weight]`, as written by save_edgelist. Missing weights default to 1.0.

**Parameters**

- **input_name** – Path to text file
- **directed** – Whether to create directed graph

py3plex.core.parsers.**parse_spin_edgelist**(*input_name: str*, *directed: bool*) → Tuple[Graph, None]

Parse SPIN format edgelist file.

SPIN format includes node pairs with edge tags and optional weights.

**File Format:**
node1 node2 tag [weight]

Each line: source_node target_node edge_tag [optional_weight]

**Parameters**

- **input_name** – Path to SPIN edgelist file
- **directed** – Whether to create a directed graph

**Returns**
(parsed_graph, None for labels)

**Return type**
Tuple[nx.Graph, None]

### Notes

- Edge tag is stored in edge 'type' attribute
- Default weight is 1 if not specified (4th column)

### Examples

```
>>> # File content:
>>> # A B protein_interaction 0.95
>>> # B C gene_regulation 0.80
>>> graph, _ = parse_spin_edgelist('spin_edges.txt', directed=False)
```

py3plex.core.parsers.**require**(**args*, ***kwargs*)

py3plex.core.parsers.**save_edgelist**(*input_network: Graph*, *output_file: str*, *attributes: bool = False*) → None

Save network to edgelist format.

For multilayer networks (where nodes are tuples of (node_id, layer)), saves in format: node1 layer1 node2 layer2

For regular networks, saves in format: node1 node2

py3plex.core.parsers.**save_gpickle**(*input_network: Any*, *output_file: str*) → None

py3plex.core.parsers.**save_multiedgelist**(*input_network: Any*, *output_file: str*, *attributes: bool = False*, *encode_with_ints: bool = False*) → Tuple[Dict[Any, str], Dict[Any, str]] | None

Save multiedgelist – as n1, l1, n2, l2, w

**Returns**

When encode_with_ints is True, returns tuple of (node_encodings, type_encodings) Otherwise returns None

py3plex.core.converters.**compute_layout**(*network: Graph*, *compute_layouts: str*, *layout_parameters: Dict[str, Any] | None*, *verbose: bool*) → Graph

Compute and normalize layout for a network.

**Parameters**

- **network** – NetworkX graph to compute layout for
- **compute_layouts** – Layout algorithm to use ('force', 'random', 'custom_coordinates')
- **layout_parameters** – Optional parameters for layout algorithms
- **verbose** – Whether to print verbose output

**Returns**

Network with 'pos' attribute added to nodes

**Contracts:**

- Precondition: network must not be None and must have at least one node
- Precondition: compute_layouts must be a valid algorithm name

- Postcondition: all nodes have 'pos' attribute (layout preserves nodes)

py3plex.core.converters.**ensure**(**args*, ***kwargs*)

py3plex.core.converters.**prepare_for_parsing**(*multinet*)

Compute layout for a hairball visualization

**Parameters**
**param1** (*obj*) – multilayer object

**Returns**
(names, prepared network)

**Return type**
tuple

py3plex.core.converters.**prepare_for_visualization**(*multinet: Graph*, *network_type: str = 'multilayer'*, *compute_layouts: str | None = 'force'*, *layout_parameters: Dict[str, Any] | None = None*, *verbose: bool = True*, *multiplex: bool = False*) → Tuple[List[Any], List[Graph], Any]

This functions takes a multilayer object and returns individual layers, their names, as well as multilayer edges spanning over multiple layers.

**Parameters**

- **multinet** – multilayer network object
- **network_type** – "multilayer" or "multiplex"
- **compute_layouts** – Layout algorithm ('force', 'random', etc.) or None to skip layout computation
- **layout_parameters** – Optional layout parameters
- **verbose** – Whether to print progress information
- **multiplex** – Whether to treat as multiplex network

**Returns**

**(layer_names, layer_networks_list, multiedges)**

- layer_names: List of layer names
- layer_networks_list: List of NetworkX graph objects for each layer
- multiedges: Dictionary of edges spanning multiple layers

**Return type**
tuple

py3plex.core.converters.**prepare_for_visualization_hairball**(*multinet*, *compute_layouts=False*)

Compute layout for a hairball visualization

**Parameters**
**param1** (*obj*) – multilayer object

**Returns**
(names, prepared network)

**Return type**
tuple

py3plex.core.converters.**require**(**args*, ***kwargs*)

**class** py3plex.core.random_generators.**SBMMetadata**(*block_memberships: ndarray*, *block_matrix: ndarray*, *node_ids: List[str]*, *layer_names: List[str]*)

Bases: `object`

Ground-truth information for a multilayer SBM sample.

**block_memberships**

Array of shape (n_nodes,) with integer block labels in {0, ..., n_blocks-1}. Shared across layers in this simple multiplex model.

**Type**
np.ndarray

**block_matrix**

Array of shape (n_blocks, n_blocks) with edge probabilities p_ab. Same for all layers in this simple model.

**Type**
np.ndarray

**node_ids**

Node identifiers used in the resulting multi_layer_network (e.g. "v0", "v1", ...).

**Type**
List[str]

**layer_names**

Names of the layers used in the multi_layer_network (e.g. "L0", "L1", ...).

**Type**
List[str]

**block_matrix: ndarray**

**block_memberships: ndarray**

**layer_names: List[str]**

**node_ids: List[str]**

py3plex.core.random_generators.**ensure**(**args*, ***kwargs*)

py3plex.core.random_generators.**random_multilayer_ER**(*n: int*, *l: int*, *p: float*, *directed: bool = False*) → Any

Generate random multilayer Erdős-Rényi network.

This generator creates a general multilayer network where each base node is assigned to exactly one layer. As a result, the number of node-layer pairs is typically `n` rather than `n * l`.

**Parameters**

- **n** – Number of nodes (must be positive)
- **l** – Number of layers (must be positive)

- **p** – Edge probability in [0, 1]
- **directed** – If True, generate directed network

**Returns**
multi_layer_network object

**Contracts:**

- Precondition: n > 0 - must have at least one node
- Precondition: l > 0 - must have at least one layer
- Precondition: 0 <= p <= 1 - probability must be valid
- Postcondition: result is not None - must return valid network

py3plex.core.random_generators.**random_multilayer_SBM**(*n_layers: int*, *n_nodes: int*, *n_blocks: int*, *p_in: float*, *p_out: float*, *coupling: float = 0.0*, *directed: bool = False*, *seed: int | None = None*) → Tuple[multi_layer_network, SBMMetadata]

Generate a simple multiplex multilayer stochastic block model (SBM) network.

This function creates a multilayer network with:

- *n_nodes* nodes shared across all layers,
- *n_layers* layers,
- *n_blocks* latent communities (blocks),
- within-block edge probability *p_in*,
- between-block edge probability *p_out*,
- optional diagonal inter-layer coupling with probability *coupling* between replicas of the same node across layers.

The block memberships are shared across layers in this simple model, and the same block probability matrix is used for all layers.

**Parameters**

- **n_layers** (*int*) – Number of layers.
- **n_nodes** (*int*) – Number of nodes (shared across all layers).
- **n_blocks** (*int*) – Number of blocks (communities).
- **p_in** (*float*) – Edge probability for edges within the same block.
- **p_out** (*float*) – Edge probability for edges between different blocks.
- **coupling** (*float, optional*) – Probability of inter-layer edges between replicas of the same node in different layers. Defaults to 0.0 (no inter-layer edges).
- **directed** (*bool, optional*) – Whether to generate directed layers. Defaults to False.
- **seed** (*int, optional*) – Random seed for reproducibility.

**Returns**

- **network** (*multi_layer_network*) – The generated multilayer network.
- **metadata** (*SBMMetadata*) – Ground-truth metadata containing block memberships and block matrix.

### Notes

- Node IDs are strings "v0", "v1", ..., "v{n_nodes-1}".
- Layer names are strings "L0", "L1", ..., "L{n_layers-1}".
- Edges are inserted via the py3plex list-based API: [src_node, src_layer, dst_node, dst_layer, weight].

### Examples

```
>>> from py3plex.core.random_generators import random_multilayer_SBM
>>> net, meta = random_multilayer_SBM(
...     n_layers=3, n_nodes=20, n_blocks=2,
...     p_in=0.5, p_out=0.05, coupling=0.1, seed=42
... )
>>> print(len(meta.block_memberships))
20
>>> print(len(meta.layer_names))
3
```

py3plex.core.random_generators.**random_multiplex_ER**(*n: int*, *l: int*, *p: float*, *directed: bool = False*) → Any

Generate random multiplex Erdős-Rényi network.

This generator creates a strict multiplex network where all `n` base nodes are instantiated in every one of the `l` layers. Consequently, the node-layer replica set has size `n * l` and the supra-adjacency matrix has shape `(n * l, n * l)`.

**Parameters**

- `n` – Number of nodes (must be positive)
- `l` – Number of layers (must be positive)
- `p` – Edge probability in [0, 1]
- `directed` – If True, generate directed network

**Returns**

multi_layer_network object

**Contracts:**

- Precondition: n > 0 - must have at least one node
- Precondition: l > 0 - must have at least one layer
- Precondition: 0 <= p <= 1 - probability must be valid
- Postcondition: result is not None - must return valid network

py3plex.core.random_generators.**random_multiplex_generator**(*n: int*, *m: int*, *d: float = 0.9*) → MultiGraph

Generate a multiplex network from a random bipartite graph.

**Parameters**

- **n** – Number of nodes (must be positive)
- **m** – Number of layers (must be positive)
- **d** – Layer dropout to avoid cliques, range [0..1] (default: 0.9)

**Returns**

Generated multiplex network as a MultiGraph

**Contracts:**

- Precondition: n > 0 - must have at least one node
- Precondition: m > 0 - must have at least one layer
- Precondition: 0 <= d <= 1 - dropout must be valid probability
- Postcondition: result is not None
- Postcondition: result is a NetworkX MultiGraph

py3plex.core.random_generators.**require**(**args*, ***kwargs*)

Supporting methods for parsers and converters.

This module provides utility functions for network parsing and conversion, including layer splitting, multiplex edge addition, and GAF parsing.

py3plex.core.supporting.**add_mpx_edges**(*input_network: Graph*) → Graph

Add multiplex edges between corresponding nodes across layers.

Multiplex edges connect nodes representing the same entity across different layers of a multilayer network.

**Parameters**

**input_network** – NetworkX graph with multilayer structure.

**Returns**

Network with added multiplex edges between corresponding nodes.

**Contracts:**

- Precondition: input_network must not be None
- Precondition: input_network must be a NetworkX graph
- Postcondition: result is a NetworkX graph

### Example

```
>>> network = nx.Graph()
>>> network.add_node(('A', 'layer1'))
>>> network.add_node(('A', 'layer2'))
>>> network = add_mpx_edges(network)
```

py3plex.core.supporting.**ensure**(**args*, ***kwargs*)

py3plex.core.supporting.**parse_gaf_to_uniprot_GO**(*gaf_mappings: str*, *filter_terms: int | None = None*) → Dict[str, List[str]]

Parse Gene Association File (GAF) to map UniProt IDs to GO terms.

**Parameters**

- **gaf_mappings** – Path to GAF file.
- **filter_terms** – Optional minimum occurrence threshold for GO terms.

**Returns**

Dictionary mapping UniProt IDs to lists of associated GO terms.

### Example

```
>>> mappings = parse_gaf_to_uniprot_GO("gaf_file.gaf", filter_terms=5)
```

py3plex.core.supporting.**require**(**args*, ***kwargs*)

py3plex.core.supporting.**split_to_layers**(*input_network: Graph*) → Dict[Any, Graph]

Split a multilayer network into separate layer subgraphs.

**Parameters**

**input_network** – NetworkX graph containing nodes from multiple layers.

**Returns**

Dictionary mapping layer names to their corresponding subgraphs.

**Contracts:**

- Precondition: input_network must not be None
- Precondition: input_network must be a NetworkX graph
- Postcondition: result is a dictionary
- Postcondition: all values are NetworkX graphs

### Example

```
>>> network = nx.Graph()
>>> network.add_node(('A', 'layer1'))
>>> network.add_node(('B', 'layer2'))
>>> layers = split_to_layers(network)
```

NetworkX compatibility layer for py3plex. This module provides compatibility functions for different NetworkX versions.

py3plex.core.nx_compat.**ensure**(**args*, ***kwargs*)

py3plex.core.nx_compat.**is_string_like**(*obj: Any*) → bool

Check if obj is string-like (compatible with NetworkX < 3.0).

**Parameters**

**obj** – Object to check

**Returns**

True if string-like

**Return type**

bool

py3plex.core.nx_compat.**nx_from_scipy_sparse_matrix**(*A: Any*, *parallel_edges: bool = False*, *create_using: Graph | None = None*, *edge_attribute: str = 'weight'*) → Graph

Create a graph from scipy sparse matrix (compatible with NetworkX < 3.0 and >= 3.0).

**Parameters**

- **A** – scipy sparse matrix
- **parallel_edges** – Whether to create parallel edges (ignored in NetworkX 3.0+)
- **create_using** – Graph type to create
- **edge_attribute** – Edge attribute name for weights

**Returns**
NetworkX graph

**Contracts:**

- Precondition: A must not be None
- Precondition: edge_attribute must be a non-empty string
- Postcondition: returns a NetworkX graph

py3plex.core.nx_compat.**nx_info**(*G: Graph*) → str

Get network information (compatible with NetworkX < 3.0 and >= 3.0).

**Parameters**
**G** – NetworkX graph

**Returns**
Network information

**Return type**
str

**Contracts:**

- Precondition: G must not be None and must be a NetworkX graph
- Postcondition: returns a non-empty string

py3plex.core.nx_compat.**nx_read_gpickle**(*path: str*) → Graph

Read a graph from a pickle file (compatible with NetworkX < 3.0 and >= 3.0).

**Parameters**
**path** – File path

**Returns**
NetworkX graph

**Contracts:**

- Precondition: path must be a non-empty string
- Postcondition: returns a NetworkX graph

py3plex.core.nx_compat.**nx_to_scipy_sparse_matrix**(*G: Graph*, *nodelist: list | None = None*, *dtype: Any | None = None*, *weight: str = 'weight'*, *format: str = 'csr'*) → Any

Convert graph to scipy sparse matrix (compatible with NetworkX < 3.0 and >= 3.0).

**Parameters**

- **G** – NetworkX graph
- **nodelist** – List of nodes
- **dtype** – Data type
- **weight** – Edge weight attribute
- **format** – Sparse matrix format

**Returns**

scipy sparse matrix

**Contracts:**

- Precondition: G must not be None and must be a NetworkX graph
- Precondition: weight must be a non-empty string
- Precondition: format must be a non-empty string
- Postcondition: returns a non-None sparse matrix

py3plex.core.nx_compat.**nx_write_gpickle**(*G: Graph*, *path: str*) → None

Write a graph to a pickle file (compatible with NetworkX < 3.0 and >= 3.0).

**Parameters**

- **G** – NetworkX graph
- **path** – File path

**Contracts:**

- Precondition: G must not be None and must be a NetworkX graph
- Precondition: path must be a non-empty string

py3plex.core.nx_compat.**require**(**args*, ***kwargs*)

### HINMINE Network Decomposition

Modules for heterogeneous information network (HINMINE) decomposition, including I/O helpers.

py3plex.core.HINMINE.decomposition.**aggregate_sum**(*input_thing*, *classes*, *universal_set*)

py3plex.core.HINMINE.decomposition.**aggregate_weighted_sum**(*input_thing*, *classes*, *universal_set*)

py3plex.core.HINMINE.decomposition.**calculate_importance_chi**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using chi-squared weighing. :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_delta**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using delta-idf weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_gr**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using the GR (gain ratio) :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_idf**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using idf weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_ig**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using IG (information gain) weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_okapi**(*classes*, *universal_set*, *linked_nodes*, *n*, *degrees=None*, *avgdegree=None*)

py3plex.core.HINMINE.decomposition.**calculate_importance_rf**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using rf weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_tf**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

Calculates importance of a single midpoint using term frequency weighing. :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set

:return: List of weights of the midpoint for each label in class

py3plex.core.HINMINE.decomposition.**calculate_importance_w2w**(*classes*, *universal_set*, *linked_nodes*, *n*, ***kwargs*)

py3plex.core.HINMINE.decomposition.**calculate_importances**(*midpoints*, *classes*, *universal_set*, *method*, *degrees=None*, *avgdegree=None*)

py3plex.core.HINMINE.decomposition.**chi_value**(*actual_pos_num*, *predicted_pos_num*, *tp*, *n*)

py3plex.core.HINMINE.decomposition.**get_aggregation_method**(*method_name*)

py3plex.core.HINMINE.decomposition.**get_calculation_method**(*method_name*)

py3plex.core.HINMINE.decomposition.**gr_value**(*actual_pos_num*, *predicted_pos_num*, *tp*, *n*)

py3plex.core.HINMINE.decomposition.**hinmine_decompose**(*network*, *heuristic*, *cycle=None*, *parallel=False*)

py3plex.core.HINMINE.decomposition.**hinmine_get_cycles**(*network*, *cycle=None*)

py3plex.core.HINMINE.decomposition.**ig_value**(*actual_pos_num*, *predicted_pos_num*, *tp*, *n*)

py3plex.core.HINMINE.decomposition.**np_calculate_importance_chi**(*predicted*, *label_matrix*, *actual_pos_nums*)

py3plex.core.HINMINE.decomposition.**np_calculate_importance_tf**(*predicted*, *label_matrix*)

py3plex.core.HINMINE.decomposition.**rf_value**(*predicted_pos_num*, *tp*)

py3plex.core.HINMINE.IO.**load_hinmine_object**(*infile*, *label_delimiter='---'*, *weight_tag=False*, *targets=True*)

### Configuration and Utilities

Global configuration objects, shared utilities, and base exceptions.

Centralized configuration for py3plex.

This module provides default settings for visualization, layout algorithms, and other configurable aspects of the library. Users can override these settings by modifying the values after import.

### Example

```
>>> from py3plex import config
>>> config.DEFAULT_NODE_SIZE = 15
>>> config.DEFAULT_EDGE_ALPHA = 0.5
```

py3plex.config.**get_color_palette**(*name: str | None = None*) → List[str]

Get a color palette by name.

**Parameters**
**name** – Palette name. If None, returns the default palette.

**Returns**
List of color hex codes.

**Raises**
**ValueError** – If palette name is not recognized.

### Example

```
>>> from py3plex.config import get_color_palette
>>> colors = get_color_palette("rainbow")
>>> print(colors[0])
'#FF6B6B'
```

py3plex.config.**reset_to_defaults**() → None

Reset all configuration values to their defaults.

This is useful for testing or when you want to start fresh.

### Example

```
>>> from py3plex import config
>>> config.DEFAULT_NODE_SIZE = 20
>>> config.reset_to_defaults()
>>> print(config.DEFAULT_NODE_SIZE)
10
```

Utility functions for py3plex.

This module provides common utilities used across the library, including random state management for reproducibility and deprecation warnings.

py3plex.utils.**deprecated**(*reason: str*, *version: str | None = None*, *alternative: str | None = None*) → Callable[[Callable], Callable]

Decorator to mark functions/methods as deprecated.

This decorator will issue a DeprecationWarning when the decorated function is called, providing information about why it's deprecated and what to use instead.

**Parameters**

- **reason** – Explanation of why the function is deprecated
- **version** – Version in which the function was deprecated (optional)
- **alternative** – Suggested alternative function/method (optional)

**Returns**
Decorator function

### Example

```
>>> @deprecated(
...     reason="This function is obsolete",
...     version="0.95a",
```

```
...     alternative="new_function()"
... )
... def old_function():
...     pass
```

py3plex.utils.**ensure**(**args*, ***kwargs*)

py3plex.utils.**get_data_path**(*relative_path: str*) → str

Get the absolute path to a data file in the repository.

This function searches for data files in multiple locations to support both: - Running examples from a cloned repository - Running scripts/notebooks from any directory with datasets locally available

Search order: 1. Relative to the calling script's directory (for examples in cloned repo) 2. Relative to current working directory (for notebooks/user scripts) 3. Relative to py3plex package location (for editable installs)

**Parameters**

**relative_path** – Path relative to repository root (e.g., "datasets/intact02.gpickle")

**Returns**

Absolute path to the file

**Return type**

str

**Raises**

**Py3plexIOError** – If the file cannot be found in any search location

### Examples

```
>>> from py3plex.utils import get_data_path
>>> path = get_data_path("datasets/intact02.gpickle")
>>> os.path.exists(path)
True
```

**Note**

When py3plex is installed via pip, datasets are not included in the package. Users should either: - Clone the repository and run examples from there - Download datasets separately and place them relative to their scripts - Use current working directory with datasets folder

py3plex.utils.**get_dataset_path**(*filename: str*) → str

Get the absolute path to a dataset file.

Convenience wrapper around get_data_path() specifically for dataset files.

**Parameters**

**filename** – Name or relative path of the dataset file

**Returns**

Absolute path to the dataset file

**Return type**
str

### Examples

```
>>> from py3plex.utils import get_dataset_path
>>> path = get_dataset_path("intact02.gpickle")
>>> os.path.exists(path)
True
```

py3plex.utils.**get_example_image_path**(*filename: str*) → str

Get the absolute path to an example image file.

Convenience wrapper around get_data_path() specifically for example image files.

**Parameters**
**filename** – Name or relative path of the image file

**Returns**
Absolute path to the example image file

**Return type**
str

### Examples

```
>>> from py3plex.utils import get_example_image_path
>>> path = get_example_image_path("intact_10_BH.png")
```

py3plex.utils.**get_layer_names**(*network*) → list

Extract layer names from a multilayer network.

This utility function extracts layer identifiers from network nodes, which are typically stored as (node_id, layer) tuples in multilayer networks.

**Parameters**
**network** – A multilayer network object with a core_network attribute

**Returns**
Sorted list of unique layer names found in the network

**Return type**
list

### Examples

```
>>> from py3plex.utils import get_layer_names
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> net.add_nodes([{'source': 'B', 'type': 'layer2'}])
>>> layers = get_layer_names(net)
>>> 'layer1' in layers and 'layer2' in layers
True
```

**Note**

This function is used by both workflows.py and pipeline.py to avoid code duplication. It safely handles networks where nodes may not follow the expected tuple format.

py3plex.utils.**get_multilayer_dataset_path**(*relative_path: str*) → str

Get the absolute path to a multilayer dataset file.

Convenience wrapper around get_data_path() specifically for multilayer dataset files.

**Parameters**
**relative_path** – Relative path within multilayer_datasets directory

**Returns**
Absolute path to the multilayer dataset file

**Return type**
str

### Examples

```
>>> from py3plex.utils import get_multilayer_dataset_path
>>> path = get_multilayer_dataset_path("MLKing/MLKing2013_multiplex.edges")
```

py3plex.utils.**get_rng**(*seed: int | Generator | None = None*) → Generator

Get a NumPy random number generator with optional seed.

This provides a unified interface for random state management across the library, ensuring reproducibility when a seed is provided.

**Parameters**
**seed** – Random seed for reproducibility. Can be: - None: Use default unseeded generator - int: Seed value for the generator - np.random.Generator: Pass through existing generator

**Returns**
Initialized random number generator

**Return type**
np.random.Generator

### Examples

```
>>> rng = get_rng(42)
>>> rng.random()  # Reproducible random number
0.7739560485559633
```

```
>>> rng1 = get_rng(42)
>>> rng2 = get_rng(42)
>>> rng1.random() == rng2.random()
True
```

```
>>> existing_rng = np.random.default_rng(123)
>>> rng = get_rng(existing_rng)
```

```
>>> rng is existing_rng
True
```

**Contracts:**

- Postcondition: result is a NumPy random Generator

**Note**

Uses numpy.random.Generator (modern API introduced in NumPy 1.17) rather than the legacy numpy.random.RandomState API.

Negative seeds are converted to positive values by taking absolute value to ensure compatibility with NumPy's SeedSequence.

py3plex.utils.**require**(**args*, ***kwargs*)

py3plex.utils.**validate_multilayer_input**(*network_data: Any*) → None

Validate multilayer network input data.

Performs sanity checks on multilayer network structures to catch common errors early.

**Parameters**

**network_data** – Network data to validate (can be various formats)

**Raises**

**ValueError** – If the network data is invalid

**Contracts:**

- Precondition: network_data must not be None

**Example**

```
>>> from py3plex.utils import validate_multilayer_input
>>> validate_multilayer_input(my_network)
```

py3plex.utils.**warn_if_deprecated**(*feature_name: str*, *reason: str*, *alternative: str | None = None*) → None

Issue a deprecation warning for a feature.

This is useful for deprecating specific usage patterns or parameter combinations rather than entire functions.

**Parameters**

- **feature_name** – Name of the deprecated feature
- **reason** – Explanation of why it's deprecated
- **alternative** – Suggested alternative (optional)

### Example

```
>>> def my_function(old_param=None, new_param=None):
...     if old_param is not None:
...         warn_if_deprecated(
...             "old_param",
...             "This parameter is no longer used",
...             "new_param"
...         )
```

Custom exception types for the py3plex library.

This module defines domain-specific exceptions to provide clear error messages and enable better error handling throughout the library.

Py3plex follows Rust's approach to error messages: - Clear, descriptive error messages - Error codes (e.g., PX101, PX201) - Helpful suggestions for fixing issues - "Did you mean?" suggestions for typos - Context showing the relevant location in files

**Example usage:**

```
>>> from py3plex.exceptions import InvalidLayerError
>>> raise InvalidLayerError(
...     "social",
...     available_layers=["work", "family", "social_media"],
...     suggestion="Did you mean 'social_media'?"
... )
```

**exception** py3plex.exceptions.**AlgorithmError**(*message: str*, ***, *algorithm_name: str | None = None*, *valid_algorithms: List[str] | None = None*, ***kwargs*)

Bases: `Py3plexException`

Exception raised when an algorithm execution fails.

Error code: PX301

**default_code: str = 'PX301'**

**exception** py3plex.exceptions.**BenchmarkError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: `Py3plexException`

Exception raised when benchmark execution fails.

Error code: PX302

**default_code: str = 'PX302'**

**exception** py3plex.exceptions.**CentralityComputationError**(*message: str*, ***, *algorithm_name: str | None = None*, *valid_algorithms: List[str] | None = None*, ***kwargs*)

Bases: `AlgorithmError`

Exception raised when centrality computation fails.

Error code: PX301

**default_code: str = 'PX301'**

**exception** py3plex.exceptions.**CommunityDetectionError**(*message: str*, *, *algorithm_name: str | None = None*, *valid_algorithms: List[str] | None = None*, ***kwargs*)

Bases: AlgorithmError

Exception raised when community detection fails.

Error code: PX301

**default_code: str = 'PX301'**

**exception** py3plex.exceptions.**ConversionError**(*message: str*, *, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when format conversion fails.

Error code: PX501

**default_code: str = 'PX501'**

**exception** py3plex.exceptions.**DecompositionError**(*message: str*, *, *algorithm_name: str | None = None*, *valid_algorithms: List[str] | None = None*, ***kwargs*)

Bases: AlgorithmError

Exception raised when network decomposition fails.

Error code: PX301

**default_code: str = 'PX301'**

**exception** py3plex.exceptions.**EmbeddingError**(*message: str*, *, *algorithm_name: str | None = None*, *valid_algorithms: List[str] | None = None*, ***kwargs*)

Bases: AlgorithmError

Exception raised when embedding generation fails.

Error code: PX301

**default_code: str = 'PX301'**

**exception** py3plex.exceptions.**ExternalToolError**(*message: str*, *, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when external tool execution fails.

Error code: PX001

**default_code: str = 'PX001'**

**exception** py3plex.exceptions.**IncompatibleNetworkError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when network format is incompatible with an operation.

Error code: PX304

**default_code: str = 'PX304'**

**exception** py3plex.exceptions.**InvalidEdgeError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when an invalid edge is specified.

Error code: PX203

**default_code: str = 'PX203'**

**exception** py3plex.exceptions.**InvalidLayerError**(*layer_name: str*, ***, *available_layers: List[str] | None = None*, ***kwargs*)

Bases: Py3plexException

Exception raised when an invalid layer is specified.

Error code: PX201

### Example

```
>>> raise InvalidLayerError(
...     "social",
...     available_layers=["work", "family"],
... )
```

**default_code: str = 'PX201'**

**exception** py3plex.exceptions.**InvalidNodeError**(*node_id: str*, ***, *available_nodes: List[str] | None = None*, ***kwargs*)

Bases: Py3plexException

Exception raised when an invalid node is specified.

Error code: PX202

**default_code: str = 'PX202'**

**exception** py3plex.exceptions.**MetaAnalysisError**(*message: str*, *, *hint: str | None = None*, ***kwargs*)

Bases: Py3plexException

Exception raised during meta-analysis operations.

Error code: PX701

Examples of meta-analysis errors: - Missing effect column in query results - Missing standard error without unweighted opt-in - Group-by mismatch across networks - Missing network metadata for subgroup/regression

**default_code: str = 'PX701'**

**exception** py3plex.exceptions.**NetworkConstructionError**(*message: str*, *, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when network construction fails.

Error code: PX208

**default_code: str = 'PX208'**

**exception** py3plex.exceptions.**ParsingError**(*message: str*, *, *file_path: str | None = None*, *line_number: int | None = None*, *expected: str | None = None*, *got: str | None = None*, ***kwargs*)

Bases: Py3plexException

Exception raised when parsing input data fails.

Error code: PX105

**default_code: str = 'PX105'**

**exception** py3plex.exceptions.**Py3plexException**(*message: str*, *, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Exception

Base exception class for all py3plex-specific exceptions.

**code**

Error code (e.g., "PX001")

**suggestions**

List of suggestions for fixing the error

**notes**

Additional context/notes

**did_you_mean**

Suggested correction for typos

**default_code: str = 'PX001'**

**format_message**(*use_color: bool = True*) → str

Format the exception with Rust-style error formatting.

**Parameters**

**use_color** – Whether to use ANSI colors

**Returns**

Formatted error message string

**exception** py3plex.exceptions.**Py3plexFormatError**(*message: str*, *, *valid_formats: List[str] | None = None*, *input_format: str | None = None*, ***kwargs*)

Bases: Py3plexException

Exception raised when input format is invalid or cannot be parsed.

Error code: PX103

**default_code: str = 'PX103'**

**exception** py3plex.exceptions.**Py3plexIOError**(*message: str*, *, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when I/O operations fail (file reading, writing, etc.).

Error code: PX101

**default_code: str = 'PX101'**

**exception** py3plex.exceptions.**Py3plexLayoutError**(*message: str*, *, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: Py3plexException

Exception raised when layout computation or visualization positioning fails.

Error code: PX402

**default_code: str = 'PX402'**

**exception** py3plex.exceptions.**Py3plexMatrixError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: `Py3plexException`

Exception raised when matrix operations fail or matrix is invalid.

Error code: PX001

**default_code: str = 'PX001'**

**exception** py3plex.exceptions.**SemiringError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: `Py3plexException`

Base exception for semiring-related errors.

Error code: PX601

**default_code: str = 'PX601'**

**exception** py3plex.exceptions.**SemiringExecutionError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: `SemiringError`

Exception raised during semiring algorithm execution.

Error code: PX603

**default_code: str = 'PX603'**

**exception** py3plex.exceptions.**SemiringValidationError**(*message: str*, ***, *counterexample: Dict[str, Any] | None = None*, ***kwargs*)

Bases: `SemiringError`

Exception raised when semiring validation fails.

Includes counterexample details when algebraic laws are violated.

Error code: PX602

**default_code: str = 'PX602'**

**exception** py3plex.exceptions.**VisualizationError**(*message: str*, ***, *code: str | None = None*, *suggestions: List[str] | None = None*, *notes: List[str] | None = None*, *did_you_mean: str | None = None*, *context: Dict[str, Any] | None = None*)

Bases: `Py3plexException`

Exception raised when visualization operations fail.

Error code: PX401

**default_code: str = 'PX401'**

### Domain-Specific Language (DSL)

SQL-like query language for selecting and computing properties on multilayer networks.

DSL v2 for Multilayer Network Queries.

This module provides a Domain-Specific Language (DSL) version 2 for querying and analyzing multilayer networks. DSL v2 introduces:

1. Unified AST representation
2. Pythonic builder API (Q, L, Param)
3. Multilayer-specific abstractions (layer algebra, intralayer/interlayer)
4. Improved ergonomics (ORDER BY, LIMIT, EXPLAIN, rich results)
5. Optional progress logging for debugging and monitoring

DSL Extensions (v2.1): 6. Network comparison (C.compare()) 7. Null models (N.model()) 8. Path queries (P.shortest(), P.random_walk()) 9. Plugin system for user-defined operators (@dsl_operator)

**Example Usage:**

```
>>> from py3plex.dsl import Q, L, Param, C, N, P
>>>
>>> # Build a query using the builder API
>>> q = (
...     Q.nodes()
...      .from_layers(L["social"] + L["work"])
...      .where(intralayer=True, degree__gt=5)
...      .compute("betweenness_centrality", alias="bc")
...      .order_by("bc", desc=True)
...      .limit(20)
... )
>>>
>>> # Execute the query (progress logging enabled by default)
>>> result = q.execute(network, k=5)
>>> df = result.to_pandas()
>>>
>>> # Execute without progress logging
>>> result = q.execute(network, progress=False)
>>>
>>> # Compare two networks
>>> comparison = C.compare("baseline", "treatment").using("multiplex_jaccard").execute(
>>>
>>> # Generate null models
>>> nullmodels = N.configuration().samples(100).seed(42).execute(network)
>>>
```

```
>>> # Find paths
>>> paths = P.shortest("Alice", "Bob").crossing_layers().execute(network)
>>>
>>> # Define custom operators
>>> @dsl_operator("my_measure")
... def my_custom_measure(context, param: float = 1.0):
...     # Use context.graph, context.current_layers, etc.
...     return result
```

A separate legacy string syntax also exists (see `py3plex.dsl_legacy` / `execute_query()`), but it is a distinct hand-written parser and executor, not compiled onto this module's AST – it does not support layer set algebra, intralayer/interlayer predicates, per-layer grouping, or coverage. A unified string syntax that compiles onto this AST and is executed by the same engine as the builder API (ensuring consistent behavior across both frontends) is planned but not yet implemented.

**class** py3plex.dsl.**ApproximationSpec**(*enabled: bool = False*, *method: str | None = None*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*, *diagnostics: bool = False*)

Bases: `object`

Specification for approximate computation of a measure.

This allows fast, approximate versions of expensive centrality algorithms.

**enabled**

Whether approximation is enabled

**Type**
bool

**method**

Approximation method name (e.g., "sampling", "landmarks", "power_iteration")

**Type**
str | None

**params**

Method-specific parameters (e.g., n_samples, n_landmarks, tol, max_iter, seed)

**Type**
Dict[str, Any]

**diagnostics**

Whether to compute per-node diagnostic information (e.g., stderr)

**Type**
bool

**diagnostics: bool = False**

**enabled: bool = False**

**method: str | None = None**

**params: Dict[str, Any]**

**class** py3plex.dsl.**AttrType**(*value*)

Bases: `Enum`

Attribute type for static analysis.

`BOOLEAN = 'boolean'`

`CATEGORICAL = 'categorical'`

`DATETIME = 'datetime'`

`EDGE_REF = 'edge_ref'`

`LAYER_REF = 'layer_ref'`

`NODE_REF = 'node_ref'`

`NUMERIC = 'numeric'`

`UNKNOWN = 'unknown'`

`is_comparable`(*other: AttrType*) → bool

Check if this type can be compared with another type.

`supports_operator`(*op: str*) → bool

Check if this type supports a given operator.

`class py3plex.dsl.AutoCommunityConfig`(*enabled: bool = False*, *kind: str = 'communities'*, *seed: int | None = None*, *fast: bool = True*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Configuration for automatic community detection.

Used by Q.communities().auto() and Q.nodes().community_auto() to specify parameters for automatic community detection algorithm selection.

`enabled`

Whether auto community detection is enabled

**Type**
bool

`kind`

Type of query - “communities” (assignment table) or “nodes_join” (annotate nodes)

**Type**
str

`seed`

Random seed for reproducibility

**Type**
int | None

`fast`

Use fast mode with smaller parameter grids

**Type**
bool

`params`

Additional parameters passed to auto_select_community

**Type**
Dict[str, Any]

`enabled: bool = False`

`fast: bool = True`

`kind: str = 'communities'`

`params: Dict[str, Any]`

`seed: int | None = None`

`class py3plex.dsl.BenchmarkAlgorithmSpec(`*algorithm: str*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*, *config_id: str | None = None*`)`

Bases: `object`

Specification for an algorithm in a benchmark.

`algorithm: str`

`config_id: str | None = None`

`params: Dict[str, Any]`

`class py3plex.dsl.BenchmarkNode(`*benchmark_type: str*, *datasets: ~typing.Any*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *algorithm_specs: ~typing.List[~py3plex.dsl.ast.BenchmarkAlgorithmSpec] = <factory>*, *metrics: ~typing.List[str] = <factory>*, *protocol: ~py3plex.dsl.ast.BenchmarkProtocol = <factory>*, *selection_mode: str = 'wins'*, *selection_weights: ~typing.Dict[str, float] | None = None*, *return_trace: bool = True*, *provenance: ~typing.Dict[str, ~typing.Any] = <factory>*`)`

Bases: `object`

AST node for benchmarking queries.

`algorithm_specs: List[BenchmarkAlgorithmSpec]`

`benchmark_type: str`

`datasets: Any`

`layer_expr: LayerExpr | None = None`

`metrics: List[str]`

`protocol: BenchmarkProtocol`

`provenance: Dict[str, Any]`

`return_trace: bool = True`

`selection_mode: str = 'wins'`

`selection_weights: Dict[str, float] | None = None`

class py3plex.dsl.BenchmarkProtocol(*repeat: int = 1*, *seed: int | None = None*, *uq_config: UQConfig | None = None*, *budget_limit_ms: float | None = None*, *budget_limit_evals: int | None = None*, *budget_per: str = 'repeat'*, *n_jobs: int = 1*)

Bases: object

Protocol specification for benchmark execution.

**budget_limit_evals:** int | None = None

**budget_limit_ms:** float | None = None

**budget_per:** str = 'repeat'

**n_jobs:** int = 1

**repeat:** int = 1

**seed:** int | None = None

**uq_config:** UQConfig | None = None

class py3plex.dsl.BenchmarkProxy

Bases: object

Proxy for creating benchmark builders.

Entry point for benchmark DSL via B namespace.

static community() → CommunityBenchmarkBuilder

Create a community detection benchmark builder.

**Returns**

CommunityBenchmarkBuilder

### Example

```
>>> from py3plex.dsl import B
>>> B.community().on(net).algorithms("louvain").execute()
```

static protocol(*repeat: int = 1*, *seed: int | None = None*, *budget_ms: float | None = None*, ***kwargs*) → BenchmarkProtocol

Create a reusable benchmark protocol.

**Parameters**

- **repeat** – Number of repeats
- **seed** – Base seed
- **budget_ms** – Time budget in milliseconds
- ****kwargs** – Additional protocol parameters

**Returns**

BenchmarkProtocol

#### Example

```
>>> protocol = B.protocol(repeat=5, seed=42, budget_ms=10_000)
>>> B.community().on(net).using(protocol).execute()
```

**class** py3plex.dsl.**BooleanExpression**(*condition: ConditionExpr*, *negated: bool = False*)

Bases: `object`

Represents a boolean expression that can be combined with & (AND), | (OR), and ~ (NOT).

This class wraps ConditionExpr AST nodes and provides operator overloading for building complex boolean logic.

**_condition**

The underlying ConditionExpr AST node

**_negated**

Whether this expression is negated

**to_condition_expr**() → ConditionExpr

Convert to AST ConditionExpr.

**Returns**

ConditionExpr that can be used in SelectStmt

**class** py3plex.dsl.**C**

Bases: `object`

Compare factory for creating CompareBuilder instances.

#### Example

```
>>> C.compare("baseline", "intervention").using("multiplex_jaccard")
```

**static compare**(*network_a: str*, *network_b: str*) → CompareBuilder

Create a comparison builder for two networks.

**class** py3plex.dsl.**CacheBackend**

Bases: `ABC`

Abstract base class for cache backends.

**abstract clear**() → None

Clear all cached entries.

**abstract get**(*key: str*) → Any | None

Get value from cache.

**Parameters**

**key** – Cache key

**Returns**

Cached value or None if not found

**abstract get_statistics**() → CacheStatistics

Get cache statistics.

**Returns**

CacheStatistics object

**abstract put**(*key: str*, *value: Any*) → None

Put value into cache.

**Parameters**

- **key** – Cache key
- **value** – Value to cache

**class py3plex.dsl.CachePlan**(*enabled: bool = False*, *entries: ~typing.Dict[int, ~py3plex.dsl.planner.CacheEntry] = <factory>*, *backend_config: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Plan for cache operations.

**enabled**

Whether caching is enabled

**Type**
bool

**entries**

Cache plan per stage (by stage index)

**Type**
Dict[int, py3plex.dsl.planner.CacheEntry]

**backend_config**

Configuration for cache backend

**Type**
Dict[str, Any]

**backend_config: Dict[str, Any]**

**enabled: bool = False**

**entries: Dict[int, CacheEntry]**

**class py3plex.dsl.CacheStatistics**(*hits: int = 0*, *misses: int = 0*, *stores: int = 0*, *evictions: int = 0*, *size_bytes: int = 0*)

Bases: object

Statistics for cache operations.

**hits**

Number of cache hits

**Type**
int

**misses**

Number of cache misses

**Type**
int

**stores**

Number of cache stores

**Type**
int

**evictions**
Number of cache evictions

**Type**
int

**size_bytes**
Estimated cache size in bytes

**Type**
int

**evictions: int = 0**

**property hit_rate: float**
Calculate cache hit rate.

**hits: int = 0**

**misses: int = 0**

**size_bytes: int = 0**

**stores: int = 0**

**to_dict()** → Dict[str, Any]
Convert to dictionary.

**class py3plex.dsl.ClaimLearnerBuilder**
Bases: object

Builder for claim learning queries.

### Example

```
>>> from py3plex.dsl import Q
>>> claims = (Q.learn_claims()
...              .from_metrics(["degree", "pagerank"])
...              .min_support(0.9)
...              .min_coverage(0.05)
...              .seed(42)
...              .execute(net))
```

**autocompute**(*enabled: bool = True*) → ClaimLearnerBuilder
Set whether to autocompute missing metrics.

Note: This is a placeholder for future functionality. Currently, all metrics specified are computed.

**Parameters**
**enabled** – Whether to autocompute (default: True)

**Returns**
Self for chaining

**cheap_metrics**(*metrics: List[str]*) → ClaimLearnerBuilder
Set which metrics to use for antecedents.

**Parameters**
**metrics** – List of metric names for antecedents

**Returns**
Self for chaining

`execute`(*network: Any*) → List[Any]

Execute claim learning.

**Parameters**
**network** – py3plex multi_layer_network object

**Returns**
List of Claim objects, sorted by rank

**Raises**
**ClaimLearningError** – If learning fails

`from_metrics`(*metrics: List[str]*) → ClaimLearnerBuilder

Set metrics to use for claim learning.

**Parameters**
**metrics** – List of metric names (e.g., [“degree”, “pagerank”])

**Returns**
Self for chaining

`layers`(*layer_expr: Any*) → ClaimLearnerBuilder

Set layers to consider.

**Parameters**
**layer_expr** – Layer expression (e.g., L[“social”] + L[“work”])

**Returns**
Self for chaining

`max_antecedents`(*n: int*) → ClaimLearnerBuilder

Set maximum antecedent terms.

MVP: Only 1 is supported. Validation happens at execution.

**Parameters**
**n** – Maximum antecedent terms (must be 1 in MVP)

**Returns**
Self for chaining

`max_claims`(*n: int*) → ClaimLearnerBuilder

Set maximum claims to return.

**Parameters**
**n** – Maximum claims (default: 20)

**Returns**
Self for chaining

`min_coverage`(*coverage: float*) → ClaimLearnerBuilder

Set minimum coverage threshold.

**Parameters**
**coverage** – Minimum coverage (default: 0.05)

**Returns**

Self for chaining

**min_support**(*support: float*) → ClaimLearnerBuilder

Set minimum support threshold.

**Parameters**
**support** – Minimum support (default: 0.9)

**Returns**
Self for chaining

**seed**(*seed: int*) → ClaimLearnerBuilder

Set random seed for determinism.

**Parameters**
**seed** – Random seed (default: 42)

**Returns**
Self for chaining

**target_metrics**(*metrics: List[str]*) → ClaimLearnerBuilder

Set which metrics to use for consequents.

**Parameters**
**metrics** – List of metric names for consequents

**Returns**
Self for chaining

**class** py3plex.dsl.**CommunityBenchmarkBuilder**

Bases: `object`

Builder for community detection benchmarks.

Supports fluent API for benchmarking community detection algorithms with fair budgets and deterministic evaluation.

### Example

```
>>> from py3plex.dsl import B, L
>>>
>>> res = (
...     B.community()
...       .on(network)
...       .layers(L["social"])
...       .algorithms(
...           ("louvain", {"grid": {"resolution": [0.8, 1.0, 1.2]}}),
...           ("autocommunity", {"mode": "pareto"}),
...       )
...       .metrics("modularity", "runtime_ms")
...       .repeat(5, seed=42)
...       .budget(runtime_ms=10_000)
...       .execute()
... )
```

**algorithms**(**algo_specs: str | Tuple[str, dict]*) → CommunityBenchmarkBuilder

Specify algorithms to benchmark.

Supports: - String: “leiden” - Tuple with params: (“leiden”, {“gamma”: 1.0}) - Tuple with grid: (“leiden”, {“grid”: {“gamma”: [0.8, 1.0, 1.2]}}) - AutoCommunity: (“autocommunity”, {“mode”: “pareto”})

**Parameters**
***algo_specs** – Algorithm specifications

**Returns**
Self for chaining

**budget**(*runtime_ms: float | None = None*, *evals: int | None = None*, *per: str = 'repeat'*) → CommunityBenchmarkBuilder

Set budget constraints.

**Parameters**

- **runtime_ms** – Time budget in milliseconds
- **evals** – Evaluation budget
- **per** – Budgeting unit (“dataset”, “repeat”)

**Returns**
Self for chaining

**execute**(***params*) → QueryResult

Execute the benchmark.

**Parameters**
****params** – Additional parameters

**Returns**
QueryResult with benchmark results

**Raises**
**ValueError** – If configuration is invalid

**layers**(*layer_expr: str | LayerExprBuilder | LayerExpr*) → CommunityBenchmarkBuilder

Set the layer expression to evaluate on.

**Parameters**
**layer_expr** – Layer expression (L[“social”], “all”, None)

**Returns**
Self for chaining

**metrics**(**metric_names: str*) → CommunityBenchmarkBuilder

Specify metrics to compute.

**Parameters**
***metric_names** – Metric names (e.g., “modularity”, “runtime_ms”)

**Returns**
Self for chaining

**n_jobs**(*n: int*) → CommunityBenchmarkBuilder

Set number of parallel jobs.

**Parameters**
**n** – Number of jobs

**Returns**

Self for chaining

`on`(*dataset: Any*) → CommunityBenchmarkBuilder

Set the dataset(s) to benchmark on.

**Parameters**
`dataset` – Single network, dataset name, dict of networks, or list

**Returns**
Self for chaining

`on_networks`(*networks: Dict[str, Any]*) → CommunityBenchmarkBuilder

Set multiple named networks to benchmark on.

**Parameters**
`networks` – Dict mapping dataset_id -> network

**Returns**
Self for chaining

`repeat`(*k: int*, *seed: int | None = None*) → CommunityBenchmarkBuilder

Set number of repeats and seed.

**Parameters**

- `k` – Number of repeats
- `seed` – Base seed for reproducibility

**Returns**
Self for chaining

`return_trace`(*enabled: bool = True*) → CommunityBenchmarkBuilder

Enable/disable trace return for AutoCommunity.

**Parameters**
`enabled` – Whether to return traces

**Returns**
Self for chaining

`select`(*mode: str | Tuple[str, Dict[str, float]] = 'wins'*) → CommunityBenchmarkBuilder

Set selection mode for winners.

**Parameters**
`mode` – Selection mode: - “wins”: Count wins across metrics - “pareto”: Pareto front - (“weighted”, {“modularity”: 0.6, “runtime_ms”: -0.4})

**Returns**
Self for chaining

`to_ast`() → BenchmarkNode

Convert to AST node.

**Returns**
BenchmarkNode

`uq`(*method: str = 'seed'*, *n_samples: int = 10*, *ci: float = 0.95*, *seed: int | None = None*, ***kwargs*) → CommunityBenchmarkBuilder

Enable uncertainty quantification.

**Parameters**

- **method** – UQ method (“seed”, “bootstrap”, “perturbation”)
- **n_samples** – Number of UQ samples
- **ci** – Confidence interval level
- **seed** – UQ seed
- ****kwargs** – Additional UQ parameters

**Returns**
Self for chaining

**using**(*protocol: BenchmarkProtocol*) → CommunityBenchmarkBuilder

Set a pre-configured protocol.

**Parameters**
**protocol** – BenchmarkProtocol instance

**Returns**
Self for chaining

**class** py3plex.dsl.**CommunityQueryBuilder**(*autocompute: bool = True*, *partition_name: str = 'default'*, *mode: str | None = None*, *fast: bool | None = None*, *uq: bool | None = None*, *uq_n_samples: int | None = None*, *uq_method: str | None = None*, *seed: int | None = None*, *write_attrs: Dict[str, str] | None = None*, ***kwargs*)

Bases: QueryBuilder

Builder for community queries.

Extends QueryBuilder with community-specific operations. Use Q.communities() to create.

**auto**(*seed: int | None = None*, *fast: bool = True*, ***kwargs*) → CommunityQueryBuilder

Automatically detect communities using the AutoCommunity meta-algorithm.

Returns an assignment table with columns: - node: Node identifier - layer: Layer name (nullable for single-layer networks) - community: Community assignment - confidence: Assignment confidence (0-1) - entropy: Assignment entropy (uncertainty measure) - margin: Margin between top two community assignments - community_size: Size of the assigned community

**Parameters**

- **seed** – Random seed for reproducibility (default: None)
- **fast** – Use fast mode with smaller parameter grids (default: True)
- ****kwargs** – Additional parameters passed to auto_select_community

**Returns**
Self for chaining (execute() will run auto community detection)

### Examples

```
>>> # Basic usage
>>> result = Q.communities().auto(seed=42).execute(network)
>>> df = result.to_pandas()
>>> print(df.columns)  # node, layer, community, confidence, entropy, margin, comm
>>>
>>> # With filtering
>>> result = (
...     Q.communities()
...       .auto(seed=42, fast=True)
...       .where(confidence__gt=0.9, community_size__gt=10)
...       .execute(network)
... )
```

### Notes

- Caching: Auto community detection runs once per execute() call
- UQ: Combine with .uq() to enable uncertainty quantification
- Filtering: All assignment table columns support predicate filters

**auto_select**(*fast: bool = True*, *max_candidates: int = 10*, *seed: int = 0*) → CommunityQueryBuilder

Automatically select best community detection algorithm.

Runs multiple community detection algorithms with parameter grids, evaluates them on multiple quality metrics (bucketed), and selects the winner using a "most wins" decision engine.

**Parameters**

- **fast** – Use fast mode with smaller parameter grids (default: True)
- **max_candidates** – Maximum number of algorithm candidates (default: 10)
- **seed** – Master random seed for reproducibility (default: 0)

**Returns**

Self for chaining (execute() will run auto-selection)

### Examples

```
>>> # Basic auto-selection
>>> result = Q.community().auto_select().execute(network)
>>> print(result.explain())
>>> network.assign_partition(result.partition)
>>>
>>> # With UQ for stability
>>> result = (
...     Q.community()
...       .auto_select(fast=True, seed=42)
...       .uq(method="seed", n_samples=50, seed=42)
...       .execute(network)
```

```
... )
>>> print(result.explain())
>>> print(result.leaderboard)
```

#### Notes

- Combine with .uq() to enable uncertainty quantification
- Returns AutoCommunityResult instead of regular QueryResult
- The winning partition is automatically assigned to the network

**boundary_edges**() → QueryBuilder

Return edges that cross community boundaries.

Returns edges where source and target are in different communities.

**Returns**
EdgeQueryBuilder for boundary edges

#### Example

```
>>> # Get inter-community edges for large communities
>>> Q.communities().where(size__gt=10).boundary_edges()
```

**execute**(*network: Any*, *progress: bool = True*, ***params*)

Execute community query or auto-selection.

**Parameters**

- **network** – Multilayer network object
- **progress** – If True, log progress messages (default: True)
- ****params** – Parameter bindings

**Returns**
AutoCommunityResult if auto_select was called, else QueryResult

**from_partition**(*name: str*) → CommunityQueryBuilder

Select which partition to query.

**Parameters**
**name** – Name of the partition (e.g., “louvain”, “infomap”, “default”)

**Returns**
Self for chaining

#### Example

```
>>> Q.communities().from_partition("louvain")
```

**members**() → QueryBuilder

Return nodes that are members of the selected communities.

This bridges from communities to nodes, allowing you to query the members of filtered communities.

**Returns**
NodeQueryBuilder for members

### Example

```
>>> # Get nodes in large communities
>>> Q.communities().where(size__gt=10).members().compute("degree")
```

**nodes**() → QueryBuilder

Trigger community detection and return a node query builder.

If auto-detection mode is configured (via mode parameter), this method will run auto_select_community, assign the results to the network, and return a QueryBuilder for nodes with community attributes.

**Returns**
QueryBuilder for nodes with community attributes

### Example

```
>>> # Auto-detect communities and query nodes
>>> result = (
...     Q.communities(mode="pareto", fast=False, uq=True, seed=42,
...                   write_attrs={"community_id": "community_id"})
...       .nodes()
...       .node_type("gene")
...       .where(degree__gt=3)
...       .execute(network)
... )
```

**class** py3plex.dsl.**CommunityRecord**(*community_id: ~typing.Any, layer_scope: str, members: ~typing.List[~typing.Tuple[~typing.Any, ~typing.Any]], size: int, intra_edges: int = 0, inter_edges: int = 0, density_intra: float = 0.0, cut_size: int = 0, boundary_edges: int = 0, modularity_contribution: float | None = None, metadata: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Canonical representation of a community in the network.

**community_id**
Stable identifier for this community (int or str)

**Type**
Any

**layer_scope**
Single layer, multi-layer, or aggregated scope

**Type**
str

**members**
List of (node_id, layer) tuples that are members

**Type**
List[Tuple[Any, Any]]

`size`
Number of nodes in the community

**Type**
int

`intra_edges`
Number of edges within the community

**Type**
int

`inter_edges`
Number of edges crossing community boundary

**Type**
int

`density_intra`
Density of connections within community

**Type**
float

`cut_size`
Number of boundary edges (same as inter_edges)

**Type**
int

`modularity_contribution`
Contribution to overall modularity (if available)

**Type**
float | None

`metadata`
Algorithm-specific fields (resolution, seed, etc.)

**Type**
Dict[str, Any]

```
boundary_edges: int = 0

community_id: Any

cut_size: int = 0

density_intra: float = 0.0

inter_edges: int = 0

intra_edges: int = 0

layer_scope: str

members: List[Tuple[Any, Any]]

metadata: Dict[str, Any]
```

**modularity_contribution:** float | None = None

**size:** int

class py3plex.dsl.**CompareBuilder**(*network_a: str*, *network_b: str*)

Bases: object

Builder for COMPARE statements.

### Example

```
>>> from py3plex.dsl import C, L
>>>
>>> result = (
...     C.compare("baseline", "intervention")
...      .using("multiplex_jaccard")
...      .on_layers(L["social"] + L["work"])
...      .measure("global_distance", "layerwise_distance")
...      .execute(networks)
... )
```

**execute**(*networks: Dict[str, Any]*) → ComparisonResult

Execute the comparison.

**Parameters**
**networks** – Dictionary mapping network names to network objects

**Returns**
ComparisonResult with comparison results

**measure**(**measures: str*) → CompareBuilder

Specify which measures to compute.

**Parameters**
***measures** – Measure names (e.g., “global_distance”, “layerwise_distance”)

**Returns**
Self for chaining

**on_layers**(*layer_expr: LayerExprBuilder*) → CompareBuilder

Filter by layers using layer algebra.

**Parameters**
**layer_expr** – Layer expression (e.g., L[“social”] + L[“work”])

**Returns**
Self for chaining

**to**(*target: str*) → CompareBuilder

Set export target.

**Parameters**
**target** – Export format (‘pandas’, ‘json’)

**Returns**
Self for chaining

`to_ast()` → CompareStmt

Export as AST CompareStmt object.

`using(`*metric: str*`)` → CompareBuilder

Set the comparison metric.

**Parameters**
`metric` – Metric name (e.g., "multiplex_jaccard")

**Returns**
Self for chaining

`class py3plex.dsl.CompareStmt(`*network_a: str*, *network_b: str*, *metric_name: str*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *measures: ~typing.List[str] = <factory>*, *export_target: str | None = None*`)`

Bases: `object`

COMPARE statement for network comparison.

**DSL Example:**
COMPARE NETWORK baseline, intervention USING multiplex_jaccard ON LAYER("offline") + LAYER("online") MEASURE global_distance TO pandas

`network_a`

Name/key for first network

**Type**
str

`network_b`

Name/key for second network

**Type**
str

`metric_name`

Comparison metric (e.g., "multiplex_jaccard")

**Type**
str

`layer_expr`

Optional layer expression for filtering

**Type**
py3plex.dsl.ast.LayerExpr | None

`measures`

List of measure types (e.g., ["global_distance", "layerwise_distance"])

**Type**
List[str]

`export_target`

Optional export format

**Type**
str | None

`export_target: str | None = None`

**layer_expr: LayerExpr | None = None**

**measures: List[str]**

**metric_name: str**

**network_a: str**

**network_b: str**

**class py3plex.dsl.Comparison**(*left: str*, *op: str*, *right: str | float | int | ParamRef*)

Bases: `object`

A comparison expression.

**left**

Attribute name (e.g., "degree", "layer")

**Type**
str

**op**

Comparison operator ('>', '>=', '<', '<=', '=', '!=')

**Type**
str

**right**

Value to compare against

**Type**
str | float | int | py3plex.dsl.ast.ParamRef

**left: str**

**op: str**

**right: str | float | int | ParamRef**

**class py3plex.dsl.ComputeItem**(*name: str*, *alias: str | None = None*, *uncertainty: bool = False*, *method: str | None = None*, *n_samples: int | None = None*, *ci: float | None = None*, *bootstrap_unit: str | None = None*, *bootstrap_mode: str | None = None*, *n_null: int | None = None*, *null_model: str | None = None*, *random_state: int | None = None*, *approx: ApproximationSpec | None = None*, *kind: str | None = None*, *uncertainty_explicit: bool | None = None*)

Bases: `object`

A measure to compute.

**name**

Measure name (e.g., 'betweenness_centrality')

**Type**
str

**alias**

Optional alias for the result (e.g., 'bc')

**Type**
str | None

**uncertainty**

Whether to compute uncertainty for this measure

**Type**
bool

**method**

Uncertainty estimation method (e.g., 'bootstrap', 'perturbation', 'null_model')

**Type**
str | None

**n_samples**

Number of samples for uncertainty estimation

**Type**
int | None

**ci**

Confidence interval level (e.g., 0.95 for 95% CI)

**Type**
float | None

**bootstrap_unit**

What to resample for bootstrap: "edges", "nodes", or "layers"

**Type**
str | None

**bootstrap_mode**

Resampling mode: "resample" or "permute"

**Type**
str | None

**n_null**

Number of null model replicates

**Type**
int | None

**null_model**

Null model type: "degree_preserving", "erdos_renyi", "configuration"

**Type**
str | None

**random_state**

Random seed for reproducibility

**Type**
int | None

**approx**

Optional approximation specification for fast approximate computation

**Type**
py3plex.dsl.ast.ApproximationSpec | None

**alias: str | None = None**

**approx:** `ApproximationSpec | None = None`

**bootstrap_mode:** `str | None = None`

**bootstrap_unit:** `str | None = None`

**ci:** `float | None = None`

**kind:** `str | None = None`

**method:** `str | None = None`

**n_null:** `int | None = None`

**n_samples:** `int | None = None`

**name:** `str`

**null_model:** `str | None = None`

**random_state:** `int | None = None`

**property result_name:** `str`

Get the name to use in results (alias or original name).

**uncertainty:** `bool = False`

**uncertainty_explicit:** `bool | None = None`

**class** py3plex.dsl.**ComputeMetric**(*measures: List[str] | None = None*)

Bases: `PhysicalOp`

Wrapper operator — actual computation is handled by the executor.

This stub records *which* metrics were requested so they are visible in plan/explain output.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Execute this operator step and return the resulting rows.

**Parameters**

- **ctx** – Shared execution context.
- **rows** – Input row sequence from the previous operator.

**Returns**

Output row sequence.

**op_name:** `str = 'ComputeMetric'`

Human-readable operator name used in explain output.

**class** py3plex.dsl.**ComputePolicy**(*value*)

Bases: `Enum`

Policy for determining which measures to compute.

**ALL** = `'all'`

**EXPLICIT** = `'explicit'`

**MINIMAL** = `'minimal'`

**exception** py3plex.dsl.**ComputedFieldMisuseError**(*field: str*, *stage: str = 'where'*, *ast_summary: str | None = None*)

Bases: `DSLCompileError`

Exception raised when filtering on a computed field before it's computed.

**field**
The computed field being misused

**stage**
Stage where the misuse occurred

class py3plex.dsl.**ConditionAtom**(*comparison: Comparison | None = None*, *function: FunctionCall | None = None*, *special: SpecialPredicate | None = None*)

Bases: `object`

A single atomic condition.

Exactly one of comparison, function, or special should be non-None.

**comparison**
Simple comparison (e.g., degree > 5)

**Type**
py3plex.dsl.ast.Comparison | None

**function**
Function call (e.g., reachable_from("Alice"))

**Type**
py3plex.dsl.ast.FunctionCall | None

**special**
Special predicate (e.g., intralayer)

**Type**
py3plex.dsl.ast.SpecialPredicate | None

**comparison: Comparison | None = None**

**function: FunctionCall | None = None**

**property is_comparison: bool**

**property is_function: bool**

**property is_special: bool**

**special: SpecialPredicate | None = None**

class py3plex.dsl.**ConditionExpr**(*atoms: ~typing.List[~py3plex.dsl.ast.ConditionAtom] = <factory>*, *ops: ~typing.List[str] = <factory>*)

Bases: `object`

Compound condition expression.

Represents conditions joined by logical operators (AND, OR).

**atoms**
List of condition atoms

**Type**
List[py3plex.dsl.ast.ConditionAtom]

**ops**

List of logical operators ('AND', 'OR') between atoms

**Type**
List[str]

**atoms:** `List[ConditionAtom]`

**ops:** `List[str]`

**class py3plex.dsl.CounterexampleBuilder**

Bases: `object`

Builder for counterexample queries.

### Example

```
>>> from py3plex.dsl import Q
>>> cex = (Q.counterexample()
...           .claim("degree__ge(k) -> pagerank__rank_gt(r)")
...           .params(k=10, r=50)
...           .seed(42)
...           .execute(net))
```

**budget**(*max_tests: int = 200*, *max_witness_size: int = 500*) → CounterexampleBuilder

Set resource budgets.

**Parameters**

- **max_tests** – Maximum violation tests during minimization
- **max_witness_size** – Maximum witness size (nodes)

**Returns**
Self for chaining

**claim**(*claim_str: str*) → CounterexampleBuilder

Set the claim to check.

**Parameters**
**claim_str** – Claim string (e.g., "degree___ge(k) -> pagerank___rank_gt(r)")

**Returns**
Self for chaining

**execute**(*network: Any*) → Any | None

Execute counterexample search.

**Parameters**
**network** – py3plex multi_layer_network object

**Returns**
Counterexample object if found, None otherwise

**Raises**
**CounterexampleNotFound** – If no violation exists

`find_minimal`(*minimal: bool = True*) → CounterexampleBuilder

Set whether to minimize witness.

**Parameters**
`minimal` – Whether to find minimal witness

**Returns**
Self for chaining

`initial_radius`(*radius: int*) → CounterexampleBuilder

Set initial ego subgraph radius.

**Parameters**
`radius` – Ego subgraph radius (default: 2)

**Returns**
Self for chaining

`layers`(*layer_expr: Any*) → CounterexampleBuilder

Set layers to consider.

**Parameters**
`layer_expr` – Layer expression (e.g., L[“social”] + L[“work”])

**Returns**
Self for chaining

`params`(***kwargs*) → CounterexampleBuilder

Set parameter bindings for the claim.

**Parameters**
`**kwargs` – Parameter bindings (e.g., k=10, r=50)

**Returns**
Self for chaining

`seed`(*seed: int*) → CounterexampleBuilder

Set random seed for determinism.

**Parameters**
`seed` – Random seed

**Returns**
Self for chaining

class py3plex.dsl.`CrossingLayersSpec`(*mode: str = 'allowed'*, *penalty: float | None = None*)

Bases: `object`

Specification for cross-layer edge handling.

`mode`

Crossing mode (“allowed”, “forbidden”, “penalty”)

**Type**
str

`penalty`

Optional penalty value (for “penalty” mode)

**Type**
float | None

```
mode: str = 'allowed'

penalty: float | None = None
```

class py3plex.dsl.D

Bases: object

Dynamics simulation factory for DSL-level dynamics.

This provides an alternative entry point to Q.dynamics() with a more simulation-oriented naming convention. Both Q.dynamics() and D.simulate() produce the same DynamicsBuilder instances.

### Example

```
>>> from py3plex.dsl import D, L
>>> from py3plex.dynamics import SIRModel
>>>
>>> # Using model instance
>>> sim = (
...     D.simulate(SIRModel(beta=0.3, gamma=0.1))
...       .on_layers(L["social"] + L["work"])
...       .seed_infections(fraction=0.01)
...       .steps(100)
...       .execute(network)
... )
>>>
>>> # Using model name
>>> sim = (
...     D.simulate("SIS", beta=0.3, mu=0.1)
...       .on_layers(L["contacts"])
...       .seed_infections(0.01)
...       .run(steps=100, replicates=10)
...       .execute(network)
... )
```

**static simulate**(*model_or_name: str | Any*, ***params*) → DynamicsBuilder

Create a dynamics simulation builder.

**Parameters**

- **model_or_name** – Either a model name string (e.g., “SIS”, “SIR”) or a model instance with parameters
- ****params** – Model parameters (only used if model_or_name is a string)

**Returns**

DynamicsBuilder for configuring and running simulations

### Example

```
>>> # From model name
>>> D.simulate("SIS", beta=0.3, mu=0.1).steps(100).execute(net)
```

```
>>> # From model instance (if supported in future)
>>> model = SIRModel(beta=0.3, gamma=0.1)
>>> D.simulate(model).steps(200).execute(net)
```

**exception** py3plex.dsl.**DSLCompileError**(*message: str*, *stage: str | None = None*, *field: str | None = None*, *suggestion: str | None = None*, *ast_summary: str | None = None*, *expected: str | None = None*, *actual: str | None = None*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslError`

Exception raised for DSL compile-time errors with rich diagnostics.

This error provides compiler-quality error messages with: - Stage identification (where error occurred in DSL pipeline) - Field-specific context - Actionable suggestions - AST summary for debugging

**message**
: Error message

**stage**
: DSL stage where error occurred (e.g., 'where', 'compute', 'join')

**field**
: Field name related to the error, if applicable

**suggestion**
: Actionable suggestion to fix the error

**ast_summary**
: Compact AST summary for context

**expected**
: Expected value/type, if applicable

**actual**
: Actual value/type received, if applicable

**class** py3plex.dsl.**DSLExecutionContext**(*graph: ~typing.Any*, *current_layers: ~typing.List[str] | None = None*, *current_nodes: ~typing.List[~typing.Any] | None = None*, *params: ~typing.Mapping[str, ~typing.Any] | None = None*, *cache: ~typing.Dict[str, ~typing.Any] = <factory>*, *debug_counters: ~typing.Dict[str, int] = <factory>*)

Bases: `object`

Execution context passed to DSL operators.

This context object provides operators with access to the network, current selection state, and query parameters.

**graph**
: The underlying multilayer network object

  **Type**
  : Any

**current_layers**

Currently selected layers (None = all layers)

**Type**
List[str] | None

**current_nodes**

Currently selected nodes (None = all nodes)

**Type**
List[Any] | None

**params**

Query parameters (e.g., from Param() in builder API)

**Type**
Mapping[str, Any]

**cache**

Cache for expensive operations (e.g., auto community detection)

**Type**
Dict[str, Any]

**debug_counters**

Counters for testing and debugging

**Type**
Dict[str, int]

**cache: Dict[str, Any]**

**current_layers: List[str] | None = None**

**current_nodes: List[Any] | None = None**

**debug_counters: Dict[str, int]**

**graph: Any**

**params: Mapping[str, Any] = None**

**exception** py3plex.dsl.**DSLExecutionError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: DslExecutionError

Exception raised for DSL execution errors.

**class** py3plex.dsl.**DSLOperator**(*name: str*, *func: Callable[[...], Any]*, *description: str | None = None*, *category: str | None = None*)

Bases: object

Metadata for a registered DSL operator.

**name**

Operator name (normalized)

**Type**
str

**func**

Python callable implementing the operator

**Type**
Callable[[...], Any]

**description**

Optional human-readable description

**Type**
str | None

**category**

Optional category (e.g., "centrality", "dynamics", "io")

**Type**
str | None

**category: str | None = None**

**description: str | None = None**

**func: Callable[[...], Any]**

**name: str**

**exception** py3plex.dsl.**DSLSyntaxError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: DslSyntaxError

Exception raised for DSL syntax errors.

**class** py3plex.dsl.**Diagnostic**(*code: str*, *severity: Literal['error', 'warning', 'info', 'hint']*, *message: str*, *span: Tuple[int, int]*, *suggested_fix: SuggestedFix | None = None*)

Bases: object

A linting diagnostic (error, warning, info, or hint).

**code**

Diagnostic code (e.g., "DSL001", "PERF301")

**Type**
str

**severity**

Severity level

**Type**
Literal['error', 'warning', 'info', 'hint']

**message**

Human-readable message

**Type**
str

**span**

Tuple of (start_index, end_index) in the query string

**Type**
Tuple[int, int]

`suggested_fix`
Optional suggested fix

**Type**
py3plex.dsl.lint.diagnostic.SuggestedFix | None

`code: str`

`message: str`

`severity: Literal['error', 'warning', 'info', 'hint']`

`span: Tuple[int, int]`

`suggested_fix: SuggestedFix | None = None`

`class py3plex.dsl.DistanceSpec(`*name: str = 'js_divergence'*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*`)`

Bases: `object`

Distance/similarity function specification for layer reduction.

`name: str = 'js_divergence'`

`params: Dict[str, Any]`

`exception py3plex.dsl.DslError(`*message: str*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*, *diagnostic: Diagnostic | None = None*`)`

Bases: `Exception`

Base exception for all DSL errors.

`query`
Query string that caused the error

`line`
Line number in query (if applicable)

`column`
Column number in query (if applicable)

`diagnostic`
Optional Diagnostic object with structured error info

`format_message()` → str
Format the error message with context.

`to_diagnostic()` → Diagnostic | None
Get the diagnostic object for this error.

**Returns**
Diagnostic object if available, None otherwise

`exception py3plex.dsl.DslExecutionError(`*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*`)`

Bases: `DslError`

Exception raised for DSL execution errors.

Enhanced with pedagogical guidance for runtime issues.

**exception** py3plex.dsl.**DslMissingMetricError**(*metric: str*, *required_by: str | None = None*, *autocompute_enabled: bool = True*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslError`

Exception raised when a required metric is missing and cannot be autocomputed.

This error occurs when: - A query references a metric that hasn't been computed - Autocompute is disabled or the metric is not autocomputable - The metric is required for an operation (e.g., top_k, where clause)

**metric**

The missing metric name

**required_by**

The operation that requires the metric

**autocompute_enabled**

Whether autocompute was enabled

**exception** py3plex.dsl.**DslSyntaxError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `DslError`

Exception raised for DSL syntax errors.

Enhanced with pedagogical guidance: - What the user likely intended - Why the syntax is invalid - Corrected query examples - Common pitfall notes

**class** py3plex.dsl.**DynamicsBuilder**(*process_name: str*, ***params*)

Bases: `object`

Builder for DYNAMICS statements.

### Example

```
>>> from py3plex.dsl import Q, L
>>>
>>> result = (
...     Q.dynamics("SIS", beta=0.3, mu=0.1)
...      .on_layers(L["contacts"] + L["travel"])
...      .seed(Q.nodes().where(degree__gt=10))
...      .parameters_per_layer({
...          "contacts": {"beta": 0.4},
...          "travel": {"beta": 0.2}
...      })
...      .run(steps=100, replicates=10)
```

```
...       .execute(network)
... )
```

**execute**(*network: Any*) → Any

Execute dynamics simulation.

**Parameters**
**network** – Multilayer network

**Returns**
DynamicsResult with simulation outputs

**on_layers**(*layer_expr: LayerExprBuilder*) → DynamicsBuilder

Filter by layers using layer algebra.

**Parameters**
**layer_expr** – Layer expression (e.g., L[“social”] + L[“work”])

**Returns**
Self for chaining

**parameters_per_layer**(*layer_params: Dict[str, Dict[str, Any]]*) → DynamicsBuilder

Set per-layer parameter overrides.

**Parameters**
**layer_params** – Dictionary mapping layer names to parameter dictionaries

**Returns**
Self for chaining

### Example

```
>>> builder.parameters_per_layer({
...     "contacts": {"beta": 0.3},
...     "travel": {"beta": 0.1}
... })
```

**random_seed**(*seed: int*) → DynamicsBuilder

Set random seed for reproducibility.

**Parameters**
**seed** – Random seed

**Returns**
Self for chaining

**run**(*steps: int = 100*, *replicates: int = 1*, *track: str | List[str] | None = None*, *n_jobs: int = 1*) → DynamicsBuilder

Set execution parameters.

**Parameters**

- **steps** – Number of time steps to simulate
- **replicates** – Number of independent runs
- **track** – Measures to track (“all” or list of specific measures)

- **n_jobs** – Number of parallel jobs (1 = sequential, -1 = all cores). Parallel execution is deterministic: same seed produces identical results regardless of n_jobs value.

**Returns**
Self for chaining

### Example

```
>>> # Sequential execution
>>> builder.run(steps=100, replicates=20, n_jobs=1)
```

```
>>> # Parallel execution (deterministic with same seed)
>>> builder.run(steps=100, replicates=20, n_jobs=4)
```

**seed**(*query_or_fraction: float | QueryBuilder*) → DynamicsBuilder

Set initial seeding for the dynamics.

**Parameters**
**query_or_fraction** – Either a fraction (e.g., 0.01 for 1%) or a QueryBuilder for selecting specific nodes to seed

**Returns**
Self for chaining

### Examples

```
>>> # Seed 1% randomly
>>> builder.seed(0.01)
```

```
>>> # Seed high-degree nodes
>>> builder.seed(Q.nodes().where(degree__gt=10))
```

**seed_infections**(*fraction: float | None = None, nodes: List[Tuple[Any, str]] | None = None*) → DynamicsBuilder

Set initial infections for epidemic models (convenience alias for .seed()).

**Parameters**

- **fraction** – Fraction of nodes to infect randomly (e.g., 0.01 for 1%)
- **nodes** – Specific nodes to infect as list of (node_id, layer) tuples

**Returns**
Self for chaining

### Examples

```
>>> # Infect 1% randomly
>>> builder.seed_infections(fraction=0.01)
```

```
>>> # Infect specific nodes
>>> builder.seed_infections(nodes=[('Alice', 'social'), ('Bob', 'work')])
```

**starting_nodes**(*nodes: List[Tuple[Any, str]]*) → DynamicsBuilder

Set starting nodes for random walk dynamics.

**Parameters**

**nodes** – List of (node_id, layer) tuples to start from

**Returns**

Self for chaining

**Example**

```
>>> builder.starting_nodes([('Alice', 'social'), ('Bob', 'work')])
```

**steps**(*n: int*) → DynamicsBuilder

Set number of simulation steps (convenience alias for .run(steps=n)).

**Parameters**

**n** – Number of time steps

**Returns**

Self for chaining

**Example**

```
>>> builder.steps(100)
```

**to**(*target: str*) → DynamicsBuilder

Set export target.

**Parameters**

**target** – Export format

**Returns**

Self for chaining

**to_ast**() → DynamicsStmt

Export as AST DynamicsStmt object.

**uq**(*ci_level: float = 0.95*, *method: str = 'dynamics_mc'*) → DynamicsBuilder

Enable uncertainty quantification for dynamics results.

Wraps per-replicate summary statistics with confidence intervals.

**Parameters**

- **ci_level** – Confidence interval level (default: 0.95 for 95% CI)
- **method** – UQ method (default: "dynamics_mc" for Monte Carlo across replicates)

**Returns**

Self for chaining

**Example**

```
>>> result = (
...     Q.dynamics("SIS", beta=0.3, mu=0.1)
...      .seed_infections(fraction=0.01)
```

```
...       .run(steps=100, replicates=50)
...       .uq(ci_level=0.95, method="dynamics_mc")
...       .execute(network)
... )
>>> # Summary stats now include CI bounds:
>>> result.mean_peak_time  # {'mean': 45.2, 'ci_low': 40.1, 'ci_high': 50.3}
```

**with_states**(***state_mapping*) → DynamicsBuilder

Explicitly define state labels (optional).

**Parameters**

****state_mapping** – State labels (e.g., S="susceptible", I="infected")

**Returns**

Self for chaining

**Note**

This is optional metadata and doesn't affect execution, but helps with documentation and trajectory queries.

**class** py3plex.dsl.**DynamicsStmt**(*process_name: str*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *seed_query: ~py3plex.dsl.ast.SelectStmt | None = None*, *seed_fraction: float | None = None*, *layer_params: ~typing.Dict[str, ~typing.Dict[str, ~typing.Any]] = <factory>*, *steps: int = 100*, *replicates: int = 1*, *track: ~typing.List[str] = <factory>*, *seed: int | None = None*, *export_target: str | None = None*)

Bases: `object`

DYNAMICS statement for declarative process simulation.

**DSL Example:**

DYNAMICS SIS WITH beta=0.3, mu=0.1 ON LAYER("contacts") + LAYER("travel") SEED FROM nodes WHERE degree > 10 PARAMETERS PER LAYER contacts: {beta=0.4}, travel: {beta=0.2} RUN FOR 100 STEPS, 10 REPLICATES TRACK prevalence, incidence

**process_name**

Name of the process (e.g., "SIS", "SIR", "RANDOM_WALK")

**Type**

str

**params**

Global process parameters (e.g., {"beta": 0.3, "mu": 0.1})

**Type**

Dict[str, Any]

**layer_expr**

Optional layer expression for filtering

**Type**

py3plex.dsl.ast.LayerExpr | None

**seed_query**

Optional SELECT query for seeding initial conditions

**Type**
py3plex.dsl.ast.SelectStmt | None

**seed_fraction**

Optional fraction for random seeding (e.g., 0.01 for 1%)

**Type**
float | None

**layer_params**

Optional per-layer parameter overrides

**Type**
Dict[str, Dict[str, Any]]

**steps**

Number of simulation steps

**Type**
int

**replicates**

Number of independent runs

**Type**
int

**track**

List of measures to track (e.g., [“prevalence”, “incidence”])

**Type**
List[str]

**seed**

Optional random seed for reproducibility

**Type**
int | None

**export_target**

Optional export format

**Type**
str | None

```
export_target:  str | None = None

layer_expr:  LayerExpr | None = None

layer_params:  Dict[str, Dict[str, Any]]

params:  Dict[str, Any]

process_name:  str

replicates:  int = 1
```

`seed: int | None = None`

`seed_fraction: float | None = None`

`seed_query: SelectStmt | None = None`

`steps: int = 100`

`track: List[str]`

`class py3plex.dsl.EdgeFeatureSpec(`*kind: str = 'hadamard'*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*`)`

Bases: `object`

Edge feature operator for embedding pipelines.

`kind: str = 'hadamard'`

`params: Dict[str, Any]`

`class py3plex.dsl.EdgeLayerConstraint(`*kind: str*, *src_layer: str | None = None*, *dst_layer: str | None = None*, *layer: str | None = None*`)`

Bases: `object`

Layer constraint for an edge.

`kind`

Type of constraint ("within", "between", "any")

**Type**
str

`src_layer`

Source layer constraint (for "between")

**Type**
str | None

`dst_layer`

Destination layer constraint (for "between")

**Type**
str | None

`layer`

Layer constraint (for "within")

**Type**
str | None

`static any_layer()` → EdgeLayerConstraint

Create constraint that accepts any edge.

`static between(`*src_layer: str*, *dst_layer: str*`)` → EdgeLayerConstraint

Create constraint for edges between two layers.

`dst_layer: str | None = None`

`kind: str`

**layer:** `str | None = None`

**matches**(*src_layer: str*, *dst_layer: str*) → bool

Check if an edge satisfies this constraint.

**src_layer:** `str | None = None`

**to_dict**() → Dict[str, Any]

Convert to dictionary for serialization.

**static within**(*layer: str*) → EdgeLayerConstraint

Create constraint for edges within a single layer.

**class py3plex.dsl.EntityRef**(*entity_type: str*, *layer: str | None = None*, *attribute: str | None = None*)

Bases: `object`

Reference to an entity (node/edge) in the schema.

**class py3plex.dsl.EvalSpec**(*metrics: ~typing.List[str] = <factory>*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Evaluation metrics specification.

**metrics:** `List[str]`

**params:** `Dict[str, Any]`

**class py3plex.dsl.ExecutionContext**(*network: Any*, *params: Dict[str, Any] | None = None*, *planner_config: Dict[str, Any] | None = None*)

Bases: `object`

Lightweight container for state shared across operators in one plan.

**network**

The multilayer network being queried.

**params**

Bound query parameters.

**planner_config**

Active planner configuration dict.

**attributes**

Mutable dict used to accumulate computed metric values across operators (analogous to a row-set).

**class py3plex.dsl.ExecutionPlan**(*steps: ~typing.List[~py3plex.dsl.ast.PlanStep] = <factory>*, *warnings: ~typing.List[str] = <factory>*)

Bases: `object`

Execution plan for EXPLAIN queries.

**steps**

List of execution steps

**Type**

List[py3plex.dsl.ast.PlanStep]

**warnings**

List of performance or correctness warnings

**Type**
List[str]

**steps: List[PlanStep]**

**warnings: List[str]**

class py3plex.dsl.**ExplainResult**(*ast_summary: str*, *type_info: Dict[str, str]*, *cost_estimate: str*, *diagnostics: List[Diagnostic]*, *plan_steps: List[str]*)

Bases: object

Result of an EXPLAIN query with linting information.

**ast_summary**

Human-readable summary of the AST

**Type**
str

**type_info**

Dictionary mapping node IDs to inferred types

**Type**
Dict[str, str]

**cost_estimate**

Rough cost classification

**Type**
str

**diagnostics**

List of diagnostics from linting

**Type**
List[py3plex.dsl.lint.diagnostic.Diagnostic]

**plan_steps**

List of execution plan steps

**Type**
List[str]

**ast_summary: str**

**cost_estimate: str**

**diagnostics: List[Diagnostic]**

**plan_steps: List[str]**

**type_info: Dict[str, str]**

class py3plex.dsl.**ExplainSpec**(*include: ~typing.List[str] = <factory>, exclude: ~typing.List[str] = <factory>, neighbors_top: int = 10, neighbors_cfg: ~typing.Dict[str, ~typing.Any] = <factory>, community_cfg: ~typing.Dict[str, ~typing.Any] = <factory>, layer_footprint_cfg: ~typing.Dict[str, ~typing.Any] = <factory>, attribution_cfg: ~typing.Dict[str, ~typing.Any] = <factory>, cache: bool = True, as_columns: bool = True, prefix: str = ''*)

Bases: `object`

Specification for attaching explanations to query results.

Explanations provide additional context for each result row (typically nodes), such as community membership, top neighbors, layer footprint, and attribution.

**include**

List of explanation blocks to compute (e.g., ["community", "top_neighbors", "attribution"])

**Type**
List[str]

**exclude**

List of explanation blocks to exclude from defaults

**Type**
List[str]

**neighbors_top**

Maximum number of neighbors to include in top_neighbors

**Type**
int

**neighbors_cfg**

Configuration for neighbor selection (metric, scope, direction)

**Type**
Dict[str, Any]

**community_cfg**

Configuration for community explanations

**Type**
Dict[str, Any]

**layer_footprint_cfg**

Configuration for layer footprint explanations

**Type**
Dict[str, Any]

**attribution_cfg**

Configuration for attribution explanations (Shapley values)

**Type**
Dict[str, Any]

**cache**

Whether to cache intermediate computations (default: True)

**Type**
bool

**as_columns**

Store explanations as top-level columns (default: True)

**Type**
bool

**prefix**

Optional prefix for explanation column names (default: “”)

**Type**
str

### Examples

```
>>> # Basic usage with defaults
>>> ExplainSpec(include=["community", "top_neighbors"])
```

```
>>> # Custom neighbor count
>>> ExplainSpec(include=["top_neighbors"], neighbors_top=5)
```

```
>>> # With attribution
>>> ExplainSpec(
...     include=["attribution"],
...     attribution_cfg={"metric": "pagerank", "levels": ["layer"], "seed": 42}
... )
```

**as_columns:** `bool = True`

**attribution_cfg:** `Dict[str, Any]`

**cache:** `bool = True`

**community_cfg:** `Dict[str, Any]`

**exclude:** `List[str]`

**include:** `List[str]`

**layer_footprint_cfg:** `Dict[str, Any]`

**neighbors_cfg:** `Dict[str, Any]`

**neighbors_top:** `int = 10`

**prefix:** `str = ''`

class py3plex.dsl.**ExportSpec**(*path: str*, *fmt: str = 'csv'*, *columns: ~typing.List[str] | None = None*, *options: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Specification for exporting query results to a file.

Used to declaratively export results as part of the DSL pipeline.

**path**

Output file path

**Type**
str

**fmt**

Format type ('csv', 'json', 'tsv', etc.)

**Type**
str

**columns**

Optional list of columns to include/order

**Type**
List[str] | None

**options**

Additional format-specific options (e.g., delimiter, orient)

**Type**
Dict[str, Any]

### Example

ExportSpec(path='results.csv', fmt='csv', columns=['node', 'score']) ExportSpec(path='output.json', fmt='json', options={'orient': 'records'})

**columns:** `List[str] | None = None`

**fmt:** `str = 'csv'`

**options:** `Dict[str, Any]`

**path:** `str`

class py3plex.dsl.**ExportTarget**(*value*)

Bases: `Enum`

Export target for query results.

**ARROW** = 'arrow'

**NETWORKX** = 'networkx'

**PANDAS** = 'pandas'

class py3plex.dsl.**ExtendedQuery**(*kind: str*, *explain: bool = False*, *select: SelectStmt | None = None*, *compare: CompareStmt | None = None*, *nullmodel: NullModelStmt | None = None*, *path: PathStmt | None = None*, *dynamics: DynamicsStmt | None = None*, *trajectories: TrajectoriesStmt | None = None*, *semiring: SemiringStmt | None = None*, *dsl_version: str = '2.0'*)

Bases: `object`

Extended query supporting multiple statement types.

This extends the basic Query to support COMPARE, NULLMODEL, PATH, DYNAMICS, and TRAJECTORIES statements in addition to SELECT statements.

**kind**

Query type ("select", "compare", "nullmodel", "path", "dynamics", "trajectories", "semiring")

**Type**
str

**explain**

If True, return execution plan instead of results

**Type**
bool

**select**

SELECT statement (if kind == "select")

**Type**
py3plex.dsl.ast.SelectStmt | None

**compare**

COMPARE statement (if kind == "compare")

**Type**
py3plex.dsl.ast.CompareStmt | None

**nullmodel**

NULLMODEL statement (if kind == "nullmodel")

**Type**
py3plex.dsl.ast.NullModelStmt | None

**path**

PATH statement (if kind == "path")

**Type**
py3plex.dsl.ast.PathStmt | None

**dynamics**

DYNAMICS statement (if kind == "dynamics")

**Type**
py3plex.dsl.ast.DynamicsStmt | None

**trajectories**

TRAJECTORIES statement (if kind == "trajectories")

**Type**
py3plex.dsl.ast.TrajectoriesStmt | None

**semiring**

SEMIRING statement (if kind == "semiring")

**Type**
py3plex.dsl.ast.SemiringStmt | None

**dsl_version**

DSL version for compatibility

**Type**
str

`compare: CompareStmt | None = None`

`dsl_version: str = '2.0'`

`dynamics: DynamicsStmt | None = None`

`explain: bool = False`

`kind: str`

`nullmodel: NullModelStmt | None = None`

`path: PathStmt | None = None`

`select: SelectStmt | None = None`

`semiring: SemiringStmt | None = None`

`trajectories: TrajectoriesStmt | None = None`

`class py3plex.dsl.FieldExpression(`*field: str*`)`

Bases: `object`

Represents a field reference that can be compared to values.

This class implements operator overloading to create comparison expressions that are then converted to AST ConditionExpr objects.

`_field`

The field name being referenced

`class py3plex.dsl.FieldProxy`

Bases: `object`

Proxy for creating field expressions via F.field_name syntax.

**This allows for intuitive syntax like:**
F.degree > 5 F.layer == “social” F.is_infected

`class py3plex.dsl.FunctionCall(`*name: str*, *args: ~typing.List[str | float | int | ~py3plex.dsl.ast.ParamRef] = <factory>*`)`

Bases: `object`

A function call in a condition.

`name`

Function name (e.g., “reachable_from”)

**Type**
str

`args`

List of arguments

**Type**
List[str | float | int | py3plex.dsl.ast.ParamRef]

`args: List[str | float | int | ParamRef]`

**name: str**

**exception** py3plex.dsl.**GroupingError**(*message: str*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: DslError

Exception raised when a grouping operation is used incorrectly.

This error is raised when operations that require active grouping (like coverage) are called without proper grouping context.

**class** py3plex.dsl.**InMemoryCacheBackend**(*max_entries: int = 100*, *max_bytes: int = 104857600*)

Bases: CacheBackend

In-memory cache backend with LRU eviction.

This is the default cache backend. It uses an OrderedDict for LRU eviction and tracks cache statistics.

**Parameters**

- **max_entries** – Maximum number of entries (default: 100)
- **max_bytes** – Maximum cache size in bytes (default: 100MB)

**clear**() → None

Clear all cached entries.

**get**(*key: str*) → Any | None

Get value from cache.

**get_statistics**() → CacheStatistics

Get cache statistics.

**put**(*key: str*, *value: Any*) → None

Put value into cache.

**exception** py3plex.dsl.**InvalidGroupAggregateError**(*field: str*, *available_aggregates: List[str] | None = None*, *ast_summary: str | None = None*)

Bases: DSLCompileError

Exception raised when filtering on raw fields after grouping.

**field**

The field being filtered

**available_aggregates**

List of available aggregate functions

**exception** py3plex.dsl.**InvalidJoinKeyError**(*missing_keys: List[str]*, *available_fields: List[str]*, *side: str = 'left'*, *ast_summary: str | None = None*)

Bases: DSLCompileError

Exception raised when join keys are invalid or missing from schema.

**missing_keys**

List of keys not found in the schema

`available_fields`
: List of available fields in the schema

`side`
: Which side of the join has the issue ('left' or 'right')

`class py3plex.dsl.JoinBuilder(`*left: QueryBuilder | QueryResult*, *right: QueryBuilder | QueryResult*, *on: Tuple[str, ...]*, *how: str*, *suffixes: Tuple[str, str]*`)`

Bases: `object`

Builder for join operations.

This class wraps a JoinNode and provides a chainable interface for operations after a join. Join execution is deferred until .execute() is called.

Do not instantiate directly - use QueryBuilder.join() or QueryResult.join().

`compute(`**measures: str*, ***kwargs*`)` → JoinBuilder

Compute measures on joined results.

**Parameters**

- `*measures` – Measure names
- `**kwargs` – Additional compute options (alias, uncertainty, etc.)

**Returns**
Self for chaining

`execute(`*network: Any*, *progress: bool = True*, *explain_plan: bool = False*, *planner: Dict[str, Any] | None = None*, ***params*`)` → QueryResult

Execute the join query.

**Parameters**

- `network` – Multilayer network object
- `progress` – If True, log progress messages
- `explain_plan` – If True, populate result.meta["plan"] with execution plan (default: False)
- `planner` – Optional planner configuration dict (compute_policy, enable_cache, etc.)
- `**params` – Parameter bindings

**Returns**
QueryResult with joined data and provenance

`limit(`*n: int*`)` → JoinBuilder

Limit joined results.

**Parameters**
`n` – Maximum number of results

**Returns**
Self for chaining

`order_by(`**keys: str*, *desc: bool = False*`)` → JoinBuilder

Order joined results.

**Parameters**

- ***keys** – Attribute names to order by
- **desc** – Default sort direction

**Returns**
Self for chaining

**where**(***kwargs*) → JoinBuilder

Filter joined results.

**Parameters**
****kwargs** – Conditions (same as QueryBuilder.where)

**Returns**
Self for chaining

**class py3plex.dsl.JoinNode**(*left: SelectStmt | JoinNode*, *right: SelectStmt | JoinNode*, *on: Tuple[str, ...]*, *how: str = 'inner'*, *suffixes: Tuple[str, str] = ('', '_r')*)

Bases: object

A join operation between two query results.

Represents a relational join between two query pipelines. Joins are row-wise relational joins, not graph merges.

**left**

Left query (SelectStmt or JoinNode)

**Type**
py3plex.dsl.ast.SelectStmt | py3plex.dsl.ast.JoinNode

**right**

Right query (SelectStmt or JoinNode)

**Type**
py3plex.dsl.ast.SelectStmt | py3plex.dsl.ast.JoinNode

**on**

Tuple of column names to join on

**Type**
Tuple[str, ...]

**how**

Join type ('inner', 'left', 'right', 'outer', 'semi', 'anti')

**Type**
str

**suffixes**

Tuple of suffixes for name collisions (left_suffix, right_suffix)

**Type**
Tuple[str, str]

### Example

```
>>> # Join nodes with communities
>>> JoinNode(
...     left=SelectStmt(target=Target.NODES, ...),
...     right=SelectStmt(target=Target.COMMUNITIES, ...),
...     on=("node", "layer"),
...     how="left",
...     suffixes=("", "_comm")
... )
```

**how:** `str = 'inner'`

**left:** `SelectStmt | JoinNode`

**on:** `Tuple[str, ...]`

**provides_fields**(*left_fields: set*, *right_fields: set*) → set

Get fields provided by this join.

**Parameters**

- **left_fields** – Fields from left query
- **right_fields** – Fields from right query

**Returns**

Set of field names after join

**requires_fields**() → set

Get fields required by this join (join keys).

**right:** `SelectStmt | JoinNode`

**suffixes:** `Tuple[str, str] = ('', '_r')`

class py3plex.dsl.**LayerConstraint**(*kind: str*, *value: str | Set[str] | Callable | None = None*)

Bases: `object`

Layer constraint for a node.

**kind**

Type of constraint ("one", "set", "wildcard", "predicate")

**Type**

str

**value**

Layer name, set of layer names, or predicate function

**Type**

str | Set[str] | Callable | None

**kind:** `str`

**matches**(*layer: str*) → bool

Check if a layer satisfies this constraint.

**static one**(*layer: str*) → LayerConstraint

Create constraint for a specific layer.

**static set_of**(*layers: Set[str]*) → LayerConstraint

Create constraint for a set of layers.

**to_dict**() → Dict[str, Any]

Convert to dictionary for serialization.

**value: str | Set[str] | Callable | None = None**

**static wildcard**() → LayerConstraint

Create wildcard constraint (any layer).

**class** py3plex.dsl.**LayerExpr**(*terms: ~typing.List[~py3plex.dsl.ast.LayerTerm] = <factory>*, *ops: ~typing.List[str] = <factory>*)

Bases: object

Layer expression with optional algebra operations.

**Supports:**

- Union: LAYER("a") + LAYER("b")
- Difference: LAYER("a") - LAYER("b")
- Intersection: LAYER("a") & LAYER("b")

**terms**

List of layer terms

**Type**
List[py3plex.dsl.ast.LayerTerm]

**ops**

List of operators between terms ('+', '-', '&')

**Type**
List[str]

**get_layer_names**() → List[str]

Get all layer names referenced in this expression.

**ops: List[str]**

**terms: List[LayerTerm]**

**class** py3plex.dsl.**LayerExprBuilder**(*term: str*)

Bases: object

Builder for layer expressions.

**Supports layer algebra:**

- Union: L["social"] + L["work"]
- Difference: L["social"] - L["bots"]
- Intersection: L["social"] & L["work"]

**class** py3plex.dsl.**LayerFilter**(*allowed_layers: List[str] | None = None*)

Bases: PhysicalOp

Filters rows to only those whose layer is in *allowed_layers*.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Execute this operator step and return the resulting rows.

**Parameters**

- **ctx** – Shared execution context.
- **rows** – Input row sequence from the previous operator.

**Returns**

Output row sequence.

`op_name: str = 'LayerFilter'`

Human-readable operator name used in explain output.

`class py3plex.dsl.LayerProxy`

Bases: `object`

Proxy for creating layer expressions via L["name"] syntax.

**Supports both simple layer names and advanced string expressions:**

- L["social"] -> single layer (backward compatible)
- L["social", "work"] -> union of layers (backward compatible)
- L["* - coupling"] -> string expression with algebra (NEW)
- L["(ppi | gene) & disease"] -> complex expression (NEW)

The proxy automatically detects whether to use the old LayerExprBuilder (for simple names) or the new LayerSet (for expressions with operators).

`static clear_groups()` → None

Clear all defined layer groups.

`static define(`*name: str*, *layer_expr: LayerExprBuilder | LayerSet*`)` → None

Define a named layer group for reuse.

**Parameters**

- **name** – Group name
- **layer_expr** – LayerExprBuilder or LayerSet to associate with the name

### Example

```
>>> bio = L["ppi"] | L["gene"] | L["disease"]
>>> L.define("bio", bio)
>>>
>>> # Later use the group
>>> result = Q.nodes().from_layers(L["bio"]).execute(net)
```

`static list_groups()` → Dict[str, Any]

List all defined layer groups.

**Returns**

Dictionary mapping group names to layer expressions

`class py3plex.dsl.LayerReductionBuilder(`*method: str | None = None*`)`

Bases: `object`

Builder for multilayer layer reduction workflows.

`aggregate(`*mode: str*`)` → LayerReductionBuilder

**criterion**(*name: str*) → LayerReductionBuilder

**distance**(*name: str*, ***kwargs*) → LayerReductionBuilder

**execute**(*network: Any*)

**method**(*name: str*) → LayerReductionBuilder

**min_similarity**(*x: float*) → LayerReductionBuilder

**preserve_interlayer**(*keep: bool = True*) → LayerReductionBuilder

**report**(*level: str = 'full'*) → LayerReductionBuilder

**seed**(*seed: int*) → LayerReductionBuilder

**target_k**(*k: int*) → LayerReductionBuilder

**to_ast**() → ReduceStmt

exception py3plex.dsl.**LayerReductionError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: DslExecutionError

Raised when layer reduction execution fails.

class py3plex.dsl.**LayerReductionResult**(*network: multi_layer_network*, *layer_mapping: dict[str, str]*, *merge_tree: list[Any] | None = None*, *loss_curve: list[dict[str, Any]] | None = None*, *similarity_matrix: ndarray | None = None*, *meta: dict[str, Any] | None = None*, *replay_fn: Callable[[], LayerReductionResult] | None = None*)

Bases: object

Result object for layer reduction workflows.

**property is_replayable: bool**

**property provenance: dict[str, Any] | None**

**replay**() → LayerReductionResult

**report**() → str

**to_dict**() → dict[str, Any]

**to_pandas**()

class py3plex.dsl.**LayerReductionSpec**(*method: str = 'hierarchical_js'*, *target_k: int = 1*, *distance: ~py3plex.dsl.ast.DistanceSpec | None = None*, *criterion: str | None = None*, *min_similarity: float | None = None*, *preserve_interlayer: bool = True*, *aggregate: str = 'sum'*, *report_level: str = 'full'*, *seed: int | None = None*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Layer reduction workflow specification.

**aggregate:** str = 'sum'

**criterion:** str | None = None

**distance:** DistanceSpec | None = None

**method:** str = 'hierarchical_js'

**min_similarity:** float | None = None

**params:** Dict[str, Any]

**preserve_interlayer:** bool = True

**report_level:** str = 'full'

**seed:** int | None = None

**target_k:** int = 1

class py3plex.dsl.**LayerSet**(*name_or_expr: str | LayerExpr*)

Bases: object

Represents an unevaluated layer expression.

LayerSet objects are immutable and composable. They maintain an internal AST representation that is only evaluated when resolve() is called.

**expr**

Internal expression AST

### Example

```
>>> # Create from layer name
>>> social = LayerSet("social")
>>>
>>> # Set operations
>>> both = social | LayerSet("work")
>>> non_coupling = LayerSet("*") - LayerSet("coupling")
>>>
>>> # Parse from string
>>> layers = LayerSet.parse("* - coupling - transport")
>>>
>>> # Resolve to actual layer names
>>> active = layers.resolve(network)
>>> print(active)  # {'social', 'work', 'hobby'}
```

**static clear_groups**() → None

Clear all defined layer groups.

Useful for testing or resetting state.

**static define_group**(*name: str*, *layer_set: LayerSet*) → None

Define a named layer group for reuse.

**Parameters**

- **name** – Group name
- **layer_set** – LayerSet to associate with the name

### Example

```
>>> bio = LayerSet.parse("ppi | gene | disease")
>>> LayerSet.define_group("bio", bio)
>>>
>>> # Later, reference the group
>>> layers = LayerSet("bio") & LayerSet("*")
```

**explain**(*network: Any | None = None*) → str

Generate human-readable explanation of the layer expression.

**Parameters**

**network** – Optional network to resolve against (shows actual layers)

**Returns**

Formatted explanation string

### Example

```
>>> layers = LayerSet("*") - LayerSet("coupling")
>>> print(layers.explain())
LayerSet:
  difference(
    all_layers("*"),
    layer("coupling")
  )
```

**static list_groups**() → Dict[str, LayerSet]

List all defined layer groups.

**Returns**

Dictionary mapping group names to LayerSet objects

**static parse**(*expr_str: str*) → LayerSet

Parse a layer expression from string.

**Supports:**

- Layer names: “social”, “work”
- Wildcard: “*”
- Union: “social | work” or “social + work”
- Intersection: “social & work”
- Difference: “social - work”
- Complement: “~social” (future)
- Parentheses: “(social | work) & ~coupling”
- Named groups: “bio” (if defined via define_group)

**Parameters**
**expr_str** – Expression string to parse

**Returns**
LayerSet object

**Raises**
**DslSyntaxError** – If expression is invalid

### Example

```
>>> layers = LayerSet.parse("* - coupling - transport")
>>> layers = LayerSet.parse("(ppi | gene) & disease")
```

**resolve**(*network: Any*, ***, *strict: bool = False*, *warn_empty: bool = True*) → Set[str]

Resolve the layer expression to a set of actual layer names.

This is where late evaluation happens. The expression is evaluated against the network's actual layers.

**Parameters**

- **network** – Multilayer network object
- **strict** – If True, raise error for unknown layers (default: False)
- **warn_empty** – If True, warn when result is empty (default: True)

**Returns**
Set of layer names (as strings)

**Raises**
**UnknownLayerError** – If strict=True and a referenced layer doesn't exist

### Example

```
>>> layers = LayerSet("social") | LayerSet("work")
>>> active = layers.resolve(network)
>>> print(active)  # {'social', 'work'}
```

**py3plex.dsl.LayerSetExpr**

alias of LayerExpr

**class py3plex.dsl.LayerTerm**(*name: str*)

Bases: object

A single layer reference in a layer expression.

**name**

Layer name (e.g., "social", "work")

**Type**
str

**name: str**

**class py3plex.dsl.Limit**(*n: int = 0*)

Bases: PhysicalOp

Truncates rows to at most *n* items.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Execute this operator step and return the resulting rows.

**Parameters**

- **ctx** – Shared execution context.
- **rows** – Input row sequence from the previous operator.

**Returns**

Output row sequence.

**op_name: str = 'Limit'**

Human-readable operator name used in explain output.

class py3plex.dsl.**LinkPredictionBuilder**

Bases: object

Builder for predictive link prediction workflows.

**by_layer_holdout**(*test_frac: float = 0.2*, *seed: int | None = None*) → LinkPredictionBuilder

**classifier**(*name: str = 'logreg'*, ***kwargs*) → LinkPredictionBuilder

**edge_features**(*kind: str = 'hadamard'*, ***kwargs*) → LinkPredictionBuilder

**eval**(*metrics: List[str]*) → LinkPredictionBuilder

**evaluate**(*metrics: List[str]*) → LinkPredictionBuilder

**execute**(*network: Any*)

**model**(*name: str*, ***kwargs*) → LinkPredictionBuilder

**negative_sampling**(*strategy: str = 'uniform'*, *ratio: float = 1.0*, *seed: int | None = None*, ***kwargs*) → LinkPredictionBuilder

**random_holdout**(*test_frac: float = 0.2*, *seed: int | None = None*) → LinkPredictionBuilder

**scope**(*layers: Any | None = None*, ***kwargs*) → LinkPredictionBuilder

**seed**(*seed: int*) → LinkPredictionBuilder

**split**(*strategy: str = 'random_holdout'*, ***kwargs*) → LinkPredictionBuilder

**temporal_holdout**(*test_frac: float = 0.2*) → LinkPredictionBuilder

**to_ast**() → PredictStmt

**top_k**(*k: int*) → LinkPredictionBuilder

**uq**(*method: str = 'bootstrap'*, *n_samples: int = 50*, *ci: float = 0.95*, *seed: int | None = None*, ***kwargs*) → LinkPredictionBuilder

class py3plex.dsl.LinkPredictionSpec(*layers_expr: ~typing.Any | None = None*, *scope: ~typing.Dict[str, ~typing.Any] = <factory>*, *split: ~py3plex.dsl.ast.SplitSpec = <factory>*, *negative_sampling: ~py3plex.dsl.ast.NegativeSamplingSpec = <factory>*, *model: ~py3plex.dsl.ast.ModelSpec = <factory>*, *edge_features: ~py3plex.dsl.ast.EdgeFeatureSpec = <factory>*, *classifier: ~py3plex.dsl.ast.ModelSpec | None = None*, *eval: ~py3plex.dsl.ast.EvalSpec = <factory>*, *top_k: int | None = None*, *uq_config: ~py3plex.dsl.ast.UQConfig | None = None*, *seed: int | None = None*)

Bases: `object`

Link prediction workflow specification.

**classifier: ModelSpec | None = None**

**edge_features: EdgeFeatureSpec**

**eval: EvalSpec**

**layers_expr: Any | None = None**

**model: ModelSpec**

**negative_sampling: NegativeSamplingSpec**

**scope: Dict[str, Any]**

**seed: int | None = None**

**split: SplitSpec**

**top_k: int | None = None**

**uq_config: UQConfig | None = None**

class py3plex.dsl.LogicalPlan(*ops: ~typing.Tuple[str, ...] = <factory>*, *metadata: ~typing.Mapping[str, ~typing.Any] = <factory>*)

Bases: `object`

An ordered sequence of logical operators.

**ops**

Tuple of logical operator name strings, in execution order.

**Type**
Tuple[str, ...]

**metadata**

Arbitrary key/value metadata about the plan (e.g. `{"required_measures": [...], "cost_hint": "expensive"}`).

**Type**
Mapping[str, Any]

**metadata: Mapping[str, Any]**

**ops:** `Tuple[str, ...]`

**class** py3plex.dsl.**MatchRow**(*bindings: ~typing.Dict[str, ~typing.Any] = <factory>*, *edge_bindings: ~typing.Dict[str, ~typing.Tuple[~typing.Any, ~typing.Any]] | None = None*)

Bases: `object`

Represents a single match result.

**bindings**

Dictionary mapping variable names to node IDs

**Type**
Dict[str, Any]

**edge_bindings**

Optional dictionary mapping edge vars to edge tuples

**Type**
Dict[str, Tuple[Any, Any]] | None

**bindings:** `Dict[str, Any]`

**edge_bindings:** `Dict[str, Tuple[Any, Any]] | None = None`

**to_dict**() → Dict[str, Any]

Convert to dictionary.

**class** py3plex.dsl.**MetaBuilder**(*name: str | None = None*)

Bases: `object`

Builder for meta-analysis queries.

Execution contract (NON-NEGOTIABLE): - .on_networks(...) MUST be called exactly once - .run(...) MUST be called exactly once - .execute() without both MUST raise MetaAnalysisError - Re-calling .on_networks() or .run() overwrites previous values but emits warning

Default model is random-effects unless .model("fixed") is specified.

**allow_unweighted**(*allow: bool = True*) → MetaBuilder

Allow unweighted pooling if SE not available.

**Parameters**
**allow** – Whether to allow unweighted pooling

**Returns**
Self for chaining

**execute**() → MetaResult

Execute meta-analysis.

**Returns**
MetaResult with pooled effects and provenance

**Raises**
**MetaAnalysisError** – If execution contract violated

**meta_regress**(*formula: str*) → MetaBuilder

Enable meta-regression (v1 constrained scope).

**Parameters**
**formula** – Regression formula (e.g., "y ~ x1 + x2")

**Returns**
Self for chaining

**Raises**
**NotImplementedError** – If formula contains unsupported features

**model**(*model_type: str*) → MetaBuilder

Specify meta-analysis model.

**Parameters**
**model_type** – "fixed" or "random"

**Returns**
Self for chaining

**Raises**
**MetaAnalysisError** – If model_type is invalid

**on_networks**(*networks: Dict[str, Any] | List[Any]*) → MetaBuilder

Specify networks for meta-analysis.

**Parameters**
**networks** – Dictionary mapping name -> network, or list of networks

**Returns**
Self for chaining

**Raises**
**MetaAnalysisError** – If networks is empty

**preserve_order**(*preserve: bool = True*) → MetaBuilder

Preserve input order for networks dict (don't sort by key).

**Parameters**
**preserve** – Whether to preserve order

**Returns**
Self for chaining

**run**(*query: Any*, *effect: str*, *se: str | None = None*, *group_by: List[str] | None = None*) → MetaBuilder

Specify query and effect to pool.

**Parameters**

- **query** – DSL v2 QueryBuilder or legacy DSL string
- **effect** – Name of effect column to pool
- **se** – Standard error specification (column name or expression)
- **group_by** – Optional list of grouping columns for node-level pooling

**Returns**
Self for chaining

**seed**(*seed: int*) → MetaBuilder

Set seed for any stochastic operations (optional).

Only fills missing seeds in queries. Never overrides explicit seeds.

**Parameters**
**seed** – Random seed

**Returns**
Self for chaining

**subgroup**(*by: str*) → MetaBuilder
Enable subgroup meta-analysis.

**Parameters**
**by** – Field in network metadata to partition by

**Returns**
Self for chaining

**with_network_meta**(*meta: Dict[str, Dict[str, Any]]*) → MetaBuilder
Provide network-level metadata for subgroup analysis or meta-regression.

**Parameters**
**meta** – Dictionary mapping network_name -> metadata dict

**Returns**
Self for chaining

class py3plex.dsl.**MetaResult**(*pooled_effects: DataFrame*, *network_effects: DataFrame*, *meta_provenance: Dict[str, Any]*, *model: str*, *subgroup_column: str | None = None*)

Bases: object

Result of meta-analysis pooling.

**pooled_effects**
DataFrame with pooled results

**Type**
pandas.core.frame.DataFrame

**network_effects**
DataFrame with per-network effects

**Type**
pandas.core.frame.DataFrame

**meta_provenance**
Aggregated provenance dictionary

**Type**
Dict[str, Any]

**model**
“fixed” or “random”

**Type**
str

**subgroup_column**
Column used for subgroup analysis (if any)

**Type**
str | None

**meta_provenance: Dict[str, Any]**

**model: str**

**network_effects: DataFrame**

**network_table()** → DataFrame

Return per-network effects table.

**Returns**

- network_name
- effect
- se
- weight_fixed
- weight_random (if model is random)
- group keys (if any)
- network metadata fields (if any)
- provenance pointers (ast_hash, etc.)

**Return type**

DataFrame with columns

**pooled_effects: DataFrame**

**subgroup_column: str | None = None**

**to_dict()** → Dict[str, Any]

Convert to JSON-serializable dictionary.

**Returns**

Dictionary with pooled_effects, network_effects, and meta_provenance

**to_pandas()** → DataFrame

Return pooled effects as tidy DataFrame.

**Returns**

- group keys (if any)
- model
- pooled_effect
- pooled_se
- ci_low
- ci_high
- k (number of studies)
- tau2 (random-effects only)
- Q (Cochran's Q)
- I2 (I-squared percentage)
- H (H statistic)
- subgroup (if subgroup analysis)

**Return type**

DataFrame with columns

class py3plex.dsl.MetricSpec(*name: str*, *target: ~typing.Literal['nodes', 'edges', 'any'] = 'nodes'*, *output_type: str = 'float'*, *requires: ~typing.Tuple[str, ...] = <factory>*, *cost_class: ~typing.Literal['constant', 'linear', 'near_linear', 'quadratic', 'cubic', 'unknown'] = 'unknown'*, *supports_uq: bool = False*, *supports_approx: bool = False*, *deterministic: bool = True*, *aliases: ~typing.Tuple[str, ...] = <factory>*)

Bases: `object`

Metadata for a single DSL compute metric.

**name**

Canonical metric name (e.g. `"betweenness_centrality"`).

**Type**
str

**target**

Which query target the metric can be applied to.

**Type**
Literal[‘nodes’, ‘edges’, ‘any’]

**output_type**

Numeric output type descriptor (`"float"` etc.).

**Type**
str

**requires**

Tuple of other metric names that must be computed first.

**Type**
Tuple[str, . . . ]

**cost_class**

Computational cost class.

**Type**
Literal[‘constant’, ‘linear’, ‘near_linear’, ‘quadratic’, ‘cubic’, ‘unknown’]

**supports_uq**

Whether the metric supports uncertainty quantification.

**Type**
bool

**supports_approx**

Whether an approximate variant is available.

**Type**
bool

**deterministic**

Whether the metric always produces the same result on the same graph.

**Type**
bool

**aliases**

Additional names that resolve to this spec.

**Type**
Tuple[str, ...]

**aliases:** `Tuple[str, ...]`

**cost_class:** `Literal['constant', 'linear', 'near_linear', 'quadratic', 'cubic', 'unknown'] = 'unknown'`

**deterministic:** `bool = True`

**is_expensive**() → bool
Return `True` when the metric has quadratic or worse cost.

**name:** `str`

**output_type:** `str = 'float'`

**requires:** `Tuple[str, ...]`

**supports_approx:** `bool = False`

**supports_uq:** `bool = False`

**target:** `Literal['nodes', 'edges', 'any'] = 'nodes'`

class py3plex.dsl.**ModelSpec**(*name: str*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Predictive model specification.

**name:** `str`

**params:** `Dict[str, Any]`

class py3plex.dsl.**N**

Bases: `object`

NullModel factory for creating NullModelBuilder instances.

### Example

```
>>> N.model("configuration").samples(100).seed(42)
```

**static configuration**() → NullModelBuilder
Create a configuration model builder.

**static edge_swap**() → NullModelBuilder
Create an edge swap model builder.

**static erdos_renyi**() → NullModelBuilder
Create an Erdős-Rényi model builder.

**static layer_shuffle**() → NullModelBuilder
Create a layer shuffle model builder.

**static model**(*model_type: str*) → NullModelBuilder
Create a null model builder.

**exception** py3plex.dsl.**NegativeSamplingError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `PredictionTaskError`

Raised for negative sampling failures in predictive tasks.

**class** py3plex.dsl.**NegativeSamplingSpec**(*strategy: str = 'uniform'*, *ratio: float = 1.0*, *seed: int | None = None*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Negative sampling strategy for predictive tasks.

**params:** `Dict[str, Any]`

**ratio:** `float = 1.0`

**seed:** `int | None = None`

**strategy:** `str = 'uniform'`

**class** py3plex.dsl.**NetworkSchemaProvider**(*network: Any*)

Bases: `object`

Schema provider backed by a py3plex multilayer network.

**get_attribute_type**(*entity_ref: EntityRef*, *attr: str*) → AttrType | None

Get attribute type by sampling nodes/edges.

**get_edge_count**(*layer: str | None = None*) → int

Get edge count.

**get_node_count**(*layer: str | None = None*) → int

Get node count.

**list_edge_types**(*layer: str | None = None*) → List[str]

Get list of edge types.

**list_layers**() → List[str]

Get list of all layers.

**list_node_types**(*layer: str | None = None*) → List[str]

Get list of node types.

**class** py3plex.dsl.**NullModelBuilder**(*model_type: str*)

Bases: `object`

Builder for NULLMODEL statements.

### Example

```
>>> from py3plex.dsl import N, L
>>>
>>> result = (
...     N.model("configuration")
...      .on_layers(L["social"])
```

```
...       .with_params(preserve_degree=True)
...       .samples(100)
...       .seed(42)
...       .execute(network)
... )
```

**execute**(*network: Any*) → NullModelResult

Execute null model generation.

**Parameters**
**network** – Multilayer network

**Returns**
NullModelResult with generated samples

**on_layers**(*layer_expr: LayerExprBuilder*) → NullModelBuilder

Filter by layers using layer algebra.

**Parameters**
**layer_expr** – Layer expression

**Returns**
Self for chaining

**samples**(*n: int*) → NullModelBuilder

Set number of samples to generate.

**Parameters**
**n** – Number of samples

**Returns**
Self for chaining

**seed**(*seed: int*) → NullModelBuilder

Set random seed.

**Parameters**
**seed** – Random seed

**Returns**
Self for chaining

**to**(*target: str*) → NullModelBuilder

Set export target.

**Parameters**
**target** – Export format

**Returns**
Self for chaining

**to_ast**() → NullModelStmt

Export as AST NullModelStmt object.

**with_params**(***params*) → NullModelBuilder

Set model parameters.

**Parameters**
****params** – Model parameters

**Returns**

Self for chaining

class py3plex.dsl.NullModelStmt(*model_type: str*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*, *num_samples: int = 1*, *seed: int | None = None*, *export_target: str | None = None*)

Bases: `object`

NULLMODEL statement for generating randomized networks.

**DSL Example:**

NULLMODEL configuration ON LAYER("social") + LAYER("work") WITH preserve_degree=True, preserve_layer_sizes=True SAMPLES 100 SEED 42

`model_type`

Type of null model (e.g., "configuration", "erdos_renyi", "layer_shuffle")

**Type**

str

`layer_expr`

Optional layer expression for filtering

**Type**

py3plex.dsl.ast.LayerExpr | None

`params`

Model parameters

**Type**

Dict[str, Any]

`num_samples`

Number of samples to generate

**Type**

int

`seed`

Optional random seed

**Type**

int | None

`export_target`

Optional export format

**Type**

str | None

`export_target: str | None = None`

`layer_expr: LayerExpr | None = None`

`model_type: str`

`num_samples: int = 1`

`params: Dict[str, Any]`

**seed:** *int | None = None*

**class py3plex.dsl.OrderBy**(*key: str = ''*, *descending: bool = False*)

Bases: PhysicalOp

Sorts *rows* by a named attribute stored in `ctx.attributes`.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Execute this operator step and return the resulting rows.

**Parameters**

- **ctx** – Shared execution context.
- **rows** – Input row sequence from the previous operator.

**Returns**

Output row sequence.

**op_name:** *str = 'OrderBy'*

Human-readable operator name used in explain output.

**class py3plex.dsl.OrderItem**(*key: str*, *desc: bool = False*)

Bases: object

Ordering specification.

**key**

Attribute or computed value to order by

**Type**

str

**desc**

True for descending order, False for ascending

**Type**

bool

**desc:** *bool = False*

**key:** *str*

**class py3plex.dsl.P**

Bases: object

Path factory for creating PathBuilder instances.

### Example

```
>>> P.shortest("Alice", "Bob").crossing_layers()
>>> P.random_walk("Alice").with_params(steps=100, teleport=0.1)
```

**static all_paths**(*source: Any*, *target: Any*) → PathBuilder

Create an all-paths query builder.

**static flow**(*source: Any*, *target: Any*) → PathBuilder

Create a flow analysis query builder.

**static random_walk**(*source: Any*) → PathBuilder

Create a random walk query builder.

**static shortest**(*source: Any*, *target: Any*) → PathBuilder

Create a shortest path query builder.

**class** py3plex.dsl.**Param**

Bases: `object`

Factory for parameter references.

Parameters are placeholders in queries that are bound at execution time.

### Example

```
>>> q = Q.nodes().where(degree__gt=Param.int("k"))
>>> result = q.execute(network, k=5)
```

**static float**(*name: str*) → ParamRef

Create a float parameter reference.

**static int**(*name: str*) → ParamRef

Create an integer parameter reference.

**static ref**(*name: str*) → ParamRef

Create a parameter reference without type hint.

**static str**(*name: str*) → ParamRef

Create a string parameter reference.

**class** py3plex.dsl.**ParamRef**(*name: str*, *type_hint: str | None = None*)

Bases: `object`

Reference to a query parameter.

Parameters are placeholders in queries that are bound at execution time.

**name**

Parameter name (e.g., "k" for :k in DSL)

**Type**
str

**type_hint**

Optional type hint for validation

**Type**
str | None

**name: str**

**type_hint: str | None = None**

**exception** py3plex.dsl.**ParameterMissingError**(*parameter: str*, *provided_params: List[str] | None = None*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslError`

Exception raised when a required parameter is not provided.

**parameter**

The missing parameter name

**provided_params**

List of provided parameter names

**class py3plex.dsl.PathBuilder**(*path_type: str*, *source: Any*, *target: Any | None = None*)

Bases: `object`

Builder for PATH statements.

### Example

```
>>> from py3plex.dsl import P, L
>>>
>>> result = (
...     P.shortest("Alice", "Bob")
...       .on_layers(L["social"] + L["work"])
...       .crossing_layers()
...       .execute(network)
... )
```

**crossing_layers**(*allow: bool = True*) → PathBuilder

Allow or disallow cross-layer paths.

**Parameters**
**allow** – Whether to allow cross-layer paths

**Returns**
Self for chaining

**execute**(*network: Any*) → PathResult

Execute path query.

**Parameters**
**network** – Multilayer network

**Returns**
PathResult with found paths

**limit**(*n: int*) → PathBuilder

Limit number of results.

**Parameters**
**n** – Maximum number of results

**Returns**
Self for chaining

**on_layers**(*layer_expr: LayerExprBuilder*) → PathBuilder

Filter by layers using layer algebra.

**Parameters**
**layer_expr** – Layer expression

**Returns**
Self for chaining

**to**(*target: str*) → PathBuilder

Set export target.

**Parameters**

**target** – Export format

**Returns**
Self for chaining

`to_ast()` → PathStmt

Export as AST PathStmt object.

`with_params(**params)` → PathBuilder

Set additional parameters.

**Parameters**
****params** – Additional parameters

**Returns**
Self for chaining

`class py3plex.dsl.PathStmt(path_type: str, source: str | ~py3plex.dsl.ast.ParamRef, target: str | ~py3plex.dsl.ast.ParamRef | None = None, layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None, cross_layer: bool = False, params: ~typing.Dict[str, ~typing.Any] = <factory>, limit: int | None = None, export_target: str | None = None)`

Bases: `object`

PATH statement for path queries and flow analysis.

**DSL Example:**
PATH SHORTEST FROM "Alice" TO "Bob" ON LAYER("social") + LAYER("work") CROSSING LAYERS LIMIT 10

`path_type`

Type of path query ("shortest", "all", "random_walk", "flow")

**Type**
str

`source`

Source node identifier

**Type**
str | py3plex.dsl.ast.ParamRef

`target`

Optional target node identifier

**Type**
str | py3plex.dsl.ast.ParamRef | None

`layer_expr`

Optional layer expression for filtering

**Type**
py3plex.dsl.ast.LayerExpr | None

`cross_layer`

Whether to allow cross-layer paths

**Type**
bool

**params**

Additional parameters (e.g., max_length, teleport probability)

**Type**
Dict[str, Any]

**limit**

Optional limit on results

**Type**
int | None

**export_target**

Optional export format

**Type**
str | None

**cross_layer: bool = False**

**export_target: str | None = None**

**layer_expr: LayerExpr | None = None**

**limit: int | None = None**

**params: Dict[str, Any]**

**path_type: str**

**source: str | ParamRef**

**target: str | ParamRef | None = None**

class py3plex.dsl.**PatternEdge**(*src: str*, *dst: str*, *directed: bool = False*, *etype: str | None = None*, *predicates: ~typing.List[~py3plex.dsl.patterns.ir.Predicate] = <factory>*, *layer_constraint: ~py3plex.dsl.patterns.ir.EdgeLayerConstraint | None = None*)

Bases: object

Represents an edge between two node variables in a pattern.

**src**

Source variable name

**Type**
str

**dst**

Destination variable name

**Type**
str

**directed**

Whether the edge is directed

**Type**
bool

**etype**
Optional edge type/relation

**Type**
str | None

**predicates**
List of predicates for filtering

**Type**
List[py3plex.dsl.patterns.ir.Predicate]

**layer_constraint**
Optional layer constraint

**Type**
py3plex.dsl.patterns.ir.EdgeLayerConstraint | None

**directed:** `bool = False`

**dst:** `str`

**etype:** `str | None = None`

**layer_constraint:** `EdgeLayerConstraint | None = None`

**predicates:** `List[Predicate]`

**src:** `str`

**to_dict**() → Dict[str, Any]
Convert to dictionary for serialization.

**class** py3plex.dsl.**PatternEdgeBuilder**(*parent: PatternQueryBuilder*, *edge: PatternEdge*)
Bases: `object`

Builder for configuring a pattern edge.

**any_layer**() → PatternQueryBuilder
Allow edge to be in any layer.

**between_layers**(*src_layer: str*, *dst_layer: str*) → PatternQueryBuilder
Constrain edge to be between two specific layers.

**where**(***kwargs*) → PatternQueryBuilder
Add predicates to the edge.

**within_layer**(*layer: str*) → PatternQueryBuilder
Constrain edge to be within a single layer.

**class** py3plex.dsl.**PatternGraph**(*nodes: ~typing.Dict[str, ~py3plex.dsl.patterns.ir.PatternNode] = <factory>, edges: ~typing.List[~py3plex.dsl.patterns.ir.PatternEdge] = <factory>, constraints: ~typing.List[~py3plex.dsl.patterns.ir.NotEqualConstraint | ~py3plex.dsl.patterns.ir.AllDistinctConstraint] = <factory>, return_vars: ~typing.List[str] | None = None*)
Bases: `object`

Represents a complete pattern query.

**nodes**

Dictionary mapping variable names to PatternNode objects

**Type**
Dict[str, py3plex.dsl.patterns.ir.PatternNode]

**edges**

List of PatternEdge objects

**Type**
List[py3plex.dsl.patterns.ir.PatternEdge]

**constraints**

List of global constraints (e.g., all-different)

**Type**
List[py3plex.dsl.patterns.ir.NotEqualConstraint | py3plex.dsl.patterns.ir.AllDistinctConstraint]

**return_vars**

List of variables to return (defaults to all)

**Type**
List[str] | None

**add_constraint**(*constraint: str*) → None

Add a global constraint. Parses string form into structured constraint.

**add_edge**(*edge: PatternEdge*) → None

Add an edge to the pattern.

**add_node**(*node: PatternNode*) → None

Add a node to the pattern.

**constraints: List[NotEqualConstraint | AllDistinctConstraint]**

**edges: List[PatternEdge]**

**get_return_vars**() → List[str]

Get the list of variables to return.

**nodes: Dict[str, PatternNode]**

**return_vars: List[str] | None = None**

**to_dict**() → Dict[str, Any]

Convert to dictionary for serialization.

**class** py3plex.dsl.**PatternNode**(*var: str*, *labels: ~typing.Set[str] | None = None*, *predicates: ~typing.List[~py3plex.dsl.patterns.ir.Predicate] = <factory>*, *layer_constraint: ~py3plex.dsl.patterns.ir.LayerConstraint | None = None*)

Bases: object

Represents a node variable in a pattern.

**var**

Variable name (e.g., “a”, “b”)

**Type**
str

**labels**

Optional semantic labels (metadata only in v1)

**Type**
Set[str] | None

**predicates**

List of predicates for filtering

**Type**
List[py3plex.dsl.patterns.ir.Predicate]

**layer_constraint**

Optional layer constraint

**Type**
py3plex.dsl.patterns.ir.LayerConstraint | None

**labels: Set[str] | None = None**

**layer_constraint: LayerConstraint | None = None**

**predicates: List[Predicate]**

**to_dict()** → Dict[str, Any]

Convert to dictionary for serialization.

**var: str**

**class py3plex.dsl.PatternNodeBuilder**(*parent: PatternQueryBuilder*, *var: str*, *labels: str | List[str] | None = None*)

Bases: object

Builder for configuring a pattern node variable.

**in_layers**(*layers: str | List[str]*) → PatternQueryBuilder

Specify layer constraint for the node.

**label**(**labels: str*) → PatternNodeBuilder

Add labels to the node.

**where**(***kwargs*) → PatternQueryBuilder

Add predicates to the node.

**class py3plex.dsl.PatternPlan**(*pattern: ~py3plex.dsl.patterns.ir.PatternGraph*, *root_var: str*, *join_order: ~typing.List[~py3plex.dsl.patterns.compiler.JoinStep] = <factory>*, *variable_plans: ~typing.Dict[str, ~py3plex.dsl.patterns.compiler.VariablePlan] = <factory>*, *estimated_complexity: int = -1*, *injective: bool = True*)

Bases: object

Complete execution plan for a pattern.

**estimated_complexity: int = -1**

**injective: bool = True**

**join_order: List[JoinStep]**

**pattern:** `PatternGraph`

**root_var:** `str`

**to_dict**() → Dict[str, Any]

Convert to dictionary for serialization/display.

**variable_plans:** `Dict[str, VariablePlan]`

**class** py3plex.dsl.**PatternQueryBuilder**

Bases: `object`

Main builder for pattern queries.

**constraint**(*expr: str*) → PatternQueryBuilder

Add a global constraint.

**Supported forms:**

- "a != b"
- "all_distinct(a, b, c)"

**edge**(*src: str*, *dst: str*, *directed: bool = False*, *etype: str | None = None*) → PatternEdgeBuilder

Add an edge between two node variables.

**execute**(*network: Any*, *backend: str = 'native'*, *max_matches: int | None = None*, *timeout: float | None = None*, *injective: bool = True*) → PatternQueryResult

Execute the pattern query on a network.

**explain**() → Dict[str, Any]

Generate and return the compilation plan.

**limit**(*n: int*) → PatternQueryBuilder

Limit the number of matches.

**node**(*var: str*, *labels: str | List[str] | None = None*) → PatternNodeBuilder

Add a node variable to the pattern.

**order_by**(*key: str*, *desc: bool = False*) → PatternQueryBuilder

Order matches by a computed attribute (future enhancement).

**path**(*vars: List[str] | Tuple[str, ...]*, *directed: bool = False*, *etype: str | None = None*, *length: int | None = None*) → PatternQueryBuilder

Add a path pattern.

**returning**(**vars: str*) → PatternQueryBuilder

Specify which variables to return in results.

**triangle**(*a: str*, *b: str*, *c: str*, *directed: bool = False*) → PatternQueryBuilder

Add a triangle motif.

**class** py3plex.dsl.**PatternQueryResult**(*pattern: PatternGraph*, *matches: List[MatchRow]*, *meta: Dict[str, Any] | None = None*)

Bases: `object`

Result container for pattern matching queries.

Provides access to matches with multiple export formats and projections.

**pattern**
Original pattern graph

**matches**
List of MatchRow objects

**meta**
Metadata about the query execution

**property count: int**
Get number of matches.

**Returns**
Number of matches

**filter**(*predicate*) → PatternQueryResult
Filter matches using a predicate function.

**Parameters**
**predicate** – Function that takes a MatchRow and returns bool

**Returns**
New PatternQueryResult with filtered matches

**limit**(*n: int*) → PatternQueryResult
Limit the number of matches.

**Parameters**
**n** – Maximum number of matches

**Returns**
New PatternQueryResult with limited matches

**property rows: List[Dict[str, Any]]**
Get matches as list of dictionaries.

**Returns**
List of dictionaries mapping variable names to node IDs

**to_edges**(*var_pairs: List[Tuple[str, str]] | None = None*) → List[Tuple[Any, Any]]
Extract edges from matches.

Infers edges from pairs of node variables in the pattern.

**Parameters**
**var_pairs** – Optional list of (src_var, dst_var) tuples to extract If None, uses pattern edges

**Returns**
List of (src_node, dst_node) tuples

**to_nodes**(*vars: List[str] | None = None*, *unique: bool = True*) → List[Any] | Set[Any]
Extract node IDs from matches.

**Parameters**

- **vars** – Optional list of variables to include (defaults to all)
- **unique** – If True, return unique nodes as a set

**Returns**
List or set of node IDs

`to_pandas`(*include_meta: bool = False*)

Export matches to pandas DataFrame.

**Parameters**
`include_meta` – If True, include metadata columns

**Returns**
pandas.DataFrame with matches

**Raises**
`ImportError` – If pandas is not available

`to_subgraph`(*network: Any*, *per_match: bool = False*) → Any

Extract induced subgraph(s) from matches.

**Parameters**

- `network` – Original network object
- `per_match` – If True, return list of subgraphs (one per match) If False, return single subgraph with all matched nodes

**Returns**
NetworkX graph or list of graphs

`class py3plex.dsl.PhysicalOp`

Bases: `ABC`

Abstract base for physical operators.

Sub-classes implement `execute()` which takes an `ExecutionContext` and an input *rows* sequence (list of node/edge identifiers or similar items) and returns a (possibly transformed) output sequence.

`abstract execute`(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Execute this operator step and return the resulting rows.

**Parameters**

- `ctx` – Shared execution context.
- `rows` – Input row sequence from the previous operator.

**Returns**
Output row sequence.

`op_name: str = 'PhysicalOp'`

Human-readable operator name used in explain output.

`class py3plex.dsl.PhysicalPlan`(*ops: ~typing.Tuple[str, ...] = <factory>*, *metadata: ~typing.Mapping[str, ~typing.Any] = <factory>*)

Bases: `object`

An ordered sequence of physical operators.

`ops`

Tuple of physical operator name strings, in execution order.

**Type**
Tuple[str, ...]

`metadata`

Arbitrary key/value metadata (planner config, timings, ...).

**Type**
Mapping[str, Any]

`metadata: Mapping[str, Any]`

`ops: Tuple[str, ...]`

`class py3plex.dsl.PlanStep(`*description: str*, *estimated_complexity: str | None = None*`)`

Bases: `object`

A step in the execution plan.

`description`

Human-readable description of the step

**Type**
str

`estimated_complexity`

Estimated time complexity (e.g., `"O(n)"` where n is node count)

**Type**
str | None

`description: str`

`estimated_complexity: str | None = None`

`class py3plex.dsl.PlannedQuery(`*planned_stages: ~typing.List[~py3plex.dsl.planner.Stage] = <factory>*, *required_measures: ~typing.Set[str] = <factory>*, *cache_plan: ~py3plex.dsl.planner.CachePlan = <factory>*, *plan_meta: ~typing.Dict[str, ~typing.Any] = <factory>*, *ast_hash: str = ''*, *plan_hash: str = ''*`)`

Bases: `object`

A fully planned query ready for execution.

`planned_stages`

Ordered list of execution stages

**Type**
List[py3plex.dsl.planner.Stage]

`required_measures`

Set of measures that must be computed

**Type**
Set[str]

`cache_plan`

Cache operations plan

**Type**
py3plex.dsl.planner.CachePlan

`plan_meta`

Metadata about the plan (costs, rewrites, warnings)

**Type**
Dict[str, Any]

**ast_hash**
Hash of original AST

**Type**
str

**plan_hash**
Hash of planned stages

**Type**
str

**ast_hash: str = ''**

**cache_plan: CachePlan**

**plan_hash: str = ''**

**plan_meta: Dict[str, Any]**

**planned_stages: List[Stage]**

**required_measures: Set[str]**

**to_dict()** → Dict[str, Any]
Convert to dictionary for metadata storage.

class py3plex.dsl.**Predicate**(*attr: str*, *op: str*, *value: Any*)
Bases: object

A predicate for filtering nodes or edges.

**attr**
Attribute name (e.g., “degree”, “weight”)

**Type**
str

**op**
Comparison operator (“>”, “>=”, “<”, “<=”, “=”, “!=”)

**Type**
str

**value**
Value to compare against

**Type**
Any

**attr: str**

**op: str**

**to_dict()** → Dict[str, Any]
Convert to dictionary for serialization.

**value: Any**

class py3plex.dsl.**PredicateFilter**(*predicate: Any | None = None*, *description: str = ''*)
Bases: PhysicalOp

Filters rows using a Python callable predicate.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Execute this operator step and return the resulting rows.

**Parameters**

- **ctx** – Shared execution context.
- **rows** – Input row sequence from the previous operator.

**Returns**

Output row sequence.

**op_name: str = 'PredicateFilter'**

Human-readable operator name used in explain output.

class py3plex.dsl.**PredictStmt**(*task: str = 'links'*, *spec: LinkPredictionSpec | None = None*)

Bases: object

Top-level predictive task statement.

**spec: LinkPredictionSpec | None = None**

**task: str = 'links'**

class py3plex.dsl.**PredictionResult**(*metrics: dict[str, float]*, *predictions: list[dict[str, Any]]*, *meta: dict[str, Any] | None = None*, *replay_fn: Callable[[], PredictionResult] | None = None*)

Bases: object

Result object for link prediction workflows.

**property is_replayable: bool**

**property provenance: dict[str, Any] | None**

**replay**() → PredictionResult

**report**() → str

**to_dict**() → dict[str, Any]

**to_pandas**()

exception py3plex.dsl.**PredictionTaskError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: DslExecutionError

Raised when predictive task DSL execution fails.

class py3plex.dsl.**Q**

Bases: object

Query factory for creating QueryBuilder instances.

### Example

```
>>> Q.nodes().where(layer="social").compute("degree")
>>> Q.edges().where(intralayer=True)
>>> Q.nodes(autocompute=False).where(degree__gt=5)  # Disable autocompute
>>> Q.dynamics("SIS", beta=0.3).run(steps=100)  # Dynamics simulation
>>> Q.trajectories("sim_result").at(50)  # Query trajectories
```

**static assert_disjoint**(*query1*, *query2*, *network=None*, *identity: str = 'by_replica'*) → bool

Assert that two queries have no overlapping results.

This verification method checks that query results are disjoint (no shared items). Useful for validating partitioning and exclusive filtering.

**Parameters**

- **query1** – First query or QueryResult
- **query2** – Second query or QueryResult
- **network** – Network to execute queries on (required if queries not executed)
- **identity** – Identity strategy - "by_id" or "by_replica" (default: "by_replica")

**Returns**

True if results are disjoint

**Raises**

- **AssertionError** – If results overlap
- **IncompatibleQueryError** – If queries have incompatible targets

### Example

```
>>> # Verify that layer filters are exclusive
>>> social = Q.nodes().from_layers(L["social"])
>>> work = Q.nodes().from_layers(L["work"])
>>> Q.assert_disjoint(social, work, network, identity="by_replica")
True
```

**static assert_nonempty**(*query*, *network=None*, *message: str | None = None*) → bool

Assert that a query returns non-empty results.

This verification method ensures that queries produce at least one result. Useful for: - Validating that intersections are meaningful - Ensuring filters don't eliminate all items - Regression testing

**Parameters**

- **query** – Query or QueryResult to check
- **network** – Network to execute query on (required if query not executed)
- **message** – Custom error message

**Returns**

True if query produces non-empty results

**Raises**

**AssertionError** – If query returns empty results

### Example

```
>>> # Verify that intersection is meaningful
>>> social = Q.nodes().from_layers(L["social"])
>>> high_degree = Q.nodes().where(degree__gt=5)
>>> intersection = social & high_degree
>>> Q.assert_nonempty(intersection, network)
True
```

```
>>> # Ensure filters don't eliminate everything
>>> filtered = Q.nodes().where(degree__gt=5, betweenness__gt=0.1)
>>> assert Q.assert_nonempty(filtered, network)
```

**static assert_subset**(*subset_query*, *superset_query*, *network=None*, *identity: str = 'by_replica'*) → bool

Assert that one query result is a subset of another.

This verification method checks algebraic subset relationships between queries. Can be used for: - Regression testing (ensuring query results remain consistent) - Monotonicity checks (verifying that filtering reduces results) - Scientific claims validation

**Parameters**

- **subset_query** – Query or QueryResult that should be subset
- **superset_query** – Query or QueryResult that should be superset
- **network** – Network to execute queries on (required if queries not executed)
- **identity** – Identity strategy - “by_id” or “by_replica” (default: “by_replica”)

**Returns**

True if subset_query  superset_query, False otherwise

**Raises**

- **AssertionError** – If subset condition is violated
- **IncompatibleQueryError** – If queries have incompatible targets

### Example

```
>>> # Verify that high-degree nodes are subset of all nodes
>>> all_nodes = Q.nodes()
>>> high_degree = Q.nodes().where(degree__gt=5)
>>> Q.assert_subset(high_degree, all_nodes, network)
True
```

```
>>> # Regression test: ensure filtering doesn't add nodes
>>> filtered = Q.nodes().where(layer="social")
```

```
>>> unfiltered = Q.nodes()
>>> assert Q.assert_subset(filtered, unfiltered, network)
```

**static communities**(*autocompute: bool = True*, *partition: str = 'default'*, *mode: str | None = None*, *fast: bool | None = None*, *uq: bool | None = None*, *uq_n_samples: int | None = None*, *uq_method: str | None = None*, *seed: int | None = None*, *write_attrs: Dict[str, str] | None = None*, ***kwargs*) → CommunityQueryBuilder

Create a query builder for communities.

**Parameters**

- **autocompute** – Whether to automatically compute missing metrics (default: True)
- **partition** – Name of the partition to query (default: "default")
- **mode** – Auto-detection mode - "pareto" or "wins" (if provided, enables auto-detection)
- **fast** – Use fast mode with smaller grids/samples (default: True)
- **uq** – Enable uncertainty quantification (default: False)
- **uq_n_samples** – Number of UQ samples (default: 10)
- **uq_method** – UQ method - "seed", "perturbation", or "bootstrap" (default: "seed")
- **seed** – Master random seed for reproducibility
- **write_attrs** – Dict mapping attribute names to write (e.g., {"community_id": "community_id"})
- ****kwargs** – Additional parameters passed to auto_select_community

**Returns**

CommunityQueryBuilder for communities

### Example

```
>>> # Query large communities
>>> Q.communities().where(size__gt=10).compute("conductance")
```

```
>>> # Query from a specific algorithm
>>> Q.communities(partition="louvain").where(density_intra__gt=0.5)
```

```
>>> # Get members of large communities
>>> Q.communities().where(size__gt=10).members().compute("degree")
```

```
>>> # Auto-detection with community assignment
>>> Q.communities(mode="pareto", fast=False, uq=True, seed=42).nodes()
```

**static counterexample**() → CounterexampleBuilder

Create a counterexample query builder.

**Returns**

CounterexampleBuilder for finding network counterexamples

### Example

```
>>> cex = (
...     Q.counterexample()
...      .claim("degree__ge(k) -> pagerank__rank_gt(r)")
...      .params(k=10, r=50)
...      .seed(42)
...      .execute(network)
... )
```

**static dynamics**(*process_name: str*, ***params*) → DynamicsBuilder

Create a dynamics simulation builder.

**Parameters**

- **process_name** – Name of the process (e.g., "SIS", "SIR", "RANDOM_WALK")
- ****params** – Process parameters (e.g., beta=0.3, mu=0.1)

**Returns**

DynamicsBuilder for configuring and running simulations

### Example

```
>>> sim = (
...     Q.dynamics("SIS", beta=0.3, mu=0.1)
...      .on_layers(L["contacts"])
...      .seed(0.01)
...      .run(steps=100, replicates=10, track="all")
...      .execute(network)
... )
```

**static edges**(*autocompute: bool = True*) → QueryBuilder

Create a query builder for edges.

**Parameters**

**autocompute** – Whether to automatically compute missing metrics (default: True)

**Returns**

QueryBuilder for edges

**static from_ast**(*query_ast: Query*) → QueryBuilder

Reconstruct a QueryBuilder from an AST.

This is the inverse of QueryBuilder.to_ast(). It enables: - AST round-trip: builder → ast → builder → ast - Query replay from stored AST - AST-based query transformations

**Parameters**

**query_ast** – Query AST to reconstruct from

**Returns**

QueryBuilder that will produce an equivalent AST

**Raises**

- **ValueError** – If AST schema version is incompatible
- **ValueError** – If AST is invalid or incomplete
- **TypeError** – If required AST fields are missing

### Example

```
>>> # Round-trip a query
>>> original = Q.nodes().where(degree__gt=5).compute("betweenness")
>>> ast = original.to_ast()
>>> reconstructed = Q.from_ast(ast)
>>> # reconstructed produces equivalent AST
>>> from py3plex.dsl.ast import ast_equals
>>> ast_equals(original.to_ast(), reconstructed.to_ast())  # True
```

**static learn_claims()** → ClaimLearnerBuilder

Create a claim learning query builder.

**Returns**

ClaimLearnerBuilder for learning claims from network data

### Example

```
>>> claims = (
...     Q.learn_claims()
...       .from_metrics(["degree", "pagerank", "betweenness"])
...       .min_support(0.9)
...       .min_coverage(0.05)
...       .max_claims(20)
...       .seed(42)
...       .execute(network)
... )
```

**static nodes**(*autocompute: bool = True*) → QueryBuilder

Create a query builder for nodes.

**Parameters**

**autocompute** – Whether to automatically compute missing metrics (default: True)

**Returns**

QueryBuilder for nodes

**static pattern()** → PatternQueryBuilder

Create a pattern matching query builder.

**Returns**

PatternQueryBuilder for constructing pattern queries

### Example

```
>>> pq = (
...     Q.pattern()
...       .node("a").where(layer="social", degree__gt=3)
```

```
...       .node("b").where(layer="social")
...       .edge("a", "b", directed=False).where(weight__gt=0.2)
...       .returning("a", "b")
... )
>>> matches = pq.execute(network)
>>> df = matches.to_pandas()
```

**predict = <py3plex.dsl.builder._PredictNamespace object>**

**reduce = <py3plex.dsl.builder._ReduceNamespace object>**

**static trajectories**(*process_ref: str*) → TrajectoriesBuilder

Create a trajectories query builder.

**Parameters**

**process_ref** – Reference to a simulation result or process name

**Returns**

TrajectoriesBuilder for querying simulation outputs

### Example

```
>>> result = (
...     Q.trajectories("sim_result")
...       .at(50)
...       .measure("peak_time", "final_state")
...       .execute(context)
... )
```

**class uncertainty**

Bases: object

Global defaults for uncertainty estimation.

This class provides a way to configure default parameters for uncertainty estimation that will be used when uncertainty=True is passed to compute() but specific parameters are omitted.

### Example

```
>>> from py3plex.dsl import Q
>>>
>>> # Set global defaults
>>> Q.uncertainty.defaults(
...     enabled=True,
...     n_boot=200,
...     ci=0.95,
...     bootstrap_unit="edges",
...     bootstrap_mode="resample",
...     random_state=42
... )
>>>
>>> # Now compute() will use these defaults
>>> Q.nodes().compute("degree", uncertainty=True).execute(net)
```

```
>>> # Reset to defaults
>>> Q.uncertainty.reset()
```

**classmethod defaults**(***kwargs*) → None

Set global defaults for uncertainty estimation.

**Parameters**

- **enabled** – Whether uncertainty is enabled by default (default: False)
- **n_boot** – Number of bootstrap replicates (default: 50)
- **n_samples** – Alias for n_boot (default: 50)
- **ci** – Confidence interval level (default: 0.95)
- **bootstrap_unit** – What to resample - "edges", "nodes", or "layers" (default: "edges")
- **bootstrap_mode** – Resampling mode - "resample" or "permute" (default: "resample")
- **method** – Uncertainty estimation method - "bootstrap", "perturbation", "seed" (default: "bootstrap")
- **random_state** – Random seed for reproducibility (default: None)
- **n_null** – Number of null model replicates (default: 200)
- **null_model** – Null model type - "degree_preserving", "erdos_renyi", "configuration" (default: "degree_preserving")

**Example**

```
>>> Q.uncertainty.defaults(
...     enabled=True,
...     n_boot=500,
...     ci=0.95,
...     bootstrap_unit="edges"
... )
```

**classmethod get**(*key: str*, *default: Any | None = None*) → Any

Get a default value.

**Parameters**

- **key** – Parameter name
- **default** – Value to return if key not found

**Returns**

Default value for the parameter

**classmethod get_all**() → Dict[str, Any]

Get all current defaults as a dictionary.

**Returns**

Dictionary of all default values

**classmethod reset**() → None

Reset all defaults to their initial values.

**class py3plex.dsl.Query**(*explain: bool*, *select: SelectStmt*, *dsl_version: str = '2.0'*)

Bases: `object`

Top-level query representation.

**explain**

If True, return execution plan instead of results

**Type**
bool

`select`

The SELECT statement

**Type**
py3plex.dsl.ast.SelectStmt

`dsl_version`

DSL version for compatibility

**Type**
str

`dsl_version: str = '2.0'`

`explain: bool`

`select: SelectStmt`

`summary()` → Dict[str, Any]

Return a stable structural summary with an AST hash.

The `ast_hash` field is a 16-character hex prefix of the SHA-256 of the canonical JSON-serialised AST. Identical queries always produce the same hash; different queries almost always produce different hashes.

**Returns**
Dictionary with at minimum an `ast_hash` key plus human-readable structural information (target, layers, compute, etc.).

**Return type**
dict

`class py3plex.dsl.QueryBuilder(`*target: Target*, *autocompute: bool = True*`)`

Bases: `object`

Chainable query builder.

Use Q.nodes() or Q.edges() to create a builder, then chain methods to construct the query.

`after(`*t: float*`)` → QueryBuilder

Add temporal constraint for edges/nodes after a specific time.

Convenience method equivalent to *.during(t, None)*. Filters to only include edges/nodes active after (and at) time t.

**Parameters**
`t` – Lower bound timestamp (inclusive)

**Returns**
Self for chaining

### Examples

```
>>> # Get all edges after time 100
>>> Q.edges().after(100.0).execute(network)
```

```
>>> # Nodes active after 2024-01-01
>>> Q.nodes().after(1704067200.0).execute(network)
```

**aggregate**(***aggregations*) → QueryBuilder

Aggregate columns with support for lambdas and builtin functions.

This method computes aggregations over the result set. It supports: - Built-in aggregation functions: mean(), sum(), min(), max(), std(), var(), median(), quantile(attr, p), count() - Direct attribute references for last/first value - Lambda functions for custom aggregations

The aggregations are computed after grouping if active, otherwise globally.

**Parameters**

****aggregations** – Named aggregations. Each key is an output column name. Each value is a string expression like `"mean(degree)"`, `"quantile(degree, 0.95)"`, `"count()"`, a plain attribute name string, or a callable lambda function.

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().per_layer().aggregate(
...     avg_degree="mean(degree)",
...     max_bc="max(betweenness_centrality)",
...     median_bc="median(betweenness_centrality)",
...     q95_degree="quantile(degree, 0.95)",
...     node_count="count()",
...     layer_name="layer"  # Direct attribute
... )
```

```
>>> # With lambda
>>> Q.nodes().aggregate(
...     community_size=lambda n: network.community_sizes[network.get_partition(n)]
... )
```

```
>>> # Edge aggregations
>>> Q.edges().per_layer_pair().aggregate(
...     avg_weight="mean(weight)",
...     total_edges="count()",
...     max_src_degree="max(src_degree)"
... )
```

**arrange**(**columns: str*, *desc: bool = False*) → QueryBuilder

Sort results by specified columns (dplyr-style alias for order_by).

This is a convenience method that provides dplyr-style syntax. Columns can be prefixed with "-" to indicate descending order.

**Parameters**

- ***columns** – Column names to sort by (prefix with "-" for descending)
- **desc** – Default sort direction (only used if column has no prefix)

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().compute("degree").arrange("degree")  # ascending
>>> Q.nodes().compute("degree").arrange("-degree")  # descending
>>> Q.nodes().compute("degree", "betweenness").arrange("degree", "-betweenness")
```

**at**(*t: float*) → QueryBuilder

Add temporal snapshot constraint (AT clause).

Filters edges to only those active at a specific point in time. For point-in-time edges (with 't' attribute), includes edges where t_edge == t. For interval edges (with 't_start', 't_end'), includes edges where t is in [t_start, t_end].

**Parameters**

**t** – Timestamp for snapshot

**Returns**

Self for chaining

### Examples

```
>>> # Snapshot at specific time
>>> Q.edges().at(150.0).execute(network)
```

**before**(*t: float*) → QueryBuilder

Add temporal constraint for edges/nodes before a specific time.

Convenience method equivalent to *.during(None, t)*. Filters to only include edges/nodes active before (and at) time t.

**Parameters**

**t** – Upper bound timestamp (inclusive)

**Returns**

Self for chaining

### Examples

```
>>> # Get all edges before time 100
>>> Q.edges().before(100.0).execute(network)
```

```
>>> # Nodes active before 2024-01-01
>>> Q.nodes().before(1704067200.0).execute(network)
```

`canonical`(*scope: str = 'relational'*) → QueryBuilder

Return a new QueryBuilder whose AST is in canonical normal form.

The canonical form is: * deterministic and idempotent: `q.canonical().canonical()` `q.canonical()` * stable predicate normalization (AND terms sorted + deduplicated) * compute items sorted alphabetically * scope controls additional rules (see below)

**Parameters**

`scope` – "relational" (default) — treats ordering as semantic (py3plex order_by affects results today). "strict" — same plus deduplication of order_by items.

**Returns**

New QueryBuilder with canonicalized AST

Example:

```
q = Q.nodes().where(degree__gt=5).compute("betweenness", "degree")
q_canon = q.canonical()
assert q_canon.canonical().canonical_ast_hash == q_canon.canonical_ast_hash
```

`property canonical_ast_hash: str`

Stable hex hash of the canonical AST under "relational" scope.

Equivalent queries share the same `canonical_ast_hash`. The hash is the first 16 hex characters of SHA-256 over the deterministic serialization of the canonical AST.

Note: This hash is distinct from the existing `ast_hash` in provenance metadata (which uses its own serialization).

Example:

```
q1 = Q.nodes().compute("betweenness", "degree")
q2 = Q.nodes().compute("degree", "betweenness")
assert q1.canonical_ast_hash == q2.canonical_ast_hash
```

`centrality`(**metrics: str*, ***aliases: str*) → QueryBuilder

Compute centrality metrics (convenience wrapper for compute).

This is a domain-specific convenience method for computing common centrality measures. It's equivalent to calling compute() with the metric names.

**Supported metrics:**

- degree
- betweenness (or betweenness_centrality)
- closeness (or closeness_centrality)
- eigenvector (or eigenvector_centrality)
- pagerank
- clustering (or clustering_coefficient)

**Parameters**

- `*metrics` – Centrality metric names
- `**aliases` – Optional aliases for metrics (alias=metric_name)

**Returns**
Self for chaining

### Example

```
>>> Q.nodes().centrality("degree", "betweenness", "pagerank")
>>> Q.nodes().centrality("degree", bc="betweenness_centrality")
```

`collect()` → QueryBuilder

Explicitly collect results (no-op for builder).

This method exists for API compatibility with graph_ops.NodeFrame but is a no-op in the builder pattern since execute() handles collection. Included for completeness and to reduce cognitive load when switching between APIs.

**Returns**
Self for chaining

### Example

```
>>> result = Q.nodes().filter(degree__gt=5).collect().execute(net)
```

`community`(*method: str = 'leiden', gamma: float | Dict[Any, float] = 1.0, omega: float | ndarray = 1.0, n_iterations: int = 2, random_state: int | None = None, partition_name: str = 'default', **kwargs*) → QueryBuilder

Run community detection and attach partition to network.

This operator runs community detection on the network and attaches the resulting partition so it can be queried or used in subsequent operations.

**Supported algorithms:**

- "leiden": Multilayer Leiden algorithm (production-ready with UQ)
- "louvain": Multilayer Louvain algorithm
- "infomap": Infomap (if available)
- "label_propagation_supra": Supra-graph label propagation (hard labels)
- "label_propagation_consensus": Multiplex consensus label propagation (hard labels)

For Leiden specifically, use in combination with .uq() for uncertainty quantification. Label propagation algorithms support deterministic execution with random_state for reproducibility.

**Parameters**

- **`method`** – Community detection algorithm (default: "leiden")
- **`gamma`** – Resolution parameter(s). Higher -> more communities. (default: 1.0) Not used for label_propagation algorithms.
- **`omega`** – Interlayer coupling strength. Higher -> stronger coupling. (default: 1.0) For label_propagation_supra: weight of identity links between layers. Not used for label_propagation_consensus.
- **`n_iterations`** – Number of iterations for iterative methods (default: 2) For label_propagation_supra: maps to max_iter (default: 100)

For label_propagation_consensus: maps to max_iter (default: 25)

- **random_state** – Random seed for reproducibility. If None, uses 0. (default: None)
- **partition_name** – Name to assign to this partition (default: "default")
- ****kwargs** – Additional algorithm-specific parameters: - projection: "none" or "majority" (label_propagation_supra only) - inner_max_iter: int (label_propagation_consensus only, default: 50) - max_iter: int (override default iterations for label propagation)

**Returns**

Self for chaining

### Examples

```
>>> # Basic Leiden community detection
>>> result = (
...     Q.nodes()
...       .community(method="leiden", gamma=1.2, random_state=42)
...       .execute(network)
... )
>>>
>>> # Supra-graph label propagation with layer coupling
>>> result = (
...     Q.nodes()
...       .community(method="label_propagation_supra", omega=0.7,
...                  projection="none", random_state=42)
...       .execute(network)
... )
>>>
>>> # Consensus label propagation (node-level)
>>> result = (
...     Q.nodes()
...       .community(method="label_propagation_consensus",
...                  max_iter=25, inner_max_iter=50, random_state=42)
...       .execute(network)
... )
>>>
>>> # Leiden with uncertainty quantification
>>> result = (
...     Q.nodes()
...       .community(method="leiden", gamma=1.2, omega=0.8, random_state=42)
...       .uq(method="ensemble", n_samples=50, seed=42)
...       .execute(network)
... )
>>> print(f"Consensus partition: {result.meta['consensus_partition']}")
>>> print(f"Score CI: {result.meta['score_ci']}")
>>>
>>> # Query communities after detection
>>> result = (
...     Q.nodes()
```

```
...       .community(method="leiden", partition_name="my_leiden")
...       .execute(network)
... )
>>> communities = Q.communities(partition="my_leiden").execute(network)
```

### Notes

- Results are attached to the network under partition_name
- Combining with .uq() enables probabilistic community detection
- Default random_state=None becomes 0 for determinism
- For large networks, consider tuning omega and gamma for performance
- Label propagation algorithms use hard labels with random tie-breaking

**community_auto**(*seed: int | None = None*, *fast: bool = True*, ***kwargs*) → QueryBuilder

Automatically detect communities and join annotations to nodes.

Extends the node query with community assignment annotations: - community: Community assignment - confidence: Assignment confidence (0-1) - entropy: Assignment entropy (uncertainty measure) - margin: Margin between top two community assignments - community_size: Size of the assigned community - layer: Layer name (nullable for single-layer networks)

These annotations can be filtered with .where() like any other attribute.

**Parameters**

- **seed** – Random seed for reproducibility (default: None)
- **fast** – Use fast mode with smaller parameter grids (default: True)
- ****kwargs** – Additional parameters passed to auto_select_community

**Returns**

Self for chaining (execute() will run auto detection and join)

### Examples

```
>>> # Basic usage
>>> result = (
...     Q.nodes()
...       .community_auto(seed=42)
...       .execute(network)
... )
>>> df = result.to_pandas()
>>> print(df[['node', 'community', 'confidence', 'community_size']])
>>>
>>> # With filtering and computation
>>> result = (
...     Q.nodes()
...       .community_auto(seed=42, fast=True)
...       .where(community_size__gt=10, confidence__gt=0.8)
...       .compute("pagerank")
```

```
...       .execute(network)
... )
```

### Notes

- Caching: Auto community detection runs once per execute() call
- Filtering: All annotation fields support predicate filters
- Computation: Can chain .compute() after community_auto()

**compile()** → GraphProgram

Compile query into a GraphProgram.

This is a public alias for `to_program()` and mirrors the executable program workflow:

`Q...compile().execute(network)`.

**compute**(**measures: str*, *alias: str | None = None*, *aliases: Dict[str, str] | None = None*, *kind: str | None = None*, *uncertainty: bool | None = None*, *method: str | None = None*, *n_samples: int | None = None*, *ci: float | None = None*, *bootstrap_unit: str | None = None*, *bootstrap_mode: str | None = None*, *n_boot: int | None = None*, *n_null: int | None = None*, *null_model: str | None = None*, *random_state: int | None = None*, *approx: bool | None = None*, *approx_method: str | None = None*, *approx_diagnostics: bool = False*, ***kwargs*) → QueryBuilder

Add measures to compute with optional uncertainty estimation and/or approximation.

**Parameters**

- ***measures** – Measure names to compute
- **alias** – Alias for single measure
- **aliases** – Dictionary mapping measure names to aliases
- **kind** – Optional metric semantic kind (e.g., degree kind: aggregate|intra|inter)
- **uncertainty** – Whether to compute uncertainty for these measures. If None, uses Q.uncertainty defaults or the global uncertainty context.
- **method** – Uncertainty estimation method ('bootstrap', 'perturbation', 'seed', 'null_model')
- **n_samples** – Approximation budget when approx=True; otherwise UQ samples. With uncertainty=True and approx=True, used for both budgets.
- **ci** – Confidence interval level (default: from Q.uncertainty.defaults)
- **bootstrap_unit** – What to resample - "edges", "nodes", or "layers" (default: from Q.uncertainty.defaults)
- **bootstrap_mode** – Resampling mode - "resample" or "permute" (default: from Q.uncertainty.defaults)
- **n_boot** – Alias for n_samples (for bootstrap)
- **n_null** – Number of null model replicates (default: from Q.uncertainty.defaults)

- **null_model** – Null model type - “degree_preserving”, “erdos_renyi”, “configuration” (default: from Q.uncertainty.defaults)
- **random_state** – Random seed for reproducibility (default: from Q.uncertainty.defaults)
- **approx** – Whether to use fast approximate computation (default: False)
- **approx_method** – Approximation method name (“sampling”, “landmarks”, “power_iteration”) If approx=True and approx_method=None, defaults are: - betweenness_centrality -> “sampling” - closeness_centrality -> “landmarks” - pagerank -> “power_iteration”
- **approx_diagnostics** – Whether to compute per-node diagnostic info (e.g., stderr)
- ****kwargs** – Additional approximation parameters (e.g., n_samples, n_landmarks, tol, max_iter, seed)

**Returns**

Self for chaining

### Example

```
>>> # Without uncertainty or approximation
>>> Q.nodes().compute("degree", "betweenness_centrality")
```

```
>>> # With uncertainty using explicit parameters
>>> Q.nodes().compute(
...     "degree", "betweenness_centrality",
...     uncertainty=True,
...     method="bootstrap",
...     n_samples=500,
...     ci=0.95
... )
```

```
>>> # With approximation
>>> Q.nodes().compute(
...     "betweenness_centrality",
...     approx=True,
...     approx_method="sampling",
...     n_samples=512,
...     seed=42
... )
```

```
>>> # With both UQ and approximation
>>> Q.nodes().compute(
...     "betweenness_centrality",
...     approx=True,
...     n_samples=256,
...     seed=42
... ).uq(method="bootstrap", n_samples=50, seed=42).execute(net)
```

`contract`(*contract: Robustness*) → QueryBuilder

Attach a robustness contract to the query (certification-grade).

Contracts ensure that query conclusions are stable under structural perturbations. This is distinct from uncertainty quantification: - UQ quantifies uncertainty of estimates (error bars) - Contracts certify robustness of conclusions (stable/unstable)

Contracts provide: - **Typed failure modes**: Clear classification of why a contract might fail - **Auto-inference**: Sensible defaults for perturbation, predicates, samples - **Repair mechanisms**: Stable cores, tiers, stable nodes - **Determinism**: Default seed=0 ensures reproducibility - **Provenance**: Full replay capability

**Parameters**

`contract` – Robustness contract object from py3plex.contracts

**Returns**

Self for chaining

### Examples

```
>>> from py3plex.contracts import Robustness
>>>
>>> # Minimal usage - all defaults
>>> result = (Q.nodes()
...           .compute("pagerank")
...           .top_k(20, "pagerank")
...           .contract(Robustness())
...           .execute(net))
>>>
>>> if result.contract_ok:
...     print("Top-20 PageRank is stable!")
... else:
...     print(f"Contract failed: {result.failure_mode}")
...     print("Stable core:", result.stable_core)
>>>
>>> # Override defaults
>>> result = (Q.nodes()
...           .compute("degree")
...           .top_k(10, "degree")
...           .contract(Robustness(n_samples=100, p_max=0.2))
...           .execute(net))
>>>
>>> # Ranking stability
>>> result = (Q.nodes()
...           .compute("betweenness_centrality")
...           .order_by("betweenness_centrality", desc=True)
...           .contract(Robustness())
...           .execute(net))
>>>
>>> # Community stability
>>> result = (Q.nodes()
...           .community()
...           .contract(Robustness())
```

```
...             .execute(net))
```

**Note**

Only one contract per query is supported. Calling .contract() multiple times will replace the previous contract.

**counterfactualize**(*network: Any*, *spec: InterventionSpec*, *repeats: int = 100*, *seed: int = 42*, ***params*) → CounterfactualResult

Execute query under custom counterfactual intervention (ADVANCED).

This is the advanced interface for counterfactual analysis, allowing full control over intervention specifications. Most users should use robustness_check() instead.

**Parameters**

- **network** – Multilayer network to analyze
- **spec** – Custom InterventionSpec (from py3plex.counterfactual.spec)
- **repeats** – Number of counterfactual runs (default: 100)
- **seed** – Random seed for reproducibility (default: 42)
- ****params** – Parameter bindings for the query

**Returns**

CounterfactualResult with full details of baseline and counterfactuals

### Example

```
>>> from py3plex.counterfactual import RemoveEdgesSpec
>>>
>>> # Custom intervention
>>> spec = RemoveEdgesSpec(proportion=0.1, mode="targeted")
>>> result = (Q.nodes()
...           .compute("degree")
...           .counterfactualize(net, spec, repeats=100))
>>>
>>> # Convert to report for analysis
>>> report = result.to_report()
>>> report.show()
```

**coverage**(*mode: str = 'all'*, *k: int | None = None*, *threshold: int | None = None*, *p: float | None = None*, *group: str | None = None*, *id_field: str = 'id'*) → QueryBuilder

Configure coverage filtering across groups.

Coverage determines which items appear in the final result based on how many groups they appear in after grouping and top_k filtering.

**Parameters**

- **mode** – Coverage mode: - “all”: Keep items that appear in ALL groups - “any”: Keep items that appear in AT LEAST ONE group - “at_least”: Keep items that appear in at least k groups (requires

k/threshold parameter) - “exact”: Keep items that appear in exactly k groups (requires k/threshold parameter) - “fraction”: Keep items that appear in at least p fraction (0-1) of groups (requires p parameter)

- **k** – Threshold for “at_least” or “exact” modes
- **threshold** – Alias for k parameter
- **p** – Fraction threshold (0.0-1.0) for “fraction” mode. E.g., p=0.67 means at least 67% of groups
- **group** – Group attribute for coverage (defaults to primary grouping context)
- **id_field** – Field to use for identity matching (default: “id” for nodes)

**Returns**

Self for chaining

**Raises**

- **ValueError** – If mode is invalid or required parameters are missing
- **ValueError** – If called without prior grouping

### Example

```
>>> # Nodes that are top-5 hubs in ALL layers
>>> Q.nodes().per_layer().top_k(5, "betweenness").coverage(mode="all")
```

```
>>> # Nodes that are top-5 in at least 2 layers
>>> Q.nodes().per_layer().top_k(5, "degree").coverage(mode="at_least", k=2)
>>> # Or equivalently:
>>> Q.nodes().per_layer().top_k(5, "degree").coverage(mode="at_least", threshold=2)
```

```
>>> # Nodes in top-10 in at least 70% of layers (0.7 fraction)
>>> Q.nodes().per_layer().top_k(10, "degree").coverage(mode="fraction", p=0.7)
```

**distinct**(**columns: str*) → QueryBuilder

Return unique rows based on specified columns.

If columns are specified, deduplicates based on those columns only. If no columns are specified, deduplicates based on all columns.

**Parameters**

***columns** – Optional column names to use for uniqueness check

**Returns**

Self for chaining

### Example

```
>>> # Unique (node, layer) pairs
>>> Q.nodes().distinct()
```

```
>>> # Unique communities per layer
>>> Q.nodes().distinct("community", "layer")
```

**drop**(**columns: str*) → QueryBuilder

Remove specified columns from the result.

This operation filters out the specified columns from the output. Complementary to select() - use drop() when it's easier to specify what to remove rather than what to keep.

**Parameters**

***columns** – Column names to remove from the result

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().compute("degree", "betweenness", "closeness").drop("closeness")
```

**during**(*t0: float | None = None*, *t1: float | None = None*) → QueryBuilder

Add temporal range constraint (DURING clause).

Filters edges to only those active during a time range [t0, t1]. For point-in-time edges, includes edges where t is in [t0, t1]. For interval edges, includes edges where the interval overlaps [t0, t1].

**Parameters**

- **t0** – Start of time range (None means -infinity)
- **t1** – End of time range (None means +infinity)

**Returns**

Self for chaining

### Examples

```
>>> # Time range query
>>> Q.edges().during(100.0, 200.0).execute(network)
```

```
>>> # Open-ended ranges
>>> Q.edges().during(100.0, None).execute(network)  # From 100 onwards
>>> Q.edges().during(None, 200.0).execute(network)  # Up to 200
```

**embed**(*method: str = 'netmf'*, *dim: int = 128*, *dimensions: int | None = None*, *multilayer: str = 'supra'*, *layers: List[str] | None = None*, *window: int = 10*, *negative: float = 1.0*, *approx: str = 'randomized_svd'*, *gamma: float = 1.0*, *seed: int | None = None*, *backend: str = 'auto'*, *cache: bool = True*, *cache_namespace: str = 'dsl_embeddings'*, *link_op: str | None = None*, *metapaths: List[List[str]] | None = None*, *walk_length: int = 80*, *num_walks: int = 10*, *p: float = 1.0*, *q: float = 1.0*, *window_size: int = 10*, *negative_samples: int = 5*, *workers: int = 1*, *order: int = 2*) → QueryBuilder

Attach a node (or edge) embedding computation to this query.

The embedding is computed after layer-filtering and WHERE filtering so that it operates on the induced subgraph of the selected items.

**Parameters**

- **method** – Embedding method. `"netmf"` (default) or `"metapath2vec"`.
- **dim** – Embedding dimensionality.
- **dimensions** – Optional alias for `dim`.
- **multilayer** – How to aggregate layers before embedding. `"supra"` (default) builds a supra-adjacency matrix; `"union"` collapses all layers into one graph.
- **layers** – Optional list of layer names to include. Defaults to all layers in the query's layer selection.
- **window** – Context window size (NetMF parameter).
- **negative** – Negative sampling ratio (NetMF parameter).
- **approx** – SVD approximation backend. `"randomized_svd"` (default) or `"eigsh"`.
- **gamma** – Interlayer coupling weight used in supra mode.
- **seed** – Random seed for reproducibility.
- **backend** – Compute backend. `"auto"` picks JAX if available, otherwise scipy.
- **cache** – Whether to cache the resulting embeddings.
- **cache_namespace** – Prefix for cache keys.
- **link_op** – For edge queries, the binary operator used to combine source and target node embeddings (e.g. `"hadamard"`).
- **metapaths** – List of metapaths for MetaPath2Vec. Each metapath is a list of layer names, e.g. `[["author", "paper", "author"]]`.
- **walk_length** – Random walk length for MetaPath2Vec.
- **num_walks** – Number of walks per node for MetaPath2Vec.

**Returns**

Self for chaining.

Example:

```
result = (
    Q.nodes()
     .from_layers(L["social"])
     .embed(method="netmf", dim=64, seed=42)
     .execute(net)
)
emb = result.to_numpy("embedding")  # shape (n, 64)

# MetaPath2Vec example::

result = (
    Q.nodes()
     .embed(
         method="metapath2vec",
         metapaths=[["author", "paper", "author"]],
         dim=32,
```

```
        seed=42,
    )
    .execute(net)
)
```

**end_grouping**() → QueryBuilder

Marker for the end of grouping configuration.

This is purely for API readability and has no effect on execution. It helps visually separate grouping operations from post-grouping operations.

**Returns**

Self for chaining

### Example

```
>>> (Q.nodes()
...   .per_layer()
...     .top_k(5, "degree")
...   .end_grouping()
...   .coverage(mode="all"))
```

**equivalent_to**(*other: QueryBuilder*, *scope: str = 'relational'*, *explain: bool = False*)

Check whether this query is semantically equivalent to *other*.

Equivalence is determined by comparing canonical AST forms under the given scope.

**Parameters**

- **other** – Another QueryBuilder to compare against
- **scope** – "relational" or "strict" (see `canonical()`)
- **explain** – If True, return `(bool, proof_dict)` instead of bool. *proof_dict* contains the rule steps for both queries and the first-mismatch path (if any).

**Returns**

bool — or `(bool, dict)` when *explain=True*

Example:

```
q1 = Q.nodes().where(degree__gt=5, layer="social")
q2 = Q.nodes().where(layer="social", degree__gt=5)
assert q1.equivalent_to(q2)

q1 = Q.nodes().limit(5).limit(3)
q2 = Q.nodes().limit(3)
assert q1.equivalent_to(q2)
```

**execute**(*network: Any*, *progress: bool = True*, *explain_plan: bool = False*, *planner: Dict[str, Any] | None = None*, *optimize: bool | None = None*, ***params*) → QueryResult

Execute the query.

**Parameters**

- **network** – Multilayer network object
- **progress** – If True, log progress messages during query execution (default: True)
- **explain_plan** – If True, populate result.meta["plan"] with execution plan (default: False)
- **planner** – Optional planner configuration dict (compute_policy, enable_cache, etc.)
- **optimize** – Override the configured DSL optimizer setting for this execution.
- ****params** – Parameter bindings

**Returns**

QueryResult with results and metadata

**explain**(*neighbors_top: int | None = None*, *include: List[str] | None = None*, *exclude: List[str] | None = None*, *neighbors: Dict[str, Any] | None = None*, *community: Dict[str, Any] | None = None*, *layer_footprint: Dict[str, Any] | None = None*, *attribution: Dict[str, Any] | None = None*, *cache: bool = True*, *as_columns: bool = True*, *prefix: str = ''*, *include_community: bool | None = None*) → QueryBuilder | ExplainQuery

Attach explanations to results OR get execution plan.

**Two modes:**

1. **Execution Plan Mode** (no arguments): Returns an ExplainQuery that shows the execution plan when executed. This is like SQL EXPLAIN - it shows what the query will do.
2. **Explanations Mode** (with arguments): Attaches explanations to each result row (typically nodes), such as: - Community membership and size - Top neighbors by weight/degree - Layer footprint (which layers the node appears in) - Attribution (Shapley value explanations for why items are ranked/scored highly)

**Parameters**

- **neighbors_top** – Maximum number of neighbors to include in top_neighbors (default: 10). If None and no other args, returns execution plan (mode 1).
- **include** – List of explanation blocks to compute. If None, uses defaults: ["community", "top_neighbors", "layer_footprint"] To include attribution, explicitly add "attribution" to the list.
- **exclude** – List of explanation blocks to exclude from include list
- **neighbors** – Optional configuration for neighbor selection: - "metric": "weight" or "degree" (default: "weight") - "scope": "layer" (per-layer) or "global" (default: "layer") - "direction": "out", "in", or "both" (default: "both")
- **community** – Optional configuration for community explanations (reserved)
- **layer_footprint** – Optional configuration for layer footprint (reserved)

- **attribution** – Optional configuration for attribution explanations (Shapley values): - “metric”: str (which metric to explain, auto-detected if None) - “objective”: “value” or “rank” (default: “value”) - “levels”: [“layer”, “edge”] (default: [“layer”]) - “method”: “shapley”, “shapley_mc”, or “influence” (default: “shapley_mc”) - “seed”: int (random seed for reproducibility) - “n_permutations”: int (Monte Carlo samples, default: 128) - “max_edges”: int (max candidate edges, default: 40) - See AttributionConfig for full options
- **cache** – Whether to cache neighbor lookups (default: True)
- **as_columns** – Store explanations as top-level columns in result (default: True)
- **prefix** – Optional prefix for explanation column names (default: “”)
- **include_community** – Optional shorthand to explicitly exclude community info (use False). By default, community info is included automatically when available. Set to False to exclude it. Setting to True is redundant (default behavior).

**Returns**

QueryBuilder (self) for chaining when in explanations mode ExplainQuery when in execution plan mode

**Raises**

- **ValueError** – If include contains unknown explanation blocks
- **ValueError** – If neighbors_top < 1

### Examples

```
>>> # Execution plan mode (no arguments)
>>> plan = Q.nodes().compute("degree").explain().execute(network)
>>> print(plan.steps)
```

```
>>> # Explanations mode with attribution
>>> result = (
...     Q.nodes()
...       .compute("pagerank")
...       .order_by("-pagerank")
...       .limit(10)
...       .explain(
...           include=["attribution"],
...           attribution={"metric": "pagerank", "levels": ["layer"], "seed": 42}
...       )
...       .execute(network)
... )
>>> df = result.to_pandas(expand_explanations=True)
>>> # df now has attribution column with layer contributions
```

```
>>> # Explicitly exclude community info if not needed
>>> result = (
...     Q.nodes()
```

```
...       .compute("betweenness")
...       .explain(neighbors_top=5, include_community=False)
...       .execute(network)
... )
```

**explain_plan**() → QueryBuilder

Enable plan explanation in the next execute() call.

When enabled, the execution plan will be included in result.meta["plan"] showing stage order, costs, and optimization rewrites.

**Returns**

Self for chaining.

### Example

```
>>> result = Q.nodes().compute("degree").explain_plan().execute(net)
>>> print(result.meta["plan"]["rewrite_summary"])
```

**explain_plan_json**() → Dict[str, Any]

Return machine-readable plan/introspection information.

**export**(*path: str*, *fmt: str = 'csv'*, *columns: List[str] | None = None*, ***options*) → QueryBuilder

Attach a file export specification to the query.

This adds a side-effect to write query results to a file when executed. The query will still return the QueryResult as normal.

**Parameters**

- **path** – Output file path (string)
- **fmt** – Format type ('csv', 'json', 'tsv')
- **columns** – Optional list of column names to include/order
- ****options** – Format-specific options (e.g., delimiter=';', orient='records')

**Returns**

Self for chaining

**Raises**

**ValueError** – If format is not supported

### Example

```
>>> q = (
...     Q.nodes()
...       .compute("degree")
...       .export("results.csv", fmt="csv", columns=["node", "degree"])
... )
```

**export_csv**(*path: str*, *columns: List[str] | None = None*, *delimiter: str = ','*, ***options*) → QueryBuilder

Export query results to CSV file.

Convenience wrapper around .export() for CSV format.

**Parameters**

- **path** – Output CSV file path
- **columns** – Optional list of columns to include/order
- **delimiter** – CSV delimiter (default: ‘,’)
- ****options** – Additional CSV-specific options

**Returns**
Self for chaining

**export_json**(*path: str*, *columns: List[str] | None = None*, *orient: str = 'records'*, ***options*) → QueryBuilder

Export query results to JSON file.

Convenience wrapper around .export() for JSON format.

**Parameters**

- **path** – Output JSON file path
- **columns** – Optional list of columns to include/order
- **orient** – JSON orientation (‘records’, ‘split’, ‘index’, ‘columns’, ‘values’)
- ****options** – Additional JSON-specific options

**Returns**
Self for chaining

**filter**(**args*, ***kwargs*) → QueryBuilder

Filter results using a predicate (dplyr-style alias for where).

This is an alias for the where() method, providing the traditional dplyr naming convention. Supports the same syntax as where(): - Keyword arguments for simple conditions - Expression objects using F

**Parameters**

- ***args** – BooleanExpression objects from F
- ****kwargs** – Conditions as keyword arguments

**Returns**
Self for chaining

### Example

```
>>> Q.nodes().filter(degree__gt=5)
>>> Q.nodes().filter(F.degree > 5, layer="social")
```

**filter_expr**(*expr: str*) → QueryBuilder

Filter results using a string expression.

Provides string-based filtering similar to pandas query() or graph_ops.NodeFrame.filter_expr(). Uses safe expression evaluation to prevent code injection.

**Parameters**

**expr** – Expression string like “degree > 10 and layer == ‘social’”

**Returns**
Self for chaining

### Example

```
>>> Q.nodes().compute("degree").filter_expr("degree > 10 and layer == 'ppi'")
>>> Q.edges().filter_expr("weight > 0.5")
```

**Note**

This method creates a post-processing filter that evaluates the expression on each result row. For better performance, use where() with keyword arguments when possible.

**first**() → QueryBuilder

Return only the first result.

Terminal-style operation that limits results to 1. Note: This still returns a QueryBuilder for chaining with execute().

**Returns**
Self for chaining

### Example

```
>>> result = Q.nodes().arrange("-degree").first().execute(net)
>>> df = result.to_pandas()  # Will have at most 1 row
```

**from_layers**(*layer_expr: LayerExprBuilder | LayerSet | list*) → QueryBuilder

Filter by layers using layer algebra.

Supports both LayerExprBuilder (backward compatible) and LayerSet (new). Also accepts a plain list of layer name strings for convenience.

**Parameters**
**layer_expr** – Layer expression (e.g., L[“social”] + L[“work”], L[”* - coupling”], or [“social”, “work”])

**Returns**
Self for chaining

### Example

```
>>> # Old style (still works)
>>> Q.nodes().from_layers(L["social"] + L["work"])
>>>
>>> # New style with string expressions
>>> Q.nodes().from_layers(L["* - coupling"])
>>> Q.nodes().from_layers(L["(ppi | gene) & disease"])
>>>
>>> # Plain list of strings
```

```
>>> Q.nodes().from_layers(["social", "work"])
>>> Q.nodes().from_layers([])  # empty → no layer filter
```

**group_by**(**fields: str*) → QueryBuilder

Group result items by given fields.

This is the low-level grouping primitive used by per_layer(). Once grouping is established, you can apply per-group operations like top_k().

**Parameters**

***fields** – Attribute names to group by (e.g., "layer")

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().group_by("layer").top_k(5, "degree")
```

**has_community**(*predicate*) → QueryBuilder

Filter nodes based on a community-related predicate.

This method filters nodes based on their community membership or community-related attributes. The predicate can be: - A callable: Called with each node tuple, should return bool - A value: Direct equality check against "community" attribute

**Parameters**

**predicate** – Either a callable(node_tuple) -> bool or a value to match

**Returns**

Self for chaining

### Example

```
>>> # Filter by community ID
>>> Q.nodes().has_community(3)
```

```
>>> # Filter by custom predicate
>>> Q.nodes().has_community(
...     lambda n: network.get_node_attribute(n, "disease_enriched") is True
... )
```

**head**(*n: int = 5*) → QueryBuilder

Keep only the first n results (dplyr-style).

Equivalent to limit(n) but with more intuitive dplyr-style naming.

**Parameters**

**n** – Number of results to keep (default: 5)

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().compute("degree").head(10)
>>> Q.nodes().arrange("-degree").head(5)
```

**hint**() → QueryBuilder

Display context-aware hints for next query-building steps.

This non-invasive discoverability mechanism suggests relevant methods based on the current query state. It prints suggestions to stdout but does NOT modify query behavior or auto-execute anything.

Suggestions adapt to: - Whether layers are selected - Whether filters are applied - Whether computations are added - Whether grouping is active - Whether UQ is enabled

**Returns**

Self for chaining (unmodified)

### Example

```
>>> # After where() -> suggests compute(), per_layer(), uq()
>>> Q.nodes().where(degree__gt=3).hint()
```

```
>>> # After per_layer() -> suggests aggregate(), coverage()
>>> Q.nodes().per_layer().hint()
```

```
>>> # After compute() -> suggests order_by(), limit(), explain()
>>> Q.nodes().compute("degree").hint()
```

**Note**

This is purely informational. The query remains unchanged. Call .execute() when ready to run the query.

**join**(*right: QueryBuilder | QueryResult*, *on: str | List[str]*, *how: str = 'inner'*, *suffixes: Tuple[str, str] = ('', '_r')*) → QueryBuilder

Join this query with another query or result.

Performs a relational join between two query results. Joins are row-wise relational joins, not graph merges. The join is lazy and will be executed when .execute() is called.

**Parameters**

- **right** – Right query (QueryBuilder or QueryResult to join with)
- **on** – Column name(s) to join on. Can be a single string or list of strings. All keys must exist in both left and right schemas.
- **how** – Join type. One of: - 'inner': Only rows with matching keys in both sides - 'left': All left rows, matching right rows (nulls for non-matches) - 'right': All right rows, matching left rows (nulls for non-matches) - 'outer': All rows from both sides (nulls for non-matches)

- 'semi': Left rows that have a match in right (left columns only) - 'anti': Left rows that have NO match in right (left columns only)

- **suffixes** – Tuple of (left_suffix, right_suffix) for name collisions. Default: ("", "_r")

**Returns**

JoinBuilder wrapping the join operation for further chaining

**Raises**

- **ValueError** – If join type is invalid
- **InvalidJoinKeyError** – If join keys don't exist in schemas (at execute time)

### Examples

```
>>> # Join nodes with communities
>>> result = (
...     Q.nodes()
...       .compute("degree")
...       .join(
...           Q.communities(mode="leiden").members(),
...           on=["node", "layer"],
...           how="left",
...           suffixes=("", "_comm")
...       )
...       .where(degree__gt=3)
...       .execute(network)
... )
```

```
>>> # Join with pre-computed result
>>> communities = Q.communities().members().execute(network)
>>> result = (
...     Q.nodes()
...       .compute("degree")
...       .join(communities, on=["node", "layer"], how="inner")
...       .execute(network)
... )
```

```
>>> # Self-join (same query, different filters)
>>> high_degree = Q.nodes().where(degree__gt=10)
>>> high_bc = Q.nodes().compute("betweenness_centrality").where(betweenness_centra
>>> result = high_degree.join(high_bc, on=["node", "layer"], how="inner").execute(
```

`last()` → QueryBuilder

Return only the last result.

Returns the last result after all other operations are applied.

**Returns**

Self for chaining

#### Example

```
>>> result = Q.nodes().arrange("degree").last().execute(net)
```

**limit**(*n: int*) → QueryBuilder

Limit number of results.

**Parameters**
**n** – Maximum number of results

**Returns**
Self for chaining

**mutate**(***transformations*) → QueryBuilder

Add or transform columns using expressions or lambda functions.

This method creates new columns or modifies existing ones by applying transformations to each row. It corresponds to dplyr::mutate and is applied row-by-row (not aggregated like summarize/aggregate).

Transformations can be: - String expressions referencing existing attributes - Lambda functions that receive a dict with item attributes - Simple values (applied to all rows)

**Parameters**
****transformations** – Named transformations. Each key is an output column name. Each value is a lambda receiving a dict of item attributes, a string expression such as `"degree * 2"`, or a constant value applied to all rows.

**Returns**
Self for chaining

#### Example

```
>>> # Create new columns from existing attributes
>>> Q.nodes().compute("degree", "clustering").mutate(
...     normalized_degree=lambda row: row.get("degree", 0) / 10.0,
...     log_degree=lambda row: np.log1p(row.get("degree", 0)),
...     score=lambda row: row.get("degree", 0) * row.get("clustering", 0)
... )
```

```
>>> # Simple transformations
>>> Q.nodes().mutate(
...     category="hub",  # Constant value
...     doubled_degree=lambda row: row.get("degree", 0) * 2
... )
```

**name**(*query_name: str*) → QueryBuilder

Assign a name to this query for provenance tracking.

Named queries appear in provenance metadata and explanations, making it easier to understand complex composed queries.

**Parameters**
**query_name** – Descriptive name for this query

**Returns**

Self for chaining

**Example**

```
>>> hubs = Q.nodes().where(degree__gt=5).name("hubs")
>>> stable = Q.nodes().uq().name("stable_nodes")
>>> combined = hubs & stable
```

**node_type**(*node_type: str*) → QueryBuilder

Filter nodes by node_type attribute.

This is a convenience method that adds a WHERE condition filtering by the "node_type" attribute. Equivalent to .where(node_type=node_type).

**Parameters**

**node_type** – Node type to filter by (e.g., "gene", "protein", "drug")

**Returns**

Self for chaining

**Example**

```
>>> Q.nodes().node_type("gene").compute("degree")
```

**order_by**(**keys: str*, *desc: bool = False*) → QueryBuilder

Add ORDER BY clause.

**Parameters**

- ***keys** – Attribute names to order by (prefix with "-" for descending)
- **desc** – Default sort direction

**Returns**

Self for chaining

**per_community**() → QueryBuilder

Group results by community (sugar for group_by("community")).

Similar to per_layer(), but groups by community attribute. Useful after community detection has been run and community assignments are stored in node attributes.

**Returns**

Self for chaining

**Example**

```
>>> # Find top nodes per community
>>> Q.nodes().per_community().top_k(5, "betweenness_centrality")
```

**per_layer**() → QueryBuilder

Group results by layer (sugar for group_by("layer")).

This is the most common grouping operation for multilayer queries. After calling this, you can apply per-layer operations like top_k().

Note: Only valid for node queries. For edge queries, use per_layer_pair().

**Returns**
Self for chaining

**Raises**
**DslExecutionError** – If called on an edge query

### Example

```
>>> Q.nodes().per_layer().top_k(5, "betweenness_centrality")
```

**per_layer_pair()** → QueryBuilder

Group edge results by (src_layer, dst_layer) pair.

This is the grouping operation for edge queries in multilayer networks. After calling this, you can apply per-layer-pair operations like top_k().

Note: Only valid for edge queries. For node queries, use per_layer().

**Returns**
Self for chaining

**Raises**
**DslExecutionError** – If called on a node query

### Example

```
>>> Q.edges().per_layer_pair().top_k(5, "edge_betweenness_centrality")
```

**planner**(*compute_policy: str = 'explicit'*, *enable_cache: bool = True*, ***kwargs*) → QueryBuilder

Configure query planner behavior.

**Parameters**

- **compute_policy** – Compute policy - “explicit” (default), “minimal”, or “all”
- **enable_cache** – Whether to enable result caching (default: True)
- ****kwargs** – Additional planner configuration

**Returns**
Self for chaining

### Example

```
>>> # Use minimal compute policy to optimize performance
>>> result = Q.nodes().compute("degree", "betweenness").where(degree__gt=5)\
...     .planner(compute_policy="minimal").execute(net)
```

```
>>> # Disable caching for one-off queries
>>> result = Q.nodes().compute("degree")\
...     .planner(enable_cache=False).execute(net)
```

**pluck**(*field: str*) → QueryBuilder

Extract values for a single field/column.

Convenience method that selects only one column. Equivalent to select(field) but with more intuitive naming for single-column extraction.

**Parameters**
`field` – Field name to extract

**Returns**
Self for chaining

### Example

```
>>> result = Q.nodes().compute("degree").pluck("degree").execute(net)
>>> df = result.to_pandas()  # Will have only 'degree' column (plus id/layer)
```

**provenance**(*mode: str = 'replayable'*, *capture: str = 'auto'*, *max_bytes: int | None = None*, *seed: int | None = None*) → QueryBuilder

Configure provenance tracking for this query.

Enables replayable provenance that captures sufficient information to deterministically reproduce the query result.

**Parameters**

- `mode` – Provenance mode ("log" or "replayable"). Default: "replayable"
- `capture` – Network capture method: - "auto": Automatically decide based on network size - "fingerprint": Only capture metadata (node/edge counts) - "snapshot": Always capture full network snapshot - "delta": Capture delta from base (if available)
- `max_bytes` – Maximum bytes for inline snapshot (default: 10MB)
- `seed` – Base random seed for reproducibility (default: None)

**Returns**
Self for chaining

### Example

```
>>> result = (Q.nodes()
...           .provenance(mode="replayable", capture="auto", seed=42)
...           .compute("degree")
...           .execute(net))
>>> result.is_replayable  # True
>>> result.replay()  # Deterministically reproduce the result
>>> result.export_bundle("result.json.gz")  # Save for later
```

**rank_by**(*attr: str*, *method: str = 'dense'*) → QueryBuilder

Add rank column based on specified attribute.

Computes ranks within the current grouping context. If grouping is active, ranks are computed per group. Otherwise, ranks are global.

The rank column will be named "{attr}_rank".

**Parameters**

- `attr` – Attribute to rank by

- **method** – Ranking method - “dense”, “min”, “max”, “average”, “first” (follows pandas.Series.rank semantics)

**Returns**
Self for chaining

**Example**

```
>>> # Global ranking
>>> Q.nodes().compute("degree").rank_by("degree")
```

```
>>> # Per-layer ranking
>>> Q.nodes().compute("degree").per_layer().rank_by("degree", "dense")
```

**rename**(***mapping: str*) → QueryBuilder

Rename columns in the result.

Provide keyword arguments where the key is the new name and the value is the old name to rename.

**Parameters**
****mapping** – Mapping from new names to old names (new=old)

**Returns**
Self for chaining

**Example**

```
>>> Q.nodes().compute("degree", "betweenness_centrality").rename(
...     deg="degree", bc="betweenness_centrality"
... )
```

**reproducible**(*enabled: bool = True*, *capture: str = 'auto'*, *seed: int | None = None*) → QueryBuilder

Enable reproducible execution (sugar for provenance).

Convenience method that sets up replayable provenance with a simpler interface.

**Parameters**

- **enabled** – Whether to enable reproducible mode
- **capture** – Network capture method (see provenance())
- **seed** – Base random seed

**Returns**
Self for chaining

**Example**

```
>>> result = Q.nodes().reproducible(True, seed=42).compute("degree").execute(net)
>>> result.is_replayable  # True
```

**resolve**(*identity: str | None = None*, *conflicts: str | None = None*) → QueryBuilder

Configure algebra resolution strategies for this query.

This method sets strategies for: - Identity comparison (by_id vs by_replica) - Attribute conflict resolution

**Parameters**

- **identity** – Identity strategy - “by_id” or “by_replica”
- **conflicts** – Conflict resolution - “prefer_left”, “prefer_right”, “mean”, “max”, “min”, “keep_both”, “error”

**Returns**

Self for chaining

### Example

```
>>> q1 = Q.nodes().compute("degree")
>>> q2 = Q.nodes().compute("pagerank")
>>> combined = (q1 & q2).resolve(identity="by_id", conflicts="mean")
```

**robustness_check**(*network: Any*, *strength: str = 'medium'*, *shake: str = 'degree_safe'*, *repeats: int = 30*, *seed: int | None = None*, *targets: Any | None = None*, ***params*) → RobustnessReport

Test robustness of query results to network perturbations.

This is the PRIMARY interface for counterfactual analysis. It runs the query on the original network (baseline) and on multiple perturbed versions to test whether analytical conclusions remain stable.

**Parameters**

- **network** – Multilayer network to analyze
- **strength** – Perturbation strength - “light”, “medium”, or “heavy”
- **shake** – Perturbation strategy preset (default: “degree_safe”) Options: “quick”, “degree_safe”, “layer_safe”, “weight_only”, “targeted”
- **repeats** – Number of counterfactual runs (default: 30)
- **seed** – Random seed for reproducibility (default: None)
- **targets** – Optional target specification for targeted perturbations
- ****params** – Parameter bindings for the query

**Returns**

RobustnessReport with human-friendly summary and analysis methods

### Example

```
>>> # Quick robustness check
>>> report = (Q.nodes()
...           .compute("pagerank")
...           .robustness_check(net))
>>> report.show()
>>>
>>> # Check top-k stability
>>> stable = report.stable_top_k(k=10, threshold=0.8)
```

```
>>> fragile = report.fragile(n=5)
```

**sample**(*n: int = 5*, *seed: int | None = None*) → QueryBuilder

Randomly sample n results.

Randomly selects n items from the result set. Useful for quick exploration or testing on a subset of data.

**Parameters**

- **n** – Number of results to sample (default: 5)
- **seed** – Optional random seed for reproducibility

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().sample(10, seed=42)
>>> Q.nodes().from_layers(L["social"]).sample(5)
```

**select**(**columns: str*) → QueryBuilder

Keep only specified columns in the result.

This operation filters the output columns, keeping only the ones specified. Useful for reducing result size and focusing on specific attributes.

**Parameters**

***columns** – Column names to keep in the result

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().compute("degree", "betweenness_centrality").select("id", "degree")
```

**sensitivity**(*perturb: str*, *grid: List[float] | None = None*, *n_samples: int = 30*, *seed: int | None = None*, *metrics: List[str] | None = None*, *scope: str = 'global'*, ***kwargs*) → QueryBuilder

Set query-scoped sensitivity analysis configuration.

Sensitivity analysis tests the robustness of query CONCLUSIONS (rankings, sets, communities) under controlled perturbations. This is DISTINCT from uncertainty quantification (.uq()):

- UQ (.uq()): Estimates uncertainty of metric VALUES (mean, std, CI)
- Sensitivity (.sensitivity()): Assesses stability of CONCLUSIONS

The sensitivity analysis runs the query on multiple perturbed versions of the network and measures how much the conclusions change using stability metrics like Jaccard@k and Kendall-.

**Parameters**

- **perturb** – Perturbation method: - 'edge_drop': Randomly drop edges - 'degree_preserving_rewire': Rewire while preserving degrees

- **grid** – Perturbation strength grid (default: [0.0, 0.05, 0.1, 0.15, 0.2]) Interpretation depends on perturbation method (e.g., fraction of edges)
- **n_samples** – Number of samples per grid point for averaging (default: 30)
- **seed** – Random seed for reproducibility (default: None)
- **metrics** – Stability metrics to compute (default: ['kendall_tau']) Options: 'jaccard_at_k(k)', 'kendall_tau', 'variation_of_information'
- **scope** – Analysis scope (default: 'global') - 'global': Overall stability curves - 'per_node': Node-level influence scores - 'per_layer': Layer-level influence scores
- ****kwargs** – Additional perturbation-specific parameters: - layer_aware: Whether to preserve layer structure (default: True) - max_attempts: Max rewiring attempts (for rewiring methods)

**Returns**

Self for chaining

### Examples

```
>>> # Sensitivity of top-k centrality rankings to edge removal
>>> result = (
...     Q.nodes()
...       .compute("betweenness_centrality")
...       .order_by("-betweenness_centrality")
...       .limit(20)
...       .sensitivity(
...           perturb="edge_drop",
...           grid=[0.0, 0.05, 0.1, 0.15, 0.2],
...           n_samples=30,
...           metrics=["jaccard_at_k(20)", "kendall_tau"],
...           seed=42
...       )
...       .execute(network)
... )
>>>
>>> # Access stability curves
>>> print(result.sensitivity_curves)
>>>
>>> # Export to pandas
>>> df = result.to_pandas(expand_sensitivity=True)
```

```
>>> # Community detection robustness
>>> result = (
...     Q.communities()
...       .detect("louvain")
...       .sensitivity(
...           perturb="degree_preserving_rewire",
...           grid=[0.0, 0.1, 0.2, 0.3],
```

```
...           metrics=["variation_of_information"],
...           seed=42
...       )
...       .execute(network)
... )
```

### Notes

- Sensitivity results include stability curves (metric vs perturbation)
- Unlike UQ, sensitivity does NOT report mean/std/CI for values
- Sensitivity analyzes how CONCLUSIONS change, not value uncertainty
- Results can identify tipping points where conclusions collapse
- Per-node/per-layer scope provides local influence attribution

**slice**(*start: int*, *end: int | None = None*) → QueryBuilder

Slice results from start to end index.

Returns a slice of the result set using Python's slicing semantics.

**Parameters**

- **start** – Starting index (0-based, inclusive)
- **end** – Ending index (exclusive). If None, slices to the end.

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().slice(5, 15)  # rows 5-14
>>> Q.nodes().slice(10)     # rows 10 to end
```

**sort**(*by: str*, *descending: bool = False*) → QueryBuilder

Sort results by a column (convenience alias for order_by).

This provides a more intuitive API matching common data analysis patterns (e.g., pandas DataFrame.sort_values).

**Parameters**

- **by** – Column name to sort by
- **descending** – If True, sort in descending order (default: False)

**Returns**

Self for chaining

### Example

```
>>> Q.nodes().compute("degree").sort(by="degree", descending=True)
```

**summarise**(***aggs: str*) → QueryBuilder

Aggregate grouped results using summary expressions.

This method is used with grouping (e.g., per_layer(), per_layer_pair()) to compute aggregate statistics per group.

**Parameters**

****aggs** – Named aggregation expressions: - count="n" - Count of items in group - mean_w="mean(weight)" - Mean of weight attribute - sum_w="sum(weight)" - Sum of weight attribute - n_layers="n_unique(layer)" - Count of unique layer values

**Returns**

Self for chaining

### Example

```
>>> Q.edges().per_layer_pair().summarise(count="n", mean_w="mean(weight)").end_grou
```

**summarize**(***aggregations: str*) → QueryBuilder

Aggregate over the current grouping context.

Computes summary statistics per group when grouping is active, or globally if no grouping is set. Aggregation expressions are strings like "mean(degree)", "max(degree)", "n()".

**Supported aggregations:**

- n() / count() : count of items
- mean(attr) : mean value
- median(attr) : median value
- sum(attr) : sum of values
- min(attr) : minimum value
- max(attr) : maximum value
- std(attr) : standard deviation
- var(attr) : variance
- quantile(attr, p) : p-th quantile (e.g., quantile(degree, 0.95))

**Parameters**

****aggregations** – Named aggregations (name=expression)

**Returns**

Self for chaining

**Raises**

**ValueError** – If aggregation expression is invalid

### Example

```
>>> Q.nodes().from_layers(L["*"]).compute("degree").per_layer().summarize(
...     mean_degree="mean(degree)",
...     max_degree="max(degree)",
...     median_degree="median(degree)",
...     q95_degree="quantile(degree, 0.95)",
...     n="n()"
```

```
... )
```

`tail`(*n: int = 5*) → QueryBuilder

Keep only the last n results.

Returns the last n results after all other operations are applied. Useful for finding the bottom-k items when combined with sorting.

**Parameters**
`n` – Number of results to keep from the end (default: 5)

**Returns**
Self for chaining

**Example**

```
>>> Q.nodes().compute("degree").arrange("degree").tail(5)  # 5 lowest degree nodes
```

`take`(*n: int = 5*) → QueryBuilder

Keep only the first n results (alias for head).

SQL-style alias for head() method.

**Parameters**
`n` – Number of results to keep (default: 5)

**Returns**
Self for chaining

**Example**

```
>>> Q.nodes().take(10)
```

`to`(*target: str*) → QueryBuilder

Set export target.

**Parameters**
`target` – Export format ('pandas', 'networkx', 'arrow')

**Returns**
Self for chaining

`to_ast()` → Query

Export as AST Query object.

Each call returns an independent snapshot of the current builder state. Mutating the builder after calling `to_ast()` does not affect previously returned objects, and two calls to `to_ast()` do not share mutable state.

**Returns**
Query AST node (deep copy of current builder state)

`to_dsl()` → str

Export as DSL string.

**Returns**
DSL query string

`to_program()` → GraphProgram

Convert query to a GraphProgram object without executing.

This creates an immutable program object that can be: - Composed with other programs - Optimized with rewrite rules - Explained with cost estimates - Diffed against other programs - Cached for reproducibility - Executed later

**Returns**

GraphProgram object

### Example

```
>>> from py3plex.dsl import Q, L
>>> program = (Q.nodes()
...     .from_layers(L["social"])
...     .compute("degree", "betweenness_centrality")
...     .top_k(10, "degree")
...     .to_program())
>>>
>>> # Inspect without executing
>>> print(program.hash())
>>> print(program.explain())
>>>
>>> # Optimize and execute
>>> optimized = program.optimize(budget="10s")
>>> result = optimized.execute(network)
```

`top_k`(*k: int*, *key: str | None = None*) → QueryBuilder

Keep the top-k items per group, ordered by the given key.

Requires that group_by() or per_layer() has been called first.

**Parameters**

- **k** – Number of items to keep per group
- **key** – Attribute/measure to sort by (descending). If None, uses existing order_by.

**Returns**

Self for chaining

**Raises**

**ValueError** – If called without prior grouping

### Example

```
>>> Q.nodes().per_layer().top_k(5, "betweenness_centrality")
```

`top_k_across_layers`(*k: int*, *key: str*, ***, *coverage_mode: str = 'at_least'*, *coverage_k: int = 2*) → QueryBuilder

Explicit across-layer top-k helper with coverage semantics.

`top_k_global`(*k: int*, *key: str*) → QueryBuilder

Explicit global top-k helper for machine-facing clarity.

**top_k_per_layer**(*k: int*, *key: str*) → QueryBuilder

Explicit per-layer top-k helper for machine-facing clarity.

**try_strengths**(*network: Any*, *repeats: int = 30*, *seed: int | None = None*, ***params*) → DataFrame

Compare light/medium/heavy perturbation strengths.

This convenience method runs robustness checks at three different perturbation strengths and returns a summary table for comparison.

**Parameters**

- **network** – Multilayer network to analyze
- **repeats** – Number of counterfactual runs per strength (default: 30)
- **seed** – Random seed for reproducibility (default: None)
- ****params** – Parameter bindings for the query

**Returns**

DataFrame with summary statistics for each strength level

### Example

```
>>> # Compare strengths
>>> summary = (Q.nodes()
...            .compute("betweenness_centrality")
...            .try_strengths(net))
>>> print(summary)
```

**uncertainty**(*method: str | None = 'perturbation'*, *n_samples: int | None = 50*, *ci: float | None = 0.95*, *seed: int | None = None*, ***kwargs*) → QueryBuilder

Alias for uq() - set query-scoped uncertainty configuration.

See uq() for full documentation.

**uq**(*method: str | None = None*, *n_samples: int | None = None*, *ci: float | None = None*, *seed: int | None = None*, *mode: str = 'summarize_only'*, *keep_samples: bool | None = None*, *reduce: str = 'empirical'*, ***kwargs*) → QueryBuilder

Set query-scoped uncertainty quantification configuration.

This method establishes uncertainty defaults for all metrics computed in this query, unless overridden on a per-metric basis in compute().

When parameters are None, they will be resolved from: - Global defaults (set via set_global_uq_defaults()) - Library defaults (perturbation, n_samples=50, ci=0.95)

**Parameters**

- **method** – Uncertainty estimation method ('bootstrap', 'perturbation', 'seed', 'null_model') If None, uses global or library default.
- **n_samples** – Number of samples for uncertainty estimation If None, uses global or library default (50).
- **ci** – Confidence interval level (e.g., 0.95 for 95% CI) If None, uses global or library default (0.95).
- **seed** – Random seed for reproducibility (default: None)

- **mode** – UQ execution mode (default: 'summarize_only') - 'summarize_only': Current behavior, UQ computed per metric - 'propagate': Execute entire query per replicate, combine results
- **keep_samples** – Whether to keep raw samples in results (default: None = auto) Auto defaults to True for propagate mode, False for summarize_only
- **reduce** – Reduction method (default: 'empirical') - 'empirical': Store full sample statistics - 'gaussian': Reduce to mean + std Gaussian approximation
- ****kwargs** – Additional method-specific parameters (e.g., bootstrap_unit='edges', bootstrap_mode='resample', null_model='configuration')

**Returns**

Self for chaining

### Example

```
>>> # Set uncertainty with explicit parameters (summarize_only)
>>> (Q.nodes()
...   .uq(method="perturbation", n_samples=100, ci=0.95, seed=42)
...   .compute("betweenness_centrality")
...   .where(betweenness_centrality__mean__gt=0.1)
...   .execute(net))
```

```
>>> # Use propagate mode for end-to-end uncertainty
>>> (Q.nodes()
...   .compute("pagerank")
...   .order_by("pagerank", desc=True)
...   .limit(3)
...   .uq(method="perturbation", n_samples=25, seed=42, mode="propagate")
...   .execute(net))
```

```
>>> # Use global defaults
>>> set_global_uq_defaults(method="bootstrap", n_samples=200, seed=42)
>>> (Q.nodes()
...   .uq()  # Uses global defaults
...   .compute("degree")
...   .execute(net))
```

```
>>> # Use UQ profile (see UQ class for presets)
>>> (Q.nodes()
...   .uq(UQ.fast(seed=7))
...   .compute("degree")
...   .execute(net))
```

```
>>> # Disable query-level uncertainty
>>> Q.nodes().uq(method=None).compute("degree").execute(net)
```

**validate**() → Dict[str, Any]

Return structured machine-readable validation diagnostics.

**where**(**args*, ***kwargs*) → QueryBuilder

Add WHERE conditions.

Supports two styles:

1. **Keyword arguments:**
   - layer=”social” -> equality
   - degree___gt=5 -> comparison (gt, ge, lt, le, eq, ne)
   - intralayer=True -> intralayer predicate
   - interlayer=(“social”,”work”) -> interlayer predicate
2. **Expression objects (using F):**
   - where(F.degree > 5)
   - where((F.degree > 5) & (F.layer == “social”))
   - where((F.degree > 10) | (F.clustering < 0.5))

**Can mix both styles:**

- where(F.degree > 5, layer=”social”)

**Parameters**

- ***args** – BooleanExpression objects from F
- ****kwargs** – Conditions as keyword arguments

**Returns**
Self for chaining

**window**(*window_size: float | str*, *step: float | str | None = None*, *start: float | None = None*, *end: float | None = None*, *aggregation: str = 'list'*) → QueryBuilder

Add sliding window specification for temporal analysis.

Enables queries that operate over sliding time windows, useful for streaming algorithms and temporal pattern analysis.

**Parameters**

- **window_size** – Size of each window. Can be: - Numeric: treated as timestamp units - String: duration like “7d”, “1h”, “30m”
- **step** – Step size between windows (defaults to window_size for non-overlapping). Same format as window_size.
- **start** – Optional start time for windowing (defaults to network’s first timestamp)
- **end** – Optional end time for windowing (defaults to network’s last timestamp)
- **aggregation** – How to aggregate results across windows: - “list”: Return list of per-window results - “concat”: Concatenate DataFrames - “avg”: Average numeric columns

**Returns**
Self for chaining

#### Examples

```
>>> # Non-overlapping windows of size 100
>>> Q.nodes().compute("degree").window(100.0).execute(tnet)
```

```
>>> # Overlapping windows: size 100, step 50
>>> Q.nodes().compute("degree").window(100.0, step=50.0).execute(tnet)
```

```
>>> # Duration strings (for datetime timestamps)
>>> Q.edges().window("7d", step="1d").execute(tnet)
```

**Note**

Window queries require a TemporalMultiLayerNetwork instance. For regular multi_layer_network, an error will be raised.

**zscore**(**attrs: str*) → QueryBuilder

Compute z-scores for specified attributes.

For each attribute, computes the z-score (standardized value) within the current grouping context. If grouping is active, z-scores are computed per group. Otherwise, they are global.

Creates new columns named "{attr}_zscore".

**Parameters**

***attrs** – Attribute names to compute z-scores for

**Returns**

Self for chaining

#### Example

```
>>> # Global z-scores
>>> Q.nodes().compute("degree", "betweenness").zscore("degree", "betweenness")
```

```
>>> # Per-layer z-scores
>>> Q.nodes().compute("degree").per_layer().zscore("degree")
```

**class py3plex.dsl.QueryPlanner**(*config: Dict[str, Any] | None = None*)

Bases: object

Query planner for DSL v2.

The planner analyzes AST queries and produces optimized execution plans.

**plan**(*ast: Query | SelectStmt*, *network: Any*, *params: Dict[str, Any] | None = None*) → PlannedQuery

Create an execution plan for a query.

**Parameters**

- **ast** – Query or SelectStmt AST
- **network** – Multilayer network

- **params** – Bound parameter values

**Returns**
PlannedQuery with optimized stage order

class py3plex.dsl.**QueryResult**(*target: str*, *items: List[Any]*, *attributes: Dict[str, List[Any] | Dict[Any, Any]] | None = None*, *meta: Dict[str, Any] | None = None*, *computed_metrics: set | None = None*)

Bases: `object`

Rich result object from DSL query execution.

Provides access to query results with multiple export formats and execution metadata.

**target**
'nodes' or 'edges'

**items**
Sequence of node/edge identifiers

**attributes**
Dictionary of computed attributes (column -> values or dict)

**meta**
Metadata about the query execution

**computed_metrics**
Set of metrics that were computed during query execution

**sensitivity_result**
Optional sensitivity analysis results (SensitivityResult)

**property benchmark**
Get benchmark result helper.

Provides convenient access to benchmark-specific views like leaderboards, summaries, and traces.

**Returns**
BenchmarkResultHelper instance

**Raises**
**ValueError** – If result does not contain benchmark data

### Example

```
>>> res = B.community().on(net).algorithms("louvain").execute()
>>> leaderboard = res.benchmark.leaderboard()
>>> summary = res.benchmark.summary()
>>> trace = res.benchmark.trace("autocommunity")
```

**canonical_export_dict**() → Dict[str, Any]
Return canonical, stable serialization for agent handoff.

**property count: int**
Get number of items in result.

**counterexample**(*claim: str*, *params: Dict[str, Any] | None = None*, *seed: int = 42*, *find_minimal: bool = True*, *budget_max_tests: int = 200*, *budget_max_witness_size: int = 500*, *initial_radius: int = 2*) → Any | None

Find counterexample for a claim using query result.

This is a convenience method that builds a counterexample query from the current result's network context.

**Parameters**

- **claim** – Claim string (e.g., "degree___ge(k) -> pagerank___rank_gt(r)")
- **params** – Parameter bindings (e.g., {"k": 10, "r": 50})
- **seed** – Random seed for determinism
- **find_minimal** – Whether to minimize witness
- **budget_max_tests** – Maximum violation tests during minimization
- **budget_max_witness_size** – Maximum witness size (nodes)
- **initial_radius** – Ego subgraph radius

**Returns**

Counterexample object if found, None otherwise

**Raises**

- **ValueError** – If network context is not available
- **CounterexampleNotFound** – If no violation exists

### Example

```
>>> result = Q.nodes().compute("degree", "pagerank").execute(net)
>>> cex = result.counterexample(
...     claim="degree__ge(k) -> pagerank__rank_gt(r)",
...     params={"k": 10, "r": 50},
...     seed=42
... )
```

**debug**() → str

Generate detailed debug information about query execution.

Shows: - Full AST structure - Execution plan - Cache hits/misses - Backend calls - Timing per stage - Randomness sources

**Returns**

Formatted debug information

### Example

```
>>> result = Q.nodes().compute("degree").execute(net)
>>> print(result.debug())
```

**property edges: List[Any]**

Get edges (raises if target is not 'edges').

**explain**() → str

Generate human-readable explanation of the query execution.

Shows: - Query summary - Resolved layers - Measures computed - Aggregations applied - Diagnostics (warnings + info) - Suggested next queries

**Returns**

Formatted explanation string

### Example

```
>>> result = Q.nodes().compute("degree").execute(net)
>>> print(result.explain())
```

**explain_plan**(*analyze: bool = False*) → dict

Return a structured description of the optimizer execution plan.

**Parameters**

**analyze** – When `True` the returned dict will include `actual_cost` values recorded during execution (if available).

**Returns**

- *dict with keys*
- * `logical_plan` – textual summary of the logical operator tree
- * `physical_plan` – serialised physical plan (or `None`)
- * `estimated_cost` – optimizer's pre-execution cost estimate
- * `actual_cost` – post-execution actual cost (if *analyze* and – available)
- * `applied_rules` – list of rule names applied by the optimizer
- * `rule_ids_applied` – stable rewrite-rule identifiers
- * `original_logical_plan` / `optimized_logical_plan` – stable – rendered plans when explanation was requested during execution
- * `optimization_trace` – structured rewrite events

### Example

```
>>> result = Q.nodes().compute("degree").execute(net)
>>> plan = result.explain_plan()
>>> print(plan["applied_rules"])
```

**export_bundle**(*path: str | Path*, *compress: bool = True*, *include_results: bool = True*) → None

Export this result with provenance as a portable bundle.

Creates a file or directory containing: - Provenance metadata (query, network, seeds, environment) - Optionally, the query results - Network snapshot if needed for replay

**Parameters**

- **path** – Output file path (will add .json or .json.gz extension)
- **compress** – Whether to compress the bundle with gzip
- **include_results** – Whether to include result data in bundle

**Raises**
**BundleError** – If export fails

**group_summary()**

Return a summary DataFrame with one row per group.

Returns a pandas DataFrame containing: - Grouping key columns (e.g., “layer”, “src_layer”, “dst_layer”) - n_items: Number of items (nodes/edges) in each group - Any group-level coverage metrics if available

This method only uses information already present in the result and does not recompute expensive measures.

**Returns**
pandas.DataFrame with one row per group

**Raises**

- **ImportError** – If pandas is not available
- **ValueError** – If result does not have grouping metadata

**property has_sensitivity: bool**

Check if this result includes sensitivity analysis.

**Returns**
True if sensitivity results are available

**has_uq(***column: str***)** → bool

Check if a column contains UQ information.

**Parameters**
**column** – Name of the column to check

**Returns**
True if the column contains UQ dicts, False otherwise

**inspect_json()** → str

Return stable JSON payload suitable for machine handoff.

**property is_replayable: bool**

Check if this result has replayable provenance.

**Returns**
True if result can be replayed deterministically

**join(***right: QueryResult | Any***,** *on: str | List[str]***,** *how: str = 'inner'***,** *suffixes: Tuple[str, str] = ('', '_r')***)** → Any

Join this result with another result or query.

This is the escape hatch for joining pre-executed results. For most cases, prefer using QueryBuilder.join() which is lazy and planner-aware.

**Parameters**

- **right** – Right result or query to join with
- **on** – Column name(s) to join on
- **how** – Join type (‘inner’, ‘left’, ‘right’, ‘outer’, ‘semi’, ‘anti’)
- **suffixes** – Tuple of suffixes for name collisions

**Returns**

JoinBuilder for further chaining and execution

### Examples

```
>>> # Join two pre-computed results
>>> nodes = Q.nodes().compute("degree").execute(network)
>>> communities = Q.communities().members().execute(network)
>>> joined = nodes.join(communities, on=["node", "layer"], how="left")
>>> result = joined.execute(network)
```

**property nodes: List[Any]**

Get nodes (raises if target is not 'nodes').

**physical_nodes()** → List[Any]

Return unique physical nodes for node results.

**property provenance: Dict[str, Any] | None**

Get provenance information from metadata.

**Returns**

Provenance dictionary if available, None otherwise

**record_as_experiment**(*, *notes: str | None = None*, *tags: list | None = None*, *store=None*, *save_table: bool = True*)

Record this query result as a trackable experiment.

Wraps the result in an `Experiment` and persists it to the experiment registry store. The returned object can be used to retrieve, reproduce, or export the experiment later.

**Parameters**

- **notes** – Free-text annotation attached to the experiment.
- **tags** – List of string tags for filtering (e.g. `["baseline", "v2"]`).
- **store** – Optional `ExperimentStore` instance. If *None* the global default store is used.
- **save_table** – If `True` (default) the result table is serialised to an Arrow/Parquet or CSV artifact alongside the metadata.

**Returns**

`Experiment` with a stable deterministic `id`.

**Raises**

**ArtifactError** – If the artifact cannot be persisted.

Example:

```
result = Q.nodes().compute("degree").execute(net)
exp = result.record_as_experiment(tags=["demo"])
print(exp.id)
```

**replay**(*backend: Any | None = None*, *strict: bool = True*) → QueryResult

Replay this query to reproduce the result.

Reconstructs the network and query from provenance, then re-executes to produce a new QueryResult. With replayable provenance, the new result should match this

one deterministically.

**Parameters**

- **backend** – Optional backend override (not currently used)
- **strict** – If True, enforce strict version compatibility checks

**Returns**
New QueryResult from replayed query

**Raises**

- **ValueError** – If result is not replayable
- **ReplayError** – If replay fails

**replica_nodes()** → List[Any]

Return replica nodes (node, layer pairs) for node results.

**property sensitivity_curves: Dict[str, Any] | None**

Get sensitivity curves if sensitivity analysis was run.

**Returns**
Dictionary of stability curves keyed by metric name, or None

**summary_dict()** → Dict[str, Any]

Return a compact, machine-readable summary for agent workflows.

**to_arrow()**

Export results to Apache Arrow table.

**Returns**
pyarrow.Table with items and computed attributes

**Raises**
**ImportError** – If pyarrow is not available

**to_csv**(*path: str | Path*, ***kwargs*) → None

Write result rows to a CSV file.

Keyword arguments are passed to `pandas.DataFrame.to_csv()`. The DataFrame index is omitted by default because result fields already identify each row.

**to_dict()** → Dict[str, Any]

Export results as a dictionary.

**Returns**
Dictionary with target, items, attributes, and metadata

**to_json**(*path: str | Path | None = None*, ***kwargs*) → str

Serialize the complete result to JSON, optionally writing it to *path*.

The JSON object contains `target`, `items`, `attributes`, and `meta` fields. Keyword arguments are passed to `json.dumps()`.

**to_networkx**(*network: Any | None = None*)

Export results to NetworkX graph.

For node queries: Returns subgraph containing the selected nodes For edge queries: Returns subgraph containing the selected edges and their endpoints

**Parameters**
**network** – Optional source network to extract subgraph from

**Returns**
NetworkX subgraph containing result items, preserving the supplied network's graph type (including directed multigraphs).

**Raises**
`ImportError` – If networkx is not available

`to_numpy`(*kind: str = 'embedding'*) → np.ndarray

Return embedding matrix aligned to `self.items`.

**Parameters**
`kind` – Currently only `"embedding"` is supported.

**Returns**
numpy.ndarray of shape `(n_items, dim)`.

**Raises**
- `KeyError` – If no embeddings have been computed.
- `ValueError` – If *kind* is not recognised.

`to_pandas`(*multiindex: bool = False*, *include_grouping: bool = True*, *expand_uncertainty: bool = False*, *expand_explanations: bool = False*, *expand_samples: bool = False*, *expand_embeddings: bool = False*, *embedding_prefix: str = 'emb_'*)

Export results to pandas DataFrame.

For node queries: returns `id` column plus computed attribute columns. For edge queries: returns `source`, `target`, `source_layer`, `target_layer`, `weight` columns plus computed attribute columns.

**Parameters**
- `multiindex` – If True and grouping metadata is present, set DataFrame index to the grouping keys (e.g., ["layer"] or ["src_layer", "dst_layer"])
- `include_grouping` – If True and grouping metadata is present, ensure grouping key columns are included in the DataFrame
- `expand_uncertainty` – If True, expand uncertainty metrics into multiple columns: - metric (point estimate/mean) - metric_std (standard deviation) - metric_ci95_low (95% CI lower bound) - metric_ci95_high (95% CI upper bound) - metric_ci95_width (CI width) - p_present (if available from propagate mode) - p_selected (if available from propagate mode with selection) - rank_uq_* (if available from propagate mode with selection)
- `expand_explanations` – If True, expand explanation fields into columns. Explanations are attached via .explain() in the query. Fields like top_neighbors (list) are converted to JSON strings.
- `expand_samples` – If True, include raw sample arrays from UQ results as JSON strings. Only applies when expand_uncertainty=True. Default False to avoid overwhelming output.

**Returns**
pandas.DataFrame with items and computed attributes

**Raises**

**ImportError** – If pandas is not available

**Example**

```
>>> result = Q.nodes().uq(UQ.fast()).compute("degree").execute(net)
>>> df = result.to_pandas(expand_uncertainty=True)
>>> # df now has columns: id, layer, degree, degree_std, degree_ci95_low, degree_c
```

```
>>> result = Q.nodes().explain().compute("degree").execute(net)
>>> df = result.to_pandas(expand_explanations=True)
>>> # df now has explanation columns: community_id, community_size, top_neighbors,
```

**to_parquet**(*path: str*)

Export results to Parquet file.

**Parameters**

**path** – Output file path

**Raises**

**ImportError** – If pyarrow is not available

**property uq_columns: List[str]**

Get list of columns that contain UQ information.

**Returns**

List of column names containing UQ dicts

**class py3plex.dsl.ReduceStmt**(*target: str = 'layers'*, *spec: LayerReductionSpec | None = None*)

Bases: `object`

Top-level reduce/simplify statement.

**spec: LayerReductionSpec | None = None**

**target: str = 'layers'**

**exception py3plex.dsl.ReductionMethodError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `DslExecutionError`

Raised when an unknown or invalid reduction method is requested.

**class py3plex.dsl.ResolvedUQConfig**(*method: str*, *n_samples: int*, *ci: float*, *seed: int | None*, *mode: str = 'summarize_only'*, *keep_samples: bool = False*, *reduce: str = 'empirical'*, *kwargs: ~typing.Dict[str, ~typing.Any] = <factory>*, *provenance: ~typing.Dict[str, str] = <factory>*, *context: str = 'unknown'*, *enabled: bool = True*)

Bases: `object`

Fully resolved and materialized UQ configuration.

This represents the final UQ configuration after applying resolution order. It includes provenance information about where each setting came from.

**method**
UQ method to use

**Type**
str

**n_samples**
Number of samples for uncertainty estimation

**Type**
int

**ci**
Confidence interval level

**Type**
float

**seed**
Random seed for reproducibility

**Type**
int | None

**mode**
UQ execution mode ('summarize_only' or 'propagate')

**Type**
str

**keep_samples**
Whether to keep raw samples

**Type**
bool

**reduce**
Reduction method ('empirical' or 'gaussian')

**Type**
str

**kwargs**
Additional method-specific parameters

**Type**
Dict[str, Any]

**provenance**
Dict mapping each config key to its source

**Type**
Dict[str, str]

**context**
Where this config applies (metric name, query, global)

**Type**
str

**enabled**
Whether UQ is enabled

**Type**
bool

`ci: float`

`context: str = 'unknown'`

`disable_inside_replicate()` → ResolvedUQConfig

Create a copy with UQ disabled for use inside replicate execution.

Returns a copy of this config with enabled=False to prevent nested UQ loops.

`enabled: bool = True`

`keep_samples: bool = False`

`kwargs: Dict[str, Any]`

`method: str`

`mode: str = 'summarize_only'`

`n_samples: int`

`provenance: Dict[str, str]`

`reduce: str = 'empirical'`

`seed: int | None`

`property structural_hash: str`

Compute a stable hash for determinism checks.

This hash includes all configuration parameters that affect execution. Used to verify that replicate execution is deterministic.

`to_dict()` → Dict[str, Any]

Convert to dictionary for storage.

`validate()` → None

Validate the resolved configuration.

**Raises**
`UQResolutionError` – If configuration is invalid

`class py3plex.dsl.S`

Bases: `object`

Semiring algebra factory for creating semiring builders.

### Example

```
>>> S.paths().from_node("A").to_node("B").semiring("min_plus")
>>> S.closure().semiring("boolean").from_layers(L["social"])
```

`static closure()` → SemiringClosureBuilder

Create a semiring closure query builder.

`static paths()` → SemiringPathBuilder

Create a semiring path query builder.

**class** py3plex.dsl.**ScanEdges**

Bases: PhysicalOp

Materialises all edges from the network.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Return all edge tuples from *ctx.network*.

**op_name: str = 'ScanEdges'**

Human-readable operator name used in explain output.

**class** py3plex.dsl.**ScanNodes**

Bases: PhysicalOp

Materialises all node replicas from the network.

**execute**(*ctx: ExecutionContext*, *rows: List[Any]*) → List[Any]

Return all `(node, layer)` tuples from *ctx.network*.

If *rows* is non-empty it is returned unchanged (allows chaining after an upstream scan).

**op_name: str = 'ScanNodes'**

Human-readable operator name used in explain output.

**class** py3plex.dsl.**SchemaProvider**(**args*, ***kwargs*)

Bases: Protocol

Protocol for schema providers.

Schema providers allow the linter to query information about available layers, node/edge types, and attributes in a network.

**get_attribute_type**(*entity_ref: EntityRef*, *attr: str*) → AttrType | None

Get the type of an attribute for a given entity.

**Parameters**

- **entity_ref** – Reference to the entity (node/edge + layer)
- **attr** – Attribute name

**Returns**

Attribute type or None if unknown

**get_edge_count**(*layer: str | None = None*) → int

Get approximate edge count, optionally for a specific layer.

**get_node_count**(*layer: str | None = None*) → int

Get approximate node count, optionally for a specific layer.

**list_edge_types**(*layer: str | None = None*) → List[str]

Get list of edge types, optionally filtered by layer.

**list_layers**() → List[str]

Get list of all layers in the network.

**list_node_types**(*layer: str | None = None*) → List[str]

Get list of node types, optionally filtered by layer.

class py3plex.dsl.**SelectStmt**(*target: ~py3plex.dsl.ast.Target*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *layer_set: ~typing.Any | None = None*, *where: ~py3plex.dsl.ast.ConditionExpr | None = None*, *compute: ~typing.List[~py3plex.dsl.ast.ComputeItem] = <factory>*, *order_by: ~typing.List[~py3plex.dsl.ast.OrderItem] = <factory>*, *limit: int | None = None*, *export: ~py3plex.dsl.ast.ExportTarget | None = None*, *file_export: ~py3plex.dsl.ast.ExportSpec | None = None*, *temporal_context: ~py3plex.dsl.ast.TemporalContext | None = None*, *window_spec: ~py3plex.dsl.ast.WindowSpec | None = None*, *group_by: ~typing.List[str] = <factory>*, *limit_per_group: int | None = None*, *coverage_mode: str | None = None*, *coverage_k: int | None = None*, *coverage_p: float | None = None*, *coverage_group: str | None = None*, *coverage_id_field: str = 'id'*, *select_cols: ~typing.List[str] | None = None*, *drop_cols: ~typing.List[str] | None = None*, *rename_map: ~typing.Dict[str, str] | None = None*, *summarize_aggs: ~typing.Dict[str, str] | None = None*, *distinct_cols: ~typing.List[str] | None = None*, *rank_specs: ~typing.List[~typing.Tuple[str, str]] | None = None*, *zscore_attrs: ~typing.List[str] | None = None*, *post_filters: ~typing.List[~typing.Dict[str, ~typing.Any]] | None = None*, *aggregate_specs: ~typing.Dict[str, ~typing.Any] | None = None*, *mutate_specs: ~typing.Dict[str, ~typing.Any] | None = None*, *autocompute: bool = True*, *uq_config: ~py3plex.dsl.ast.UQConfig | None = None*, *explain_spec: ~py3plex.dsl.ast.ExplainSpec | None = None*, *counterfactual_spec: ~py3plex.dsl.ast.CounterfactualSpec | None = None*, *sensitivity_spec: ~py3plex.dsl.ast.SensitivitySpec | None = None*, *contract_spec: ~py3plex.dsl.ast.ContractSpec | None = None*, *auto_community_config: ~py3plex.dsl.ast.AutoCommunityConfig | None = None*, *embedding_spec: ~py3plex.dsl.ast.EmbeddingSpec | None = None*)

Bases: `object`

A SELECT statement.

**target**

What to select (nodes or edges)

**Type**
py3plex.dsl.ast.Target

**layer_expr**

Optional layer expression for filtering

**Type**
py3plex.dsl.ast.LayerExpr | None

**where**

Optional WHERE conditions

**Type**
py3plex.dsl.ast.ConditionExpr | None

`compute`
List of measures to compute

**Type**
List[py3plex.dsl.ast.ComputeItem]

`order_by`
List of ordering specifications

**Type**
List[py3plex.dsl.ast.OrderItem]

`limit`
Optional limit on results

**Type**
int | None

`export`
Optional export target (for result format conversion)

**Type**
py3plex.dsl.ast.ExportTarget | None

`file_export`
Optional file export specification (for writing to files)

**Type**
py3plex.dsl.ast.ExportSpec | None

`temporal_context`
Optional temporal context for time-based queries

**Type**
py3plex.dsl.ast.TemporalContext | None

`window_spec`
Optional window specification for sliding window analysis

**Type**
py3plex.dsl.ast.WindowSpec | None

`group_by`
List of attribute names to group by (e.g., [“layer”])

**Type**
List[str]

`limit_per_group`
Optional per-group limit for top-k filtering

**Type**
int | None

`coverage_mode`
Coverage filtering mode (“all”, “any”, “at_least”, “exact”, “fraction”)

**Type**
str | None

`coverage_k`
: Threshold for “at_least” or “exact” coverage modes

  **Type**
  int | None

`coverage_p`
: Fraction threshold for “fraction” coverage mode

  **Type**
  float | None

`coverage_group`
: Group attribute for coverage (defaults to primary grouping)

  **Type**
  str | None

`coverage_id_field`
: Field to use for coverage identity (default: “id”)

  **Type**
  str

`select_cols`
: Optional list of columns to keep (for select() operation)

  **Type**
  List[str] | None

`drop_cols`
: Optional list of columns to drop (for drop() operation)

  **Type**
  List[str] | None

`rename_map`
: Optional mapping of old column names to new names

  **Type**
  Dict[str, str] | None

`summarize_aggs`
: Optional dict of name -> aggregation expression for summarize()

  **Type**
  Dict[str, str] | None

`distinct_cols`
: Optional list of columns for distinct operation

  **Type**
  List[str] | None

`rank_specs`
: Optional list of (attr, method) tuples for rank_by()

  **Type**
  List[Tuple[str, str]] | None

`zscore_attrs`
: Optional list of attributes to compute z-scores for

**Type**
List[str] | None

`post_filters`
Optional list of filter specifications to apply after computation

**Type**
List[Dict[str, Any]] | None

`aggregate_specs`
Optional dict of name -> aggregation spec for aggregate()

**Type**
Dict[str, Any] | None

`mutate_specs`
Optional dict of name -> transformation spec for mutate()

**Type**
Dict[str, Any] | None

`autocompute`
Whether to automatically compute missing metrics (default: True)

**Type**
bool

`uq_config`
Optional query-scoped uncertainty quantification configuration

**Type**
py3plex.dsl.ast.UQConfig | None

`counterfactual_spec`
Optional counterfactual robustness specification

**Type**
py3plex.dsl.ast.CounterfactualSpec | None

```
aggregate_specs: Dict[str, Any] | None = None

auto_community_config: AutoCommunityConfig | None = None

autocompute: bool = True

compute: List[ComputeItem]

contract_spec: ContractSpec | None = None

counterfactual_spec: CounterfactualSpec | None = None

coverage_group: str | None = None

coverage_id_field: str = 'id'

coverage_k: int | None = None

coverage_mode: str | None = None

coverage_p: float | None = None

distinct_cols: List[str] | None = None
```

drop_cols: List[str] | None = None

embedding_spec: EmbeddingSpec | None = None

explain_spec: ExplainSpec | None = None

export: ExportTarget | None = None

file_export: ExportSpec | None = None

group_by: List[str]

layer_expr: LayerExpr | None = None

layer_set: Any | None = None

limit: int | None = None

limit_per_group: int | None = None

mutate_specs: Dict[str, Any] | None = None

order_by: List[OrderItem]

post_filters: List[Dict[str, Any]] | None = None

rank_specs: List[Tuple[str, str]] | None = None

rename_map: Dict[str, str] | None = None

select_cols: List[str] | None = None

sensitivity_spec: SensitivitySpec | None = None

summarize_aggs: Dict[str, str] | None = None

target: Target

temporal_context: TemporalContext | None = None

uq_config: UQConfig | None = None

where: ConditionExpr | None = None

window_spec: WindowSpec | None = None

zscore_attrs: List[str] | None = None

class py3plex.dsl.SemiringClosureBuilder

Bases: object

Builder for SEMIRING CLOSURE statements.

### Example

```
>>> from py3plex.dsl import S, L
>>>
>>> result = (
...     S.closure()
...      .semiring("boolean")
...      .from_layers(L["social"])
```

```
...       .method("auto")
...       .execute(network)
... )
```

**backend**(*name: str*) → SemiringClosureBuilder

Set backend.

**Parameters**
**name** – Backend name

**Returns**
Self for chaining

**execute**(*network: Any*, ***params*) → QueryResult

Execute closure query.

**Parameters**

- **network** – Multilayer network
- ****params** – Parameter bindings

**Returns**
QueryResult with closure data

**from_layers**(*layer_expr: LayerExprBuilder*) → SemiringClosureBuilder

Filter by layers.

**Parameters**
**layer_expr** – Layer expression

**Returns**
Self for chaining

**lift**(*attr: str | None = None*, *transform: str | None = None*, *default: Any = 1.0*, *on_missing: str = 'default'*) → SemiringClosureBuilder

Set weight lifting specification.

**Parameters**

- **attr** – Edge attribute name
- **transform** – Optional transformation
- **default** – Default value
- **on_missing** – Behavior on missing

**Returns**
Self for chaining

**method**(*name: str*) → SemiringClosureBuilder

Set closure method.

**Parameters**
**name** – Method name (“auto”, “floyd_warshall”, “iterative”)

**Returns**
Self for chaining

**semiring**(*name: str*) → SemiringClosureBuilder

Set semiring.

**Parameters**
**name** – Semiring name

**Returns**
Self for chaining

`to_ast()` → SemiringClosureStmt

Export as AST.

`class py3plex.dsl.SemiringClosureStmt(`*semiring_spec: ~py3plex.dsl.ast.SemiringSpecNode = <factory>, lift_spec: ~py3plex.dsl.ast.WeightLiftSpecNode = <factory>, layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None, crossing_layers: ~py3plex.dsl.ast.CrossingLayersSpec = <factory>, method: str = 'auto', backend: str = 'graph', output_format: str = 'sparse'*`)`

Bases: `object`

SEMIRING CLOSURE statement for transitive closure.

**DSL Example:**

**S.closure()**
.semiring("boolean") .lift(attr="weight") .from_layers(L["social"]) .backend("graph")

`semiring_spec`

Semiring specification

**Type**
py3plex.dsl.ast.SemiringSpecNode

`lift_spec`

Weight lifting specification

**Type**
py3plex.dsl.ast.WeightLiftSpecNode

`layer_expr`

Optional layer expression

**Type**
py3plex.dsl.ast.LayerExpr | None

`crossing_layers`

Cross-layer edge handling

**Type**
py3plex.dsl.ast.CrossingLayersSpec

`method`

Closure method ("auto", "floyd_warshall", "iterative")

**Type**
str

`backend`

Backend selection

**Type**
str

`output_format`
Output format (“sparse”, “dense”)

**Type**
str

`backend: str = 'graph'`

`crossing_layers: CrossingLayersSpec`

`layer_expr: LayerExpr | None = None`

`lift_spec: WeightLiftSpecNode`

`method: str = 'auto'`

`output_format: str = 'sparse'`

`semiring_spec: SemiringSpecNode`

`class py3plex.dsl.SemiringFixedPointStmt(`*operator: str = 'closure'*, *semiring_spec: ~py3plex.dsl.ast.SemiringSpecNode = <factory>*, *lift_spec: ~py3plex.dsl.ast.WeightLiftSpecNode = <factory>*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *max_iters: int = 100*, *tol: float | None = None*`)`

Bases: `object`

SEMIRING FIXED_POINT statement for iterative computation.

**DSL Example:**

**S.fixed_point()**
.operator(“closure”) .semiring(“boolean”) .max_iters(100) .tol(1e-6)

`operator`
Operator to iterate (“closure”, custom)

**Type**
str

`semiring_spec`
Semiring specification

**Type**
py3plex.dsl.ast.SemiringSpecNode

`lift_spec`
Weight lifting specification

**Type**
py3plex.dsl.ast.WeightLiftSpecNode

`layer_expr`
Optional layer expression

**Type**
py3plex.dsl.ast.LayerExpr | None

**max_iters**
Maximum iterations

**Type**
int

**tol**
Optional tolerance for convergence

**Type**
float | None

**layer_expr:** `LayerExpr | None = None`

**lift_spec:** `WeightLiftSpecNode`

**max_iters:** `int = 100`

**operator:** `str = 'closure'`

**semiring_spec:** `SemiringSpecNode`

**tol:** `float | None = None`

**class** py3plex.dsl.**SemiringPathBuilder**

Bases: `object`

Builder for SEMIRING PATH statements.

### Example

```
>>> from py3plex.dsl import S, L
>>>
>>> result = (
...     S.paths()
...       .from_node("Alice")
...       .to_node("Bob")
...       .semiring("min_plus")
...       .lift(attr="weight", default=1.0)
...       .from_layers(L["social"] + L["work"])
...       .crossing_layers(mode="allowed")
...       .max_hops(5)
...       .execute(network)
... )
```

**backend**(*name: str*) → SemiringPathBuilder

Set backend selection.

**Parameters**
**name** – Backend name (“graph”, “matrix”)

**Returns**
Self for chaining

**crossing_layers**(*mode: str = 'allowed'*, *penalty: float | None = None*) → SemiringPathBuilder

Set cross-layer edge handling.

**Parameters**

- **mode** – Crossing mode (“allowed”, “forbidden”, “penalty”)
- **penalty** – Optional penalty value (for “penalty” mode)

**Returns**
Self for chaining

**execute**(*network: Any*, ***params*) → QueryResult

Execute semiring path query.

**Parameters**

- **network** – Multilayer network
- ****params** – Parameter bindings

**Returns**
QueryResult with path data

**from_layers**(*layer_expr: LayerExprBuilder*) → SemiringPathBuilder

Filter by layers using layer algebra.

**Parameters**
**layer_expr** – Layer expression (e.g., L[“social”] + L[“work”])

**Returns**
Self for chaining

**from_node**(*source: str | ParamRef*) → SemiringPathBuilder

Set source node.

**Parameters**
**source** – Source node identifier or parameter reference

**Returns**
Self for chaining

**k_best**(*k: int*) → SemiringPathBuilder

Find k best paths.

**Parameters**
**k** – Number of best paths to find

**Returns**
Self for chaining

**lift**(*attr: str | None = None*, *transform: str | None = None*, *default: Any = 1.0*, *on_missing: str = 'default'*) → SemiringPathBuilder

Set weight lifting specification.

**Parameters**

- **attr** – Edge attribute name
- **transform** – Optional transformation (“log”)
- **default** – Default value if missing
- **on_missing** – Behavior on missing (“default”, “fail”, “drop”)

**Returns**
Self for chaining

`max_hops`(*n: int*) → SemiringPathBuilder

Set maximum path length.

**Parameters**
`n` – Maximum number of hops

**Returns**
Self for chaining

`semiring`(*name_or_spec: str | Dict[str, str]*) → SemiringPathBuilder

Set semiring specification.

**Parameters**
`name_or_spec` – Either a semiring name (e.g., "min_plus") or a dict mapping layer names to semiring names

**Returns**
Self for chaining

`to_ast`() → SemiringPathStmt

Export as AST SemiringPathStmt object.

`to_node`(*target: str | ParamRef*) → SemiringPathBuilder

Set target node (optional).

**Parameters**
`target` – Target node identifier or parameter reference

**Returns**
Self for chaining

`uq`(*mode: str | None = None*, *samples: int | None = None*, *method: str | None = None*, *seed: int | None = None*, ***kwargs*) → SemiringPathBuilder

Enable uncertainty quantification.

**Parameters**

- `mode` – UQ mode ("bootstrap", "seed")
- `samples` – Number of samples
- `method` – Resampling method
- `seed` – Random seed
- `**kwargs` – Additional UQ parameters

**Returns**
Self for chaining

`witness`(*enabled: bool = True*) → SemiringPathBuilder

Enable/disable witness tracking for path reconstruction.

**Parameters**
`enabled` – Whether to track witnesses

**Returns**
Self for chaining

class py3plex.dsl.SemiringPathStmt(*source: str | ~py3plex.dsl.ast.ParamRef*, *target: str | ~py3plex.dsl.ast.ParamRef | None = None*, *semiring_spec: ~py3plex.dsl.ast.SemiringSpecNode = <factory>*, *lift_spec: ~py3plex.dsl.ast.WeightLiftSpecNode = <factory>*, *layer_expr: ~py3plex.dsl.ast.LayerExpr | None = None*, *crossing_layers: ~py3plex.dsl.ast.CrossingLayersSpec = <factory>*, *max_hops: int | None = None*, *k_best: int | None = None*, *witness: bool = False*, *backend: str = 'graph'*, *uq_config: ~py3plex.dsl.ast.UQConfig | None = None*)

Bases: object

SEMIRING PATH statement for path queries using semiring algebra.

**DSL Example:**

**S.paths()**
.from_node("Alice") .to_node("Bob") .semiring("min_plus") .lift(attr="weight", default=1.0) .from_layers(L["social"] + L["work"]) .crossing_layers(mode="allowed") .max_hops(5) .k_best(3) .witness(True) .backend("graph")

**source**

Source node identifier

**Type**
str | py3plex.dsl.ast.ParamRef

**target**

Optional target node identifier

**Type**
str | py3plex.dsl.ast.ParamRef | None

**semiring_spec**

Semiring specification

**Type**
py3plex.dsl.ast.SemiringSpecNode

**lift_spec**

Weight lifting specification

**Type**
py3plex.dsl.ast.WeightLiftSpecNode

**layer_expr**

Optional layer expression for filtering

**Type**
py3plex.dsl.ast.LayerExpr | None

**crossing_layers**

Cross-layer edge handling

**Type**
py3plex.dsl.ast.CrossingLayersSpec

**max_hops**

Optional maximum path length

**Type**
int | None

**k_best**

Optional number of best paths to find

**Type**
int | None

**witness**

Whether to track path witnesses

**Type**
bool

**backend**

Backend selection (“graph”, “matrix”)

**Type**
str

**uq_config**

Optional uncertainty quantification config

**Type**
py3plex.dsl.ast.UQConfig | None

**backend:** `str = 'graph'`

**crossing_layers:** `CrossingLayersSpec`

**k_best:** `int | None = None`

**layer_expr:** `LayerExpr | None = None`

**lift_spec:** `WeightLiftSpecNode`

**max_hops:** `int | None = None`

**semiring_spec:** `SemiringSpecNode`

**source:** `str | ParamRef`

**target:** `str | ParamRef | None = None`

**uq_config:** `UQConfig | None = None`

**witness:** `bool = False`

**class** py3plex.dsl.**SemiringSpecNode**(*name: str | None = None*, *per_layer: Dict[str, str] | None = None*, *combine_strategy: str | None = None*)

Bases: `object`

Semiring specification for algebra operations.

Supports: - Single semiring: name or per_layer dict - Combined semirings: product or lexicographic combination

**name**

Semiring name (e.g., “min_plus”, “boolean”, “max_times”)

**Type**
str | None

**per_layer**

Optional dict mapping layer -> semiring name

**Type**
Dict[str, str] | None

**combine_strategy**

How to combine per-layer semirings (“product”, “lexicographic”)

**Type**
str | None

**combine_strategy: str | None = None**

**name: str | None = None**

**per_layer: Dict[str, str] | None = None**

**class** py3plex.dsl.**SemiringStmt**(*operation: str*, *paths: SemiringPathStmt | None = None*, *closure: SemiringClosureStmt | None = None*, *fixed_point: SemiringFixedPointStmt | None = None*)

Bases: `object`

Top-level SEMIRING statement (union of path/closure/fixed_point).

**operation**

Operation type (“paths”, “closure”, “fixed_point”)

**Type**
str

**paths**

Path statement (if operation == “paths”)

**Type**
py3plex.dsl.ast.SemiringPathStmt | None

**closure**

Closure statement (if operation == “closure”)

**Type**
py3plex.dsl.ast.SemiringClosureStmt | None

**fixed_point**

Fixed-point statement (if operation == “fixed_point”)

**Type**
py3plex.dsl.ast.SemiringFixedPointStmt | None

**closure: SemiringClosureStmt | None = None**

**fixed_point: SemiringFixedPointStmt | None = None**

**operation: str**

**paths:** `SemiringPathStmt | None = None`

**class** py3plex.dsl.**SpecialPredicate**(*kind: str*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Special multilayer predicates.

**Supported kinds:**

- 'intralayer': Edges within the same layer
- 'interlayer': Edges between specific layers
- 'motif': Motif pattern matching
- 'reachable_from': Cross-layer reachability

**kind**

Predicate type

**Type**
str

**params**

Additional parameters for the predicate

**Type**
Dict[str, Any]

**kind:** `str`

**params:** `Dict[str, Any]`

**class** py3plex.dsl.**SplitSpec**(*strategy: str = 'random_holdout'*, *test_frac: float = 0.2*, *seed: int | None = None*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Data split strategy for predictive tasks.

**params:** `Dict[str, Any]`

**seed:** `int | None = None`

**strategy:** `str = 'random_holdout'`

**test_frac:** `float = 0.2`

**exception** py3plex.dsl.**SplitStrategyError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `PredictionTaskError`

Raised for invalid or failed predictive split strategies.

**class** py3plex.dsl.**Stage**(*stage_type: ~py3plex.dsl.planner.StageType*, *requires_fields: ~typing.Set[str] = <factory>*, *provides_fields: ~typing.Set[str] = <factory>*, *cost_estimate: int = 1*, *name: str = ''*, *params: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Base class for execution stages.

**stage_type**

Type of stage

**Type**
py3plex.dsl.planner.StageType

**requires_fields**

Fields that must exist before this stage

**Type**
Set[str]

**provides_fields**

Fields provided by this stage

**Type**
Set[str]

**cost_estimate**

Relative cost estimate (1-100 scale)

**Type**
int

**name**

Human-readable stage name

**Type**
str

**params**

Stage-specific parameters

**Type**
Dict[str, Any]

**cost_estimate: int = 1**

**name: str = ''**

**params: Dict[str, Any]**

**provides_fields: Set[str]**

**requires_fields: Set[str]**

**stage_type: StageType**

class py3plex.dsl.**StageType**(*value*)

Bases: Enum

Types of execution stages in the planner.

**AGGREGATE = 'aggregate'**

**COMPUTE = 'compute'**

**COVERAGE = 'coverage'**

**EXPLAIN = 'explain'**

EXPORT = 'export'

FILTER_LAYERS = 'filter_layers'

FILTER_WHERE = 'filter_where'

GET_ITEMS = 'get_items'

GROUP = 'group'

LIMIT = 'limit'

ORDER_BY = 'order_by'

class py3plex.dsl.SuggestedFix(*replacement: str*, *span: Tuple[int, int]*)

Bases: object

A suggested fix for a diagnostic.

replacement

The replacement text

**Type**
str

span

Tuple of (start_index, end_index) in the query string

**Type**
Tuple[int, int]

replacement: str

span: Tuple[int, int]

class py3plex.dsl.Target(*value*)

Bases: str, Enum

Query target - what to select from the network.

COMMUNITIES = 'communities'

EDGES = 'edges'

NODES = 'nodes'

class py3plex.dsl.TemporalContext(*kind: str*, *t0: float | None = None*, *t1: float | None = None*, *range_name: str | None = None*)

Bases: object

Temporal context for time-based queries.

This represents temporal constraints on a query, specified via AT or DURING clauses.

kind

Type of temporal constraint ("at" for point-in-time, "during" for interval)

**Type**
str

t0

Start time for interval queries (None for point-in-time)

**Type**
float | None

**t1**

End time for interval queries (None for point-in-time)

**Type**
float | None

**range_name**

Optional named range reference (e.g., "Q1_2023")

**Type**
str | None

### Examples

```
>>> # Point-in-time: AT 1234567890
>>> TemporalContext(kind="at", t0=1234567890.0, t1=1234567890.0)
```

```
>>> # Time range: DURING [100, 200]
>>> TemporalContext(kind="during", t0=100.0, t1=200.0)
```

```
>>> # Named range: DURING RANGE "Q1_2023"
>>> TemporalContext(kind="during", range_name="Q1_2023")
```

**kind:** `str`

**range_name:** `str | None = None`

**t0:** `float | None = None`

**t1:** `float | None = None`

**class** py3plex.dsl.**TrajectoriesBuilder**(*process_ref: str*)

Bases: `object`

Builder for TRAJECTORIES statements.

### Example

```
>>> from py3plex.dsl import Q
>>>
>>> result = (
...     Q.trajectories("sim_result")
...       .where(replicate=5)
...       .at(50)
...       .measure("peak_time", "final_state")
...       .order_by("node_id")
...       .limit(100)
...       .execute()
... )
```

**at**(*t: float*) → TrajectoriesBuilder

Filter to specific time point.

**Parameters**
`t` – Timestamp

**Returns**
Self for chaining

`during`(*t0: float*, *t1: float*) → TrajectoriesBuilder

Filter to time range.

**Parameters**

- `t0` – Start time
- `t1` – End time

**Returns**
Self for chaining

`execute`(*context: Any | None = None*) → Any

Execute trajectory query.

**Parameters**
`context` – Optional context containing simulation results

**Returns**
QueryResult with trajectory data

`limit`(*n: int*) → TrajectoriesBuilder

Limit number of results.

**Parameters**
`n` – Maximum number of results

**Returns**
Self for chaining

`measure`(**measures: str*) → TrajectoriesBuilder

Add trajectory measures to compute.

**Parameters**
`*measures` – Measure names

**Returns**
Self for chaining

`order_by`(*key: str*, *desc: bool = False*) → TrajectoriesBuilder

Add ordering specification.

**Parameters**

- `key` – Attribute to order by
- `desc` – If True, descending order

**Returns**
Self for chaining

`to`(*target: str*) → TrajectoriesBuilder

Set export target.

**Parameters**
`target` – Export format

**Returns**

Self for chaining

`to_ast()` → TrajectoriesStmt

Export as AST TrajectoriesStmt object.

`where(**kwargs)` → TrajectoriesBuilder

Add WHERE conditions on trajectories.

**Parameters**
**`**kwargs`** – Conditions (e.g., replicate=5, node="Alice")

**Returns**
Self for chaining

`class py3plex.dsl.TrajectoriesStmt(`*process_ref: str*, *where: ~py3plex.dsl.ast.ConditionExpr | None = None*, *temporal_context: ~py3plex.dsl.ast.TemporalContext | None = None*, *measures: ~typing.List[str] = <factory>*, *order_by: ~typing.List[~py3plex.dsl.ast.OrderItem] = <factory>*, *limit: int | None = None*, *export_target: str | None = None*`)`

Bases: `object`

TRAJECTORIES statement for querying simulation results.

**DSL Example:**
TRAJECTORIES FROM process_result WHERE replicate = 5 AT time = 50 MEASURE peak_time, final_state ORDER BY node_id LIMIT 100

`process_ref`

Reference to a dynamics process or result

**Type**
str

`where`

Optional WHERE conditions on trajectories

**Type**
py3plex.dsl.ast.ConditionExpr | None

`temporal_context`

Optional temporal filtering (at specific time, during range)

**Type**
py3plex.dsl.ast.TemporalContext | None

`measures`

List of trajectory measures to compute

**Type**
List[str]

`order_by`

List of ordering specifications

**Type**
List[py3plex.dsl.ast.OrderItem]

**limit**

Optional limit on results

**Type**
int | None

**export_target**

Optional export format

**Type**
str | None

**export_target:** `str | None = None`

**limit:** `int | None = None`

**measures:** `List[str]`

**order_by:** `List[OrderItem]`

**process_ref:** `str`

**temporal_context:** `TemporalContext | None = None`

**where:** `ConditionExpr | None = None`

**class** py3plex.dsl.**TypeEnvironment**

Bases: `object`

Type environment for DSL queries.

Tracks types of attributes, computed values, and other entities in a query context.

**add_layer**(*layer: str*)

Add a layer reference.

**get_attribute_type**(*name: str*) → AttrType

Get the type of an attribute.

**get_computed_type**(*name: str*) → AttrType

Get the type of a computed value.

**has_layer**(*layer: str*) → bool

Check if a layer is known.

**set_attribute_type**(*name: str*, *attr_type: AttrType*)

Set the type of an attribute.

**set_computed_type**(*name: str*, *attr_type: AttrType*)

Set the type of a computed value.

**exception** py3plex.dsl.**TypeMismatchError**(*attribute: str*, *expected_type: str*, *actual_type: str*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslError`

Exception raised when there's a type mismatch.

**attribute**

The attribute with the type mismatch

**expected_type**
Expected type

**actual_type**
Actual type received

**class py3plex.dsl.UQ**

Bases: object

Uncertainty quantification profiles for ergonomic one-liners.

This class provides convenient presets for common uncertainty estimation scenarios. Each profile returns a UQConfig that can be passed to .uq().

### Example

```
>>> from py3plex.dsl import Q, UQ
>>>
>>> # Fast exploratory analysis
>>> Q.nodes().uq(UQ.fast(seed=42)).compute("degree").execute(net)
>>>
>>> # Default balanced settings
>>> Q.nodes().uq(UQ.default()).compute("betweenness_centrality").execute(net)
>>>
>>> # Publication-quality with more samples
>>> Q.nodes().uq(UQ.paper(seed=123)).compute("closeness").execute(net)
```

**static default**(*seed: int | None = None*) → UQConfig

Default balanced profile.

Settings: perturbation, n=50, ci=0.95

Use this for general-purpose uncertainty estimation with reasonable computational cost.

**Parameters**
**seed** – Random seed for reproducibility (default: None)

**Returns**
UQConfig with default settings

### Example

```
>>> Q.nodes().uq(UQ.default()).compute("betweenness_centrality").execute(net)
```

**static fast**(*seed: int | None = None*) → UQConfig

Fast exploratory profile with minimal samples.

Settings: perturbation, n=25, ci=0.95

Use this for quick exploratory analysis when speed matters more than precision.

**Parameters**
**seed** – Random seed for reproducibility (default: None)

**Returns**
UQConfig with fast settings

#### Example

```
>>> Q.nodes().uq(UQ.fast(seed=0)).compute("degree").execute(net)
```

**static paper**(*seed: int | None = None*) → UQConfig

Publication-quality profile with thorough sampling.

Settings: bootstrap, n=300, ci=0.95

Use this for publication-quality results where precision is critical and computational cost is acceptable.

**Parameters**
**seed** – Random seed for reproducibility (default: None)

**Returns**
UQConfig with publication-quality settings

#### Example

```
>>> Q.nodes().uq(UQ.paper(seed=123)).compute("closeness").execute(net)
```

**class py3plex.dsl.UQConfig**(*method: str | None = None*, *n_samples: int | None = None*, *ci: float | None = None*, *seed: int | None = None*, *mode: str = 'summarize_only'*, *keep_samples: bool | None = None*, *reduce: str = 'empirical'*, *kwargs: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Query-scoped uncertainty quantification configuration.

This dataclass stores uncertainty estimation settings at the query level, providing defaults for all metrics computed in the query unless explicitly overridden on a per-metric basis.

**method**

Uncertainty estimation method ('bootstrap', 'perturbation', 'seed', 'null_model', 'stratified_perturbation')

**Type**
str | None

**n_samples**

Number of samples for uncertainty estimation

**Type**
int | None

**ci**

Confidence interval level (e.g., 0.95 for 95% CI)

**Type**
float | None

**seed**

Random seed for reproducibility

**Type**
int | None

**mode**

UQ execution mode ('summarize_only', 'propagate') - 'summarize_only': Current behavior, UQ computed per metric - 'propagate': Execute entire query per replicate, combine results

**Type**
str

**keep_samples**

Whether to keep raw samples in UQValue (None = auto)

**Type**
bool | None

**reduce**

Reduction method ('empirical', 'gaussian') - 'empirical': Store full sample statistics - 'gaussian': Reduce to mean + std Gaussian approximation

**Type**
str

**kwargs**

Additional method-specific parameters (e.g., bootstrap_unit, bootstrap_mode, strata, bins for stratified_perturbation)

**Type**
Dict[str, Any]

### Example

```
>>> uq = UQConfig(method="perturbation", n_samples=100, ci=0.95, seed=42)
>>> uq = UQConfig(method="bootstrap", n_samples=200, ci=0.95,
...               kwargs={"bootstrap_unit": "edges", "bootstrap_mode": "resample"})
>>> uq = UQConfig(method="stratified_perturbation", n_samples=100, ci=0.95, seed=42,
...               kwargs={"strata": ["degree", "layer"], "bins": {"degree": 5}})
>>> uq = UQConfig(method="perturbation", n_samples=50, ci=0.95, seed=42,
...               mode="propagate", keep_samples=True, reduce="empirical")
```

**ci:** `float | None = None`

**keep_samples:** `bool | None = None`

**kwargs:** `Dict[str, Any]`

**method:** `str | None = None`

**mode:** `str = 'summarize_only'`

**n_samples:** `int | None = None`

**reduce:** `str = 'empirical'`

**seed:** `int | None = None`

**exception** py3plex.dsl.**UQResolutionError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `DslExecutionError`

Error raised when UQ configuration conflicts or is invalid.

**exception** py3plex.dsl.**UQSchemaValidationError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `DslExecutionError`

Error raised when UQ result doesn't conform to canonical schema.

**exception** py3plex.dsl.**UQUnsupportedError**(*message: str*, *intent: str | None = None*, *why_failed: str | None = None*, *examples: List[str] | None = None*, *pitfall: str | None = None*, ***kwargs*)

Bases: `DslExecutionError`

Error raised when UQ is requested for unsupported context.

**exception** py3plex.dsl.**UnknownAttributeError**(*attribute: str*, *known_attributes: List[str] | None = None*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslExecutionError`

Exception raised when an unknown attribute is referenced.

**attribute**
: The unknown attribute name

**known_attributes**
: List of valid attribute names

**suggestion**
: Suggested alternative, if any

**exception** py3plex.dsl.**UnknownLayerError**(*layer: str*, *known_layers: List[str] | None = None*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslError`

Exception raised when an unknown layer is referenced.

**layer**
: The unknown layer name

**known_layers**
: List of valid layer names

**suggestion**
: Suggested alternative, if any

**exception** py3plex.dsl.**UnknownMeasureError**(*measure: str*, *known_measures: List[str] | None = None*, *query: str | None = None*, *line: int | None = None*, *column: int | None = None*)

Bases: `DslExecutionError`

Exception raised when an unknown measure is referenced.

**measure**

The unknown measure name

**known_measures**

List of valid measure names

**suggestion**

Suggested alternative, if any

class py3plex.dsl.**WeightLiftSpecNode**(*attr: str | None = None*, *transform: str | None = None*, *default: Any = 1.0*, *on_missing: str = 'default'*)

Bases: `object`

Weight lifting specification for edge attribute extraction.

**attr**

Edge attribute name (e.g., "weight", "cost")

**Type**
str | None

**transform**

Optional transformation ("log", custom callable reference)

**Type**
str | None

**default**

Default value if attribute missing

**Type**
Any

**on_missing**

Behavior on missing attribute ("default", "fail", "drop")

**Type**
str

**attr:  str | None = None**

**default:  Any = 1.0**

**on_missing:  str = 'default'**

**transform:  str | None = None**

py3plex.dsl.**build_community_records**(*network: Any*, *partition: Dict[Tuple[Any, Any], Any]*, *name: str = 'default'*, *metadata: Dict[str, Any] | None = None*) → List[CommunityRecord]

Build CommunityRecord objects from a partition.

**Parameters**

- **network** – Multilayer network object
- **partition** – Dict mapping (node, layer) tuples to community IDs
- **name** – Name of the partition (e.g., "louvain", "infomap")

- **metadata** – Algorithm-specific metadata

**Returns**
List of CommunityRecord objects

py3plex.dsl.clear_cache() → None

Clear the global cache.

py3plex.dsl.compile_pattern(*pattern: PatternGraph*, *network: Any | None = None*, *injective: bool = True*) → PatternPlan

Compile a pattern graph into an execution plan.

py3plex.dsl.compute_centrality_for_layer(*network: Any*, *layer: str*, *centrality: str = 'betweenness_centrality'*) → Dict[Any, float]

Compute centrality for all nodes in a layer.

**Parameters**

- **network** – Multilayer network object
- **layer** – Layer identifier
- **centrality** – Centrality measure name

**Returns**
Dictionary mapping nodes to centrality values

py3plex.dsl.compute_community_metric(*record: CommunityRecord*, *metric: str*, *network: Any | None = None*) → Any

Compute a metric for a single community.

**Parameters**

- **record** – CommunityRecord object
- **metric** – Metric name (e.g., "size", "conductance", "hub_nodes_top_k")
- **network** – Network object (needed for some metrics)

**Returns**
Computed metric value

py3plex.dsl.create_cache_key(*network_fingerprint: Dict[str, Any]*, *ast_hash: str*, *measure_name: str*, *params: Dict[str, Any] | None = None*, *seed: int | None = None*, *uq_method: str | None = None*, *n_samples: int | None = None*) → str

Create a deterministic cache key.

**Parameters**

- **network_fingerprint** – Network fingerprint dict
- **ast_hash** – AST hash
- **measure_name** – Measure name
- **params** – Query parameters
- **seed** – Random seed if applicable
- **uq_method** – Uncertainty quantification method if applicable
- **n_samples** – Number of samples for UQ if applicable

**Returns**
Cache key (hex string)

py3plex.dsl.**create_degenerate_uq_result**(*value: float*, *method: str = 'deterministic'*, *n_samples: int = 1*, *seed: int | None = None*) → Dict[str, Any]

Create a degenerate UQ result for deterministic metrics.

This is used when UQ is requested but the metric is deterministic (e.g., degree, which doesn't vary with random seeds).

**Parameters**

- **value** – The deterministic value
- **method** – Method name (default: "deterministic")
- **n_samples** – Number of samples (default: 1)
- **seed** – Random seed (optional)

**Returns**
Dictionary conforming to canonical UQ schema with std=0

py3plex.dsl.**describe_operator**(*name: str*) → Dict[str, Any] | None

Get detailed information about a registered operator.

**Parameters**
**name** – Operator name

**Returns**
Dictionary with operator metadata, or None if not found

### Example

```
>>> @dsl_operator("layer_resilience", description="Compute resilience score for current
... def layer_resilience_op(context: DSLExecutionContext, alpha: float = 0.1):
...     return 42.0
>>> info = describe_operator("layer_resilience")
>>> info["description"]
'Compute resilience score for current layers.'
```

py3plex.dsl.**detect_communities**(*network: Any*, *layer: str | None = None*) → Dict[str, Any]

Detect communities in the network using Louvain algorithm via DSL.

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**

- 'partition': Dict mapping nodes to community IDs
- 'num_communities': Number of communities detected
- 'community_sizes': Dict mapping community ID to size
- 'biggest_community': Tuple (community_id, size)
- 'smallest_community': Tuple (community_id, size)

- 'size_distribution': List of community sizes

**Return type**
Dictionary containing

### Example

```
>>> from py3plex.core import multinet
>>> from py3plex.dsl import detect_communities
>>>
>>> # Create a sample network
>>> network = multinet.multi_layer_network()
>>> # ... add nodes and edges ...
>>>
>>> # Detect communities
>>> result = detect_communities(network)
>>> print(f"Found {result['num_communities']} communities")
>>> print(f"Biggest community has {result['biggest_community'][1]} nodes")
```

py3plex.dsl.**dsl_operator**(*name: str | None = None*, ***, *description: str | None = None*, *category: str | None = None*, *overwrite: bool = False*)

Decorator to register a Python function as a DSL operator.

This decorator allows users to define custom operators that can be used in DSL queries. The decorated function should accept a DSLExecutionContext as its first argument, followed by any keyword arguments.

**Parameters**

- **name** – Operator name (defaults to function name if not provided)
- **description** – Human-readable description (defaults to function docstring)
- **category** – Optional category for organization (e.g., "centrality", "dynamics")
- **overwrite** – If True, allow replacing existing operators

**Returns**
Decorator function that registers the operator

### Example

```
>>> @dsl_operator("layer_resilience", category="dynamics", overwrite=True)
... def layer_resilience_op(context: DSLExecutionContext, alpha: float = 0.1):
...     '''Compute resilience score for current layers.'''
...     # Access context.graph, context.current_layers, etc.
...     return 42.0
```

```
>>> # Use in DSL:
>>> # measure layer_resilience(alpha=0.2) on layers ["infra", "power"]
```

py3plex.dsl.**execute_ast**(*network: Any*, *query: Query*, *params: Dict[str, Any] | None = None*, *progress: bool = True*, *explain_plan: bool = False*, *planner_config: Dict[str, Any] | None = None*, *optimize: bool | None = None*) → QueryResult | ExecutionPlan

Execute an AST query on a multilayer network.

**Parameters**

- **network** – Multilayer network object
- **query** – Query AST
- **params** – Parameter bindings
- **progress** – If True, log progress messages during query execution (default: True)
- **explain_plan** – If True, populate result.meta["plan"] with execution plan
- **planner_config** – Optional planner configuration dict
- **optimize** – Override the configured DSL optimizer setting for this execution

**Returns**

QueryResult or ExecutionPlan (if explain=True)

py3plex.dsl.**execute_query**(*network: Any*, *query: str*) → Dict[str, Any]

Execute a DSL query on a multilayer network.

Supports both SELECT and MATCH queries:

**SELECT queries:**

SELECT nodes WHERE layer="transport" AND degree > 5 SELECT * FROM nodes IN LAYER 'ppi' WHERE degree > 10 SELECT id, degree FROM nodes IN LAYERS ('ppi', 'coexpr') WHERE color = 'red'

**MATCH queries (Cypher-like):**

MATCH (g:Gene)-[r:REGULATES]->(t:Gene) IN LAYER 'reg' WHERE g.degree > 10 RETURN g, t

**Parameters**

- **network** – Multilayer network object (multi_layer_network instance)
- **query** – DSL query string

**Returns**

- 'nodes' or 'edges' or 'bindings': List of selected items / pattern matches
- 'computed': Dictionary of computed measures (if COMPUTE used)
- 'query': Original query string

**Return type**

Dictionary containing

**Raises**

- **DSLSyntaxError** – If query syntax is invalid
- **DSLExecutionError** – If query cannot be executed

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'transport'}])
>>> net.add_nodes([{'source': 'B', 'type': 'transport'}])
>>> net.add_nodes([{'source': 'C', 'type': 'social'}])
>>> net.add_edges([
...     {'source': 'A', 'target': 'B', 'source_type': 'transport', 'target_type': 'tran
...     {'source': 'B', 'target': 'C', 'source_type': 'social', 'target_type': 'social'
... ])
>>>
>>> # Select all nodes in "transport" layer
>>> result = execute_query(net, 'SELECT nodes WHERE layer="transport"')
>>> result['count'] >= 0
True
>>>
>>> # Select high-degree nodes and compute centrality
>>> result = execute_query(net, 'SELECT nodes WHERE degree > 0 COMPUTE betweenness_cent
>>> 'computed' in result
True
>>>
>>> # Complex query with multiple conditions
>>> result = execute_query(net, 'SELECT nodes WHERE layer="social" AND degree >= 0')
>>> result['count'] >= 0
True
```

py3plex.dsl.**explain**(*query: Query*, *graph: Any | None = None*, *schema: SchemaProvider | None = None*) → ExplainResult

Explain a DSL query with type information and cost estimates.

Provides detailed information about: - Query structure (AST) - Inferred types for all expressions - Estimated execution cost - Potential issues (via linting)

**Parameters**

- **query** – Query AST to explain
- **graph** – Optional py3plex network for schema extraction
- **schema** – Optional schema provider

**Returns**

ExplainResult with detailed query information

### Example

```
>>> result = explain(query, graph=network)
>>> print(result.ast_summary)
>>> print(f"Cost: {result.cost_estimate}")
>>> for diag in result.diagnostics:
...     print(diag)
```

py3plex.dsl.**export_result**(*result: Any*, *spec: ExportSpec*) → None

Export query result to a file according to the export specification.

This is the main entry point for file exports. It normalizes the result into a tabular format and dispatches to format-specific writers.

**Parameters**

- **result** – Query result (QueryResult, dict, or other supported type)
- **spec** – Export specification with path, format, columns, and options

**Raises**

**DslExecutionError** – If export fails or result type is not supported

py3plex.dsl.**filter_communities**(*records: List[CommunityRecord]*, ***filters*) → List[CommunityRecord]

Filter community records based on predicates.

**Parameters**

- **records** – List of CommunityRecord objects
- ****filters** – Filter predicates (e.g., size__gt=10, density_intra__lt=0.5)

**Returns**

Filtered list of CommunityRecord objects

py3plex.dsl.**format_result**(*result: Dict[str, Any]*, *limit: int = 10*) → str

Format query result as human-readable string.

**Parameters**

- **result** – Result dictionary from execute_query
- **limit** – Maximum number of items to display

**Returns**

Formatted string representation

py3plex.dsl.**get_biggest_community**(*network: Any*, *layer: str | None = None*) → Tuple[int, int, List[Any]]

Get the largest community in the network.

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**

Tuple of (community_id, size, list_of_nodes)

### Example

```
>>> community_id, size, nodes = get_biggest_community(network)
>>> print(f"Community {community_id} has {size} nodes")
>>> print(f"Nodes: {nodes}")
```

py3plex.dsl.**get_cache_statistics**() → Dict[str, Any]

Get global cache statistics.

**Returns**

Dictionary with cache statistics

py3plex.dsl.**get_community_partition**(*network: Any*, *layer: str | None = None*) → Dict[Any, int]

Get community partition (mapping of nodes to community IDs).

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**

Dictionary mapping nodes to community IDs

### Example

```
>>> partition = get_community_partition(network)
>>> for node, community_id in partition.items():
...     print(f"Node {node} is in community {community_id}")
```

py3plex.dsl.**get_community_size_distribution**(*network: Any*, *layer: str | None = None*) → List[int]

Get the distribution of community sizes.

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**

List of community sizes sorted in descending order

### Example

```
>>> distribution = get_community_size_distribution(network)
>>> print(f"Largest community: {distribution[0]} nodes")
>>> print(f"Smallest community: {distribution[-1]} nodes")
>>> print(f"Average size: {sum(distribution) / len(distribution):.1f}")
```

py3plex.dsl.**get_community_sizes**(*network: Any*, *layer: str | None = None*) → Dict[int, int]

Get the size of each community in the network.

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**

Dictionary mapping community ID to its size

### Example

```
>>> sizes = get_community_sizes(network)
>>> for comm_id, size in sizes.items():
...     print(f"Community {comm_id}: {size} nodes")
```

py3plex.dsl.**get_global_cache**() → CacheBackend

Get or create the global cache instance.

**Returns**
Global CacheBackend instance

py3plex.dsl.**get_global_uq_defaults**() → Dict[str, Any]

Get current global UQ defaults.

py3plex.dsl.**get_metric**(*name: str*) → MetricSpec

Return the `MetricSpec` for *name*, raising if unknown.

Handles canonical names as well as registered aliases.

**Parameters**
**name** – Metric name to look up.

**Returns**
`MetricSpec`

**Raises**
**ValueError** – If *name* is not a known canonical name or alias.

py3plex.dsl.**get_num_communities**(*network: Any*, *layer: str | None = None*) → int

Get the number of communities in the network.

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**
Number of communities detected

### Example

```
>>> num_communities = get_num_communities(network)
>>> print(f"Found {num_communities} communities")
```

py3plex.dsl.**get_operator**(*name: str*) → DSLOperator | None

Get a registered operator by name from the global registry.

This is a convenience function that delegates to operator_registry.get().

**Parameters**
**name** – Operator name (will be normalized)

**Returns**
DSLOperator instance or None if not found

py3plex.dsl.**get_smallest_community**(*network: Any*, *layer: str | None = None*) → Tuple[int, int, List[Any]]

Get the smallest community in the network.

**Parameters**

- **network** – Multilayer network object
- **layer** – Optional layer to filter nodes by

**Returns**

Tuple of (community_id, size, list_of_nodes)

**Example**

```
>>> community_id, size, nodes = get_smallest_community(network)
>>> print(f"Community {community_id} has {size} nodes")
>>> print(f"Nodes: {nodes}")
```

py3plex.dsl.**is_known_metric**(*name: str*) → bool

Return `True` if *name* is a known metric (canonical or alias).

**Parameters**

**name** – Metric name to check.

**Returns**

bool

py3plex.dsl.**lint**(*query: Query*, *graph: Any | None = None*, *schema: SchemaProvider | None = None*) → List[Diagnostic]

Lint a DSL query.

Analyzes the query for potential issues including: - Unknown layers or attributes - Type mismatches - Unsatisfiable or redundant predicates - Performance issues

**Parameters**

- **query** – Query AST to lint
- **graph** – Optional py3plex network for schema extraction
- **schema** – Optional schema provider (auto-created from graph if not provided)

**Returns**

List of diagnostics found

**Example**

```
>>> from py3plex.dsl import Q, L, lint
>>> from py3plex.core import multinet
>>>
>>> network = multinet.multi_layer_network()
>>> # ... build network ...
>>>
>>> query = Q.nodes().from_layers(L["social"]).where(degree__gt=5).build()
>>> diagnostics = lint(query, graph=network)
>>>
>>> for diag in diagnostics:
...     print(f"{diag.severity}: {diag.message}")
```

py3plex.dsl.**list_operators**() → Dict[str, DSLOperator]

List all registered operators from the global registry.

This is a convenience function that delegates to operator_registry.list_operators().

**Returns**

Dictionary mapping operator names to DSLOperator instances

py3plex.dsl.load_from_parquet(*path: str*) → Any

Load query result from Parquet file.

**Parameters**

**path** – Input file path

**Returns**

pandas DataFrame with the loaded data

**Raises**

- **ImportError** – If pyarrow or pandas is not available
- **DslExecutionError** – If file cannot be read

py3plex.dsl.match_pattern(*network: Any*, *pattern: PatternGraph*, *plan: PatternPlan*, *limit: int | None = None*, *timeout: float | None = None*) → List[MatchRow]

Execute pattern matching on a network.

py3plex.dsl.plan_query(*ast: Query | SelectStmt*, *network: Any*, *params: Dict[str, Any] | None = None*, *config: Dict[str, Any] | None = None*) → PlannedQuery

Plan a query for execution.

This is the main entry point for query planning.

**Parameters**

- **ast** – Query or SelectStmt AST
- **network** – Multilayer network
- **params** – Bound parameter values
- **config** – Optional planner configuration

**Returns**

PlannedQuery with optimized execution plan

### Example

```
>>> from py3plex.dsl import Q
>>> q = Q.nodes().compute("degree").order_by("degree")
>>> plan = plan_query(q._ast, network)
>>> print(plan.plan_meta["rewrite_summary"])
```

py3plex.dsl.register_operator(*name: str*, *func: Callable[[...], Any] | None = None*, *description: str | None = None*, *category: str | None = None*, *overwrite: bool = False*) → Callable[[...], Any]

Register a DSL operator with the global registry.

This is a convenience function that delegates to operator_registry.register().

**Parameters**

- **name** – Operator name (will be normalized)
- **func** – Python callable implementing the operator
- **description** – Optional description
- **category** – Optional category for organization

- **overwrite** – If True, allow replacing existing operators

py3plex.dsl.**reset_global_uq_defaults**() → None

Reset global UQ defaults to empty.

py3plex.dsl.**resolve_uq_config**(*compute_item: ComputeItem*, *query_uq_config: UQConfig | None*, *metric_name: str*) → ResolvedUQConfig | None

Resolve UQ configuration following the priority order.

**Resolution order:**

1. Metric-level (from compute_item parameters)
2. Query-level (from .uq() call)
3. Global defaults (from UQ.defaults())
4. Library defaults

**Parameters**

- **compute_item** – ComputeItem with potential metric-level config
- **query_uq_config** – Query-level UQConfig (from .uq() call)
- **metric_name** – Name of the metric being computed

**Returns**

ResolvedUQConfig if UQ is enabled, None otherwise

**Raises**

**UQResolutionError** – If conflicting configs at same priority level

py3plex.dsl.**save_to_parquet**(*result: Any*, *path: str*, *columns: List[str] | None = None*) → None

Save query result to Parquet file.

This is a convenience function for saving QueryResult or similar objects to Parquet format.

**Parameters**

- **result** – Query result (QueryResult or compatible type)
- **path** – Output file path
- **columns** – Optional column selection

**Raises**

- **ImportError** – If pyarrow is not available
- **DslExecutionError** – If result type is not supported

py3plex.dsl.**select_high_degree_nodes**(*network: Any*, *min_degree: int*, *layer: str | None = None*) → List[Any]

Select nodes with degree greater than threshold.

**Parameters**

- **network** – Multilayer network object
- **min_degree** – Minimum degree threshold (exclusive - nodes must have degree > min_degree)
- **layer** – Optional layer to filter by

**Returns**
List of nodes with degree > min_degree

py3plex.dsl.**select_nodes_by_layer**(*network: Any*, *layer: str*) → List[Any]

Select all nodes in a specific layer.

**Parameters**

- **network** – Multilayer network object
- **layer** – Layer identifier

**Returns**
List of nodes in the specified layer

py3plex.dsl.**set_global_cache**(*cache: CacheBackend | None*) → None

Set the global cache instance.

**Parameters**
**cache** – CacheBackend instance or None to disable

py3plex.dsl.**set_global_uq_defaults**(***kwargs*) → None

Set global UQ defaults.

These defaults are used when no query-level or metric-level config is provided.

**Parameters**
****kwargs** – UQ parameters (method, n_samples, ci, seed, etc.)

py3plex.dsl.**unregister_operator**(*name: str*) → None

Unregister an operator from the global registry.

This is a convenience function that delegates to operator_registry.unregister(). Primarily useful for testing and cleanup.

**Parameters**
**name** – Operator name (will be normalized)

py3plex.dsl.**validate_uq_result_schema**(*result: Dict[str, Any]*, *metric_name: str*, *allow_degenerate: bool = False*) → None

Validate that a UQ result conforms to the canonical schema.

**Parameters**

- **result** – UQ result dictionary to validate
- **metric_name** – Name of the metric (for error messages)
- **allow_degenerate** – If True, allow degenerate UQ (std=0, no quantiles)

**Raises**
**UQSchemaValidationError** – If result doesn't conform to schema

py3plex.dsl.**wrap_deterministic_as_uq**(*value: Any*, *resolved_config: ResolvedUQConfig*) → Dict[str, Any]

Wrap a deterministic value with UQ scaffolding.

This is used when UQ is requested but the computation is deterministic.

**Parameters**

- **value** – The deterministic value (scalar or dict)
- **resolved_config** – The resolved UQ configuration

**Returns**
Dictionary conforming to canonical UQ schema

### Uncertainty Quantification

Uncertainty estimation tools and supporting types for ranking and inference tasks.

First-class uncertainty support for py3plex.

This module provides types and utilities for representing statistics with uncertainty information. The core idea is that every statistic is an object that can carry a distribution:

- Deterministic mode: object has mean with std=None (certainty 1.0)
- Uncertainty mode: object has mean, std, quantiles populated

This makes uncertainty "first-class" - it's baked into how numbers exist in the library, not bolted on later.

### Examples

```
>>> from py3plex.uncertainty import StatSeries
>>> # Deterministic result
>>> result = StatSeries(
...     index=['a', 'b', 'c'],
...     mean=np.array([1.0, 2.0, 3.0])
... )
>>> result.is_deterministic
True
>>> result.certainty
1.0
```

```
>>> # Uncertain result
>>> result_unc = StatSeries(
...     index=['a', 'b', 'c'],
...     mean=np.array([1.0, 2.0, 3.0]),
...     std=np.array([0.1, 0.2, 0.15]),
...     quantiles={
...         0.025: np.array([0.8, 1.6, 2.7]),
...         0.975: np.array([1.2, 2.4, 3.3])
...     }
... )
>>> result_unc.is_deterministic
False
>>> np.array(result_unc)  # Backward compat - gives mean
array([1., 2., 3.])
```

**class** py3plex.uncertainty.**CoAssignmentReducer**(*n_nodes: int*, *sparse: bool = True*, *topk: int = 50*, *threshold: float = 0.0*)

Bases: `PartitionReducer`

Compute co-assignment probability matrix.

P(i, j) = probability that nodes i and j are in the same community.

This can be computed in sparse mode (top-k neighbors only) to save memory.

**Parameters**

- **n_nodes** (*int*) – Number of nodes
- **sparse** (*bool, default=True*) – If True, store only top-k co-assignments per node
- **topk** (*int, default=50*) – Number of top co-assignments to keep per node (if sparse=True)
- **threshold** (*float, default=0.0*) – Minimum co-assignment probability to store (if sparse=True)

### Examples

```
>>> reducer = CoAssignmentReducer(n_nodes=4, sparse=False)
>>> reducer.update(np.array([0, 0, 1, 1]))
>>> reducer.update(np.array([0, 0, 0, 1]))
>>> coassoc = reducer.finalize()
>>> coassoc.shape
(4, 4)
```

**finalize**() → ndarray

Compute co-assignment probabilities.

**Returns**

Co-assignment matrix, shape (n_nodes, n_nodes) If sparse=True, only top-k entries per row are non-zero

**Return type**

np.ndarray

**reset**()

Reset reducer.

**update**(*partition: ndarray*, *weight: float = 1.0*)

Update with partition sample.

**Parameters**

- **partition** (*np.ndarray*) – Partition labels, shape (n_nodes,)
- **weight** (*float*) – Sample weight

class py3plex.uncertainty.**CommunityDistribution**(*partitions: List[ndarray]*, *nodes: List[Any]*, *weights: ndarray | None = None*, *meta: Dict[str, Any] | None = None*)

Bases: object

Distribution over community partitions with lazy evaluation.

Represents a collection of community detection runs, providing methods to compute summary statistics like co-association matrices, consensus partitions, and node-level confidence metrics.

All expensive computations (co-association, consensus, alignment) are performed lazily and cached for efficiency.

**Parameters**

- **partitions** (*list of np.ndarray*) – List of partition arrays. Each array has shape (n_nodes,) with integer community labels. All arrays must have the same length and correspond to the same node ordering.
- **nodes** (*list*) – Ordered list of node identifiers corresponding to partition indices.
- **weights** (*np.ndarray, optional*) – Per-partition weights (e.g., modularity scores), shape (n_partitions,). If None, all partitions are weighted equally.
- **meta** (*dict, optional*) – Metadata about the distribution generation: - method: Algorithm name (e.g., "multilayer_louvain") - n_runs: Number of partitions - resampling: Resampling strategy ("seed", "bootstrap", "perturbation") - gamma: Resolution parameter(s) used - seed: Base random seed - n_jobs: Number of parallel jobs - layers: List of layer names (for multilayer networks)

**n_nodes**

Number of nodes in the network.

**Type**
int

**n_partitions**

Number of partitions in the distribution.

**Type**
int

### Examples

```
>>> partitions = [
...     np.array([0, 0, 1]),
...     np.array([0, 0, 1]),
...     np.array([0, 1, 1]),
... ]
>>> dist = CommunityDistribution(
...     partitions=partitions,
...     nodes=['A', 'B', 'C'],
...     meta={'method': 'louvain', 'n_runs': 3}
... )
>>> dist.n_nodes
3
>>> coassoc = dist.coassociation(mode='dense')
>>> coassoc.shape
(3, 3)
```

**align_labels**(*reference: str | ndarray = 'medoid', metric: str = 'overlap'*)

Align partition labels to a reference using Hungarian matching.

After alignment, membership probabilities P(node_i in community_k) can be computed meaningfully. Alignment solves the label correspondence problem (community 0 in partition A may correspond to community 3 in B).

This method modifies internal state and caches aligned partitions.

**Parameters**

- **reference** (*{'medoid', 'first'} or np.ndarray*) – Reference partition for alignment: - 'medoid': Use medoid partition as reference - 'first': Use first partition as reference - np.ndarray: Use provided partition as reference
- **metric** (*{'overlap', 'nmi'}*) – Metric for computing cost matrix for Hungarian algorithm: - 'overlap': Maximize overlap (intersection) between communities - 'nmi': Maximize normalized mutual information

**Returns**

Returns self for method chaining.

**Return type**

self

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> dist.align_labels(reference='medoid')
>>> probs = dist.node_membership_probs()
```

**coassociation**(*mode: str = 'auto'*, *topk: int = 50*, *sample_pairs: int | None = None*) → ndarray | Dict[int, List[Tuple[int, float]]]

Compute co-association matrix: P(node_i and node_j in same community).

The co-association matrix measures how often each pair of nodes appears in the same community across all partitions. This is label-permutation invariant (does not require alignment).

**Parameters**

- **mode** (*{'auto', 'dense', 'sparse'}*) – Output mode: - 'dense': Return full n×n numpy array - 'sparse': Return dict mapping node_idx to list of (neighbor_idx, prob) - 'auto': Choose based on n_nodes (sparse if > 2000)
- **topk** (*int, default=50*) – For sparse mode, keep only top-k highest co-association neighbors per node. Set to -1 to keep all non-zero entries.
- **sample_pairs** (*int, optional*) – For very large graphs, approximate co-association by sampling this many node pairs uniformly. If None, compute exactly.

**Returns**

If mode='dense': Array of shape (n_nodes, n_nodes) with co-assoc probs. If mode='sparse': Dict mapping node_idx to list of (neighbor_idx, prob).

**Return type**

np.ndarray or dict

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> coassoc_dense = dist.coassociation(mode='dense')
>>> coassoc_dense[0, 1]  # P(node 0 and 1 in same community)
0.75
>>>
>>> coassoc_sparse = dist.coassociation(mode='sparse', topk=10)
>>> coassoc_sparse[0]  # Top 10 most co-associated nodes with node 0
[(2, 0.9), (5, 0.85), ...]
```

**consensus_partition**(*method: str = 'medoid'*, ***kwargs*) → ndarray

Compute consensus partition from the distribution.

The consensus partition is a representative partition that best summarizes the distribution. Multiple methods are supported.

**Parameters**

- **method** (*str*) – Consensus method. Use `'medoid'` to choose the partition with minimum VI distance to all others (parameter-free), or `'cluster_coassoc'` to cluster the co-association matrix using spectral or hierarchical clustering (requires scipy).
- ****kwargs** – Additional arguments: `metric` (str, for `'medoid'`; `'vi'` or `'pair_agreement'`), or `n_clusters` (int, for `'cluster_coassoc'`).

**Returns**

Consensus partition array, shape (n_nodes,)

**Return type**

np.ndarray

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> consensus = dist.consensus_partition(method='medoid')
>>> consensus.shape
(100,)
```

**property n_nodes: int**

Number of nodes.

**property n_partitions: int**

Number of partitions in the distribution.

**node_confidence**(*consensus: ndarray | None = None*) → ndarray

Compute per-node confidence scores.

Confidence measures how consistently a node is assigned to its consensus community across all partitions. Computed as the mean co-association with other nodes in the same consensus community.

**Parameters**

**consensus** (*np.ndarray, optional*) – Consensus partition to use. If None, computed automatically.

**Returns**
Confidence scores, shape (n_nodes,). Values in [0, 1] where 1 = always in same community, 0 = never consistent.

**Return type**
np.ndarray

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> confidence = dist.node_confidence()
>>> low_conf_nodes = np.where(confidence < 0.7)[0]
```

**node_entropy**(*aligned: bool = False*, *base: float = 2.0*) → ndarray

Compute per-node entropy across partitions.

Entropy measures the uncertainty in community assignment. Higher entropy indicates more disagreement across partitions.

**Parameters**

- **aligned** (*bool, default=False*) – If True, use aligned membership probabilities (requires alignment). If False, compute proxy entropy from co-association.
- **base** (*float, default=2.0*) – Logarithm base for entropy (2.0 = bits, e = nats).

**Returns**
Entropy values, shape (n_nodes,). Higher = more uncertain.

**Return type**
np.ndarray

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> entropy = dist.node_entropy()
>>> uncertain_nodes = np.where(entropy > 1.5)[0]
```

**node_margin**(*consensus: ndarray | None = None*) → ndarray

Compute per-node margin (confidence gap).

Margin is the difference between the two most frequent community assignments. Requires label alignment if using membership probs.

If alignment not available, uses consensus confidence as proxy.

**Parameters**
**consensus** (*np.ndarray, optional*) – Consensus partition to use for proxy margin.

**Returns**
Margin values, shape (n_nodes,). Higher = more confident.

**Return type**
np.ndarray

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> margin = dist.node_margin()
>>> boundary_nodes = np.where(margin < 0.2)[0]
```

**node_membership_probs**() → ndarray

Get per-node membership probabilities P(node_i in community_k).

Requires label alignment via align_labels(). Returns array of shape (n_nodes, n_communities) where entry [i, k] is the probability that node i belongs to community k across all aligned partitions.

**Returns**
Membership probability matrix, shape (n_nodes, n_communities).

**Return type**
np.ndarray

**Raises**
**AlgorithmError** – If labels have not been aligned. Call align_labels() first.

### Examples

```
>>> dist = CommunityDistribution([...], nodes=[...])
>>> dist.align_labels(reference='medoid')
>>> probs = dist.node_membership_probs()
>>> probs.shape
(100, 5)  # 100 nodes, 5 communities
>>> probs[0]  # Membership probabilities for node 0
array([0.8, 0.15, 0.05, 0.0, 0.0])
```

**property nodes: List[Any]**

Ordered list of node identifiers.

**property partitions: List[ndarray]**

List of partition arrays (copies for safety).

**to_dict**(*node_id: Any*) → Dict[str, Any]

Get community assignment info for a specific node.

**Parameters**
**node_id** (*any*) – Node identifier.

**Returns**
Dictionary with keys: - 'consensus': Consensus community label - 'confidence': Node confidence score - 'entropy': Node entropy score - 'margin': Node margin score - 'membership_probs': Membership probabilities (if aligned)

**Return type**
dict

### Examples

```
>>> dist = CommunityDistribution([...], nodes=['A', 'B', 'C'])
>>> info = dist.to_dict('A')
>>> info['consensus']
0
>>> info['confidence']
0.85
```

**property weights: ndarray**

Normalized partition weights.

**class py3plex.uncertainty.CommunityStats**(*labels: ~typing.Dict[~typing.Any, int], modularity: float | None = None, modularity_std: float | None = None, coassoc: ~py3plex.uncertainty.types.StatMatrix | None = None, stability: ~typing.Dict[~typing.Any, float] | None = None, n_communities: int = 0, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Statistics from community detection with optional uncertainty.

Wraps cluster labels, modularity, co-association matrix, and stability indices computed from multiple runs.

**Parameters**

- **labels** (*dict[Any, int]*) – Node -> community ID mapping (from deterministic run or consensus).
- **modularity** (*float or None*) – Modularity score (mean if multiple runs).
- **modularity_std** (*float or None*) – Standard deviation of modularity across runs.
- **coassoc** (*StatMatrix or None*) – Co-association matrix (probability nodes are in same community).
- **stability** (*dict[Any, float] or None*) – Per-node stability index (how often node stays in same cluster).
- **n_communities** (*int*) – Number of communities detected.
- **meta** (*dict[str, Any]*) – Optional metadata.

### Examples

```
>>> cs = CommunityStats(
...     labels={'a': 0, 'b': 0, 'c': 1},
...     modularity=0.42,
...     n_communities=2
... )
>>> cs.is_deterministic
True
```

```
>>> cs.labels['a']
0
```

**property certainty: float**

Return certainty level (1.0 if deterministic, 0.0 otherwise).

**coassoc: StatMatrix | None = None**

**property is_deterministic: bool**

Return True if no uncertainty info is present.

**labels: Dict[Any, int]**

**meta: Dict[str, Any]**

**modularity: float | None = None**

**modularity_std: float | None = None**

**n_communities: int = 0**

**stability: Dict[Any, float] | None = None**

**class** py3plex.uncertainty.**ConsensusPartitionReducer**(*marginal_reducer: NodeMarginalReducer*)

Bases: `PartitionReducer`

Compute consensus partition using node marginals (UQ spine version).

This reducer depends on NodeMarginalReducer to provide marginal distributions. It extracts the consensus labels from the marginals.

This is a thin wrapper that extracts consensus from NodeMarginalReducer output, following the design principle that ConsensusReducer uses marginals from NodeMarginalReducer.

**Parameters**

**marginal_reducer** (*NodeMarginalReducer*) – NodeMarginalReducer instance to use for marginals

### Examples

```
>>> marginal_reducer = NodeMarginalReducer(n_nodes=3)
>>> consensus_reducer = ConsensusPartitionReducer(marginal_reducer)
>>> # Updates are handled by marginal_reducer
>>> consensus = consensus_reducer.finalize()
```

**finalize()** → ndarray

Compute consensus partition from marginals.

**Returns**

Consensus partition, shape (n_nodes,)

**Return type**

np.ndarray

**reset()**

Reset is delegated to marginal_reducer.

**update**(*sample_output: Any*, *weight: float = 1.0*)

Update is delegated to marginal_reducer.

**Parameters**

- **sample_output** (*PartitionOutput or np.ndarray*) – Partition sample
- **weight** (*float*) – Sample weight

**class** py3plex.uncertainty.**ConsensusReducer**(*n_nodes: int*)

Bases: `PartitionReducer`

Compute consensus partition (mode of labels per node).

The consensus partition assigns each node to its most frequent community label across samples.

**Parameters**

**n_nodes** (*int*) – Number of nodes

### Examples

```
>>> reducer = ConsensusReducer(n_nodes=4)
>>> reducer.update(np.array([0, 0, 1, 1]))
>>> reducer.update(np.array([0, 0, 0, 1]))
>>> reducer.update(np.array([0, 0, 0, 1]))
>>> consensus = reducer.finalize()
>>> consensus
array([0, 0, 0, 1])
```

**finalize**() → ndarray

Compute consensus partition.

**Returns**

Consensus partition, shape (n_nodes,)

**Return type**

np.ndarray

**reset**()

Reset reducer.

**update**(*partition: ndarray*, *weight: float = 1.0*)

Update with partition sample.

**Parameters**

- **partition** (*np.ndarray*) – Partition labels
- **weight** (*float*) – Sample weight

**class** py3plex.uncertainty.**EdgeDrop**(*p: float*, *preserve_connectivity: bool = False*)

Bases: `NoiseModel`

Drop edges uniformly at random.

This noise model removes a fraction p of edges from the network, chosen uniformly at random. Useful for testing robustness to missing edges.

**Parameters**

- **p** (*float*) – Probability of dropping each edge (0 < p < 1)
- **preserve_connectivity** (*bool, default=False*) – If True, only drop edges that don't disconnect the network

### Examples

```
>>> noise = EdgeDrop(p=0.1)
>>> perturbed = noise.apply(network, seed=42)
```

**apply**(*network: Any*, *seed: int | None = None*) → Any

Apply edge dropping to network.

**Parameters**

- **network** (*multi_layer_network*) – Input network
- **seed** (*int, optional*) – Random seed

**Returns**

Network with edges dropped

**Return type**

multi_layer_network

**p: float**

**preserve_connectivity: bool = False**

**to_dict**() → Dict[str, Any]

Serialize to dict.

**class** py3plex.uncertainty.**GroupedReducer**(*reducer_class: type*, *reducer_kwargs: Dict | None = None*)

Bases: SelectionReducer

Wrapper to maintain separate reducers per group.

**Parameters**

- **reducer_class** (*type*) – Reducer class to instantiate per group
- **reducer_kwargs** (*dict, optional*) – Keyword arguments for reducer instantiation

**finalize**() → Dict[Tuple, Dict[str, Any]]

Finalize all group reducers.

**Returns**

Maps group_key -> reducer results

**Return type**

dict

**update**(*selection: SelectionOutput*) → None

Update the appropriate grouped reducer.

**class** py3plex.uncertainty.**InclusionReducer**

Bases: SelectionReducer

Track item inclusion probabilities across samples.

Maintains count of how many times each item appears in selections.

**n_samples**

Total number of samples processed

**Type**
int

**item_counts**

Count of appearances per item

**Type**
dict

**finalize**() → Dict[str, Any]

Compute inclusion probabilities.

**Returns**
Keys: - present_prob : dict mapping item -> probability - n_samples : int - items_universe : list of all items seen

**Return type**
dict

**update**(*selection: SelectionOutput*) → None

Update with a new selection.

**class** py3plex.uncertainty.**LayerDrop**(*p: float | None = None*, *layers: List[str] | None = None*)

Bases: NoiseModel

Drop entire layers from multilayer network.

This noise model removes layers from the network, useful for testing robustness to missing data modalities.

**Parameters**

- **p** (*float, optional*) – Probability of dropping each layer (0 < p < 1)
- **layers** (*list of str, optional*) – Specific layers to drop (mutually exclusive with p)

### Examples

```
>>> # Drop each layer with 20% probability
>>> noise = LayerDrop(p=0.2)
>>> perturbed = noise.apply(network, seed=42)
>>>
>>> # Drop specific layers
>>> noise = LayerDrop(layers=["twitter", "facebook"])
>>> perturbed = noise.apply(network, seed=42)
```

**apply**(*network: Any*, *seed: int | None = None*) → Any

Apply layer dropping to network.

**Parameters**

- **network** (*multi_layer_network*) – Input network
- **seed** (*int, optional*) – Random seed

**Returns**
Network with layers dropped

**Return type**
multi_layer_network

`layers: List[str] | None = None`

`p: float | None = None`

`to_dict()` → Dict[str, Any]
Serialize to dict.

`class py3plex.uncertainty.NoNoise`

Bases: `NoiseModel`

No-op noise model for deterministic execution.

This noise model returns an unmodified copy of the network. It is useful for seed-based UQ strategies where the network structure is fixed and only algorithm randomness varies.

### Examples

```
>>> noise = NoNoise()
>>> net_copy = noise.apply(network, seed=42)
>>> # net_copy is a deep copy of network with no perturbations
```

`apply(network: Any, seed: int | None = None)` → Any
Return an unmodified copy of the network.

**Parameters**

- **network** (*multi_layer_network*) – Input network
- **seed** (*int, optional*) – Random seed (unused, for interface compatibility)

**Returns**
Deep copy of the network

**Return type**
multi_layer_network

`to_dict()` → Dict[str, Any]
Serialize to dict.

`class py3plex.uncertainty.NodeEntropyReducer(n_nodes: int, node_ids: List[Any] | None = None)`

Bases: `PartitionReducer`

Compute per-node entropy across partitions.

Entropy H(node) = -sum_k p_k * log(p_k) where p_k is the probability that node is in community k.

This requires tracking community label frequencies per node.

**Parameters**

- **n_nodes** (*int*) – Number of nodes
- **node_ids** (*list, optional*) – Node identifiers (for mapping back to

nodes)

### Examples

```
>>> reducer = NodeEntropyReducer(n_nodes=3)
>>> reducer.update(np.array([0, 0, 1]))
>>> reducer.update(np.array([0, 1, 1]))
>>> entropy = reducer.finalize()
>>> entropy.shape
(3,)
```

**finalize**() → ndarray

Compute per-node entropy.

**Returns**

Entropy values, shape (n_nodes,)

**Return type**

np.ndarray

**get_max_membership**() → Tuple[ndarray, ndarray]

Get maximum membership probability per node.

**Returns**

- **labels** (*np.ndarray*) – Most frequent label per node
- **probs** (*np.ndarray*) – Probability of most frequent label

**get_membership_probs**() → ndarray

Get membership probability matrix.

**Returns**

Probability matrix, shape (n_nodes, n_communities) where entry [i, k] is P(node i in community k)

**Return type**

np.ndarray

**reset**()

Reset reducer.

**update**(*partition: ndarray*, *weight: float = 1.0*)

Update with partition sample.

**Parameters**

- **partition** (*np.ndarray*) – Partition labels, shape (n_nodes,)
- **weight** (*float*) – Sample weight

**class** py3plex.uncertainty.**NodeMarginalReducer**(*n_nodes: int*, *node_ids: List[Any] | None = None*)

Bases: PartitionReducer

Compute per-node marginal membership distributions (UQ spine version).

This reducer is the canonical implementation for the UQ execution spine. It tracks the frequency of each community label for each node across samples and computes: - Node entropy: H(node) = -sum_k p_k * log(p_k) - Maximum membership probability: max_c p(node, c) - Consensus labels: argmax_c p(node, c)

This replaces NodeEntropyReducer as the primary reducer for partition UQ.

**Parameters**

- **n_nodes** (*int*) – Number of nodes
- **node_ids** (*list, optional*) – Node identifiers (for mapping back to nodes)

### Examples

```
>>> reducer = NodeMarginalReducer(n_nodes=3)
>>> from py3plex.uncertainty.partition_types import PartitionOutput
>>> reducer.update(PartitionOutput(labels={'A': 0, 'B': 0, 'C': 1}))
>>> reducer.update(PartitionOutput(labels={'A': 0, 'B': 1, 'C': 1}))
>>> result = reducer.finalize()
>>> result['entropy'].shape
(3,)
```

**finalize**() → Dict[str, ndarray]

Compute node-level statistics.

**Returns**

Dictionary with keys: - 'entropy': Node entropy, shape (n_nodes,) - 'p_max': Maximum membership probability, shape (n_nodes,) - 'consensus_labels': Most frequent label per node, shape (n_nodes,)

**Return type**

dict

**reset**()

Reset reducer.

**update**(*sample_output: Any*, *weight: float = 1.0*)

Update with partition sample.

**Parameters**

- **sample_output** (*PartitionOutput or np.ndarray*) – If PartitionOutput: dict mapping node_id -> community_id If np.ndarray: partition labels, shape (n_nodes,)
- **weight** (*float*) – Sample weight

**class py3plex.uncertainty.NoiseModel**

Bases: ABC

Base class for network noise models.

A NoiseModel defines a stochastic transformation of a network that preserves the general structure but introduces controlled perturbations.

All subclasses must implement: - apply(network, seed) -> perturbed_network - to_dict() -> serializable representation

**abstract apply**(*network: Any*, *seed: int | None = None*) → Any

Apply noise model to network.

**Parameters**

- **network** (*multi_layer_network*) – Input network to perturb

- **seed** (*int , optional*) – Random seed for reproducibility

**Returns**
Perturbed copy of the network

**Return type**
multi_layer_network

**abstract to_dict()** → Dict[str, Any]

Serialize noise model to dictionary.

**Returns**
Serializable representation including all parameters

**Return type**
dict

**class py3plex.uncertainty.PartitionDistanceReducer**(*metric: str = 'vi'*, *store_samples: bool = False*)

Bases: `PartitionReducer`

Compute pairwise partition distances (VI, NMI).

This reducer accumulates all pairwise distances between partitions to compute statistics (mean, std) of partition stability.

**Parameters**

- **metric** (*str , default="vi"*) – Distance metric: "vi" or "nmi"
- **store_samples** (*bool , default=False*) – If True, store all partition samples for later analysis

### Examples

```
>>> reducer = PartitionDistanceReducer(metric="vi")
>>> reducer.update(np.array([0, 0, 1, 1]))
>>> reducer.update(np.array([0, 0, 0, 1]))
>>> stats = reducer.finalize()
>>> stats.keys()
dict_keys(['vi_mean', 'vi_std', 'vi_min', 'vi_max'])
```

**finalize()** → Dict[str, float]

Compute distance statistics.

**Returns**
Statistics with keys: - {metric}_mean: Mean distance - {metric}_std: Standard deviation of distance - {metric}_min: Minimum distance - {metric}_max: Maximum distance

**Return type**
dict

**reset()**

Reset reducer.

**update**(*partition: ndarray*, *weight: float = 1.0*)

Update with partition sample.

Computes distances to all previously seen partitions.

**Parameters**

- **partition** (*np.ndarray*) – Partition labels
- **weight** (*float*) – Sample weight (currently unused, for future weighting)

class py3plex.uncertainty.PartitionOutput(*labels: Dict[Any, Any]*)

Bases: object

Internal representation of a partition (community assignment).

This is the unified output format returned by base_callable that gets fed into reducers for uncertainty quantification.

PartitionOutput is a thin adapter that wraps community detection results into a standard format. Community algorithms return node → community assignments, and this class normalizes that representation.

**labels**

Mapping from node IDs to community IDs. Node IDs can be any hashable type (str, int, tuple). Community IDs are typically integers but can be any hashable type.

**Type**
dict[node_id, community_id]

### Examples

```
>>> # From a community detection algorithm
>>> partition = PartitionOutput(labels={
...     'A': 0, 'B': 0, 'C': 1, 'D': 1
... })
>>>
>>> # Check assignment
>>> partition.labels['A']
0
```

### Notes

- Community algorithms MUST be adapted to return PartitionOutput
- This adaptation can be done in executor glue code
- DO NOT modify algorithms to return PartitionOutput directly

**labels: Dict[Any, Any]**

class py3plex.uncertainty.PartitionReducer

Bases: ABC

Base class for partition reducers.

A reducer accumulates statistics from a stream of partitions without storing all partitions in memory.

**abstract finalize()** → Any

Finalize and return accumulated statistics.

**Returns**
Reducer-specific statistics

**Return type**
Any

**abstract reset()**

Reset reducer to initial state.

**abstract update**(*partition: ndarray*, *weight: float = 1.0*)

Update reducer with a new partition sample.

**Parameters**

- **partition** (*np.ndarray*) – Partition array, shape (n_nodes,) with integer labels
- **weight** (*float, default=1.0*) – Weight for this sample (for weighted aggregation)

**class** py3plex.uncertainty.**PartitionUQ**(*node_ids: ~typing.List[~typing.Any], n_samples: int, consensus_partition: ~numpy.ndarray, membership_entropy: ~numpy.ndarray, p_max_membership: ~numpy.ndarray, vi_mean: float = 0.0, vi_std: float = 0.0, nmi_mean: float = 1.0, nmi_std: float = 0.0, coassoc_matrix: ~numpy.ndarray | None = None, store_mode: str = 'sketch', samples: ~typing.List[~numpy.ndarray] | None = None, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Uncertainty quantification for community partitions.

This class stores canonical UQ outputs for partitions without requiring storage of all samples.

**node_ids**

Ordered list of node identifiers

**Type**
list

**n_samples**

Number of partition samples used

**Type**
int

**consensus_partition**

Consensus partition (mode or medoid), shape (n_nodes,)

**Type**
np.ndarray

**membership_entropy**

Per-node entropy, shape (n_nodes,). Higher = more uncertain.

**Type**
np.ndarray

**p_max_membership**

Maximum membership probability per node, shape (n_nodes,)

**Type**
np.ndarray

**coassoc_matrix**
Co-assignment probability matrix (sparse or dense)

**Type**
np.ndarray, optional

**vi_mean**
Mean variation of information between partitions

**Type**
float

**vi_std**
Std of variation of information

**Type**
float

**nmi_mean**
Mean normalized mutual information

**Type**
float

**nmi_std**
Std of normalized mutual information

**Type**
float

**store_mode**
Storage mode: “none”, “samples”, or “sketch”

**Type**
str

**samples**
Raw partition samples (only if store_mode=”samples”)

**Type**
list of np.ndarray, optional

**meta**
Additional metadata (algorithm, parameters, provenance)

**Type**
dict

### Examples

```
>>> uq = PartitionUQ.from_samples(
...     partitions=[...],
...     node_ids=['A', 'B', 'C'],
...     store="sketch"
... )
```

```
>>>
>>> # Access consensus
>>> print(uq.consensus_partition)
>>>
>>> # Check uncertain nodes
>>> uncertain = uq.boundary_nodes(threshold=0.5)
>>>
>>> # Get stability summary
>>> summary = uq.stability_summary()
```

**boundary_nodes**(*threshold: float = 0.5*, *metric: str = 'entropy'*) → List[Any]

Get boundary nodes (high uncertainty).

**Parameters**

- **threshold** (*float*) – Threshold for boundary classification
- **metric** (*str*) – Metric to use: “entropy” or “confidence”

**Returns**

Node IDs with high uncertainty

**Return type**

list

### Examples

```
>>> boundary = uq.boundary_nodes(threshold=0.5, metric="entropy")
```

**coassoc_matrix: ndarray | None = None**

**consensus_partition: ndarray**

**classmethod from_samples**(*partitions: List[ndarray]*, *node_ids: List[Any]*, *store: str = 'sketch'*, *sparse_topk: int = 50*, *sparse_threshold: float = 0.7*, *weights: ndarray | None = None*, *meta: Dict[str, Any] | None = None*) → PartitionUQ

Create PartitionUQ from partition samples.

This is the main constructor that processes samples through reducers.

**Parameters**

- **partitions** (*list of np.ndarray*) – List of partition samples, each shape (n_nodes,)
- **node_ids** (*list*) – Ordered node identifiers
- **store** (*str, default="sketch"*) – Storage mode: - “none”: Only summary statistics - “samples”: Store all samples - “sketch”: Store sparse co-assignment
- **sparse_topk** (*int, default=50*) – Top-k neighbors to keep in sparse co-assignment
- **sparse_threshold** (*float, default=0.7*) – Minimum co-assignment probability to store
- **weights** (*np.ndarray, optional*) – Per-sample weights (default:

uniform)

- **meta** (*dict , optional*) – Metadata to attach

**Returns**
Constructed PartitionUQ object

**Return type**
PartitionUQ

### Examples

```
>>> partitions = [
...     np.array([0, 0, 1, 1]),
...     np.array([0, 0, 0, 1]),
... ]
>>> uq = PartitionUQ.from_samples(
...     partitions=partitions,
...     node_ids=['A', 'B', 'C', 'D'],
...     store="sketch"
... )
```

**classmethod from_uq_result**(*uq_result*, *node_ids: List[Any]*, *meta: Dict[str, Any] | None = None*) → PartitionUQ

Create PartitionUQ from UQResult (UQ spine integration).

This factory method creates a PartitionUQ from a UQResult produced by run_uq() with the appropriate reducers. It expects: - NodeMarginalReducer output (required) - StabilityReducer output (optional)

### Note: Co-assignment matrix computation

By default, the co-assignment matrix is NOT computed when using the UQ spine approach. This is intentional for memory efficiency. If you need the co-assignment matrix, you must: 1. Add CoAssignmentReducer to your UQPlan.reducers list 2. Extract the result from uq_result.reducer_outputs['CoAssignmentReducer'] 3. Manually set it on the PartitionUQ object after construction

**param uq_result**
Result from run_uq() execution

**type uq_result**
UQResult

**param node_ids**
Ordered node identifiers

**type node_ids**
list

**param meta**
Additional metadata to attach

**type meta**
dict, optional

**returns**
Constructed PartitionUQ object

**rtype**
PartitionUQ

### Examples

```
>>> from py3plex.uncertainty.runner import run_uq
>>> from py3plex.uncertainty.plan import UQPlan
>>> from py3plex.uncertainty.partition_reducers import NodeMarginalReducer
>>>
>>> # Create plan with reducers
>>> marginal_reducer = NodeMarginalReducer(n_nodes=4, node_ids=['A', 'B', 'C', 'D']
>>> plan = UQPlan(
...     base_callable=my_algorithm,
...     strategy="seed",
...     noise_model=NoNoise(),
...     n_samples=50,
...     seed=42,
...     reducers=[marginal_reducer]
... )
>>>
>>> # Execute UQ
>>> result = run_uq(plan, network)
>>>
>>> # Create PartitionUQ from result
>>> partition_uq = PartitionUQ.from_uq_result(
...     uq_result=result,
...     node_ids=['A', 'B', 'C', 'D']
... )
```

**membership_entropy: ndarray**

**meta: Dict[str, Any]**

**property n_communities: int**
Number of communities in consensus partition.

**property n_nodes: int**
Number of nodes.

**n_samples: int**

**nmi_mean: float = 1.0**

**nmi_std: float = 0.0**

**node_ids: List[Any]**

**node_summary**(*node_id: Any*) → Dict[str, Any]
Get UQ summary for a specific node.

**Parameters**
**node_id** (*any*) – Node identifier

**Returns**
Summary with keys: - consensus: Consensus community label - en-

tropy: Node entropy - confidence: Maximum membership probability - coassoc_neighbors: Top co-associated nodes (if available)

**Return type**
dict

`p_max_membership: ndarray`

`samples: List[ndarray] | None = None`

`stability_summary()` → Dict[str, float]

Get partition stability summary.

**Returns**
Statistics: - vi_mean, vi_std: Variation of information - nmi_mean, nmi_std: Normalized mutual information - n_samples: Number of samples - n_communities: Number of communities in consensus - mean_entropy: Mean node entropy - mean_confidence: Mean node confidence

**Return type**
dict

`store_mode: str = 'sketch'`

`to_dict()` → Dict[str, Any]

Serialize to dictionary.

**Returns**
Serializable representation

**Return type**
dict

`vi_mean: float = 0.0`

`vi_std: float = 0.0`

`class py3plex.uncertainty.ProbabilisticCommunityResult(`*distribution: CommunityDistribution*, *consensus_method: str = 'medoid'*`)`

Bases: `object`

Probabilistic community detection result with backward compatibility.

This class wraps CommunityDistribution to provide: 1. Backward-compatible hard labels (deterministic view) 2. Probabilistic membership distributions (uncertainty view) 3. Node-level uncertainty metrics (entropy, confidence) 4. Community-level stability metrics 5. Partition-space variability metrics

**Parameters**

- **distribution** (*CommunityDistribution*) – The underlying distribution of community partitions.
- **consensus_method** (*str, default='medoid'*) – Method for computing consensus partition ('medoid' or 'cluster_coassoc').

`n_nodes`

Number of nodes in the network.

**Type**
int

`n_partitions`
Number of partitions in the ensemble.

**Type**
int

`is_deterministic`
True if only one partition (no uncertainty).

**Type**
bool

### Examples

```
>>> # Create from CommunityDistribution
>>> dist = CommunityDistribution(partitions=[...], nodes=[...])
>>> result = ProbabilisticCommunityResult(dist)
>>>
>>> # Access deterministic labels (backward compatible)
>>> labels = result.labels  # Dict[node, community_id]
>>>
>>> # Access probabilistic information (if n_partitions > 1)
>>> if not result.is_deterministic:
...     probs = result.probs  # Dict[node, Dict[comm_id, prob]]
...     entropy = result.entropy  # Dict[node, float]
...     confidence = result.confidence  # Dict[node, float]
```

`property community_stability: Dict[int, Dict[str, float]]`
Per-community stability metrics.

For each community in the consensus partition, computes: - persistence: Fraction of partitions where community exists - size_mean: Mean community size across partitions - size_std: Standard deviation of community size - size_cv: Coefficient of variation (std/mean)

**Returns**
Nested dict: {community_id: {metric: value}}

**Return type**
dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> stability = result.community_stability
>>> stability[0]
{'persistence': 0.95, 'size_mean': 12.5, 'size_std': 1.2, 'size_cv': 0.096}
```

`property confidence: Dict[Any, float]`
Per-node confidence (max membership probability).

Confidence is the probability of the most likely community. For deterministic results, confidence is 1.0.

**Returns**
Mapping from node to confidence value in [0, 1].

**Return type**
dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> confidence = result.confidence
>>> confidence[('A', 'social')]
0.85  # 85% probability of consensus community
```

**property entropy: Dict[Any, float]**

Per-node entropy H(community assignment).

Higher entropy indicates more uncertainty in community assignment. For deterministic results, entropy is 0.0.

**Returns**
Mapping from node to entropy value (in bits).

**Return type**
dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> entropy = result.entropy
>>> entropy[('A', 'social')]
0.75  # Some uncertainty in assignment
```

**property is_deterministic: bool**

True if this is a deterministic result (single partition).

**property labels: Dict[Any, int]**

Hard community labels (consensus partition).

This provides backward-compatible deterministic labels. For n_partitions=1, returns the single partition. For n_partitions>1, returns consensus (medoid or cluster-based).

**Returns**
Mapping from node to community ID.

**Return type**
dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> labels = result.labels
>>> labels[('A', 'social')]
0
```

**property margin: Dict[Any, float]**

Per-node margin (difference between top 2 membership probabilities).

Margin measures how much more likely the top community is compared to the second most likely. Higher margin = more confident assignment. For deterministic results, margin is 1.0.

**Returns**

Mapping from node to margin value in [0, 1].

**Return type**

dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> margin = result.margin
>>> margin[('A', 'social')]
0.65  # Top community 65% more likely than second
```

**property n_nodes: int**

Number of nodes.

**property n_partitions: int**

Number of partitions in the ensemble.

**property nodes: List[Any]**

List of node identifiers.

**property partition_metrics: Dict[str, Any]**

Partition-space variability metrics.

Computes distribution of partition similarity metrics: - vi_mean, vi_std: Variation of Information statistics - ari_mean, ari_std: Adjusted Rand Index statistics - nmi_mean, nmi_std: Normalized Mutual Information statistics

**Returns**

Dictionary with summary statistics for partition similarities.

**Return type**

dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> metrics = result.partition_metrics
>>> metrics['vi_mean']
0.25  # Average VI distance between partitions
```

**property probs: Dict[Any, Dict[int, float]]**

Membership probabilities P(node in community).

Requires n_partitions > 1. For deterministic results (n_partitions=1), returns probabilities of 1.0 for assigned community.

**Returns**

Nested dict: {node: {community_id: probability}}

**Return type**
dict

**Raises**
**AlgorithmError** – If label alignment has not been performed for n_partitions > 1.

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> probs = result.probs
>>> probs[('A', 'social')]
{0: 0.8, 1: 0.15, 2: 0.05}
```

**to_dict**(*include_probs: bool = False*) → Dict[str, Any]

Export result as dictionary.

**Parameters**
**include_probs** (*bool, default=False*) – If True, include full membership probability distributions.

**Returns**
Dictionary with community information.

**Return type**
dict

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> d = result.to_dict(include_probs=True)
>>> d.keys()
dict_keys(['labels', 'entropy', 'confidence', 'probs', ...])
```

**to_pandas**(*expand_uncertainty: bool = False*) → DataFrame

Export result as pandas DataFrame.

**Parameters**
**expand_uncertainty** (*bool, default=False*) – If True, expand uncertainty information into separate columns: - community_id: Hard label - community_confidence: Max probability - membership_entropy: Entropy - membership_margin: Margin - p_comm_0, p_comm_1, ...: Top-k membership probabilities

**Returns**
DataFrame with one row per node.

**Return type**
pd.DataFrame

### Examples

```
>>> result = ProbabilisticCommunityResult(dist)
>>> df = result.to_pandas(expand_uncertainty=True)
>>> df.columns
```

```
Index(['id', 'community_id', 'community_confidence',
       'membership_entropy', 'membership_margin',
       'p_comm_0', 'p_comm_1', ...])
```

**class** py3plex.uncertainty.**RankReducer**(*k: int | None = None*, *store_samples: bool = False*)

Bases: `SelectionReducer`

Track rank statistics for items when ranking is available.

Maintains online mean/variance of ranks per item, and p_in_topk.

**n_samples**

Number of samples

**Type**
int

**k**

Top-k parameter if relevant

**Type**
int, optional

**rank_sums**

Sum of ranks per item (when present)

**Type**
dict

**rank_sq_sums**

Sum of squared ranks per item

**Type**
dict

**rank_counts**

Count of samples where item had a rank

**Type**
dict

**topk_counts**

Count of times item was in top-k

**Type**
dict

**rank_samples**

Raw rank arrays per item if storing samples

**Type**
dict, optional

**finalize**() → Dict[str, Any]

Compute rank statistics.

**Returns**
Keys per item: - rank_mean : dict - rank_std : dict - p_in_topk : dict (if k is set) - rank_ci_low : dict (if samples stored) - rank_ci_high :

dict (if samples stored)

**Return type**
dict

`update(`*selection: SelectionOutput*`)` → None

Update with a new selection.

`class py3plex.uncertainty.Reducer`

Bases: `ABC`

Base protocol for online reducers.

Reducers follow the online aggregation pattern: 1. Initialize state in ___init___ 2. update(sample_output) - accumulate from one sample 3. finalize() - compute final result 4. reset() - clear state for reuse

Reducers must be: - Online: Process samples one at a time - Composable: Multiple reducers can run in parallel - Memory-efficient: Don't store all samples (unless explicitly needed) - Independent: No dependencies on DSL or QueryResult

### Notes

- Reducers MUST NOT control execution flow
- Reducers MUST NOT depend on sample order (unless explicitly designed to)
- Reducers SHOULD be deterministic given the same sample sequence

`abstract finalize()` → Any

Finalize and return accumulated statistics.

This method is called once after all samples have been processed. It should compute the final result from accumulated state.

**Returns**
Reducer-specific output (e.g., dict, array, scalar). The return type is specific to each reducer implementation.

**Return type**
Any

`abstract reset()`

Reset reducer to initial state.

This method allows reusing the same reducer instance for multiple UQ runs. It should clear all accumulated state.

`abstract update(`*sample_output: Any*`)`

Update reducer with a new sample.

This method is called once per sample during UQ execution. It should accumulate whatever statistics are needed without storing the full sample (unless storage is required).

**Parameters**
**sample_output** (*Any*) – Output from base_callable for this sample. Type depends on the algorithm (e.g., PartitionOutput for communities).

`class py3plex.uncertainty.ResamplingStrategy(`*value*`)`

Bases: `Enum`

Strategy for estimating uncertainty via resampling.

`SEED`

Run with different random seeds (Monte Carlo).

**Type**
str

`BOOTSTRAP`

Bootstrap resampling of nodes or edges.

**Type**
str

`JACKKNIFE`

Leave-one-out jackknife resampling.

**Type**
str

`PERTURBATION`

Add noise/perturbations to network structure or parameters.

**Type**
str

`STRATIFIED_PERTURBATION`

Stratified perturbation that preserves key structural distributions (degree bins, layer densities, edge weight bins, layer-pair frequencies). Reduces estimator variance without increasing sample count.

**Type**
str

`BOOTSTRAP = 'bootstrap'`

`JACKKNIFE = 'jackknife'`

`PERTURBATION = 'perturbation'`

`SEED = 'seed'`

`STRATIFIED_PERTURBATION = 'stratified_perturbation'`

`class py3plex.uncertainty.SelectionOutput(`*items: List[Any]*, *scores: Dict[Any, float] | None = None*, *ranks: Dict[Any, int] | None = None*, *k: int | None = None*, *target: str = 'nodes'*, *group_key: Tuple | None = None*`)`

Bases: `object`

Internal representation of a selection query result.

This is the unified output format returned by base_callable that gets fed into reducers for uncertainty quantification.

`items`

Selected item IDs (node IDs or edge tuples)

**Type**
list

**scores**

Scores per item if ranked query (e.g., centrality values)

**Type**
dict, optional

**ranks**

Exact rank per item if ranking is defined (1-indexed)

**Type**
dict, optional

**k**

Top-k parameter if relevant

**Type**
int, optional

**target**

Type of items: "nodes" or "edges"

**Type**
str

**group_key**

Grouping key for per_layer/per_layer_pair queries

**Type**
tuple, optional

### Notes

If .top_k(k, ...) was used, ranks must be defined for returned items. If only filtering (no ordering), scores/ranks may be None.

```
group_key:  Tuple | None = None

items:  List[Any]

k:  int | None = None

ranks:  Dict[Any, int] | None = None

scores:  Dict[Any, float] | None = None

target:  str = 'nodes'
```

**class** py3plex.uncertainty.**SelectionReducer**

Bases: ABC

Base class for selection reducers.

Reducers process SelectionOutput samples incrementally and compute summary statistics without storing all samples.

**abstract finalize**() → Dict[str, Any]

Compute final statistics.

**Returns**
Summary statistics computed from all updates

**Return type**
dict

**abstract update**(*selection: SelectionOutput*) → None

Update reducer with a new selection sample.

**Parameters**
**selection** (*SelectionOutput*) – A single selection query result

**class** py3plex.uncertainty.**SelectionUQ**(*n_samples: int, items_universe: ~typing.List[~typing.Any], samples_seen: int, present_prob: ~typing.Dict[~typing.Any, float], present_ci_low: ~typing.Dict[~typing.Any, float] = <factory>, present_ci_high: ~typing.Dict[~typing.Any, float] = <factory>, size_stats: ~typing.Dict[str, float] = <factory>, stability_stats: ~typing.Dict[str, float] = <factory>, rank_mean: ~typing.Dict[~typing.Any, float] | None = None, rank_std: ~typing.Dict[~typing.Any, float] | None = None, rank_ci_low: ~typing.Dict[~typing.Any, float] | None = None, rank_ci_high: ~typing.Dict[~typing.Any, float] | None = None, p_in_topk: ~typing.Dict[~typing.Any, float] | None = None, topk_overlap_stats: ~typing.Dict[str, float] | None = None, consensus_items: set = <factory>, borderline_items: ~typing.List[~typing.Any] = <factory>, target: str = 'nodes', k: int | None = None, store_mode: str = 'sketch', ci_method: str = 'wilson', meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Uncertainty quantification for selection queries.

This class stores canonical UQ outputs for selections (top-k, filters) without requiring storage of all samples.

**n_samples**
Number of selection samples used

**Type**
int

**items_universe**
All items seen across samples

**Type**
list

**samples_seen**
Effective number of samples (same as n_samples usually)

**Type**
int

`present_prob`
: Inclusion probability per item: Pr(item  result)

  **Type**
  : dict

`present_ci_low`
: Lower CI bound for present_prob

  **Type**
  : dict, optional

`present_ci_high`
: Upper CI bound for present_prob

  **Type**
  : dict, optional

`size_stats`
: Selection set size statistics (mean, std, quantiles)

  **Type**
  : dict

`stability_stats`
: Stability metrics (Jaccard vs consensus)

  **Type**
  : dict

`rank_mean`
: Mean rank per item (if ranking exists)

  **Type**
  : dict, optional

`rank_std`
: Std of rank per item

  **Type**
  : dict, optional

`rank_ci_low`
: Lower CI bound for rank

  **Type**
  : dict, optional

`rank_ci_high`
: Upper CI bound for rank

  **Type**
  : dict, optional

`p_in_topk`
: Pr(rank  k) per item

  **Type**
  : dict, optional

`topk_overlap_stats`
: Top-k overlap statistics

**Type**
dict, optional

**consensus_items**
Items in consensus selection (present_prob  threshold)

**Type**
set

**borderline_items**
Items with uncertain inclusion (near threshold)

**Type**
list

**target**
Type of items: “nodes” or “edges”

**Type**
str

**k**
Top-k parameter if relevant

**Type**
int, optional

**store_mode**
Storage mode: “none”, “sketch”, or “samples”

**Type**
str

**ci_method**
Method used for confidence intervals

**Type**
str

**meta**
Additional metadata (method, noise_model, provenance)

**Type**
dict

### Examples

```
>>> from py3plex.uncertainty.selection_uq import SelectionUQ
>>> # Created internally by execute_selection_uq
>>> uq = SelectionUQ(...)
>>>
>>> # Access inclusion probabilities
>>> print(uq.present_prob)
>>>
>>> # Check borderline items
>>> print(uq.borderline_items)
>>>
>>> # Get summary
```

```
>>> summary = uq.summary()
```

**borderline_items:** `List[Any]`

**ci_method:** `str = 'wilson'`

**consensus_items:** `set`

**classmethod from_reducers**(*inclusion_result: Dict[str, Any]*, *size_result: Dict[str, Any]*, *stability_result: Dict[str, Any]*, *rank_result: Dict[str, Any] | None = None*, *topk_overlap_result: Dict[str, Any] | None = None*, *consensus_threshold: float = 0.5*, *borderline_epsilon: float = 0.1*, *ci_method: str = 'wilson'*, *ci_alpha: float = 0.05*, *target: str = 'nodes'*, *k: int | None = None*, *store_mode: str = 'sketch'*, *meta: Dict[str, Any] | None = None*) → SelectionUQ

Construct SelectionUQ from reducer results.

**Parameters**

- **inclusion_result** (*dict*) – Output from InclusionReducer.finalize()
- **size_result** (*dict*) – Output from SizeReducer.finalize()
- **stability_result** (*dict*) – Output from StabilityReducer.finalize()
- **rank_result** (*dict, optional*) – Output from RankReducer.finalize()
- **topk_overlap_result** (*dict, optional*) – Output from TopKOverlapReducer.finalize()
- **consensus_threshold** (*float*) – Threshold for consensus inclusion (default: 0.5)
- **borderline_epsilon** (*float*) – Tolerance for borderline detection (default: 0.1)
- **ci_method** (*str*) – CI method: “wilson” or “clopper-pearson”
- **ci_alpha** (*float*) – Significance level for CI (default: 0.05)
- **target** (*str*) – “nodes” or “edges”
- **k** (*int, optional*) – Top-k parameter
- **store_mode** (*str*) – Storage mode used
- **meta** (*dict, optional*) – Additional metadata

**Returns**

Constructed instance

**Return type**

SelectionUQ

**get_top_stable**(*n: int = 10*) → List[Tuple[Any, float]]

Get top n most stable items by present_prob.

**Parameters**

**n** (*int*) – Number of items to return

**Returns**

Sorted by present_prob descending

**Return type**
list of (item, present_prob)

`get_uncertain(`*threshold: float = 0.2*`)` → List[Tuple[Any, float]]

Get items with uncertain inclusion (low present_prob).

**Parameters**
**threshold** (*float*) – Items with present_prob < threshold

**Returns**
Sorted by present_prob ascending

**Return type**
list of (item, present_prob)

`items_universe: List[Any]`

`k: int | None = None`

`meta: Dict[str, Any]`

`n_samples: int`

`p_in_topk: Dict[Any, float] | None = None`

`present_ci_high: Dict[Any, float]`

`present_ci_low: Dict[Any, float]`

`present_prob: Dict[Any, float]`

`rank_ci_high: Dict[Any, float] | None = None`

`rank_ci_low: Dict[Any, float] | None = None`

`rank_mean: Dict[Any, float] | None = None`

`rank_std: Dict[Any, float] | None = None`

`samples_seen: int`

`size_stats: Dict[str, float]`

`stability_stats: Dict[str, float]`

`store_mode: str = 'sketch'`

`summary()` → Dict[str, Any]

Generate human-readable summary.

**Returns**
Compact summary with key statistics

**Return type**
dict

`target: str = 'nodes'`

`to_dict()` → Dict[str, Any]

Serialize to dictionary.

**Returns**
Serializable representation

**Return type**
dict

`to_pandas(`*expand: bool = True*`)` → DataFrame

Convert to tidy pandas DataFrame.

**Parameters**
**expand** (*bool*) – If True, include all UQ columns (probabilities, CIs, ranks)

**Returns**
Tidy table with one row per item

**Return type**
pd.DataFrame

`topk_overlap_stats: Dict[str, float] | None = None`

`class py3plex.uncertainty.SizeReducer(`*store_samples: bool = False*`)`

Bases: `SelectionReducer`

Track selection set size distribution.

Maintains online mean/variance of selection size.

`n_samples`

Number of samples

**Type**
int

`size_sum`

Sum of sizes

**Type**
float

`size_sq_sum`

Sum of squared sizes

**Type**
float

`sizes`

Raw sizes if storing samples

**Type**
list, optional

`finalize()` → Dict[str, Any]

Compute size statistics.

**Returns**
Keys: - mean : float - std : float - min : int (if samples stored) - max : int (if samples stored) - q05 : float (if samples stored) - q95 : float (if samples stored)

**Return type**
dict

**update**(*selection: SelectionOutput*) → None

Update with a new selection.

**class py3plex.uncertainty.StabilityReducer**(*consensus_threshold: float = 0.5*, *store_samples: bool = False*)

Bases: `SelectionReducer`

Track stability via Jaccard similarity to consensus selection.

Maintains a running consensus and computes Jaccard similarities.

**n_samples**

Number of samples

**Type**
int

**item_counts**

Count of appearances per item

**Type**
dict

**jaccard_sum**

Sum of Jaccard similarities

**Type**
float

**jaccard_sq_sum**

Sum of squared Jaccard similarities

**Type**
float

**jaccards**

Raw Jaccard values if storing samples

**Type**
list, optional

**finalize**() → Dict[str, Any]

Compute stability statistics.

**Returns**
Keys: - jaccard_mean : float - jaccard_std : float - q05 : float (if samples stored) - q95 : float (if samples stored)

**Return type**
dict

**update**(*selection: SelectionOutput*) → None

Update with a new selection.

**class py3plex.uncertainty.StatMatrix**(*index: ~typing.List[~typing.Any], mean: ~numpy.ndarray, std: ~numpy.ndarray | None = None, quantiles: ~typing.Dict[float, ~numpy.ndarray] | None = None, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

A matrix of statistics with optional uncertainty.

Used for adjacency matrices, co-association matrices, distance matrices, etc.

**Parameters**

- **index** (*list[Any]*) – Row/column labels (assumed square matrix for simplicity).
- **mean** (*np.ndarray*) – The mean matrix, shape (n, n).
- **std** (*np.ndarray or None*) – The standard deviation matrix, shape (n, n), or None.
- **quantiles** (*dict[float, np.ndarray] or None*) – Quantile matrices, e.g., {0.025: (n, n), 0.975: (n, n)}.
- **meta** (*dict[str, Any]*) – Optional metadata.

### Examples

```
>>> import numpy as np
>>> m = StatMatrix(
...     index=['a', 'b', 'c'],
...     mean=np.array([[0, 1, 0], [1, 0, 1], [0, 1, 0]], dtype=float)
... )
>>> m.is_deterministic
True
>>> np.array(m).shape
(3, 3)
```

**property certainty: float**

Return certainty level (1.0 if deterministic, 0.0 otherwise).

**index: List[Any]**

**property is_deterministic: bool**

Return True if this is a deterministic result.

**mean: ndarray**

**meta: Dict[str, Any]**

**quantiles: Dict[float, ndarray] | None = None**

**std: ndarray | None = None**

**class** py3plex.uncertainty.**StatSeries**(*index: ~typing.List[~typing.Any], mean: ~numpy.ndarray, std: ~numpy.ndarray | None = None, quantiles: ~typing.Dict[float, ~numpy.ndarray] | None = None, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

A series of statistics with optional uncertainty information.

This is the canonical result type for statistics that return a value per node, time point, or other index.

In deterministic mode (uncertainty=False): - mean contains the single-run values - std = None - quantiles = None - certainty = 1.0

In uncertain mode (uncertainty=True): - mean contains the average across runs - std contains the standard deviation - quantiles contains percentile arrays (e.g., {0.025: arr, 0.975: arr}) - certainty < 1.0

**Parameters**

- **index** (*list[Any]*) – The index labels (e.g., node IDs, time points).
- **mean** (*np.ndarray*) – The mean values, shape (n,).
- **std** (*np.ndarray or None*) – The standard deviations, shape (n,), or None if deterministic.
- **quantiles** (*dict[float, np.ndarray] or None*) – Quantile arrays, e.g., {0.025: (n,), 0.975: (n,)}, or None.
- **meta** (*dict[str, Any]*) – Optional metadata (e.g., algorithm parameters, run info).

### Examples

```
>>> import numpy as np
>>> # Deterministic
>>> s = StatSeries(
...     index=['a', 'b', 'c'],
...     mean=np.array([1.0, 2.0, 3.0])
... )
>>> s.is_deterministic
True
>>> s.certainty
1.0
>>> np.array(s)
array([1., 2., 3.])
```

```
>>> # With uncertainty
>>> s_unc = StatSeries(
...     index=['a', 'b', 'c'],
...     mean=np.array([1.0, 2.0, 3.0]),
...     std=np.array([0.1, 0.2, 0.15]),
...     quantiles={0.025: np.array([0.8, 1.6, 2.7]),
...                0.975: np.array([1.2, 2.4, 3.3])}
... )
>>> s_unc.is_deterministic
False
>>> s_unc.certainty
0.0
```

**property certainty: float**

Return certainty level.

Returns 1.0 if deterministic, 0.0 otherwise. In the future, this could return a richer metric.

**index: List[Any]**

**property is_deterministic: bool**

Return True if this is a deterministic result (no uncertainty).

**mean: ndarray**

**meta: Dict[str, Any]**

**quantiles: Dict[float, ndarray] | None = None**

**std: ndarray | None = None**

**to_dict()** → Dict[Any, Dict[str, Any]]

Convert to dictionary mapping index -> stats dict.

**Returns**

Dictionary with keys from index, values are dicts with 'mean', optionally 'std' and 'quantiles'.

**Return type**

dict

**class py3plex.uncertainty.StratificationSpec**(*strata: ~typing.List[str] = <factory>, bins: ~typing.Dict[str, int] = <factory>, seed: int | None = None*)

Bases: object

Configuration for stratified resampling.

**strata**

List of stratification dimensions to apply. Supported: "degree", "layer", "layer_pair", "weight"

**Type**

List[str]

**bins**

Number of bins for each continuous dimension. E.g., {"degree": 5, "weight": 3}

**Type**

Dict[str, int]

**seed**

Random seed for reproducibility.

**Type**

Optional[int]

**bins: Dict[str, int]**

**seed: int | None = None**

**strata: List[str]**

**to_dict()** → Dict[str, Any]

Convert to dictionary for serialization.

**class py3plex.uncertainty.TemporalWindowBootstrap**(*window_size: float, n_windows: int | None = None*)

Bases: NoiseModel

Bootstrap temporal windows for temporal networks.

This noise model resamples time windows with replacement, useful for testing stability of temporal community detection.

**Parameters**

- **window_size** (*float*) – Size of each time window
- **n_windows** (*int, optional*) – Number of windows to sample (default: same as original)

### Examples

```
>>> noise = TemporalWindowBootstrap(window_size=100.0)
>>> perturbed = noise.apply(temporal_network, seed=42)
```

**apply**(*network: Any*, *seed: int | None = None*) → Any

Apply temporal bootstrap to network.

**Parameters**

- **network** (*multi_layer_network or TemporalMultiLayerNetwork*) – Input network with temporal edges
- **seed** (*int, optional*) – Random seed

**Returns**

Network with bootstrapped time windows

**Return type**

multi_layer_network

**n_windows**: int | None = None

**to_dict**() → Dict[str, Any]

Serialize to dict.

**window_size**: float

class py3plex.uncertainty.**TopKOverlapReducer**(*store_samples: bool = False*)

Bases: SelectionReducer

Track overlap between top-k sets across samples.

Computes expected overlap size and distribution.

**n_pairs**

Number of pairs compared

**Type**

int

**overlap_sum**

Sum of overlap sizes

**Type**

float

**overlap_sq_sum**

Sum of squared overlap sizes

**Type**
float

`overlaps`

Raw overlap values if storing samples

**Type**
list, optional

`finalize()` → Dict[str, Any]

Compute top-k overlap statistics.

**Returns**
Keys: - overlap_mean : float - overlap_std : float - q05 : float (if samples stored) - q95 : float (if samples stored)

**Return type**
dict

`update(`*selection: SelectionOutput*`)` → None

Update with a new selection.

`class py3plex.uncertainty.UQPlan(`*base_callable: Callable[[Any, Any], Any]*, *strategy: Literal['seed', 'perturbation', 'bootstrap', 'jackknife']*, *noise_model: NoiseModel | None*, *n_samples: int*, *seed: int*, *reducers: List[Reducer]*, *storage_mode: Literal['none', 'sketch', 'samples'] = 'sketch'*, *backend: Literal['python', 'jax'] = 'python'*`)`

Bases: `object`

Canonical specification for UQ execution.

UQPlan encapsulates all information needed to execute uncertainty quantification for any algorithm. It is: - Strategy-agnostic (seed, perturbation, bootstrap, jackknife) - Reducer-driven (reducers compute statistics online) - Reproducible (seed controls all randomness) - Memory-efficient (storage_mode controls what to keep)

`base_callable`

Function that executes exactly one algorithm run. Must be deterministic given RNG. Signature: (network, rng) -> SampleOutput

**Type**
Callable[[Network, RNG], SampleOutput]

`strategy`

Resampling strategy: - “seed”: Multiple runs with different RNG seeds - “perturbation”: Apply noise_model before each run - “bootstrap”: Bootstrap resampling (not yet implemented) - “jackknife”: Leave-one-out (not yet implemented)

**Type**
Literal[“seed”, “perturbation”, “bootstrap”, “jackknife”]

`noise_model`

Noise model for perturbation strategy (required if strategy=”perturbation”) Must be None for seed/bootstrap/jackknife strategies.

**Type**
NoiseModel | None

**n_samples**

Number of samples to generate

**Type**
int

**seed**

Master random seed for reproducibility

**Type**
int

**reducers**

List of reducer instances to accumulate statistics. Reducers must be provided at construction time.

**Type**
list[Reducer]

**storage_mode**

Controls what samples are stored: - “none”: No samples stored (only reducer outputs) - “sketch”: Store summary statistics (default) - “samples”: Store all sample outputs (memory-intensive)

**Type**
Literal[“none”, “sketch”, “samples”]

**backend**

Execution backend (only “python” currently supported)

**Type**
Literal[“python”, “jax”]

### Examples

```
>>> from py3plex.uncertainty.plan import UQPlan
>>> from py3plex.uncertainty.noise_models import NoNoise
>>> from py3plex.uncertainty.partition_reducers import NodeEntropyReducer
>>>
>>> def my_algorithm(net, rng):
...     # Run community detection
...     return PartitionOutput(labels={...})
>>>
>>> plan = UQPlan(
...     base_callable=my_algorithm,
...     strategy="seed",
...     noise_model=NoNoise(),
...     n_samples=50,
...     seed=42,
...     reducers=[NodeEntropyReducer(n_nodes=10)],
...     storage_mode="sketch",
...     backend="python"
... )
```

**backend:** `Literal['python', 'jax'] = 'python'`

**base_callable:** Callable[[Any, Any], Any]

**n_samples:** int

**noise_model:** NoiseModel | None

**reducers:** List[Reducer]

**seed:** int

**storage_mode:** Literal['none', 'sketch', 'samples'] = 'sketch'

**strategy:** Literal['seed', 'perturbation', 'bootstrap', 'jackknife']

class py3plex.uncertainty.**UQResult**(*n_samples: int*, *reducer_outputs: dict = <factory>*, *samples: ~typing.List[~typing.Any] | None = None*, *provenance: dict = <factory>*)

Bases: object

Result container for UQ execution.

**n_samples**

Number of samples executed

**Type**
int

**reducer_outputs**

Dictionary mapping reducer class name to finalized output

**Type**
dict

**samples**

Raw sample outputs (only if storage_mode="samples")

**Type**
list, optional

**provenance**

Execution metadata (seed, strategy, noise_model, etc.)

**Type**
dict

**n_samples:** int

**provenance:** dict

**reducer_outputs:** dict

**samples:** List[Any] | None = None

class py3plex.uncertainty.**UncertaintyConfig**(*mode: UncertaintyMode = UncertaintyMode.OFF*, *default_n_runs: int = 50*, *default_resampling: ResamplingStrategy = ResamplingStrategy.SEED*)

Bases: object

Configuration for uncertainty estimation.

**mode**

The current uncertainty mode.

**Type**
UncertaintyMode

**default_n_runs**

Default number of runs for uncertainty estimation.

**Type**
int

**default_resampling**

Default resampling strategy.

**Type**
ResamplingStrategy

**default_n_runs: int = 50**

**default_resampling: ResamplingStrategy = 'seed'**

**mode: UncertaintyMode = 'off'**

class py3plex.uncertainty.**UncertaintyMode**(*value*)

Bases: Enum

Global mode for uncertainty computation.

**OFF**

Always deterministic, std=None.

**Type**
str

**ON**

Try to compute uncertainty when supported.

**Type**
str

**AUTO**

Only do it if explicitly requested by a function.

**Type**
str

**AUTO = 'auto'**

**OFF = 'off'**

**ON = 'on'**

class py3plex.uncertainty.**WeightNoise**(*dist: str*, *sigma: float*, *clip_min: float = 0.0*, *clip_max: float | None = None*)

Bases: NoiseModel

Add multiplicative noise to edge weights.

This noise model multiplies each edge weight by a random factor drawn from a specified distribution. Useful for testing sensitivity to weight uncertainty.

**Parameters**

- **dist** (*str*) – Distribution type: “lognormal”, “uniform”, or “normal”
- **sigma** (*float*) – Scale parameter for the distribution - lognormal: standard deviation of log-weights - uniform: half-width of interval - normal: standard deviation
- **clip_min** (*float, optional*) – Minimum weight value (default: 0.0)
- **clip_max** (*float, optional*) – Maximum weight value (default: None)

### Examples

```
>>> # Lognormal noise with sigma=0.2
>>> noise = WeightNoise(dist="lognormal", sigma=0.2)
>>> perturbed = noise.apply(network, seed=42)
>>>
>>> # Uniform noise in [0.8, 1.2] range
>>> noise = WeightNoise(dist="uniform", sigma=0.2)
>>> perturbed = noise.apply(network, seed=42)
```

**apply**(*network: Any*, *seed: int | None = None*) → Any

Apply weight noise to network.

**Parameters**

- **network** (*multi_layer_network*) – Input network
- **seed** (*int, optional*) – Random seed

**Returns**

Network with noisy weights

**Return type**

multi_layer_network

**clip_max:** **float | None = None**

**clip_min:** **float = 0.0**

**dist:** **str**

**sigma:** **float**

**to_dict**() → Dict[str, Any]

Serialize to dict.

py3plex.uncertainty.**adjusted_rand_index**(*partition1: ndarray*, *partition2: ndarray*) → float

Compute Adjusted Rand Index between two partitions.

ARI measures the similarity between two clusterings adjusted for chance.

ARI is: - Symmetric - Bounded by 1 (perfect agreement) - Expected value 0 for random partitions - Can be negative

**Parameters**

- **partition1** (*np.ndarray*) – First partition, shape (n,) with integer labels

- **partition2** (*np.ndarray*) – Second partition, shape (n,) with integer labels

**Returns**
Adjusted Rand Index

**Return type**
float

### Examples

```
>>> p1 = np.array([0, 0, 1, 1])
>>> p2 = np.array([0, 0, 0, 1])
>>> ari = adjusted_rand_index(p1, p2)
>>> -1 <= ari <= 1
True
```

py3plex.uncertainty.**ari**(*partition1: ndarray*, *partition2: ndarray*) → float

Compute Adjusted Rand Index between two partitions.

ARI measures the similarity between two clusterings adjusted for chance.

ARI is: - Symmetric - Bounded by 1 (perfect agreement) - Expected value 0 for random partitions - Can be negative

**Parameters**

- **partition1** (*np.ndarray*) – First partition, shape (n,) with integer labels
- **partition2** (*np.ndarray*) – Second partition, shape (n,) with integer labels

**Returns**
Adjusted Rand Index

**Return type**
float

### Examples

```
>>> p1 = np.array([0, 0, 1, 1])
>>> p2 = np.array([0, 0, 0, 1])
>>> ari = adjusted_rand_index(p1, p2)
>>> -1 <= ari <= 1
True
```

py3plex.uncertainty.**auto_select_strata**(*target: str*) → List[str]

Automatically select stratification dimensions based on query target.

**Parameters**
**target** (*str*) – Query target type (“nodes” or “edges”)

**Returns**
Recommended stratification dimensions

**Return type**
List[str]

py3plex.uncertainty.**binomial_proportion_ci**(*successes: int*, *n_trials: int*, *alpha: float = 0.05*, *method: str = 'wilson'*) → Tuple[float, float]

Compute confidence interval for a binomial proportion.

**Parameters**

- **successes** (*int*) – Number of successes
- **n_trials** (*int*) – Total number of trials
- **alpha** (*float, default=0.05*) – Significance level
- **method** (*str, default="wilson"*) – Method to use: "wilson" or "clopper-pearson"

**Returns**

Lower and upper bounds of the confidence interval

**Return type**

tuple of (float, float)

**Raises**

**ValueError** – If method is not recognized

py3plex.uncertainty.**bootstrap_metric**(*graph: multi_layer_network*, *metric_fn: Callable[[multi_layer_network], Dict[Any, float]]*, *n_boot: int = 50*, *unit: str = 'edges'*, *mode: str = 'resample'*, *ci: float = 0.95*, *random_state: int | None = None*, *n_jobs: int | None = None*) → Dict[str, ndarray]

Bootstrap a metric for uncertainty estimation.

This function resamples the graph by unit (edges, nodes, or layers) and recomputes the metric on each bootstrap sample to estimate uncertainty.

**Parameters**

- **graph** (*multi_layer_network*) – The multilayer network to analyze.
- **metric_fn** (*callable*) – Function that takes a network and returns a dict mapping items (usually nodes) to metric values. Must have signature: metric_fn(network) -> Dict[item_id, float]
- **n_boot** (*int, default=200*) – Number of bootstrap replicates.
- **unit** (*str, default="edges"*) – What to resample: "edges", "nodes", or "layers".
- **mode** (*str, default="resample"*) – Resampling mode: - "resample": Sample with replacement (classic bootstrap) - "permute": Permute the units (permutation test)
- **ci** (*float, default=0.95*) – Confidence interval level (e.g., 0.95 for 95% CI).
- **random_state** (*int, optional*) – Random seed for reproducibility.
- **n_jobs** (*int, optional*) – Number of parallel jobs. If None, uses config.DEFAULT_N_JOBS. If 1, runs serially. If >1, runs in parallel.

**Returns**

Dictionary with keys: - "mean": np.ndarray of shape (n_items,) with

mean values - “std”: np.ndarray of shape (n_items,) with standard errors - “ci_low”: np.ndarray of shape (n_items,) with lower CI bounds - “ci_high”: np.ndarray of shape (n_items,) with upper CI bounds - “index”: List of item IDs - “n_boot”: Number of bootstrap replicates used - “method”: String describing the bootstrap method

**Return type**
dict

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.uncertainty.bootstrap import bootstrap_metric
>>>
>>> # Create a network
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([["a", "L0", "b", "L0", 1.0]], input_type="list")
>>>
>>> # Define metric function
>>> def degree_metric(network):
...     result = {}
...     for node in network.get_nodes():
...         result[node] = network.core_network.degree(node)
...     return result
>>>
>>> # Bootstrap
>>> boot_result = bootstrap_metric(
...     net, degree_metric, n_boot=100, unit="edges"
... )
>>> boot_result["mean"]  # Mean degree values
>>> boot_result["ci_low"]  # Lower CI bounds
```

### Notes

- For “edges” unit: resamples edges with replacement
- For “nodes” unit: resamples nodes with replacement
- For “layers” unit: resamples layers with replacement
- The metric_fn must be able to handle graphs with different structure

py3plex.uncertainty.**bootstrap_network_edges**(*network: multi_layer_network*, ***, *seed: int | None = None*, *preserve_layers: bool = True*) → multi_layer_network

Bootstrap resample network edges with replacement.

Creates a new network by sampling edges with replacement from the original network. The new network has approximately the same number of edges, but some may be duplicated and some omitted.

**Never modifies the input network** - returns a new copy.

**Parameters**

- **network** (*multi_layer_network*) – Input network to resample (will

not be modified).

- **seed** (*int , optional*) – Random seed for reproducibility.
- **preserve_layers** (*bool , default=True*) – If True, preserve layer labels and structure.

**Returns**
New network with bootstrapped edges.

**Return type**
multi_layer_network

### Notes

For undirected networks, each undirected edge is treated as a single unit for resampling (not resampled twice as two directed edges).

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.uncertainty.resampling_graph import bootstrap_network_edges
>>>
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
... ], input_type='list')
>>>
>>> boot = bootstrap_network_edges(net, seed=42)
>>> # Some edges may be duplicated, some omitted
```

py3plex.uncertainty.**clopper_pearson_interval**(*successes: int*, *n_trials: int*, *alpha: float = 0.05*) → Tuple[float, float]

Compute Clopper-Pearson exact confidence interval for a binomial proportion.

This is a more conservative (wider) interval than Wilson score, guaranteed to have at least the nominal coverage. Useful when exactness is critical.

**Parameters**

- **successes** (*int*) – Number of successes
- **n_trials** (*int*) – Total number of trials
- **alpha** (*float , default=0.05*) – Significance level (e.g., 0.05 for 95% CI)

**Returns**
Lower and upper bounds of the confidence interval

**Return type**
tuple of (float, float)

### Notes

Based on the beta distribution relationship to the binomial.

### Examples

```
>>> clopper_pearson_interval(50, 100, alpha=0.05)
(0.39..., 0.60...)
```

py3plex.uncertainty.**compute_composite_strata**(*network: multi_layer_network*, *spec: StratificationSpec*, *target: str = 'nodes'*) → Dict[Tuple, List[Any]]

Compute composite stratification by crossing multiple dimensions.

**Parameters**

- **network** (*multi_layer_network*) – The network to stratify.
- **spec** (*StratificationSpec*) – Stratification specification.
- **target** (*str*) – Target type (“nodes” or “edges”)

**Returns**
Mapping from composite stratum key to list of items.

**Return type**
Dict[Tuple, List[Any]]

py3plex.uncertainty.**compute_variance_reduction_ratio**(*baseline_std: ndarray*, *stratified_std: ndarray*) → float

Compute variance reduction ratio.

**Parameters**

- **baseline_std** (*np.ndarray*) – Standard deviations from baseline (unstratified) estimation.
- **stratified_std** (*np.ndarray*) – Standard deviations from stratified estimation.

**Returns**
Variance reduction ratio: (baseline_var - stratified_var) / baseline_var Values $> 0$ indicate variance reduction, $< 0$ indicate increase.

**Return type**
float

py3plex.uncertainty.**estimate_uncertainty**(*network: multi_layer_network*, *metric_fn: Callable[[multi_layer_network], Dict[Any, float] | float | ndarray]*, ***, *n_runs: int | None = None*, *resampling: ResamplingStrategy | None = None*, *random_seed: int | None = None*, *perturbation_params: Dict[str, Any] | None = None*) → StatSeries | float

Estimate uncertainty for a network statistic.

This is the main entry point for adding uncertainty to any statistic. It runs the metric function multiple times with different random seeds or network perturbations, then computes mean, std, and quantiles.

**Parameters**

- **network** (*multi_layer_network*) – The network to analyze.
- **metric_fn** (*callable*) – Function that computes a statistic. Must accept a network and return: - dict[node, float] for per-node statistics - float for scalar statistics - np.ndarray for array statistics
- **n_runs** (*int , optional*) – Number of runs for uncertainty estimation. If None, uses the default from the current uncertainty config.
- **resampling** (*ResamplingStrategy , optional*) – Strategy for resampling. If None, uses default from config.
- **random_seed** (*int , optional*) – Random seed for reproducibility.
- **perturbation_params** (*dict , optional*) – Parameters for perturbation strategies. For example: {“edge_drop_p”: 0.05, “node_drop_p”: 0.02}

**Returns**

If metric_fn returns a dict: StatSeries with uncertainty info If metric_fn returns a scalar: float (mean value) The returned object has mean, std, and quantiles populated.

**Return type**

StatSeries or float

### Examples

```
>>> from py3plex.uncertainty import estimate_uncertainty, ResamplingStrategy
>>> from py3plex.core import multinet
>>>
>>> # Create a simple network
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([["a", "L0", "b", "L0", 1.0]], input_type="list")
>>>
>>> # Define a metric function
>>> def my_metric(network):
...     # Return per-node degree
...     degrees = {}
...     for node in network.get_nodes():
...         degrees[node] = network.core_network.degree(node)
...     return degrees
>>>
>>> # Estimate uncertainty
>>> result = estimate_uncertainty(
...     net,
...     my_metric,
...     n_runs=50,
...     resampling=ResamplingStrategy.PERTURBATION,
...     perturbation_params={"edge_drop_p": 0.1}
... )
>>> result.mean  # Mean degree values
>>> result.std   # Std deviation of degrees
>>> result.quantiles  # Confidence intervals
```

### Notes

- For SEED strategy: runs the metric with different random seeds
- For PERTURBATION strategy: applies edge/node drops then recomputes
- For BOOTSTRAP: resamples nodes/edges with replacement (not yet implemented)
- For JACKKNIFE: leave-one-out resampling (not yet implemented)

py3plex.uncertainty.**execute_selection_uq**(*base_callable: Callable[[Any, Dict], SelectionOutput]*, *network: Any*, *params: Dict | None = None*, *method: str = 'perturbation'*, *n_samples: int = 50*, *seed: int | None = None*, *noise_model: NoiseModel | None = None*, *ci: float = 0.95*, *consensus_threshold: float = 0.5*, *borderline_epsilon: float = 0.1*, *store_mode: str = 'sketch'*, *max_items_tracked: int | None = None*, *grouped: bool = False*, ***kwargs*) → SelectionUQ

Execute a selection query with uncertainty quantification.

**Parameters**

- **base_callable** (*callable*) – Function that takes (network, params) and returns SelectionOutput
- **network** (*multi_layer_network*) – Input network
- **params** (*dict, optional*) – Parameters for base_callable
- **method** (*str*) – UQ method: “perturbation”, “seed”, “bootstrap”, “jackknife”
- **n_samples** (*int*) – Number of samples to generate
- **seed** (*int, optional*) – Random seed for reproducibility
- **noise_model** (*NoiseModel, optional*) – Noise model for perturbation method
- **ci** (*float*) – Confidence interval level (e.g., 0.95)
- **consensus_threshold** (*float*) – Threshold for consensus selection (default: 0.5)
- **borderline_epsilon** (*float*) – Epsilon for borderline detection (default: 0.1)
- **store_mode** (*str*) – Storage mode: “none”, “sketch”, “samples”
- **max_items_tracked** (*int, optional*) – Maximum number of items to track (memory cap)
- **grouped** (*bool*) – Whether query has grouping (per_layer/per_layer_pair)
- ****kwargs** – Additional method-specific parameters

**Returns**

If grouped=False: SelectionUQ instance If grouped=True: dict mapping group_key -> SelectionUQ

**Return type**
SelectionUQ or dict

### Examples

```
>>> from py3plex.uncertainty.noise_models import EdgeDrop
>>> from py3plex.uncertainty.selection_execution import execute_selection_uq
>>>
>>> def my_query(net, params):
...     # Run query and return SelectionOutput
...     return SelectionOutput(items=['a', 'b'], target='nodes')
>>>
>>> uq = execute_selection_uq(
...     base_callable=my_query,
...     network=net,
...     method="perturbation",
...     noise_model=EdgeDrop(p=0.05),
...     n_samples=100,
...     seed=42
... )
```

py3plex.uncertainty.**generate_community_ensemble**(*network: Any*, *algorithm: str = 'louvain'*, *method: str = 'seed'*, *n_samples: int = 50*, *seed: int | None = None*, *perturbation_rate: float = 0.1*, *bootstrap_unit: str = 'edges'*, *algorithm_params: Dict[str, Any] | None = None*, *n_jobs: int = 1*, *verbose: bool = False*) → CommunityDistribution

Generate an ensemble of community partitions with UQ.

This is the main entry point for probabilistic community detection. It generates multiple partitions using the specified resampling strategy and returns a CommunityDistribution for analysis.

**Parameters**

- **network** (*multi_layer_network*) – The network to analyze.
- **algorithm** (*str, default='louvain'*) – Community detection algorithm. Supported: - 'louvain': Louvain modularity optimization - 'label_propagation': Label propagation - 'infomap': Infomap (requires infomap package)
- **method** (*str, default='seed'*) – Resampling method for uncertainty estimation: - 'seed': Run algorithm with different random seeds (Monte Carlo) - 'perturbation': Perturb edges before each run - 'bootstrap': Bootstrap resample nodes or edges
- **n_samples** (*int, default=50*) – Number of partitions to generate in the ensemble.
- **seed** (*int, optional*) – Base random seed for reproducibility. If provided, generates deterministic sequence of seeds for parallel runs.

- **perturbation_rate** (*float, default=0.1*) – For method='perturbation': fraction of edges to perturb (0.0 to 1.0).
- **bootstrap_unit** (*str, default='edges'*) – For method='bootstrap': what to resample ('edges', 'nodes', 'layers').
- **algorithm_params** (*dict, optional*) – Additional parameters to pass to the community detection algorithm.
- **n_jobs** (*int, default=1*) – Number of parallel jobs. Currently not implemented (sequential).
- **verbose** (*bool, default=False*) – If True, print progress information.

**Returns**

Distribution over community partitions with metadata.

**Return type**

CommunityDistribution

**Raises**

**AlgorithmError** – If algorithm or method is not supported, or if n_samples < 1.

### Examples

```
>>> from py3plex.uncertainty import generate_community_ensemble
>>> from py3plex.core import multinet
>>>
>>> net = multinet.multi_layer_network(directed=False)
>>> # ... create network ...
>>>
>>> # Seed-based ensemble (pure algorithmic stochasticity)
>>> dist = generate_community_ensemble(
...     net, algorithm='louvain', method='seed',
...     n_samples=100, seed=42
... )
>>>
>>> # Perturbation-based ensemble (structural uncertainty)
>>> dist = generate_community_ensemble(
...     net, algorithm='louvain', method='perturbation',
...     n_samples=50, perturbation_rate=0.05, seed=42
... )
>>>
>>> # Get probabilistic result
>>> from py3plex.uncertainty import ProbabilisticCommunityResult
>>> result = ProbabilisticCommunityResult(dist)
>>> labels = result.labels
>>> probs = result.probs
>>> entropy = result.entropy
```

py3plex.uncertainty.**get_uncertainty_config**() → UncertaintyConfig

Get the current uncertainty configuration.

**Returns**

The current configuration from the context.

**Return type**
UncertaintyConfig

### Examples

```
>>> from py3plex.uncertainty import get_uncertainty_config
>>> cfg = get_uncertainty_config()
>>> cfg.mode
<UncertaintyMode.OFF: 'off'>
>>> cfg.default_n_runs
50
```

py3plex.uncertainty.**nmi**(*partition1: ndarray*, *partition2: ndarray*, *method: str = 'arithmetic'*) → float

Compute Normalized Mutual Information between two partitions.

NMI normalizes mutual information by the arithmetic or geometric mean of the entropies.

NMI is: - Symmetric - In range [0, 1] - 1 iff partitions are identical (up to label permutation) - 0 if independent

**Parameters**

- **partition1** (*np.ndarray*) – First partition, shape (n,) with integer labels
- **partition2** (*np.ndarray*) – Second partition, shape (n,) with integer labels
- **method** (*str, default="arithmetic"*) – Normalization method: - “arithmetic”: 2*I(X,Y) / (H(X) + H(Y)) - “geometric”: I(X,Y) / sqrt(H(X) * H(Y)) - “max”: I(X,Y) / max(H(X), H(Y)) - “min”: I(X,Y) / min(H(X), H(Y))

**Returns**
Normalized mutual information in [0, 1]

**Return type**
float

### Examples

```
>>> p1 = np.array([0, 0, 1, 1])
>>> p2 = np.array([0, 0, 0, 1])
>>> nmi_val = normalized_mutual_information(p1, p2)
>>> 0 <= nmi_val <= 1
True
```

py3plex.uncertainty.**noise_model_from_dict**(*data: Dict[str, Any]*) → NoiseModel

Deserialize noise model from dictionary.

**Parameters**
**data** (*dict*) – Dictionary with “type” key and model-specific parameters

**Returns**
Instantiated noise model

**Return type**
NoiseModel

### Examples

```
>>> data = {"type": "EdgeDrop", "p": 0.1}
>>> noise = noise_model_from_dict(data)
>>> isinstance(noise, EdgeDrop)
True
```

py3plex.uncertainty.**normalized_mutual_information**(*partition1: ndarray*, *partition2: ndarray*, *method: str = 'arithmetic'*) → float

Compute Normalized Mutual Information between two partitions.

NMI normalizes mutual information by the arithmetic or geometric mean of the entropies.

NMI is: - Symmetric - In range [0, 1] - 1 iff partitions are identical (up to label permutation) - 0 if independent

**Parameters**

- **partition1** (*np.ndarray*) – First partition, shape (n,) with integer labels
- **partition2** (*np.ndarray*) – Second partition, shape (n,) with integer labels
- **method** (*str, default="arithmetic"*) – Normalization method: - “arithmetic”: 2*I(X,Y) / (H(X) + H(Y)) - “geometric”: I(X,Y) / sqrt(H(X) * H(Y)) - “max”: I(X,Y) / max(H(X), H(Y)) - “min”: I(X,Y) / min(H(X), H(Y))

**Returns**
Normalized mutual information in [0, 1]

**Return type**
float

### Examples

```
>>> p1 = np.array([0, 0, 1, 1])
>>> p2 = np.array([0, 0, 0, 1])
>>> nmi_val = normalized_mutual_information(p1, p2)
>>> 0 <= nmi_val <= 1
True
```

py3plex.uncertainty.**null_model_metric**(*graph: multi_layer_network*, *metric_fn: Callable[[multi_layer_network], Dict[Any, float]]*, *n_null: int = 200*, *model: str = 'degree_preserving'*, *random_state: int | None = None*, *n_jobs: int | None = None*) → Dict[str, ndarray]

Compute metric on null models for statistical significance testing.

This function generates null models of the network (e.g., via degree-preserving rewiring)

and computes the metric on each null network. It then computes z-scores and p-values for the observed metric values.

**Parameters**

- **graph** (*multi_layer_network*) – The multilayer network to analyze.
- **metric_fn** (*callable*) – Function that takes a network and returns a dict mapping items (usually nodes) to metric values. Must have signature: metric_fn(network) -> Dict[item_id, float]
- **n_null** (*int, default=200*) – Number of null model replicates to generate.
- **model** (*str, default="degree_preserving"*) – Null model type: - "degree_preserving": Rewire edges while preserving degree sequence - "erdos_renyi": Random graph with same density - "configuration": Configuration model matching degree distribution
- **random_state** (*int, optional*) – Random seed for reproducibility.
- **n_jobs** (*int, optional*) – Number of parallel jobs. If None, uses config.DEFAULT_N_JOBS. If 1, runs serially. If >1, runs in parallel.

**Returns**

Dictionary with keys: - "observed": np.ndarray of shape (n_items,) with observed metric values - "mean_null": np.ndarray of shape (n_items,) with mean null values - "std_null": np.ndarray of shape (n_items,) with std of null values - "zscore": np.ndarray of shape (n_items,) with z-scores - "pvalue": np.ndarray of shape (n_items,) with two-tailed p-values - "index": List of item IDs - "n_null": Number of null replicates used - "model": String describing the null model

**Return type**

dict

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.uncertainty.null_models import null_model_metric
>>>
>>> # Create a network
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([["a", "L0", "b", "L0", 1.0]], input_type="list")
>>>
>>> # Define metric function
>>> def degree_metric(network):
...     result = {}
...     for node in network.get_nodes():
...         result[node] = network.core_network.degree(node)
...     return result
>>>
>>> # Compute null model statistics
>>> null_result = null_model_metric(
...     net, degree_metric, n_null=100, model="degree_preserving"
... )
>>> null_result["zscore"]  # Z-scores for each node
```

```
>>> null_result["pvalue"]  # P-values for each node
```

### Notes

- Z-scores indicate how many standard deviations the observed value is from the null distribution mean
- P-values are two-tailed by default: `P(|Z| >= |Z_obs|)` under the null. For one-tailed tests, users can compute p_one_sided = p_two_sided / 2 and check the sign of z-score to determine the direction.
- High z-score magnitude and low p-value indicate statistical significance

py3plex.uncertainty.**pairwise_partition_distances**(*partitions: list[ndarray]*, *metric: str = 'vi'*) → ndarray

Compute pairwise distances between partitions.

**Parameters**

- **partitions** (*list of np.ndarray*) – List of partition arrays
- **metric** (*str, default="vi"*) – Distance metric: “vi”, “nmi”, or “ari”

**Returns**
Distance matrix, shape (n_partitions, n_partitions)

**Return type**
np.ndarray

### Examples

```
>>> partitions = [
...     np.array([0, 0, 1, 1]),
...     np.array([0, 0, 0, 1]),
...     np.array([0, 1, 0, 1]),
... ]
>>> D = pairwise_partition_distances(partitions, metric="vi")
>>> D.shape
(3, 3)
>>> np.allclose(D.diagonal(), 0)  # Diagonal is zero
True
```

py3plex.uncertainty.**partition_array_to_dict**(*partition_array: ndarray*, *node_index: List[Any]*) → Dict[Any, int]

Convert partition array to dictionary.

**Parameters**

- **partition_array** (*np.ndarray*) – Partition labels, shape (n_nodes,).
- **node_index** (*list*) – Ordered list of node identifiers.

**Returns**
Mapping from node to community ID.

**Return type**
dict

### Examples

```
>>> arr = np.array([0, 0, 1])
>>> nodes = [('A', 'L1'), ('B', 'L1'), ('C', 'L1')]
>>> d = partition_array_to_dict(arr, nodes)
>>> d[('A', 'L1')]
0
```

py3plex.uncertainty.**partition_dict_to_array**(*partition_dict: Dict[Any, int]*, *node_index: List[Any]*) → ndarray

Convert partition dictionary to array with canonical node ordering.

**Parameters**

- **partition_dict** (*dict*) – Mapping from node (or node-layer tuple) to community ID.
- **node_index** (*list*) – Ordered list of nodes defining the canonical ordering.

**Returns**

Partition array with shape (len(node_index),).

**Return type**

np.ndarray

### Examples

```
>>> partition = {('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1}
>>> nodes = [('A', 'L1'), ('B', 'L1'), ('C', 'L1')]
>>> arr = partition_dict_to_array(partition, nodes)
>>> arr
array([0, 0, 1])
```

py3plex.uncertainty.**perturb_network_edges**(*network: multi_layer_network*, ***, *edge_drop_p: float*, *seed: int | None = None*, *preserve_layers: bool = True*) → multi_layer_network

Perturb network by randomly dropping edges.

Creates a new network with a fraction of edges randomly removed. Useful for structural uncertainty quantification via edge perturbation.

**Never modifies the input network** - returns a new copy.

**Parameters**

- **network** (*multi_layer_network*) – Input network to perturb (will not be modified).
- **edge_drop_p** (*float*) – Probability of dropping each edge, in [0, 1]. - 0.0: No edges dropped (returns exact copy) - 1.0: All edges dropped (returns empty network)
- **seed** (*int, optional*) – Random seed for reproducibility.
- **preserve_layers** (*bool, default=True*) – If True, preserve layer labels and structure. If False, allow layer simplification (not recom-

mended).

**Returns**
New network with randomly dropped edges.

**Return type**
multi_layer_network

**Raises**
**AlgorithmError** – If edge_drop_p is not in [0, 1].

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.uncertainty.resampling_graph import perturb_network_edges
>>>
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
...     ['C', 'L1', 'A', 'L1', 1],
... ], input_type='list')
>>>
>>> # Drop 20% of edges
>>> perturbed = perturb_network_edges(net, edge_drop_p=0.2, seed=42)
>>> # Original net unchanged
>>> len(list(net.get_edges()))
3
>>> len(list(perturbed.get_edges()))  # ~2-3 edges
2
```

py3plex.uncertainty.**rank_ci_from_samples**(*ranks: ndarray*, *alpha: float = 0.05*) → Tuple[float, float]

Compute empirical confidence interval for ranks from samples.

**Parameters**

- **ranks** (*np.ndarray*) – Array of rank values from different samples
- **alpha** (*float , default=0.05*) – Significance level

**Returns**
Lower and upper bounds of the confidence interval (quantiles)

**Return type**
tuple of (float, float)

py3plex.uncertainty.**resample_network_nodes**(*network: multi_layer_network*, ***, *seed: int | None = None*, *preserve_layers: bool = True*) → multi_layer_network

Bootstrap resample network nodes with replacement.

Creates a new network by sampling nodes with replacement, including their incident edges. Less commonly used than edge resampling for community detection UQ.

**Never modifies the input network** - returns a new copy.

**Parameters**

- **network** (*multi_layer_network*) – Input network to resample (will not be modified).
- **seed** (*int, optional*) – Random seed for reproducibility.
- **preserve_layers** (*bool, default=True*) – If True, preserve layer labels.

**Returns**

New network with bootstrapped nodes and their edges.

**Return type**

multi_layer_network

### Notes

This resampling can significantly change network structure and is less suitable for community detection than edge resampling.

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.uncertainty.resampling_graph import resample_network_nodes
>>>
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
... ], input_type='list')
>>>
>>> boot = resample_network_nodes(net, seed=42)
```

py3plex.uncertainty.**run_uq**(*plan: UQPlan*, *network: Any*) → UQResult

Execute uncertainty quantification according to plan.

This is the canonical UQ execution function. It implements the following strict execution semantics:

1. Initialize RNG sequence from plan.seed
2. **For i in range(n_samples):**
   a. Derive per-sample RNG
   b. Apply NoiseModel (or NoNoise)
   c. Call base_callable(net_i, rng_i)
   d. Pass output to ALL reducers
3. After loop, call finalize() on reducers
4. Assemble UQResult
5. Attach centralized provenance

ABSOLUTE RULES: - No reducer may control execution - No algorithm may loop internally for UQ - No sample outputs stored unless storage_mode == “samples”

**Parameters**

- **plan** (*UQPlan*) – UQ execution plan specifying all parameters

- **network** (*multi_layer_network*) – Input network to analyze

**Returns**
Result container with reducer outputs and provenance

**Return type**
UQResult

### Examples

```
>>> from py3plex.uncertainty.plan import UQPlan
>>> from py3plex.uncertainty.runner import run_uq
>>> from py3plex.uncertainty.noise_models import EdgeDrop
>>> from py3plex.uncertainty.partition_reducers import (
...     NodeEntropyReducer, ConsensusReducer
... )
>>>
>>> plan = UQPlan(
...     base_callable=lambda net, rng: my_community_detection(net, seed=int(rng.integer
...     strategy="perturbation",
...     noise_model=EdgeDrop(p=0.1),
...     n_samples=100,
...     seed=42,
...     reducers=[
...         NodeEntropyReducer(n_nodes=network.number_of_nodes()),
...         ConsensusReducer(n_nodes=network.number_of_nodes())
...     ],
...     storage_mode="sketch"
... )
>>>
>>> result = run_uq(plan, network)
>>> print(result.reducer_outputs.keys())
```

py3plex.uncertainty.**set_uncertainty_config**(*config: UncertaintyConfig*) → Token

Set the uncertainty configuration.

**Parameters**
**config** (*UncertaintyConfig*) – The new configuration to set.

**Returns**
A token that can be used to reset the configuration.

**Return type**
Token

### Examples

```
>>> from py3plex.uncertainty import set_uncertainty_config, UncertaintyConfig
>>> from py3plex.uncertainty import UncertaintyMode
>>> cfg = UncertaintyConfig(mode=UncertaintyMode.ON, default_n_runs=100)
>>> token = set_uncertainty_config(cfg)
>>> # ... do work ...
>>> _uncertainty_ctx.reset(token)  # restore previous config
```

py3plex.uncertainty.**uncertainty_enabled**(*, *n_runs: int | None = None*, *resampling: ResamplingStrategy | None = None*)

Context manager to enable uncertainty estimation.

Within this context, all supported functions will compute uncertainty by default (unless explicitly disabled with uncertainty=False).

**Parameters**

- **n_runs** (*int , optional*) – Number of runs for uncertainty estimation. If None, uses the default from the current config.
- **resampling** (*ResamplingStrategy , optional*) – Resampling strategy to use. If None, uses the default from the current config.

**Yields**

*None*

### Examples

```
>>> from py3plex.uncertainty import uncertainty_enabled
>>> from py3plex.algorithms.centrality_toolkit import multilayer_pagerank
>>>
>>> # Without uncertainty
>>> result = multilayer_pagerank(network)
>>> result.is_deterministic
True
>>>
>>> # With uncertainty
>>> with uncertainty_enabled(n_runs=100):
...     result = multilayer_pagerank(network)
>>> result.is_deterministic
False
>>> result.std is not None
True
```

### Notes

This uses contextvars, so it's thread-safe and async-safe. Each context gets its own configuration.

py3plex.uncertainty.**variation_of_information**(*partition1: ndarray*, *partition2: ndarray*, *normalized: bool = False*) → float

Compute Variation of Information between two partitions.

VI(X, Y) = H(X) + H(Y) - 2*I(X, Y)

where H is entropy and I is mutual information.

VI is: - Symmetric: VI(X, Y) = VI(Y, X) - Non-negative: VI(X, Y) >= 0 - Zero iff partitions are identical (up to label permutation) - Bounded by 2*log(n) where n is number of elements

**Parameters**

- **partition1** (*np.ndarray*) – First partition, shape (n,) with integer labels

- **partition2** (*np.ndarray*) – Second partition, shape (n,) with integer labels
- **normalized** (*bool, default=False*) – If True, normalize by log(n) to get value in [0, 2]

**Returns**
Variation of information distance

**Return type**
float

### Examples

```
>>> p1 = np.array([0, 0, 1, 1])
>>> p2 = np.array([0, 0, 0, 1])
>>> vi = variation_of_information(p1, p2)
>>> vi > 0  # Different partitions
True
>>>
>>> p3 = np.array([5, 5, 7, 7])  # Same structure, different labels
>>> vi2 = variation_of_information(p1, p3)
>>> abs(vi2) < 1e-10  # Should be zero (label-invariant)
True
```

py3plex.uncertainty.vi(*partition1: ndarray*, *partition2: ndarray*, *normalized: bool = False*) → float

Compute Variation of Information between two partitions.

VI(X, Y) = H(X) + H(Y) - 2*I(X, Y)

where H is entropy and I is mutual information.

VI is: - Symmetric: VI(X, Y) = VI(Y, X) - Non-negative: VI(X, Y) >= 0 - Zero iff partitions are identical (up to label permutation) - Bounded by 2*log(n) where n is number of elements

**Parameters**

- **partition1** (*np.ndarray*) – First partition, shape (n,) with integer labels
- **partition2** (*np.ndarray*) – Second partition, shape (n,) with integer labels
- **normalized** (*bool, default=False*) – If True, normalize by log(n) to get value in [0, 2]

**Returns**
Variation of information distance

**Return type**
float

### Examples

```
>>> p1 = np.array([0, 0, 1, 1])
>>> p2 = np.array([0, 0, 0, 1])
>>> vi = variation_of_information(p1, p2)
>>> vi > 0  # Different partitions
True
>>>
>>> p3 = np.array([5, 5, 7, 7])  # Same structure, different labels
>>> vi2 = variation_of_information(p1, p3)
>>> abs(vi2) < 1e-10  # Should be zero (label-invariant)
True
```

py3plex.uncertainty.**wilson_score_interval**(*successes: int*, *n_trials: int*, *alpha: float = 0.05*) → Tuple[float, float]

Compute Wilson score confidence interval for a binomial proportion.

The Wilson score interval is robust for small sample sizes and probabilities near 0 or 1. It is preferred over the normal approximation (Wald interval).

**Parameters**

- **successes** (*int*) – Number of successes (e.g., number of times item appeared in selection)
- **n_trials** (*int*) – Total number of trials (e.g., number of UQ samples)
- **alpha** (*float, default=0.05*) – Significance level (e.g., 0.05 for 95% CI)

**Returns**

Lower and upper bounds of the confidence interval

**Return type**

tuple of (float, float)

### Notes

Based on: Wilson, E.B. (1927). "Probable inference, the law of succession, and statistical inference". Journal of the American Statistical Association.

For a CI level of (1-alpha), we use z = ^(-1)(1 - alpha/2).

### Examples

```
>>> wilson_score_interval(50, 100, alpha=0.05)
(0.4..., 0.6...)
```

```
>>> wilson_score_interval(0, 100, alpha=0.05)
(0.0, 0.036...)
```

```
>>> wilson_score_interval(100, 100, alpha=0.05)
(0.963..., 1.0)
```

Core statistic types for first-class uncertainty representation.

This module defines the fundamental types that wrap statistics with optional uncertainty information (standard deviations, confidence intervals, quantiles).

**class** py3plex.uncertainty.types.**CommunityStats**(*labels: ~typing.Dict[~typing.Any, int], modularity: float | None = None, modularity_std: float | None = None, coassoc: ~py3plex.uncertainty.types.StatMatrix | None = None, stability: ~typing.Dict[~typing.Any, float] | None = None, n_communities: int = 0, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Statistics from community detection with optional uncertainty.

Wraps cluster labels, modularity, co-association matrix, and stability indices computed from multiple runs.

**Parameters**

- **labels** (*dict[Any, int]*) – Node -> community ID mapping (from deterministic run or consensus).
- **modularity** (*float or None*) – Modularity score (mean if multiple runs).
- **modularity_std** (*float or None*) – Standard deviation of modularity across runs.
- **coassoc** (*StatMatrix or None*) – Co-association matrix (probability nodes are in same community).
- **stability** (*dict[Any, float] or None*) – Per-node stability index (how often node stays in same cluster).
- **n_communities** (*int*) – Number of communities detected.
- **meta** (*dict[str, Any]*) – Optional metadata.

### Examples

```
>>> cs = CommunityStats(
...     labels={'a': 0, 'b': 0, 'c': 1},
...     modularity=0.42,
...     n_communities=2
... )
>>> cs.is_deterministic
True
>>> cs.labels['a']
0
```

**property certainty: float**

Return certainty level (1.0 if deterministic, 0.0 otherwise).

**coassoc: StatMatrix | None = None**

property is_deterministic: bool

Return True if no uncertainty info is present.

labels: Dict[Any, int]

meta: Dict[str, Any]

modularity: float | None = None

modularity_std: float | None = None

n_communities: int = 0

stability: Dict[Any, float] | None = None

class py3plex.uncertainty.types.ResamplingStrategy(*value*)

Bases: Enum

Strategy for estimating uncertainty via resampling.

SEED

Run with different random seeds (Monte Carlo).

**Type**
str

BOOTSTRAP

Bootstrap resampling of nodes or edges.

**Type**
str

JACKKNIFE

Leave-one-out jackknife resampling.

**Type**
str

PERTURBATION

Add noise/perturbations to network structure or parameters.

**Type**
str

STRATIFIED_PERTURBATION

Stratified perturbation that preserves key structural distributions (degree bins, layer densities, edge weight bins, layer-pair frequencies). Reduces estimator variance without increasing sample count.

**Type**
str

BOOTSTRAP = 'bootstrap'

JACKKNIFE = 'jackknife'

PERTURBATION = 'perturbation'

SEED = 'seed'

STRATIFIED_PERTURBATION = 'stratified_perturbation'

class py3plex.uncertainty.types.StatMatrix(*index: ~typing.List[~typing.Any], mean: ~numpy.ndarray, std: ~numpy.ndarray | None = None, quantiles: ~typing.Dict[float, ~numpy.ndarray] | None = None, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

A matrix of statistics with optional uncertainty.

Used for adjacency matrices, co-association matrices, distance matrices, etc.

**Parameters**

- **index** (*list[Any]*) – Row/column labels (assumed square matrix for simplicity).
- **mean** (*np.ndarray*) – The mean matrix, shape (n, n).
- **std** (*np.ndarray or None*) – The standard deviation matrix, shape (n, n), or None.
- **quantiles** (*dict[float, np.ndarray] or None*) – Quantile matrices, e.g., {0.025: (n, n), 0.975: (n, n)}.
- **meta** (*dict[str, Any]*) – Optional metadata.

### Examples

```
>>> import numpy as np
>>> m = StatMatrix(
...     index=['a', 'b', 'c'],
...     mean=np.array([[0, 1, 0], [1, 0, 1], [0, 1, 0]], dtype=float)
... )
>>> m.is_deterministic
True
>>> np.array(m).shape
(3, 3)
```

**property certainty: float**

Return certainty level (1.0 if deterministic, 0.0 otherwise).

**index: List[Any]**

**property is_deterministic: bool**

Return True if this is a deterministic result.

**mean: ndarray**

**meta: Dict[str, Any]**

**quantiles: Dict[float, ndarray] | None = None**

**std: ndarray | None = None**

class py3plex.uncertainty.types.StatSeries(*index: ~typing.List[~typing.Any], mean: ~numpy.ndarray, std: ~numpy.ndarray | None = None, quantiles: ~typing.Dict[float, ~numpy.ndarray] | None = None, meta: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

A series of statistics with optional uncertainty information.

This is the canonical result type for statistics that return a value per node, time point, or other index.

In deterministic mode (uncertainty=False): - mean contains the single-run values - std = None - quantiles = None - certainty = 1.0

In uncertain mode (uncertainty=True): - mean contains the average across runs - std contains the standard deviation - quantiles contains percentile arrays (e.g., {0.025: arr, 0.975: arr}) - certainty < 1.0

**Parameters**

- **index** (*list[Any]*) – The index labels (e.g., node IDs, time points).
- **mean** (*np.ndarray*) – The mean values, shape (n,).
- **std** (*np.ndarray or None*) – The standard deviations, shape (n,), or None if deterministic.
- **quantiles** (*dict[float, np.ndarray] or None*) – Quantile arrays, e.g., {0.025: (n,), 0.975: (n,)}, or None.
- **meta** (*dict[str, Any]*) – Optional metadata (e.g., algorithm parameters, run info).

### Examples

```
>>> import numpy as np
>>> # Deterministic
>>> s = StatSeries(
...     index=['a', 'b', 'c'],
...     mean=np.array([1.0, 2.0, 3.0])
... )
>>> s.is_deterministic
True
>>> s.certainty
1.0
>>> np.array(s)
array([1., 2., 3.])
```

```
>>> # With uncertainty
>>> s_unc = StatSeries(
...     index=['a', 'b', 'c'],
...     mean=np.array([1.0, 2.0, 3.0]),
...     std=np.array([0.1, 0.2, 0.15]),
...     quantiles={0.025: np.array([0.8, 1.6, 2.7]),
...                0.975: np.array([1.2, 2.4, 3.3])}
```

```
... )
>>> s_unc.is_deterministic
False
>>> s_unc.certainty
0.0
```

**property certainty: float**

Return certainty level.

Returns 1.0 if deterministic, 0.0 otherwise. In the future, this could return a richer metric.

**index: List[Any]**

**property is_deterministic: bool**

Return True if this is a deterministic result (no uncertainty).

**mean: ndarray**

**meta: Dict[str, Any]**

**quantiles: Dict[float, ndarray] | None = None**

**std: ndarray | None = None**

**to_dict()** → Dict[Any, Dict[str, Any]]

Convert to dictionary mapping index -> stats dict.

**Returns**

Dictionary with keys from index, values are dicts with 'mean', optionally 'std' and 'quantiles'.

**Return type**

dict

**class** py3plex.uncertainty.types.**UncertaintyConfig**(*mode: UncertaintyMode = UncertaintyMode.OFF*, *default_n_runs: int = 50*, *default_resampling: ResamplingStrategy = ResamplingStrategy.SEED*)

Bases: `object`

Configuration for uncertainty estimation.

**mode**

The current uncertainty mode.

**Type**

UncertaintyMode

**default_n_runs**

Default number of runs for uncertainty estimation.

**Type**

int

**default_resampling**

Default resampling strategy.

**Type**
ResamplingStrategy

**default_n_runs: int = 50**

**default_resampling: ResamplingStrategy = 'seed'**

**mode: UncertaintyMode = 'off'**

class py3plex.uncertainty.types.**UncertaintyMode**(*value*)

Bases: Enum

Global mode for uncertainty computation.

**OFF**

Always deterministic, std=None.

**Type**
str

**ON**

Try to compute uncertainty when supported.

**Type**
str

**AUTO**

Only do it if explicitly requested by a function.

**Type**
str

**AUTO = 'auto'**

**OFF = 'off'**

**ON = 'on'**

Context management for global uncertainty settings.

This module provides context variables and context managers for controlling uncertainty estimation globally across a pipeline or workflow.

py3plex.uncertainty.context.**get_uncertainty_config**() → UncertaintyConfig

Get the current uncertainty configuration.

**Returns**
The current configuration from the context.

**Return type**
UncertaintyConfig

### Examples

```
>>> from py3plex.uncertainty import get_uncertainty_config
>>> cfg = get_uncertainty_config()
>>> cfg.mode
<UncertaintyMode.OFF: 'off'>
>>> cfg.default_n_runs
50
```

py3plex.uncertainty.context.**set_uncertainty_config**(*config: UncertaintyConfig*) → Token

Set the uncertainty configuration.

**Parameters**
**config** (*UncertaintyConfig*) – The new configuration to set.

**Returns**
A token that can be used to reset the configuration.

**Return type**
Token

### Examples

```
>>> from py3plex.uncertainty import set_uncertainty_config, UncertaintyConfig
>>> from py3plex.uncertainty import UncertaintyMode
>>> cfg = UncertaintyConfig(mode=UncertaintyMode.ON, default_n_runs=100)
>>> token = set_uncertainty_config(cfg)
>>> # ... do work ...
>>> _uncertainty_ctx.reset(token)  # restore previous config
```

py3plex.uncertainty.context.**uncertainty_disabled**()

Context manager to disable uncertainty estimation.

Within this context, all functions will compute deterministic results (unless explicitly requested with uncertainty=True).

**Yields**
*None*

### Examples

```
>>> from py3plex.uncertainty import uncertainty_disabled, uncertainty_enabled
>>>
>>> with uncertainty_enabled():
...     # Nested context to temporarily disable
...     with uncertainty_disabled():
...         result = multilayer_pagerank(network)
>>> result.is_deterministic
True
```

py3plex.uncertainty.context.**uncertainty_enabled**(**, n_runs: int | None = None, resampling: ResamplingStrategy | None = None*)

Context manager to enable uncertainty estimation.

Within this context, all supported functions will compute uncertainty by default (unless explicitly disabled with uncertainty=False).

**Parameters**

- **n_runs** (*int , optional*) – Number of runs for uncertainty estimation. If None, uses the default from the current config.

- **resampling** (*ResamplingStrategy , optional*) – Resampling strategy to use. If None, uses the default from the current config.

**Yields**

*None*

### Examples

```
>>> from py3plex.uncertainty import uncertainty_enabled
>>> from py3plex.algorithms.centrality_toolkit import multilayer_pagerank
>>>
>>> # Without uncertainty
>>> result = multilayer_pagerank(network)
>>> result.is_deterministic
True
>>>
>>> # With uncertainty
>>> with uncertainty_enabled(n_runs=100):
...     result = multilayer_pagerank(network)
>>> result.is_deterministic
False
>>> result.std is not None
True
```

### Notes

This uses contextvars, so it's thread-safe and async-safe. Each context gets its own configuration.

Uncertainty estimation helpers.

This module provides the main helper function for estimating uncertainty in network statistics via resampling or perturbation strategies.

py3plex.uncertainty.estimation.**estimate_uncertainty**(*network: multi_layer_network*, *metric_fn: Callable[[multi_layer_network], Dict[Any, float] | float | ndarray]*, ***, *n_runs: int | None = None*, *resampling: ResamplingStrategy | None = None*, *random_seed: int | None = None*, *perturbation_params: Dict[str, Any] | None = None*) → StatSeries | float

Estimate uncertainty for a network statistic.

This is the main entry point for adding uncertainty to any statistic. It runs the metric function multiple times with different random seeds or network perturbations, then computes mean, std, and quantiles.

**Parameters**

- **network** (*multi_layer_network*) – The network to analyze.
- **metric_fn** (*callable*) – Function that computes a statistic. Must

accept a network and return: - dict[node, float] for per-node statistics - float for scalar statistics - np.ndarray for array statistics

- **n_runs** (*int , optional*) – Number of runs for uncertainty estimation. If None, uses the default from the current uncertainty config.
- **resampling** (*ResamplingStrategy , optional*) – Strategy for resampling. If None, uses default from config.
- **random_seed** (*int , optional*) – Random seed for reproducibility.
- **perturbation_params** (*dict , optional*) – Parameters for perturbation strategies. For example: {"edge_drop_p": 0.05, "node_drop_p": 0.02}

**Returns**

If metric_fn returns a dict: StatSeries with uncertainty info If metric_fn returns a scalar: float (mean value) The returned object has mean, std, and quantiles populated.

**Return type**

StatSeries or float

### Examples

```
>>> from py3plex.uncertainty import estimate_uncertainty, ResamplingStrategy
>>> from py3plex.core import multinet
>>>
>>> # Create a simple network
>>> net = multinet.multi_layer_network(directed=False)
>>> net.add_edges([["a", "L0", "b", "L0", 1.0]], input_type="list")
>>>
>>> # Define a metric function
>>> def my_metric(network):
...     # Return per-node degree
...     degrees = {}
...     for node in network.get_nodes():
...         degrees[node] = network.core_network.degree(node)
...     return degrees
>>>
>>> # Estimate uncertainty
>>> result = estimate_uncertainty(
...     net,
...     my_metric,
...     n_runs=50,
...     resampling=ResamplingStrategy.PERTURBATION,
...     perturbation_params={"edge_drop_p": 0.1}
... )
>>> result.mean  # Mean degree values
>>> result.std   # Std deviation of degrees
>>> result.quantiles  # Confidence intervals
```

#### Notes

- For SEED strategy: runs the metric with different random seeds
- For PERTURBATION strategy: applies edge/node drops then recomputes
- For BOOTSTRAP: resamples nodes/edges with replacement (not yet implemented)
- For JACKKNIFE: leave-one-out resampling (not yet implemented)

### Algorithms

Algorithm implementations grouped by task.

### Community Detection

py3plex.algorithms.community_detection.community_wrapper.**NoRC_communities**(*network*, *verbose=True*, *clustering_scheme='kmean*
*output='mapping'*, *prob_threshold=0.00*
*parallel_step=8*, *community_range=None*, *fine_range=3*)

py3plex.algorithms.community_detection.community_wrapper.**infomap_communities**(*graph: Graph*, *binary: str = './infomap'*, *edge_list_file: str = './tmp/tmpedgelis*
*multiplex: bool = False*, *verbose: bool = False*, *overlapping: bool = False*, *iterations: int = 200*, *output: str = 'mapping'*, *seed: int | None = None*) → Dict[Any, int] | Dict[Any, List[int]]

Detect communities using the Infomap algorithm.

**Parameters**

- **graph** – Input graph (NetworkX graph or multi_layer_network)
- **binary** – Path to Infomap binary (default: "./infomap")
- **edgelist_file** – Temporary file for edgelist (default: "./tmp/tmpedgelist.txt")
- **multiplex** – Whether to use multiplex mode (default: False)
- **verbose** – Whether to show verbose output (default: False)
- **overlapping** – Whether to detect overlapping communities (default: False)
- **iterations** – Number of iterations (default: 200)
- **output** – Output format - "mapping" or "partition" (default: "mapping")
- **seed** – Random seed for reproducibility (default: None) Note: Requires Infomap binary that supports –seed parameter

**Returns**

Dict mapping nodes to community IDs (if output="mapping") or Dict mapping community IDs to lists of nodes (if output="partition")

**Raises**

- **FileNotFoundError** – If Infomap binary is not found
- **PermissionError** – If Infomap binary is not executable

### Examples

```
>>> # Using with seed for reproducibility
>>> partition = infomap_communities(graph, seed=42)
>>>
>>> # Get partition format instead of mapping
>>> communities = infomap_communities(graph, output="partition")
```

py3plex.algorithms.community_detection.community_wrapper.**louvain_communities**(*network*, *output='mapping'*)

py3plex.algorithms.community_detection.community_wrapper.**parse_infomap**(*outfile*)

py3plex.algorithms.community_detection.community_wrapper.**run_infomap**(*infile: str*, *multiplex: bool = True*, *overlapping: bool = False*, *binary: str = './infomap'*, *verbose: bool = True*, *iterations: int = 1000*, *seed: int | None = None*) → None

Multilayer Modularity Maximization (Mucha et al., 2010)

This module implements multilayer modularity quality function and optimization algorithms for community detection in multilayer/multiplex networks.

### References

Mucha et al., "Community Structure in Time-Dependent, Multiscale, and Multiplex Networks", Science 328:876-878 (2010)

py3plex.algorithms.community_detection.multilayer_modularity.**build_supra_modularity_matrix**(

Build the supra-modularity matrix B for multilayer network.

The supra-modularity matrix is: B_{i,j} = (A^[]_ij - ^[] k_i^ k_j^ / 2m_) _ + _ij _

This matrix can be used for spectral community detection methods.

**Parameters**

- **network** – py3plex multi_layer_network object
- **gamma** – Resolution parameter(s)
- **omega** – Inter-layer coupling strength
- **weight** – Edge weight attribute

**Returns**

Tuple of (modularity_matrix, node_layer_list) - modularity_matrix: Supra-modularity matrix B (NL × NL) - node_layer_list: List of (node, layer) tuples corresponding to matrix indices

py3plex.algorithms.community_detection.multilayer_modularity.**louvain_multilayer**(*network: Any*, *gamma: float | Dict[Any, float] = 1.0*, *omega: float | ndarray = 1.0*, *weight: str = 'weight'*, *max_iter: int = 100*, *random_state: int | None = None*) → Dict[Tuple[Any, Any], int]

Generalized Louvain algorithm for multilayer networks.

This implements the multilayer Louvain method as described in Mucha et al. (2010), which greedily maximizes the multilayer modularity quality function.

**Complexity:**

- **Time: O(n × L × d × k) per iteration, where:**
  - n = number of nodes per layer
  - L = number of layers
  - d = average degree
  - k = number of communities
- Typical: O(n × L) iterations until convergence
- Worst case: $O((n \times L)^2)$ for dense networks
- Space: $O((n \times L)^2)$ for supra-adjacency matrix (use sparse for large networks)

**Parameters**

- **network** – py3plex multi_layer_network object
- **gamma** – Resolution parameter(s)
- **omega** – Inter-layer coupling strength
- **weight** – Edge weight attribute
- **max_iter** – Maximum number of iterations
- **random_state** – Random seed for reproducibility

**Returns**

Dictionary mapping (node, layer) tuples to community IDs

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.algorithms.community_detection.multilayer_modularity import louvain_mu
>>>
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
...     ['A', 'L2', 'C', 'L2', 1]
... ], input_type='list')
>>>
>>> communities = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
>>> print(communities)
```

**Note**

For reproducible results, always set random_state parameter.

py3plex.algorithms.community_detection.multilayer_modularity.**multilayer_modularity**(*network: Any*, *communities: Dict[Tuple, Any], int]*, *gamma: float | Dict[Any, float] = 1.0*, *omega: float | ndarray = 1.0*, *weight: str = 'weight'*) → float

Calculate multilayer modularity quality function (Mucha et al., 2010).

The multilayer modularity is defined as: Q = (1/2) _{ij} [(A^[]_ij - ^[]P^[]_ij)_ + _ij _] (g_i, g_j)

where: - A^[]_ij is the adjacency matrix of layer - P^[]_ij is the null model (e.g., Newman-Girvan: k_i^ k_j^ / 2m_) - ^[] is the resolution parameter for layer - _ is the inter-layer coupling strength - _ = 1 if =, else 0 (Kronecker delta) - _ij = 1 if i=j, else 0 - (g_i, g_j) = 1 if node i in layer and node j in layer are in same community - is the total edge weight in the supra-network

**Parameters**

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) tuples to community IDs
- **gamma** – Resolution parameter(s). Can be: - Single float: same resolution for all layers - Dict[layer, float]: layer-specific resolution parameters
- **omega** – Inter-layer coupling strength. Can be: - Single float: uniform coupling between all layer pairs - np.ndarray: layer-pair specific coupling matrix (L×L)
- **weight** – Edge weight attribute (default: “weight”)

**Returns**

Modularity value Q [-1, 1]

### Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.algorithms.community_detection.multilayer_modularity import multilayer
>>>
>>> # Create a simple multilayer network
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
...     ['A', 'L2', 'C', 'L2', 1]
... ], input_type='list')
>>>
>>> # Assign communities
>>> communities = {
...     ('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1,
...     ('A', 'L2'): 0, ('C', 'L2'): 0
... }
>>>
>>> # Calculate modularity
>>> Q = multilayer_modularity(network, communities, gamma=1.0, omega=1.0)
>>> print(f"Modularity: {Q:.3f}")
```

This module implements community detection.

**class** py3plex.algorithms.community_detection.community_louvain.**Status**

Bases: `object`

To handle several data in one struct.

Could be replaced by named tuple, but don't want to depend on python 2.6

**copy**()

Perform a deep copy of status

**degrees: dict = {}**

**gdegrees: dict = {}**

**init**(*graph*, *weight*, *part=None*)

Initialize the status of a graph with every node in one community

**internals: dict = {}**

**node2com: dict = {}**

**total_weight = 0**

py3plex.algorithms.community_detection.community_louvain.**best_partition**(*graph: Graph*, *partition: Dict | None = None*, *weight: str = 'weight'*, *resolution: float = 1.0*, *randomize: bool = False*) → Dict

Compute the partition of the graph nodes which maximises the modularity (or try..) using the Louvain heuristices

This is the partition of highest modularity, i.e. the highest partition of the dendrogram generated by the Louvain algorithm.

**Parameters**

- **graph** (*networkx.Graph*) – the networkx graph which is decomposed
- **partition** (*dict, optional*) – the algorithm will start using this partition of the nodes. It's a dictionary where keys are their nodes and values the communities
- **weight** (*str, optional*) – the key in graph to use as weight. Default to 'weight'
- **resolution** (*double, optional*) – Will change the size of the communities, default to 1. represents the time described in "Laplacian Dynamics and Multiscale Modular Structure in Networks", R. Lambiotte, J.-C. Delvenne, M. Barahona
- **randomize** (*boolean, optional*) – Will randomize the node evaluation order and the community evaluation order to get different partitions at each call

**Returns**

**partition** – The partition, with communities numbered from 0 to number of communities

**Return type**

dictionnary

**Raises**

**NetworkXError** – If the graph is not Eulerian.

**See also**

generate_dendrogram

### Notes

Uses Louvain algorithm

### References

[1] Blondel, V.D. et al. Fast unfolding of communities in large networks. J. Stat. Mech 10008, 1-12(2008).

### Examples

```
>>>  #Basic usage
>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> part = best_partition(G)
```

```
>>> #other example to display a graph with its community :
>>> #better with karate_graph() as defined in networkx examples
>>> #erdos renyi don't have true community structure
>>> G = nx.erdos_renyi_graph(30, 0.05)
>>> #first compute the best partition
>>> partition = community.best_partition(G)
>>>  #drawing
>>> size = float(len(set(partition.values())))
>>> pos = nx.spring_layout(G)
>>> count = 0.
>>> for com in set(partition.values()) :
>>>     count += 1.
>>>     list_nodes = [nodes for nodes in partition.keys()
>>>                                 if partition[nodes] == com]
>>>     nx.draw_networkx_nodes(G, pos, list_nodes, node_size = 20,
                                node_color = str(count / size))
>>> nx.draw_networkx_edges(G, pos, alpha=0.5)
>>> plt.show()
```

py3plex.algorithms.community_detection.community_louvain.**generate_dendrogram**(*graph: Graph*, *part_init: Dict | None = None*, *weight: str = 'weight'*, *resolution: float = 1.0*, *randomize: bool = False*) → List[Dict]

Find communities in the graph and return the associated dendrogram

A dendrogram is a tree and each level is a partition of the graph nodes. Level 0 is the first partition, which contains the smallest communities, and the best is len(dendrogram) - 1. The higher the level is, the bigger are the communities

**Parameters**

- **graph** (*networkx.Graph*) – the networkx graph which will be decomposed
- **part_init** (*dict, optional*) – the algorithm will start using this partition of the nodes. It's a dictionary where keys are their nodes and values the communities
- **weight** (*str, optional*) – the key in graph to use as weight. Default to 'weight'
- **resolution** (*double, optional*) – Will change the size of the communities, default to 1. represents the time described in "Laplacian Dynamics and Multiscale Modular Structure in Networks", R. Lambiotte, J.-C. Delvenne, M. Barahona

**Returns**

**dendrogram** – a list of partitions, ie dictionnaries where keys of the i+1 are the values of the i. and where keys of the first are the nodes of graph

**Return type**

list of dictionaries

**Raises**

**TypeError** – If the graph is not a networkx.Graph

**See also**

best_partition

### Notes

Uses Louvain algorithm

### References

[1] Blondel, V.D. et al. Fast unfolding of communities in large networks. J. Stat. Mech 10008, 1-12(2008).

### Examples

```
>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> dendo = generate_dendrogram(G)
>>> for level in range(len(dendo) - 1) :
>>>     print("partition at level", level,
>>>           "is", partition_at_level(dendo, level))
:param weight:
:type weight:
```

py3plex.algorithms.community_detection.community_louvain.**induced_graph**(*partition: Dict*, *graph: Graph*, *weight: str = 'weight'*) → Graph

Produce the graph where nodes are the communities

there is a link of weight w between communities if the sum of the weights of the links between their elements is w

**Parameters**

- **partition** (*dict*) – a dictionary where keys are graph nodes and values the part the node belongs to
- **graph** (*networkx.Graph*) – the initial graph
- **weight** (*str, optional*) – the key in graph to use as weight. Default to 'weight'

**Returns**

**g** – a networkx graph where nodes are the parts

**Return type**

networkx.Graph

### Examples

```
>>> n = 5
>>> g = nx.complete_graph(2*n)
>>> part = dict([])
>>> for node in g.nodes() :
>>>     part[node] = node % 2
>>> ind = induced_graph(part, g)
>>> goal = nx.Graph()
>>> goal.add_weighted_edges_from([(0,1,n*n),(0,0,n*(n-1)/2), (1, 1, n*(n-1)/2)])  # NOQ
>>> nx.is_isomorphic(int, goal)
True
```

py3plex.algorithms.community_detection.community_louvain.**load_binary**(*data*)

Load binary graph as used by the cpp implementation of this algorithm

py3plex.algorithms.community_detection.community_louvain.**modularity**(*partition: Dict*, *graph: Graph*, *weight: str = 'weight'*) → float

Compute the modularity of a partition of a graph

**Parameters**

- **partition** (*dict*) – the partition of the nodes, i.e a dictionary where keys are their nodes and values the communities
- **graph** (*networkx.Graph*) – the networkx graph which is decomposed
- **weight** (*str, optional*) – the key in graph to use as weight. Default to 'weight'

**Returns**

**modularity** – The modularity

**Return type**

float

**Raises**

- **KeyError** – If the partition is not a partition of all graph nodes
- **ValueError** – If the graph has no link
- **TypeError** – If graph is not a networkx.Graph

### References

[1] Newman, M.E.J. & Girvan, M. Finding and evaluating community structure in networks. Physical Review E 69, 26113(2004).

### Examples

```
>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> part = best_partition(G)
>>> modularity(part, G)
```

py3plex.algorithms.community_detection.community_louvain.**partition_at_level**(*dendrogram: List[Dict]*, *level: int*) → Dict

Return the partition of the nodes at the given level

A dendrogram is a tree and each level is a partition of the graph nodes. Level 0 is the first partition, which contains the smallest communities, and the best is len(dendrogram) - 1. The higher the level is, the bigger are the communities

**Parameters**

- **dendrogram** (*list of dict*) – a list of partitions, ie dictionnaries where keys of the i+1 are the values of the i.
- **level** (*int*) – the level which belongs to [0..len(dendrogram)-1]

**Returns**

**partition** – A dictionary where keys are the nodes and the values are the set it belongs to

**Return type**

dictionnary

**Raises**

**KeyError** – If the dendrogram is not well formed or the level is too high

**See also**

best_partition, generate_dendrogram

### Examples

```
>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> dendrogram = generate_dendrogram(G)
>>> for level in range(len(dendrogram) - 1) :
>>>     print("partition at level", level, "is", partition_at_level(dendrogram, level))
```

Community quality measures and metrics.

This module provides various measures for assessing the quality of community partitions in networks, including modularity, size distribution, and other statistical metrics.

py3plex.algorithms.community_detection.community_measures.**modularity**(*G: Graph*, *communities: Dict[Any, List[Any]]*, *weight: str = 'weight'*) → float

Calculate modularity of a graph partition.

**Parameters**

- **G** – NetworkX graph
- **communities** – Dictionary mapping community IDs to node lists
- **weight** – Edge weight attribute (default: "weight")

**Returns**

Modularity value

py3plex.algorithms.community_detection.community_measures.**number_of_communities**(*network_part* *Dict[Any, List[Any]]*) → int

Count number of communities in a partition.

**Parameters**

**network_partition** – Dictionary mapping community IDs to node lists

**Returns**

Number of communities

py3plex.algorithms.community_detection.community_measures.**size_distribution**(*network_partition* *Dict[Any, List[Any]]*) → ndarray

Calculate size distribution of communities.

**Parameters**

**network_partition** – Dictionary mapping community IDs to node lists

**Returns**

Array of community sizes

Multilayer Synthetic Graph Generation for Community Detection Benchmarks

This module implements synthetic multilayer/multiplex graph generators with ground-truth community structure for benchmarking community detection algorithms.

Includes: - Multilayer LFR (Lancichinetti-Fortunato-Radicchi) benchmark - Coupled/Interdependent Erdős-Rényi models - Support for overlapping communities across layers - Support for partial node presence across layers

### References

Lancichinetti et al., "Benchmark graphs for testing community detection algorithms", Phys. Rev. E 78, 046110 (2008)

Granell et al., "Benchmark model to assess community structure in evolving networks", Phys. Rev. E 92, 012805 (2015)

`py3plex.algorithms.community_detection.multilayer_benchmark.generate_coupled_er_multilayer(`

Generate coupled/interdependent Erdős-Rényi multilayer networks.

Creates random Erdős-Rényi graphs in each layer and couples nodes across layers with specified coupling strength and probability.

**Parameters**

- **n** – Number of nodes
- **layers** – List of layer names
- **p** – Edge probability per layer. Can be float (same for all layers) or list of floats per layer.
- **omega** – Inter-layer coupling strength (weight of identity links)

- **coupling_probability** – Probability that a node has inter-layer coupling. Range: [0, 1] - 1.0 = all nodes coupled (full multiplex) - <1.0 = partial coupling (interdependent networks)
- **directed** – Whether to generate directed networks
- **seed** – Random seed for reproducibility

**Returns**

py3plex multi_layer_network object

### Examples

```
>>> from py3plex.algorithms.community_detection.multilayer_benchmark import generate_co
>>>
>>> # Full multiplex ER network
>>> network = generate_coupled_er_multilayer(
...     n=100,
...     layers=['L1', 'L2', 'L3'],
...     p=0.1,
...     omega=1.0,
...     coupling_probability=1.0
... )
>>>
>>> # Partially coupled (interdependent)
>>> network = generate_coupled_er_multilayer(
...     n=100,
...     layers=['L1', 'L2'],
...     p=0.1,
...     omega=0.5,
...     coupling_probability=0.5  # Only 50% nodes coupled
... )
```

py3plex.algorithms.community_detection.multilayer_benchmark.**generate_multilayer_lfr**(*n: int*, *layers: List[str]*, *tau1: float = 2.0*, *tau2: float = 1.5*, *mu: float | List[float = 0.1*, *avg_degr float | List[float = 10.0*, *min_com int = 20*, *max_com int | None = None*, *community_per float = 1.0*, *node_ove float = 1.0*, *overlapping_nod int = 0*, *overlapping_me int = 2*,

Generate multilayer LFR benchmark networks with controllable community structure.

This extends the LFR benchmark to multilayer networks, allowing control over: - Community persistence across layers (how many nodes keep their community) - Node overlap across layers (which nodes appear in which layers) - Overlapping communities (nodes belonging to multiple communities)

**Parameters**

- **n** – Number of nodes
- **layers** – List of layer names
- **tau1** – Power-law exponent for degree distribution (typically 2-3)
- **tau2** – Power-law exponent for community size distribution (typically 1-2)
- **mu** – Mixing parameter (fraction of edges outside community). Can be float (same for all layers) or list of floats per layer. Range: [0, 1], where 0 = perfect communities, 1 = random
- **avg_degree** – Average degree per layer. Can be float (same for all layers) or list of floats per layer.
- **min_community** – Minimum community size
- **max_community** – Maximum community size (default: n/2)
- **community_persistence** – Probability that a node keeps its community from one layer to the next. Range: [0, 1] - 1.0 = identical communities across all layers - 0.0 = completely independent communities per layer
- **node_overlap** – Fraction of nodes present in all layers. Range: [0, 1] - 1.0 = all nodes in all layers (full multiplex) - <1.0 = some nodes absent from some layers
- **overlapping_nodes** – Number of nodes that belong to multiple communities within each layer
- **overlapping_membership** – Number of communities each overlapping node belongs to
- **directed** – Whether to generate directed networks
- **seed** – Random seed for reproducibility

**Returns**

Tuple of (network, ground_truth_communities) - network: py3plex multi_layer_network object - ground_truth_communities: Dict mapping (node, layer) to Set of community IDs

### Examples

```
>>> from py3plex.algorithms.community_detection.multilayer_benchmark import generate_mu
>>>
>>> # Generate with identical communities across layers
>>> network, communities = generate_multilayer_lfr(
...     n=100,
...     layers=['L1', 'L2', 'L3'],
```

```
...     mu=0.1,
...     community_persistence=1.0
... )
>>>
>>> # Generate with evolving communities
>>> network, communities = generate_multilayer_lfr(
...     n=100,
...     layers=['T0', 'T1', 'T2'],
...     mu=0.1,
...     community_persistence=0.7  # 70% nodes keep community
... )
```

py3plex.algorithms.community_detection.multilayer_benchmark.**generate_sbm_multilayer**(*n: int*, *layers: List[str]*, *communities: List[Set[*
*p_in: float | List[float* = *0.3*, *p_out: float | List[float* = *0.05*, *community_per float* = *1.0*, *directed: bool* = *False*, *seed: int | None* = *None*) → Tuple[Any, Dict[Tup str], int]]

Generate multilayer stochastic block model (SBM) networks.

Creates networks where nodes are divided into communities with different intra- and inter-community connection probabilities.

**Parameters**

- **n** – Number of nodes

- **layers** – List of layer names
- **communities** – List of node sets defining initial communities
- **p_in** – Intra-community edge probability per layer
- **p_out** – Inter-community edge probability per layer
- **community_persistence** – Probability nodes keep their community across layers
- **directed** – Whether to generate directed networks
- **seed** – Random seed for reproducibility

**Returns**

Tuple of (network, ground_truth_communities) - network: py3plex multi_layer_network object - ground_truth_communities: Dict mapping (node, layer) to community ID

### Examples

```
>>> from py3plex.algorithms.community_detection.multilayer_benchmark import generate_sb
>>>
>>> # Define initial communities
>>> communities = [
...     {0, 1, 2, 3, 4},  # Community 0
...     {5, 6, 7, 8, 9}   # Community 1
... ]
>>>
>>> network, ground_truth = generate_sbm_multilayer(
...     n=10,
...     layers=['L1', 'L2'],
...     communities=communities,
...     p_in=0.7,
...     p_out=0.1,
...     community_persistence=0.8
... )
```

## Statistics

Network statistics, enrichment tests, and topology utilities.

Multilayer Network Statistics

This module implements various statistics for multilayer and multiplex networks, following standard definitions from multilayer network analysis literature.

## References


- Kivelä et al. (2014), "Multilayer networks", J. Complex Networks 2(3), 203-271
- De Domenico et al. (2013), "Mathematical formulation of multilayer networks", PRX 3, 041022
- Mucha et al. (2010), "Community Structure in Time-Dependent, Multiscale, and Multiplex Networks", Science 328, 876-878


Authors: py3plex contributors Date: 2025

py3plex.algorithms.statistics.multilayer_statistics.**algebraic_connectivity**(*network: Any*) → float

Calculate algebraic connectivity ($_2$).

Formula: $_2()$

Second smallest eigenvalue of the supra-Laplacian (Fiedler value).

Indicates global connectivity and diffusion efficiency of the multilayer system.

**Properties:**

$_0 = 0$ always (associated with constant eigenvector) $_1 > 0$ if and only if the multilayer network is connected Larger $_1$ indicates better connectivity and faster diffusion/synchronization

**Parameters**

**network** – py3plex multi_layer_network object

**Returns**

Second smallest eigenvalue (Fiedler value)

### Examples

```
>>> alg_conn = algebraic_connectivity(network)
```

**Reference:**

Fiedler (1973), Sole-Ribalta et al. (2013)

py3plex.algorithms.statistics.multilayer_statistics.**community_participation_coefficient**(*netu Any com mu- ni- ties: Dict Any int], node Any* → float

Calculate participation coefficient for a node across community structure.

Measures how evenly a node's connections are distributed across different communities, across all layers. A node with connections to many communities has high participation.

Formula: P = 1 - $(k / k)^2$

where k is the number of connections node i has to community s, and k is the total degree of node i across all layers.

**Parameters**

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) to community ID

- **node** – Node identifier (not node-layer tuple)

**Returns**

Participation coefficient value between 0 and 1

### Examples

```
>>> communities = detect_communities(network)
>>> pc = community_participation_coefficient(network, communities, 'Alice')
>>> print(f"Participation: {pc:.3f}")
```

**Reference:**

Guimerà & Amaral (2005), "Functional cartography of complex metabolic networks"

py3plex.algorithms.statistics.multilayer_statistics.**community_participation_entropy**(*network: Any*, *communities: Dict[Tuple[Any, Any], int]*, *node: Any*) → float

Calculate participation entropy for a node across community structure.

Shannon entropy-based measure of how evenly a node distributes its connections across different communities. Higher entropy indicates more diverse community participation.

Formula: H = - (k / k) log(k / k)

where k is connections to community s, k is total degree.

**Parameters**

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) to community ID
- **node** – Node identifier (not node-layer tuple)

**Returns**

Entropy value (higher = more diverse participation)

### Examples

```
>>> entropy = community_participation_entropy(network, communities, 'Alice')
>>> print(f"Participation entropy: {entropy:.3f}")
```

**Reference:**

Based on Shannon entropy applied to community structure

py3plex.algorithms.statistics.multilayer_statistics.**compute_modularity_score**(*network: Any*, *communities: Dict[Tuple[Any, Any], int]*, *gamma: float = 1.0*, *omega: float = 1.0*) → float

Compute explicit multislice modularity score.

Direct computation of the modularity quality function for a given community partition, without running detection algorithms.

Formula: Q = (1/2) [(A - · kk/(2m)) + · ] (c, c)

**Parameters**

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) to community ID
- **gamma** – Resolution parameter (default: 1.0)
- **omega** – Inter-layer coupling strength (default: 1.0)

**Returns**

Modularity score Q (higher is better)

### Examples

```
>>> communities = {('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1}
>>> Q = compute_modularity_score(network, communities)
>>> print(f"Modularity: {Q:.3f}")
```

**Reference:**

Mucha et al. (2010), Science 328, 876-878

py3plex.algorithms.statistics.multilayer_statistics.**cross_layer_mutual_information**(*network: Any*, *layer_i: str*, *layer_j: str*, *bins: int = 10*) → float

Calculate cross-layer mutual information (I(L; L)).

Formula: I(L; L) = H(L) + H(L) - H(L, L)

Measures statistical dependence between degree distributions in two layers; quantifies how much knowing a node's degree in one layer tells us about its degree in another layer.

**Variables:**
H(L) = entropy of degree distribution in layer i H(L) = entropy of degree distribution in layer j H(L, L) = joint entropy of degree distributions

**Properties:**
I = 0 when layers are independent I > 0 indicates statistical dependence (higher = stronger) I(L; L) min(H(L), H(L))

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier
- **bins** – Number of bins for discretizing degree distributions

**Returns**
Mutual information value in bits

### Examples

```
>>> mi = cross_layer_mutual_information(network, 'L1', 'L2', bins=10)
>>> print(f"Mutual information: {mi:.3f} bits")
```

**Reference:**
Cover & Thomas (2006), "Elements of Information Theory" De Domenico et al. (2015), "Structural reducibility"

py3plex.algorithms.statistics.multilayer_statistics.**cross_layer_redundancy_entropy**(*network: Any*) → float

Calculate cross-layer redundancy entropy (H_redundancy).

Formula: H_r = - r $\log_2$(r), where r is the normalized edge overlap between layers i and j.

Measures diversity in structural redundancy across layer pairs; high entropy indicates varied overlap patterns, low entropy indicates uniform redundancy.

**Variables:**
r = edge_overlap(i,j) / edge_overlap(,) Normalized overlap proportion for each layer pair

**Parameters**
**network** – py3plex multi_layer_network object

**Returns**
Entropy value in bits

### Examples

```
>>> entropy = cross_layer_redundancy_entropy(network)
>>> print(f"Cross-layer redundancy entropy: {entropy:.3f}")
```

**Reference:**
Bianconi (2018), "Multilayer Networks: Structure and Function"

py3plex.algorithms.statistics.multilayer_statistics.**degree_vector**(*network: Any*, *node: Any*, *weighted: bool = False*) → Dict[str, float]

Calculate degree vector (k).

Formula: k = ($k^1$, $k^2$, ..., k)

Node degree in each layer; can be analyzed via mean, variance, or entropy to capture node versatility.

**Variables:**
k = degree of node i in layer For undirected: k = A

**Parameters**

- **network** – py3plex multi_layer_network object
- **node** – Node identifier
- **weighted** – If True, return strength instead of degree

**Returns**
Dictionary mapping layer to degree/strength

### Examples

```
>>> degrees = degree_vector(network, 'A')
>>> print(f"Degree in layer L1: {degrees['L1']}")
```

**Reference:**
Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

py3plex.algorithms.statistics.multilayer_statistics.**edge_overlap**(*network: Any*, *layer_i: str*, *layer_j: str*) → float

Calculate edge overlap ( ).

Formula:  =  |E  E| / |E  E|

Jaccard similarity of edge sets between two layers; measures structural redundancy.

**Variables:**
E = set of edges in layer  E = set of edges in layer  | · | = cardinality (number of elements)

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

**Returns**
Overlap coefficient between 0 and 1 (Jaccard similarity)

### Examples

```
>>> overlap = edge_overlap(network, 'L1', 'L2')
```

**Reference:**
Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

py3plex.algorithms.statistics.multilayer_statistics.**entropy_of_multiplexity**(*network: Any*) → float

Calculate entropy of multiplexity (H).

Formula: H = - p $\log_2$(p), where p = E /  E

Shannon entropy of layer contributions; measures layer diversity.

**Variables:**
p = proportion of edges in layer  E = number of edges in layer  $\log_2$ gives entropy in bits

**Properties:**
H = 0 when all edges are in one layer (minimum entropy/diversity) H = $\log_2$(L) when edges are uniformly distributed across L layers (maximum entropy)

**Parameters**
**network** – py3plex multi_layer_network object

**Returns**
Entropy value in bits

### Examples

```
>>> entropy = entropy_of_multiplexity(network)
```

**Reference:**
De Domenico et al. (2013), Shannon (1948)

py3plex.algorithms.statistics.multilayer_statistics.**inter_layer_assortativity**(*network: Any*, *layer_i: str*, *layer_j: str*) → float

Calculate inter-layer assortativity (r).

Formula: r^ = cov(k^, k^) / ( ) = corr(k^, k^)

Measures whether nodes with similar degrees tend to connect across different layers.

**Variables:**
k^ = degree vector in layer k^ = degree vector in layer , = standard deviations of degrees in layers and Equivalent to Pearson correlation of degree vectors

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

**Returns**
Assortativity coefficient

### Examples

```
>>> assort = inter_layer_assortativity(network, 'L1', 'L2')
```

**Reference:**
Newman (2002), Nicosia & Latora (2015)

py3plex.algorithms.statistics.multilayer_statistics.**inter_layer_coupling_strength**(*network: Any*, *layer_i: str*, *layer_j: str*) → float

Calculate inter-layer coupling strength (C^).

Formula: C^ = (1/N_) w^

Average weight of inter-layer connections between corresponding nodes in two layers. Quantifies cross-layer connectivity.

**Variables:**
N_ = number of nodes present in both layers and w^ = weight of inter-layer edge connecting node i in layer to node i in layer

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

**Returns**
Average coupling strength

### Examples

```
>>> coupling = inter_layer_coupling_strength(network, 'L1', 'L2')
```

**Reference:**
De Domenico et al. (2013), Physical Review X 3(4), 041022

**Contracts:**

- Precondition: network must not be None
- Precondition: layer_i and layer_j must be non-empty strings
- Postcondition: result is non-negative (weights are non-negative)
- Postcondition: result is not NaN

py3plex.algorithms.statistics.multilayer_statistics.**inter_layer_degree_correlation**(*network: Any*, *layer_i: str*, *layer_j: str*) → float

Calculate inter-layer degree correlation (r^).

Formula: r^ = (k - k)(k - k) / [((k - k)$^2$) ((k - k)$^2$)]

Pearson correlation of node degrees between two layers; reveals if highly connected nodes in one layer are also central in others.

**Variables:**
k = degree of node i in layer k = mean degree in layer Sum over nodes present in both layers

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

**Returns**
Pearson correlation coefficient between -1 and 1

### Examples

```
>>> corr = inter_layer_degree_correlation(network, 'L1', 'L2')
```

**Reference:**
Battiston et al. (2014), Nicosia & Latora (2015)

py3plex.algorithms.statistics.multilayer_statistics.**inter_layer_dependence_entropy**(*network: Any*, *layer_i: str*, *layer_j: str*) → float

Calculate inter-layer dependence entropy (H_dep).

Formula: H_dep = - p $\log_2$(p), where p is the proportion of inter-layer edges for each node n connecting layers i and j.

Measures heterogeneity in how nodes couple the two layers; high entropy indicates diverse coupling patterns, low entropy indicates uniform coupling.

**Variables:**
p = proportion of inter-layer edges incident to node n Total over all nodes connecting the two layers

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier

**Returns**
Entropy value in bits

### Examples

```
>>> entropy = inter_layer_dependence_entropy(network, 'L1', 'L2')
>>> print(f"Inter-layer dependence entropy: {entropy:.3f}")
```

**Reference:**
De Domenico et al. (2015), "Ranking in interconnected multilayer networks"

py3plex.algorithms.statistics.multilayer_statistics.**interdependence**(*network: Any*, *sample_size: int = 100*, *seed: int | None = None*) → float

Calculate interdependence ().

Formula:  = ⟨d⟩ / ⟨d⟩

Quantifies how much shortest-path communication depends on inter-layer connections.

**Variables:**
d = shortest path from node i to node j in the full multilayer network d = (1/L)  d is the average shortest path across individual layers ⟨ · ⟩ = average over sampled node pairs

**Interpretation:**
< 1: multilayer connectivity reduces path lengths (positive interdependence)  1: inter-layer connections provide little benefit  > 1: multilayer structure increases path lengths (rare)

**Parameters**

- **network** – py3plex multi_layer_network object
- **sample_size** – Number of node pairs to sample for estimation
- **seed** – Optional random seed for deterministic node-pair sampling

**Returns**
Interdependence ratio

### Examples

```
>>> interdep = interdependence(network, sample_size=50)
```

**Reference:**
Gomez et al. (2013), Buldyrev et al. (2010)

py3plex.algorithms.statistics.multilayer_statistics.**interlayer_degree_correlation_matrix**(*net*
*An*
→
Tu-
ple
list

Calculate inter-layer degree correlation matrix.

Computes Pearson correlation coefficients for node degrees between all pairs of layers, organized as a symmetric correlation matrix.

Formula: Matrix[,] = r^ = corr(k^, k^)

where k^ and k^ are degree vectors for layers  and  over common nodes.

**Properties:**

- Diagonal elements are 1.0 (self-correlation)
- Off-diagonal elements in [-1, 1]
- Symmetric matrix
- Positive values indicate positive degree correlation
- Negative values indicate negative degree correlation

**Parameters**
**network** – py3plex multi_layer_network object

**Returns**

- correlation_matrix: 2D numpy array of shape (num_layers, num_layers)
- layer_labels: List of layer names corresponding to matrix indices

**Return type**
Tuple of (correlation_matrix, layer_labels)

### Examples

```
>>> corr_matrix, layers = interlayer_degree_correlation_matrix(network)
>>> import matplotlib.pyplot as plt
>>> import seaborn as sns
>>> sns.heatmap(corr_matrix, annot=True, xticklabels=layers,
...             yticklabels=layers, cmap='coolwarm', center=0,
...             vmin=-1, vmax=1)
>>> plt.title('Inter-layer Degree Correlation Matrix')
>>> plt.show()
```

**Reference:**
Nicosia & Latora (2015), "Measuring and modeling correlations in multiplex networks" Battiston et al. (2014), "Structural measures for multiplex networks"

py3plex.algorithms.statistics.multilayer_statistics.**layer_connectivity_entropy**(*network: Any*, *layer: str*) → float

Calculate entropy of layer connectivity (H_connectivity).

Formula: H_c = - (k/k) $\log_2$(k/k)

Shannon entropy of degree distribution within a layer; measures heterogeneity of node connectivity patterns.

**Variables:**
k = degree of node i in the layer k = sum of all degrees (2 * edges for undirected)

**Properties:**
H_c = 0 when all nodes have the same degree (uniform distribution) H_c is maximized when degree distribution is highly uneven

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier

**Returns**
Entropy value in bits

### Examples

```
>>> from py3plex.core import multinet
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1]
... ], input_type='list')
>>> entropy = layer_connectivity_entropy(network, 'L1')
>>> print(f"Connectivity entropy: {entropy:.3f}")
```

**Reference:**
Solé-Ribalta et al. (2013), "Spectral properties of complex networks" Shannon (1948), "A Mathematical Theory of Communication"

py3plex.algorithms.statistics.multilayer_statistics.**layer_density**(*network: Any*, *layer: str*) → float

Calculate layer density ().

**Formula:** **= (2E) / (N(N - 1)) [undirected]**
= E / (N(N - 1)) [directed]

Measures the fraction of possible edges present in a specific layer, indicating how densely connected that layer is.

**Variables:**
E = number of edges in layer N = number of nodes in layer

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier

**Returns**
Density value between 0 and 1

### Examples

```
>>> from py3plex.core import multinet
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1]
... ], input_type='list')
>>> density = layer_density(network, 'L1')
>>> print(f"Layer L1 density: {density:.3f}")
```

**Reference:**
Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

**Contracts:**

- Precondition: network must not be None
- Precondition: layer must be a non-empty string

- Postcondition: result is in [0, 1] (fundamental property of density)
- Postcondition: result is not NaN

py3plex.algorithms.statistics.multilayer_statistics.**layer_influence_centrality**(*network: Any*, *layer: str*, *method: str = 'coupling'*, *sample_size: int = 100*) → float

Calculate layer influence centrality (I).

Formula (coupling): I = C^ / (L-1) Formula (flow): I = F^ / (L-1)

Quantifies how much a layer influences other layers through inter-layer connections (coupling method) or information flow (flow method).

**Variables:**

C^ = inter-layer coupling strength between layers and F^ = flow from layer to layer (random walk transition probability) L = total number of layers

**Properties:**

Higher values indicate layers that strongly influence others Useful for identifying critical layers in the multilayer structure

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier
- **method** – 'coupling' for structural influence, 'flow' for dynamic influence
- **sample_size** – Number of random walk steps for flow simulation

**Returns**

Influence centrality value

### Examples

```
>>> influence = layer_influence_centrality(network, 'L1', method='coupling')
>>> print(f"Layer L1 influence: {influence:.3f}")
```

**Reference:**

Cozzo et al. (2013), "Mathematical formulation of multilayer networks" De Domenico et al. (2014), "Identifying modular flows"

py3plex.algorithms.statistics.multilayer_statistics.**layer_redundancy_coefficient**(*network: Any*, *layer_i: str*, *layer_j: str*) → float

Calculate layer redundancy coefficient.

Measures the proportion of edges in one layer that are redundant (also present) in another layer. Values close to 1 indicate high redundancy, while values close to 0 indicate complementary layers.

Formula: R = |E  E| / |E|

where E and E are edge sets of layers  and .

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier

**Returns**

Redundancy coefficient between 0 and 1

### Examples

```
>>> redundancy = layer_redundancy_coefficient(network, 'social', 'work')
>>> print(f"Redundancy: {redundancy:.2%}")
```

**Reference:**

Nicosia & Latora (2015), "Measuring and modeling correlations in multiplex networks"

py3plex.algorithms.statistics.multilayer_statistics.**layer_similarity**(*network: Any*, *layer_i: str*, *layer_j: str*, *method: str = 'cosine'*) → float

Calculate layer similarity (S^).

Formula: S^ = ⟨A, A⟩ / (‖A‖ ‖A‖) =  AA / ((A)$^2$) ((A)$^2$)

Cosine or Jaccard similarity between adjacency matrices of two layers.

**Variables:**

A, A = adjacency matrices for layers  and  ⟨ · , · ⟩ = Frobenius inner product ‖ · ‖ = Frobenius norm

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()
- **method** – 'cosine' or 'jaccard'

**Returns**

Similarity value between 0 and 1

### Examples

```
>>> similarity = layer_similarity(network, 'L1', 'L2', method='cosine')
```

**Reference:**

De Domenico et al. (2013), Physical Review X 3(4), 041022

py3plex.algorithms.statistics.multilayer_statistics.**multilayer_betweenness_surface**(*network: Any*, *normalized: bool = True*, *weight: str | None = None*) → ndarray

Calculate multilayer betweenness surface (tensor representation).

Computes betweenness centrality for each node-layer pair and organizes the results as a 2D array (nodes × layers) that can be visualized as a heatmap or surface plot.

Formula: Surface[i,] = B

where B is the betweenness centrality of node i in layer .

**Parameters**

- **network** – py3plex multi_layer_network object
- **normalized** – Whether to normalize betweenness values
- **weight** – Edge weight attribute name (None for unweighted)

**Returns**

2D numpy array of shape (num_nodes, num_layers) containing betweenness values. Also returns tuple of (node_labels, layer_labels) for axis labeling.

### Examples

```
>>> surface, (nodes, layers) = multilayer_betweenness_surface(network)
>>> import matplotlib.pyplot as plt
>>> plt.imshow(surface, aspect='auto', cmap='viridis')
>>> plt.xlabel('Layers')
>>> plt.ylabel('Nodes')
>>> plt.xticks(range(len(layers)), layers)
>>> plt.yticks(range(len(nodes)), nodes)
>>> plt.colorbar(label='Betweenness Centrality')
>>> plt.title('Multilayer Betweenness Surface')
>>> plt.show()
```

**Reference:**

De Domenico et al. (2015), "Structural reducibility of multilayer networks"

py3plex.algorithms.statistics.multilayer_statistics.**multilayer_clustering_coefficient**(*networ Any, node: Any | None = None*) → float | Dict[A float]

Calculate multilayer clustering coefficient (C).

Formula: C = T / T

Extends transitivity to account for triangles that span multiple layers.

**Variables:**

T = number of closed triplets (triangles) involving node i across all layers T = maximum possible triplets = k(k - 1)/2 for undirected networks Average over all nodes: C = (1/N) C

**Parameters**

- **network** – py3plex multi_layer_network object
- **node** – If specified, compute for single node; otherwise compute for all

**Returns**

Clustering coefficient value or dict of values per node

### Examples

```
>>> clustering = multilayer_clustering_coefficient(network)
>>> node_clustering = multilayer_clustering_coefficient(network, node='A')
```

**Reference:**

Battiston et al. (2014), Section III.C

py3plex.algorithms.statistics.multilayer_statistics.**multilayer_modularity**(*network: Any*, *communities: Dict[Tuple[Any, Any], int]*, *gamma: float | Dict[Any, float] = 1.0*, *omega: float | ndarray = 1.0*, *weight: str = 'weight'*) → float

Calculate multilayer modularity (Q).

This is a wrapper for the existing multilayer_modularity implementation in py3plex.algorithms.community_detection.multilayer_modularity.

Formula: Q = (1/2) [(A - P) + ] (g, g)

Extension of Newman-Girvan modularity to multiplex networks (Mucha et al., 2010). Measures community quality across layers.

**Variables:**

= total edge weight in supra-network A = adjacency matrix element for layer P = kk/(2m) is the null model (configuration model) = resolution parameter for layer = inter-layer coupling strength = Kronecker delta (1 if =, 0 otherwise) = Kronecker delta (1 if i=j, 0 otherwise) (g, g) = 1 if node i in layer and node j in layer are in same community

**Parameters**

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) tuples to community IDs
- **gamma** – Resolution parameter(s)
- **omega** – Inter-layer coupling strength
- **weight** – Edge weight attribute

**Returns**

Modularity value Q

### Examples

```
>>> communities = {('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1}
>>> Q = multilayer_modularity(network, communities)
```

**Reference:**
Mucha et al. (2010), Science 328(5980), 876-878

py3plex.algorithms.statistics.multilayer_statistics.**multilayer_motif_frequency**(*network: Any*, *motif_size: int = 3*) → Dict[str, float]

Calculate multilayer motif frequency (f).

Formula: f = n / n

Frequency of recurring subgraph patterns across layers.

**Variables:**
n = count of motif type m n = total count of all motifs

Note: This is a simplified implementation counting basic patterns (intra-layer vs. inter-layer triangles). Complete multilayer motif enumeration includes many more configurations and is computationally expensive.

**Parameters**

- **network** – py3plex multi_layer_network object
- **motif_size** – Size of motifs to count (default: 3 for triangles)

**Returns**
Dictionary of motif type frequencies

### Examples

```
>>> motifs = multilayer_motif_frequency(network, motif_size=3)
```

**Reference:**
Battiston et al. (2014), Section IV

py3plex.algorithms.statistics.multilayer_statistics.**multiplex_betweenness_centrality**(*network: Any*, *normalized: bool = True*, *weight: str | None = None*) → Dict[Tuple[Any, Any], float]

Calculate multiplex betweenness centrality.

Computes betweenness centrality on the supra-graph, accounting for paths that traverse inter-layer couplings. This extends the standard betweenness definition to multiplex networks where paths can cross layers.

Formula: B = ((i) / )

where is the total number of shortest paths from s to t, and (i) is the number of those paths passing through node i in layer .

**Parameters**

- **network** – py3plex multi_layer_network object
- **normalized** – Whether to normalize by the number of node pairs
- **weight** – Edge weight attribute name (None for unweighted)

**Returns**

Dictionary mapping (node, layer) tuples to betweenness centrality values

### Examples

```
>>> betweenness = multiplex_betweenness_centrality(network)
>>> top_nodes = sorted(betweenness.items(), key=lambda x: x[1], reverse=True)[:5]
```

**Reference:**

De Domenico et al. (2015), "Structural reducibility of multilayer networks"

py3plex.algorithms.statistics.multilayer_statistics.**multiplex_closeness_centrality**(*network: Any*, *normalized: bool = True*, *weight: str | None = None*, *variant: str = 'standard'*) → Dict[Tuple[Any, Any], float]

Calculate multiplex closeness centrality.

Computes closeness centrality on the supra-graph, where shortest paths can traverse interlayer edges. This captures how quickly a node-layer can reach all other node-layers in the multiplex network.

Standard closeness formula: C = (N*L - 1) / d(i, j)

Harmonic closeness formula: HC = 1/d(i, j)

where d(i, j) is the shortest path distance from node i in layer to node j in layer , and N*L is the total number of node-layer pairs.

**Parameters**

- **network** – py3plex multi_layer_network object
- **normalized** – Whether to normalize by network size
- **weight** – Edge weight attribute name (None for unweighted)
- **variant** – Closeness variant to use. Options:
  - 'standard': Classic closeness (reciprocal of sum of distances). Can produce biased values for nodes in disconnected components.
  - 'harmonic': Harmonic closeness (sum of reciprocal distances). Recommended for disconnected multilayer networks.
  - 'auto': Automatically selects 'harmonic' if the network has multiple connected components, otherwise uses 'standard'.

  Default is 'standard' for backward compatibility.

**Returns**

Dictionary mapping (node, layer) tuples to closeness centrality values

### Examples

```
>>> closeness = multiplex_closeness_centrality(network)
>>> central_nodes = {k: v for k, v in closeness.items() if v > 0.5}
```

```
>>> # For disconnected networks, use harmonic variant
>>> closeness = multiplex_closeness_centrality(network, variant='harmonic')
```

**Reference:**

De Domenico et al. (2015), "Structural reducibility of multilayer networks" Boldi, P., & Vigna, S. (2014). Axioms for Centrality. Internet Math.

py3plex.algorithms.statistics.multilayer_statistics.**multiplex_rich_club_coefficient**(*network: Any*, *k: int*, *normalized: bool = True*) → float

Calculate multiplex rich-club coefficient.

Measures the tendency of high-degree nodes to be more densely connected to each other than expected by chance, accounting for the multiplex structure.

Formula: (k) = E(>k) / (N(>k) * (N(>k)-1) / 2)

where E(>k) is the number of edges among nodes with overlapping degree > k, and N(>k) is the number of such nodes.

**Parameters**

- **network** – py3plex multi_layer_network object
- **k** – Degree threshold
- **normalized** – Whether to normalize by random expectation

**Returns**

Rich-club coefficient value

### Examples

```
>>> rich_club = multiplex_rich_club_coefficient(network, k=10)
>>> print(f"Rich-club coefficient: {rich_club:.3f}")
```

**Reference:**

Alstott et al. (2014), "powerlaw: A Python Package for Analysis of Heavy-Tailed Distributions" Extended to multiplex networks

py3plex.algorithms.statistics.multilayer_statistics.**node_activity**(*network: Any*, *node: Any*) → float

Calculate node activity (a).

Formula: a = (1/L)  (v  V)

Fraction of layers in which node i is active (has at least one connection).

**Variables:**
L = total number of layers (v  V) = indicator function (1 if node i is active in layer , 0 otherwise) V = set of active nodes in layer

**Parameters**

- **network** – py3plex multi_layer_network object
- **node** – Node identifier

**Returns**
Activity value between 0 and 1

### Examples

```
>>> activity = node_activity(network, 'A')
```

**Reference:**
Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

**Contracts:**

- Precondition: network must not be None
- Precondition: node must not be None
- Postcondition: result is in [0, 1] (fraction of layers)
- Postcondition: result is not NaN

py3plex.algorithms.statistics.multilayer_statistics.**percolation_threshold**(*network: Any*, *removal_strategy: str = 'random'*, *trials: int = 10*) → float

Estimate percolation threshold for the multiplex network.

Determines the fraction of nodes that must be removed before the network fragments into disconnected components. Uses sampling to estimate threshold.

**Parameters**

- **network** – py3plex multi_layer_network object

- **removal_strategy** – 'random', 'degree', or 'betweenness'
- **trials** – Number of trials for averaging

**Returns**

Estimated percolation threshold (fraction of nodes)

### Examples

```
>>> threshold = percolation_threshold(network, removal_strategy='degree')
>>> print(f"Percolation threshold: {threshold:.2%}")
```

**Reference:**

Buldyrev et al. (2010), "Catastrophic cascade of failures in interdependent networks"

py3plex.algorithms.statistics.multilayer_statistics.**resilience**(*network: Any*, *perturbation_type: str = 'layer_removal'*, *perturbation_param: str | float | None = None*, *seed: int | None = None*) → float

Calculate resilience (R).

Formula: R = S' / $S_0$

Ratio of largest connected component after perturbation to original size.

**Variables:**

$S_0$ = size of largest connected component in original network S' = size of largest connected component after perturbation

**Perturbation types:**

1. Layer removal: Remove all nodes/edges in a specific layer
2. Coupling removal: Remove a fraction of inter-layer edges

**Properties:**

R = 1 indicates full resilience (no impact from perturbation) R = 0 indicates complete fragmentation 0 < R < 1 indicates partial resilience

**Parameters**

- **network** – py3plex multi_layer_network object
- **perturbation_type** – 'layer_removal' or 'coupling_removal'
- **perturbation_param** – Layer to remove or fraction of inter-layer edges
- **seed** – Optional random seed for stochastic perturbation steps

**Returns**

Resilience ratio between 0 and 1

### Examples

```
>>> r = resilience(network, 'layer_removal', perturbation_param='L1')
>>> r = resilience(network, 'coupling_removal', perturbation_param=0.5)
```

**Reference:**

Buldyrev et al. (2010), Nature 464, 1025-1028

py3plex.algorithms.statistics.multilayer_statistics.**supra_laplacian_spectrum**(*network: Any*, *k: int = 10*) → ndarray

Calculate supra-Laplacian spectrum ().

Formula: = -

Eigenvalue spectrum of the supra-Laplacian matrix; captures diffusion properties. Uses sparse eigenvalue computation when beneficial.

**Variables:**

= supra-adjacency matrix (NL × NL block matrix containing all layers and inter-layer couplings) = supra-degree matrix (diagonal matrix with row sums of ) = supra-Laplacian matrix = $\{_0, _1, \ldots, _1\}$ with $0 = _0 \ _1 \ \cdots \ _1$

**Parameters**

- **network** – py3plex multi_layer_network object
- **k** – Number of smallest eigenvalues to compute

**Returns**

Array of k smallest eigenvalues

### Examples

```
>>> spectrum = supra_laplacian_spectrum(network, k=10)
```

**Reference:**

De Domenico et al. (2013), Gomez et al. (2013)

### Notes

- Uses sparse eigsh() for sparse matrices (more efficient)
- Falls back to dense computation for small matrices (n < 100) or when k is large relative to n
- Laplacian is always symmetric for undirected graphs (PSD with smallest eigenvalue = 0)

py3plex.algorithms.statistics.multilayer_statistics.**targeted_layer_removal**(*network: Any*, *layer: str*, *return_resilience: bool = False*) → Any | Tuple[Any, float]

Simulate targeted removal of an entire layer.

Removes all edges in a specified layer and returns the modified network or resilience score.

**Parameters**

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier to remove
- **return_resilience** – If True, return resilience score instead of network

**Returns**

Modified network or resilience score

### Examples

```
>>> resilience = targeted_layer_removal(network, 'social', return_resilience=True)
>>> print(f"Resilience after removing social layer: {resilience:.3f}")
```

**Reference:**

Buldyrev et al. (2010), "Catastrophic cascade of failures"

py3plex.algorithms.statistics.multilayer_statistics.**unique_redundant_edges**(*network: Any*, *layer_i: str*, *layer_j: str*) → Tuple[int, int]

Count unique and redundant edges between two layers.

Returns the number of edges unique to the first layer and the number of edges present in both layers (redundant).

**Parameters**

- **network** – py3plex multi_layer_network object

- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier

**Returns**

Tuple of (unique_edges, redundant_edges)

### Examples

```
>>> unique, redundant = unique_redundant_edges(network, 'social', 'work')
>>> print(f"Unique: {unique}, Redundant: {redundant}")
```

py3plex.algorithms.statistics.multilayer_statistics.**versatility_centrality**(*network: Any*, *centrality_type: str = 'degree'*, *alpha: Dict[str, float] | None = None*) → Dict[Any, float]

Calculate versatility centrality (V).

Formula: V = w C

Weighted combination of node centrality values across layers; measures overall influence.

**Variables:**

w = weight for layer (typically 1/L for uniform weighting, w = 1) C = centrality of node i in layer (can be degree, betweenness, closeness, etc.)

**Parameters**

- **network** – py3plex multi_layer_network object
- **centrality_type** – Type of centrality ('degree', 'betweenness', 'closeness')
- **alpha** – Layer weights (default: uniform weights)

**Returns**

Dictionary mapping nodes to versatility centrality values

### Examples

```
>>> versatility = versatility_centrality(network, centrality_type='degree')
```

**Reference:**
De Domenico et al. (2015), Nature Communications 6, 6868

py3plex.algorithms.statistics.topology.**basic_pl_stats**(*degree_sequence: List[int]*) → Tuple[float, float]

Calculate basic power law statistics for a degree sequence.

**Parameters**
**degree_sequence** – Degree sequence of individual nodes

**Returns**
Tuple of (alpha, sigma) values

py3plex.algorithms.statistics.topology.**plot_power_law**(*degree_sequence: List[int]*, *title: str*, *xlabel: str*, *plabel: str*, *ylabel: str = 'Number of nodes'*, *formula_x: int = 70*, *formula_y: float = 0.05*, *show: bool = True*, *use_normalization: bool = False*) → Any

py3plex.algorithms.statistics.correlation_networks.**default_correlation_to_network**(*matrix: ndarray*, *input_type: str = 'matrix'*, *preprocess: str = 'standard'*) → ndarray

Convert correlation matrix to network using optimal thresholding.

**Parameters**

- **matrix** – Input data matrix
- **input_type** – Type of input (default: “matrix”)
- **preprocess** – Preprocessing method (default: “standard”)

**Returns**
Binary adjacency matrix

py3plex.algorithms.statistics.correlation_networks.**pick_threshold**(*matrix: ndarray*) → float

Pick optimal threshold for correlation network construction.

**Parameters**
**matrix** – Input data matrix

**Returns**
Optimal threshold value

py3plex.algorithms.statistics.basic_statistics.**core_network_statistics**(*G: Graph*, *labels: Any | None = None*, *name: str = 'example'*) → DataFrame

Compute core statistics for a network.

**Parameters**

- **G** – NetworkX graph to analyze
- **labels** – Optional label matrix with shape attribute
- **name** – Name identifier for the network (default: "example")

**Returns**
DataFrame containing network statistics

py3plex.algorithms.statistics.basic_statistics.**ensure**(**args*, ***kwargs*)

py3plex.algorithms.statistics.basic_statistics.**identify_n_hubs**(*G: Graph*, *top_n: int = 100*, *node_type: str | None = None*) → Dict[Any, int]

Identify the top N hub nodes in a network based on degree centrality.

**Parameters**

- **G** – NetworkX graph to analyze
- **top_n** – Number of top hubs to return (default: 100)
- **node_type** – Optional filter for specific node type

**Returns**
Dictionary mapping node identifiers to their degree values

**Contracts:**

- Precondition: top_n must be positive
- Postcondition: result has at most top_n entries

- Postcondition: all degree values are non-negative integers

py3plex.algorithms.statistics.basic_statistics.**require**(**args*, ***kwargs*)

### Multilayer Algorithms

Algorithms designed for multilayer graphs, including centrality and entanglement measures.

Multilayer/Multiplex Network Centrality Measures

This module implements various centrality measures for multilayer and multiplex networks, following standard definitions from multilayer network analysis literature.

### Weight Handling Notes:

For path-based centralities (betweenness, closeness): - Edge weights from the supra-adjacency matrix are converted to distances (inverse of weight) - NetworkX algorithms use these distances for shortest path computation - betweenness_centrality: weight parameter specifies the edge attribute for path computation - closeness_centrality: distance parameter specifies the edge attribute for path computation

For disconnected graphs: - closeness_centrality uses wf_improved parameter (Wasserman-Faust scaling) by default - When wf_improved=True, scores are normalized by reachable nodes only - When wf_improved=False, unreachable nodes contribute infinite distance

Weight Constraints: - Weights should be positive (> 0) for shortest path algorithms - Zero or negative weights will cause undefined behavior - For unweighted analysis, use weighted=False parameters

Authors: py3plex contributors Date: 2025

**class** py3plex.algorithms.multilayer_algorithms.centrality.**MultilayerCentrality**(*network: Any*)

Bases: `object`

Class for computing centrality measures on multilayer networks.

This class provides implementations of various centrality measures specifically designed for multilayer/multiplex networks, including degree-based, eigenvector-based, and path-based measures.

**accessibility_centrality**(*h=2*)

Compute accessibility centrality (entropy-based reach within h steps).

Accessibility measures the diversity of nodes reachable within h steps using entropy of the probability distribution.

Access_r = exp(H_r) where H_r is the entropy of the h-step distribution

**Parameters**
**h** – Number of steps (default: 2)

**Returns**
{(node, layer): accessibility}

**Return type**
dict

**Note**

Accessibility is measured as the effective number of h-step destinations, using the entropy of the random walk distribution.

Dangling Node Handling: For nodes with no outgoing edges (dangling nodes), this implementation uses uniform teleportation to all nodes. This differs from the original Travencolo-Costa definition which does not include teleportation. The teleportation ensures a well-defined random walk distribution but may affect accessibility values for nodes near dangling nodes compared to implementations that handle dangling nodes differently.

### References


- Travencolo, B. A. N., & Costa, L. D. F. (2008). Accessibility in complex networks. Physics Letters A, 373(1), 89-95.


`aggregate_to_node_level`(*node_layer_centralities*, *method='sum'*, *weights=None*)

Aggregate node-layer centralities to node level.

**Parameters**

- **`node_layer_centralities`** – dict with {(node, layer): value} entries
- **`method`** – 'sum', 'mean', 'max', 'weighted_sum'
- **`weights`** – dict with {layer: weight} for weighted_sum method

**Returns**

{node: aggregated_value}

**Return type**

dict

`bridging_centrality`()

Compute bridging centrality for nodes in the supra-graph.

Bridging centrality combines betweenness with the bridging coefficient, which measures how much a node connects sparse regions of the network.

Bridging(u) = B(u) * BCoeff(u) where BCoeff(u) = (1 / k_u) * sum_{v in N(u)} [1 / k_v]

**Returns**

{(node, layer): bridging_centrality}

**Return type**

dict

**Note**

Nodes with higher bridging centrality act as important bridges connecting different parts of the network.

`collective_influence`(*radius=2*)

Compute collective influence (CI_) for multiplex networks.

Collective influence identifies influential spreaders by considering not just immediate neighbors but also nodes at distance .

CI_(u) = (k_u - 1) * sum_{v in Ball_(u)} (k_v - 1)

**Parameters**
**radius** – Radius for the ball boundary (default: 2)

**Returns**
{(node, layer): collective_influence}

**Return type**
dict

**Note**

Uses overlapping degree across all layers for each physical node.

`communicability_betweenness_centrality(`*normalized=True*`)`

Compute communicability betweenness centrality on the supra-graph.

This measure quantifies how much a node contributes to the communicability between other pairs of nodes. It uses the matrix exponential to account for all walks between nodes.

**Parameters**
**normalized** – If True (default), normalize scores by dividing by the maximum value (min-max scaling to [0, 1] range). Note that this is simple rescaling, not the theoretical Estrada-Hatano normalization from the original literature.

**Returns**
{(node, layer): communicability_betweenness}

**Return type**
dict

**Note**

This implementation uses NetworkX's communicability_betweenness_centrality, which operates on unweighted graphs. Edge weights from the supra-adjacency matrix are used only to determine edge existence (weight $> 0$), not for weighted communicability computation. For truly weighted communicability analysis, a custom implementation using the weighted matrix exponential would be required.

This is computationally expensive as it requires computing the matrix exponential multiple times. For large networks, this may take significant time.

`current_flow_betweenness_centrality()`

Compute current-flow betweenness centrality via supra Laplacian pseudoinverse.

This measure is based on the electrical current flow through each node.

**Returns**
{(node, layer): current_flow_betweenness}

**Return type**
dict

`current_flow_closeness_centrality()`

Compute current-flow closeness centrality via supra Laplacian pseudoinverse.

This measure is based on the resistance distance in electrical networks.

**Returns**
{(node, layer): current_flow_closeness}

**Return type**
dict

`edge_betweenness_centrality`(*normalized=True*)

Compute edge betweenness centrality on the supra-graph.

Edge betweenness measures the fraction of shortest paths that pass through each edge.

**Returns**
{(source, target): edge_betweenness}

**Return type**
dict

**Note**

This returns edge-level centrality, not node-level.

`flow_betweenness_centrality`(*samples=100*)

Compute flow betweenness based on maximum flow sampling.

Flow betweenness measures how much flow passes through a node when maximizing flow between randomly sampled source-target pairs.

**Parameters**
**`samples`** – Number of source-target pairs to sample (default: 100)

**Returns**
{(node, layer): flow_betweenness}

**Return type**
dict

**Note**

This is a sampling-based approximation of flow betweenness. The implementation samples random pairs of supra-graph nodes and computes the maximum flow through each intermediate node.

Unlike classical Freeman flow betweenness which uses a normalization factor based on the number of nodes, this implementation returns the average flow per sampled pair without additional normalization. For multilayer networks, the number of supra-graph nodes (N * L for N physical nodes and L layers) affects the raw values.

For large networks, this sampling approach is more computationally feasible than computing exact flow betweenness for all pairs.

`harmonic_closeness_centrality`()

Compute harmonic closeness centrality on the supra-graph.

Harmonic closeness handles disconnected graphs better than standard closeness by

summing the reciprocals of distances instead of taking reciprocal of sum.

HC(u) = sum_{vu} (1 / d(u,v)) for finite distances

**Returns**
{(node, layer): harmonic_closeness}

**Return type**
dict

**Note**

This measure naturally handles disconnected components as unreachable nodes contribute 0 (instead of infinity) to the sum.

Weight Interpretation: Edge weights from the supra-adjacency matrix are interpreted as connection strengths (larger weight = stronger connection = shorter distance). They are converted to distances via 1/weight for shortest path computation. If your edge weights already represent distances, do not use this function directly—you would need to invert them first or use the distance values directly in a custom computation.

`hits_centrality`(*max_iter=1000*, *tol=1e-06*)

Compute HITS (hubs and authorities) centrality on the supra-graph.

For undirected networks, this equals eigenvector centrality. For directed networks, computes separate hub and authority scores.

**Parameters**

- `max_iter` – Maximum number of iterations.
- `tol` – Tolerance for convergence.

**Returns**

**If directed network: {'hubs': {(node, layer): score}, 'authorities': {(node, layer): score}}**
If undirected network: {(node, layer): score} (equivalent to eigenvector centrality)

**Return type**
dict

`information_centrality`()

Compute Information Centrality (Stephenson-Zelen style) on the supra-graph.

Information centrality measures the importance of a node based on the information flow through the network. It uses the inverse of a modified Laplacian matrix.

**Returns**
{(node, layer): information_centrality}

**Return type**
dict

**Note**

This implementation returns values for each node-layer pair in the supra-graph,

not aggregated physical node values. To obtain physical node-level information centrality, use the aggregate_to_node_level() method on the returned dictionary.

The implementation uses NetworkX's information_centrality for computation. Falls back to harmonic closeness if information centrality computation fails.

Information centrality is defined only for undirected graphs. For directed networks, the graph is symmetrized with a warning.

### References


- Stephenson, K., & Zelen, M. (1989). Rethinking centrality: Methods and examples. Social Networks, 11(1), 1-37.


`katz_bonacich_centrality(`*alpha=0.1*, *beta=None*`)`

Compute Katz-Bonacich centrality on the supra-graph.

z = _{t=0}^∞ ^t M^t b = (I - M)^{-1} b

**Parameters**

- **alpha** – Attenuation parameter (should be < 1/(M)). If None, automatically computes a safe value as 0.85/(M) where (M) is the spectral radius.
- **beta** – Exogenous preference vector. If None, uses vector of ones.

**Returns**
{(node, layer): centrality_value}

**Return type**
dict

`layer_degree_centrality(`*layer: str | None = None*, *weighted: bool = False*, *direction: str = 'out'*`)` → Dict[str | Tuple[str, str], float]

Compute layer-specific degree (or strength) centrality.

**For undirected networks:**
k^[]_i = _j 1(A^[]_ij > 0) [unweighted] s^[]_i = _j A^[]_ij [weighted]

**For directed networks:**
k^[,out]_i = _j 1(A^[]_ij > 0) [out-degree] k^[,in]_i = _j 1(A^[]_ji > 0) [in-degree]

**Parameters**

- **layer** – Layer to compute centrality for. If None, compute for all layers.
- **weighted** – If True, compute strength instead of degree.
- **direction** – 'out', 'in', or 'both' for directed networks.

**Returns**

**{(node, layer): centrality_value} if layer is None,**
{node: centrality_value} if layer is specified.

**Return type**
dict

`load_centrality()`

Compute load centrality (shortest-path load).

Load centrality measures the fraction of shortest paths that pass through each node, counting all paths (not just unique pairs).

Load[k] = sum over all (s,t) pairs of [number of shortest paths through k / total shortest paths]

**Returns**
{(node, layer): load_centrality}

**Return type**
dict

**Note**

This is similar to betweenness but counts all paths rather than normalizing by the number of node pairs.

`local_efficiency_centrality()`

Compute local efficiency centrality.

Local efficiency measures how efficiently information is exchanged among a node's neighbors when the node is removed. It quantifies the fault tolerance of the network.

LE(u) = (1 / (|N_u|*(|N_u|-1))) * sum_{ij in N_u} [1 / d(i,j)]

**Returns**
{(node, layer): local_efficiency}

**Return type**
dict

**Note**

For nodes with less than 2 neighbors, local efficiency is 0.

`lp_aggregated_centrality(`*layer_centralities*, *p=2*, *weights=None*, *exclude_missing=True*`)`

Compute Lp-aggregated per-layer centrality.

Aggregates per-layer centrality values using Lp norm.

C_i = (sum_{ i in L_i} w_ * |c^[i]|^p)^{1/p} for p $< \infty$ C_i = max_{ i in L_i} (w_ * |c^[i]|) for p = $\infty$

where L_i is the set of layers where node i exists.

**Parameters**

- **layer_centralities** – dict of {layer: {node: centrality}} or {(node, layer): centrality}
- **p** – Lp norm parameter (default: 2). Use float('inf') for L-infinity norm.
- **weights** – dict of {layer: weight}. If None, uniform weights are used.
- **exclude_missing** – If True (default), only aggregate over layers where

the node exists (has a centrality value). If False, treat nodes absent from a layer as having zero centrality, which may bias scores for nodes with sparse layer participation.

**Returns**
{node: aggregated_centrality}

**Return type**
dict

**Note**

This is a framework for aggregating any per-layer centrality measure. Input can be degree, PageRank, eigenvector, etc.

Sparse Layer Participation: When exclude_missing=True (default), nodes that only appear in a subset of layers are aggregated only over those layers where they exist. This prevents bias against nodes with sparse layer participation. When exclude_missing=False, missing layers contribute zero to the aggregation, which may penalize nodes that don't appear in all layers.

`multilayer_betweenness_centrality`(*normalized=True*, *endpoints=False*)

Compute betweenness centrality on the supra-graph.

For each node-layer pair (i,), computes the fraction of shortest paths between all pairs of nodes that pass through (i,).

**Parameters**

- **`normalized`** – Whether to normalize the betweenness values.
- **`endpoints`** – Whether to include endpoints in path counts.

**Returns**
{(node, layer): betweenness_centrality}

**Return type**
dict

**Note**

This is computationally expensive for large networks as it requires computing shortest paths between all pairs of nodes.

Weight handling: Edge weights from the supra-adjacency matrix are converted to distances (1/weight) for shortest path computation. Weights must be positive ($> 0$). Zero or negative weights will cause undefined behavior.

**Raises**
**`AlgorithmCompatibilityError`** – If network is incompatible with algorithm requirements

`multilayer_closeness_centrality`(*normalized=True*, *wf_improved=True*, *variant='standard'*)

Compute closeness centrality on the supra-graph.

For each node-layer pair (i,), computes:

**Standard closeness:**
C_c(i,) = (n-1) / _{(j,)} d((i,), (j,))

**Harmonic closeness (recommended for disconnected networks):**
HC(i,) = _{(j,)(i,)} 1/d((i,), (j,))

where d((i,), (j,)) is the shortest path distance in the supra-graph.

**Parameters**

- **normalized** – This parameter is kept for API compatibility but has no effect. Standard closeness is always normalized by (n-1) per the NetworkX implementation. Harmonic closeness returns unnormalized sums of reciprocal distances by definition.
- **wf_improved** – If True, use Wasserman-Faust improved closeness scaling for disconnected graphs. Default is True. This affects the magnitude and ordering of scores in graphs with multiple components (e.g., low interlayer coupling). See NetworkX documentation for details. Only used when variant='standard'.
- **variant** – Closeness variant to use. Options:
  - 'standard': Classic closeness (reciprocal of sum of distances). For disconnected graphs, uses Wasserman-Faust scaling if wf_improved=True. Can produce biased values for nodes in small or disconnected components.
  - 'harmonic': Harmonic closeness (sum of reciprocal distances). Mathematically well-defined for disconnected networks: unreachable nodes contribute 0 instead of infinity. Recommended for disconnected multilayer networks.
  - 'auto': Automatically selects 'harmonic' if the supra-graph has multiple connected components, otherwise uses 'standard'.

  Default is 'standard' for backward compatibility.

**Returns**
{(node, layer): closeness_centrality}

**Return type**
dict

**Note**

This implementation uses NetworkX's shortest path algorithms on the supra-graph representation. For large networks, this can be computationally expensive.

For disconnected multilayer graphs (e.g., layers with no inter-layer coupling, or networks with isolated components), use variant='harmonic' or variant='auto' to get mathematically consistent closeness values. The harmonic variant naturally handles unreachable nodes by summing 1/d for finite distances only (infinite distances contribute 0).

Weight Interpretation: Edge weights from the supra-adjacency matrix are interpreted as connection strengths (larger weight = stronger connection). They are converted to distances via 1/weight for shortest path computation. If your edge weights already represent distances, use them directly without this function's

weight inversion.

### Examples

```
>>> # For connected networks, standard closeness works well
>>> closeness = calc.multilayer_closeness_centrality(variant='standard')
```

```
>>> # For potentially disconnected networks, use harmonic
>>> closeness = calc.multilayer_closeness_centrality(variant='harmonic')
```

```
>>> # Let the algorithm decide based on connectivity
>>> closeness = calc.multilayer_closeness_centrality(variant='auto')
```

### References

- Wasserman, S., & Faust, K. (1994). Social Network Analysis.
- Boldi, P., & Vigna, S. (2014). Axioms for Centrality. Internet Math.
- De Domenico, M., et al. (2015). Structural reducibility of multilayer networks.

**Raises**

**AlgorithmCompatibilityError** – If network is incompatible with algorithm requirements

**multiplex_coreness()**

Alias for multiplex_k_core for compatibility.

**Returns**

{(node, layer): core_number}

**Return type**

dict

**multiplex_eigenvector_centrality**(*max_iter: int = 1000*, *tol: float = 1e-06*) → Dict

Compute multiplex eigenvector centrality (node-layer level).

x = (1/_max) * M * x where x_{i} is the centrality of node i in layer , and _max is the spectral radius of the supra-adjacency matrix M.

**Parameters**

- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

**Returns**

{(node, layer): centrality_value}

**Return type**

dict

**multiplex_eigenvector_versatility**(*max_iter: int = 1000*, *tol: float = 1e-06*) → Dict

Compute node-level eigenvector versatility.

x_i = _ x_{i}

**Parameters**

- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

**Returns**
{node: versatility_value}

**Return type**
dict

**multiplex_k_core()**

Compute multiplex k-core decomposition.

A node belongs to the k-core if it has at least k neighbors in the multilayer network. This implementation computes the core number for each node-layer pair.

**Returns**
{(node, layer): core_number}

**Return type**
dict

**overlapping_degree_centrality**(*weighted: bool = False*) → Dict

Compute overlapping degree/strength centrality (node level).

k^{over}_i = _ k^[]_i [unweighted] s^{over}_i = _ s^[]_i [weighted]

**Parameters**
**weighted** – If True, compute overlapping strength.

**Returns**
{node: centrality_value}

**Return type**
dict

**pagerank_centrality**(*damping=0.85*, *max_iter=1000*, *tol=1e-06*)

Compute PageRank centrality on the supra-graph.

Uses the standard PageRank algorithm on the supra-adjacency matrix representing the multilayer network. Properly handles dangling nodes (nodes with no outgoing edges) via teleportation.

This implementation preserves sparsity when possible for memory efficiency.

**Parameters**

- **damping** – Damping parameter (typically 0.85).
- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

**Returns**
{(node, layer): centrality_value}

**Return type**
dict

**Mathematical Invariants:**

- PageRank values sum to 1.0 (within tol=1e-6)

- All values are non-negative
- Converges for strongly connected components or with teleportation

**Raises**
**AlgorithmCompatibilityError** – If network is incompatible with algorithm requirements

**participation_coefficient**(*weighted: bool = False*) → Dict

Compute participation coefficient across layers.

Measures how evenly a node's degree is distributed across layers: P_i = 1 - _ (k^[]_i / k^{over}_i)^2

Set P_i = 0 if k^{over}_i = 0.

**Parameters**
**weighted** – If True, use strength instead of degree.

**Returns**
{node: participation_coefficient}

**Return type**
dict

**percolation_centrality**(*edge_activation_prob=0.5*, *trials=100*)

Compute percolation centrality using bond percolation Monte Carlo simulation.

This implementation measures the average relative component size that a node belongs to across multiple bond percolation realizations, providing an estimate of a node's importance for network connectivity under random edge failures.

**Parameters**

- **edge_activation_prob** – Probability that an edge is active (default: 0.5)
- **trials** – Number of Monte Carlo trials (default: 100)

**Returns**
{(node, layer): percolation_centrality}

**Return type**
dict

**Note**

This is a component-size-based percolation measure, not the path-based percolation betweenness from the original Piraveenan et al. literature, which requires recomputing betweenness on each percolated realization. The original percolation centrality is computationally expensive ($O(n^3)$ per trial), so this implementation provides a more efficient alternative that captures related connectivity information.

Values are normalized to [0, 1] range where higher values indicate nodes that tend to belong to larger connected components across percolation realizations.

### References

- Piraveenan, M., Prokopenko, M., & Hossain, L. (2013). Percolation centrality: Quantifying graph-theoretic impact of nodes during percolation in networks. PloS one, 8(1), e53095.

**spreading_centrality**(*beta=0.2*, *mu=0.1*, *trials=50*, *steps=100*)

Compute spreading (epidemic) centrality using SIR model.

Spreading centrality measures how influential a node is in spreading information or disease through the network, based on Monte Carlo simulations of discrete-time SIR dynamics.

**Parameters**

- **beta** – Infection rate per edge per time step (default: 0.2)
- **mu** – Recovery rate per time step (default: 0.1)
- **trials** – Number of simulation trials per node (default: 50)
- **steps** – Maximum simulation steps (default: 100)

**Returns**
{(node, layer): spreading_centrality}

**Return type**
dict

**Note**

This measures the average outbreak size (fraction of nodes ever infected) when seeding the epidemic from each node. Values are normalized by the total number of nodes, producing scores in the range [1/n, 1] where n is the number of supra-graph nodes.

This is an empirical simulation-based measure, not normalized by theoretical epidemic threshold or branching factor as in some literature definitions. The normalization allows comparison of relative spreading power within the network but may not be directly comparable across networks of different sizes or structures.

### References

- Kitsak, M., et al. (2010). Identification of influential spreaders in complex networks. Nature physics, 6(11), 888-893.

**subgraph_centrality**()

Compute subgraph centrality via matrix exponential of the supra-adjacency matrix.

Subgraph centrality counts closed walks of all lengths starting and ending at each node. SC_i = (e^A)_ii where A is the adjacency matrix.

**Returns**
{(node, layer): subgraph_centrality}

**Return type**
dict

**supra_degree_centrality**(*weighted: bool = False*) → Dict

Compute supra degree/strength centrality (node-layer level).

k_{i} = _{j,} 1(M_{(i,),(j,)} > 0) [unweighted] s_{i} = _{j,} M_{(i,),(j,)} [weighted]

**Parameters**
**weighted** – If True, compute strength instead of degree.

**Returns**
{(node, layer): centrality_value}

**Return type**
dict

`total_communicability()`

Compute total communicability via matrix exponential.

Total communicability is the row sum of the matrix exponential: TC_i = sum_j (e^A)_ij

**Returns**
{(node, layer): total_communicability}

**Return type**
dict

`py3plex.algorithms.multilayer_algorithms.centrality.communicability_centrality(`*supra_matrix: sp-ma-trix, nor-mal-ize: bool = True, use_sparse: bool = True, max_iter: int = 100, tol: float = 1e-06*`)` → ndarray

Compute communicability centrality for each node-layer pair.

Communicability centrality measures the weighted sum of all walks between nodes, with exponentially decaying weights for longer walks. It is computed as the row sum of the matrix exponential:

c_i = sum_j (exp(A))_ij

This implementation uses scipy.sparse.linalg.expm_multiply for efficient sparse matrix exponential computation.

**Parameters**

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **normalize** – If True, normalize output to sum to 1.
- **use_sparse** – If True, use sparse matrix operations. Falls back to dense for matrices smaller than 10000 elements.
- **max_iter** – Maximum number of iterations for sparse approximation (currently unused).
- **tol** – Tolerance for convergence (currently unused).

**Returns**

Communicability centrality scores for each node-layer pair (n,).

**Return type**

np.ndarray

**Raises**

**Py3plexMatrixError** – If matrix is invalid (non-square, empty, etc.).

### Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> comm = communicability_centrality(A)
>>> print(f"Communicability centrality computed for {len(comm)} node-layer pairs")
```

py3plex.algorithms.multilayer_algorithms.centrality.**compute_all_centralities**(*network*, *include_path_based=*, *include_advanced=*, *include_extended=*, *preset=None*, *wf_improved=Tr*, *closeness_variant='st*

Compute all available centrality measures for a multilayer network.

**Parameters**

- **network** – py3plex multi_layer_network object
- **include_path_based** – Whether to include computationally expensive path-based measures (betweenness, closeness). Default: False
- **include_advanced** – Whether to include advanced measures (HITS, current-flow, communicability, k-core). Default: False
- **include_extended** – Whether to include extended measures (information, accessibility, percolation, spreading, collective influence, load, flow

betweenness, harmonic, bridging, local efficiency). Default: False

- **preset** – Convenience parameter to set all inclusion flags at once. Options:
  - 'basic': Only degree and eigenvector-based measures (default behavior)
  - 'standard': Includes path-based measures
  - 'advanced': Includes path-based and advanced measures
  - 'all': Includes all measures (path-based, advanced, and extended)
  - None: Use individual flags (default)
- **wf_improved** – If True, use Wasserman-Faust improved scaling for closeness centrality in disconnected graphs. Default: True. Only used when closeness_variant='standard'.
- **closeness_variant** – Variant of closeness centrality to use. Options:
  - 'standard': Classic closeness (reciprocal of sum of distances). Uses Wasserman-Faust scaling if wf_improved=True.
  - 'harmonic': Harmonic closeness (sum of reciprocal distances). Recommended for disconnected multilayer networks.
  - 'auto': Automatically selects 'harmonic' if the network has disconnected components, otherwise uses 'standard'.

  Default: 'standard' for backward compatibility.

**Returns**

**Dictionary containing all computed centrality measures with keys:**

- Degree-based: "layer_degree", "layer_strength", "supra_degree", "supra_strength", "overlapping_degree", "overlapping_strength", "participation_coefficient", "participation_coefficient_strength"
- Eigenvector-based: "multiplex_eigenvector", "eigenvector_versatility", "katz_bonacich", "pagerank"
- Path-based (if include_path_based=True): "closeness", "betweenness"
- Advanced (if include_advanced=True): "hits", "current_flow_closeness", "current_flow_betweenness", "subgraph_centrality", "total_communicability", "multiplex_k_core"
- Extended (if include_extended=True): "information", "communicability_betweenness", "accessibility", "harmonic_closeness", "local_efficiency", "edge_betweenness", "bridging", "percolation", "spreading", "collective_influence", "load", "flow_betweenness"

**Return type**

dict

**Note**

Path-based, advanced, and extended measures are computationally expensive for large networks. Use flags or presets to control which measures are computed.

For disconnected multilayer graphs (e.g., networks without inter-layer coupling, or with isolated components), use closeness_variant='harmonic' or 'auto' to get mathematically consistent closeness values.

### Examples

```
>>> # Compute only basic measures (fast)
>>> results = compute_all_centralities(network)
```

```
>>> # Use preset for standard analysis
>>> results = compute_all_centralities(network, preset='standard')
```

```
>>> # Compute everything
>>> results = compute_all_centralities(network, preset='all')
```

```
>>> # Fine-grained control
>>> results = compute_all_centralities(
...     network,
...     include_path_based=True,
...     include_extended=True
... )
```

```
>>> # For disconnected multilayer networks, use harmonic closeness
>>> results = compute_all_centralities(
...     network,
...     include_path_based=True,
...     closeness_variant='harmonic'
... )
```

py3plex.algorithms.multilayer_algorithms.centrality.**katz_centrality**(*supra_matrix: spmatrix*, *alpha: float | None = None*, *beta: float = 1.0*, *tol: float = 1e-06*) → ndarray

Compute Katz centrality for each node-layer pair.

Katz centrality measures node influence by accounting for all paths with exponentially decaying weights. It is computed as:

x = (I - alpha * A)^{-1} * beta * 1

where alpha < 1/lambda_max(A) to ensure convergence. If alpha is not provided, it defaults to 0.85 / lambda_max(A).

**Parameters**

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).

- **alpha** – Attenuation parameter. Must be less than 1/spectral_radius(A). If None, defaults to 0.85 / lambda_max(A).
- **beta** – Weight of exogenous influence (typically 1.0).
- **tol** – Tolerance for eigenvalue computation.

**Returns**

**Katz centrality scores for each node-layer pair (n,).**
Normalized to sum to 1.

**Return type**
np.ndarray

**Raises**
**Py3plexMatrixError** – If matrix is invalid or alpha is out of valid range.

### Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> katz = katz_centrality(A)
>>> print(f"Katz centrality computed for {len(katz)} node-layer pairs")
>>> # With custom alpha
>>> katz_custom = katz_centrality(A, alpha=0.05)
```

Built-in multilayer centrality toolkit.

Implements core multilayer variants of centrality measures: - Multilayer PageRank - Multilayer betweenness centrality - Multilayer eigenvector centrality - Multiplex degree centrality

These algorithms properly account for the multilayer structure of networks.

All centrality functions now support first-class uncertainty estimation via the *uncertainty* parameter.

Authors: py3plex contributors Date: 2025

py3plex.algorithms.centrality_toolkit.**aggregate_centrality_across_layers**(*centrality_dict: Dict[Tuple, float]*, *aggregation: str = 'sum'*) → Dict[Any, float]

Aggregate node centrality values across layers.

Given centrality scores for (node, layer) tuples, aggregate to get per-node scores.

**Parameters**

- **centrality_dict** – Dictionary mapping (node, layer) -> score
- **aggregation** – Aggregation method ('sum', 'mean', 'max', 'min')

**Returns**

Dictionary mapping node_id -> aggregated score

### Example

```
>>> scores = {('A', 'L1'): 0.5, ('A', 'L2'): 0.3, ('B', 'L1'): 0.7}
>>> aggregate_centrality_across_layers(scores, 'mean')
{'A': 0.4, 'B': 0.7}
```

py3plex.algorithms.centrality_toolkit.**multilayer_betweenness_centrality**(*network: Any*, *normalized: bool = True*, *weight: str | None = None*) → Dict[Tuple, float]

Compute multilayer betweenness centrality.

Computes betweenness centrality on the supra-graph, where shortest paths can traverse multiple layers.

**Parameters**

- **network** – Multilayer network object
- **normalized** – Whether to normalize by number of pairs
- **weight** – Edge attribute to use as weight (None for unweighted)

**Returns**

Dictionary mapping (node, layer) tuples to betweenness scores

**Algorithm:**

For each pair of nodes (s,t), count the fraction of shortest paths passing through each node v:

BC(v) = _{svt} _{st}(v) / _{st}

where _{st} is the number of shortest paths from s to t, and _{st}(v) is the number passing through v.

### References


- De Domenico, M., et al. (2015). "Ranking in interconnected multilayer networks reveals versatile nodes." Nature Communications, 6, 6868.

py3plex.algorithms.centrality_toolkit.**multilayer_eigenvector_centrality**(*network: Any*, *max_iter: int = 100*, *tol: float = 1e-06*) → Dict[Tuple, float]

Compute multilayer eigenvector centrality.

Computes the principal eigenvector of the supra-adjacency matrix. Nodes are important if connected to other important nodes across layers.

**Parameters**

- **network** – Multilayer network object
- **max_iter** – Maximum number of power iteration steps
- **tol** – Convergence tolerance

**Returns**

Dictionary mapping (node, layer) tuples to eigenvector centrality scores

**Algorithm:**

Find the principal eigenvector of the supra-adjacency matrix:

A * x =  * x

where x is the eigenvector with largest eigenvalue .

### References

- Solá, L., et al. (2013). "Eigenvector centrality of nodes in multiplex networks." Chaos, 23(3), 033131.

py3plex.algorithms.centrality_toolkit.**multilayer_pagerank**(*network: Any*, *alpha: float = 0.85*, *max_iter: int = 100*, *tol: float = 1e-06*, *personalization: Dict | None = None*, *uncertainty: bool = False*, *n_runs: int | None = None*, *resampling: ResamplingStrategy | None = None*, *random_seed: int | None = None*) → Dict[Tuple, float] | StatSeries

Compute multilayer PageRank centrality with optional uncertainty estimation.

Implements PageRank on the supra-adjacency matrix, accounting for random walks across layers.

**Parameters**

- **network** – Multilayer network object
- **alpha** – Damping factor (teleportation probability = 1-alpha)
- **max_iter** – Maximum number of iterations
- **tol** – Convergence tolerance
- **personalization** – Optional personalization vector (node -> weight)
- **uncertainty** – If True, estimate uncertainty via resampling
- **n_runs** – Number of runs for uncertainty estimation (default from config)
- **resampling** – Resampling strategy (default from config)
- **random_seed** – Random seed for reproducibility

**Returns**

StatSeries with deterministic values (std=None) If uncertainty=True: StatSeries with mean, std, quantiles

**Return type**

If uncertainty=False

**Algorithm:**

PR = (1-)/N +  * A^T * PR

where A is the column-normalized supra-adjacency matrix

### References

- Halu, A., et al. (2013). "Multiplex PageRank." PLoS ONE, 8(10), e78293.

### Examples

```
>>> # Deterministic
>>> result = multilayer_pagerank(network)
>>> result[('A', 'L1')]  # Dict-like access
{'mean': 0.25}
>>> np.array(result)  # Array access (backward compat)
```

```
>>> # With uncertainty
>>> result = multilayer_pagerank(network, uncertainty=True, n_runs=50)
>>> result.mean  # Average PageRank values
>>> result.std   # Standard deviations
>>> result.quantiles  # Confidence intervals
```

py3plex.algorithms.centrality_toolkit.**multiplex_degree_centrality**(*network: Any*, *normalized: bool = True*, *consider_interlayer: bool = True*) → Dict[Tuple, float]

Compute multiplex degree centrality.

Sums degree across all layers for each node. For multiplex networks where nodes exist in all layers.

**Parameters**

- **network** – Multiplex network object
- **normalized** – Whether to normalize by maximum possible degree
- **consider_interlayer** – Whether to count inter-layer edges

**Returns**

Dictionary mapping (node, layer) tuples to degree centrality scores

**Algorithm:**

For node i: DC(i) = _ k_i^

where k_i^ is the degree of node i in layer .

### References

- Battiston, F., et al. (2014). "Structural measures for multiplex networks." Physical Review E, 89(3), 032804.

py3plex.algorithms.centrality_toolkit.**versatility_score**(*centrality_dict: Dict[Tuple, float]*, *normalized: bool = True*) → Dict[Any, float]

Compute versatility score for each node.

Measures how evenly a node's centrality is distributed across layers. High versatility means the node is important in multiple layers.

**Parameters**

- **centrality_dict** – Dictionary mapping (node, layer) -> centrality score
- **normalized** – Whether to normalize to [0, 1]

**Returns**

Dictionary mapping node_id -> versatility score

**Algorithm:**

V(i) = 1 - _ (c_i^ / c_i^total)^2

where c_i^ is centrality of node i in layer . This is similar to the Herfindahl-Hirschman index.

### References

- Battiston, F., et al. (2014). "Structural measures for multiplex networks." Physical Review E, 89(3), 032804.

MultiXRank: Random Walk with Restart on Universal Multilayer Networks

This module implements the MultiXRank algorithm described in: Baptista et al. (2022), "Universal multilayer network exploration by random walk with restart", Communications Physics,

5, 170.

MultiXRank performs random walk with restart (RWR) on a supra-heterogeneous adjacency matrix built from multiple multiplexes connected by bipartite blocks.

### References

- Paper: https://doi.org/10.1038/s42005-022-00937-9
- arXiv: https://arxiv.org/abs/2106.07869
- Package: https://github.com/anthbapt/multixrank
- Docs: https://multixrank-doc.readthedocs.io/

**class** py3plex.algorithms.multilayer_algorithms.multixrank.**MultiXRank**(*restart_prob: float = 0.4*, *epsilon: float = 1e-06*, *max_iter: int = 100000*, *verbose: bool = True*)

Bases: `object`

MultiXRank: Universal multilayer network exploration by random walk with restart.

This class implements the MultiXRank algorithm for node prioritization and ranking in universal multilayer networks. It builds a supra-heterogeneous adjacency matrix from multiple multiplexes and bipartite inter-multiplex connections, then performs random walk with restart.

**multiplexes**

Dictionary of multiplex supra-adjacency matrices

**Type**
Dict[str, sp.spmatrix]

**bipartite_blocks**

Inter-multiplex connection matrices

**Type**
Dict[Tuple[str, str], sp.spmatrix]

**restart_prob**

Restart probability (r) for RWR

**Type**
float

**epsilon**

Convergence threshold

**Type**
float

**max_iter**

Maximum number of iterations

**Type**
int

**node_order**

Node ordering for each multiplex

**Type**
Dict[str, List]

**supra_matrix**

Built supra-heterogeneous adjacency matrix

**Type**
sp.spmatrix

**transition_matrix**

Column-stochastic transition matrix

**Type**
sp.spmatrix

**add_bipartite_block**(*multiplex_from: str*, *multiplex_to: str*, *bipartite_matrix: spmatrix | ndarray*, *weight: float = 1.0*)

Add a bipartite block connecting two multiplexes.

**Parameters**

- **multiplex_from** – Name of source multiplex
- **multiplex_to** – Name of target multiplex
- **bipartite_matrix** – Matrix of connections (rows: from, cols: to)
- **weight** – Optional weight to scale this bipartite block

**add_multiplex**(*name: str*, *supra_adjacency: spmatrix | ndarray*, *node_order: List | None = None*)

Add a multiplex to the universal multilayer network.

**Parameters**

- **name** – Unique identifier for this multiplex
- **supra_adjacency** – Supra-adjacency matrix for the multiplex (can be from multi_layer_network.get_supra_adjacency_matrix())
- **node_order** – Optional list of node IDs in the order they appear in the matrix. If None, uses integer indices.

**aggregate_scores**(*scores: ndarray*, *aggregation: str = 'sum'*) → Dict[str, Dict]

Aggregate scores per multiplex and optionally per physical node.

**Parameters**

- **scores** – Probability vector from RWR (length = total supra-matrix dimension)
- **aggregation** – How to aggregate scores ('sum', 'mean', 'max')

**Returns**
Dictionary mapping multiplex names to dictionaries of node scores

**build_supra_heterogeneous_matrix**(*block_weights: Dict[str | Tuple[str, str], float] | None = None*)

Build the supra-heterogeneous adjacency matrix S.

This constructs the universal multilayer network matrix by: 1. Placing each multiplex supra-adjacency on the block diagonal 2. Adding bipartite blocks for inter-multiplex connections

**Parameters**

**block_weights** – Optional dictionary to weight blocks. Keys can be: - Multiplex names (to weight within-multiplex edges) - Tuple (multiplex_from, multiplex_to) for bipartite blocks

**Returns**

The supra-heterogeneous adjacency matrix

**column_normalize**(*handle_dangling: str = 'uniform'*) → spmatrix

Column-normalize the supra-heterogeneous matrix to create a stochastic transition matrix.

This ensures each column sums to 1, making the matrix suitable for RWR.

**Parameters**

**handle_dangling** – How to handle dangling nodes (columns with zero sum): - 'uniform': Distribute mass uniformly across all nodes - 'self': Add self-loop (mass stays at the node) - 'ignore': Leave as zero (not recommended for RWR)

**Returns**

Column-stochastic transition matrix

**get_top_ranked**(*scores: ndarray*, *k: int = 10*, *multiplex: str | None = None*, *exclude_seeds: bool = True*, *seed_nodes: List[int] | Dict[str, List] | None = None*) → List[Tuple]

Get top-k ranked nodes from RWR scores.

**Parameters**

- **scores** – Probability vector from RWR
- **k** – Number of top nodes to return
- **multiplex** – If specified, only return nodes from this multiplex
- **exclude_seeds** – Whether to exclude seed nodes from results
- **seed_nodes** – Seed nodes to exclude (if exclude_seeds=True)

**Returns**

List of (global_index, score) tuples, sorted by score descending

**random_walk_with_restart**(*seed_nodes: List[int] | Dict[str, List] | ndarray*, *seed_weights: ndarray | None = None*, *multiplex_name: str | None = None*) → ndarray

Perform Random Walk with Restart (RWR) from seed nodes.

**Parameters**

- **seed_nodes** – Seed node specification. Can be: - List of global indices in the supra-matrix - Dict mapping multiplex names to lists of local node indices - NumPy array of global indices

- **seed_weights** – Optional weights for seed nodes (must match length of seed_nodes). If None, uniform weights are used.
- **multiplex_name** – If seed_nodes is a list/array of local indices, specify which multiplex they belong to

**Returns**

Steady-state probability vector (length = total nodes across all multiplexes)

py3plex.algorithms.multilayer_algorithms.multixrank.**multixrank_from_py3plex_networks**(*network*
*Dict[str*
*multi-*
*net.mul*
*bi-*
*par-*
*tite_co*
*Dict[Tu*
*str],*
*sp-*
*ma-*
*trix*
*|*
*ndar-*
*ray]*
*|*
*None*
*=*
*None*,
*seed_no*
*Dict[str*
*List]*
*|*
*None*
*=*
*None*,
*restart_*
*float*
*=*
*0.4*,
*ep-*
*silon:*
*float*
*=*
*1e-*
*06*,
*max_it*
*int*
*=*
*100000*
*ver-*
*bose:*
*bool*
*=*
*True*)
→
Tu-
ple[Mul
ndarray

Convenience function to run MultiXRank on py3plex multi_layer_network objects.

**Parameters**

- **networks** – Dictionary mapping names to multi_layer_network objects
- **bipartite_connections** – Optional dict of inter-network connection matrices
- **seed_nodes** – Dict mapping network names to lists of seed node IDs
- **restart_prob** – Restart probability for RWR
- **epsilon** – Convergence threshold
- **max_iter** – Maximum iterations
- **verbose** – Whether to log progress

**Returns**

Tuple of (MultiXRank object, scores array)

### Example

```
>>> from py3plex.core import multinet
>>> net1 = multinet.multi_layer_network()
>>> net1.load_network('network1.edgelist', ...)
>>> net2 = multinet.multi_layer_network()
>>> net2.load_network('network2.edgelist', ...)
>>>
>>> networks = {'net1': net1, 'net2': net2}
>>> seed_nodes = {'net1': ['node1', 'node2']}
>>>
>>> mxr, scores = multixrank_from_py3plex_networks(
...     networks, seed_nodes=seed_nodes
... )
>>>
>>> # Get aggregated scores per network
>>> aggregated = mxr.aggregate_scores(scores)
```

Authors: Benjamin Renoust (github.com/renoust) Date: 2018/02/13 Description: Loads a Detangler JSON format graph and compute unweighted entanglement analysis with Py3Plex

py3plex.algorithms.multilayer_algorithms.entanglement.**build_occurrence_matrix**(*network: Any*) → Tuple[ndarray, List[Any]]

Build occurrence matrix from multilayer network.

**Parameters**

**network** – Multilayer network object

**Returns**

Tuple of (c_matrix, layers) where c_matrix is the normalized occurrence matrix and layers is the list of layer names

py3plex.algorithms.multilayer_algorithms.entanglement.**compute_blocks**(*c_matrix: ndarray*) → Tuple[List[List[int]], List[ndarray]]

Compute block decomposition of occurrence matrix.

**Parameters**
**c_matrix** – Occurrence matrix

**Returns**
Tuple of (indices, blocks) where indices are the layer indices in each block and blocks are the submatrices for each block

py3plex.algorithms.multilayer_algorithms.entanglement.**compute_entanglement**(*block_matrix: ndarray*) → Tuple[List[float], List[float]]

Compute entanglement metrics for a block.

**Parameters**
**block_matrix** – Block submatrix

**Returns**
Tuple of ([intensity, homogeneity, normalized_homogeneity], gamma_layers)

py3plex.algorithms.multilayer_algorithms.entanglement.**compute_entanglement_analysis**(*network: Any*) → List[Dict[str, Any]]

Compute full entanglement analysis for a multilayer network.

**Parameters**
**network** – Multilayer network object

**Returns**
List of block analysis dictionaries with entanglement metrics

Supra Matrix Function Centralities for Multilayer Networks.

This module implements centrality measures based on matrix functions of the supra-adjacency matrix, including: - Communicability Centrality (Estrada & Hatano, 2008) - Katz Centrality (Katz, 1953)

These measures operate directly on the sparse supra-adjacency matrix obtained from multi_layer_network.get_supra_adjacency_matrix().

### References


- Estrada, E., & Hatano, N. (2008). Communicability in complex networks. Physical Review E, 77(3), 036111.
- Katz, L. (1953). A new status index derived from sociometric analysis. Psychometrika, 18(1), 39-43.


Authors: py3plex contributors Date: October 2025 (Phase II)

**py3plex.algorithms.multilayer_algorithms.supra_matrix_function_centrality.communicability_ce**

Compute communicability centrality for each node-layer pair.

Communicability centrality measures the weighted sum of all walks between nodes, with exponentially decaying weights for longer walks. It is computed as the row sum of the matrix exponential:

c_i = sum_j (exp(A))_ij

This implementation uses scipy.sparse.linalg.expm_multiply for efficient sparse matrix exponential computation.

**Parameters**

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **normalize** – If True, normalize output to sum to 1.
- **use_sparse** – If True, use sparse matrix operations. Falls back to dense for matrices smaller than 10000 elements.
- **max_iter** – Maximum number of iterations for sparse approximation (currently unused).
- **tol** – Tolerance for convergence (currently unused).

**Returns**
Communicability centrality scores for each node-layer pair (n,).

**Return type**
np.ndarray

**Raises**
**Py3plexMatrixError** – If matrix is invalid (non-square, empty, etc.).

### Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> comm = communicability_centrality(A)
>>> print(f"Communicability centrality computed for {len(comm)} node-layer pairs")
```

py3plex.algorithms.multilayer_algorithms.supra_matrix_function_centrality.**katz_centrality**(*su*
*sp*
*m*
*tr*
*ai*
*pi*
*fl*
*/*
*N*
*=*
*N*
*be*
*fl*
*=*
*1.*
*to*
*fl*
*=*
*1e*
*0(*
*—*
*no*

Compute Katz centrality for each node-layer pair.

Katz centrality measures node influence by accounting for all paths with exponentially decaying weights. It is computed as:

x = (I - alpha * A)^{-1} * beta * 1

where alpha < 1/lambda_max(A) to ensure convergence. If alpha is not provided, it defaults to 0.85 / lambda_max(A).

**Parameters**

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **alpha** – Attenuation parameter. Must be less than 1/spectral_radius(A). If None, defaults to 0.85 / lambda_max(A).
- **beta** – Weight of exogenous influence (typically 1.0).

- **tol** – Tolerance for eigenvalue computation.

**Returns**

**Katz centrality scores for each node-layer pair (n,).**
Normalized to sum to 1.

**Return type**
np.ndarray

**Raises**
**Py3plexMatrixError** – If matrix is invalid or alpha is out of valid range.

### Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> katz = katz_centrality(A)
>>> print(f"Katz centrality computed for {len(katz)} node-layer pairs")
>>> # With custom alpha
>>> katz_custom = katz_centrality(A, alpha=0.05)
```

Multiplex Participation Coefficient (MPC) for multiplex networks.

This module implements the Multiplex Participation Coefficient metric for multiplex networks (networks with identical node sets across all layers).

Authors: py3plex contributors Date: 2025

py3plex.algorithms.multicentrality.**ensure**(**args*, ***kwargs*)

py3plex.algorithms.multicentrality.**multiplex_participation_coefficient**(*multinet: Any*, *normalized: bool = True*, *check_multiplex: bool = True*) → Dict[Any, float]

Compute the Multiplex Participation Coefficient (MPC) for multiplex networks. MPC measures how evenly a node participates across layers.

**Parameters**

- **multinet** (*py3plex.core.multinet.multi_layer_network*) – Multiplex network object (same node set across layers).
- **normalized** (*bool, optional*) – Whether to normalize MPC to [0,1]. Default: True.
- **check_multiplex** (*bool, optional*) – Validate that all layers share the same node set.

**Returns**
Node → MPC value mapping.

**Return type**
dict

### Notes

**The MPC is computed as:**
MPC(i) = 1 - sum_ (k_i^ / k_i^total)^2

Where: - k_i^ is the degree of node i in layer - k_i^total is the total degree of node i across all layers

When normalized=True, the result is multiplied by L/(L-1) to normalize to [0,1] range, where L is the number of layers.

### References


- Battiston, F., et al. (2014). "Structural measures for multiplex networks." Physical Review E, 89(3), 032804.
- De Domenico, M., et al. (2015). "Identifying modular flows on multilayer networks reveals highly overlapping organization in interconnected systems." Physical Review X, 5(1), 011027.
- Harooni, M., et al. (2025). "Centrality in Multilayer Networks: Accurate Measurements with MultiNetPy." The Journal of Supercomputing, 81(1), 92. DOI: 10.1007/s11227-025-07197-8


**Contracts:**

- Precondition: multinet must not be None
- Precondition: normalized and check_multiplex must be booleans
- Postcondition: returns a dictionary with numeric values

`py3plex.algorithms.multicentrality.require(*args, **kwargs)`

## General Algorithms

General-purpose algorithms such as random walkers and benchmarking helpers.

Random walk primitives for graph-based algorithms.

This module provides foundation for higher-level algorithms like Node2Vec, DeepWalk, and diffusion processes. Implements both basic and second-order (biased) random walks with proper edge weight handling and multilayer support.

**Key Features:**

- Basic random walks with weighted edge sampling
- Second-order (Node2Vec-style) biased random walks with p/q parameters
- Multiple simultaneous walks with deterministic reproducibility
- Support for directed, weighted, and multilayer networks
- Efficient sparse adjacency handling

### References


- Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. KDD '16. https://doi.org/10.1145/2939672.2939754
- Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). DeepWalk: Online learning of social representations. KDD '14. https://doi.org/10.1145/2623330.2623732


py3plex.algorithms.general.walkers.**basic_random_walk**(*G: Graph*, *start_node: int | str*, *walk_length: int*, *weighted: bool = True*, *seed: int | None = None*) → List[int | str]

Perform a basic random walk on a graph with proper edge weight handling.

The next step is sampled proportionally to the normalized edge weights of the current node. For unweighted graphs, transitions are uniform.

**Parameters**

- **G** – NetworkX graph (directed or undirected, weighted or unweighted)
- **start_node** – Node to start the walk from
- **walk_length** – Number of steps in the walk
- **weighted** – Whether to use edge weights (default: True)
- **seed** – Random seed for reproducibility (default: None)

**Returns**

List of nodes representing the walk path (includes start_node)

**Raises**

**ValueError** – If start_node not in graph or walk_length < 1

### Examples

```
>>> G = nx.Graph()
>>> G.add_weighted_edges_from([(0, 1, 1.0), (1, 2, 2.0), (1, 3, 1.0)])
>>> walk = basic_random_walk(G, 0, walk_length=3, seed=42)
>>> len(walk)
4
>>> walk[0]
0
```

**Note**

- Handles disconnected nodes by terminating walk early
- Edge weights must be positive for weighted walks
- Sum of transition probabilities from any node equals 1.0

py3plex.algorithms.general.walkers.**general_random_walk**(*G*, *start_node*, *iterations=1000*, *teleportation_prob=0*)

Legacy random walk with teleportation (for backward compatibility).

Deprecated since version 0.95a: Use `basic_random_walk()` or `node2vec_walk()` instead. This function will be removed in version 1.0.

**Parameters**

- **G** – NetworkX graph
- **start_node** – Starting node
- **iterations** – Number of steps
- **teleportation_prob** – Probability of teleporting to random visited node

**Returns**

List of visited nodes (excluding start_node)

py3plex.algorithms.general.walkers.**generate_walks**(*G: Graph*, *num_walks: int*, *walk_length: int*, *start_nodes: List[int | str] | None = None*, *p: float = 1.0*, *q: float = 1.0*, *weighted: bool = True*, *return_edges: bool = False*, *seed: int | None = None*) → List[List[int | str]] | List[List[Tuple[int | str, int | str]]]

Generate multiple random walks from specified or all nodes.

This interface supports multiple simultaneous walks with deterministic reproducibility under fixed RNG seed. Can return either node sequences or edge sequences.

**Parameters**

- **G** – NetworkX graph
- **num_walks** – Number of walks to generate per start node
- **walk_length** – Number of steps in each walk
- **start_nodes** – Nodes to start walks from (if None, uses all nodes)
- **p** – Return parameter for Node2Vec (1.0 = no bias)
- **q** – In-out parameter for Node2Vec (1.0 = no bias)
- **weighted** – Whether to use edge weights
- **return_edges** – Return edge sequences instead of node sequences
- **seed** – Random seed for reproducibility

**Returns**

- List of nodes (if return_edges=False)
- List of edges as tuples (if return_edges=True)

**Return type**

List of walks, where each walk is either

### Examples

```
>>> G = nx.karate_club_graph()
>>> # Generate 10 walks from each node
>>> walks = generate_walks(G, num_walks=10, walk_length=5, seed=42)
>>> len(walks)
340
```

```
>>> # Generate walks with Node2Vec bias
>>> walks = generate_walks(G, num_walks=5, walk_length=10, p=0.5, q=2.0, seed=42)
```

```
>>> # Get edge sequences
>>> edge_walks = generate_walks(G, num_walks=3, walk_length=5, return_edges=True, seed=
```

**Note**

- With same seed, generates identical walks across runs
- If p == q == 1.0, uses basic random walk (faster)
- Empty walks are included if node has no neighbors

py3plex.algorithms.general.walkers.**layer_specific_random_walk**(*G: Graph*, *start_node: int | str*, *walk_length: int*, *layer: str | None = None*, *cross_layer_prob: float = 0.0*, *weighted: bool = True*, *seed: int | None = None*) → List[int | str]

Perform random walk with layer constraints for multilayer networks.

In a multilayer network represented in py3plex format (where node names include layer information), this function can constrain walks to specific layers with occasional inter-layer transitions.

**Parameters**

- **G** – NetworkX graph (multilayer network in py3plex format)
- **start_node** – Node to start walk from (may include layer info)
- **walk_length** – Number of steps in the walk
- **layer** – Target layer to constrain walk to (None = no constraint)
- **cross_layer_prob** – Probability of crossing to different layer (0-1)
- **weighted** – Whether to use edge weights
- **seed** – Random seed for reproducibility

**Returns**

List of nodes representing the walk path

### Examples

```
>>> # Multilayer network with layer-specific walks
>>> from py3plex.core import multinet
>>> network = multinet.multi_layer_network()
>>> network.add_layer("social")
>>> network.add_layer("biological")
>>> # ... add nodes and edges ...
>>> walk = layer_specific_random_walk(
...     network.core_network,
...     "nodeA---social",
...     walk_length=10,
...     layer="social",
...     cross_layer_prob=0.1
... )
```

**Note**

- If layer is None, behaves like basic_random_walk
- cross_layer_prob controls inter-layer transitions
- Node format should follow py3plex convention: "nodeID—layerID"

py3plex.algorithms.general.walkers.**node2vec_walk**(*G: Graph*, *start_node: int | str*, *walk_length: int*, *p: float = 1.0*, *q: float = 1.0*, *weighted: bool = True*, *seed: int | None = None*) → List[int | str]

Perform a second-order (biased) random walk following Node2Vec logic.

When transitioning from node t → v → x, the probability of choosing x is biased by parameters p (return) and q (in-out): - If x == t (return to previous): weight / p - If x is neighbor of t (stay close): weight / 1 - If x is not neighbor of t (explore): weight / q

**Parameters**

- **G** – NetworkX graph (directed or undirected, weighted or unweighted)
- **start_node** – Node to start the walk from
- **walk_length** – Number of steps in the walk
- **p** – Return parameter (higher p = less likely to return to previous node)
- **q** – In-out parameter (higher q = less likely to explore further)
- **weighted** – Whether to use edge weights (default: True)
- **seed** – Random seed for reproducibility (default: None)

**Returns**

List of nodes representing the walk path (includes start_node)

**Raises**

**ValueError** – If p <= 0 or q <= 0 or start_node not in graph

### Examples

```
>>> G = nx.Graph()
>>> G.add_edges_from([(0, 1), (1, 2), (1, 3), (0, 2)])
>>> # Low p, high q: tends to backtrack
>>> walk = node2vec_walk(G, 0, walk_length=5, p=0.1, q=10.0, seed=42)
>>> # High p, low q: tends to explore outward
>>> walk2 = node2vec_walk(G, 0, walk_length=5, p=10.0, q=0.1, seed=42)
```

**Note**

- First step is always a basic random walk (no previous node)
- Properly normalizes probabilities at each step
- Handles disconnected nodes by terminating early

### References

Grover & Leskovec (2016), node2vec: Scalable feature learning for networks

Benchmark algorithms for node classification performance evaluation.

This module provides algorithms for benchmarking node classification performance, including oracle-based F1 score evaluation.

py3plex.algorithms.general.benchmark_classification.**evaluate_oracle_F1**(*probs: ndarray*, *Y_real: ndarray*) → tuple[float, float]

Evaluate oracle F1 scores for multi-label classification.

This function computes micro and macro F1 scores by selecting the top-k predictions for each sample, where k is determined by the ground truth number of labels.

**Parameters**

- **probs** – Predicted probability matrix of shape (n_samples, n_labels).
- **Y_real** – Ground truth binary label matrix of shape (n_samples, n_labels).

**Returns**

- micro: Micro-averaged F1 score
- macro: Macro-averaged F1 score

**Return type**

A tuple containing

### Example

```
>>> probs = np.array([[0.9, 0.1, 0.2], [0.1, 0.8, 0.7]])
>>> Y_real = np.array([[1, 0, 0], [0, 1, 1]])
>>> micro, macro = evaluate_oracle_F1(probs, Y_real)
```

## Node Ranking

Node ranking routines for multilayer and single-layer graphs.

py3plex.algorithms.node_ranking.node_ranking.**authority_matrix**(*graph: Graph*) → ndarray

Get the authority matrix representation of a graph.

Computes the matrix A = A.T @ A where A is the adjacency matrix. The authority matrix is used in HITS algorithm computation.

**Parameters**
**graph** – NetworkX graph

**Returns**
Authority matrix (N x N) where N is number of nodes

**Return type**
np.ndarray

### Notes

- For directed graphs: A[i,j] = number of nodes that point to both i and j
- Used internally by HITS algorithm to compute authority scores

**See also**

hub_matrix: Complementary hub matrix hubs_and_authorities: Compute actual hub/authority scores

py3plex.algorithms.node_ranking.node_ranking.**hub_matrix**(*graph: Graph*) → ndarray

Get the hub matrix representation of a graph.

Computes the matrix H = A @ A.T where A is the adjacency matrix. The hub matrix is used in HITS algorithm computation.

**Parameters**
**graph** – NetworkX graph

**Returns**
Hub matrix (N x N) where N is number of nodes

**Return type**
np.ndarray

### Notes

- For directed graphs: H[i,j] = number of nodes pointed to by both i and j
- Used internally by HITS algorithm to compute hub scores

**See also**

authority_matrix: Complementary authority matrix hubs_and_authorities: Compute actual hub/authority scores

py3plex.algorithms.node_ranking.node_ranking.**hubs_and_authorities**(*graph: Graph*) → Tuple[dict, dict]

Compute HITS (Hubs and Authorities) scores for all nodes in a graph.

Implements the Hyperlink-Induced Topic Search (HITS) algorithm to identify hub nodes (nodes that point to many authorities) and authority nodes (nodes pointed to by many hubs) in a network.

**Parameters**
**graph** – NetworkX graph (directed or undirected)

**Returns**

**(hub_scores, authority_scores)**

- hub_scores: Dictionary mapping node -> hub score
- authority_scores: Dictionary mapping node -> authority score

**Return type**
Tuple[dict, dict]

### Notes

- Uses an internal power-iteration fallback compatible with NetworkX 3.x
- Scores are normalized so that the sum of squares equals 1
- For undirected graphs, hub and authority scores are identical
- Converges using power iteration method

### Examples

```
>>> import networkx as nx
>>> G = nx.DiGraph([(0, 1), (0, 2), (1, 2)])
>>> hubs, authorities = hubs_and_authorities(G)
>>> # Node 0 has high hub score (points to others)
>>> # Node 2 has high authority score (pointed to by others)
```

**See also**

hub_matrix: Get the hub matrix representation authority_matrix: Get the authority matrix representation

py3plex.algorithms.node_ranking.node_ranking.**modularity**(*G: Graph*, *communities: List[List[Any]]*, *weight: str = 'weight'*) → float

Calculate modularity of a graph partition.

**Parameters**

- **G** – NetworkX graph
- **communities** – List of communities (each community is a list of nodes)
- **weight** – Edge weight attribute name

**Returns**

Modularity value

py3plex.algorithms.node_ranking.node_ranking.**sparse_page_rank**(*matrix: spmatrix*, *start_nodes: List[int] | range*, *epsilon: float = 1e-06*, *max_steps: int = 100000*, *damping: float = 0.5*, *spread_step: int = 10*, *spread_percent: float = 0.3*, *try_shrink: bool = True*) → ndarray

Compute personalized PageRank using sparse matrix operations.

Implements an efficient personalized PageRank algorithm with adaptive sparsification to reduce memory usage and computation time. The algorithm uses a power iteration method with early stopping based on convergence criteria.

**Parameters**

- **matrix** – Column-stochastic sparse adjacency matrix (use stochastic_normalization first)
- **start_nodes** – List or range of starting nodes for personalized PageRank
- **epsilon** – Convergence threshold for L1 norm difference (default: 1e-6)
- **max_steps** – Maximum number of iterations (default: 100000)
- **damping** – Damping factor / teleportation probability (default: 0.5) Higher values (e.g., 0.85) favor network structure over random jumps
- **spread_step** – Number of steps to check for sparsity pattern (default: 10)
- **spread_percent** – Maximum fraction of nodes to consider for shrinkage (default: 0.3)
- **try_shrink** – Enable adaptive shrinkage to reduce computation (default: True)

**Returns**

PageRank scores for all nodes, with start_nodes set to 0

**Return type**
np.ndarray

### Notes

- Assumes matrix is column-stochastic (use stochastic_normalization first)
- Adaptive shrinkage identifies nodes unreachable from start_nodes and excludes them from computation for efficiency
- Convergence is measured by L1 norm of rank vector difference
- Start nodes are zeroed out in the final result to avoid self-importance

**Complexity:**

- Time: O(k * E) where k is iterations and E is edges
- Space: O(N) for rank vectors, plus matrix storage

### Examples

```
>>> import scipy.sparse as sp
>>> # Create and normalize adjacency matrix
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>> adj_norm = stochastic_normalization(adj)
>>> # Compute PageRank from node 0
>>> pr = sparse_page_rank(adj_norm, [0], damping=0.85)
>>> pr  # PageRank scores (node 0 will be 0)
array([0.  , 0.5, 0.5])
```

**Raises**
**AssertionError** – If start_nodes is empty

**See also**

stochastic_normalization: Required preprocessing step run_PPR: Parallel wrapper for computing multiple PageRank vectors

py3plex.algorithms.node_ranking.node_ranking.**stochastic_normalization**(*matrix: spmatrix*) → spmatrix

Normalize a sparse matrix to stochastic form (column-stochastic).

Converts an adjacency matrix to a stochastic matrix where each column sums to 1. This normalization is required for PageRank-style random walk algorithms.

**Parameters**
**matrix** – Sparse adjacency matrix to normalize

**Returns**
Column-stochastic sparse matrix where each column sums to 1

**Return type**
sp.spmatrix

### Notes

- Removes self-loops (sets diagonal to 0) before normalization
- Handles zero-degree nodes by leaving corresponding columns as zeros
- Preserves sparsity structure for memory efficiency

### Examples

```
>>> import scipy.sparse as sp
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>> stoch_adj = stochastic_normalization(adj)
>>> stoch_adj.sum(axis=1).A1  # Column sums should be 1
array([1., 1., 1.])
```

py3plex.algorithms.node_ranking.node_ranking.**stochastic_normalization_hin**(*matrix: spmatrix*) → spmatrix

Normalize a heterogeneous information network matrix stochastically.

**Parameters**

**matrix** – Sparse matrix to normalize

**Returns**

Stochastically normalized sparse matrix

## Network Classification

Label propagation and related network classification helpers.

py3plex.algorithms.network_classification.label_propagation.**label_propagation**(*graph_matrix: sparse matrix*, *class_matrix: ndarray*, *alpha: float = 0.001*, *epsilon: float = 1e-12*, *max_steps: int = 100000*, *normalization: str | List[str] = 'freq'*) → ndarray

Propagate labels through a graph.

**Parameters**

- **graph_matrix** – Sparse graph adjacency matrix
- **class_matrix** – Initial class label matrix
- **alpha** – Propagation weight parameter
- **epsilon** – Convergence threshold
- **max_steps** – Maximum number of iterations
- **normalization** – Normalization scheme(s) to apply

**Returns**

Propagated label matrix

py3plex.algorithms.network_classification.label_propagation.**label_propagation_normalization**(

Normalize a matrix for label propagation.

**Parameters**
**matrix** – Sparse matrix to normalize

**Returns**
Normalized sparse matrix

py3plex.algorithms.network_classification.label_propagation.**label_propagation_tf**() → None

TensorFlow-based label propagation (TODO: implement).

Placeholder for future TensorFlow implementation.

py3plex.algorithms.network_classification.label_propagation.**normalize_amplify_freq**(*mat: ndarray*) → ndarray

Normalize and amplify matrix by frequency.

**Parameters**
**mat** – Matrix to normalize

**Returns**
Normalized and amplified matrix

py3plex.algorithms.network_classification.label_propagation.**normalize_exp**(*mat: ndarray*) → ndarray

Apply exponential normalization.

**Parameters**
**mat** – Matrix to normalize

**Returns**
Exponentially normalized matrix

py3plex.algorithms.network_classification.label_propagation.**normalize_initial_matrix_freq**(*m*
*n*
*r*
→
n

Normalize matrix by frequency.

**Parameters**
**mat** – Matrix to normalize

**Returns**

Normalized matrix

py3plex.algorithms.network_classification.label_propagation.**normalize_none**(*mat: ndarray*) → ndarray

No normalization (identity function).

**Parameters**
**mat** – Matrix to return unchanged

**Returns**
Original matrix

py3plex.algorithms.network_classification.label_propagation.**validate_label_propagation**(*core_sp-matrix, labels: ndarray | sp-matrix, datas str = 'test' repe-titions. int = 5, nor-mal-iza-tion_ str | List[s = 'ba-sic', al-pha_ float = 0.001 ran-dom_ int = 123, ver-bose: bool = False* → Datal

Validate label propagation with cross-validation.

**Parameters**

- **core_network** – Sparse network adjacency matrix
- **labels** – Label matrix
- **dataset_name** – Name of the dataset
- **repetitions** – Number of repetitions
- **normalization_scheme** – Normalization scheme to use
- **alpha_value** – Alpha parameter for propagation
- **random_seed** – Random seed for reproducibility
- **verbose** – Whether to print progress

**Returns**

DataFrame with validation results

### Visualization

Plotting utilities for multilayer networks and layouts.

py3plex.visualization.multilayer.**draw_multiedges**(*network_list: List[Graph] | Dict[Any, Graph]*, *multi_edge_tuple: List[Any]*, *input_type: str = 'nodes'*, *linepoints: str = '-.'*, *alphachannel: float = 0.3*, *linecolor: str = 'black'*, *curve_height: float = 1*, *style: str = 'curve2_bezier'*, *linewidth: float = 1*, *invert: bool = False*, *linmod: str = 'both'*, *resolution: float = 0.001*, *ax: Any | None = None*) → Any

Draw edges connecting multiple layers.

Draws curved or straight edges that connect nodes across different layers in a multilayer network visualization. Typically used after draw_multilayer_default to add inter-layer connections.

**Parameters**

- **network_list** – List of NetworkX graphs (layers) or dict of layer_name -> graph
- **multi_edge_tuple** – List of tuples specifying edges to draw, e.g. [(node1, node2), ...]
- **input_type** – Type of input (“nodes” or other)
- **linepoints** – Line style (e.g., “-.”, “–”, “-“)
- **alphachannel** – Transparency level (0.0 to 1.0)
- **linecolor** – Color of the lines
- **curve_height** – Height of curved edges
- **style** – Style of edges (“curve2_bezier”, “line”, “curve3_bezier”, “curve3_fit”, “piramidal”)

- **linewidth** – Width of lines
- **invert** – Whether to invert drawing direction
- **linmod** – Line modification mode
- **resolution** – Resolution for curve drawing
- **ax** – Matplotlib Axes to draw on. If None, uses current axes (plt.gca())

**Returns**

Matplotlib Axes object containing the visualization.

### Example

```
>>> import matplotlib.pyplot as plt
>>> from py3plex.visualization import draw_multilayer_default, draw_multiedges
>>> fig, ax = plt.subplots(figsize=(10, 10))
>>> ax = draw_multilayer_default(graphs, ax=ax)
>>> ax = draw_multiedges(graphs, edges, ax=ax)
>>> plt.savefig("multilayer_with_edges.png")
```

py3plex.visualization.multilayer.**draw_multilayer_default**(*network_list: List[Graph] | Dict[Any, Graph]*, *display: bool = False*, *node_size: int = 10*, *alphalevel: float = 0.13*, *rectanglex: float = 1*, *rectangley: float = 1*, *background_shape: str = 'circle'*, *background_color: str = 'rainbow'*, *networks_color: str = 'rainbow'*, *labels: bool = False*, *arrowsize: float = 0.5*, *label_position: int = 1*, *verbose: bool = False*, *remove_isolated_nodes: bool = False*, *ax: Any | None = None*, *edge_size: float = 1*, *node_labels: bool = False*, *node_font_size: int = 5*, *scale_by_size: bool = False*, ***, *axis: Any | None = None*) → Any

Core multilayer drawing method.

Draws a diagonal multilayer network visualization where each layer is offset to create a 3D-like effect. Nodes within each layer are drawn with their positions, and background shapes indicate layer boundaries.

**Parameters**

- **network_list** – List of NetworkX graphs to visualize (or dict of

layer_name -> graph)

- **display** – If True, calls plt.show() after drawing. Default is False to let the caller control rendering.
- **node_size** – Base size of nodes
- **alphalevel** – Transparency level for background shapes
- **rectanglex** – Width of rectangular backgrounds
- **rectangley** – Height of rectangular backgrounds
- **background_shape** – Background shape type ("circle" or "rectangle")
- **background_color** – Background color scheme ("default", "rainbow", or None)
- **networks_color** – Network color scheme ("rainbow" or "black")
- **labels** – Layer labels to display
- **arrowsize** – Size of edge arrows
- **label_position** – Position offset for layer labels
- **verbose** – Whether to log network information
- **remove_isolated_nodes** – Whether to remove isolated nodes
- **ax** – Matplotlib Axes to draw on. If None, uses current axes (plt.gca())
- **edge_size** – Width of edges
- **node_labels** – Whether to display node labels
- **node_font_size** – Font size for node labels
- **scale_by_size** – Whether to scale node size by degree
- **axis** – Deprecated. Use ax instead.

**Returns**

Matplotlib Axes object containing the visualization.

### Example

```
>>> import matplotlib.pyplot as plt
>>> from py3plex.visualization import draw_multilayer_default
>>> # Create figure and get axes
>>> fig, ax = plt.subplots(figsize=(10, 10))
>>> # Draw on the axes (returns the axes)
>>> ax = draw_multilayer_default(graphs, ax=ax)
>>> # Caller controls when to display
>>> plt.savefig("multilayer.png")  # or plt.show()
```

py3plex.visualization.multilayer.**draw_multilayer_flow**(*graphs: List[Graph]*, *multilinks: Dict[str, List[Tuple]]*, *labels: List[str] | None = None*, *node_activity: Dict[Any, float] | None = None*, *ax: Any | None = None*, *display: bool = True*, *layer_gap: float = 3.0*, *node_size: float = 30*, *node_cmap: str = 'viridis'*, *flow_alpha: float = 0.3*, *flow_min_width: float = 0.2*, *flow_max_width: float = 4.0*, *aggregate_by: Tuple[str, ...] = ('u', 'v', 'layer_u', 'layer_v')*, ***kwargs*) → Any

Draw multilayer network as layered flow visualization (alluvial-style).

Shows each layer as a horizontal band with nodes positioned along the x-axis. Intra-layer activity is encoded as node color/size, and inter-layer edges are shown as thick flow ribbons (Bezier curves) where width encodes edge weight.

**Parameters**

- **graphs** – List of NetworkX graphs, one per layer (from multi_layer_network.get_layers())
- **multilinks** – Dictionary mapping edge_type -> list of multi-layer edges
- **labels** – Optional list of layer labels. If None, uses layer indices
- **node_activity** – Optional dict mapping node_id -> activity value. If None, computes intra-layer degree
- **ax** – Matplotlib axes to draw on. If None, creates new figure
- **display** – If True, calls plt.show() at the end
- **layer_gap** – Vertical distance between layer bands
- **node_size** – Base marker size for nodes
- **node_cmap** – Matplotlib colormap name for node activity coloring
- **flow_alpha** – Base transparency for flow ribbons
- **flow_min_width** – Minimum line width for flows
- **flow_max_width** – Maximum line width for flows
- **aggregate_by** – Tuple of keys for aggregating flows (currently not used, for future extension)
- ****kwargs** – Reserved for future extensions

**Returns**

Matplotlib axes object

### Examples

```
>>> network = multi_layer_network()
>>> network.load_network("data.txt", input_type="multiedgelist")
>>> labels, graphs, multilinks = network.get_layers()
>>> draw_multilayer_flow(graphs, multilinks, labels=labels)
```

py3plex.visualization.multilayer.**generate_random_multiedges**(*network_list: List[Graph]*, *random_edges: int*, *style: str = 'line'*, *linepoints: str = '-.'*, *upper_first: int = 2*, *lower_first: int = 0*, *lower_second: int = 2*, *inverse_tag: bool = False*, *pheight: float = 1*) → None

Generate and draw random multi-layer edges.

**Parameters**

- **network_list** – List of NetworkX graphs (layers)
- **random_edges** – Number of random edges to generate
- **style** – Style of edges to draw
- **linepoints** – Line style
- **upper_first** – Upper bound for first layer
- **lower_first** – Lower bound for first layer
- **lower_second** – Lower bound for second layer
- **inverse_tag** – Whether to invert drawing
- **pheight** – Height parameter for curves

py3plex.visualization.multilayer.**generate_random_networks**(*number_of_networks: int*) → List[Graph]

Generate random networks for testing.

**Parameters**

**number_of_networks** – Number of random networks to generate

**Returns**

List of NetworkX graphs with random layouts

py3plex.visualization.multilayer.**hairball_plot**(*g: Graph | Any*, *color_list: List[str] | List[int] | None = None*, *display: bool = False*, *node_size: float = 1*, *text_color: str = 'black'*, *node_sizes: List[float] | None = None*, *layout_parameters: dict | None = None*, *legend: Any | None = None*, *scale_by_size: bool = True*, *layout_algorithm: str = 'force'*, *edge_width: float = 0.01*, *alpha_channel: float = 0.5*, *labels: List[str] | None = None*, *draw: bool = True*, *label_font_size: int = 2*, *ax: Any | None = None*, *skip_layout: bool = False*) → Any | None

Draw a force-directed "hairball" visualization of a network.

Creates a force-directed layout visualization where nodes are colored by type/layer and sized by degree. This is a common visualization for showing the overall structure of a network.

**Parameters**

- **g** – NetworkX graph to visualize
- **color_list** – List of colors for nodes. If None, colors are assigned based on node types.
- **display** – If True, calls plt.show() after drawing. Default is False to let the caller control rendering.
- **node_size** – Base size of nodes
- **text_color** – Color for node labels
- **node_sizes** – Custom list of node sizes (overrides node_size and scale_by_size)
- **layout_parameters** – Parameters for the layout algorithm (e.g., {"pos": {...}})
- **legend** – If True, display a legend mapping colors to node types
- **scale_by_size** – If True, scale node sizes by log(degree)
- **layout_algorithm** – Layout algorithm to use. Options: - "force": Force-directed layout (spring layout) - "random": Random layout - "custom_coordinates": Use positions from layout_parameters["pos"] - "custom_coordinates_initial_force": Use custom positions as initial layout
- **edge_width** – Width of edges
- **alpha_channel** – Transparency level (0.0 to 1.0)
- **labels** – List of node labels to display (None for no labels)
- **draw** – If True, draw the network. If False, only compute layout and return data.
- **label_font_size** – Font size for node labels

- **ax** – Matplotlib Axes to draw on. If None, uses current axes (plt.gca())
- **skip_layout** – If True, skip all layout computation and visualization, returning immediately. This is useful for examples that only need to test the API without actually rendering or when you want to avoid expensive force-directed layout calculations. When True, returns None immediately. Default is False.

**Returns**

Matplotlib Axes object containing the visualization - If draw=False: Tuple of (graph, node_sizes, color_mapping, positions)

**Return type**

- If draw=True

### Example

```
>>> import matplotlib.pyplot as plt
>>> from py3plex.visualization import hairball_plot
>>> fig, ax = plt.subplots(figsize=(10, 10))
>>> ax = hairball_plot(network.core_network, ax=ax, legend=True)
>>> plt.savefig("hairball.png")
```

py3plex.visualization.multilayer.**interactive_diagonal_plot**(*network_list: List[Graph] | Dict[Any, Graph]*, *layer_labels: List[str] | None = None*, *layout_algorithm: str = 'force'*, *layer_gap: float = 4.0*, *node_size_base: int = 8*, *layer_colors: List[str] | None = None*, *show_interlayer_edges: bool = True*, *interlayer_edges: List[Tuple[Any, Any]] | None = None*) → bool | Any

Create an interactive 2.5D diagonal multilayer plot using Plotly.

This function creates an interactive version of the diagonal multilayer visualization, mimicking the traditional 2D diagonal layout but in an interactive 3D environment. Each layer is positioned diagonally with clear visual separation, similar to the static diagonal visualization.

**Parameters**

- **network_list** – List of NetworkX graphs (layers) or dict of layer_name -> graph
- **layer_labels** – Optional labels for each layer
- **layout_algorithm** – Layout algorithm for nodes (“force”, “circular”,

"random")

- **layer_gap** – Distance between layers in diagonal direction (default: 2.5)
- **node_size_base** – Base size for nodes (default: 8)
- **layer_colors** – Optional list of colors for each layer (HTML color names or hex)
- **show_interlayer_edges** – Whether to show inter-layer edges
- **interlayer_edges** – List of tuples (node1, node2) for inter-layer connections

**Returns**

False if plotly not available, otherwise plotly figure object

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.load_network("network.txt", input_type="multiedgelist")
>>> labels, graphs, multilinks = net.get_layers("diagonal")
>>> fig = interactive_diagonal_plot(graphs, layer_labels=labels)
```

py3plex.visualization.multilayer.**interactive_hairball_plot**(*G: Graph*, *nsizes: List[float]*, *final_color_mapping: dict*, *pos: dict*, *colorscale: str = 'Rainbow'*) → bool | Any

Create an interactive 3D hairball plot using Plotly.

**Parameters**

- **G** – NetworkX graph to visualize
- **nsizes** – Node sizes
- **final_color_mapping** – Mapping of nodes to colors
- **pos** – Node positions
- **colorscale** – Color scale to use

**Returns**

False if plotly not available, otherwise plotly figure object

py3plex.visualization.multilayer.**onclick**(*event: Any*) → None

Handle mouse click events on plots.

**Parameters**

**event** – Matplotlib event object

py3plex.visualization.multilayer.**plot_edge_colored_projection**(*multilayer_network*, *layout: str = 'spring'*, *node_size: int = 50*, *layer_colors: Dict[Any, str] | None = None*, *aggregate_multilayer_edges: bool = True*, *figsize: Tuple[float, float] = (12, 9)*, *edge_alpha: float = 0.7*, ***kwargs*)

Create an aggregated projection where edge colors indicate layer membership.

This visualization projects all layers onto a single 2D graph, using edge colors to distinguish which layer each edge belongs to. Useful for seeing the overall structure while maintaining layer information.

**Parameters**

- **multilayer_network** – A MultiLayerNetwork instance
- **layout** – Layout algorithm to use ("spring", "circular", "random", "kamada_kawai")
- **node_size** – Size of nodes
- **layer_colors** – Optional dict mapping layer names to colors; if None, auto-generated
- **aggregate_multilayer_edges** – If True, show edges from all layers with distinct colors
- **figsize** – Figure size as (width, height) tuple
- **edge_alpha** – Transparency level for edges (0-1)
- ****kwargs** – Additional arguments for customization

**Returns**
The created figure

**Return type**
matplotlib.figure.Figure

py3plex.visualization.multilayer.**plot_ego_multilayer**(*multilayer_network*, *ego*, *layers: List[Any] | None = None*, *max_depth: int = 1*, *layout: str = 'spring'*, *figsize: Tuple[float, float] | None = None*, *max_cols: int = 3*, *node_size: int = 500*, *ego_node_size: int = 1200*, ***kwargs*)

Create an ego-centric multilayer visualization.

This visualization focuses on a single node (ego) and shows its neighborhood across dif-

ferent layers, highlighting the ego node's position in each layer.

**Parameters**

- **multilayer_network** – A MultiLayerNetwork instance
- **ego** – The ego node to focus on
- **layers** – Optional list of specific layers to visualize; if None, uses all layers
- **max_depth** – Maximum depth of neighborhood to include (number of hops)
- **layout** – Layout algorithm for each ego graph
- **figsize** – Optional figure size; if None, auto-calculated
- **max_cols** – Maximum columns in subplot grid
- **node_size** – Size of regular nodes
- **ego_node_size** – Size of the ego node (highlighted)
- ****kwargs** – Additional drawing parameters

**Returns**
The created figure

**Return type**
matplotlib.figure.Figure

py3plex.visualization.multilayer.**plot_radial_layers**(*multilayer_network*, *base_radius: float = 1.0*, *radius_step: float = 1.0*, *node_size: int = 500*, *draw_inter_layer_edges: bool = True*, *figsize: Tuple[float, float] = (12, 12)*, *edge_alpha: float = 0.5*, *draw_layer_bands: bool = True*, *band_alpha: float = 0.25*, ***kwargs*)

Create a radial/concentric visualization with layers as rings.

This visualization arranges layers as concentric circles, with nodes positioned on rings based on their layer. Inter-layer edges appear as radial connections.

**Parameters**

- **multilayer_network** – A MultiLayerNetwork instance
- **base_radius** – Radius of the innermost layer
- **radius_step** – Distance between consecutive layer rings
- **node_size** – Size of nodes (default: 500 for better visibility)
- **draw_inter_layer_edges** – If True, draw edges between layers
- **figsize** – Figure size as (width, height) tuple
- **edge_alpha** – Transparency for edges
- **draw_layer_bands** – If True, draw semi-transparent circular bands around layers

- **band_alpha** – Transparency for layer bands (default: 0.25)
- ****kwargs** – Additional drawing parameters

**Returns**
The created figure

**Return type**
matplotlib.figure.Figure

py3plex.visualization.multilayer.**plot_small_multiples**(*multilayer_network*, *layout: str = 'spring'*, *max_cols: int = 3*, *node_size: int = 50*, *shared_layout: bool = True*, *show_layer_titles: bool = True*, *figsize: Tuple[float, float] | None = None*, ***kwargs*)

Create a small multiples visualization with one subplot per layer.

This visualization shows each layer as a separate subplot in a grid layout, making it easy to compare the structure of different layers side-by-side.

**Parameters**

- **multilayer_network** – A MultiLayerNetwork instance
- **layout** – Layout algorithm to use ("spring", "circular", "random", "kamada_kawai")
- **max_cols** – Maximum number of columns in the subplot grid
- **node_size** – Size of nodes in each subplot
- **shared_layout** – If True, compute one layout and reuse for all layers; if False, compute independent layouts per layer
- **show_layer_titles** – If True, show layer names as subplot titles
- **figsize** – Optional figure size as (width, height) tuple
- ****kwargs** – Additional arguments passed to nx.draw_networkx

**Returns**
The created figure

**Return type**
matplotlib.figure.Figure

py3plex.visualization.multilayer.**plot_supra_adjacency_heatmap**(*multilayer_network*, *include_inter_layer: bool = False*, *inter_layer_weight: float = 1.0*, *node_order: List[Any] | None = None*, *cmap: str = 'viridis'*, *figsize: Tuple[float, float] = (10, 10)*, ***kwargs*)

Create a supra-adjacency matrix heatmap visualization.

This visualization shows the multilayer network as a block matrix where each block represents the adjacency matrix of one layer. Optionally includes inter-layer connections.

**Parameters**

- **multilayer_network** – A MultiLayerNetwork instance
- **include_inter_layer** – If True, include inter-layer edges/couplings
- **inter_layer_weight** – Default weight for inter-layer connections
- **node_order** – Optional list specifying node ordering; if None, uses sorted order
- **cmap** – Colormap name for the heatmap
- **figsize** – Figure size as (width, height) tuple
- ****kwargs** – Additional arguments for imshow

**Returns**

The created figure

**Return type**

matplotlib.figure.Figure

py3plex.visualization.multilayer.**supra_adjacency_matrix_plot**(*matrix: ndarray*, *display: bool = False*, *ax: Any | None = None*, *cmap: str = 'binary'*) → Any

Plot a supra-adjacency matrix as a heatmap.

Visualizes the supra-adjacency matrix of a multilayer network, where the matrix shows both intra-layer and inter-layer connections. The matrix is displayed as a heatmap with configurable colormap.

**Parameters**

- **matrix** – Supra-adjacency matrix to plot (numpy ndarray or scipy sparse matrix)
- **display** – If True, calls plt.show() after drawing. Default is False to let the caller control rendering.
- **ax** – Matplotlib Axes to draw on. If None, uses current axes (plt.gca())
- **cmap** – Colormap to use for the heatmap (default: “binary”)

**Returns**

Matplotlib Axes object containing the visualization.

### Example

```
>>> import matplotlib.pyplot as plt
>>> from py3plex.visualization import supra_adjacency_matrix_plot
>>> fig, ax = plt.subplots(figsize=(8, 8))
>>> ax = supra_adjacency_matrix_plot(supra_matrix, ax=ax, cmap="viridis")
>>> plt.colorbar(ax.images[0])
>>> plt.savefig("supra_matrix.png")
```

py3plex.visualization.multilayer.**visualize_multilayer_network**(*multilayer_network*, *visualization_type: str = 'diagonal'*, ***kwargs*)

High-level function to visualize multilayer networks with multiple visualization modes.

This function provides a unified interface for various multilayer network visualization techniques, making it easy to switch between different visual representations.

**Parameters**

- **multilayer_network** – A MultiLayerNetwork instance from py3plex.core.multinet
- **visualization_type** – Type of visualization to use. Options: - "diagonal": Default layer-centric diagonal layout (existing behavior) - "small_multiples": One subplot per layer with shared or independent layouts - "edge_colored_projection": Aggregate projection with edge colors by layer - "supra_adjacency_heatmap": Matrix representation of multilayer structure - "radial_layers": Concentric circles for layers with radial inter-layer edges - "ego_multilayer": Ego-centric view focused on a specific node
- ****kwargs** – Additional keyword arguments specific to each visualization type. See individual plot functions for details.

**Returns**

The created figure object

**Return type**

matplotlib.figure.Figure

**Raises**

**ValueError** – If visualization_type is not recognized

### Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.load_network("network.txt", input_type="multiedgelist")
>>>
>>> # Use default diagonal visualization
>>> fig = visualize_multilayer_network(net)
>>>
>>> # Use small multiples view
>>> fig = visualize_multilayer_network(net, visualization_type="small_multiples")
>>>
>>> # Use edge-colored projection
>>> fig = visualize_multilayer_network(net, visualization_type="edge_colored_projection
```

py3plex.visualization.drawing_machinery.**draw**(*G*, *pos=None*, *ax=None*, ***kwds*)

Draw the graph G with Matplotlib.

Draw the graph as a simple representation with no node labels or edge labels and using the full Matplotlib figure area and no axis labels by default. See draw_networkx() for more full-featured drawing that allows title, axis labels etc.

**Parameters**

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary, optional*) – A dictionary with nodes as keys and positions as values. If not specified a spring layout positioning will be computed. See networkx.drawing.layout for functions that compute node positions.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in specified Matplotlib axes.
- **kwds** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords.

### Examples

```
>>> G = nx.dodecahedral_graph()
>>> nx.draw(G)
>>> nx.draw(G, pos=nx.spring_layout(G))  # use spring layout
```

**See also**

draw_networkx, draw_networkx_nodes, draw_networkx_edges, draw_networkx_labels, draw_networkx_edge_labels

### Notes

This function has the same name as pylab.draw and pyplot.draw so beware when using

```
>>> from networkx import *
```

since you might overwrite the pylab.draw function.

With pyplot use

```
>>> import matplotlib.pyplot as plt
>>> import networkx as nx
>>> G = nx.dodecahedral_graph()
>>> nx.draw(G)  # networkx draw()
>>> plt.draw()  # pyplot draw()
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

py3plex.visualization.drawing_machinery.**draw_circular**(*G*, ***kwargs*)

Draw the graph G with a circular layout.

**Parameters**

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_kamada_kawai**(*G*, ***kwargs*)

Draw the graph G with a Kamada-Kawai force-directed layout.

**Parameters**

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_networkx**(*G*, *pos=None*, *arrows=True*, *with_labels=True*, ***kwds*)

Draw the graph G using Matplotlib.

Draw the graph with Matplotlib with options for node positions, labeling, titles, and many other drawing features. See draw() for simple drawing without labels or axes.

**Parameters**

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary, optional*) – A dictionary with nodes as keys and positions as values. If not specified a spring layout positioning will be computed. See `networkx.drawing.layout` for functions that compute node positions.
- **arrows** (*bool, optional (default=True)*) – For directed graphs, if True draw arrowheads. Note: Arrows will be the same color as edges.
- **arrowstyle** (*str, optional (default='-|>')*) – For directed graphs, choose the style of the arrowsheads. See :py:class: *matplotlib.patches.ArrowStyle* for more options.
- **arrowsize** (*int, optional (default=10)*) – For directed graphs, choose the size of the arrow head head's length and width. See :py:class: *matplotlib.patches.FancyArrowPatch* for attribute *mutation_scale* for more info.
- **with_labels** (*bool, optional (default=True)*) – Set to True to draw labels on the nodes.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **nodelist** (*list, optional (default G.nodes())*) – Draw only specified nodes
- **edgelist** (*list, optional (default=G.edges())*) – Draw only specified edges
- **node_size** (*scalar or array, optional (default=300)*) – Size of nodes. If an array is specified it must be the same length as nodelist.
- **node_color** (*color string, or array of floats, (default='r')*) – Node color. Can be a single color format string, or a sequence of colors with the same length as nodelist. If numeric values are specified they will be mapped to colors using the cmap and vmin,vmax parameters. See matplotlib.scatter for more details.
- **node_shape** (*string, optional (default='o')*) – The shape of

the node. Specification is as matplotlib.scatter marker, one of 'so^>v<dph8'.

- **alpha** (*float, optional (default=1.0)*) – The node and edge transparency
- **cmap** (*Matplotlib colormap, optional (default=None)*) – Colormap for mapping intensities of nodes
- **vmin** (*float, optional (default=None)*) – Minimum and maximum for node colormap scaling
- **vmax** (*float, optional (default=None)*) – Minimum and maximum for node colormap scaling
- **linewidths** (*[None | scalar | sequence]*) – Line width of symbol border (default =1.0)
- **width** (*float, optional (default=1.0)*) – Line width of edges
- **edge_color** (*color string, or array of floats (default='r')*) – Edge color. Can be a single color format string, or a sequence of colors with the same length as edgelist. If numeric values are specified they will be mapped to colors using the edge_cmap and edge_vmin,edge_vmax parameters.
- **edge_cmap** (*Matplotlib colormap, optional (default=None)*) – Colormap for mapping intensities of edges
- **edge_vmin** (*floats, optional (default=None)*) – Minimum and maximum for edge colormap scaling
- **edge_vmax** (*floats, optional (default=None)*) – Minimum and maximum for edge colormap scaling
- **style** (*string, optional (default='solid')*) – Edge line style (solid|dashed|dotted,dashdot)
- **labels** (*dictionary, optional (default=None)*) – Node labels in a dictionary keyed by node of text labels
- **font_size** (*int, optional (default=12)*) – Font size for text labels
- **font_color** (*string, optional (default='k' black)*) – Font color string
- **font_weight** (*string, optional (default='normal')*) – Font weight
- **font_family** (*string, optional (default='sans-serif')*) – Font family
- **label** (*string, optional*) – Label for graph legend

### Notes

For directed graphs, arrows are drawn at the head end. Arrows can be turned off with keyword arrows=False.

### Examples

```
>>> G = nx.dodecahedral_graph()
>>> nx.draw(G)
>>> nx.draw(G, pos=nx.spring_layout(G))  # use spring layout
```

```
>>> import matplotlib.pyplot as plt
>>> limits = plt.axis('off')  # turn of axis
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

**See also**

draw, draw_networkx_nodes, draw_networkx_edges, draw_networkx_labels, draw_networkx_edge_labels

py3plex.visualization.drawing_machinery.**draw_networkx_edge_labels**(*G*, *pos*, *edge_labels=None*, *label_pos=0.5*, *font_size=10*, *font_color='k'*, *font_family='sans-serif'*, *font_weight='normal'*, *alpha=1.0*, *bbox=None*, *ax=None*, *rotate=True*, ***kwds*)

Draw edge labels.

**Parameters**

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **alpha** (*float*) – The text transparency (default=1.0)
- **edge_labels** (*dictionary*) – Edge labels in a dictionary keyed by edge two-tuple of text labels (default=None). Only labels for the keys in the dictionary are drawn.
- **label_pos** (*float*) – Position of edge label along edge (0=head, 0.5=center, 1=tail)

- **font_size** (*int*) – Font size for text labels (default=12)
- **font_color** (*string*) – Font color string (default='k' black)
- **font_weight** (*string*) – Font weight (default='normal')
- **font_family** (*string*) – Font family (default='sans-serif')
- **bbox** (*Matplotlib bbox*) – Specify text box shape and colors.
- **clip_on** (*bool*) – Turn on clipping at axis boundaries (default=True)

**Returns**
*dict* of labels keyed on the edges

**Return type**
dict

### Examples

```
>>> G = nx.dodecahedral_graph()
>>> edge_labels = nx.draw_networkx_edge_labels(G, pos=nx.spring_layout(G))
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

**See also**

draw, draw_networkx, draw_networkx_nodes, draw_networkx_edges, draw_networkx_labels

py3plex.visualization.drawing_machinery.**draw_networkx_edges**(*G*, *pos*, *edgelist=None*, *width=1.0*, *edge_color='k'*, *style='solid'*, *alpha=1.0*, *arrowstyle='-|>'*, *arrowsize=10*, *edge_cmap=None*, *edge_vmin=None*, *edge_vmax=None*, *ax=None*, *arrows=True*, *label=None*, *node_size=300*, *nodelist=None*, *node_shape='o'*, ***kwds*)

Draw the edges of the graph G.

This draws only the edges of the graph G.

**Parameters**

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as

values. Positions should be sequences of length 2.

- **edgelist** (*collection of edge tuples*) – Draw only specified edges(default=G.edges())
- **width** (*float, or array of floats*) – Line width of edges (default=1.0)
- **edge_color** (*color string, or array of floats*) – Edge color. Can be a single color format string (default='r'), or a sequence of colors with the same length as edgelist. If numeric values are specified they will be mapped to colors using the edge_cmap and edge_vmin,edge_vmax parameters.
- **style** (*string*) – Edge line style (default='solid') (solid|dashed|dotted,dashdot)
- **alpha** (*float*) – The edge transparency (default=1.0)
- **cmap** (*edge*) – Colormap for mapping intensities of edges (default=None)
- **edge_vmin** (*floats*) – Minimum and maximum for edge colormap scaling (default=None)
- **edge_vmax** (*floats*) – Minimum and maximum for edge colormap scaling (default=None)
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **arrows** (*bool, optional (default=True)*) – For directed graphs, if True draw arrowheads. Note: Arrows will be the same color as edges.
- **arrowstyle** (*str, optional (default='-|>')*) – For directed graphs, choose the style of the arrow heads. See :py:class: *matplotlib.patches.ArrowStyle* for more options.
- **arrowsize** (*int, optional (default=10)*) – For directed graphs, choose the size of the arrow head head's length and width. See :py:class: *matplotlib.patches.FancyArrowPatch* for attribute *mutation_scale* for more info.
- **label** (*[None | string]*) – Label for legend

**Returns**

- *matplotlib.collection.LineCollection* – *LineCollection* of the edges
- *list of matplotlib.patches.FancyArrowPatch* – *FancyArrowPatch* instances of the directed edges
- *Depending whether the drawing includes arrows or not.*

### Notes

For directed graphs, arrows are drawn at the head end. Arrows can be turned off with keyword arrows=False. Be sure to include *node_size* as a keyword argument; arrows are drawn considering the size of nodes.

### Examples

```
>>> G = nx.dodecahedral_graph()
>>> edges = nx.draw_networkx_edges(G, pos=nx.spring_layout(G))
```

```
>>> G = nx.DiGraph()
>>> G.add_edges_from([(1, 2), (1, 3), (2, 3)])
>>> arcs = nx.draw_networkx_edges(G, pos=nx.spring_layout(G))
>>> alphas = [0.3, 0.4, 0.5]
>>> for i, arc in enumerate(arcs):  # change alpha values of arcs
...     arc.set_alpha(alphas[i])
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

**See also**

draw, draw_networkx, draw_networkx_nodes, draw_networkx_labels, draw_networkx_edge_labels

py3plex.visualization.drawing_machinery.**draw_networkx_labels**(*G*, *pos*, *labels=None*, *font_size=1*, *font_color='k'*, *font_family='sans-serif'*, *font_weight='normal'*, *alpha=1.0*, *bbox=None*, *ax=None*, ***kwds*)

Draw node labels on the graph G.

**Parameters**

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **labels** (*dictionary, optional (default=None)*) – Node labels in a dictionary keyed by node of text labels Node-keys in labels should appear as keys in *pos*. If needed use: *{n:lab for n,lab in labels.items() if n in pos}*
- **font_size** (*int*) – Font size for text labels (default=12)
- **font_color** (*string*) – Font color string (default='k' black)
- **font_family** (*string*) – Font family (default='sans-serif')
- **font_weight** (*string*) – Font weight (default='normal')
- **alpha** (*float*) – The text transparency (default=1.0)
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.

**Returns**

*dict* of labels keyed on the nodes

**Return type**
dict

### Examples

```
>>> G = nx.dodecahedral_graph()
>>> labels = nx.draw_networkx_labels(G, pos=nx.spring_layout(G))
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

**See also**

`draw`, `draw_networkx`, `draw_networkx_nodes`, `draw_networkx_edges`, `draw_networkx_edge_labels`

py3plex.visualization.drawing_machinery.**draw_networkx_nodes**(*G*, *pos*, *nodelist=None*, *node_size=300*, *node_color='r'*, *node_shape='o'*, *alpha=1.0*, *cmap=None*, *vmin=None*, *vmax=None*, *ax=None*, *linewidths=None*, *edgecolors=None*, *label=None*, ***kwds*)

Draw the nodes of the graph G.

This draws only the nodes of the graph G.

**Parameters**

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **nodelist** (*list, optional*) – Draw only specified nodes (default G.nodes())
- **node_size** (*scalar or array*) – Size of nodes (default=300). If an array is specified it must be the same length as nodelist.
- **node_color** (*color string, or array of floats*) – Node color. Can be a single color format string (default='r'), or a sequence of colors with the same length as nodelist. If numeric values are specified they will be mapped to colors using the cmap and vmin,vmax parameters. See matplotlib.scatter for more details.
- **node_shape** (*string*) – The shape of the node. Specification is as matplotlib.scatter marker, one of 'so^>v<dph8' (default='o').

- **alpha** (*float or array of floats*) – The node transparency. This can be a single alpha value (default=1.0), in which case it will be applied to all the nodes of color. Otherwise, if it is an array, the elements of alpha will be applied to the colors in order (cycling through alpha multiple times if necessary).
- **cmap** (*Matplotlib colormap*) – Colormap for mapping intensities of nodes (default=None)
- **vmin** (*floats*) – Minimum and maximum for node colormap scaling (default=None)
- **vmax** (*floats*) – Minimum and maximum for node colormap scaling (default=None)
- **linewidths** (*[None | scalar | sequence]*) – Line width of symbol border (default =1.0)
- **edgecolors** (*[None | scalar | sequence]*) – Colors of node borders (default = node_color)
- **label** (*[None | string]*) – Label for legend

**Returns**
*PathCollection* of the nodes.

**Return type**
matplotlib.collections.PathCollection

### Examples

```
>>> G = nx.dodecahedral_graph()
>>> nodes = nx.draw_networkx_nodes(G, pos=nx.spring_layout(G))
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

**See also**

`draw`, `draw_networkx`, `draw_networkx_edges`, `draw_networkx_labels`, `draw_networkx_edge_labels`

py3plex.visualization.drawing_machinery.**draw_random**(*G*, ***kwargs*)

Draw the graph G with a random layout.

**Parameters**

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_shell**(*G*, ***kwargs*)

Draw networkx graph with shell layout.

**Parameters**

- **G** (*graph*) – A networkx graph

- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_spectral**(*G*, ***kwargs*)

Draw the graph G with a spectral layout.

**Parameters**

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_spring**(*G*, ***kwargs*)

Draw the graph G with a spring layout.

**Parameters**

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

Color utilities for py3plex visualization.

py3plex.visualization.colors.**RGB_to_hex**(*RGB: List[int]*) → str

Convert RGB list to hex color.

**Parameters**
**RGB** – RGB values as [R, G, B] list

**Returns**
Hex color string like "#FFFFFF"

py3plex.visualization.colors.**color_dict**(*gradient: List[List[int]]*) → Dict[str, List]

Takes in a list of RGB sub-lists and returns dictionary of colors in RGB and hex form for use in a graphing function defined later on.

**Parameters**
**gradient** – List of RGB color values

**Returns**
Dictionary with 'hex', 'r', 'g', 'b' keys

py3plex.visualization.colors.**hex_to_RGB**(*hex: str*) → List[int]

Convert hex color to RGB list.

**Parameters**
**hex** – Hex color string like "#FFFFFF"

**Returns**
RGB values as [R, G, B] list

py3plex.visualization.colors.**linear_gradient**(*start_hex: str*, *finish_hex: str = '#FFFFFF'*, *n: int = 10*) → Dict[str, List]

Returns a gradient list of (n) colors between two hex colors.

**Parameters**

- **start_hex** – Starting color as six-digit hex string (e.g., "#FFFFFF")
- **finish_hex** – Ending color as six-digit hex string (default: "#FFFFFF")
- **n** – Number of colors in gradient (default: 10)

**Returns**

Dictionary with 'hex', 'r', 'g', 'b' keys containing gradient colors

py3plex.visualization.layout_algorithms.**compute_force_directed_layout**(*g: Graph*, *layout_parameters: Dict[str, Any] | None = None*, *verbose: bool = True*, *gravity: float = 0.2*, *strongGravityMode: bool = False*, *barnesHutTheta: float = 1.2*, *edgeWeightInfluence: float = 1*, *scalingRatio: float = 2.0*, *forceImport: bool = True*, *seed: int | None = None*) → Dict[Any, ndarray]

Compute force-directed layout for a graph using ForceAtlas2 or NetworkX spring layout.

**Parameters**

- **g** – NetworkX graph to layout
- **layout_parameters** – Optional parameters to pass to layout algorithm
- **verbose** – Whether to print progress information

- **gravity** – Attraction force towards the center (must be non-negative)
- **strongGravityMode** – Use strong gravity mode
- **barnesHutTheta** – Barnes-Hut approximation parameter
- **edgeWeightInfluence** – Influence of edge weights on layout
- **scalingRatio** – Scaling factor for the layout (must be positive)
- **forceImport** – Whether to use ForceAtlas2 (if available)
- **seed** – Random seed for reproducibility in fallback spring layout

**Returns**

Dictionary mapping nodes to 2D position arrays

**Note**

For large networks (>1000 nodes), this may be slow. Consider using faster layouts (circular, random, spectral) or matrix visualization.

**Contracts:**

- Precondition: graph must not be None and be a NetworkX graph
- Precondition: graph must have at least one node
- Precondition: gravity must be non-negative
- Precondition: scalingRatio must be positive
- Postcondition: result is a dictionary
- Postcondition: result has positions for all nodes

py3plex.visualization.layout_algorithms.**compute_random_layout**(*g: Graph*, *seed: int | None = None*) → Dict[Any, ndarray]

Compute a random layout for the graph.

**Parameters**

- **g** – NetworkX graph
- **seed** – Random seed for reproducibility

**Returns**

Dictionary mapping nodes to 2D positions

**Contracts:**

- Precondition: graph must not be None and be a NetworkX graph
- Precondition: graph must have at least one node
- Postcondition: result is a dictionary
- Postcondition: result has positions for all nodes

py3plex.visualization.layout_algorithms.**ensure**(**args*, ***kwargs*)

py3plex.visualization.layout_algorithms.**require**(**args*, ***kwargs*)

py3plex.visualization.bezier.**bezier_calculate_dfy**(*mp_y: float*, *path_height: float*, *x0: float*, *midpoint_x: float*, *x1: float*, *y0: float*, *y1: float*, *dfx: ndarray*, *mode: str = 'upper'*) → ndarray

Calculate y-coordinates for bezier curve.

**Parameters**

- **mp_y** – Midpoint y-coordinate
- **path_height** – Height of the path
- **x0** – Start x-coordinate
- **midpoint_x** – Midpoint x-coordinate
- **x1** – End x-coordinate
- **y0** – Start y-coordinate
- **y1** – End y-coordinate
- **dfx** – Array of x-coordinates
- **mode** – Mode for curve calculation (“upper” or “bottom”)

**Returns**

Array of y-coordinates

py3plex.visualization.bezier.**draw_bezier**(*total_size: int*, *p1: Tuple[float, float]*, *p2: Tuple[float, float]*, *mode: str = 'quadratic'*, *inversion: bool = False*, *path_height: float = 2*, *linemode: str = 'both'*, *resolution: float = 0.1*) → Tuple[ndarray, ndarray]

Draw bezier curve between two points.

**Parameters**

- **total_size** – Total size of the drawing area
- **p1** – First point coordinates (x0, x1)
- **p2** – Second point coordinates (y0, y1)
- **mode** – Drawing mode (default: “quadratic”)
- **inversion** – Whether to invert the curve
- **path_height** – Height of the path
- **linemode** – Line drawing mode (“upper”, “bottom”, or “both”)
- **resolution** – Resolution for curve sampling

**Returns**

Tuple of (x-coordinates, y-coordinates) arrays

py3plex.visualization.benchmark_visualizations.**generic_grouping**(*fname: DataFrame*, *score_name: str*, *threshold: float = 1.0*, *percentages: bool = True*) → DataFrame

py3plex.visualization.benchmark_visualizations.**plot_core_macro**(*fname: DataFrame*) → int

A very simple visualization of the results..

py3plex.visualization.benchmark_visualizations.**plot_core_macro_box**(*fname: str*) → int

py3plex.visualization.benchmark_visualizations.**plot_core_macro_gg**(*fnamex: DataFrame*) → None

py3plex.visualization.benchmark_visualizations.**plot_core_micro**(*fname: DataFrame*) → int

A very simple visualization of the results..

py3plex.visualization.benchmark_visualizations.**plot_core_micro_gg**(*fnamex: DataFrame*) → None

py3plex.visualization.benchmark_visualizations.**plot_core_micro_grid**(*fname: str*) → None

py3plex.visualization.benchmark_visualizations.**plot_core_time**(*fnamex: DataFrame*) → int

py3plex.visualization.benchmark_visualizations.**plot_core_time_gg**(*fname: str*) → None

py3plex.visualization.benchmark_visualizations.**plot_core_variability**(*fname: str*) → None

py3plex.visualization.benchmark_visualizations.**plot_critical_distance**(*fname: str*, *num_algo: int = 14*) → None

py3plex.visualization.benchmark_visualizations.**plot_mean_times**(*fn: DataFrame*) → None

py3plex.visualization.benchmark_visualizations.**plot_robustness**(*infile: DataFrame*) → None

py3plex.visualization.benchmark_visualizations.**table_to_latex**(*fname*, *outfolder='../final_results/tables/'*, *threshold=1*)

### Wrappers

Convenience wrappers for embedding and benchmarking workflows.

class py3plex.wrappers.benchmark_nodes.**TopKRanker**(*estimator*, *, *n_jobs=None*, *verbose=0*)

Bases: OneVsRestClassifier

**predict**(*X: ndarray*, *top_k_list: List[int]*) → List[List]

Predict top K labels for each sample.

**Parameters**

- **X** – Feature matrix
- **top_k_list** – List of K values for each sample

**Returns**

List of predicted labels for each sample

**set_partial_fit_request**(*, *classes: bool | None | str = '$UNCHANGED$'*) → TopKRanker

Configure whether metadata should be requested to be passed to the partial_fit method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with enable_metadata_routing=True (see sklearn.set_config()). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- True: metadata is requested, and passed to partial_fit if provided. The request is ignored if metadata is not provided.
- False: metadata is not requested and the meta-estimator will not pass it to partial_fit.
- None: metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- str: metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (sklearn.utils.metadata_routing.UNCHANGED) retains the existing request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

**Parameters**

**classes** (*str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*) – Metadata routing for classes parameter in partial_fit.

**Returns**

**self** – The updated object.

**Return type**

object

**set_predict_request**(*, *top_k_list: bool | None | str = '$UNCHANGED$'*) → TopKRanker

Configure whether metadata should be requested to be passed to the `predict` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config()`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True`: metadata is requested, and passed to `predict` if provided. The request is ignored if metadata is not provided.
- `False`: metadata is not requested and the meta-estimator will not pass it to `predict`.
- `None`: metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str`: metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

**Parameters**
`top_k_list` (*str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*) – Metadata routing for `top_k_list` parameter in `predict`.

**Returns**
**self** – The updated object.

**Return type**
object

`set_score_request`(*, *sample_weight: bool | None | str = '$UNCHANGED$'*) → TopKRanker

Configure whether metadata should be requested to be passed to the `score` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config()`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True`: metadata is requested, and passed to `score` if provided. The request is ignored if metadata is not provided.
- `False`: metadata is not requested and the meta-estimator will not pass it to `score`.
- `None`: metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str`: metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing

request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

**Parameters**

**sample_weight** (*str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*) – Metadata routing for `sample_weight` parameter in `score`.

**Returns**

**self** – The updated object.

**Return type**

object

`py3plex.wrappers.benchmark_nodes.benchmark_node_classification`(*path: str*, *core_network: Any*, *labels_matrix: Any*, *percent: Any = 'all'*) → Dict[Any, Any]

Benchmark node classification using embeddings.

**Parameters**

- **path** – Path to embeddings file
- **core_network** – Network adjacency matrix
- **labels_matrix** – Labels for nodes
- **percent** – Training percentage or "all" for multiple percentages

**Returns**

Dictionary of classification results

`py3plex.wrappers.benchmark_nodes.main`()

`py3plex.wrappers.benchmark_nodes.sparse2graph`(*x: spmatrix*) → Dict[str, List[str]]

Convert sparse matrix to graph dictionary.

**Parameters**

**x** – Sparse matrix representation of graph

**Returns**

Dictionary mapping node IDs to lists of neighbor IDs

### I/O Operations

File readers, writers, and related helpers.

### Aggregation and Network Operations

Layer aggregation and network-level transformations.

Vectorized multiplex aggregation for multilayer networks.

This module provides optimized implementations for aggregating edges across multiple layers using vectorized NumPy and SciPy sparse operations, replacing slower Python loops with efficient matrix operations.

Performance targets: - 3× speedup for 1M edges across 4 layers compared to legacy loop-based methods - Memory-efficient sparse matrix output by default - Float tolerance of 1e-6 for numerical equivalence with legacy methods

Author: py3plex development team License: MIT

py3plex.multinet.aggregation.**aggregate_layers**(*edges: ndarray | list*, *weight_col: str | int = 'w'*, *reducer: Literal['sum', 'mean', 'max'] = 'sum'*, *to_sparse: bool = True*) → csr_matrix | ndarray

Aggregate edge weights across multiple layers using vectorized operations.

This function replaces Python loops with efficient NumPy and SciPy sparse matrix operations for superior performance on large multilayer networks.

**Complexity**: O(E) where E is the number of edges **Memory**: O(E) for sparse output, $O(N^2)$ for dense output

**Parameters**

- **edges** – Edge data as ndarray with shape (E, >=3) containing (layer, src, dst, [weight]) columns. If no weight column, assumes weight=1.0 for all edges. Can also accept list of lists.
- **weight_col** – Column name or index for weights (default "w"). If edges is ndarray, must be int index (0-based). Column 3 is used if it exists, else weights default to 1.
- **reducer** – Aggregation method - one of: - "sum": Sum weights for edges appearing in multiple layers (default) - "mean": Average weights across layers - "max": Take maximum weight across layers
- **to_sparse** – If True, return scipy.sparse.csr_matrix (default, memory-efficient). If False, return dense numpy.ndarray.

**Returns**

Aggregated adjacency matrix in requested format (sparse CSR or dense). Shape is (N, N) where N is the maximum node ID + 1.

**Raises**

- **ValueError** – If edges array has wrong shape or reducer is invalid.
- **TypeError** – If edges is not ndarray or list-like.

### Examples

```
>>> import numpy as np
>>> # Create edge list: (layer, src, dst, weight)
>>> edges = np.array([
...     [0, 0, 1, 1.0],
...     [0, 1, 2, 2.0],
...     [1, 0, 1, 0.5],  # Same edge in layer 1
...     [1, 2, 3, 1.5],
... ])
>>> mat = aggregate_layers(edges, reducer="sum")
>>> mat.shape
(4, 4)
>>> mat[0, 1]  # Sum of weights from both layers
```

```
1.5
```

```
>>> # With mean aggregation
>>> mat_mean = aggregate_layers(edges, reducer="mean", to_sparse=False)
>>> mat_mean[0, 1]  # Average of 1.0 and 0.5
0.75
```

#### Notes

- Node IDs are assumed to be integers starting from 0
- Self-loops are supported
- For directed graphs, (i,j) and (j,i) are different edges
- Sparse output recommended for large networks (N > 1000)
- Deterministic output for fixed input order

**Performance:**

Achieves 3× speedup vs loop-based aggregation on 1M edges: - Legacy loop: ~2.5s for 1M edges, 4 layers - Vectorized: ~0.8s for same dataset (measured on standard hardware)

py3plex.multinet.aggregation.**ensure**(**args*, ***kwargs*)

py3plex.multinet.aggregation.**require**(**args*, ***kwargs*)

### Profiling Utilities

Timing and profiling utilities for performance analysis.

Performance profiling utilities for py3plex.

This module provides decorators and utilities for tracking function execution time, memory usage, and performance metrics. These tools enable performance regression detection and optimization efforts.

#### Example

```
>>> from py3plex.profiling import profile_performance
>>>
>>> @profile_performance
>>> def slow_function():
...     # ... computation
...     pass
>>>
>>> slow_function()  # Logs execution time automatically
```

**class** py3plex.profiling.**PerformanceMonitor**

Bases: `object`

Global performance monitoring registry.

Tracks execution times and call counts for profiled functions. Can be used to generate performance reports and detect regressions.

**enabled**

Whether profiling is enabled globally

**stats**

Dictionary mapping function names to performance statistics

**clear()**

Clear all collected statistics.

**get_report()** → str

Generate a performance report.

**Returns**

String containing formatted performance statistics

**record**(*func_name: str*, *elapsed: float*, *memory_delta: float | None = None*)

Record performance metrics for a function call.

**Parameters**

- **func_name** – Name of the function
- **elapsed** – Execution time in seconds
- **memory_delta** – Memory usage change in MB (optional)

py3plex.profiling.**benchmark**(*func: Callable*, *iterations: int = 100*, *warmup: int = 10*, *args: tuple = ()*, *kwargs: dict | None = None*) → Dict[str, float]

Benchmark a function with multiple iterations.

Runs the function multiple times and collects statistics about execution time. Includes a warmup phase to allow JIT compilation and caching.

**Parameters**

- **func** – Function to benchmark
- **iterations** – Number of iterations to run (default: 100)
- **warmup** – Number of warmup iterations (default: 10)
- **args** – Positional arguments for the function
- **kwargs** – Keyword arguments for the function

**Returns**

- mean: Average execution time (seconds)
- median: Median execution time (seconds)
- min: Minimum execution time (seconds)
- max: Maximum execution time (seconds)
- std: Standard deviation (seconds)
- total: Total time for all iterations (seconds)

**Return type**

Dictionary containing benchmark statistics

### Example

```
>>> from py3plex.profiling import benchmark
>>>
>>> def my_function(n):
...     return sum(range(n))
>>>
>>> stats = benchmark(my_function, iterations=1000, args=(1000,))
>>> print(f"Average time: {stats['mean']*1000:.3f}ms")
```

py3plex.profiling.**get_monitor**() → PerformanceMonitor

Get the global performance monitor instance.

**Returns**
Global performance monitoring instance

**Return type**
PerformanceMonitor

### Example

```
>>> from py3plex.profiling import get_monitor
>>> monitor = get_monitor()
>>> print(monitor.get_report())
```

py3plex.profiling.**profile_performance**(*func: Callable | None = None*, ***, *log_args: bool = False*, *track_memory: bool = False*) → Callable

Decorator to track function execution time and optionally memory usage.

This decorator measures the wall-clock time taken by a function and logs it. It can also track memory usage changes if requested. Performance metrics are stored in the global performance monitor for later analysis.

**Parameters**

- **func** – Function to decorate (when used without arguments)
- **log_args** – If True, log function arguments (default: False)
- **track_memory** – If True, track memory usage (default: False)

**Returns**
Decorated function that logs execution time

### Examples

Basic usage: >>> @profile_performance ... def my_function(x, y): ... return x + y

With options: >>> @profile_performance(log_args=True, track_memory=True) ... def expensive_function(data): ... # ... expensive computation ... return result

Manual wrapping: >>> def my_function(): ... pass >>> profiled_func = profile_performance(my_function)

**Note**

Memory tracking requires the tracemalloc module and adds overhead. Use it only when investigating memory issues.

py3plex.profiling.**timed_section**(*name: str*, *log_level: str = 'info'*)

Context manager for timing code blocks.

Useful for timing specific sections of code without wrapping entire functions.

**Parameters**

- **name** – Name of the code section being timed
- **log_level** – Logging level ('debug', 'info', 'warning', 'error')

### Example

```
>>> from py3plex.profiling import timed_section
>>>
>>> with timed_section("data loading"):
...     data = load_large_dataset()
>>>
>>> with timed_section("computation", log_level="debug"):
...     result = complex_calculation()
```

**Yields**

None

## Logging Configuration

Default logging setup for py3plex.

Logging configuration for py3plex.

This module provides centralized logging configuration for the py3plex library.

py3plex.logging_config.**get_logger**(*name: str | None = None*, *level: int = 20*) → Logger

Get or create a logger for py3plex modules.

**Parameters**

- **name** – Logger name. If None, returns the root py3plex logger.
- **level** – Logging level (default: INFO)

**Returns**

Configured logger instance

### Example

```
>>> from py3plex.logging_config import get_logger
>>> logger = get_logger(__name__)
>>> logger.info("Processing network...")
```

py3plex.logging_config.**setup_logging**(*level: int = 20*, *format_string: str | None = None*) → Logger

Configure logging for py3plex with custom settings.

**Parameters**

- **level** – Logging level (default: INFO)
- **format_string** – Custom format string (optional)

**Returns**
Root py3plex logger

### Example

```
>>> from py3plex.logging_config import setup_logging
>>> logger = setup_logging(level=logging.DEBUG)
```

## I/O Schema and Validation

Schema definitions and validation helpers for I/O routines.

Schema definitions for multilayer graphs using dataclasses.

This module provides dataclass representations of multilayer graph components with built-in validation and serialization support.

**class** py3plex.io.schema.**Edge**(*src: ~typing.Hashable*, *dst: ~typing.Hashable*, *src_layer: ~typing.Hashable*, *dst_layer: ~typing.Hashable*, *key: int = 0*, *attributes: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: `object`

Represents an edge in a multilayer network.

**src**
Source node ID

**Type**
Hashable

**dst**
Destination node ID

**Type**
Hashable

**src_layer**
Source layer ID

**Type**
Hashable

**dst_layer**
Destination layer ID

**Type**
Hashable

**key**
Optional edge key for multigraphs (default: 0)

**Type**
int

**attributes**
Dictionary of edge attributes (must be JSON-serializable)

**Type**
Dict[str, Any]

`attributes: Dict[str, Any]`

`dst: Hashable`

`dst_layer: Hashable`

`edge_tuple()` → Tuple[Hashable, Hashable, Hashable, Hashable, int]

Return edge as a tuple for uniqueness checking.

**Returns**
Tuple of (src, dst, src_layer, dst_layer, key)

`classmethod from_dict(`*data: Dict[str, Any]*`)` → Edge

Create edge from dictionary.

**Parameters**
**data** – Dictionary containing edge data

**Returns**
Edge instance

`key: int = 0`

`src: Hashable`

`src_layer: Hashable`

`to_dict()` → Dict[str, Any]

Convert edge to dictionary.

**Returns**
Dictionary representation of the edge

`class py3plex.io.schema.Layer(`*id: ~typing.Hashable*, *attributes: ~typing.Dict[str, ~typing.Any] = <factory>*`)`

Bases: `object`

Represents a layer in a multilayer network.

`id`

Unique identifier for the layer

**Type**
Hashable

`attributes`

Dictionary of layer attributes (must be JSON-serializable)

**Type**
Dict[str, Any]

`attributes: Dict[str, Any]`

`classmethod from_dict(`*data: Dict[str, Any]*`)` → Layer

Create layer from dictionary.

**Parameters**
**data** – Dictionary containing layer data

**Returns**
Layer instance

`id: Hashable`

`to_dict() → Dict[str, Any]`

Convert layer to dictionary.

**Returns**
Dictionary representation of the layer

`class py3plex.io.schema.MultiLayerGraph(`*nodes: ~typing.Dict[~typing.Hashable, ~py3plex.io.schema.Node] = <factory>, layers: ~typing.Dict[~typing.Hashable, ~py3plex.io.schema.Layer] = <factory>, edges: ~typing.List[~py3plex.io.schema.Edge] = <factory>, directed: bool = True, attributes: ~typing.Dict[str, ~typing.Any] = <factory>*`)`

Bases: `object`

Represents a complete multilayer graph.

`nodes`

Dictionary mapping node IDs to Node objects

**Type**
Dict[Hashable, py3plex.io.schema.Node]

`layers`

Dictionary mapping layer IDs to Layer objects

**Type**
Dict[Hashable, py3plex.io.schema.Layer]

`edges`

List of Edge objects

**Type**
List[py3plex.io.schema.Edge]

`directed`

Whether the graph is directed

**Type**
bool

`attributes`

Dictionary of graph-level attributes

**Type**
Dict[str, Any]

`add_edge(`*edge: Edge*`)`

Add an edge to the graph.

**Parameters**
`edge` – Edge to add

**Raises**

- **ReferentialIntegrityError** – If edge references non-existent nodes or layers
- **SchemaValidationError** – If edge is duplicate

**add_layer**(*layer: Layer*)

Add a layer to the graph.

**Parameters**
**layer** – Layer to add

**Raises**
**SchemaValidationError** – If layer ID already exists

**add_node**(*node: Node*)

Add a node to the graph.

**Parameters**
**node** – Node to add

**Raises**
**SchemaValidationError** – If node ID already exists

**attributes: Dict[str, Any]**

**directed: bool = True**

**edges: List[Edge]**

**classmethod from_dict**(*data: Dict[str, Any]*) → MultiLayerGraph

Create graph from dictionary.

**Parameters**
**data** – Dictionary containing graph data

**Returns**
MultiLayerGraph instance

**layers: Dict[Hashable, Layer]**

**nodes: Dict[Hashable, Node]**

**to_dict**() → Dict[str, Any]

Convert graph to dictionary.

**Returns**
Dictionary representation of the graph

**class py3plex.io.schema.Node**(*id: ~typing.Hashable*, *attributes: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Represents a node in a multilayer network.

**id**

Unique identifier for the node

**Type**
Hashable

**attributes**

Dictionary of node attributes (must be JSON-serializable)

**Type**
Dict[str, Any]

`attributes: Dict[str, Any]`

`classmethod from_dict(`*data: Dict[str, Any]*`)` → Node

Create node from dictionary.

**Parameters**
**data** – Dictionary containing node data

**Returns**
Node instance

`id: Hashable`

`to_dict()` → Dict[str, Any]

Convert node to dictionary.

**Returns**
Dictionary representation of the node

Public API for reading and writing multilayer graphs.

This module provides the main entry points for I/O operations with format detection and a registry system for extensibility.

`py3plex.io.api.ensure(`**args*, ***kwargs*`)`

`py3plex.io.api.read(`*filepath: str | Path*, *format: str | None = None*, ***kwargs*`)` → MultiLayerGraph

Read a multilayer graph from a file.

**Parameters**

- **filepath** – Path to the input file
- **format** – Format name (e.g., 'json', 'csv'). If None, auto-detected from extension
- ****kwargs** – Additional arguments passed to the format-specific reader

**Returns**
MultiLayerGraph instance

**Raises**

- **FormatUnsupportedError** – If format is not supported or cannot be detected
- **FileNotFoundError** – If file does not exist

### Example

```
>>> graph = read('network.json')
>>> graph = read('network.csv', format='csv')
```

`py3plex.io.api.register_reader(`*format_name: str*, *reader_func: Callable[[...], MultiLayerGraph]*`)` → None

Register a reader function for a specific format.

**Parameters**

- **format_name** – Name of the format (e.g., 'json', 'csv', 'graphml')
- **reader_func** – Function that takes (filepath, **kwargs) and returns MultiLayerGraph

#### Example

```
>>> def my_reader(filepath, **kwargs):
...     # Custom reading logic
...     return MultiLayerGraph(...)
>>> register_reader('myformat', my_reader)
```

**Contracts:**

- Precondition: format_name must be a non-empty string
- Precondition: reader_func must be callable
- Postcondition: reader is registered in _READERS

py3plex.io.api.**register_writer**(*format_name: str*, *writer_func: Callable[[...], None]*) → None

Register a writer function for a specific format.

**Parameters**

- **format_name** – Name of the format (e.g., 'json', 'csv', 'graphml')
- **writer_func** – Function that takes (graph, filepath, **kwargs) and writes to file

#### Example

```
>>> def my_writer(graph, filepath, **kwargs):
...     # Custom writing logic
...     pass
>>> register_writer('myformat', my_writer)
```

**Contracts:**

- Precondition: format_name must be a non-empty string
- Precondition: writer_func must be callable
- Postcondition: writer is registered in _WRITERS

py3plex.io.api.**require**(**args*, ***kwargs*)

py3plex.io.api.**supported_formats**(*read: bool = True*, *write: bool = True*) → Dict[str, List[str]]

Get list of supported formats for read and/or write operations.

**Parameters**

- **read** – Include formats that support reading
- **write** – Include formats that support writing

**Returns**

Dictionary with 'read' and/or 'write' keys containing lists of format names

#### Example

```
>>> formats = supported_formats()
>>> print(formats)
{'read': ['json', 'jsonl', 'csv'], 'write': ['json', 'jsonl', 'csv']}
```

**Contracts:**

- Postcondition: result is a dictionary
- Postcondition: result contains 'read' key when read=True
- Postcondition: result contains 'write' key when write=True

py3plex.io.api.**write**(*graph: MultiLayerGraph*, *filepath: str | Path*, *format: str | None = None*, ***kwargs*) → None

Write a multilayer graph to a file.

**Parameters**

- **graph** – MultiLayerGraph to write
- **filepath** – Path to the output file
- **format** – Format name (e.g., 'json', 'csv'). If None, auto-detected from extension
- ****kwargs** – Additional arguments passed to the format-specific writer

**Raises**

**FormatUnsupportedError** – If format is not supported or cannot be detected

#### Example

```
>>> write(graph, 'network.json')
>>> write(graph, 'network.csv', format='csv', deterministic=True)
```

### Hedwig Rule Learning

Hedwig inductive logic programming components and helpers.

py3plex.algorithms.hedwig.**build_graph**(*kwargs: Dict[str, Any]*) → Any

py3plex.algorithms.hedwig.**generate_rules_report**(*kwargs: ~typing.Dict[str, ~typing.Any]*, *rules_per_target: ~typing.List[~typing.Tuple[~typing.Any, ~typing.List[~typing.Any]]]*, *human: ~typing.Callable[[~typing.Any, ~typing.Any], ~typing.Any] = <function <lambda>>*) → str

py3plex.algorithms.hedwig.**rule_kernel**(*target: Any*) → Tuple[Any, List[Any]]

py3plex.algorithms.hedwig.**run**(*kwargs: Dict[str, Any]*, *cli: bool = True*, *generator_tag: bool = False*, *num_threads: str | int = 'all'*) → List[Tuple[Any, List[Any]]]

py3plex.algorithms.hedwig.**run_learner**(*kwargs: Dict[str, Any]*, *kb: ExperimentKB*, *validator: Validate*, *generator: bool = False*, *num_threads: str | int = 'all'*) → List[Tuple[Any, List[Any]]]

Example-related classes.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.core.example.**Example**(*id*, *label*, *score*, *annotations=None*, *weights=None*)

Bases: `object`

Represents an example with its score, label, id and annotations.

**ClassLabeled = 'class'**

**Ranked = 'ranked'**

Predicate-related classes.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.core.predicate.**BinaryPredicate**(*label*, *pairs*, *kb*, *producer_pred=None*)

Bases: `Predicate`

A binary predicate.

**class** py3plex.algorithms.hedwig.core.predicate.**Predicate**(*label*, *kb*, *producer_pred*)

Bases: `object`

Represents a predicate as a member of a certain rule.

**i = -1**

**class** py3plex.algorithms.hedwig.core.predicate.**UnaryPredicate**(*label*, *members*, *kb*, *producer_pred=None*, *custom_var_name=None*, *negated=False*)

Bases: `Predicate`

A unary predicate.

The rule class.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.core.rule.**Rule**(*kb*, *predicates=None*, *target=None*)

Bases: `object`

Represents a rule, along with its description, examples and statistics.

**clone**()

Returns a clone of this rule. The predicates themselves are NOT cloned.

**clone_append**(*predicate_label*, *producer_pred*, *bin=False*)

Returns a copy of this rule where 'predicate_label' is appended to the rule.

**clone_negate**(*target_pred*)

Returns a copy of this rule where 'taget_pred' is negated.

**clone_swap_with_subclass**(*target_pred*, *child_pred_label*)

Returns a copy of this rule where 'target_pred' is swapped for 'child_pred_label'.

**examples**(*positive_only=False*)

Returns the covered examples.

**property positives**

**precision**()

**rule_report**(*show_uris=False*, *latex=False*)

Rule as string with some statistics.

**static ruleset_examples_json**(*rules_per_target*, *show_uris=False*)

**static ruleset_report**(*rules*, *show_uris=False*, *latex=False*, *human=<function Rule.<lambda>>*)

**similarity**(*rule*)

Calculates the similarity between this rule and 'rule'.

**size**()

Returns the number of conjunts.

**static to_json**(*rules_per_target*, *show_uris=False*)

### Force Atlas 2 Visualization

ForceAtlas2 layout implementation and utilities.

**class** py3plex.visualization.fa2.forceatlas2.**ForceAtlas2**(*outboundAttractionDistribution=False*, *linLogMode=False*, *adjustSizes=False*, *edgeWeightInfluence=1.0*, *jitterTolerance=1.0*, *barnesHutOptimize=True*, *barnesHutTheta=1.2*, *multiThreaded=False*, *scalingRatio=2.0*, *strongGravityMode=False*, *gravity=1.0*, *verbose=True*)

Bases: object

**forceatlas2**(*G*, *pos=None*, *iterations=30*)

**forceatlas2_igraph_layout**(*G*, *pos=None*, *iterations=100*, *weight_attr=None*)

**forceatlas2_networkx_layout**(*G*, *pos=None*, *iterations=100*)

**init**(*G*, *pos=None*)

**class** py3plex.visualization.fa2.forceatlas2.**Timer**(*name='Timer'*)

Bases: object

**display**()

**start**()

**stop**()

class py3plex.visualization.fa2.fa2util.**Edge**

Bases: object

class py3plex.visualization.fa2.fa2util.**Node**

Bases: object

class py3plex.visualization.fa2.fa2util.**Region**(*nodes*)

Bases: object

**applyForce**(*n*, *theta*, *coefficient=0*)

**applyForceOnNodes**(*nodes*, *theta*, *coefficient=0*)

**buildSubRegions**()

**updateMassAndGeometry**()

py3plex.visualization.fa2.fa2util.**adjustSpeedAndApplyForces**(*nodes*, *speed*, *speedEfficiency*, *jitterTolerance*)

py3plex.visualization.fa2.fa2util.**apply_attraction**(*nodes*, *edges*, *distributedAttraction*, *coefficient*, *edgeWeightInfluence*)

py3plex.visualization.fa2.fa2util.**apply_gravity**(*nodes*, *gravity*, *useStrongGravity=False*)

py3plex.visualization.fa2.fa2util.**apply_repulsion**(*nodes*, *coefficient*)

py3plex.visualization.fa2.fa2util.**linAttraction**(*n1*, *n2*, *e*, *distributedAttraction*, *coefficient=0*)

py3plex.visualization.fa2.fa2util.**linGravity**(*n*, *g*)

py3plex.visualization.fa2.fa2util.**linRepulsion**(*n1*, *n2*, *coefficient=0*)

py3plex.visualization.fa2.fa2util.**linRepulsion_region**(*n*, *r*, *coefficient=0*)

py3plex.visualization.fa2.fa2util.**strongGravity**(*n*, *g*, *coefficient=0*)

### Embedding Visualization

Helpers for visualizing graph embeddings.

py3plex.visualization.embedding_visualization.embedding_visualization.**visualize_embedding**(*m*
*la*
*be*
*ve*
*bo*

### Network Generation and Benchmarking

Synthetic network generation and benchmarking utilities.

### Additional Statistics and Analysis

Supplementary statistical tests and information-theoretic utilities.

py3plex.algorithms.statistics.bayesiantests.**correlated_ttest**(*x*, *rope*, *runs=1*, *verbose=False*, *names=('C1', 'C2')*)

py3plex.algorithms.statistics.bayesiantests.**correlated_ttest_MC**(*x*, *rope*, *runs=1*, *nsamples=50000*)

See correlated_ttest module for explanations

py3plex.algorithms.statistics.bayesiantests.**heaviside**(*X*)

Compute the Heaviside step function.

The Heaviside function returns 1 for positive values, 0.5 for zero, and 0 for negative values. This is used in signed-rank tests for Bayesian comparisons.

**Parameters**

**X** – Input array or scalar

**Returns**

**Array with same shape as X containing:**

- 1.0 where X > 0
- 0.5 where X == 0
- 0.0 where X < 0

**Return type**

np.ndarray

#### Examples

```
>>> heaviside(np.array([-1, 0, 1, 2]))
array([0. , 0.5, 1. , 1. ])
```

py3plex.algorithms.statistics.bayesiantests.**hierarchical**(*diff*, *rope*, *rho*, *upperAlpha=2*, *lowerAlpha=1*, *lowerBeta=0.01*, *upperBeta=0.1*, *std_upper_bound=1000*, *verbose=False*, *names=('C1', 'C2')*)

Perform hierarchical Bayesian test for comparing algorithms across multiple datasets.

This test accounts for the hierarchical structure of the data (multiple datasets, each with multiple folds) and correlations due to overlapping training sets.

**Parameters**

- **diff** – Array of differences between classifier scores

- **rope** – Width of the region of practical equivalence (ROPE)
- **rho** – Correlation between folds (typically around 1/n_folds)
- **upperAlpha** – Upper bound for alpha parameter of Gamma prior (default: 2)
- **lowerAlpha** – Lower bound for alpha parameter of Gamma prior (default: 1)
- **lowerBeta** – Lower bound for beta parameter of Gamma prior (default: 0.01)
- **upperBeta** – Upper bound for beta parameter of Gamma prior (default: 0.1)
- **std_upper_bound** – Upper bound multiplier for standard deviation prior (default: 1000) Posterior is insensitive to this if large enough (>100)
- **verbose** – Whether to print probability results (default: False)
- **names** – Tuple of classifier names for verbose output (default: ("C1", "C2"))

**Returns**

**(p_left, p_rope, p_right)**

- p_left: Probability that first classifier is worse
- p_rope: Probability that classifiers are practically equivalent
- p_right: Probability that first classifier is better

**Return type**
Tuple[float, float, float]

### Notes

- The Gamma distribution parameters control the prior on degrees of freedom
- The hierarchical structure models between-dataset and within-dataset variance
- Use when comparing algorithms across multiple datasets with cross-validation

### References

- Benavoli, A., Corani, G., & Mangili, F. (2016). Should we really use post-hoc tests based on mean-ranks? The Journal of Machine Learning Research.

**See also**

hierarchical_MC: Monte Carlo sampling version correlated_ttest: Simpler test for single dataset comparisons

py3plex.algorithms.statistics.bayesiantests.**hierarchical_MC**(*diff*, *rope*, *rho*, *upperAlpha=2*, *lowerAlpha=1*, *lowerBeta=0.01*, *upperBeta=0.1*, *std_upper_bound=1000*, *names=('C1', 'C2')*)

Monte Carlo sampling for hierarchical Bayesian test.

Generates Monte Carlo samples from the posterior distribution for hierarchical comparison of algorithms across multiple datasets with cross-validation.

**Parameters**

- **diff** – Array of differences between classifier scores (shape: n_datasets x n_folds)
- **rope** – Width of the region of practical equivalence (ROPE)
- **rho** – Correlation between folds due to overlapping training sets
- **upperAlpha** – Upper bound for alpha parameter of Gamma prior (default: 2)
- **lowerAlpha** – Lower bound for alpha parameter of Gamma prior (default: 1)
- **lowerBeta** – Lower bound for beta parameter of Gamma prior (default: 0.01)
- **upperBeta** – Upper bound for beta parameter of Gamma prior (default: 0.1)
- **std_upper_bound** – Upper bound multiplier for standard deviation prior (default: 1000)
- **names** – Tuple of classifier names for identification (default: ("C1", "C2"))

**Returns**

**Monte Carlo samples with shape (n_samples, 3)**
Each row contains [p_left, p_rope, p_right] for one MC sample

**Return type**
np.ndarray

### Notes

- Uses PyStan for Bayesian inference with hierarchical model
- Data is rescaled by mean standard deviation for numerical stability
- Hierarchical structure captures both within-dataset and between-dataset variance
- Requires pystan package to be installed

**Implementation:**

- Uses Stan's NUTS sampler with 4 chains
- Each chain runs 100 iterations (including warmup)
- Total posterior samples: ~200 after warmup

**See also**

hierarchical: Main function that processes MC samples correlated_ttest_MC: Simpler MC version for single dataset

**Raises**

**ImportError** – If pystan is not installed

py3plex.algorithms.statistics.bayesiantests.**plot_posterior**(*samples*, *names=('C1', 'C2')*, *proba_triplet=None*)

**Parameters**

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **names** (*pair of str*) – the names of the two classifiers

**Returns**

matplotlib.pyplot.figure

py3plex.algorithms.statistics.bayesiantests.**plot_simplex**(*points*, *names=('C1', 'C2')*, *proba_triplet=None*)

py3plex.algorithms.statistics.bayesiantests.**signrank**(*x*, *rope*, *prior_strength=0.6*, *prior_place=1*, *nsamples=50000*, *verbose=False*, *names=('C1', 'C2')*)

**Parameters**

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 0.6)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples
- **verbose** (*bool*) – report the computed probabilities
- **names** (*pair of str*) – the names of the two classifiers

**Returns**

p_left, p_rope, p_right

py3plex.algorithms.statistics.bayesiantests.**signrank_MC**(*x*, *rope*, *prior_strength=0.6*, *prior_place=1*, *nsamples=50000*)

**Parameters**

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope

- **prior_strength** (*float*) – prior strength (default: 0.6)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples

**Returns**

2-d array with rows corresponding to samples and columns to probabilities *[p_left, p_rope, p_right]*

py3plex.algorithms.statistics.bayesiantests.**signtest**(*x*, *rope*, *prior_strength=1*, *prior_place=1*, *nsamples=50000*, *verbose=False*, *names=('C1', 'C2')*)

**Parameters**

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 1)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples
- **verbose** (*bool*) – report the computed probabilities
- **names** (*pair of str*) – the names of the two classifiers

**Returns**

p_left, p_rope, p_right

py3plex.algorithms.statistics.bayesiantests.**signtest_MC**(*x*, *rope*, *prior_strength=1*, *prior_place=1*, *nsamples=50000*)

**Parameters**

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 1)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples

**Returns**

2-d array with rows corresponding to samples and columns to probabilities *[p_left, p_rope, p_right]*

### Community Detection Advanced

Additional community detection and ranking algorithms.

Node Ranking and Clustering (NoRC) module for community detection.

This module implements algorithms for node ranking and hierarchical clustering in networks, including parallel PageRank computation and hierarchical merging.

py3plex.algorithms.community_detection.NoRC.**NoRC_communities_main**(*input_graph*, *clustering_scheme='hierarchical'*, *max_com_num=100*, *verbose=False*, *parallel_step=None*, *prob_threshold=0.0005*, *community_range=None*, *fine_range=3*, *lag_threshold=10*)

py3plex.algorithms.community_detection.NoRC.**page_rank_kernel**(*index_row*)

Compute normalized PageRank vector for a given node.

**Parameters**

**index_row** – Node index for which to compute PageRank

**Returns**

Tuple of (node_index, normalized_pagerank_vector)

Community-based node ranking framework.

This module implements a framework for ranking nodes within and across communities using PageRank-based metrics and hierarchical clustering.

py3plex.algorithms.community_detection.community_ranking.**create_tree**(*centers: ndarray*) → Dict[int, Dict[str, List]]

py3plex.algorithms.community_detection.community_ranking.**page_rank_kernel**(*index_row: int*) → Tuple[int, ndarray]

py3plex.algorithms.community_detection.community_ranking.**return_infomap_communities**(*network: Any*) → List[List[

### Node Ranking and Classification

Combined node ranking and classification utilities.

Node ranking algorithms for multilayer networks.

This module provides various node ranking algorithms including PageRank variants, HITS (Hubs and Authorities), and personalized PageRank (PPR) for network analysis.

**Key Functions:**

- sparse_page_rank: Compute PageRank scores using sparse matrix operations
- run_PPR: Run Personalized PageRank in parallel across multiple cores
- hubs_and_authorities: Compute HITS scores for nodes
- stochastic_normalization: Normalize adjacency matrix to stochastic form

### Notes

The PageRank implementations use sparse matrices for memory efficiency and support parallel computation for large-scale networks.

`py3plex.algorithms.node_ranking.authority_matrix(`*graph: Graph*`)` → ndarray

Get the authority matrix representation of a graph.

Computes the matrix A = A.T @ A where A is the adjacency matrix. The authority matrix is used in HITS algorithm computation.

**Parameters**
**graph** – NetworkX graph

**Returns**
Authority matrix (N x N) where N is number of nodes

**Return type**
np.ndarray

#### Notes

- For directed graphs: A[i,j] = number of nodes that point to both i and j
- Used internally by HITS algorithm to compute authority scores

**See also**

hub_matrix: Complementary hub matrix hubs_and_authorities: Compute actual hub/authority scores

`py3plex.algorithms.node_ranking.damping_hyper: float`

`py3plex.algorithms.node_ranking.hub_matrix(`*graph: Graph*`)` → ndarray

Get the hub matrix representation of a graph.

Computes the matrix H = A @ A.T where A is the adjacency matrix. The hub matrix is used in HITS algorithm computation.

**Parameters**
**graph** – NetworkX graph

**Returns**
Hub matrix (N x N) where N is number of nodes

**Return type**
np.ndarray

### Notes

- For directed graphs: H[i,j] = number of nodes pointed to by both i and j
- Used internally by HITS algorithm to compute hub scores

**See also**

authority_matrix: Complementary authority matrix hubs_and_authorities: Compute actual hub/authority scores

py3plex.algorithms.node_ranking.**hubs_and_authorities**(*graph: Graph*) → Tuple[dict, dict]

Compute HITS (Hubs and Authorities) scores for all nodes in a graph.

Implements the Hyperlink-Induced Topic Search (HITS) algorithm to identify hub nodes (nodes that point to many authorities) and authority nodes (nodes pointed to by many hubs) in a network.

**Parameters**
**graph** – NetworkX graph (directed or undirected)

**Returns**

**(hub_scores, authority_scores)**

- hub_scores: Dictionary mapping node -> hub score
- authority_scores: Dictionary mapping node -> authority score

**Return type**
Tuple[dict, dict]

### Notes

- Uses an internal power-iteration fallback compatible with NetworkX 3.x
- Scores are normalized so that the sum of squares equals 1
- For undirected graphs, hub and authority scores are identical
- Converges using power iteration method

### Examples

```
>>> import networkx as nx
>>> G = nx.DiGraph([(0, 1), (0, 2), (1, 2)])
>>> hubs, authorities = hubs_and_authorities(G)
>>> # Node 0 has high hub score (points to others)
>>> # Node 2 has high authority score (pointed to by others)
```

**See also**

hub_matrix: Get the hub matrix representation authority_matrix: Get the authority matrix representation

py3plex.algorithms.node_ranking.**page_rank_kernel**(*index_row: int*) → Tuple[int, ndarray]

Compute PageRank vector for a single starting node (multiprocessing kernel).

This function is designed to be called in parallel via multiprocessing.Pool.map(). It computes the personalized PageRank vector starting from a single node.

**Parameters**

**index_row** – Index of the starting node for personalized PageRank

**Returns**

**(node_index, normalized_pagerank_vector)**

- node_index: The input index (for tracking results)
- pagerank_vector: L2-normalized PageRank scores for all nodes

**Return type**

Tuple[int, np.ndarray]

### Notes

- Accesses global variables: ___graph_matrix, damping_hyper, spread_step_hyper, spread_percent_hyper (set by run_PPR before parallel execution)
- Returns zero vector if normalization fails
- L2 normalization ensures comparable magnitudes across different starting nodes

**See also**

run_PPR: Main function that sets up parallel execution sparse_page_rank: Core PageRank computation

py3plex.algorithms.node_ranking.**run_PPR**(*network: spmatrix*, *cores: int | None = None*, *jobs: List[Sequence[int]] | None = None*, *damping: float = 0.85*, *spread_step: int = 10*, *spread_percent: float = 0.3*, *targets: List[int] | None = None*, *parallel: bool = True*) → Generator[Tuple[int, ndarray] | List[Tuple[int, ndarray]], None, None]

Run Personalized PageRank (PPR) in parallel for multiple starting nodes.

Computes personalized PageRank vectors for multiple nodes using parallel processing. This is useful for creating node embeddings or analyzing node importance from different perspectives in the network.

**Parameters**

- **network** – Sparse adjacency matrix (will be automatically normalized to stochastic form)

- **cores** – Number of CPU cores to use (default: all available cores)
- **jobs** – Custom job batches as list of ranges (default: auto-generated)
- **damping** – Damping factor for PageRank (default: 0.85) Higher values (0.85-0.99) emphasize network structure
- **spread_step** – Steps to check spread pattern for optimization (default: 10)
- **spread_percent** – Max node fraction for shrinkage optimization (default: 0.3)
- **targets** – Specific node indices to compute PPR for (default: all nodes)
- **parallel** – Enable parallel processing (default: True) Set to False for debugging or single-core execution

**Yields**

*Union[Tuple[int, np.ndarray], List[Tuple[int, np.ndarray]]]* – - If parallel=True: Lists of (node_index, pagerank_vector) tuples (batched) - If parallel=False: Individual (node_index, pagerank_vector) tuples

### Notes

- Automatically normalizes input matrix to column-stochastic form
- Uses multiprocessing.Pool for parallel execution
- Global variables are used to share the graph matrix across processes
- Results are yielded incrementally (generator pattern) to save memory
- Each pagerank_vector is L2-normalized for comparability

### Examples

```
>>> import scipy.sparse as sp
>>> # Create a small network
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>>
>>> # Compute PPR for all nodes in parallel
>>> for batch in run_PPR(adj, cores=2, parallel=True):
...     for node_idx, pr_vector in batch:
...         print(f"Node {node_idx}: {pr_vector}")
>>>
>>> # Compute PPR for specific nodes without parallelism
>>> for node_idx, pr_vector in run_PPR(adj, targets=[0, 1], parallel=False):
...     print(f"Node {node_idx}: {pr_vector}")
```

**Performance:**

- Parallel speedup scales with number of cores (up to ~0.8 * cores efficiency)
- Memory usage: O(N * N_targets) for storing results
- For large networks (>100K nodes), consider processing targets in batches

**See also**

sparse_page_rank: Core PageRank computation page_rank_kernel: Worker function for parallel execution stochastic_normalization: Matrix normalization (called internally)

py3plex.algorithms.node_ranking.**sparse_page_rank**(*matrix: spmatrix*, *start_nodes: List[int] | range*, *epsilon: float = 1e-06*, *max_steps: int = 100000*, *damping: float = 0.5*, *spread_step: int = 10*, *spread_percent: float = 0.3*, *try_shrink: bool = True*) → ndarray

Compute personalized PageRank using sparse matrix operations.

Implements an efficient personalized PageRank algorithm with adaptive sparsification to reduce memory usage and computation time. The algorithm uses a power iteration method with early stopping based on convergence criteria.

**Parameters**

- **matrix** – Column-stochastic sparse adjacency matrix (use stochastic_normalization first)
- **start_nodes** – List or range of starting nodes for personalized PageRank
- **epsilon** – Convergence threshold for L1 norm difference (default: 1e-6)
- **max_steps** – Maximum number of iterations (default: 100000)
- **damping** – Damping factor / teleportation probability (default: 0.5) Higher values (e.g., 0.85) favor network structure over random jumps
- **spread_step** – Number of steps to check for sparsity pattern (default: 10)
- **spread_percent** – Maximum fraction of nodes to consider for shrinkage (default: 0.3)
- **try_shrink** – Enable adaptive shrinkage to reduce computation (default: True)

**Returns**
PageRank scores for all nodes, with start_nodes set to 0

**Return type**
np.ndarray

### Notes

- Assumes matrix is column-stochastic (use stochastic_normalization first)
- Adaptive shrinkage identifies nodes unreachable from start_nodes and excludes them from computation for efficiency
- Convergence is measured by L1 norm of rank vector difference
- Start nodes are zeroed out in the final result to avoid self-importance

**Complexity:**

- Time: O(k * E) where k is iterations and E is edges
- Space: O(N) for rank vectors, plus matrix storage

### Examples

```
>>> import scipy.sparse as sp
>>> # Create and normalize adjacency matrix
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>> adj_norm = stochastic_normalization(adj)
>>> # Compute PageRank from node 0
>>> pr = sparse_page_rank(adj_norm, [0], damping=0.85)
>>> pr  # PageRank scores (node 0 will be 0)
array([0.  , 0.5, 0.5])
```

**Raises**

**AssertionError** – If start_nodes is empty

**See also**

stochastic_normalization: Required preprocessing step run_PPR: Parallel wrapper for computing multiple PageRank vectors

py3plex.algorithms.node_ranking.**spread_percent_hyper: float**

py3plex.algorithms.node_ranking.**spread_step_hyper: int**

py3plex.algorithms.node_ranking.**stochastic_normalization**(*matrix: spmatrix*) → spmatrix

Normalize a sparse matrix to stochastic form (column-stochastic).

Converts an adjacency matrix to a stochastic matrix where each column sums to 1. This normalization is required for PageRank-style random walk algorithms.

**Parameters**

**matrix** – Sparse adjacency matrix to normalize

**Returns**

Column-stochastic sparse matrix where each column sums to 1

**Return type**

sp.spmatrix

### Notes

- Removes self-loops (sets diagonal to 0) before normalization
- Handles zero-degree nodes by leaving corresponding columns as zeros
- Preserves sparsity structure for memory efficiency

### Examples

```
>>> import scipy.sparse as sp
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>> stoch_adj = stochastic_normalization(adj)
>>> stoch_adj.sum(axis=1).A1  # Column sums should be 1
array([1., 1., 1.])
```

py3plex.algorithms.network_classification.**benchmark_classification**(*matrix: spmatrix*, *labels: ndarray*, *alpha_value: float = 0.85*, *iterations: int = 30*, *normalization_scheme: str = 'freq'*, *dataset_name: str = 'example'*, *verbose: bool = False*, *test_size: float | None = None*) → DataFrame

py3plex.algorithms.network_classification.**label_propagation**(*graph_matrix: spmatrix*, *class_matrix: ndarray*, *alpha: float = 0.001*, *epsilon: float = 1e-12*, *max_steps: int = 100000*, *normalization: str | List[str] = 'freq'*) → ndarray

Propagate labels through a graph.

**Parameters**

- **graph_matrix** – Sparse graph adjacency matrix
- **class_matrix** – Initial class label matrix
- **alpha** – Propagation weight parameter
- **epsilon** – Convergence threshold
- **max_steps** – Maximum number of iterations
- **normalization** – Normalization scheme(s) to apply

**Returns**

Propagated label matrix

py3plex.algorithms.network_classification.**label_propagation_normalization**(*matrix: spmatrix*) → spmatrix

Normalize a matrix for label propagation.

**Parameters**
**matrix** – Sparse matrix to normalize

**Returns**
Normalized sparse matrix

py3plex.algorithms.network_classification.**label_propagation_tf**() → None

py3plex.algorithms.network_classification.**normalize_amplify_freq**(*mat: ndarray*) → ndarray

Normalize and amplify matrix by frequency.

**Parameters**
**mat** – Matrix to normalize

**Returns**
Normalized and amplified matrix

py3plex.algorithms.network_classification.**normalize_exp**(*mat: ndarray*) → ndarray

Apply exponential normalization.

**Parameters**
**mat** – Matrix to normalize

**Returns**
Exponentially normalized matrix

py3plex.algorithms.network_classification.**normalize_initial_matrix_freq**(*mat: ndarray*) → ndarray

Normalize matrix by frequency.

**Parameters**
**mat** – Matrix to normalize

**Returns**
Normalized matrix

py3plex.algorithms.network_classification.**validate_label_propagation**(*core_network: spmatrix*, *labels: ndarray | spmatrix*, *dataset_name: str = 'test'*, *repetitions: int = 5*, *normalization_scheme: str | List[str] = 'basic'*, *alpha_value: float = 0.001*, *random_seed: int = 123*, *verbose: bool = False*) → DataFrame

Validate label propagation with cross-validation.

**Parameters**

- **core_network** – Sparse network adjacency matrix
- **labels** – Label matrix
- **dataset_name** – Name of the dataset
- **repetitions** – Number of repetitions
- **normalization_scheme** – Normalization scheme to use
- **alpha_value** – Alpha parameter for propagation
- **random_seed** – Random seed for reproducibility
- **verbose** – Whether to print progress

**Returns**

DataFrame with validation results

### Embeddings and Wrappers

Embedding primitives and wrapper scripts for training node representations.

### Core Embedding APIs

Base embedding interfaces.

class py3plex.ml.embedding.base.**BaseEmbedding**

Bases: ABC

Base interface for first-class embedding models.

**abstract fit**(*network: Any*) → BaseEmbedding

Fit embedding model.

**fit_transform**(*network: Any*) → EmbeddingResult

Fit and transform in one call.

**abstract get_embedding**(*node: Any*) → ndarray

Get vector for a single node.

**name: str = 'base'**

**abstract to_numpy**() → ndarray

Convert embeddings to NumPy matrix.

**abstract to_pandas**()

Convert embeddings to pandas DataFrame.

**abstract transform**(*nodes: List[Any] | None = None*) → EmbeddingResult

Transform nodes into embedding vectors.

class py3plex.ml.embedding.base.**EmbeddingResult**(*matrix: ndarray*, *item_ids: List[Any]*, *method: str*, *meta: Dict[str, Any] | None = None*)

Bases: object

Container for node or edge embedding vectors.

**build_index**(*method: str = 'hnsw'*, ***kwargs: Any*) → EmbeddingResult

Build optional ANN index for faster KNN lookup.

**cluster**(*method: str = 'kmeans'*, *k: int = 10*) → Dict[Any, int]

Cluster embedding vectors and return node -> cluster mapping.

**property dim: int**

**property dimension: int**

Alias for embedding dimensionality.

**distance**(*node_a: Any*, *node_b: Any*, *metric: str = 'euclidean'*) → float

Distance convenience wrapper.

**expand_layers**() → List[tuple]

Return identifiers in canonical **(node, layer)** form.

**flatten_nodes**() → List[Any]

Return base node identifiers without layer information.

**get_embedding**(*item: Any*) → ndarray

Get vector for item id.

**group_by_layer**() → Dict[Any, List[Any]]

Group identifiers by layer id.

**group_by_node**() → Dict[Any, List[Any]]

Group identifiers by base node id.

**info**() → Dict[str, Any]

Return structured provenance metadata.

**knn**(*node: Any*, *k: int = 10*, *metric: str = 'cosine'*) → List[tuple]

Return k nearest neighbors as (node_id, score).

**property layers: List[Any]**

Unique layer ids in stable order.

**classmethod load**(*path: str*) → EmbeddingResult

Load persisted embedding result.

**most_similar**(*node: Any*, *k: int = 10*) → List[tuple]

Alias for `knn()` with cosine similarity.

**property n_items: int**

**property nodes: List[Any]**

Alias for item identifiers.

**normalize**(*inplace: bool = False*) → EmbeddingResult

L2-normalize all vectors.

**norms**() → ndarray

**reduce**(*method: str = 'pca'*, *dim: int = 2*, ***kwargs: Any*) → EmbeddingResult

Reduce vectors to lower dimension.

**reorder**(*ids: List[Any]*) → EmbeddingResult

**reproduce**(*network: Any*) → EmbeddingResult

Replay embedding computation from stored metadata.

**save**(*path: str*) → None

Persist embedding result to parquet/arrow/npz.

**similarity**(*node_a: Any*, *node_b: Any*, *metric: str = 'cosine'*) → float

Compute pairwise similarity/distance between two vectors.

**similarity_matrix**(*nodes: Sequence[Any] | None = None*, *metric: str = 'cosine'*) → ndarray

Compute pairwise similarity/distance matrix.

**subset**(*nodes: Sequence[Any]*) → EmbeddingResult

Return a subset of embeddings by id.

**to_arrow**()

Convert embeddings to an Arrow table.

**to_numpy**() → ndarray

Return embedding matrix.

**to_pandas**()

Convert embeddings to a pandas DataFrame.

**to_parquet**(*path: str*) → None

Persist embeddings to parquet.

**validate**(*network: Any | None = None*) → Dict[str, bool]

Validate embedding consistency checks.

**property vectors: Dict[Any, ndarray]**

Dictionary mapping item id -> embedding vector.

Reusable embedding training utilities.

**class** py3plex.ml.embedding.trainer.**EmbeddingTrainer**(*, *backend: str = 'numpy'*, *seed: int | None = None*)

Bases: object

Utility class for random-walk + skipgram-style training.

**generate_walks**(*network: Any*, *, *nodes: Sequence[Any] | None = None*, *walk_length: int = 40*, *num_walks: int = 10*, *p: float = 1.0*, *q: float = 1.0*, *biased: bool = False*) → List[List[Any]]

Generate random walks over graph nodes.

**optimize_embeddings**(*network: Any*, *, *dimensions: int = 128*, *walk_length: int = 40*, *num_walks: int = 10*, *window_size: int = 10*, *p: float = 1.0*, *q: float = 1.0*, *biased: bool = False*, *item_ids: List[Any] | None = None*, *method: str = 'skipgram'*) → EmbeddingResult

End-to-end train embeddings.

**train_skipgram**(*walks: Sequence[Sequence[Any]]*, *, *dimensions: int = 128*, *window_size: int = 10*, *method: str = 'skipgram'*, *item_ids: List[Any] | None = None*) → EmbeddingResult

Train a lightweight skipgram proxy via co-occurrence + SVD.

py3plex.ml.embedding.trainer.**generate_walks**(*network: Any*, ***kwargs*) → List[List[Any]]

Functional helper around EmbeddingTrainer.generate_walks.

py3plex.ml.embedding.trainer.**train_skipgram**(*walks: Sequence[Sequence[Any]]*, ***kwargs*) → EmbeddingResult

Functional helper around EmbeddingTrainer.train_skipgram.

Node2Vec embedding model.

**class** py3plex.ml.embedding.node2vec.**Node2VecEmbedding**(*dimensions: int = 128*, *walk_length: int = 80*, *num_walks: int = 10*, *p: float = 1.0*, *q: float = 1.0*, *window_size: int = 10*, *negative_samples: int = 5*, *workers: int = 1*, *backend: str = 'numpy'*, *seed: int | None = None*)

Bases: BaseEmbedding

Biased random-walk node embedding.

**fit**(*network: Any*) → Node2VecEmbedding

Fit embedding model.

**fit_transform**(*network: Any*) → EmbeddingResult

Fit and transform in one call.

**get_embedding**(*node: Any*) → ndarray

Get vector for a single node.

**name: str = 'node2vec'**

**to_numpy**() → ndarray

Convert embeddings to NumPy matrix.

**to_pandas**()

Convert embeddings to pandas DataFrame.

**transform**(*nodes: List[Any] | None = None*) → EmbeddingResult

Transform nodes into embedding vectors.

DeepWalk embedding model.

**class** py3plex.ml.embedding.deepwalk.**DeepWalkEmbedding**(*dimensions: int = 128*, *walk_length: int = 80*, *num_walks: int = 10*, *window_size: int = 10*, *negative_samples: int = 5*, *workers: int = 1*, *backend: str = 'numpy'*, *seed: int | None = None*)

Bases: Node2VecEmbedding

DeepWalk implementation as an unbiased Node2Vec variant.

**name: str = 'deepwalk'**

NetMF embedding wrapper.

**class** py3plex.ml.embedding.netmf.**NetMFEmbedding**(*dimensions: int = 128*, *window: int = 10*, *negative: float = 1.0*, *multilayer: str = 'supra'*, *gamma: float = 1.0*, *approx: str = 'randomized_svd'*, *seed: int | None = None*)

Bases: BaseEmbedding

Spectral DeepWalk approximation using NetMF.

**fit**(*network: Any*) → NetMFEmbedding

Fit embedding model.

**fit_transform**(*network: Any*) → EmbeddingResult

Fit and transform in one call.

**get_embedding**(*node: Any*) → ndarray

Get vector for a single node.

**name: str = 'netmf'**

**to_numpy**() → ndarray

Convert embeddings to NumPy matrix.

**to_pandas**()

Convert embeddings to pandas DataFrame.

**transform**(*nodes: List[Any] | None = None*) → EmbeddingResult

Transform nodes into embedding vectors.

LINE embedding model.

**class** py3plex.ml.embedding.line.**LINEEmbedding**(*dimensions: int = 128*, *order: int = 2*, *negative_samples: int = 5*, *lr: float = 0.025*, *epochs: int = 5*, *seed: int | None = None*)

Bases: `BaseEmbedding`

Lightweight LINE approximation via sparse adjacency factorization.

**fit**(*network: Any*) → LINEEmbedding

Fit embedding model.

**fit_transform**(*network: Any*) → EmbeddingResult

Fit and transform in one call.

**get_embedding**(*node: Any*) → ndarray

Get vector for a single node.

**name: str = 'line'**

**to_numpy**() → ndarray

Convert embeddings to NumPy matrix.

**to_pandas**()

Convert embeddings to pandas DataFrame.

**transform**(*nodes: List[Any] | None = None*) → EmbeddingResult

Transform nodes into embedding vectors.

MetaPath2Vec embedding wrapper.

**class** py3plex.ml.embedding.metapath2vec.**MetaPath2VecEmbedding**(*metapaths: List[List[str]]*, *dimensions: int = 128*, *walk_length: int = 40*, *num_walks: int = 10*, *window_size: int = 10*, *negative_samples: int = 5*, *epochs: int = 5*, *seed: int | None = None*)

Bases: `BaseEmbedding`

Meta-path guided embedding for multilayer networks.

**fit**(*network: Any*) → MetaPath2VecEmbedding

Fit embedding model.

**fit_transform**(*network: Any*) → EmbeddingResult

Fit and transform in one call.

**get_embedding**(*node: Any*) → ndarray

Get vector for a single node.

**name: str = 'metapath2vec'**

**to_numpy**() → ndarray

Convert embeddings to NumPy matrix.

**to_pandas**()

Convert embeddings to pandas DataFrame.

**transform**(*nodes: List[Any] | None = None*) → EmbeddingResult

Transform nodes into embedding vectors.

Multiplex-aware and supra-graph embedding variants.

**class** py3plex.ml.embedding.multiplex.**BaseMultiLayerEmbedding**(*config: MultiLayerEmbeddingConfig | None = None*)

Bases: BaseEmbedding

Base class for state-node-first multilayer embedding models.

**fit_transform**(*network: Any*) → EmbeddingResult

Fit and transform in one call.

**get_embedding**(*node: Any*) → ndarray

Get vector for a single node.

**name: str = 'multilayer_base'**

**node_embeddings**() → EmbeddingResult | None

**to_numpy**() → ndarray

Convert embeddings to NumPy matrix.

**to_pandas**()

Convert embeddings to pandas DataFrame.

**transform**(*nodes: List[Any] | None = None*) → EmbeddingResult

Transform nodes into embedding vectors.

**class** py3plex.ml.embedding.multiplex.**LayerRegularizedEmbedding**(*dimensions: int = 128*, *alpha: float = 0.5*, *seed: int | None = None*)

Bases: NetMFEmbedding

Layer-regularized embedding that blends per-layer and supra vectors.

**fit**(*network: Any*) → LayerRegularizedEmbedding

Fit embedding model.

**name: str = 'layer_regularized'**

class py3plex.ml.embedding.multiplex.**MELLEmbedding**(*dimensions: int = 128*, *directed: bool = False*, *negative_ratio: int = 5*, *lambda_nodes: float = 0.0001*, *beta_variance: float = 1.0*, *gamma_layers: float = 0.0001*, *epochs: int = 50*, *lr: float = 0.001*, *seed: int | None = None*, *target: Literal['state', 'node', 'both'] = 'state'*, *node_reduce: Literal['mean', 'sum', 'max', 'attention'] = 'mean'*)

Bases: `BaseMultiLayerEmbedding`

Multiplex embedding with layer vectors and variance regularization.

**fit**(*network: Any*) → MELLEmbedding

Fit embedding model.

**name: str = 'mell'**

class py3plex.ml.embedding.multiplex.**MNEEmbedding**(*dimensions_common: int = 128*, *dimensions_relation: int = 16*, *window_size: int = 10*, *negative_samples: int = 5*, *layer_weights: Dict[Hashable, float] | None = None*, *transform_norm_bound: float = 1000.0*, *walk_length: int = 80*, *num_walks: int = 10*, *seed: int | None = None*, *optimizer: str = 'adam'*, *lr: float = 0.01*, *epochs: int = 5*, *target: Literal['state', 'node', 'both'] = 'state'*, *node_reduce: Literal['mean', 'sum', 'max', 'attention'] = 'mean'*)

Bases: `SupraNode2VecEmbedding`

Multiplex embedding with shared + relation-specific factors.

**fit**(*network: Any*) → MNEEmbedding

Fit embedding model.

**name: str = 'mne'**

class py3plex.ml.embedding.multiplex.**MultiLayerEmbeddingConfig**(*dimensions: int = 128*, *seed: int | None = None*, *target: Literal['state', 'node', 'both'] = 'state'*, *node_reduce: Literal['mean', 'sum', 'max', 'attention'] = 'mean'*, *include_interlayer_edges: bool = True*, *coupling_weight_multiplier: float = 1.0*, *coupling_edge_type: str = 'identity'*)

Bases: `object`

Shared multilayer embedding configuration.

**coupling_edge_type:** `str = 'identity'`

**coupling_weight_multiplier:** `float = 1.0`

**dimensions:** `int = 128`

**include_interlayer_edges:** `bool = True`

**node_reduce:** `Literal['mean', 'sum', 'max', 'attention'] = 'mean'`

**seed:** `int | None = None`

**target:** `Literal['state', 'node', 'both'] = 'state'`

class py3plex.ml.embedding.multiplex.**MultiLayerGNNEmbedding**(*dimensions: int = 128*, *layers: int = 2*, *hidden_dim: int = 128*, *dropout: float = 0.1*, *model: Literal['mgnn', 'mpxgat', 'gatne'] = 'mgnn'*, *objective: Literal['lp', 'nc', 'contrastive'] = 'lp'*, *epochs: int = 50*, *lr: float = 0.001*, *weight_decay: float = 0.0*, *batch_size: int = 512*, *negative_samples: int = 5*, *backend: Literal['dgl', 'torch_sparse'] = 'torch_sparse'*, *seed: int | None = None*, *device: str = 'cpu'*, *target: Literal['state', 'node', 'both'] = 'state'*, *node_reduce: Literal['mean', 'sum', 'max', 'attention'] = 'mean'*)

Bases: `BaseMultiLayerEmbedding`

Experimental deep backend extension point for multilayer embeddings.

**fit**(*network: Any*) → MultiLayerGNNEmbedding

Fit embedding model.

**name: str = 'multilayer_gnn'**

class py3plex.ml.embedding.multiplex.**MultiplexNode2Vec**(**args*, *layer_weight: float = 1.0*, ***kwargs*)

Bases: `SupraNode2VecEmbedding`

Backward-compatible multiplex Node2Vec alias.

**fit**(*network: Any*) → MultiplexNode2Vec

Fit embedding model.

**name: str = 'multiplex_node2vec'**

class py3plex.ml.embedding.multiplex.**NodeLayerIndexer**(*state_nodes: List[Tuple[Hashable, Hashable]]*, *to_index: Dict[Tuple[Hashable, Hashable], int]*, *to_state: List[Tuple[Hashable, Hashable]]*)

Bases: `object`

Deterministic state-node indexing with layer-major ordering.

**classmethod from_nodes**(*nodes: Iterable[Any]*) → NodeLayerIndexer

**index_of**(*state_node: Tuple[Hashable, Hashable]*) → int

**state_nodes: List[Tuple[Hashable, Hashable]]**

**state_of**(*index: int*) → Tuple[Hashable, Hashable]

**to_index: Dict[Tuple[Hashable, Hashable], int]**

**to_state: List[Tuple[Hashable, Hashable]]**

class py3plex.ml.embedding.multiplex.**SupraAdjacencyEmbedding**(**args*, *gamma: float = 1.0*, ***kwargs*)

Bases: `SupraNetMFEmbedding`

Backward-compatible name for NetMF over supra adjacency.

**name: str = 'supra_adjacency'**

class py3plex.ml.embedding.multiplex.**SupraNetMFEmbedding**(*dimensions: int = 128*, *window: int = 10*, *negative: float = 1.0*, *multilayer: str = 'supra'*, *approx: str = 'randomized_svd'*, *gamma: float = 1.0*, *seed: int | None = None*)

Bases: `NetMFEmbedding`

NetMF with explicit supra-graph defaults.

**name: str = 'supra_netmf'**

class py3plex.ml.embedding.multiplex.**SupraNode2VecEmbedding**(*dimensions: int = 128*, *walk_length: int = 80*, *num_walks: int = 10*, *p: float = 1.0*, *q: float = 1.0*, *window_size: int = 10*, *negative_samples: int = 5*, *workers: int = 1*, *cross_layer_prob: float | None = None*, *coupling_weight_multiplier: float = 1.0*, *seed: int | None = None*, *backend: str = 'numpy'*, *target: Literal['state', 'node', 'both'] = 'state'*, *node_reduce: Literal['mean', 'sum', 'max', 'attention'] = 'mean'*, *include_interlayer_edges: bool = True*, *negative_sampling_domain: Literal['same_layer', 'all_state_nodes', 'aligned_nodes'] = 'all_state_nodes'*, *coupling_edge_type: str = 'identity'*)

Bases: `BaseMultiLayerEmbedding`

Node2Vec over a supra-graph of *(node, layer)* state nodes.

**fit**(*network: Any*) → SupraNode2VecEmbedding

Fit embedding model.

**name: str = 'supra_node2vec'**

class py3plex.ml.embedding.multiplex.**SupraSpectralEmbedding**(*dimensions: int = 128*, *laplacian: Literal['sym', 'unnorm', 'rw'] = 'sym'*, *solver: str = 'eigsh'*, *which: str = 'SM'*, *tol: float = 1e-05*, *maxiter: int = 2000*, *seed: int | None = None*, *target: Literal['state', 'node', 'both'] = 'state'*, *node_reduce: Literal['mean', 'sum', 'max', 'attention'] = 'mean'*, *include_interlayer_edges: bool = True*, *coupling_weight_multiplier: float = 1.0*, *coupling_edge_type: str = 'identity'*)

Bases: `BaseMultiLayerEmbedding`

Deterministic spectral embedding on supra-graph Laplacians.

**fit**(*network: Any*) → SupraSpectralEmbedding

Fit embedding model.

**name: str = 'supra_spectral'**

Embedding evaluation utilities.

py3plex.ml.embedding.evaluation.**build_edge_features**(*embedding: EmbeddingResult*, *edges: Sequence[tuple[Any, Any]]*, *operator: str = 'hadamard'*) → ndarray

Build edge feature matrix from an embedding and edge list.

py3plex.ml.embedding.evaluation.**edge_operator**(*u: ndarray*, *v: ndarray*, *operator: str = 'hadamard'*) → ndarray

Combine two node embeddings into an edge feature vector.

py3plex.ml.embedding.evaluation.**evaluate_clustering**(*y_true: Sequence[int]*, *y_pred: Sequence[int]*) → dict

Evaluate clustering with NMI and ARI.

py3plex.ml.embedding.evaluation.**evaluate_link_prediction**(*y_true: Sequence[int]*, *y_score: Sequence[float]*) → float

Evaluate link prediction with ROC-AUC.

py3plex.ml.embedding.evaluation.**evaluate_link_prediction_embeddings**(*embedding: EmbeddingResult*, *positive_edges: Sequence[tuple[Any, Any]]*, *negative_edges: Sequence[tuple[Any, Any]]*, *operator: str = 'hadamard'*, *random_state: int = 42*) → Dict[str, float]

Evaluate link prediction with logistic probe on edge features.

py3plex.ml.embedding.evaluation.**evaluate_node_classification**(*X: ndarray*, *y: Sequence[int]*, *test_size: float = 0.3*, *random_state: int = 42*) → float

Evaluate node classification with macro-F1.

py3plex.ml.embedding.evaluation.**evaluate_node_classification_report**(*X: ndarray*, *y: Sequence[int]*, *test_size: float = 0.3*, *random_state: int = 42*) → Dict[str, float]

Evaluate node classification with linear probe metrics.

Similarity helpers for embedding vectors.

py3plex.ml.embedding.similarity.**cosine_similarity**(*a: ndarray*, *b: ndarray*) → float

Compute cosine similarity.

py3plex.ml.embedding.similarity.**dot_similarity**(*a: ndarray*, *b: ndarray*) → float

Compute dot-product similarity.

py3plex.ml.embedding.similarity.**euclidean_distance**(*a: ndarray*, *b: ndarray*) → float

Compute Euclidean distance.

### Wrapper Entry Points

py3plex.wrappers.train_node2vec_embedding.**call_node2vec_binary**(*input_graph: str*, *output_graph: str*, *p: float = 1*, *q: float = 1*, *dimension: int = 128*, *directed: bool = False*, *weighted: bool = True*, *binary: str | None = None*, *timeout: int = 300*) → None

Call the Node2Vec C++ binary with specified parameters.

**Parameters**

- **input_graph** – Path to input graph file
- **output_graph** – Path to output embedding file
- **p** – Return parameter
- **q** – In-out parameter
- **dimension** – Embedding dimension
- **directed** – Whether graph is directed
- **weighted** – Whether graph is weighted
- **binary** – Path to node2vec binary (defaults to PY3PLEX_NODE2VEC_BINARY env var or "./node2vec")
- **timeout** – Maximum execution time in seconds

**Raises**

**ExternalToolError** – If binary is not found or execution fails

py3plex.wrappers.train_node2vec_embedding.**learn_embedding**(*core_network: Any*, *labels: List[Any] | None = None*, *ssize: float = 0.5*, *embedding_outfile: str = 'out.emb'*, *p: float = 0.1*, *q: float = 0.1*, *binary_path: str | None = None*, *parameter_range: str = '[0.25,0.50,1,2,4]'*, *timeout: int = 300*) → Tuple[str, float]

Learn node embeddings for a network.

**Parameters**

- **core_network** – NetworkX graph
- **labels** – Node labels
- **ssize** – Sample size

- **embedding_outfile** – Output file for embeddings
- **p** – Return parameter
- **q** – In-out parameter
- **binary_path** – Path to node2vec binary (defaults to PY3PLEX_NODE2VEC_BINARY env var or "./node2vec")
- **parameter_range** – String representation of parameter range list
- **timeout** – Maximum execution time in seconds per call

**Returns**

Tuple of (method_name, elapsed_time)

py3plex.wrappers.train_node2vec_embedding.**n2v_embedding**(*G: Any*, *targets: Any*, *verbose: bool = False*, *sample_size: float = 0.5*, *outfile_name: str = 'test.emb'*, *p: float | None = None*, *q: float | None = None*, *binary_path: str | None = None*, *parameter_range: List[float] | None = None*, *embedding_dimension: int = 128*, *timeout: int = 300*) → None

Train Node2Vec embeddings with parameter optimization.

**Parameters**

- **G** – NetworkX graph
- **targets** – Target labels for nodes
- **verbose** – Whether to print verbose output
- **sample_size** – Sample size for training
- **outfile_name** – Output embedding file name
- **p** – Return parameter (None triggers grid search)
- **q** – In-out parameter (None triggers grid search)
- **binary_path** – Path to node2vec binary (defaults to PY3PLEX_NODE2VEC_BINARY env var or "./node2vec")
- **parameter_range** – Range of parameters to search
- **embedding_dimension** – Dimension of embeddings
- **timeout** – Maximum execution time in seconds per call

### Network Motifs and Patterns

Motif detection and pattern discovery tools.

### HINMINE Data Structures

Typed data structures used by HINMINE components.

**class** py3plex.core.HINMINE.dataStructures.**Class**(*lab_id*, *name*, *members*)

Bases: object

**class** py3plex.core.HINMINE.dataStructures.**HeterogeneousInformationNetwork**(*network*, *label_delimiter*, *weight_tag=False*, *target_tag=True*)

Bases: object

**add_label**(*node*, *label_id*, *label_name=None*)

**calculate_decomposition_candidates**(*max_decomposition_length=10*)

**calculate_schema**()

**create_label_matrix**(*weights=None*)

**decompose_from_iterator**(*name*, *weighing*, *summing*, *generator=None*, *degrees=None*, *parallel=False*, *pool=None*)

**midpoint_generator**(*node_sequence*, *edge_sequence*)

**process_network**(*label_delimiter*)

**split_to_indices**(*train_indices=()*, *validate_indices=()*, *test_indices=()*)

**split_to_parts**(*lst*, *n*)

### Command-Line Interface

CLI entry points and argument parsing.

Command-line interface for py3plex.

This module provides a comprehensive CLI tool for multilayer network analysis with full coverage of main algorithms.

py3plex.cli.**cmd_aggregate**(*args: Namespace*) → int

Aggregate multilayer network into single layer.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_capabilities**(*args: Namespace*) → int

Show the runtime capability report.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_centrality**(*args: Namespace*) → int

Compute node centrality measures.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_check**(*args: Namespace*) → int

Lint and validate a graph data file.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success, non-zero for errors)

py3plex.cli.**cmd_community**(*args: Namespace*) → int

Detect communities in the network.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_convert**(*args: Namespace*) → int

Convert network between different formats.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_create**(*args: Namespace*) → int

Create a new multilayer network.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_dsl_lint**(*args: Namespace*) → int

Lint and analyze DSL queries.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for no errors, 1 for errors found, 2 for command error)

py3plex.cli.**cmd_dynamics**(*args: Namespace*) → int

Run epidemic dynamics simulations on a network.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_embed**(*args: Namespace*) → int

Learn node embeddings from a network.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_help**(*args: Namespace*) → int

Show detailed help information.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_load**(*args: Namespace*) → int

Load and inspect a network.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_query**(*args: Namespace*) → int

Execute DSL queries on networks with Unix piping support.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_quickstart**(*args: Namespace*) → int

Run quickstart demo - creates a tiny demo graph and demonstrates basic operations.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_run_config**(*args: Namespace*) → int

Run workflow from configuration file.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_selftest**(*args: Namespace*) → int

Run self-test to verify installation and core functionality.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_stats**(*args: Namespace*) → int

Compute multilayer network statistics.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_tutorial**(*args: Namespace*) → int

Run interactive tutorial mode to learn py3plex step by step.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**cmd_visualize**(*args: Namespace*) → int

Visualize the multilayer network.

**Parameters**
**args** – Parsed command-line arguments

**Returns**
Exit code (0 for success)

py3plex.cli.**create_parser**() → ArgumentParser

Create and configure the argument parser for py3plex CLI.

**Returns**
Configured ArgumentParser instance

py3plex.cli.**main**(*argv: List[str] | None = None*) → int

Main entry point for the CLI.

**Parameters**
**argv** – Command-line arguments (defaults to sys.argv)

**Returns**
Exit code (0 for success, non-zero for errors)

### Validation Utilities

Validation helpers for input data and configuration.

Input validation utilities for Py3plex.

This module provides pre-validation functions to catch common input errors early and provide clear, actionable error messages to users.

py3plex.validation.**validate_csv_columns**(*file_path: str*, *required_columns: List[str]*, *optional_columns: List[str] | None = None*) → None

Validate that a CSV file has required columns.

**Parameters**

- **file_path** – Path to CSV file

- **required_columns** – List of column names that must be present
- **optional_columns** – List of column names that are optional

**Raises**

**ParsingError** – If required columns are missing

**Contracts:**

- Precondition: file_path must be a non-empty string
- Precondition: required_columns must be a non-empty list of strings

py3plex.validation.**validate_edgelist_format**(*file_path: str*, *delimiter: str | None = None*) → None

Validate simple edgelist file format (source target weight).

**Parameters**

- **file_path** – Path to edgelist file
- **delimiter** – Optional delimiter (default: whitespace)

**Raises**

**ParsingError** – If file format is invalid

**Contracts:**

- Precondition: file_path must be a non-empty string
- Precondition: delimiter must be None or a string

py3plex.validation.**validate_file_exists**(*file_path: str*) → None

Validate that a file exists and is readable.

**Parameters**

**file_path** – Path to file to validate

**Raises**

**ParsingError** – If file doesn't exist or isn't readable

**Contracts:**

- Precondition: file_path must be a non-empty string

py3plex.validation.**validate_input_type**(*input_type: str*, *valid_types: Set[str] | None = None*) → None

Validate that input_type is recognized.

**Parameters**

- **input_type** – The input type string to validate
- **valid_types** – Optional set of valid types (uses default if None)

**Raises**

**ParsingError** – If input_type is not valid

**Contracts:**

- Precondition: input_type must be a non-empty string
- Precondition: valid_types must be None or a set

py3plex.validation.**validate_multiedgelist_format**(*file_path: str*, *delimiter: str | None = None*) → None

Validate multiedgelist file format (source target layer weight).

**Parameters**

- **file_path** – Path to multiedgelist file
- **delimiter** – Optional delimiter (default: whitespace)

**Raises**

**ParsingError** – If file format is invalid

**Contracts:**

- Precondition: file_path must be a non-empty string
- Precondition: delimiter must be None or a string

py3plex.validation.**validate_network_data**(*file_path: str*, *input_type: str*) → None

Validate network data before parsing.

This is the main entry point for validation. It performs appropriate validation based on the input type.

**Parameters**

- **file_path** – Path to network file
- **input_type** – Type of input file

**Raises**

**ParsingError** – If validation fails

**Contracts:**

- Precondition: file_path must be a string
- Precondition: input_type must be a non-empty string

### Network Comparison and Testing

Network distance measures, slicing, and comparison helpers.

### Hedwig Learning Algorithms

Learning strategies within the Hedwig framework.

Main learner class.

@author: anze.vavpetic@ijs.si

class py3plex.algorithms.hedwig.learners.learner.**Learner**(*kb*, *n=None*, *min_sup=1*, *sim=1*, *depth=4*, *target=None*, *use_negations=False*, *optimal_subclass=False*)

Bases: object

Learner class, supporting various types of induction from the knowledge base.

**Default = 'default'**

**Improvement = 'improvement'**

**Similarity = 'similarity'**

**can_specialize**(*rule*)

Is the rule good enough to be further refined?

**extend**(*rules*, *specializations*)

Extends the ruleset in the given way.

**extend_replace_worst**(*rules*, *specializations*)

Extends the list by replacing the worst rules.

**extend_with_similarity**(*rules*, *specializations*)

Extends the list based on how similar is 'new_rule' to the rules contained in 'rules'.

**get_subclasses**(*pred*)

**get_superclasses**(*pred*)

**group_score**(*rules*)

Calculates the score of the whole list of rules.

**induce**()

Induces rules for the given knowledge base.

**is_implicit_root**(*pred*)

**non_redundant**(*rule*, *new_rule*)

Is the rule non-redundant compared to its immediate generalization?

**specialize**(*rule*)

Returns a list of all specializations of 'rule'.

**specialize_add_relation**(*rule*)

Specialize with new binary relation.

Main learner class.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.learners.bottomup.**BottomUpLearner**(*kb*, *n=None*, *min_sup=1*, *sim=1*, *depth=4*, *target=None*, *use_negations=False*)

Bases: `object`

Bottom-up learner.

**Default = 'default'**

**Improvement = 'improvement'**

**Similarity = 'similarity'**

**bottom_clause**()

**get_subclasses**(*pred*)

**get_superclasses**(*pred*)

**induce**()

Induces rules for the given knowledge base.

**is_implicit_root**(*pred*)

Main learner class.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.learners.optimal.**OptimalLearner**(*kb*, *n=None*, *min_sup=1*, *sim=1*, *depth=4*, *target=None*, *use_negations=False*, *optimal_subclass=True*)

Bases: Learner

Finds the optimal top-k rules.

**induce**()

Induces rules for the given knowledge base.

### Hedwig Statistics and Scoring

Scoring functions and validation routines for Hedwig learners.

Score function definitions.

@author: anze.vavpetic@ijs.si

py3plex.algorithms.hedwig.stats.scorefunctions.**chisq**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**enrichment_score**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**interesting**(*rule*)

Checks if a given rule is interesting for the given score function

py3plex.algorithms.hedwig.stats.scorefunctions.**kaplan_meier_AUC**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**leverage**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**lift**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**precision**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**t_score**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**wracc**(*rule*)

py3plex.algorithms.hedwig.stats.scorefunctions.**z_score**(*rule*)

Module for ruleset validation.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.stats.validate.**Validate**(*kb*, *significance_test=<function apply_fisher>*, *adjustment=<function fdr>*)

Bases: object

**test**(*ruleset*, *alpha=0.05*, *q=0.01*)

Tests the given ruleset and returns the significant rules.

### Hedwig Core Components

Core converters, knowledge base structures, and settings used by Hedwig.

py3plex.algorithms.hedwig.core.converters.**convert_mapping_to_rdf**(*input_mapping_file*, *extract_subnode_info=False*, *split_node_by=':'*, *keep_index=1*, *layer_type='uniprotkb'*, *annotation_mapping_file='test.gaf'*, *go_identifier='GO:'*, *prepend_string=None*)

py3plex.algorithms.hedwig.core.converters.**obo2n3**(*obofile*, *n3out*, *gaf_file*)

Knowledge-base class.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.core.kb.**ExperimentKB**(*triplets*, *score_fun*, *instances_as_leaves=True*)

Bases: `object`

The knowledge base for one specific experiment.

**add_sub_class**(*sub*, *obj*)

Adds the resource 'sub' as a subclass of 'obj'.

**bits_to_indices**(*bits*)

Converts the bitset to a set of indices.

**get_domains**(*predicate*)

Returns the bitsets for input and outputexamples of the binary predicate 'predicate'.

**get_empty_domain**()

Returns a bitset covering no examples.

**get_examples**()

Returns all examples for this experiment.

**get_full_domain**()

Returns a bitset covering all examples.

**get_members**(*predicate*, *bit=True*)

Returns the examples for this predicate, either as a bitset or a set of ids.

**get_reverse_members**(*predicate*, *bit=True*)

Returns the examples for this predicate, either as a bitset or a set of ids.

**get_root**()

Root predicate, which covers all examples.

**get_score**(*ex_idx*)

Returns the score for example id 'ex_idx'.

**get_subclasses**(*predicate*, *producer_pred=None*)

Returns a list of subclasses (as predicate objects) for 'predicate'.

**indices_to_bits**(*indices*)

Converts the indices to a bitset.

**is_discrete_target**()

**n_examples**()

Returns the number of examples.

**n_members**(*predicate*)

**super_classes**(*pred*)

Returns all super classes of pred (with transitivity).

Global settings file.

@author: anze.vavpetic@ijs.si

**class** py3plex.algorithms.hedwig.core.settings.**Defaults**

Bases: object

**ADJUST** = 'fwer'

**ALPHA** = 0.05

**BEAM_SIZE** = 20

**COVERED** = None

**DEPTH** = 5

**FDR_Q** = 0.05

**FORMAT** = 'n3'

**LEARNER** = 'heuristic'

**LEAVES** = False

**MODE** = 'subgroups'

**NEGATIONS** = False

**NO_CACHE** = False

**OPTIMAL_SUBCLASS** = False

**OUTPUT** = None

**SCORE** = 'lift'

**SUPPORT** = 0.1

**TARGET** = None

**URIS** = False

**VERBOSE** = False

### Time Series and Temporal Analysis

Temporal analysis utilities.

### Advanced Visualization

Specialized visualizations such as hairball and Sankey diagrams.

Sankey diagram visualization for multilayer networks.

This module provides inter-layer flow visualization to show connection strength between layers in multilayer networks. The visualization displays flows as arrows with widths proportional to the number of inter-layer connections.

Note: This uses a simplified flow diagram approach rather than matplotlib's Sankey class, as the Sankey class is designed for more complex flow networks and doesn't map directly to multilayer network inter-layer connections.

py3plex.visualization.sankey.**draw_multilayer_sankey**(*graphs: List[Graph]*, *multilinks: Dict[str, List[Tuple]]*, *labels: List[str] | None = None*, *ax: Any | None = None*, *display: bool = False*, ***kwargs*) → Any

Draw inter-layer flow diagram showing connection strength in multilayer networks.

Creates a flow visualization where: - Each layer is represented in the diagram - Flows between layers show the strength (number) of inter-layer connections - Flow width/text indicates the number of inter-layer edges

**Parameters**

- **graphs** – List of NetworkX graphs, one per layer
- **multilinks** – Dictionary mapping edge_type -> list of multi-layer edges
- **labels** – Optional list of layer labels. If None, uses layer indices
- **ax** – Matplotlib axes to draw on. If None, creates new figure
- **display** – If True, calls plt.show() after drawing. Default is False to let the caller control rendering.
- ****kwargs** – Reserved for future extensions

**Returns**

Matplotlib axes object

### Examples

```
>>> import matplotlib.pyplot as plt
>>> from py3plex.visualization import draw_multilayer_sankey
>>> network = multi_layer_network()
>>> network.load_network("data.txt", input_type="multiedgelist")
>>> labels, graphs, multilinks = network.get_layers()
>>> fig, ax = plt.subplots(figsize=(12, 8))
>>> ax = draw_multilayer_sankey(graphs, multilinks, labels=labels, ax=ax)
>>> plt.savefig("sankey.png")
```

**Note**

This visualization is most effective for networks with 2-5 layers. For networks with many layers, the diagram may become cluttered. The implementation uses a simplified flow visualization approach.

#### Link Prediction

Link prediction algorithms.

### 3.5.7 Configuration & Environment

This page collects the practical knobs for running py3plex: configuration files, environment variables, supported I/O formats, and common algorithm parameters.

#### Configuration Files

py3plex supports configuration files for reproducible workflows. Provide them as YAML or JSON; omit sections you do not need.

**Network Configuration (YAML):**

```yaml
# network_config.yaml
network:
  input_file: "data/multilayer.edges"
  input_type: "multiedgelist"  # one of: multiedgelist, edgelist, json, graphml, parquet
  directed: false
  weighted: true

# field meanings:
# input_file: path to your dataset
# input_type: parser to use (match the file format)
# directed/weighted: toggle graph properties

# Load with:
# network.load_network(**config['network'])
```

**Algorithm Parameters:**

```yaml
# algorithm_config.yaml
community_detection:
  algorithm: "louvain_multilayer"
  params:
    gamma: 1.0          # resolution; higher favors smaller communities
    omega: 0.5          # inter-layer coupling strength
    random_state: 42

node2vec:
  dimensions: 128
  walk_length: 80
  num_walks: 10
  p: 1.0               # return hyperparameter (p > 1 biases against revisiting)
```

```
  q: 1.0                # in-out hyperparameter (q < 1 encourages outward exploration)

dynamics:
  model: "SIR"
  beta: 0.3             # infection probability per contact
  gamma: 0.1            # recovery probability
  initial_infected: 0.05  # fraction of nodes initially infected (0-1)
  steps: 100
```

Section meanings:

- `community_detection` — choose algorithm and pass algorithm-specific params
- `node2vec` — structural embedding hyperparameters (`p`/`q` control walk bias)
- `dynamics` — compartmental model parameters; `initial_infected` expects a fraction

**Visualization Settings:**

```
# viz_config.yaml
visualization:
  layout: "force_directed"
  node_size: 50
  edge_width: 2
  dpi: 300
  output_format: "png"
  color_by_layer: true
```

**Loading configuration:**

```
import yaml
from py3plex.core import multinet

# Load config
with open('config.yaml', 'r') as f:
    config = yaml.safe_load(f)

# Use config
network = multinet.multi_layer_network(
    directed=config['network']['directed']
)
network.load_network(
    config['network']['input_file'],
    input_type=config['network']['input_type']
)
```

### Environment Variables

py3plex respects the following environment variables for customization. Override them in your shell before running scripts or set them in code via `os.environ`.

**Data Directories:**

- `PY3PLEX_DATA_DIR` — Override default data directory location
  - Default: `~/.py3plex/data`

  - Used for: Cached datasets, downloaded networks
- `PY3PLEX_CACHE_DIR` — Cache directory for computed results
  - Default: `~/.py3plex/cache`
  - Used for: Embedding caches, computation results

**Performance Tuning:**

- `PY3PLEX_NUM_WORKERS` — Suggested number of parallel workers for algorithms
  - Default recommendation: `os.cpu_count()` when you wire this through to APIs
  - Effect: only applied by functions/scripts that read the variable
- `PY3PLEX_MEMORY_LIMIT` — Optional memory budget for large-scale operations (in GB)
  - Default: unset (falls back to library defaults)
  - Effect: honored only by routines that support explicit memory caps

**Setting environment variables:**

```
# Linux/macOS
export PY3PLEX_DATA_DIR="/path/to/data"
export PY3PLEX_NUM_WORKERS=8

# Or in Python
import os
os.environ['PY3PLEX_DATA_DIR'] = '/path/to/data'
```

**Debugging and Development:**

- `PY3PLEX_DEBUG` — Enable debug mode (verbose output)
  - Values: `0` (off), `1` (on)
  - Default: `0`
- `PY3PLEX_PROFILE` — Enable profiling for performance analysis
  - Values: `0` (off), `1` (on)
  - Default: `0`

### Input Formats

Supported file formats:

- `multiedgelist` — Multilayer edge list format with layer annotations
- `edgelist` — Standard edge list (single layer)
- `json` — JSON network format produced by py3plex tools
- `graphml` — GraphML format for interoperability with NetworkX/Gephi
- `parquet` — Apache Parquet for columnar, high-performance I/O

See *How to Load and Build Networks* for format details.

### Output Formats

Networks can be exported to:

- Pickle (Python serialization)
- JSON
- CSV/TSV
- GraphML
- Parquet

See *How to Export and Serialize Networks* for export details.

### Algorithm Configuration

Many algorithms accept configuration parameters. The snippets below show common defaults and what each parameter controls.

### Community Detection

```
from py3plex.algorithms.community_detection import multilayer_louvain

communities = multilayer_louvain(
    network,
    resolution=1.0,  # Higher → more, smaller communities
    omega=0.5        # Inter-layer coupling strength
)
```

### Node2Vec

```
from py3plex.wrappers import train_node2vec

embeddings = train_node2vec(
    network,
    dimensions=128,      # Embedding size
    walk_length=80,      # Steps per walk
    num_walks=10,        # Walks per node
    p=1.0,               # Return parameter (larger discourages backtracking)
    q=1.0,               # In-out parameter (smaller explores outward)
    workers=4            # Parallel workers
)
```

### Dynamics

```
from py3plex.dynamics import SIRDynamics

sir = SIRDynamics(
    network,
    beta=0.3,            # Infection probability per contact
    gamma=0.1,           # Recovery probability per step
```

```
    initial_infected=0.05  # Fraction of nodes initially infected (0-1)
)
```

See *Algorithm Roadmap* for complete parameter documentation.

### Visualization Configuration

Customize visualization; adjust node/edge sizes to match network scale. Set `show=False` for headless rendering:

```
network.visualize_network(
    output_file='network.png',
    layout_algorithm='force_directed',
    node_size=50,
    edge_width=2,
    dpi=300,
    show=False
)
```

See *How to Visualize Multilayer Networks* for visualization options.

### Logging Configuration

py3plex uses Python's standard `logging` module. Configure logging for debugging and monitoring:

**Basic Logging Setup:**

```
import logging

# Set log level
logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s'
)

# Now py3plex operations will log to console
from py3plex.core import multinet
network = multinet.multi_layer_network()
# ... logs will appear
```

**Log Levels:**

- `DEBUG` — Detailed information, typically of interest only when diagnosing problems
- `INFO` — Confirmation that things are working as expected (default)
- `WARNING` — An indication that something unexpected happened
- `ERROR` — A more serious problem, the software has not been able to perform some function
- `CRITICAL` — A serious error, the program may be unable to continue running

**Log to File:**

```
import logging
```

```
# Configure file handler
logging.basicConfig(
    level=logging.DEBUG,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s',
    handlers=[
        logging.FileHandler('py3plex.log'),
        logging.StreamHandler()  # Also log to console
    ]
)
```

**Log File Location:**

By default, logs are written to:

- Console (stdout) when using `StreamHandler`
- Custom file when using `FileHandler`
- Environment variable `PY3PLEX_LOG_FILE` can override default location

**Component-Specific Logging:**

```
import logging

# Enable debug logging only for specific components
logging.getLogger('py3plex.algorithms').setLevel(logging.DEBUG)
logging.getLogger('py3plex.dsl').setLevel(logging.INFO)
logging.getLogger('py3plex.dynamics').setLevel(logging.WARNING)
```

**Custom Logging Handlers:**

For advanced use cases (e.g., sending logs to external systems):

```
import logging
from logging.handlers import RotatingFileHandler

# Rotating file handler (10MB max, keep 5 backups)
handler = RotatingFileHandler(
    'py3plex.log',
    maxBytes=10*1024*1024,
    backupCount=5
)
handler.setLevel(logging.DEBUG)
handler.setFormatter(
    logging.Formatter('%(asctime)s - %(name)s - %(levelname)s - %(message)s')
)

# Add to py3plex logger
logger = logging.getLogger('py3plex')
logger.addHandler(handler)
logger.setLevel(logging.DEBUG)
```

#### Performance Tuning

Quick levers:

- Match `PY3PLEX_NUM_WORKERS` to available cores when parallelism is supported.
- Use `parquet` I/O for large networks to reduce load time.
- Set `PY3PLEX_MEMORY_LIMIT` when supported to avoid oversubscription on shared machines.

See *Benchmarking & Performance* for performance optimization strategies.

#### Next Steps

- **Load networks:** *How to Load and Build Networks*
- **Export networks:** *How to Export and Serialize Networks*
- **Algorithm reference:** *Algorithm Roadmap*

---

## 3.6 Part VI: Examples, Recipes & Embeddings

Complete working examples, analysis recipes, and embedding workflows.

### 3.6.1 Examples & Recipes

The runnable examples live in the repository examples directory[15]. Clone the repository before running these scripts; installing the Python package does not install this source directory.

#### Start Here

From the repository root, with py3plex installed:

```
python examples/getting_started/01_basic_query.py
python examples/getting_started/dsl_patterns_quick_reference.py
python examples/getting_started/tutorial_10min.py
```

The first script creates a small network and queries node degrees. The second covers common builder patterns. The tutorial adds loading, communities, and visualization; it takes longer and writes plots to `example_images/`.

For a notebook environment, use the linked notebooks in *How to Query Multilayer Graphs with the SQL-like DSL* and *How to Run Community Detection on Multilayer Networks*.

#### Browse by Task

- `getting_started/` — Basic construction, datasets, queries, and tutorials.
- `io_and_data/` — File loading, conversion, and exports.
- `network_analysis/` — Statistics, community detection, and DSL workflows.
- `dsl_zoo/` — Composable DSL query examples and helper datasets.
- `visualization/` — Layouts, layer plots, and interactive figures.
- `advanced/` — Temporal networks, uncertainty, dynamics, and specialized methods.

[15] https://github.com/SkBlaz/py3plex/tree/master/examples

- `pipelines/` — Reusable analysis pipelines.
- `out_of_core/` — Streaming and out-of-core workflows.

Individual examples may require optional extras or additional datasets. Read the script's prerequisites before running it. For feature installation commands, see *Installation and Setup*.

#### Continue Learning

- *Quick Start Tutorial* — First network and analysis.
- ../user_guide/dsl — Query language and builder API.
- *Analysis Recipes & Workflows* — Reusable recipes.
- ../user_guide/visualization — Visualization concepts and examples.

### 3.6.2 Random Walk Algorithms

Py3plex provides comprehensive random walk primitives for graph-based algorithms, including Node2Vec, DeepWalk, and diffusion processes. These implementations support weighted, directed, and multilayer networks with deterministic reproducibility.

#### Overview

Random walks are fundamental building blocks for many graph algorithms:

- **Node2Vec**: Learn node embeddings using biased random walks
- **DeepWalk**: Generate node embeddings through uniform random walks
- **Personalized PageRank**: Compute node importance via random walks with restart
- **Diffusion processes**: Model information or disease propagation
- **Community detection**: Identify network structure through walk patterns

#### Key Features

- [OK] Basic random walks with proper edge weight handling
- [OK] Second-order (biased) random walks following Node2Vec logic
- [OK] Multiple simultaneous walks with deterministic seeding
- [OK] Support for directed, weighted, and multigraphs
- [OK] Multilayer network-aware walks with layer constraints
- [OK] Efficient sparse adjacency matrix handling
- [OK] Comprehensive test suite validating correctness properties

#### Basic Random Walk

The basic random walk samples the next node proportionally to normalized edge weights.

#### Simple Example

```
from py3plex.algorithms.general.walkers import basic_random_walk
import networkx as nx
```

```
# Create a simple graph
G = nx.karate_club_graph()

# Perform a random walk
walk = basic_random_walk(
    G,
    start_node=0,
    walk_length=10,
    seed=42  # for reproducibility
)

print(f"Walk: {walk}")
print(f"Length: {len(walk)}")  # 11 nodes (includes start)
```

### Weighted Graphs

Edge weights are automatically used when `weighted=True` (default):

```
from py3plex.algorithms.general.walkers import basic_random_walk
import networkx as nx

# Create weighted graph
G = nx.Graph()
G.add_weighted_edges_from([
    (0, 1, 10.0),  # high weight -> more likely
    (0, 2, 1.0),   # low weight -> less likely
    (1, 2, 5.0),
])

# Weighted walk (respects edge weights)
walk = basic_random_walk(G, 0, walk_length=20, weighted=True, seed=42)

# Unweighted walk (uniform transitions)
walk_uniform = basic_random_walk(G, 0, walk_length=20, weighted=False, seed=42)
```

### Directed Graphs

Directed edges are handled correctly:

```
import networkx as nx
from py3plex.algorithms.general.walkers import basic_random_walk

# Create directed graph
G = nx.DiGraph()
G.add_edges_from([(0, 1), (1, 2), (2, 0), (1, 3)])

# Walk respects edge direction
walk = basic_random_walk(G, 0, walk_length=10, seed=42)

# Verify all transitions follow directed edges
```

```
for i in range(len(walk) - 1):
    assert G.has_edge(walk[i], walk[i+1])
```

### Node2Vec Biased Random Walk

Second-order random walks with return parameter `p` and in-out parameter `q` following the Node2Vec algorithm (Grover & Leskovec, 2016).

### Parameters

When transitioning from node `t` → `v` → `x`, the probability of choosing `x` is biased:

- **If x == t** (return to previous): `weight / p`
- **If x is neighbor of t** (stay close): `weight / 1`
- **If x is not neighbor of t** (explore): `weight / q`

**Parameter Effects:**

- **p > 1**: Less likely to return to previous node (explore forward)
- **p < 1**: More likely to return (backtrack)
- **q > 1**: Less likely to explore distant nodes (stay local, BFS-like)
- **q < 1**: More likely to explore distant nodes (venture far, DFS-like)
- **p = q = 1**: Equivalent to basic random walk

### Basic Usage

```
from py3plex.algorithms.general.walkers import node2vec_walk
import networkx as nx

G = nx.karate_club_graph()

# Balanced walk (p=1, q=1)
walk_balanced = node2vec_walk(G, 0, walk_length=20, p=1.0, q=1.0, seed=42)

# BFS-like walk (stay local)
walk_bfs = node2vec_walk(G, 0, walk_length=20, p=1.0, q=2.0, seed=42)

# DFS-like walk (explore outward)
walk_dfs = node2vec_walk(G, 0, walk_length=20, p=1.0, q=0.5, seed=42)
```

### Exploring vs Backtracking

```
from py3plex.algorithms.general.walkers import node2vec_walk
import networkx as nx

# Triangle graph
G = nx.Graph()
G.add_edges_from([(0, 1), (1, 2), (2, 0)])
```

```
# Low p, high q: tends to backtrack
walk_backtrack = node2vec_walk(G, 0, walk_length=20, p=0.1, q=10.0, seed=42)

# High p, low q: tends to explore outward
walk_explore = node2vec_walk(G, 0, walk_length=20, p=10.0, q=0.1, seed=42)

# Count backtracking patterns
backtracks = sum(1 for i in range(2, len(walk_backtrack))
                 if walk_backtrack[i] == walk_backtrack[i-2])
print(f"Backtracks: {backtracks}")
```

### Multiple Walk Generation

Generate multiple walks efficiently with deterministic seeding.

### Generate Walks from All Nodes

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate 10 walks from each node
walks = generate_walks(
    G,
    num_walks=10,
    walk_length=10,
    seed=42
)

print(f"Total walks: {len(walks)}")  # 340 (34 nodes × 10 walks)
print(f"First walk: {walks[0]}")
```

### Generate from Specific Nodes

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate walks only from nodes 0, 1, 2
walks = generate_walks(
    G,
    num_walks=5,
    walk_length=15,
    start_nodes=[0, 1, 2],
    seed=42
)

print(f"Total walks: {len(walks)}")  # 15 (3 nodes × 5 walks)
```

### Node2Vec-Style Walks

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate biased walks with p=0.5, q=2.0
walks = generate_walks(
    G,
    num_walks=10,
    walk_length=20,
    p=0.5,
    q=2.0,
    seed=42
)
```

### Edge Sequences

Return walks as edge sequences instead of node sequences:

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate edge sequences
edge_walks = generate_walks(
    G,
    num_walks=5,
    walk_length=10,
    return_edges=True,
    seed=42
)

# First walk is a list of edges
print(edge_walks[0])  # [(0, 3), (3, 2), (2, 1), ...]
```

### Multilayer Network Walks

Perform random walks on multilayer networks with layer constraints.

### Layer-Constrained Walks

```
from py3plex.algorithms.general.walkers import layer_specific_random_walk
from py3plex.core import multinet

# Create multilayer network
```

```
network = multinet.multi_layer_network()
network.add_layer("social")
network.add_layer("biological")

network.add_nodes([
    ("A", "social"), ("B", "social"), ("C", "social"),
    ("A", "biological"), ("B", "biological")
])

network.add_edges([
    (("A", "social"), ("B", "social")),
    (("B", "social"), ("C", "social")),
    (("A", "biological"), ("B", "biological")),
])

# Walk constrained to social layer
walk = layer_specific_random_walk(
    network.core_network,
    start_node="A---social",
    walk_length=10,
    layer="social",
    cross_layer_prob=0.0,  # never cross layers
    seed=42
)

# All nodes in walk are in social layer
print(walk)
```

### Cross-Layer Transitions

Allow occasional transitions between layers:

```
from py3plex.algorithms.general.walkers import layer_specific_random_walk

# Same network as above

# Walk with 10% probability of crossing layers
walk = layer_specific_random_walk(
    network.core_network,
    start_node="A---social",
    walk_length=20,
    layer="social",
    cross_layer_prob=0.1,  # 10% chance to cross
    seed=42
)

# May contain nodes from both layers
print(walk)
```

### Reproducibility

All walk functions support deterministic seeding for reproducibility.

### Fixed Seed

```
from py3plex.algorithms.general.walkers import basic_random_walk
import networkx as nx

G = nx.karate_club_graph()

# Same seed produces identical walks
walk1 = basic_random_walk(G, 0, 50, seed=42)
walk2 = basic_random_walk(G, 0, 50, seed=42)

assert walk1 == walk2  # Identical
```

### Different Seeds

```
# Different seeds produce different walks
walk1 = basic_random_walk(G, 0, 50, seed=42)
walk2 = basic_random_walk(G, 0, 50, seed=123)

assert walk1 != walk2  # Different
```

### Batch Reproducibility

```
from py3plex.algorithms.general.walkers import generate_walks

G = nx.karate_club_graph()

# Same seed produces identical walk sets
walks1 = generate_walks(G, num_walks=100, walk_length=10, seed=42)
walks2 = generate_walks(G, num_walks=100, walk_length=10, seed=42)

assert walks1 == walks2
```

### Performance Tips

### Large Graphs

For large sparse graphs (>100k nodes), use basic walks for better performance:

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

# Large sparse graph
G = nx.erdos_renyi_graph(100000, 0.0001, seed=42)

# Use basic walk (p=1, q=1) for speed
```

```
walks = generate_walks(
    G,
    num_walks=10,
    walk_length=100,
    p=1.0,  # No bias = faster
    q=1.0,
    seed=42
)
```

### Parallel Processing

For generating many walks, consider parallel processing:

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx
from multiprocessing import Pool

G = nx.karate_club_graph()

def generate_batch(seed):
    return generate_walks(G, num_walks=10, walk_length=20, seed=seed)

# Generate walks in parallel
with Pool(4) as pool:
    batch_walks = pool.map(generate_batch, range(10))

# Flatten results
all_walks = [walk for batch in batch_walks for walk in batch]
```

### Correctness Validation

The implementation has been validated against the following properties:

### Uniformity Test

On unweighted regular graphs, transition probabilities are uniform:

```
import networkx as nx
from py3plex.algorithms.general.walkers import basic_random_walk
from collections import Counter

# Complete graph (regular)
G = nx.complete_graph(10)

# Count visits to neighbors from node 0
visits = Counter()
for i in range(10000):
    walk = basic_random_walk(G, 0, 1, weighted=False, seed=i)
    if len(walk) > 1:
        visits[walk[1]] += 1
```

```
# All neighbors should be visited roughly equally
print(visits)
# Expected: ~1111 visits to each of 9 neighbors
```

### Conservation Test

Sum of transition probabilities equals 1.0 (within machine epsilon):

```
import networkx as nx
import numpy as np

G = nx.karate_club_graph()

# For each node, verify probability conservation
for node in G.nodes():
    neighbors = list(G.neighbors(node))
    if len(neighbors) > 0:
        weights = np.array([G[node][n].get('weight', 1.0) for n in neighbors])
        probs = weights / weights.sum()
        assert abs(probs.sum() - 1.0) < 1e-10  # Machine epsilon
```

### Bias Consistency Test

Node2Vec parameters p and q produce expected biases:

```
import networkx as nx
from py3plex.algorithms.general.walkers import node2vec_walk

# Triangle graph
G = nx.Graph()
G.add_edges_from([(0, 1), (1, 2), (2, 0)])

# Low p should encourage backtracking
backtracks_low_p = 0
for i in range(1000):
    walk = node2vec_walk(G, 0, 10, p=0.1, q=1.0, seed=i)
    for j in range(2, len(walk)):
        if walk[j] == walk[j-2]:
            backtracks_low_p += 1

# High p should discourage backtracking
backtracks_high_p = 0
for i in range(1000):
    walk = node2vec_walk(G, 0, 10, p=10.0, q=1.0, seed=i)
    for j in range(2, len(walk)):
        if walk[j] == walk[j-2]:
            backtracks_high_p += 1

assert backtracks_low_p > backtracks_high_p * 2  # Significantly more
```

### Edge Weight Test

Visit frequency matches edge weight ratios:

```
import networkx as nx
from py3plex.algorithms.general.walkers import basic_random_walk
from collections import Counter

G = nx.Graph()
G.add_weighted_edges_from([
    (0, 1, 1.0),
    (0, 2, 2.0),
    (0, 3, 3.0),
])

visits = Counter()
for i in range(10000):
    walk = basic_random_walk(G, 0, 1, weighted=True, seed=i)
    if len(walk) > 1:
        visits[walk[1]] += 1

# Expected ratio: 1:2:3
total = sum(visits.values())
ratios = [visits[1]/total, visits[2]/total, visits[3]/total]
expected = [1/6, 2/6, 3/6]

# Within 5% tolerance
for obs, exp in zip(ratios, expected):
    assert abs(obs - exp) < 0.05
```

### Complete Embedding Workflow

This section provides a complete, end-to-end workflow for generating node embeddings from a multilayer network and using them for downstream machine learning tasks.

### Step 1: Prepare the Network

```
from py3plex.core import multinet
import numpy as np

# Set random seed for reproducibility
np.random.seed(42)

# Load your multilayer network
network = multinet.multi_layer_network().load_network(
    "your_network.multiedgelist",
    input_type="multiedgelist",
    directed=False
)

# Verify the network
```

```
network.basic_stats()

# Get the core NetworkX graph
G = network.core_network
```

### Step 2: Generate Random Walks

```
from py3plex.algorithms.general.walkers import generate_walks

# Generate walks with Node2Vec-style biased sampling
# Recommended starting parameters
walks = generate_walks(
    G,
    num_walks=10,        # 10 walks per node (more = better embeddings, slower)
    walk_length=80,      # 80 steps per walk (adjust based on graph diameter)
    p=1.0,               # Return parameter (1.0 = balanced)
    q=1.0,               # In-out parameter (1.0 = balanced)
    seed=42              # For reproducibility
)

print(f"Generated {len(walks)} walks")
print(f"Sample walk: {walks[0][:10]}...")  # First 10 nodes of first walk
```

### Step 3: Train Embedding Model

```
# Requires gensim: pip install gensim
from gensim.models import Word2Vec

# Convert walks to strings (Word2Vec expects string tokens)
walks_str = [[str(node) for node in walk] for walk in walks]

# Train the embedding model
model = Word2Vec(
    walks_str,
    vector_size=128,     # Embedding dimension (64-256 typical)
    window=5,            # Context window size
    min_count=1,         # Include nodes that appear at least once
    sg=1,                # Use skip-gram (1) vs CBOW (0)
    workers=4,           # Parallel training threads
    epochs=5,            # Training epochs
    seed=42              # Reproducibility
)

print(f"Trained embeddings for {len(model.wv)} nodes")
print(f"Embedding dimension: {model.wv.vector_size}")
```

### Step 4: Extract and Use Embeddings

```
import numpy as np

# Get all node IDs
node_ids = list(model.wv.index_to_key)

# Create embedding matrix
embeddings = np.array([model.wv[node_id] for node_id in node_ids])
print(f"Embedding matrix shape: {embeddings.shape}")

# Get embedding for a specific node
sample_node = node_ids[0]
node_embedding = model.wv[sample_node]
print(f"Embedding for {sample_node}: {node_embedding[:5]}... (dim={len(node_embedding)})")

# Find similar nodes
similar_nodes = model.wv.most_similar(sample_node, topn=5)
print(f"\nMost similar to {sample_node}:")
for node, similarity in similar_nodes:
    print(f"  {node}: {similarity:.4f}")
```

### Step 5: Downstream Tasks

**Node Clustering:**

```
from sklearn.cluster import KMeans
import numpy as np

# Cluster nodes based on embeddings
n_clusters = 5
kmeans = KMeans(n_clusters=n_clusters, random_state=42)
clusters = kmeans.fit_predict(embeddings)

# Assign clusters to nodes
node_clusters = dict(zip(node_ids, clusters))

print("Cluster distribution:")
from collections import Counter
for cluster, count in sorted(Counter(clusters).items()):
    print(f"  Cluster {cluster}: {count} nodes")
```

**Node Classification:**

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# Assume you have labels for nodes
# labels = {node_id: label for node_id, label in your_label_data}
```

```
# For demonstration, create synthetic labels based on layer
labels = {}
for node_id in node_ids:
    # Extract layer from node-layer tuple string representation
    # Adjust this based on your actual node ID format
    if 'layer1' in str(node_id):
        labels[node_id] = 0
    elif 'layer2' in str(node_id):
        labels[node_id] = 1
    else:
        labels[node_id] = 2

# Prepare data
X = embeddings
y = np.array([labels[node_id] for node_id in node_ids])

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train classifier
clf = RandomForestClassifier(n_estimators=100, random_state=42)
clf.fit(X_train, y_train)

# Evaluate
y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Classification accuracy: {accuracy:.4f}")
```

**Link Prediction:**

```
from sklearn.linear_model import LogisticRegression
import numpy as np

def edge_embedding(node1, node2, method='hadamard'):
    """Compute edge embedding from node embeddings."""
    emb1 = model.wv[str(node1)]
    emb2 = model.wv[str(node2)]

    if method == 'hadamard':
        return emb1 * emb2
    elif method == 'average':
        return (emb1 + emb2) / 2
    elif method == 'concat':
        return np.concatenate([emb1, emb2])
    else:
        raise ValueError(f"Unknown method: {method}")

# Get existing edges (positive examples)
existing_edges = list(G.edges())[:100]  # Sample for demo
```

```
# Generate negative examples (non-edges)
nodes = list(G.nodes())
negative_edges = []
while len(negative_edges) < len(existing_edges):
    n1 = nodes[np.random.randint(len(nodes))]
    n2 = nodes[np.random.randint(len(nodes))]
    if n1 != n2 and not G.has_edge(n1, n2):
        negative_edges.append((n1, n2))

# Create training data
X_edges = []
y_edges = []

for edge in existing_edges:
    try:
        X_edges.append(edge_embedding(edge[0], edge[1]))
        y_edges.append(1)  # Positive
    except KeyError:
        pass  # Skip if node not in vocabulary

for edge in negative_edges:
    try:
        X_edges.append(edge_embedding(edge[0], edge[1]))
        y_edges.append(0)  # Negative
    except KeyError:
        pass

X_edges = np.array(X_edges)
y_edges = np.array(y_edges)

# Train link prediction model
X_train, X_test, y_train, y_test = train_test_split(
    X_edges, y_edges, test_size=0.2, random_state=42
)

clf = LogisticRegression(random_state=42)
clf.fit(X_train, y_train)

accuracy = clf.score(X_test, y_test)
print(f"Link prediction accuracy: {accuracy:.4f}")
```

### Step 6: Save and Load Embeddings

```
# Save embeddings in Word2Vec format
model.wv.save_word2vec_format("node_embeddings.txt", binary=False)
print("Embeddings saved to node_embeddings.txt")

# Save as NumPy arrays (for sklearn compatibility)
np.save("embedding_matrix.npy", embeddings)
```

```
np.save("node_ids.npy", np.array(node_ids))
print("Embeddings saved as NumPy arrays")

# Load embeddings later
from gensim.models import KeyedVectors
loaded_embeddings = KeyedVectors.load_word2vec_format("node_embeddings.txt")
print(f"Loaded embeddings for {len(loaded_embeddings)} nodes")
```

### Parameter Tuning Guide

#### Walk Length Guidelines

| Network Size | walk_length | Reasoning | Trade-off |
|---|---|---|---|
| Small (<1K nodes) | 40-80 | Shorter walks sufficient | Speed vs coverage |
| Medium (1K-10K) | 80-120 | Standard choice | Balanced |
| Large (>10K) | 80-100 | Longer walks costly | Memory/time vs quality |

**Rule of thumb:** `walk_length` should be at least 2-3× the network diameter.

#### Number of Walks Guidelines

| Task | num_walks | Reasoning | Trade-off |
|---|---|---|---|
| Exploration | 5-10 | Quick results | Speed vs quality |
| Standard | 10-20 | Good balance | Recommended |
| High-quality | 20-50 | Better embeddings | Time vs quality |
| Research | 50-100 | Maximum quality | Very slow |

**Rule of thumb:** More walks = better embeddings, but with diminishing returns after ~20.

#### P and Q Parameter Effects

The p and q parameters control the walk's exploration behavior:

**Return parameter p:**

- **p < 1:** More likely to backtrack (return to previous node)
  - Use when: Local structure is important; you want to capture tight clusters
- **p = 1:** Balanced (equivalent to basic random walk for backtracking)
  - Use when: No prior about network structure
- **p > 1:** Less likely to backtrack (move forward)
  - Use when: You want walks to explore broadly

**In-out parameter q:**

- **q < 1:** More likely to explore distant nodes (DFS-like)
  - Use when: You want to capture global structure, find communities

- **q = 1:** Balanced exploration
  - Use when: No prior about network structure
- **q > 1:** More likely to stay in local neighborhood (BFS-like)
  - Use when: You want to capture local structure, homophily

**Common presets:**

```
# Local/homophily-focused (similar to BFS)
p, q = 1.0, 2.0

# Global/structural (similar to DFS)
p, q = 1.0, 0.5

# Balanced (DeepWalk-like)
p, q = 1.0, 1.0

# Strong local focus
p, q = 0.5, 2.0

# Strong exploration
p, q = 2.0, 0.5
```

### Common Failure Modes

#### Disconnected Graph

**Problem:** Walk gets stuck or terminates early because node has no neighbors.

**Symptoms:**

- Walks shorter than expected
- Some nodes never appear in walks
- Embeddings missing for some nodes

**Solutions:**

```
import networkx as nx

# Check connectivity
if not nx.is_connected(G.to_undirected()):
    print("Graph is disconnected!")

    # Option 1: Use largest connected component
    largest_cc = max(nx.connected_components(G.to_undirected()), key=len)
    G_connected = G.subgraph(largest_cc).copy()

    # Option 2: Add small random edges to connect components
    # (use with caution - changes graph structure)
```

### Highly Skewed Degree Distribution

**Problem:** High-degree hub nodes dominate walks; low-degree nodes underrepresented.

**Symptoms:**

- Most walks pass through the same few nodes
- Embeddings for peripheral nodes are poor quality
- Similarity to hub nodes is overestimated

**Solutions:**

```
# Option 1: Use more walks per node
walks = generate_walks(G, num_walks=30, walk_length=80, seed=42)

# Option 2: Add node-specific walks for low-degree nodes
low_degree_nodes = [n for n in G.nodes() if G.degree(n) < 5]
extra_walks = generate_walks(
    G,
    num_walks=20,
    walk_length=80,
    start_nodes=low_degree_nodes,
    seed=42
)
walks.extend(extra_walks)

# Option 3: Use weighted walks that downweight high-degree nodes
# (requires custom implementation)
```

### Isolated Nodes

**Problem:** Nodes with degree 0 cannot be walked from.

**Symptoms:**

- KeyError when looking up embeddings
- Missing nodes in similarity queries

**Solutions:**

```
# Remove isolated nodes before generating walks
isolated = list(nx.isolates(G))
G_filtered = G.copy()
G_filtered.remove_nodes_from(isolated)

# Generate walks on filtered graph
walks = generate_walks(G_filtered, num_walks=10, walk_length=80, seed=42)

# Handle isolated nodes separately (e.g., assign zero embedding)
```

### Very Small Graph

**Problem:** With few nodes, walks become repetitive quickly.

**Symptoms:**

- Walks cycle through the same nodes repeatedly
- All nodes have similar embeddings
- Poor discriminative power

**Solutions:**

```
# Use shorter walks for small graphs
if G.number_of_nodes() < 100:
    walk_length = min(40, G.number_of_nodes() * 2)

# Use fewer walks (avoid redundancy)
num_walks = min(10, G.number_of_nodes())

# Consider direct matrix-based methods for very small graphs
# (spectral embeddings, Laplacian eigenvectors)
```

### Memory Issues with Large Graphs

**Problem:** Storing all walks in memory causes OOM errors.

**Symptoms:**

- MemoryError during walk generation
- System becomes unresponsive

**Solutions:**

```
# Option 1: Generate walks in batches
all_walks = []
nodes = list(G.nodes())
batch_size = 1000

for i in range(0, len(nodes), batch_size):
    batch_nodes = nodes[i:i+batch_size]
    batch_walks = generate_walks(
        G,
        num_walks=10,
        walk_length=80,
        start_nodes=batch_nodes,
        seed=42+i
    )
    all_walks.extend(batch_walks)
    print(f"Processed {i+batch_size}/{len(nodes)} nodes")

# Option 2: Stream walks directly to Word2Vec
# (requires custom iterator implementation)

# Option 3: Use disk-based storage
```

```
# Write walks to file, then load for training
```

### Native Multilayer Embedding API

Beyond low-level walker primitives, py3plex exposes first-class embedding models for multilayer and multiplex networks in `py3plex.ml.embedding` and via `multi_layer_network.embed(...)`.

Available multilayer-focused methods include:

- `supra_node2vec` (`py3plex.ml.embedding.SupraNode2VecEmbedding`)
- `supra_spectral` (`py3plex.ml.embedding.SupraSpectralEmbedding`)
- `supra_netmf` (`py3plex.ml.embedding.SupraNetMFEmbedding`)
- `mne` (`py3plex.ml.embedding.MNEEmbedding`)
- `mell` (`py3plex.ml.embedding.MELLEmbedding`)
- `multilayer_gnn` (`py3plex.ml.embedding.MultiLayerGNNEmbedding`, experimental scaffold)

These models use `(node, layer)` state nodes as the canonical internal target. Physical-node embeddings can be derived with `target="both"` and `node_reduce` set to one of `"mean"`, `"sum"`, or `"max"`.

```
from py3plex.ml.embedding import SupraNode2VecEmbedding

model = SupraNode2VecEmbedding(
    dimensions=128,
    walk_length=80,
    num_walks=10,
    cross_layer_prob=0.2,
    target="both",
    node_reduce="mean",
    seed=42,
)
result_state = model.fit_transform(net)
result_node = model.node_embeddings()
```

You can access the same models through `multi_layer_network.embed`:

```
result = net.embed(
    method="supra_node2vec",
    dimensions=128,
    walk_length=80,
    num_walks=10,
    seed=42,
)
```

### References

**Node2Vec:**
Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 855-864. https://doi.org/10.1145/2939672.2939754

**DeepWalk:**
Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). DeepWalk: Online learning of social representations. *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 701-710. https://doi.org/10.1145/2623330.2623732

**Random Walks on Graphs:**
Lovász, L. (1993). Random walks on graphs: A survey. *Combinatorics, Paul Erdős is Eighty*, 2, 1-46.

### API Reference

See `py3plex.algorithms.general.walkers` for complete API documentation.

### Examples

Complete examples can be found in:

- `examples/advanced/example_random_walks.py` - Basic usage examples
- `tests/test_random_walks.py` - Comprehensive test suite

Repository: https://github.com/SkBlaz/py3plex

## 3.6.3 Analysis Recipes & Workflows

This guide provides **practical recipes** for **common analysis workflows** in py3plex. Each recipe is a **complete, ready-to-use solution** for a real-world task.

**DSL for Rapid Analysis**

Many recipes can be expressed concisely using the DSL:

```
from py3plex.dsl import Q, L

# Quick hub identification
hubs = (
    Q.nodes()
     .where(degree__gt=5)
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(10)
     .execute(network)
)

# Multi-layer comparison
for layer_name in ["social", "work", "family"]:
    result = (
        Q.nodes()
         .from_layers(L[layer_name])
```

```
        .compute("degree", "clustering")
        .execute(network)
)
print(f"{layer_name}: {result.count} nodes")
```

See the DSL recipes section below for more patterns!

**Related Example Scripts:**

Many recipes have corresponding runnable example scripts in the repository:

- I/O examples: `examples/io_and_data/example_IO.py`, `examples/io_and_data/example_new_io.py`
- Community detection: `examples/communities/example_community_detection.py`
- Network statistics: `examples/network_analysis/example_multilayer_statistics.py`
- Visualization: `examples/visualization/example_multilayer_visualization.py`
- DSL queries: `examples/network_analysis/example_dsl_queries.py`
- Complete pipelines: `examples/pipelines/example_6_complex_pipeline.py`

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### Data Import & Setup

#### Recipe 1: Load Network from Multiple File Formats

**Task:** Load **multilayer networks** from various **file formats** (edgelist, GraphML, CSV).

**Script:** `examples/io_and_data/example_IO.py`

```
from py3plex.core import multinet

# From multiedgelist (node1, layer1, node2, layer2, weight)
network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist",
    directed=False
)

# From GraphML
network = multinet.multi_layer_network().load_network(
```

```
    "data/network.graphml",
    input_type="graphml"
)

# From edges list programmatically
network = multinet.multi_layer_network()
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1.0],
    ['B', 'layer1', 'C', 'layer1', 1.0],
    ['A', 'layer2', 'C', 'layer2', 0.5],
], input_type="list")

# Verify the network
network.basic_stats()
```

**Expected Output:**

```
Number of nodes: 3
Number of edges: 3
Number of unique nodes (as node-layer tuples): 5
Number of unique node IDs (across all layers): 3
Nodes per layer:
  Layer 'layer1': 3 nodes
  Layer 'layer2': 2 nodes
```

**When to use:** Starting any **analysis project** with external data.

### Recipe 2: Create Synthetic Benchmark Networks

**Task:** Generate **synthetic multilayer networks** for **testing algorithms**.

**Script:** `examples/getting_started/example_random_generator.py`

```
from py3plex.core.random_generators import random_multilayer_ER
import numpy as np

# Set random seed for reproducibility
np.random.seed(42)

# Create a random Erdős-Rényi multilayer network
# Parameters: n_nodes, n_layers, edge_probability
network = random_multilayer_ER(
    n=100,        # 100 nodes
    L=3,          # 3 layers
    p=0.1,        # 10% edge probability
    directed=False
)

network.basic_stats()
```

**When to use:** **Algorithm validation**, **testing**, **benchmarking**, or when real data is unavailable.

### Recipe 3: Convert Between Network Formats

**Task:** Export networks to **different formats** for use in **other tools**.

```
from py3plex.core import multinet

# Load network
network = multinet.multi_layer_network().load_network(
    "input.txt",
    input_type="multiedgelist"
)

# Save as GraphML (standard format, compatible with Gephi, Cytoscape)
network.save_network("output.graphml", output_type="graphml")

# Save as edge list with node mapping
inverse_map = network.serialize_to_edgelist("output_edges.txt")

# Save as pickle (preserves all Python objects)
import pickle
with open("network.pkl", "wb") as f:
    pickle.dump(network, f)
```

**When to use:** Sharing data with collaborators, importing to visualization tools, or archiving results.

### Network Exploration & Analysis

### Recipe 4: Quick Network Summary

**Task:** Get comprehensive overview of network structure.

```
from py3plex.core import multinet

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Basic statistics
print("=== Basic Statistics ===")
network.basic_stats()

# Layer information
layers = network.get_layers()
print(f"\nLayers: {layers}")
print(f"Number of layers: {len(layers)}")

# Node and edge counts
nodes = list(network.get_nodes())
edges = list(network.get_edges())
print(f"Total nodes: {len(nodes)}")
```

```
print(f"Total edges: {len(edges)}")

# Get unique nodes (nodes may appear in multiple layers)
unique_nodes = set([n[0] for n in nodes])
print(f"Unique nodes: {len(unique_nodes)}")
```

**Expected Output:**

```
=== Basic Statistics ===
Number of nodes: 250
Number of edges: 180
Number of unique nodes (as node-layer tuples): 250
Number of unique node IDs (across all layers): 120
Nodes per layer:
  Layer 'collaboration': 85 nodes
  Layer 'citation': 90 nodes
  Layer 'coauthor': 75 nodes

Layers: ['collaboration', 'citation', 'coauthor']
Number of layers: 3
Total nodes: 250
Total edges: 180
Unique nodes: 120
```

**When to use:** Initial data exploration, quality checks, reporting.

#### Recipe 5: Layer-by-Layer Analysis

**Task:** Analyze each layer independently and compare properties.

```
from py3plex.core import multinet
import networkx as nx

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Analyze each layer
layers = network.get_layers()
layer_stats = {}

for layer_name in layers:
    # Extract single layer
    layer = network.subnetwork([layer_name], subset_by="layers")

    # Compute layer statistics
    stats = {}
    stats['nodes'] = len(list(layer.get_nodes()))
    stats['edges'] = len(list(layer.get_edges()))
```

```
    # NetworkX metrics on layer
    G = layer.core_network
    if len(G) > 0:  # Check if layer has nodes
        stats['density'] = nx.density(G)
        stats['avg_degree'] = sum(dict(G.degree()).values()) / len(G)

        # Check connectivity
        if not layer.directed:
            stats['connected'] = nx.is_connected(G)
            if stats['connected']:
                stats['diameter'] = nx.diameter(G)

    layer_stats[layer_name] = stats

# Display results
print("\n=== Layer-by-Layer Analysis ===")
for layer, stats in layer_stats.items():
    print(f"\nLayer: {layer}")
    for metric, value in stats.items():
        print(f"  {metric}: {value}")
```

**When to use:** Understanding layer-specific properties, identifying important layers.

### Recipe 6: Compute Multilayer Statistics

**Task:** Calculate multilayer-specific metrics that account for cross-layer structure.

**Script:** `examples/network_analysis/example_multilayer_statistics.py`

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

layers = network.get_layers()

# === Layer-Level Statistics ===
print("=== Layer Statistics ===")
for layer in layers:
    density = mls.layer_density(network, layer)
    print(f"Layer {layer} density: {density:.4f}")

# Layer diversity
entropy = mls.entropy_of_multiplexity(network)
print(f"\nLayer diversity (entropy): {entropy:.4f} bits")

# === Node-Level Statistics ===
print("\n=== Node Activity ===")
sample_nodes = list(network.get_nodes())[:5]
```

```
for node_tuple in sample_nodes:
    node = node_tuple[0]  # Extract node name
    activity = mls.node_activity(network, node)
    print(f"Node {node} active in {activity*100:.1f}% of layers")

# === Cross-Layer Analysis ===
if len(layers) >= 2:
    print("\n=== Cross-Layer Similarity ===")
    layer1, layer2 = str(layers[0]), str(layers[1])

    # Cosine similarity of adjacency matrices
    similarity = mls.layer_similarity(network, layer1, layer2, method='cosine')
    print(f"Cosine similarity ({layer1} vs {layer2}): {similarity:.4f}")

    # Jaccard similarity of edge sets
    overlap = mls.edge_overlap(network, layer1, layer2)
    print(f"Edge overlap (Jaccard): {overlap:.4f}")

# === Versatility Analysis ===
print("\n=== Node Versatility (Cross-Layer Importance) ===")
versatility = mls.versatility_centrality(network, centrality_type='degree')
top_nodes = sorted(versatility.items(), key=lambda x: x[1], reverse=True)[:5]
for node, score in top_nodes:
    print(f"  {node}: {score:.4f}")
```

**When to use:** Quantifying multilayer structure, comparing layers, identifying versatile nodes.

### Centrality & Node Importance

#### Recipe 7: Single-Layer Centrality Analysis

**Task:** Identify important nodes within individual layers.

```
from py3plex.core import multinet

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Select a layer
layers = network.get_layers()
target_layer = [str(layers[0])]
layer = network.subnetwork(target_layer, subset_by="layers")

# Compute multiple centrality measures
centrality_measures = {
    'degree': layer.monoplex_nx_wrapper("degree_centrality"),
    'betweenness': layer.monoplex_nx_wrapper("betweenness_centrality"),
    'closeness': layer.monoplex_nx_wrapper("closeness_centrality"),
    'eigenvector': layer.monoplex_nx_wrapper("eigenvector_centrality"),
```

```
}

# Display top nodes for each measure
print(f"=== Centrality Analysis for Layer {target_layer[0]} ===\n")
for measure_name, scores in centrality_measures.items():
    print(f"{measure_name.capitalize()} Centrality (Top 5):")
    top_nodes = sorted(scores.items(), key=lambda x: x[1], reverse=True)[:5]
    for node, score in top_nodes:
        print(f"  {node}: {score:.4f}")
    print()
```

**When to use:** Identifying influential nodes, hub detection, network simplification.

### Recipe 8: Multiplex Participation Coefficient

**Task:** Measure how evenly nodes participate across layers in a multiplex network.

```
from py3plex.core import multinet
from py3plex.algorithms.multicentrality import multiplex_participation_coefficient

# Load or create multiplex network (same nodes across all layers)
network = multinet.multi_layer_network().load_network(
    "data/multiplex_network.txt",
    input_type="multiedgelist"
)

# Compute MPC (requires true multiplex: identical node set per layer)
try:
    mpc = multiplex_participation_coefficient(
        network,
        normalized=True,
        check_multiplex=True
    )

    # Display results
    print("=== Multiplex Participation Coefficient ===")
    print("(0 = active in one layer only, 1 = equally active across all layers)\n")

    # Sort by MPC score
    sorted_nodes = sorted(mpc.items(), key=lambda x: x[1], reverse=True)

    print("Top 10 most versatile nodes:")
    for node, score in sorted_nodes[:10]:
        print(f"  {node}: {score:.4f}")

    print("\nTop 10 most specialized nodes:")
    for node, score in sorted_nodes[-10:]:
        print(f"  {node}: {score:.4f}")

except ValueError as e:
    print(f"Error: {e}")
```

```
    print("Network is not a true multiplex (layers have different node sets)")
```

**When to use:** Analyzing multiplex networks, identifying cross-layer hubs, understanding node specialization.

### Community Detection

#### Recipe 9: Multilayer Community Detection

**Task:** Detect communities that span multiple layers.

**Script:** `examples/communities/example_community_detection.py`

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import Counter

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Run multilayer Louvain algorithm
# gamma: resolution parameter (higher = more communities)
# omega: inter-layer coupling strength (higher = communities span layers)
partition = louvain_multilayer(
    network,
    gamma=1.0,      # Standard resolution
    omega=1.0,      # Standard coupling
    random_state=42 # For reproducibility
)

# Analyze results
num_communities = len(set(partition.values()))
print(f"Detected {num_communities} communities")

# Community size distribution
community_sizes = Counter(partition.values())
print("\nTop 10 communities by size:")
for comm_id, size in community_sizes.most_common(10):
    print(f"  Community {comm_id}: {size} nodes")

# Check which nodes are in largest community
largest_comm = community_sizes.most_common(1)[0][0]
largest_comm_nodes = [node for node, comm in partition.items()
                      if comm == largest_comm]
print(f"\nNodes in largest community: {largest_comm_nodes[:10]}...")
```

**When to use:** Understanding network organization, identifying functional modules, clustering nodes.

### Recipe 10: Compare Community Detection Algorithms

**Task:** Run multiple community detection methods and compare results.

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection import community_wrapper as cw
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import Counter
import numpy as np

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Method 1: Single-layer Louvain on aggregated network
partition_louvain = cw.louvain_communities(network)

# Method 2: Multilayer Louvain
partition_multilayer = louvain_multilayer(
    network,
    gamma=1.0,
    omega=1.0,
    random_state=42
)

# Compare results
print("=== Community Detection Comparison ===\n")
print(f"Single-layer Louvain: {len(set(partition_louvain.values()))} communities")
print(f"Multilayer Louvain: {len(set(partition_multilayer.values()))} communities")

# Calculate normalized mutual information (if sklearn available)
try:
    from sklearn.metrics import normalized_mutual_info_score

    # Align node sets
    common_nodes = set(partition_louvain.keys()) & set(partition_multilayer.keys())
    labels1 = [partition_louvain[n] for n in common_nodes]
    labels2 = [partition_multilayer[n] for n in common_nodes]

    nmi = normalized_mutual_info_score(labels1, labels2)
    print(f"\nNormalized Mutual Information: {nmi:.4f}")
    print("(1.0 = identical partitions, 0.0 = independent)")
except ImportError:
    print("\nInstall scikit-learn to compute NMI: pip install scikit-learn")
```

**When to use:** Method validation, algorithm selection, robustness analysis.

### Visualization

#### Recipe 11: Basic Network Visualization

**Task:** Create publication-ready network visualizations.

**Script:** `examples/visualization/example_multilayer_visualization.py`

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import hairball_plot
import matplotlib
matplotlib.use('Agg')  # For non-interactive environments
import matplotlib.pyplot as plt

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Get network in visualization format
network_colors, graph = network.get_layers(style="hairball")

# Create visualization
plt.figure(figsize=(12, 12))
hairball_plot(
    graph,
    network_colors,
    layout_algorithm="force",        # Options: "force", "circular", "spring"
    layout_parameters={"iterations": 100},
    node_size=5,
    edge_width=0.5,
    alpha_channel=0.8,
    scale_by_size=True
)
plt.title("Multilayer Network", fontsize=16, fontweight='bold')
plt.axis('off')
plt.tight_layout()
plt.savefig("network_viz.png", dpi=150, bbox_inches='tight',
            facecolor='white', edgecolor='none')
plt.close()
print("Visualization saved to network_viz.png")
```

**When to use:** Presentations, papers, reports, exploratory analysis.

#### Recipe 12: Visualize Communities on Network

**Task:** Color nodes by community membership in network plot.

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import hairball_plot
from py3plex.visualization.colors import colors_default
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import Counter
```

```
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use('Agg')

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Detect communities
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)

# Select top N communities to color
top_n = 10
community_counts = Counter(partition.values())
top_communities = [c for c, _ in community_counts.most_common(top_n)]

# Create color mapping
color_map = dict(zip(top_communities, colors_default[:top_n]))

# Assign colors to nodes
network_colors = [
    color_map.get(partition.get(node), "lightgray")
    for node in network.get_nodes()
]

# Get network for visualization
_, graph = network.get_layers(style="hairball")

# Create plot
plt.figure(figsize=(14, 14))
hairball_plot(
    graph,
    network_colors,
    layout_algorithm="force",
    layout_parameters={"iterations": 150},
    node_size=8,
    edge_width=0.5,
    alpha_channel=0.7,
    scale_by_size=True
)
plt.title(f"Network with {top_n} Largest Communities",
          fontsize=16, fontweight='bold')
plt.axis('off')
plt.tight_layout()
plt.savefig("network_communities.png", dpi=150,
            bbox_inches='tight', facecolor='white')
plt.close()
print("Community visualization saved to network_communities.png")
```

**When to use:** Visualizing analysis results, understanding community structure.

### Advanced Workflows

#### Recipe 13: Complete Analysis Pipeline

**Task:** Full workflow from data loading to results export.

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import Counter
import json
import numpy as np

# Set random seed
np.random.seed(42)

# === 1. Load Data ===
print("=== Loading Network ===")
network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist",
    directed=False
)
network.basic_stats()

# === 2. Compute Statistics ===
print("\n=== Computing Multilayer Statistics ===")
layers = network.get_layers()

stats = {
    'num_layers': len(layers),
    'layer_densities': {},
    'layer_diversity': mls.entropy_of_multiplexity(network),
}

for layer in layers:
    stats['layer_densities'][str(layer)] = mls.layer_density(network, layer)

# Versatility
versatility = mls.versatility_centrality(network, centrality_type='degree')
top_versatile = sorted(versatility.items(), key=lambda x: x[1], reverse=True)[:10]
stats['top_versatile_nodes'] = {str(k): float(v) for k, v in top_versatile}

# === 3. Community Detection ===
print("\n=== Detecting Communities ===")
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
community_sizes = Counter(partition.values())

stats['num_communities'] = len(set(partition.values()))
stats['community_sizes'] = dict(community_sizes.most_common(10))

# === 4. Export Results ===
```

```
print("\n=== Exporting Results ===")

# Save statistics as JSON
with open("analysis_results.json", "w") as f:
    json.dump(stats, f, indent=2)
print("Statistics saved to analysis_results.json")

# Save community assignments
with open("communities.txt", "w") as f:
    for node, comm in partition.items():
        f.write(f"{node}\t{comm}\n")
print("Communities saved to communities.txt")

# Save processed network
network.save_network("processed_network.graphml", output_type="graphml")
print("Network saved to processed_network.graphml")

print("\n=== Analysis Complete ===")
```

**When to use:** Reproducible research workflows, batch processing, automated analysis.

### Recipe 14: Network Comparison Study

**Task:** Compare multiple networks systematically.

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
import pandas as pd
import numpy as np

# List of networks to compare
network_files = [
    ("Network A", "data/network_a.txt"),
    ("Network B", "data/network_b.txt"),
    ("Network C", "data/network_c.txt"),
]

results = []

for name, filepath in network_files:
    print(f"\nAnalyzing {name}...")

    # Load network
    network = multinet.multi_layer_network().load_network(
        filepath,
        input_type="multiedgelist",
        directed=False
    )

    # Compute metrics
    layers = network.get_layers()
```

```
    nodes = list(network.get_nodes())
    edges = list(network.get_edges())

    metrics = {
        'Network': name,
        'Layers': len(layers),
        'Total Nodes': len(nodes),
        'Unique Nodes': len(set([n[0] for n in nodes])),
        'Total Edges': len(edges),
        'Layer Diversity': mls.entropy_of_multiplexity(network),
    }

    # Average layer density
    densities = [mls.layer_density(network, layer) for layer in layers]
    metrics['Avg Layer Density'] = np.mean(densities)
    metrics['Std Layer Density'] = np.std(densities)

    results.append(metrics)

# Create comparison table
df = pd.DataFrame(results)
print("\n=== Network Comparison ===")
print(df.to_string(index=False))

# Save to CSV
df.to_csv("network_comparison.csv", index=False)
print("\nComparison saved to network_comparison.csv")
```

**When to use:** Comparative studies, evaluating datasets, selecting networks for analysis.

### Recipe 15: Random Walks for Node Embeddings

**Task:** Generate node embeddings using random walks (Node2Vec approach).

**Script:** `examples/advanced/example_random_walks.py`

```
from py3plex.core import multinet
from py3plex.algorithms.general.walkers import generate_walks, node2vec_walk
import numpy as np

# Set random seed
np.random.seed(42)

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Get core NetworkX graph
G = network.core_network

# Generate random walks
```

```
# p: return parameter (higher = less likely to return to previous node)
# q: in-out parameter (higher = more BFS-like, lower = more DFS-like)
walks = generate_walks(
    G,
    num_walks=10,       # Number of walks per node
    walk_length=80,     # Length of each walk
    p=1.0,              # Return parameter
    q=1.0,              # In-out parameter
    seed=42
)

print(f"Generated {len(walks)} random walks")
print(f"First walk (truncated): {walks[0][:10]}...")

# Use walks for downstream tasks:
# 1. Train Word2Vec model (if gensim available)
try:
    from gensim.models import Word2Vec

    # Convert walks to strings for Word2Vec
    walks_str = [[str(node) for node in walk] for walk in walks]

    # Train embedding model
    model = Word2Vec(
        walks_str,
        vector_size=128,    # Embedding dimension
        window=5,           # Context window
        min_count=1,
        sg=1,               # Skip-gram
        workers=4,
        epochs=5,
        seed=42
    )

    print(f"\nTrained embedding model with {len(model.wv)} nodes")

    # Get embedding for a node
    sample_node = str(list(G.nodes())[0])
    if sample_node in model.wv:
        embedding = model.wv[sample_node]
        print(f"Embedding for node {sample_node}: {embedding[:5]}... (dim={len(embedding)}

        # Find similar nodes
        similar = model.wv.most_similar(sample_node, topn=5)
        print(f"\nMost similar nodes to {sample_node}:")
        for node, similarity in similar:
            print(f"  {node}: {similarity:.4f}")

    # Save embeddings
    model.wv.save_word2vec_format("node_embeddings.txt")
```

```
    print("\nEmbeddings saved to node_embeddings.txt")

except ImportError:
    print("\nInstall gensim for embedding generation: pip install gensim")
```

**When to use:** Node classification, link prediction, network clustering, feature extraction.

### Performance & Optimization

#### Recipe 16: Handle Large Networks Efficiently

**Task:** Process large multilayer networks without running out of memory.

```
from py3plex.core import multinet
import gc

# === 1. Load network efficiently ===
# Use generators instead of loading all at once
network = multinet.multi_layer_network()

# For very large files, load in chunks or stream
# (This is a conceptual example - actual implementation may vary)

# === 2. Process layers independently ===
layers = network.get_layers()

layer_results = {}
for layer_name in layers:
    print(f"Processing layer {layer_name}...")

    # Extract and process one layer at a time
    layer = network.subnetwork([layer_name], subset_by="layers")

    # Compute statistics for this layer
    nodes = len(list(layer.get_nodes()))
    edges = len(list(layer.get_edges()))

    layer_results[str(layer_name)] = {'nodes': nodes, 'edges': edges}

    # Clean up to free memory
    del layer
    gc.collect()

print("\n=== Results ===")
for layer, stats in layer_results.items():
    print(f"{layer}: {stats['nodes']} nodes, {stats['edges']} edges")

# === 3. Use sparse representations ===
# When computing supra-adjacency matrix
supra_adj = network.get_supra_adjacency_matrix(sparse=True)
print(f"\nSupra-adjacency matrix: {supra_adj.shape} (sparse)")
```

```
# === 4. Sample for exploratory analysis ===
# For visualization or initial exploration, work with a sample
all_nodes = list(network.get_nodes())
import random
random.seed(42)
sample_size = min(1000, len(all_nodes))
sampled_nodes = random.sample(all_nodes, sample_size)

# Create subnetwork with sampled nodes
# (Implementation depends on network structure)
print(f"\nSampled {sample_size} nodes for exploratory analysis")
```

**When to use:** Networks with >10,000 nodes, limited RAM, large-scale studies.

**Tips:** - Process layers sequentially rather than all at once - Use sparse matrices for large supra-adjacency representations - Sample networks for visualization and initial exploration - Save intermediate results to disk - Use generators and iterators instead of lists

### Recipe 17: Benchmark and Profile Your Analysis

**Task:** Measure performance and identify bottlenecks.

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
import time
import psutil
import os

def get_memory_usage():
    """Get current memory usage in MB"""
    process = psutil.Process(os.getpid())
    return process.memory_info().rss / 1024 / 1024

# Load network
print("=== Performance Benchmarking ===\n")
start_mem = get_memory_usage()

start = time.time()
network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)
load_time = time.time() - start
load_mem = get_memory_usage() - start_mem

print(f"Network loading:")
print(f"  Time: {load_time:.3f} seconds")
print(f"  Memory: {load_mem:.2f} MB")

# Benchmark statistics computation
start = time.time()
```

```
network.basic_stats()
stats_time = time.time() - start
print(f"\nBasic statistics:")
print(f"  Time: {stats_time:.3f} seconds")

# Benchmark community detection
start_mem = get_memory_usage()
start = time.time()
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
comm_time = time.time() - start
comm_mem = get_memory_usage() - start_mem

print(f"\nCommunity detection:")
print(f"  Time: {comm_time:.3f} seconds")
print(f"  Memory: {comm_mem:.2f} MB")
print(f"  Communities found: {len(set(partition.values()))}")

# Total resource usage
print(f"\n=== Total Performance ===")
print(f"Total time: {load_time + stats_time + comm_time:.3f} seconds")
print(f"Peak memory: {get_memory_usage():.2f} MB")
```

**When to use:** Optimizing workflows, comparing algorithms, capacity planning.

**Note:** Requires `psutil` package: `pip install psutil`

### Tips & Best Practices

#### General Tips

1. **Always set random seeds** for reproducibility:

   ```
   import numpy as np
   np.random.seed(42)
   ```

2. **Verify network structure** after loading:

   ```
   network.basic_stats()  # Quick sanity check
   ```

3. **Handle errors gracefully**:

   ```
   try:
       result = network.some_operation()
   except Exception as e:
       print(f"Operation failed: {e}")
   ```

4. **Save intermediate results** for long analyses:

   ```
   import pickle
   with open("checkpoint.pkl", "wb") as f:
       pickle.dump(results, f)
   ```

5. **Use appropriate file formats**:

- GraphML: Standard, compatible with other tools
- Pickle: Fast, preserves Python objects, not portable
- Edgelist: Simple, human-readable, large file size

### Performance Tips

1. **For large networks**, process layers independently
2. **Use sparse matrices** when working with supra-adjacency
3. **Sample networks** for visualization (>1000 nodes gets crowded)
4. **Profile your code** to find bottlenecks
5. **Close plots** after saving to free memory: `plt.close()`

### Visualization Tips

1. **Choose layout carefully**:
   - Force-directed: General purpose, slow for >5000 nodes
   - Circular: Fast, good for small networks
   - Spring: Balance between quality and speed
2. **Adjust node size** based on network size:

```
node_size = max(1, 100 / np.sqrt(num_nodes))
```

3. **Use alpha channel** for edge-dense networks:

```
alpha_channel=0.3  # More transparent
```

4. **Save high-resolution** for publications:

```
plt.savefig("figure.png", dpi=300)
```

### Community Detection Tips

1. **Tune resolution parameter** (gamma):
   - Higher gamma → more, smaller communities
   - Lower gamma → fewer, larger communities
2. **Tune coupling parameter** (omega) for multilayer:
   - Higher omega → communities span layers
   - Lower omega → layer-specific communities
3. **Run multiple times** with different random seeds to check stability
4. **Compare multiple algorithms** to validate findings

### Common Issues & Solutions

#### Issue: "Network appears to be empty"

**Cause:** Incorrect file format or delimiter.

**Solution:**:

```
# Check file format
with open("data.txt", "r") as f:
    print(f.readlines()[:5])  # Inspect first lines

# Try different input_type
network = multinet.multi_layer_network().load_network(
    "data.txt",
    input_type="multiedgelist"  # or "edgelist", "graphml"
)
```

#### Issue: "No module named 'community'"

**Cause:** Optional dependency not installed.

**Solution:**:

```
pip install python-louvain
```

#### Issue: "Visualization doesn't appear"

**Cause:** Using non-interactive backend or missing display.

**Solution:**:

```
import matplotlib
matplotlib.use('Agg')  # For saving files
import matplotlib.pyplot as plt

# ... create plot ...
plt.savefig("output.png")  # Save instead of show
```

#### Issue: "Memory error with large network"

**Cause:** Loading entire network into RAM at once.

**Solution:** See Recipe 16 for memory-efficient processing.

#### Issue: "Community detection is slow"

**Cause:** Large network or complex structure.

**Solution:**:

```
# Use simpler algorithm for quick results
from py3plex.algorithms.community_detection import community_wrapper as cw
partition = cw.louvain_communities(network)  # Faster than multilayer
```

### Further Reading

**Documentation:**

- Installation guide - See main documentation
- Quick start tutorial - See main documentation
- Core concepts - See main documentation
- Algorithm selection guide - See main documentation

**Examples:**

See `examples/` directory for 50+ complete, working examples covering all major functionality.

**Research:**

- Kivelä et al. (2014) - Multilayer networks theory
- De Domenico et al. (2013) - Mathematical framework
- See Citations section in main documentation for complete reference list

### DSL Query Recipes

The py3plex DSL provides powerful, concise syntax for network queries. These recipes showcase common patterns.

### Recipe 18: Hub Identification with DSL

**Task:** Find and rank influential nodes across layers using the DSL.

**DSL Example: Hub Identification**

```
from py3plex.core import multinet
from py3plex.dsl import Q, L

# Load network
network = multinet.multi_layer_network().load_network(
    "data/network.txt", input_type="multiedgelist"
)

# Find top 20 hubs by betweenness centrality
hubs = (
    Q.nodes()
     .where(degree__gt=3)  # Pre-filter by degree for efficiency
     .compute("betweenness_centrality", "degree", "clustering")
     .order_by("-betweenness_centrality")
     .limit(20)
     .export_csv("top_hubs.csv")
     .execute(network)
)

# Display results
df = hubs.to_pandas()
print(df)
```

**When to use:** Identifying key players, finding bridge nodes, influence analysis.

### Recipe 19: Multi-Layer Comparative Analysis

**Task:** Compare network structure across different layers using DSL.

**DSL Example: Layer Comparison**

```
from py3plex.dsl import Q, L
import pandas as pd

# Analyze each layer separately
layers = ["social", "work", "family"]
results = []

for layer_name in layers:
    result = (
        Q.nodes()
         .from_layers(L[layer_name])
         .compute("degree", "betweenness_centrality", "clustering")
         .execute(network)
    )

    # Calculate layer-level statistics
    df = result.to_pandas()
    layer_stats = {
        'layer': layer_name,
        'nodes': result.count,
        'avg_degree': df['degree'].mean(),
        'max_centrality': df['betweenness_centrality'].max(),
        'avg_clustering': df['clustering'].mean(),
    }
    results.append(layer_stats)

# Create comparison table
comparison_df = pd.DataFrame(results)
print(comparison_df)
```

**When to use:** Understanding layer-specific properties, detecting structural differences.

### Recipe 20: Advanced Filtering and Export

**Task:** Complex multi-criteria filtering with automated export.

**DSL Example: Advanced Filtering**

```
from py3plex.dsl import Q, L

# Find nodes that meet multiple criteria
result = (
```

```
    Q.nodes()
     # Combine layers using set operations
     .from_layers(L["social"] + L["work"] - L["bots"])
     # Multiple filter conditions
     .where(degree__gte=5, clustering__gt=0.3)
     # Compute multiple measures
     .compute("betweenness_centrality", "pagerank", "degree")
     # Sort and limit
     .order_by("-pagerank")
     .limit(50)
     # Export in multiple formats
     .export_json("filtered_nodes.json", orient="records")
     .execute(network)
)

# Also available as pandas DataFrame
df = result.to_pandas()

# Or as NetworkX subgraph
subgraph = result.to_networkx(network)
```

**When to use:** Targeted analysis, automated reporting, filtering complex networks.

### Recipe 21: Parameterized Query Templates

**Task:** Create reusable query templates for systematic analysis.

**DSL Example: Query Templates**

```
from py3plex.dsl import Q, L, Param

# Define a reusable query template
def find_top_nodes(layer_name, degree_threshold, top_n):
    """Reusable template for finding top nodes in a layer"""
    return (
        Q.nodes()
         .from_layers(L[layer_name])
         .where(degree__gt=degree_threshold)
         .compute("betweenness_centrality", "pagerank")
         .order_by("-betweenness_centrality")
         .limit(top_n)
    )

# Use the template with different parameters
social_hubs = find_top_nodes("social", 5, 10).execute(network)
work_hubs = find_top_nodes("work", 3, 20).execute(network)

# Alternative: Use Param for dynamic parameter binding
query = (
    Q.nodes()
```

```
    .from_layers(L[Param.str("layer")])
    .where(degree__gt=Param.int("min_degree"))
    .compute("betweenness_centrality")
    .limit(Param.int("top_n"))
)

# Execute with different parameters
result1 = query.execute(network, layer="social", min_degree=5, top_n=10)
result2 = query.execute(network, layer="work", min_degree=3, top_n=20)
```

**When to use:** Batch analysis, systematic parameter sweeps, reproducible workflows.

### Recipe 22: DSL with EXPLAIN Mode

**Task:** Optimize queries before execution on large networks.

**DSL Example: Query Optimization**

```
from py3plex.dsl import Q, L

# Build a potentially expensive query
query = (
    Q.nodes()
     .compute("betweenness_centrality")  # O(n^3) operation!
     .order_by("-betweenness_centrality")
     .limit(10)
)

# Check the execution plan first
plan = query.explain().execute(network)

print("Query Execution Plan:")
for step in plan.steps:
    print(f"- {step.description}")
    print(f"  Complexity: {step.estimated_complexity}")

# Check for warnings
if plan.warnings:
    print("\nWarnings:")
    for warning in plan.warnings:
        print(f"  {warning}")

# Optimized version: filter first to reduce computation
optimized_query = (
    Q.nodes()
     .where(degree__gt=5)  # Pre-filter to reduce nodes
     .compute("betweenness_centrality")
     .order_by("-betweenness_centrality")
     .limit(10)
)
```

```
# Execute the optimized query
result = optimized_query.execute(network)
```

**When to use:** Large networks, expensive centrality computations, performance optimization.

### 3.6.4 Use Cases & Case Studies

*"In theory there is no difference between theory and practice. In practice there is."* — Yogi Berra

**DSL in Case Studies**

Throughout these case studies, notice how DSL simplifies analysis:

```
from py3plex.dsl import Q, L

# Identify candidate proteins (biological networks)
candidates = (
    Q.nodes()
     .where(degree__gt=10)
     .compute("betweenness_centrality", "clustering")
     .order_by("-betweenness_centrality")
     .limit(50)
     .execute(ppi_network)
)

# Compare social media presence (social networks)
for platform in ["twitter", "facebook", "instagram"]:
    influencers = (
        Q.nodes()
         .from_layers(L[platform])
         .where(degree__gt=100)
         .execute(social_network)
    )
    print(f"{platform}: {influencers.count} influencers")
```

DSL enables rapid iteration in exploratory analysis!

This chapter provides complete, end-to-end case studies demonstrating py3plex in different research domains. Each case study includes:

1. **Problem context** — What real-world question are we answering?
2. **Data modeling decisions** — How do we represent this problem as a multilayer network?
3. **Complete code** — Working examples you can adapt
4. **Interpretation** — What do the results mean?
5. **Adaptation guide** — How to apply this template to your own data

### Why Case Studies Matter

Reading about algorithms is one thing. Applying them to real problems is another entirely.

The case studies in this chapter come from real research domains where multilayer network analysis has proven valuable. They're not toy examples—they represent the kinds of questions that researchers and practitioners actually ask:

- How do we identify proteins that are likely to be functionally important?
- Which social media influencers are genuinely cross-platform celebrities versus one-hit wonders?
- What happens to urban mobility when a major transit hub fails?
- Which academic researchers are likely to collaborate based on their publication patterns?

Each case study walks through the complete analysis workflow, from problem formulation through data modeling, analysis, and interpretation. Pay attention not just to the code, but to the reasoning behind each decision.

### Lessons from the Field

Before diving into specific cases, here are some lessons learned from applying multilayer network analysis to real problems:

**Modeling decisions matter more than algorithm choice.** Whether you use Louvain or Infomap for community detection is less important than whether you've correctly identified what should be a layer, what should be a node, and how layers should be coupled. Spend your time on modeling.

**Start simple, add complexity as needed.** It's tempting to immediately build a network with ten layers, weighted edges, temporal dynamics, and inter-layer dependencies. Resist this urge. Start with the simplest model that might answer your question, and add complexity only when you can demonstrate it improves your analysis.

**Validate with domain knowledge.** Network analysis can reveal surprising patterns—but it can also reveal artifacts of your modeling choices. Always sanity-check results against what domain experts know. If your community detection puts obviously unrelated entities in the same cluster, you have a modeling problem.

**Document your decisions.** When you return to an analysis six months later, you won't remember why you chose certain parameters. Write down your reasoning. Future-you will thank present-you.

---

### Case Study 1: Biological Network Analysis

*Identifying high-confidence protein interactions and functional modules*

### Real-World Impact

Protein-protein interaction (PPI) networks are foundational to understanding cellular function. The proteins in your body don't work alone—they form complexes, signaling cascades, and metabolic pathways. Knowing which proteins interact helps researchers:

- **Understand disease mechanisms:** Many diseases result from disrupted protein interactions
- **Identify drug targets:** Proteins central to disease networks are potential therapeutic targets

- **Predict protein function:** A protein's interactors suggest its biological role

The challenge? PPI data comes from multiple experimental methods, each with different biases and error rates. High-throughput screens find many interactions quickly but include false positives. Literature curation is accurate but incomplete. How do you combine these sources intelligently?

**This is exactly what multilayer network analysis is for.**

### Multilayer Protein-Protein Interaction Network

**Problem Context:**

You're a computational biologist studying protein interactions in yeast. You have PPI data from three experimental sources with different reliability levels:

- **Yeast two-hybrid (Y2H):** High-throughput but many false positives
- **Affinity purification mass spectrometry (AP-MS):** More reliable but biased toward stable complexes
- **Literature curated:** Highly reliable but incomplete

You want to:

1. Find functional modules (groups of proteins that work together)
2. Identify proteins that are central across evidence types (likely real interactions)
3. Discover proteins that bridge different functional modules

**Data Modeling Decisions:**

- **Node type:** Proteins (same set across layers for multiplex analysis)
- **Layers:** One per evidence type (Y2H, AP-MS, Literature)
- **Edges:** Physical protein-protein interactions
- **Network type:** Multiplex (same proteins, different interaction evidence)

**Complete Code:**

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import Counter
import numpy as np

# === 1. Create the network ===
# In practice, you'd load from files; here we create a toy example
network = multinet.multi_layer_network(network_type='multiplex')

# Simulated PPI data (protein pairs by evidence type)
# Replace with your actual data loading
ppi_data = {
    'Y2H': [
        ('CDC28', 'CLB2'), ('CDC28', 'CLB5'), ('CLB2', 'SIC1'),
        ('ACT1', 'MYO2'), ('ACT1', 'TPM1'), ('MYO2', 'TPM1'),
        ('CDC28', 'CDC6'), ('CDC6', 'ORC1'),  # cell cycle proteins
    ],
```

```
    'APMS': [
        ('CDC28', 'CLB2'), ('CDC28', 'CKS1'), ('CLB2', 'CKS1'),
        ('ACT1', 'ABP1'), ('ACT1', 'TPM1'),
        ('CDC6', 'ORC1'), ('ORC1', 'ORC2'), ('ORC2', 'ORC3'),  # origin recognition complex
    ],
    'Literature': [
        ('CDC28', 'CLB2'), ('CDC28', 'CLB5'), ('CDC28', 'CKS1'),
        ('ACT1', 'MYO2'), ('MYO2', 'TPM1'),
        ('CDC6', 'ORC1'),  # well-established interaction
    ],
}

# Add edges to network
for layer, interactions in ppi_data.items():
    for protein1, protein2 in interactions:
        network.add_edges([
            [protein1, layer, protein2, layer, 1.0]
        ], input_type="list")

# === 2. Basic exploration ===
print("=== Network Overview ===")
network.basic_stats()

# === 3. Identify high-confidence interactions ===
# Interactions that appear in multiple evidence types are more reliable
print("\n=== Evidence Overlap Analysis ===")

# Get edges from each layer
edges_by_layer = {}
for layer in network.get_layers():
    layer_subnet = network.subnetwork([layer], subset_by="layers")
    # Extract edge pairs (ignoring layer info)
    edges = set()
    for edge in layer_subnet.get_edges():
        # Normalize edge representation (sorted tuple)
        pair = tuple(sorted([edge[0][0], edge[1][0]]))
        edges.add(pair)
    edges_by_layer[layer] = edges

# Find edges in multiple layers
all_edges = set()
for edges in edges_by_layer.values():
    all_edges.update(edges)

edge_evidence_counts = {}
for edge in all_edges:
    count = sum(1 for layer_edges in edges_by_layer.values() if edge in layer_edges)
    edge_evidence_counts[edge] = count

print("High-confidence interactions (in 3/3 evidence types):")
```

```
for edge, count in sorted(edge_evidence_counts.items(), key=lambda x: x[1], reverse=True):
    if count == 3:
        print(f"  {edge[0]} -- {edge[1]}: {count} evidence types")

print("\nMedium-confidence interactions (in 2/3 evidence types):")
for edge, count in sorted(edge_evidence_counts.items(), key=lambda x: x[1], reverse=True):
    if count == 2:
        print(f"  {edge[0]} -- {edge[1]}: {count} evidence types")

# === 4. Node activity analysis ===
print("\n=== Protein Activity Across Evidence Types ===")
unique_proteins = set()
for node in network.get_nodes():
    unique_proteins.add(node[0])  # Extract protein name

protein_activity = {}
for protein in unique_proteins:
    activity = mls.node_activity(network, protein)
    protein_activity[protein] = activity

print("Proteins present in all evidence types (activity = 1.0):")
for protein, activity in sorted(protein_activity.items(), key=lambda x: x[1], reverse=True)
    if activity == 1.0:
        print(f"  {protein}")

# === 5. Community detection (functional modules) ===
print("\n=== Functional Module Detection ===")

# Lower coupling (omega=0.5) because different evidence types
# may reveal different aspects of the interactome
partition = louvain_multilayer(
    network,
    gamma=1.0,      # Standard resolution
    omega=0.5,      # Moderate coupling
    random_state=42
)

# Analyze communities
num_communities = len(set(partition.values()))
print(f"Detected {num_communities} functional modules")

# Group proteins by community
communities = {}
for node, comm in partition.items():
    protein = node[0]  # Extract protein name
    if comm not in communities:
        communities[comm] = set()
    communities[comm].add(protein)

print("\nFunctional modules found:")
```

```
for comm_id, proteins in sorted(communities.items(), key=lambda x: len(x[1]), reverse=True)
    print(f"  Module {comm_id}: {proteins}")

# === 6. Versatility analysis (cross-module bridging) ===
print("\n=== Bridge Proteins (High Versatility) ===")
versatility = mls.versatility_centrality(network, centrality_type='degree')
top_versatile = sorted(versatility.items(), key=lambda x: x[1], reverse=True)[:5]
for node, score in top_versatile:
    print(f"  {node}: versatility = {score:.4f}")
```

**Interpretation:**

The analysis reveals several insights:

1. **High-confidence interactions** (CDC28-CLB2, ACT1-TPM1) appear in all three evidence types, making them likely true interactions suitable for downstream analysis.
2. **Functional modules** correspond to known protein complexes:
   - Cell cycle regulators (CDC28, CLB2, CLB5, CKS1, SIC1)
   - Actin cytoskeleton (ACT1, MYO2, TPM1, ABP1)
   - DNA replication origin complex (CDC6, ORC1, ORC2, ORC3)
3. **Bridge proteins** with high versatility may connect different functional modules and could be interesting drug targets or key regulators.

**Adapting to Your Data:**

1. Replace the `ppi_data` dictionary with your actual data loading code
2. Adjust `omega` based on your domain: lower for diverse evidence types, higher if you expect consistency
3. Add Gene Ontology enrichment analysis to validate that communities match known functional annotations
4. Consider edge weights based on evidence quality (e.g., literature = 3.0, APMS = 2.0, Y2H = 1.0)

### Case Study 2: Multi-Platform Social Network Analysis

#### Cross-Platform Influence Analysis

**Problem Context:**

You're a social media researcher studying how influencers operate across platforms. You have data from three platforms:

- **Twitter/X:** Public follower/following relationships
- **YouTube:** Subscription relationships
- **Instagram:** Follow relationships

You want to:

1. Identify influencers who are successful across platforms (vs. platform-specific)
2. Understand how different platforms are related (do the same people follow each other?)
3. Find communities that span platforms

**Data Modeling Decisions:**

- **Node type:** Users (mapped across platforms using verified identities)
- **Layers:** One per platform (Twitter, YouTube, Instagram)
- **Edges:** Follow/subscribe relationships (directed in reality, undirected for this analysis)
- **Network type:** Multiplex (same users, different platforms)

**Complete Code:**

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
from collections import Counter
import networkx as nx

# === 1. Create the network ===
network = multinet.multi_layer_network(network_type='multiplex')

# Simulated social network data
# In practice: load from APIs or datasets with user ID mapping
social_data = {
    'Twitter': [
        ('influencer_A', 'user_1'), ('influencer_A', 'user_2'),
        ('influencer_A', 'user_3'), ('influencer_B', 'user_2'),
        ('influencer_B', 'user_4'), ('user_1', 'user_2'),
        ('user_3', 'user_5'), ('influencer_C', 'user_6'),
    ],
    'YouTube': [
        ('influencer_A', 'user_1'), ('influencer_A', 'user_3'),
        ('influencer_B', 'user_2'), ('influencer_B', 'user_3'),
        ('influencer_B', 'user_4'), ('influencer_C', 'user_6'),
        ('influencer_C', 'user_7'),
    ],
    'Instagram': [
        ('influencer_A', 'user_1'), ('influencer_A', 'user_2'),
        ('influencer_A', 'user_4'), ('influencer_B', 'user_5'),
        ('user_1', 'user_2'), ('user_2', 'user_3'),
        ('influencer_C', 'user_6'), ('influencer_C', 'user_8'),
    ],
}

for layer, connections in social_data.items():
    for user1, user2 in connections:
        network.add_edges([
            [user1, layer, user2, layer, 1.0]
        ], input_type="list")

# === 2. Basic statistics ===
print("=== Network Overview ===")
network.basic_stats()
```

```python
# === 3. Cross-platform presence ===
print("\n=== Cross-Platform Presence ===")
unique_users = set()
for node in network.get_nodes():
    unique_users.add(node[0])

for user in sorted(unique_users):
    activity = mls.node_activity(network, user)
    platforms = activity * len(network.get_layers())
    if activity == 1.0:
        print(f"  {user}: present on ALL platforms")
    elif activity > 0.5:
        print(f"  {user}: present on {platforms:.0f}/3 platforms")

# === 4. Platform-specific centrality ===
print("\n=== Centrality by Platform ===")

for platform in network.get_layers():
    layer_subnet = network.subnetwork([platform], subset_by="layers")
    G = layer_subnet.core_network

    # Compute degree centrality
    degree_cent = nx.degree_centrality(G)
    top_users = sorted(degree_cent.items(), key=lambda x: x[1], reverse=True)[:3]

    print(f"\n{platform} - Top 3 by degree:")
    for node, score in top_users:
        print(f"  {node[0]}: {score:.3f}")

# === 5. Cross-platform influence (versatility) ===
print("\n=== Cross-Platform Influencers (High Versatility) ===")
versatility = mls.versatility_centrality(network, centrality_type='degree')
top_versatile = sorted(versatility.items(), key=lambda x: x[1], reverse=True)[:5]

for node, score in top_versatile:
    print(f"  {node}: {score:.4f}")

# === 6. Layer similarity ===
print("\n=== Platform Similarity ===")
layers = list(network.get_layers())
for i in range(len(layers)):
    for j in range(i+1, len(layers)):
        layer1, layer2 = str(layers[i]), str(layers[j])

        # Edge overlap
        overlap = mls.edge_overlap(network, layer1, layer2)

        # Jaccard similarity
        similarity = mls.layer_similarity(network, layer1, layer2, method='jaccard')
```

```
        print(f"  {layer1} vs {layer2}:")
        print(f"    Edge overlap: {overlap:.3f}")
        print(f"    Jaccard similarity: {similarity:.3f}")

# === 7. Cross-platform community detection ===
print("\n=== Cross-Platform Communities ===")

# High coupling because we expect influencers to attract similar audiences
partition = louvain_multilayer(
    network,
    gamma=1.0,
    omega=1.5,      # High coupling for cross-platform consistency
    random_state=42
)

num_communities = len(set(partition.values()))
print(f"Detected {num_communities} cross-platform communities")

# Analyze which users are in which community
communities = {}
for node, comm in partition.items():
    user = node[0]
    if comm not in communities:
        communities[comm] = set()
    communities[comm].add(user)

print("\nCommunities (by unique users):")
for comm_id, users in sorted(communities.items(), key=lambda x: len(x[1]), reverse=True):
    influencers = [u for u in users if 'influencer' in u]
    regular = [u for u in users if 'influencer' not in u]
    print(f"  Community {comm_id}: {len(influencers)} influencers, {len(regular)} regular
    print(f"    Influencers: {influencers}")
    print(f"    Sample users: {regular[:3]}")
```

**Interpretation:**

1. **Cross-platform influencers** (influencer_A, influencer_B) have high versatility, indicating they successfully maintain audiences across platforms. Platform-specific influencers (influencer_C) may have niche appeal.
2. **Platform similarity** shows how audiences overlap:
   - High Twitter-Instagram similarity suggests similar user bases
   - Lower YouTube overlap might indicate different content consumption patterns
3. **Communities** reveal audience segments:
   - Some communities cluster around specific influencers
   - Cross-platform communities suggest genuine fandom that follows across platforms

**Adapting to Your Data:**

1. Map user identities across platforms (use verified accounts, linked profiles, or username matching)

2. Consider using directed edges for asymmetric follow relationships
3. Add edge weights based on interaction strength (likes, comments, shares)
4. Include temporal layers to track audience evolution over time

### Case Study 3: Multi-Modal Transportation Analysis

#### Urban Mobility and Resilience

**Problem Context:**

You're an urban transportation analyst studying a city's public transit system. You have network data for:

- **Metro/Subway:** High-capacity, fixed routes
- **Bus:** Lower capacity, more flexible routes
- **Bike-share:** Individual, first/last mile connections

You want to:

1. Identify multimodal hubs (stations serving multiple transport modes)
2. Analyze network resilience (what happens if a metro station closes?)
3. Find communities ("travel basins") that span transport modes

**Data Modeling Decisions:**

- **Node type:** Stations/stops (with geographic coordinates)
- **Layers:** One per transport mode (Metro, Bus, Bikeshare)
- **Edges:** Direct connections between stops (can walk/transfer)
- **Network type:** Multiplex with geographic coupling (same location = same node)

**Complete Code:**

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_multilayer
import networkx as nx
from collections import Counter

# === 1. Create the transportation network ===
network = multinet.multi_layer_network(network_type='multiplex')

# Simulated city transport data
# Stations are named by neighborhood/landmark
transport_data = {
    'Metro': [
        ('Central', 'Financial'), ('Financial', 'Tech_Park'),
        ('Central', 'University'), ('University', 'Hospital'),
        ('Central', 'Shopping_Mall'), ('Shopping_Mall', 'Residential_North'),
    ],
    'Bus': [
        ('Central', 'Financial'), ('Financial', 'Residential_East'),
        ('Central', 'University'), ('University', 'Residential_South'),
        ('Central', 'Shopping_Mall'), ('Shopping_Mall', 'Residential_North'),
```

```
        ('Tech_Park', 'Residential_East'),  # Bus connects where metro doesn't
        ('Hospital', 'Residential_South'),
        ('Airport', 'Central'),  # Bus to airport
    ],
    'Bikeshare': [
        ('Central', 'Financial'), ('Financial', 'Tech_Park'),
        ('Central', 'University'), ('University', 'Residential_South'),
        ('Shopping_Mall', 'Residential_North'),
        ('Tech_Park', 'Coffee_District'),  # Bike-only area
        ('University', 'Student_Housing'),  # Bike-only connection
    ],
}

for mode, connections in transport_data.items():
    for stop1, stop2 in connections:
        network.add_edges([
            [stop1, mode, stop2, mode, 1.0]
        ], input_type="list")

# === 2. Network overview ===
print("=== Transportation Network Overview ===")
network.basic_stats()

# === 3. Multimodal hub identification ===
print("\n=== Multimodal Hubs ===")
unique_stations = set()
for node in network.get_nodes():
    unique_stations.add(node[0])

station_modes = {}
for station in unique_stations:
    activity = mls.node_activity(network, station)
    num_modes = activity * len(network.get_layers())
    station_modes[station] = num_modes

print("Stations by number of transport modes:")
for station, modes in sorted(station_modes.items(), key=lambda x: x[1], reverse=True):
    mode_names = []
    for mode in network.get_layers():
        layer_subnet = network.subnetwork([mode], subset_by="layers")
        layer_nodes = [n[0] for n in layer_subnet.get_nodes()]
        if station in layer_nodes:
            mode_names.append(mode)

    if modes >= 2:
        print(f"  {station}: {modes:.0f} modes ({', '.join(mode_names)})")

# === 4. Mode-specific analysis ===
print("\n=== Transport Mode Characteristics ===")
```

```
for mode in network.get_layers():
    layer_subnet = network.subnetwork([mode], subset_by="layers")
    G = layer_subnet.core_network

    num_stops = G.number_of_nodes()
    num_connections = G.number_of_edges()
    density = mls.layer_density(network, mode)

    # Find most connected stops
    degree_cent = nx.degree_centrality(G)
    top_stop = max(degree_cent.items(), key=lambda x: x[1])

    print(f"\n{mode}:")
    print(f"  Stops: {num_stops}")
    print(f"  Connections: {num_connections}")
    print(f"  Network density: {density:.3f}")
    print(f"  Most connected: {top_stop[0][0]} (degree={top_stop[1]:.2f})")

# === 5. Resilience analysis: What if Central closes? ===
print("\n=== Resilience Analysis: Central Station Failure ===")

# Current connectivity
G_full = network.core_network
num_components_before = nx.number_connected_components(G_full.to_undirected())
print(f"Before failure: {num_components_before} connected component(s)")

# Simulate failure by removing Central from all layers
G_after = G_full.copy()
central_nodes = [n for n in G_after.nodes() if n[0] == 'Central']
G_after.remove_nodes_from(central_nodes)

num_components_after = nx.number_connected_components(G_after.to_undirected())
print(f"After removing Central: {num_components_after} connected component(s)")

# Which stations become disconnected?
if num_components_after > 1:
    components = list(nx.connected_components(G_after.to_undirected()))
    print("\nResulting network fragments:")
    for i, comp in enumerate(sorted(components, key=len, reverse=True)):
        stations = set(n[0] for n in comp)
        print(f"  Fragment {i+1} ({len(stations)} stations): {stations}")

# === 6. Travel basin detection ===
print("\n=== Travel Basins (Communities) ===")

# Moderate coupling: travel patterns may differ by mode
partition = louvain_multilayer(
    network,
    gamma=1.0,
    omega=1.0,
```

```
    random_state=42
)

num_basins = len(set(partition.values()))
print(f"Detected {num_basins} travel basins")

# Group stations by basin
basins = {}
for node, basin in partition.items():
    station = node[0]
    if basin not in basins:
        basins[basin] = set()
    basins[basin].add(station)

print("\nTravel basins:")
for basin_id, stations in sorted(basins.items(), key=lambda x: len(x[1]), reverse=True):
    print(f"  Basin {basin_id}: {stations}")
```

**Interpretation:**

1. **Multimodal hubs** (Central, Financial, University) are critical infrastructure points where passengers transfer between modes. These should be prioritized for maintenance and security.
2. **Resilience analysis** reveals that Central is a critical node—its failure fragments the network. This suggests the need for redundant connections or alternative routes.
3. **Travel basins** correspond to geographic areas where people typically travel:
   - Downtown basin: Central, Financial, Shopping_Mall
   - University/Hospital basin: University, Hospital, Student_Housing
   - Residential basins: various residential areas with their nearest transit

**Adapting to Your Data:**

1. Include geographic distance as edge weight (travel time between stops)
2. Add passenger flow data to weight edges by usage
3. Include temporal layers (peak hours, off-peak, weekend)
4. Add inter-layer edges representing walking transfers between nearby stations

### Case Study 4: Heterogeneous Academic Network

### Research Collaboration and Impact Analysis

**Problem Context:**

You're analyzing academic research patterns using data from publications. Your network includes:

- **Authors:** Researchers who write papers
- **Papers:** Research publications
- **Venues:** Conferences and journals where papers appear
- **Institutions:** Organizations where authors work

You want to:

1. Find influential authors (not just by citation count)
2. Identify research communities that span institutions
3. Discover emerging collaboration patterns

**Data Modeling Decisions:**

- **Node types:** Authors, Papers, Venues, Institutions (heterogeneous)
- **Edge types:** * Author → Paper (authorship) * Paper → Venue (publication) * Author → Institution (affiliation)
- **Network type:** Heterogeneous Information Network (HIN)

**Complete Code:**

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
import networkx as nx
from collections import Counter, defaultdict

# === 1. Create the academic network ===
network = multinet.multi_layer_network()

# Academic publication data
# In practice: load from DBLP, Semantic Scholar, or similar

# Authors write papers
authorship_data = [
    ('Dr_Smith', 'Paper_1'), ('Dr_Smith', 'Paper_2'),
    ('Dr_Jones', 'Paper_1'), ('Dr_Jones', 'Paper_3'),
    ('Dr_Chen', 'Paper_2'), ('Dr_Chen', 'Paper_4'),
    ('Dr_Kim', 'Paper_3'), ('Dr_Kim', 'Paper_4'),
    ('Dr_Garcia', 'Paper_5'), ('Dr_Lee', 'Paper_5'),
    ('Dr_Smith', 'Paper_6'), ('Dr_Lee', 'Paper_6'),  # Cross-institution collab
]

# Papers published in venues
publication_data = [
    ('Paper_1', 'ICML'), ('Paper_2', 'NeurIPS'),
    ('Paper_3', 'ICML'), ('Paper_4', 'NeurIPS'),
    ('Paper_5', 'AAAI'), ('Paper_6', 'ICML'),
]

# Authors affiliated with institutions
affiliation_data = [
    ('Dr_Smith', 'MIT'), ('Dr_Jones', 'MIT'),
    ('Dr_Chen', 'Stanford'), ('Dr_Kim', 'Stanford'),
    ('Dr_Garcia', 'Berkeley'), ('Dr_Lee', 'Berkeley'),
]

# Add to network with explicit layer/type information
for author, paper in authorship_data:
```

```
    network.add_edges([
        [author, 'authors', paper, 'papers', 1.0]
    ], input_type="list")

for paper, venue in publication_data:
    network.add_edges([
        [paper, 'papers', venue, 'venues', 1.0]
    ], input_type="list")

for author, institution in affiliation_data:
    network.add_edges([
        [author, 'authors', institution, 'institutions', 1.0]
    ], input_type="list")

# === 2. Network overview ===
print("=== Academic Network Overview ===")
network.basic_stats()

# === 3. Meta-path based analysis ===
# Meta-path: Author → Paper → Venue → Paper → Author
# This finds authors who publish in the same venues

print("\n=== Meta-Path Analysis: Authors Publishing in Same Venues ===")

G = network.core_network

# Find which authors publish where
author_venues = defaultdict(set)
for author, paper in authorship_data:
    for p, venue in publication_data:
        if paper == p:
            author_venues[author].add(venue)

print("Authors by venue:")
venue_authors = defaultdict(list)
for author, venues in author_venues.items():
    for venue in venues:
        venue_authors[venue].append(author)

for venue, authors in sorted(venue_authors.items()):
    print(f"  {venue}: {authors}")

# Find co-venue patterns (authors who could collaborate based on venue)
print("\nPotential collaborators (same venues):")
for venue, authors in venue_authors.items():
    if len(authors) >= 2:
        for i in range(len(authors)):
            for j in range(i+1, len(authors)):
                print(f"  {authors[i]} -- {authors[j]} (both publish at {venue})")
```

```
# === 4. Cross-institution collaboration analysis ===
print("\n=== Cross-Institution Collaborations ===")

# Find author affiliations
author_institution = dict(affiliation_data)

# Find collaborations (authors on same paper)
collaborations = defaultdict(set)
paper_authors = defaultdict(list)
for author, paper in authorship_data:
    paper_authors[paper].append(author)

cross_institution_collabs = []
for paper, authors in paper_authors.items():
    if len(authors) >= 2:
        for i in range(len(authors)):
            for j in range(i+1, len(authors)):
                a1, a2 = authors[i], authors[j]
                inst1 = author_institution.get(a1, 'Unknown')
                inst2 = author_institution.get(a2, 'Unknown')
                if inst1 != inst2:
                    cross_institution_collabs.append((a1, a2, paper, inst1, inst2))

print("Cross-institution collaborations:")
for a1, a2, paper, inst1, inst2 in cross_institution_collabs:
    print(f"  {a1} ({inst1}) -- {a2} ({inst2}) on {paper}")

# === 5. Author influence metrics ===
print("\n=== Author Influence ===")

# Count papers per author
author_paper_count = Counter(a for a, p in authorship_data)

# Count unique venues per author
author_venue_count = {a: len(v) for a, v in author_venues.items()}

# Count co-authors
author_coauthors = defaultdict(set)
for paper, authors in paper_authors.items():
    for author in authors:
        for coauthor in authors:
            if author != coauthor:
                author_coauthors[author].add(coauthor)

print("Author metrics:")
print(f"{'Author':<15} {'Papers':<8} {'Venues':<8} {'Coauthors':<10}")
print("-" * 45)

all_authors = set(a for a, p in authorship_data)
for author in sorted(all_authors):
```

```
    papers = author_paper_count.get(author, 0)
    venues = author_venue_count.get(author, 0)
    coauthors = len(author_coauthors.get(author, set()))
    print(f"{author:<15} {papers:<8} {venues:<8} {coauthors:<10}")

# === 6. Institution collaboration network ===
print("\n=== Institution Collaboration Network ===")

# Build institution-level collaboration graph
inst_collabs = Counter()
for a1, a2, paper, inst1, inst2 in cross_institution_collabs:
    pair = tuple(sorted([inst1, inst2]))
    inst_collabs[pair] += 1

print("Institution pairs by collaboration count:")
for pair, count in inst_collabs.most_common():
    print(f"  {pair[0]} -- {pair[1]}: {count} joint papers")
```

**Interpretation:**

1. **Meta-path analysis** reveals implicit relationships—authors who publish at the same venues but haven't collaborated yet are potential future collaborators.
2. **Cross-institution collaborations** highlight knowledge transfer patterns. The Smith-Lee collaboration bridges MIT and Berkeley.
3. **Author influence** considers multiple factors beyond raw paper count:
   - Venue diversity (publishing at multiple top venues)
   - Collaboration breadth (working with many different people)
   - Cross-institution reach

**Adapting to Your Data:**

1. Load from academic databases (DBLP XML, Semantic Scholar API)
2. Add citation edges between papers for impact analysis
3. Include temporal information for trend analysis
4. Weight edges by author position (first author, corresponding author)

### Adapting Case Studies to Your Domain

#### General Adaptation Process

1. **Identify your entities** → These become node types or layers
2. **Identify your relationships** → These become edges
3. **Decide multiplex vs. heterogeneous:**
   - Same entities, different relationship types → Multiplex
   - Different entity types → Heterogeneous
4. **Map identifiers** → Ensure same entity has same ID across layers
5. **Choose appropriate metrics** based on your research questions

6. **Validate with domain knowledge** → Do results match expectations?

### Data Loading Templates

**From CSV with explicit layers:**

```
import pandas as pd
from py3plex.core import multinet

network = multinet.multi_layer_network()

df = pd.read_csv('your_data.csv')
# Expected columns: source, target, layer, weight

for _, row in df.iterrows():
    network.add_edges([
        [row['source'], row['layer'], row['target'], row['layer'], row['weight']]
    ], input_type="list")
```

**From multiple files (one per layer):**

```
layer_files = {
    'layer1': 'layer1_edges.csv',
    'layer2': 'layer2_edges.csv',
    'layer3': 'layer3_edges.csv',
}

network = multinet.multi_layer_network()

for layer_name, filepath in layer_files.items():
    df = pd.read_csv(filepath)
    for _, row in df.iterrows():
        network.add_edges([
            [row['source'], layer_name, row['target'], layer_name, 1.0]
        ], input_type="list")
```

**From database:**

```
import sqlite3
from py3plex.core import multinet

network = multinet.multi_layer_network()

conn = sqlite3.connect('network.db')
cursor = conn.cursor()

cursor.execute('SELECT source, target, layer, weight FROM edges')
for source, target, layer, weight in cursor.fetchall():
    network.add_edges([
        [source, layer, target, layer, weight]
    ], input_type="list")
```

```
conn.close()
```

### Conclusion: Patterns Across Domains

Looking across these case studies, several patterns emerge:

**Multilayer structure reveals what single-layer analysis misses.** In the PPI network, interactions confirmed by multiple evidence types are more reliable—information you lose if you flatten into a single network. In the social network, cross-platform influencers are fundamentally different from single-platform celebrities. In transportation, failure cascades depend on inter-modal connections.

**The coupling parameter matters.** Lower coupling (omega $< 1$) finds layer-specific communities; higher coupling (omega $> 1$) finds cross-layer communities. The right choice depends on your question. If you expect the same underlying structure across layers, use higher coupling. If layers represent truly different phenomena, use lower coupling.

**Validation requires domain expertise.** Network analysis can find structure, but whether that structure is meaningful requires interpretation. Do the detected communities correspond to known functional categories? Do the identified hubs match known influential entities? Ground your analysis in domain knowledge.

**Start simple.** Each case study could be made more complex—weighted edges, temporal dynamics, more layers, heterogeneous node types. But starting simple lets you understand what's happening and catch errors before the analysis becomes opaque.

### The Meta-Lesson

These case studies demonstrate a workflow pattern:

1. **Formulate your question** in terms of network structure
2. **Model your data** as nodes, edges, and layers
3. **Apply appropriate algorithms** based on your question
4. **Interpret results** using domain knowledge
5. **Iterate** as you learn what works

This pattern applies regardless of domain. The specific proteins, users, or stations change; the analytical approach remains similar.

### Further Reading

- *Analysis Recipes & Workflows* — More ready-to-use recipes
- *Multilayer Networks 101* — Conceptual foundations
- community_detection — Detailed community detection guide
- statistics — Complete statistics reference
- *Algorithm Landscape* — Algorithm selection guide

**Academic References:**


- Kivelä et al. (2014). "Multilayer networks." *Journal of Complex Networks*.
- De Domenico et al. (2015). "Ranking in interconnected multilayer networks." *Nature Communications*.

- Battiston et al. (2017). "The physics of higher-order interactions in complex systems." *Nature Physics*.

---

## 3.7 Part VII: Project Info

Project information, contributing guidelines, and citations.

### 3.7.1 Project Info

Overview of the py3plex project, with pointers on how to contribute, cite, and stay up-to-date with development.

**This section covers:**

- *Changelog* — Release history and version notes
- *Roadmap & Vision* — Future development plans
- *Contributing to py3plex* — How to contribute to py3plex
- *Benchmarking & Performance* — Performance characteristics and optimization
- *Citation and References* — How to cite py3plex in publications

**Quick Links:**

- GitHub repository: https://github.com/SkBlaz/py3plex
- Issue tracker: https://github.com/SkBlaz/py3plex/issues
- PyPI package: https://pypi.org/project/py3plex/
- Documentation: https://skblaz.github.io/py3plex/

#### Getting Involved

**Report issues:** Found a bug or have a feature request? Open an issue on GitHub with a clear title and reproduction steps.

**Contribute code:** See *Contributing to py3plex* for branch, testing, and documentation expectations before opening a pull request.

**Join discussions:** Participate in GitHub Discussions to ask questions, share ideas, or propose new features.

**Stay updated:** Watch the repository for releases, follow the *Changelog*, and review the *Roadmap & Vision* for upcoming work.

#### License

py3plex is released under the MIT License. See the `LICENSE` file in the repository for full terms.

#### Acknowledgements

See *Citation and References* for acknowledgements, citation instructions, and BibTeX entries.

### 3.7.2 Changelog

This page summarizes the release history of py3plex. For granular, date-stamped entries, see GitHub Releases[16].

#### Release History

#### Version 1.0.0 (Current)

**Release Date:** 2024 (current stable line)

The 1.0 series focuses on a stable DSL, multilayer ergonomics, and reproducible analysis workflows.

**Major Features:**

- **DSL Query Language:** SQL-like domain-specific language for querying multilayer networks
  - String syntax (`execute_query`) for simple queries
  - Builder API (`Q.nodes()`, `Q.edges()`) for complex, chainable queries
  - Layer algebra (`L[layer]`, `L.intersection()`, `L.union()`)
  - Comprehensive filter, compute, and aggregation operations
- **First-Class Uncertainty Quantification:**
  - `StatSeries` and `StatMatrix` types for statistics with uncertainty
  - Resampling methods (SEED, PERTURBATION) for error propagation
  - Backward-compatible with existing code expecting scalars
  - Integrated across centrality, community detection, and dynamics modules
- **Temporal Network Support:**
  - Time-stamped edges and temporal queries
  - Snapshot-based analysis
  - Temporal dynamics simulation
- **Enhanced Dynamics Framework:**
  - Refactored dynamics API with consistent interface
  - SIR, SIS, and custom dynamics models
  - Layer-specific and time-varying parameters
  - Comprehensive result objects with measure extraction
- **Community Detection Improvements:**
  - Multilayer Louvain with inter-layer coupling
  - Infomap integration (with external binary)
  - Layer-specific community detection workflows
  - Community comparison and stability analysis
- **Web GUI:**
  - Interactive network exploration (FastAPI + SvelteKit)

[16] https://github.com/SkBlaz/py3plex/releases

  - Real-time visualization and query interface
  - REST API for programmatic access
- **CLI Tools:**
  - `py3plex` command-line interface
  - Community detection, statistics, DSL linting
  - Network conversion and export utilities
- **Documentation Overhaul:**
  - Comprehensive how-to guides
  - Conceptual documentation
  - API reference with examples
  - Tutorial notebooks
- **Testing & Quality:**
  - 500+ unit and integration tests
  - Continuous integration (CI/CD)
  - Type hints throughout codebase
  - Code coverage tracking

**API Changes:**

- Standardized function signatures across modules
- Consistent parameter naming (`gamma` for resolution, `omega` for coupling)
- Deprecated legacy APIs (warnings emitted, will be removed in 2.0)

**Performance Improvements:**

- Optimized tensor operations for multilayer algebra
- Sparse matrix support for large networks
- Caching for expensive computations (centrality, community detection)

**Bug Fixes:**

- Numerous edge-case fixes in community detection
- Corrected centrality calculations for directed multilayer networks
- Fixed memory leaks in long-running dynamics simulations
- Resolved NetworkX compatibility issues (version pinning and adapters)

### Previous Versions (0.x Series)

The 0.x series delivered the multilayer foundations: data structures, early DSL prototypes, and visualization primitives. Expect minor API inconsistencies compared to 1.0; migrate when possible.

**Version 0.30 - 0.35 (2020-2023)**

- Incremental feature additions
- Experimental DSL prototype
- Initial dynamics framework

- Community detection algorithms

**Version 0.20 - 0.29 (2018-2020)**

- Multilayer centrality measures
- Visualization improvements
- NetworkX integration

**Version 0.10 - 0.19 (2016-2018)**

- Initial public release
- Core multilayer network data structures
- Basic algorithms (degree, clustering)
- Example notebooks

**Version 0.1 - 0.9 (2014-2016)**

- Research prototype
- Proof-of-concept implementations
- Academic collaborations

### How to Check Your Version

Confirm the installed package version from Python (or use `pip show py3plex` in the shell):

```
import py3plex
print(py3plex.__version__)
```

**Expected output:**

```
1.0.0
```

### Upgrade Instructions

**Upgrade to latest stable release:**

```
pip install --upgrade py3plex
```

**Install a specific version:**

```
pip install py3plex==1.0.0
```

**Install development snapshot:**

```
pip install git+https://github.com/SkBlaz/py3plex.git@main
```

### Migration Guides

### Migrating from 0.x to 1.0

**Breaking Changes:**

1. **Network construction API:**
   - Old: `network.add_nodes(node_list)`

- New: `network.add_nodes([('node', 'layer'), ...])` (explicit tuples)

2. **Community detection returns:**
   - Old: `louvain_communities()` returned dict with node IDs as keys
   - New: Returns dict with `(node, layer)` tuples as keys
3. **Centrality functions:**
   - Old: Scalar returns
   - New: Can return `StatSeries` when `uncertainty=True`
   - Backward compatible: Use `result.mean` or `np.array(result)`

**Deprecated APIs (still work but emit warnings):**

- `network.get_nodes_by_layer(layer)` → Use DSL: `Q.nodes().from_layers(L[layer])`
- `network.get_layer_edges(layer)` → Use DSL: `Q.edges().from_layers(L[layer])`
- `compute_centrality(network, measure)` → Use DSL: `Q.nodes().compute(measure)`

**New Recommended Patterns:**

```
# Old way (still works)
nodes = network.get_nodes_by_layer('layer1')
for node in nodes:
    degree = network.core_network.degree(node)

# New DSL way (recommended)
from py3plex.dsl import Q, L
result = (
    Q.nodes()
     .from_layers(L['layer1'])
     .compute("degree")
     .execute(network)
)
```

**Migration Steps:**

1. Update py3plex: `pip install --upgrade py3plex`
2. Run your existing code; deprecation warnings will indicate legacy usage
3. Gradually migrate to DSL for queries and layer selection
4. Update community detection code to expect tuple `(node, layer)` keys
5. Handle uncertainty-aware results if using `uncertainty=True` (e.g., use `result.mean`)
6. Re-run tests or notebooks to confirm parity before removing legacy patterns

### API Changes and Deprecations

Use this section to triage warnings during upgrades; deprecated items will be removed in 2.0 unless noted otherwise.

**Removed in 1.0:**

- `old_louvain_function()` - replaced by `louvain_communities()`

- `legacy_centrality_wrapper()` - use centrality functions directly

**Deprecated (warnings only, will be removed in 2.0):**

- Direct core_network manipulation - use DSL or network methods
- Legacy node/edge addition formats - use dict or tuple formats

**Renamed:**

- `multinet.multi_layer_network()` - unchanged (primary API)
- `execute_dsl_query()` → `execute_query()` (alias still works)

#### Contributing to Changelog

When submitting PRs, mirror major notes here if they affect users:

- Brief description of changes
- Link to issue (if applicable)
- Breaking changes clearly marked
- Migration notes for API changes

See *Contributing to py3plex* for details.

#### Staying Updated

- **GitHub Releases:** https://github.com/SkBlaz/py3plex/releases
- **PyPI:** https://pypi.org/project/py3plex/
- **Discussions:** https://github.com/SkBlaz/py3plex/discussions

#### Next Steps

- **Upgrade guide:** See "Migration Guides" above
- **What's new:** *Roadmap & Vision* for planned features
- **Report issues:** https://github.com/SkBlaz/py3plex/issues

### 3.7.3 Roadmap & Vision

This page outlines the vision and planned development for py3plex. Roadmaps are snapshots and may shift with user feedback. For the latest updates, see GitHub Releases[17] and GitHub Issues[18].

#### Long-Term Vision

py3plex aims to be the go-to Python library for multilayer network analysis with a focus on:

**Research Goals:**

- Implementing state-of-the-art multilayer network algorithms from current research
- Providing a flexible framework for developing and comparing new multilayer methods
- Supporting reproducible research with clear documentation and runnable examples

**Production Use Cases:**

[17] https://github.com/SkBlaz/py3plex/releases
[18] https://github.com/SkBlaz/py3plex/issues

- Enabling analysis of real-world multilayer networks at scale (millions of nodes)
- Providing robust APIs for integration into production systems
- Supporting diverse data formats and interoperability with existing tools

**Community Building:**

- Growing an active community of researchers and practitioners
- Facilitating collaboration through open-source development
- Providing educational resources and workshops

### Current Focus (v1.0+)

**Established Features:**

- Core multilayer network data structures and operations
- DSL (Domain-Specific Language) for declarative network queries
- Community detection (Louvain, Infomap, multilayer variants)
- Dynamics simulation (SIR, SIS, random walks, custom models)
- Network embeddings (Node2Vec, DeepWalk)
- Centrality measures (degree, betweenness, PageRank, multilayer variants)
- Uncertainty quantification utilities
- Visualization tools (matplotlib, plotly, interactive)
- CLI tools for common operations and a web GUI for exploration

### Planned Enhancements

Planned work is grouped by expected timeframe. Items may move as priorities change.

### Near Term (Next Release)

- Performance optimizations for large-scale networks (>1M nodes)
- Additional temporal network analysis tools
- Extended DSL capabilities (aggregation, window functions)
- More community detection algorithms
- Enhanced visualization options (layer-specific layouts)

### Medium Term (1-2 Releases)

- GPU acceleration for select algorithms
- Streaming network analysis capabilities
- Advanced temporal dynamics models
- Integration with graph database systems
- Additional export formats (Neo4j, GraphML extensions)
- More comprehensive benchmarking suite

### Longer Term (Future Releases)

- Distributed computing support (Dask, Spark integration)
- Advanced machine learning on multilayer networks
- Causal inference tools for multilayer systems
- Time-varying parameter estimation
- Additional domain-specific toolkits (bioinformatics, social, transport)

**Note:** Feature priorities may shift based on community feedback and research developments. Track specific feature requests and vote on priorities through GitHub Issues[19].

### Research Directions

py3plex development is informed by ongoing research in multilayer network science:

**Current Research Areas:**

- **Multilayer centrality:** Novel measures that account for layer dependencies
- **Temporal dynamics:** Time-varying multilayer processes and intervention modeling
- **Uncertainty quantification:** Principled handling of measurement error and structural uncertainty
- **Scalability:** Algorithms that scale to networks with millions of nodes and layers
- **Domain applications:** Collaborations in biology, social science, transportation, finance

**Contributing Research:**

If you're developing new multilayer network methods, we welcome contributions! See *Contributing to py3plex* for guidelines on:

- Implementing new algorithms
- Adding test cases and benchmarks
- Contributing examples and documentation
- Citing your research in the codebase

### Community Requests

The following features have been requested by the community. For the full list and to vote on priorities, see GitHub Issues[20].

**High-Priority Requests:**

- Support for very large networks (>10M nodes) - **In progress**
- Additional temporal network analysis tools - **Planned**
- More extensive documentation and tutorials - **Ongoing**
- Enhanced performance for centrality computations - **Planned**
- Integration with popular graph databases - **Under consideration**

[19] https://github.com/SkBlaz/py3plex/issues
[20] https://github.com/SkBlaz/py3plex/issues

### Documentation Gaps Blocking New-User Adoption

After reviewing `README.md`, `AGENTS.md`, and the current docs tree, these are the highest-impact documentation gaps for first-time users:

1. **Conflicting example inventory and scale claims**
   - README says **170+** examples *(as observed on 2026-03-28 in ``README.md``)*.
   - `AGENTS.md` says **267** examples *(as observed on 2026-03-28 in ``AGENTS.md``)*.
   - `docfiles/getting_started/tutorial_10min.rst` says **26 focused examples** *(as observed on 2026-03-28 in that file's "Next Steps" section)*.

   New users cannot quickly trust which navigation map is current.

2. **Broken or stale example path references in onboarding docs**

   Several onboarding pages reference example folders that are not present in the current top-level `examples/` layout. For example:
   - `docfiles/getting_started/tutorial_10min.rst` references `examples/00_quickstart/...` and category folders such as `examples/communities/` and `examples/temporal/`.
   - `docfiles/examples/index.rst` describes a folder taxonomy that includes categories not present in the current top-level examples tree.

   This blocks copy-paste adoption because the first command often fails with "file not found".

3. **No single canonical "start here" path across README and docs**

   README, docs homepage, and tutorial pages each present different starting flows. The project has strong content depth, but first-time users lack one authoritative linear path from install -> first successful query -> next steps.

4. **README flagship snippet is advanced before a minimal validated success case**

   The first large code sample is feature-rich (community detection + UQ + grouping + explainability), which is valuable for power users but can overwhelm new adopters who need a shortest possible "first success" script first.

5. **Audience boundary between AGENTS.md and human docs is unclear**

   `AGENTS.md` is comprehensive and agent-oriented, but there is no prominent warning in beginner-facing docs clarifying that it is an expert/automation playbook rather than the recommended first learning resource for humans.

6. **No documented single source of truth for doc metadata drift-prone fields**

   Counts and path catalogs (examples totals, folder maps, quickstart file paths) are repeated in multiple places without a clearly designated canonical source, causing recurring drift.

### Vote on Features

Help prioritize development by:

- Commenting on existing feature request issues with your use case
- Opening new issues for feature requests (use the "enhancement" label)
- Using reactions to vote on priorities
- Contributing implementations for features you need

### Contributing to Development

Want to help implement these features? We welcome contributions of all kinds:

- **Code contributions:** Implement new algorithms, fix bugs, improve performance
- **Documentation:** Write tutorials, improve API docs, add examples
- **Testing:** Add test cases, report bugs, validate algorithms
- **Community:** Answer questions, review PRs, organize workshops

See *Contributing to py3plex* for detailed guidelines.

### Release Cycle

py3plex follows semantic versioning (MAJOR.MINOR.PATCH):

**Version Scheme:**

- **Major releases** (x.0.0): Breaking API changes, major new features
- **Minor releases** (1.x.0): New features, backward-compatible changes
- **Patch releases** (1.0.x): Bug fixes, documentation updates

**Development Process:**

- **Main branch:** Stable, production-ready code
- **Development branches:** Feature development and testing
- **Release candidates:** Beta testing before major releases
- **Documentation:** Updated with each release

**Release Timeline:**

- Major releases: ~1-2 per year
- Minor releases: As features are completed (typically quarterly)
- Patch releases: As needed for bug fixes

See *Changelog* for release history.

### Staying Updated

- **GitHub Releases:** https://github.com/SkBlaz/py3plex/releases
- **Issue Tracker:** https://github.com/SkBlaz/py3plex/issues
- **Discussions:** https://github.com/SkBlaz/py3plex/discussions

### Next Steps

- **Current features:** *Quick Start Tutorial*
- **Contribute:** *Contributing to py3plex*
- **Report issues:** https://github.com/SkBlaz/py3plex/issues

### 3.7.4 Contributing to py3plex

We welcome contributions to py3plex. This guide explains how to contribute effectively and what we expect before you open a pull request.

#### Quick Checklist

- Use a feature branch (`git checkout -b feature/my-topic`).
- Keep changes focused and small; split unrelated changes into separate PRs.
- Add or update tests that cover the change.
- Run `make lint` and `make test` before opening a PR.
- Update documentation (docstrings and RST) when behavior or APIs change.

#### Ways to Contribute

You can contribute in many ways:

- Report bugs - Open issues for bugs you encounter
- Suggest features - Propose new features or improvements
- Write documentation - Improve or expand documentation
- Fix bugs - Submit pull requests fixing issues
- Add features - Implement new algorithms or capabilities
- Write tests - Improve test coverage
- Review PRs - Help review pull requests from others

#### Getting Started

#### Fork and Clone

1. Fork the repository on GitHub
2. Clone your fork:

```
git clone https://github.com/YOUR_USERNAME/py3plex.git
cd py3plex
```

3. Add upstream remote:

```
git remote add upstream https://github.com/SkBlaz/py3plex.git
```

#### Development Setup

**Recommended (Makefile-based):**

Use the Makefile for a streamlined setup:

```
# Create virtual environment and install dependencies
make setup

# Install package in editable mode with dev dependencies
make dev-install
```

**Alternative (manual venv + pip):**

If you prefer manual setup:

```
# Create virtual environment
python3 -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate

# Install in editable mode with dev dependencies
pip install -e ".[dev]"
```

Both methods install:

- Core dependencies
- Testing tools (pytest, coverage)
- Linting tools (black, ruff, isort, mypy)
- Documentation tools (sphinx)

### Development Workflow

#### Create a Branch

Always create a new branch for your work:

```
git checkout -b feature/my-new-feature
# or
git checkout -b fix/issue-123
```

#### Make Changes

1. Write your code following our coding standards (see below).
2. Add or update tests that exercise the change, including edge cases.
3. Update documentation (docstrings, RST, examples) if behavior changes.
4. Run linters and tests locally to catch issues early.

#### Run Tests

```
# Run all tests
make test

# Or directly with pytest
python run_tests.py
```

#### Run Linters

```
# Run all linters
make lint

# Or individual tools
make format  # Auto-format code (black, isort)
```

```
ruff check .
mypy py3plex
```

### Commit Changes

Write clear, descriptive commit messages:

```
git add .
git commit -m "Add feature X to support Y"
git commit --amend  # optional, to refine the message before pushing
# A good body explains:
# - Why the change is needed
# - What was changed
# - Any follow-up steps
```

### Push and Create PR

```
git push origin feature/my-new-feature
```

Then create a Pull Request on GitHub with:

- Clear title describing the change
- Description of what changed and why
- Reference to related issues (e.g., "Fixes #123")
- Screenshots for UI changes

### Coding Standards

### Code Style

We follow PEP 8 with these specifics:

- Line length: 100 characters (not 80)
- Indentation: 4 spaces (no tabs)
- String quotes: Single quotes preferred ('text' not "text")
- Imports: Grouped and sorted (using isort)

Auto-format your code:

```
make format
```

This runs:

- `isort` - Sort and organize imports
- `black` - Format code consistently
- `ruff --fix` - Auto-fix linting issues

### Naming Conventions

- Functions/variables: `snake_case`
- Classes: `PascalCase`
- Constants: `UPPER_SNAKE_CASE`
- Private members: `_leading_underscore`

```
# Good
def compute_centrality(network, node_id):
    MAX_ITERATIONS = 100
    centrality_scores = {}
    return centrality_scores

class MultilayerNetwork:
    def __init__(self):
        self._internal_state = {}
```

### Docstrings

Use NumPy-style docstrings for all public functions and classes:

```
def multilayer_degree_centrality(network, layer=None, weighted=False):
    """
    Compute degree centrality for multilayer network.

    Parameters
    ----------
    network : multi_layer_network
        The multilayer network to analyze.
    layer : str, optional
        Specific layer to analyze. If None, aggregates across all layers.
    weighted : bool, default=False
        If True, compute strength (weighted degree) instead of degree.

    Returns
    -------
    dict
        Dictionary mapping node names to centrality scores.

    Examples
    --------
    >>> network = multinet.multi_layer_network()
    >>> network.add_edges([['A', 'L1', 'B', 'L1', 1]])
    >>> centrality = multilayer_degree_centrality(network)
    >>> centrality['A---L1']
    1.0

    See Also
    --------
    multilayer_betweenness_centrality : Betweenness centrality
```

```
    Notes
    -----
    For directed networks, this computes out-degree by default.

    References
    ----------
    .. [1] De Domenico, M., et al. (2013). Mathematical formulation of
           multilayer networks. Physical Review X, 3(4), 041022.
    """
    pass
```

### Type Hints

Add type hints to improve code clarity:

```
from typing import Dict, List, Optional
from py3plex.core.multinet import multi_layer_network

def compute_statistics(
    network: multi_layer_network,
    layers: Optional[List[str]] = None
) -> Dict[str, float]:
    """Compute network statistics."""
    pass
```

### Testing Guidelines

#### Test Requirements

All new code should include tests:

- Unit tests for individual functions
- Integration tests for workflows
- Edge cases for boundary conditions
- Documentation tests for examples in docstrings

### Writing Tests

Use pytest for testing:

```
# tests/test_my_feature.py
import pytest
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

def test_layer_density():
    """Test layer density calculation."""
    # Setup
    network = multinet.multi_layer_network()
```

```
    network.add_edges([
        ['A', 'L1', 'B', 'L1', 1],
        ['B', 'L1', 'C', 'L1', 1],
    ], input_type='list')

    # Execute
    density = mls.layer_density(network, 'L1')

    # Assert
    assert 0 <= density <= 1
    assert density == pytest.approx(0.333, abs=0.01)

def test_invalid_layer():
    """Test error handling for invalid layer."""
    network = multinet.multi_layer_network()

    with pytest.raises(ValueError):
        mls.layer_density(network, 'nonexistent')
```

### Test Coverage

Aim for high test coverage:

```
# Run tests with coverage
pytest --cov=py3plex --cov-report=html

# View coverage report
open htmlcov/index.html  # macOS
# or
xdg-open htmlcov/index.html  # Linux
```

Target: >80% coverage for new code

### Documentation

### Update Documentation

When adding features, update:

1. **Docstrings** in the code
2. **RST files** in `docfiles/` if adding major features
3. **Examples** in `examples/` directory
4. **README** if changing installation or basic usage

### Build Documentation

```
cd docfiles
make html

# View documentation
```

```
open _build/html/index.html
```

### Documentation Style

- Clear and concise - Use simple language
- Code examples - Include working examples
- Cross-references - Link to related documentation
- Visual aids - Add diagrams where helpful

### Pull Request Guidelines

### Before Submitting

[OK] Code follows style guide (`make lint` passes).

[OK] All tests pass (`make test` passes).

[OK] New code has tests (unit and integration when applicable).

[OK] Documentation is updated (docstrings and relevant RST pages).

[OK] Commit messages are clear and scoped to the change.

[OK] Branch is up to date with `main` (rebase if needed to avoid conflicts).

### PR Description

Include in your PR description:

- What changed
- Why the change is needed
- How you implemented it
- Testing done
- Screenshots for visual changes
- Breaking changes if any

Example PR template:

```
## Description

Implements multilayer PageRank centrality following [citation].

## Motivation

Closes #123 - needed for analyzing directed multilayer networks.

## Changes

- Add `multilayer_pagerank()` function
- Add tests with 90% coverage
- Add example in `example_centrality.py`
- Update documentation in `centrality.rst`
```

```
## Testing

- [x] Unit tests pass
- [x] Integration test on real network
- [x] Tested on Python 3.8, 3.10, 3.12
- [x] Linting passes

## Breaking Changes

None.
```

### Review Process

1. Maintainers will review your PR.
2. Address feedback promptly; keep commits focused on the requested changes.
3. Once approved, your PR will be merged.
4. Your contribution will be included in the next release.

### Reporting Issues

### Bug Reports

When reporting bugs, include:

- Description of the bug
- Steps to reproduce
- Expected behavior
- Actual behavior
- Python version (`python --version`)
- py3plex version
- Operating system
- Minimal code example

Example:

````
## Bug Description

`layer_density()` returns negative values for certain graphs.

## Steps to Reproduce

```python
network = multinet.multi_layer_network()
network.add_edges([['A', 'L1', 'B', 'L1', -1]])
density = mls.layer_density(network, 'L1')
print(density)  # -0.5
```
````

```
## Expected

Density should be between 0 and 1, or raise ValueError for negative weights.

## Actual

Returns -0.5

## Environment

- Python 3.10.5
- py3plex 0.95a
- Ubuntu 22.04
```

### Feature Requests

For feature requests, describe:

- Use case - What problem does it solve?
- Proposed solution - How should it work?
- Alternatives - Other approaches considered
- Additional context - Examples, papers, etc.

### Code of Conduct

- Be respectful and inclusive
- Welcome newcomers
- Focus on what is best for the community
- Show empathy towards other community members

Contributors are expected to follow these principles to foster a welcoming and productive community.

### License

By contributing, you agree that your contributions will be licensed under the MIT License.

### Recognition

Contributors are recognized in:

- GitHub contributors page
- Release notes
- ../acknowledgements section in the documentation

### Getting Help

- Questions: Open a GitHub Discussion
- Chat: Join our community chat (link in README)
- Email: Contact maintainers at blaz.skrlj@ijs.si

### Next Steps

- Read ../development for development workflow details
- See ../architecture for system architecture
- Browse existing issues[21]
- Check good first issues[22]

Thank you for contributing to py3plex!

## 3.7.5 Benchmarking & Performance

Performance characteristics, runtime expectations, and practical optimization strategies for py3plex.

### Network Scale Guidelines

py3plex targets research-scale multilayer networks. Use the ranges below as directional guidance rather than hard limits; structure (density, layer coupling) has a bigger impact than raw node counts.

Table 3: Network Scale Performance

| Network Size | Performance | Visualization | Recommendations |
|---|---|---|---|
| Small (<100 nodes) | Excellent | Fast, detailed | Use dense visualization mode |
| Medium (100-1k nodes) | Good | Fast, balanced | Default settings work well |
| Large (1k-10k nodes) | Good | Slower, minimal | Use sparse matrices, sampling |
| Very Large (>10k nodes) | Variable | Very slow | Sampling required |

### Performance Tips

### Use Sparse Matrices

For large networks, use sparse matrix representations:

```
from py3plex.core import multinet

network = multinet.multi_layer_network(sparse=True)
```

This typically reduces memory usage by 10-100x for moderately sparse graphs.

[21] https://github.com/SkBlaz/py3plex/issues
[22] https://github.com/SkBlaz/py3plex/issues?q=is%3Aissue+is%3Aopen+label%3A%22good+first+issue%22

### Batch Operations

Process multiple operations together instead of issuing separate passes:

```
from py3plex.dsl import Q

# Compute multiple metrics at once
result = (
    Q.nodes()
     .compute("degree", "betweenness_centrality", "clustering")
     .execute(network)
)
```

Avoid repeated single-metric computations when they can be combined.

### Use Arrow/Parquet for I/O

For large datasets, prefer columnar formats over CSV:

```
import pyarrow.parquet as pq

# Save
table = pq.write_table(edges_table, 'network.parquet')

# Load (much faster than CSV)
table = pq.read_table('network.parquet')
```

### Parallel Processing

For Node2Vec and other CPU-intensive algorithms:

```
from py3plex.wrappers import train_node2vec

embeddings = train_node2vec(
    network,
    workers=8  # Use multiple CPU cores
)
```

### Benchmark Results

Performance benchmarks for common operations on synthetic multilayer networks. Use these results for rough planning, not as guarantees.

**Test Environment**

- CPU: Intel Core i7-9700K @ 3.6GHz (8 cores)
- RAM: 32 GB DDR4
- Python: 3.10
- py3plex: v1.0.0

**Methodology & Caveats**

- Synthetic multilayer graphs with comparable density across sizes

- Single-run wall-clock timings on the hardware above
- Runtimes vary materially with density, weight usage, and layer count; rerun locally for precise estimates

### Algorithm Runtimes vs. Network Size

#### Community Detection (Louvain)

Table 4: Louvain Algorithm Runtime

| Nodes | Edges | Layers | Runtime |
|---|---|---|---|
| 100 | 500 | 3 | 0.05s |
| 1,000 | 5,000 | 3 | 0.3s |
| 10,000 | 50,000 | 3 | 4.2s |
| 100,000 | 500,000 | 3 | 58s |

#### Centrality Computation (Betweenness)

Table 5: Betweenness Centrality Runtime

| Nodes | Edges | Layers | Runtime |
|---|---|---|---|
| 100 | 500 | 3 | 0.12s |
| 1,000 | 5,000 | 3 | 8.5s |
| 10,000 | 50,000 | 3 | 1,240s (21 min) |
| 100,000 | 500,000 | 3 | N/A (too slow) |

*Note: Betweenness is $O(n^3)$; use approximation methods for large networks.*

#### Node2Vec Embeddings

Table 6: Node2Vec Runtime (128-dim, 10 walks/node)

| Nodes | Edges | Layers | Runtime |
|---|---|---|---|
| 100 | 500 | 3 | 2.3s |
| 1,000 | 5,000 | 3 | 18s |
| 10,000 | 50,000 | 3 | 245s (4 min) |
| 100,000 | 500,000 | 3 | 3,200s (53 min) |

#### Dynamics Simulation (SIR)

Table 7: SIR Simulation Runtime (100 steps)

| Nodes | Edges | Layers | Runtime |
|---|---|---|---|
| 100 | 500 | 3 | 0.8s |
| 1,000 | 5,000 | 3 | 4.5s |
| 10,000 | 50,000 | 3 | 52s |
| 100,000 | 500,000 | 3 | 680s (11 min) |

### Memory Usage Profiles

Table 8: Peak Memory Usage

| Nodes | Edges | Dense Storage | Sparse Storage |
|---|---|---|---|
| 100 | 500 | 2 MB | 0.5 MB |
| 1,000 | 5,000 | 24 MB | 2 MB |
| 10,000 | 50,000 | 2.4 GB | 18 MB |
| 100,000 | 500,000 | 240 GB | 180 MB |

**Key Insight:** Sparse storage reduces memory by 10-1000x for typical moderately sparse networks (adjacency-like storage, unweighted).

### Comparison with Other Tools

#### Community Detection: py3plex vs. NetworkX

Table 9: Louvain Performance Comparison (1k nodes, 5k edges, single layer)

| Tool | Runtime | Notes |
|---|---|---|
| py3plex | 0.3s | Multilayer-aware |
| NetworkX + python-louvain | 0.2s | Single-layer only |
| graph-tool | 0.08s | C++ backend, faster |

**Verdict:** py3plex is competitive on single-layer inputs while adding multilayer capability others lack.

#### Node Embeddings: py3plex vs. node2vec

Table 10: Node2Vec Performance (1k nodes, 128-dim, 10 walks)

| Tool | Runtime | Notes |
|---|---|---|
| py3plex | 18s | Python wrapper |
| node2vec (original) | 15s | C++ implementation |
| Gensim | 12s | Optimized Word2Vec |

**Verdict:** py3plex wraps established libraries (e.g., gensim); performance is comparable, with minor Python overhead expected.

**Benchmarking Notes**

- Results vary with network structure (density, clustering, layer coupling).
- Algorithmic scaling differs (linear vs. quadratic vs. cubic) and dominates at large sizes.
- Treat these tables as directional; rerun locally for reliable planning.
- For the most accurate estimates, run the bundled scripts on your hardware and data.

### Running Benchmarks

Reproduce or extend the tables above with the bundled scripts:

```
cd benchmarks
python run_benchmarks.py
```

### Profiling Your Code

Use Python profiling tools:

```
import cProfile
import pstats

# Profile your analysis
cProfile.run('your_analysis_function(network)', 'profile_stats')

# View results
stats = pstats.Stats('profile_stats')
stats.sort_stats('cumulative')
stats.print_stats(20)
```

### Memory Profiling

```
pip install memory_profiler
python -m memory_profiler your_script.py
```

### Optimization Strategies

#### For Large Networks

1. **Sample the network** for exploratory analysis
2. **Use layer-specific analysis** instead of full multilayer
3. **Compute metrics incrementally** rather than all at once
4. **Cache intermediate results**

#### For Repeated Analysis

1. **Precompute and save** expensive metrics
2. **Use config-driven workflows** for reproducibility
3. **Batch process** multiple networks

#### For Production

1. **Use Docker containers** for consistent environments
2. **Implement monitoring** for long-running jobs
3. **Add checkpointing** for crash recovery

See *Docker Usage Guide* for deployment best practices.

#### Hardware Recommendations

**Minimum:**

- 4 GB RAM
- 2 CPU cores
- Small networks (<1k nodes)

**Recommended:**

- 16 GB RAM
- 8 CPU cores
- Networks up to 10k nodes

**High-Performance:**

- 64+ GB RAM
- 16+ CPU cores
- Large networks (>10k nodes)

#### Next Steps

- **Optimize I/O:** *How to Export and Serialize Networks*
- **Deploy to production:** *Docker Usage Guide*
- **See deployment guide:** *Performance and Scalability Best Practices* (original full guide)

### 3.7.6 Citation and References

If you use py3plex in your research, please **cite our work**. The quickest recipe:

- Always cite the primary journal article for general usage (analysis, visualization, CLI, DSL).
- Add algorithm-specific citations when you rely on a particular method (community detection, embedding, ranking).
- Where possible, include the DOI to keep references unambiguous.

#### Primary Citation

**Recommended citation for py3plex (general use of the library and DSL):**

```
@Article{Skrlj2019,
  author    = {{\v{S}}krlj, Bla{\v{z}} and Kralj, Jan and Lavra{\v{c}}, Nada},
  title     = {Py3plex toolkit for visualization and analysis of multilayer networks},
  journal   = {Applied Network Science},
  year      = {2019},
  volume    = {4},
  number    = {1},
  pages     = {94},
  doi       = {10.1007/s41109-019-0203-7},
  url       = {https://doi.org/10.1007/s41109-019-0203-7}
}
```

**Plain text:**

Škrlj, B., Kralj, J., & Lavrač, N. (2019). Py3plex toolkit for visualization and analysis of multilayer networks. *Applied Network Science*, 4(1), 94. https://doi.org/10.1007/s41109-019-0203-7

### Conference Paper

For the **initial algorithmic work and early benchmarks**, add this citation alongside the primary one:

```
@InProceedings{Skrlj2019Conference,
  author    = {{\v{S}}krlj, Bla{\v{z}} and Kralj, Jan and Lavra{\v{c}}, Nada},
  editor    = {Aiello, Luca Maria and Cherifi, Chantal and Cherifi, Hocine
               and Lambiotte, Renaud and Li{\'o}, Pietro and Rocha, Luis M.},
  title     = {Py3plex: A Library for Scalable Multilayer Network Analysis and Visualizatio
  booktitle = {Complex Networks and Their Applications VII},
  year      = {2019},
  publisher = {Springer International Publishing},
  address   = {Cham},
  pages     = {757--768},
  isbn      = {978-3-030-05411-3},
  doi       = {10.1007/978-3-030-05411-3_60}
}
```

### Algorithm-Specific Citations

When using **specific algorithms**, please also cite the **original papers** in addition to the primary py3plex article. Cite only the methods you actively use.

### Multilayer Modularity

Use when reporting multilayer modularity community detection results.

```
@Article{Mucha2010,
  author  = {Mucha, Peter J. and Richardson, Thomas and Macon, Kevin
             and Porter, Mason A. and Onnela, Jukka-Pekka},
  title   = {Community structure in time-dependent, multiscale, and multiplex networks},
  journal = {Science},
  year    = {2010},
  volume  = {328},
  number  = {5980},
  pages   = {876--878},
  doi     = {10.1126/science.1184819}
}
```

### Node2Vec

Use when generating node embeddings with Node2Vec.

```
@InProceedings{Grover2016,
  author    = {Grover, Aditya and Leskovec, Jure},
  title     = {node2vec: Scalable Feature Learning for Networks},
  booktitle = {Proceedings of the 22nd ACM SIGKDD International Conference
```

```
                on Knowledge Discovery and Data Mining},
  year      = {2016},
  pages     = {855--864},
  doi       = {10.1145/2939672.2939754}
}
```

### Louvain Algorithm

Use when computing Louvain community detection results.

```
@Article{Blondel2008,
  author  = {Blondel, Vincent D. and Guillaume, Jean-Loup and Lambiotte, Renaud
             and Lefebvre, Etienne},
  title   = {Fast unfolding of communities in large networks},
  journal = {Journal of Statistical Mechanics: Theory and Experiment},
  year    = {2008},
  volume  = {2008},
  number  = {10},
  pages   = {P10008},
  doi     = {10.1088/1742-5468/2008/10/P10008}
}
```

### Infomap

Use when computing Infomap community detection results.

```
@Article{Rosvall2008,
  author  = {Rosvall, Martin and Bergstrom, Carl T.},
  title   = {Maps of random walks on complex networks reveal community structure},
  journal = {Proceedings of the National Academy of Sciences},
  year    = {2008},
  volume  = {105},
  number  = {4},
  pages   = {1118--1123},
  doi     = {10.1073/pnas.0706851105}
}
```

### MultiXRank

Use when running MultiXRank for multilayer ranking.

```
@Article{Baptista2022,
  author  = {Baptista, Anthony and Gonzalez, Aur{\'e}lien and Baudot, Ana{\"\i}s},
  title   = {Universal multilayer network exploration by random walk with restart},
  journal = {Communications Physics},
  year    = {2022},
  volume  = {5},
  number  = {1},
  pages   = {170},
```

```
  doi     = {10.1038/s42005-022-00937-9}
}
```

### DeepWalk

Use when generating DeepWalk embeddings.

```
@InProceedings{Perozzi2014,
  author    = {Perozzi, Bryan and Al-Rfou, Rami and Skiena, Steven},
  title     = {DeepWalk: Online Learning of Social Representations},
  booktitle = {Proceedings of the 20th ACM SIGKDD International Conference
               on Knowledge Discovery and Data Mining},
  year      = {2014},
  pages     = {701--710},
  doi       = {10.1145/2623330.2623732}
}
```

### Multilayer Network Theory

**Foundational papers** on multilayer networks. Cite when you rely on formal multilayer definitions or theoretical background.

```
@Article{Kivela2014,
  author  = {Kivel{\"a}, Mikko and Arenas, Alex and Barthelemy, Marc
             and Gleeson, James P. and Moreno, Yamir and Porter, Mason A.},
  title   = {Multilayer networks},
  journal = {Journal of Complex Networks},
  year    = {2014},
  volume  = {2},
  number  = {3},
  pages   = {203--271},
  doi     = {10.1093/comnet/cnu016}
}
```

```
@Article{Boccaletti2014,
  author  = {Boccaletti, Stefano and Bianconi, Ginestra and Criado, Regino
             and del Genio, Charo I. and G{\'o}mez-Garde{\~n}es, Jes{\'u}s
             and Romance, Miguel and Sendi{\~n}a-Nadal, Irene and Wang, Zhen
             and Zanin, Massimiliano},
  title   = {The structure and dynamics of multilayer networks},
  journal = {Physics Reports},
  year    = {2014},
  volume  = {544},
  number  = {1},
  pages   = {1--122},
  doi     = {10.1016/j.physrep.2014.07.001}
}
```

```
@Article{DeDomenico2013,
  author  = {De Domenico, Manlio and Sol{\'e}-Ribalta, Albert and Cozzo, Emanuele
```

```
            and Kivel{\"a}, Mikko and Moreno, Yamir and Porter, Mason A.
            and G{\'o}mez, Sergio and Arenas, Alex},
  title   = {Mathematical formulation of multilayer networks},
  journal = {Physical Review X},
  year    = {2013},
  volume  = {3},
  number  = {4},
  pages   = {041022},
  doi     = {10.1103/PhysRevX.3.041022}
}
```

### Complete Citation List

For algorithm citations, see the ../algorithm_guide, which includes all major algorithms with their original references and DOIs.

### Acknowledgments

### Development and Contributors

py3plex is developed and maintained by:

- **Blaž Škrlj** (lead developer) - Jožef Stefan Institute, Ljubljana, Slovenia
- **Jan Kralj** (contributor) - Jožef Stefan Institute
- **Nada Lavrač** (contributor) - Jožef Stefan Institute

### Funding and Support

This work has been **supported by**:

- **Jožef Stefan Institute**, Ljubljana, Slovenia
- **Slovenian Research Agency** (research program P2-0103)

### External Libraries

py3plex builds upon **excellent open-source libraries** (please cite them when their capabilities are central to your results):

- **NetworkX** - Graph data structures and algorithms
- **NumPy** - Numerical computing
- **SciPy** - Scientific computing
- **Matplotlib** - Visualization
- **scikit-learn** - Machine learning utilities

### Community Acknowledgments

We thank all contributors, users, and the broader network science community for their support, feedback, and contributions.

Special thanks to contributors who have submitted pull requests, reported issues, or provided feedback.

### License Information

py3plex is released under the **MIT License**.

```
MIT License

Copyright (c) 2019-2025 Blaž Škrlj, Jan Kralj, Nada Lavrač

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SOFTWARE.
```

**Note:** Some bundled algorithms may have **different licenses**:

- Infomap community detection code: **AGPLv3**
- Louvain community detection: **BSD-3-Clause**

See `LICENSE` in the repository for license compatibility details, especially if redistributing binaries that package third-party implementations.

### Contact

For questions about citing py3plex or collaboration opportunities:

- **Email:** blaz.skrlj@ijs.si
- **GitHub:** https://github.com/SkBlaz/py3plex
- **Issues:** https://github.com/SkBlaz/py3plex/issues

### Related Work

Other tools and libraries for multilayer network analysis (for context and comparison):

- **muxViz** - Multilayer network visualization (R)
- **pymnet** - Multilayer networks in Python
- **multinet** - R package for multilayer networks
- **graphistry** - GPU-accelerated graph visualization

For a comprehensive comparison and positioning of py3plex, see the original publication.

### Usage in Publications

If you've used py3plex in your research and would like to be listed here, please:

1. Open an issue or pull request
2. Provide the citation to your paper (with DOI if available)
3. Add a brief description of how py3plex was used (e.g., DSL queries, community detection, embeddings)

We maintain a list of publications using py3plex to showcase applications and build community. New additions are welcome.

### See Also

- ../acknowledgements - Full acknowledgments
- ../algorithm_guide - Complete algorithm citations with references
- py3plex GitHub[23] - Source code and issues
- Applied Network Science paper[24] - Primary publication

---

## 3.8 Additional Sections

### 3.8.1 Environments & Deployment

Use this section to run py3plex from the command line, inside Docker containers, and at scale—while keeping runs reproducible.

**This section covers:**

- *Docker Usage Guide* — Command-line interface and containerized deployment
- *Performance and Scalability Best Practices* — Memory management, optimization, large networks

### When to Use This Section

Use these chapters when you want to:

- Automate analyses that you currently run interactively
- Process networks too large for a single Python session
- Share reproducible environments with collaborators or CI
- Integrate py3plex into a production data pipeline

**CLI** gives you a scriptable interface for common operations—no Python coding required and suitable for headless automation. Start here if you already know the operations you want to run.

**Docker** keeps environments identical across machines, avoiding "works on my machine" issues. Use it when you need the same environment on laptops, CI, and servers.

The **performance** chapter covers memory management and optimization for large networks once you have a working workflow. It is most useful after you have a repeatable pipeline and want to speed it up or shrink memory use.

[23] https://github.com/SkBlaz/py3plex
[24] https://doi.org/10.1007/s41109-019-0203-7

Most readers start with *Docker Usage Guide* to script or containerize workflows, then move to *Performance and Scalability Best Practices* to tune runtime and memory as datasets grow.

**Tip**

**Deployment checklist:**

- Pin dependency versions (requirements.txt or lock file) and set a random seed for reproducibility
- Handle missing files, empty inputs, and unexpected formats explicitly
- Validate results on a small test network before scaling up
- Enable logging for reproducibility and debugging; keep logs with your outputs
- Run a dry run on the target environment (local, Docker, or cluster) before scheduling large jobs

### 3.8.2 Docker Usage Guide

This guide provides concise, step-by-step instructions for running Py3plex with Docker. It assumes Docker is installed and that you are working from the repository root unless noted otherwise.

**Table of Contents**


- *Prerequisites*
- *Quickstart*
- *Building the Image*
  - *Using Docker*
  - *Using Docker Compose*
- *Running Commands*
  - *Basic Commands*
  - *Using Docker Compose*
- *Working with Files*
  - *Creating a Data Directory*
  - *Mounting Volumes*
  - *Complete Workflow Example*
- *Advanced Usage*
  - *Interactive Shell*
  - *Using Python API*
  - *Custom Image Builds*
  - *Docker Compose with Multiple Services*
- *Helper Scripts*

  - *py3plex-docker.sh*
  - *test-docker-setup.sh*
- *Docker Configuration Files*
  - *.dockerignore*
- *Troubleshooting*
  - *Testing the Docker Setup*
  - *Permission Issues*
  - *Container Not Found*
  - *Network Issues During Build*
  - *Out of Disk Space*
  - *Debugging Build Issues*
- *Best Practices*
- *Practical Examples*
  - *Basic Network Analysis*
  - *Community Detection*
  - *Centrality Analysis*
  - *Batch Processing*
  - *Custom Analysis Script*
  - *Complete Analysis Pipeline*
  - *Comparative Network Analysis*
- *CI/CD Integration*
  - *GitHub Actions*
  - *GitLab CI*
- *Additional Resources*
- *Support*


### Prerequisites

- Docker installed and running (`docker ps` should work).
- Optional: Docker Compose (`docker compose`) if you prefer compose workflows.
- Run commands from the repository root unless a command explicitly sets another path (the `docker-compose.yml` lives here).

**Note**

This guide uses the modern `docker compose` command (Docker Compose v2+, included with Docker Desktop). If using older standalone Docker Compose v1, replace `docker compose` with `docker-compose` (hyphenated).

### Quickstart

The fastest way to get started with Py3plex using Docker. This builds the image locally and uses the `latest` tag:

```
# Clone the repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Build the Docker image
docker build -t py3plex:latest .

# Run a self-test to verify installation
docker run --rm py3plex:latest selftest

# Display help
docker run --rm py3plex:latest help
```

### Building the Image

#### Using Docker

Build directly from the repository root:

```
# Build from the repository root (default tag)
docker build -t py3plex:latest .

# Build with a specific tag
docker build -t py3plex:0.95a .

# Build without cache (if needed)
docker build --no-cache -t py3plex:latest .
```

#### Using Docker Compose

If you prefer Compose, use the included `docker-compose.yml` from the repository root:

```
# Build using Compose
docker compose build

# Force rebuild
docker compose build --no-cache
```

### Running Commands

#### Basic Commands

The image sets `py3plex` as the entrypoint, so you can run CLI commands directly. Substitute your tag if not using `latest`.

```
# Show version
docker run --rm py3plex:latest --version
```

```
# Run self-test
docker run --rm py3plex:latest selftest

# Show help
docker run --rm py3plex:latest help

# Show help for a specific command
docker run --rm py3plex:latest create --help
```

### Using Docker Compose

Compose commands are slightly more verbose but easier to repeat:

```
# Show version
docker compose run --rm py3plex --version

# Run self-test (add -v for verbose)
docker compose run --rm py3plex selftest

# Show help
docker compose run --rm py3plex help
```

### Working with Files

Persist inputs/outputs by mounting a host directory to `/data` inside the container. Use the same mount across commands so outputs accumulate in one place. Examples assume `./data` on the host; adjust path syntax for Windows shells.

### Creating a Data Directory

```
# Create a local data directory
mkdir -p data
```

### Mounting Volumes

**With docker run:**

```
DATA_DIR="$(pwd)/data"  # adjust for Windows shells

# Mount current directory's data folder (Linux/macOS)
docker run --rm -v "${DATA_DIR}":/data py3plex:latest \
  create --nodes 100 --layers 3 --output /data/network.edgelist

# Windows PowerShell
# docker run --rm -v ${PWD}/data:/data py3plex:latest create --nodes 100 --layers 3 --outpu

# Windows CMD
# docker run --rm -v %cd%/data:/data py3plex:latest create --nodes 100 --layers 3 --output
```

**With docker compose:**

The `docker-compose.yml` file maps `./data` to `/data` by default, so you can simply:

```
docker compose run --rm py3plex create --nodes 100 --layers 3 --output /data/network.edgelis
```

### Complete Workflow Example

End-to-end example using volume mounts and the default `latest` tag:

```
# Create data directory
mkdir -p data

# 1. Create a multilayer network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  create --nodes 100 --layers 3 --type random --probability 0.1 --output /data/network.edgel

# 2. Load and display network information
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  load /data/network.edgelist --info

# 3. Compute statistics
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  stats /data/network.edgelist --measure all --output /data/stats.json

# 4. Detect communities
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  community /data/network.edgelist --algorithm louvain --output /data/communities.json

# 5. Visualize the network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  visualize /data/network.edgelist --output /data/network.png --layout multilayer

# 6. View the results
ls -lh data/
```

All generated files appear in `./data` on the host.

### Advanced Usage

#### Interactive Shell

To explore the container interactively:

```
# Start an interactive shell in the container
docker run --rm -it -v $(pwd)/data:/data --entrypoint /bin/bash py3plex:latest

# Inside the container, you can run:
# py3plex --version
# py3plex selftest
# python -c "import py3plex; print(py3plex.__version__)"
```

### Using Python API

You can also use py3plex as a Python library inside the container:

```
# Create a Python script
cat > data/analyze.py << 'EOF'
from py3plex.core import multinet

# Create a network
network = multinet.multi_layer_network()
nodes = [{"source": f"node{i}", "type": "layer1"} for i in range(10)]
network.add_nodes(nodes, input_type="dict")

print(f"Created network with {network.core_network.number_of_nodes()} nodes")
EOF

# Run the script in the container
docker run --rm -v $(pwd)/data:/data --entrypoint python py3plex:latest /data/analyze.py
```

### Custom Image Builds

If you need additional Python or system packages, extend the base image. Pin the base tag you want to customize (e.g., `py3plex:latest` or a versioned tag):

```
# Create a custom Dockerfile
FROM py3plex:latest

# Install additional packages
RUN pip install --no-cache-dir pandas jupyter

# Or system packages
USER root
RUN apt-get update && apt-get install -y vim && rm -rf /var/lib/apt/lists/*
USER py3plex
```

### Docker Compose with Multiple Services

You can extend `docker-compose.yml` with related services that share the same data mount (example below adds Jupyter; start it with `docker compose up jupyter`):

```
version: '3.8'

services:
  py3plex:
    build: .
    image: py3plex:latest
    volumes:
      - ./data:/data

  jupyter:
    image: py3plex:latest
```

```
    ports:
      - "8888:8888"
    volumes:
      - ./data:/data
      - ./notebooks:/notebooks
    command: jupyter notebook --ip=0.0.0.0 --port=8888 --no-browser --allow-root
    working_dir: /notebooks
```

### Helper Scripts

The repository includes optional helper scripts to simplify Docker usage.

#### py3plex-docker.sh

A wrapper script that simplifies running py3plex commands via Docker:

```
# The script automatically handles:
# - Docker installation checks
# - Image building (if needed)
# - Data directory creation
# - Volume mounting

# Usage examples:
./py3plex-docker.sh --version
./py3plex-docker.sh selftest
./py3plex-docker.sh create --nodes 100 --layers 3 --output /data/network.edgelist
./py3plex-docker.sh load /data/network.edgelist --info
```

**Features:**

- Automatically checks if Docker is installed
- Prompts to build the image if it doesn't exist
- Creates the `data/` directory if needed
- Mounts the local `data/` directory to `/data` in the container
- Passes all arguments directly to py3plex

**Script location:** `py3plex-docker.sh` in the repository root

#### test-docker-setup.sh

A comprehensive test script to validate your Docker setup:

```
# Run the validation script
./test-docker-setup.sh
```

**What it tests:**

1. Docker installation verification
2. Dockerfile existence check
3. .dockerignore existence check
4. docker-compose.yml existence check

5. Docker image build process
6. Container version command
7. Container help command
8. Container selftest
9. Volume mounting and file creation

The script provides colored output ([OK] PASS / [X] FAIL) and a summary report.

**Script location:** `test-docker-setup.sh` in the repository root

### Docker Configuration Files

#### .dockerignore

The `.dockerignore` file controls which files are excluded from the Docker build context:

**Key exclusions:**

- **Git and version control**: `.git/`, `.github/`, `.gitignore`
- **Python artifacts**: `__pycache__/`, `*.pyc`, `dist/`, `build/`
- **Virtual environments**: `venv/`, `env/`
- **Testing and coverage**: `.pytest_cache/`, `htmlcov/`, `coverage.*`
- **Documentation builds**: `docs/_build/`, `docfiles/`
- **Examples and datasets**: `examples/`, `datasets/`, `multilayer_datasets/`
- **Development files**: `tests/`, `run_tests.py`, `validate_*.py`

**Why this matters:**

- **Faster builds**: Smaller build context means faster Docker builds
- **Smaller images**: Excludes unnecessary files from the final image
- **Security**: Prevents accidental inclusion of sensitive files
- **Efficiency**: Only includes files needed to run py3plex

The `.dockerignore` file is located in the repository root.

### Troubleshooting

#### Testing the Docker Setup

To verify your Docker setup is working correctly, use the included test script from the repository root. It builds and exercises the `py3plex:latest` image:

```
# Run the test script
./test-docker-setup.sh
```

This script will:

- Check Docker installation
- Verify all Docker-related files exist
- Build the Docker image
- Run basic py3plex commands (–version, help, selftest)

- Test volume mounting and file creation
- Clean up test artifacts

### Permission Issues

If you encounter permission issues with mounted volumes (common on Linux), run with your UID/GID so files stay writable on the host:

```
# On Linux, you might need to run with your user ID
docker run --rm -v $(pwd)/data:/data --user $(id -u):$(id -g) py3plex:latest create --output
```

For Compose, set `user: "${UID}:${GID}"` in the service definition if you hit the same issue.

### Container Not Found

If the image is missing locally:

```
# List available images
docker images | grep py3plex

# Rebuild if necessary
docker build -t py3plex:latest .
```

Use the same tag (`py3plex:latest` in these examples) when running containers.

### Network Issues During Build

If you encounter network timeouts during build, try a higher timeout or a different PyPI mirror:

```
# Increase timeout
docker build --build-arg PIP_DEFAULT_TIMEOUT=300 -t py3plex:latest .

# Or point pip to a mirror reachable from your network
docker build --build-arg PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/simple -t py3plex:
```

### Out of Disk Space

Clean up unused Docker resources:

```
# Remove unused images
docker image prune -a

# Remove all stopped containers, unused networks, and dangling images
docker system prune -a
```

These commands remove unused artifacts; review before running on shared hosts.

### Debugging Build Issues

Build with verbose output:

```
docker build --progress=plain --no-cache -t py3plex:latest .
```

### Best Practices

1. **Use Volume Mounts:** Always mount volumes for input/output files.
2. **Use –rm Flag:** Remove containers after execution with `--rm` to save space.
3. **Tag Images:** Use specific tags (e.g., `py3plex:0.95a`) for reproducibility.
4. **Keep Data Separate:** Store network files in the mounted `data` directory.
5. **Use Docker Compose:** For repeated operations, Docker Compose simplifies commands.
6. **Regular Updates:** Rebuild after pulling new commits or dependency changes to keep images current.

### Practical Examples

The following examples assume you created `./data` and will reuse it across commands.

### Basic Network Analysis

Create and analyze a multilayer network:

```
# Create data directory
mkdir -p data

# Create a network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  create --nodes 100 --layers 3 --type random --probability 0.05 \
  --output /data/network.edgelist

# Analyze the network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  load /data/network.edgelist --info --stats

# Visualize
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  visualize /data/network.edgelist --output /data/network.png
```

### Community Detection

```
# Detect communities using Louvain
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  community /data/network.edgelist --algorithm louvain \
  --output /data/communities.json

# View community statistics
cat data/communities.json | python -m json.tool | head -20
```

### Centrality Analysis

```
# Compute degree centrality
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  centrality /data/network.edgelist --measure degree \
  --top 10 --output /data/centrality.json

# Compute betweenness centrality
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  centrality /data/network.edgelist --measure betweenness \
  --top 10 --output /data/betweenness.json
```

### Batch Processing

Process multiple networks (save as `batch-process.sh`):

```
#!/bin/bash
# Process multiple networks in batch
set -e

DATA_DIR=$(pwd)/data
mkdir -p $DATA_DIR

# Create several networks
for i in {1..5}; do
  echo "Creating network $i..."
  docker run --rm -v $DATA_DIR:/data py3plex:latest \
    create --nodes $((50 * i)) --layers 3 \
    --type random --probability 0.1 \
    --output /data/network_$i.edgelist
done

# Analyze each network
for i in {1..5}; do
  echo "Analyzing network $i..."
  docker run --rm -v $DATA_DIR:/data py3plex:latest \
    stats /data/network_$i.edgelist --measure all \
    --output /data/stats_$i.json
done

echo "Done! Results in $DATA_DIR"
```

### Custom Analysis Script

Create a Python script (`analysis.py`) and run it in the container:

**Python script:**

```
# analysis.py
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
```

```
# Load network
network = multinet.multi_layer_network()
network.load_network("/data/network.edgelist", input_type="multiedgelist")

# Compute statistics
layers = ["layer1", "layer2", "layer3"]
for layer in layers:
    density = mls.layer_density(network, layer)
    print(f"{layer} density: {density:.4f}")

print("Analysis complete!")
```

**Run it:**

Adjust the layer names in the script to match your data before running:

```
# Copy script to data directory
cp analysis.py data/

# Run in container
docker run --rm -v $(pwd)/data:/data \
  --entrypoint python py3plex:latest /data/analysis.py
```

### Complete Analysis Pipeline

A comprehensive analysis pipeline:

```
#!/bin/bash
# complete-pipeline.sh
set -e

DATA_DIR=$(pwd)/data
mkdir -p $DATA_DIR

echo "Step 1: Creating network..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  create --nodes 200 --layers 3 --type ba --probability 0.05 \
  --output /data/network.edgelist

echo "Step 2: Computing statistics..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  stats /data/network.edgelist --measure all \
  --output /data/stats.json

echo "Step 3: Detecting communities..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  community /data/network.edgelist --algorithm louvain \
  --output /data/communities.json

echo "Step 4: Computing centrality..."
```

```
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  centrality /data/network.edgelist --measure pagerank \
  --top 20 --output /data/centrality.json

echo "Step 5: Visualizing..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  visualize /data/network.edgelist --output /data/network.png \
  --layout multilayer --width 16 --height 12

echo "Pipeline complete! Check $DATA_DIR for results."
```

#### Comparative Network Analysis

Compare different network types:

```
#!/bin/bash
# compare-networks.sh
set -e

DATA_DIR=$(pwd)/data
mkdir -p $DATA_DIR

TYPES=("random" "er" "ba" "ws")

for type in "${TYPES[@]}"; do
  echo "Creating $type network..."
  docker run --rm -v $DATA_DIR:/data py3plex:latest \
    create --nodes 100 --layers 3 --type $type --probability 0.1 \
    --output /data/network_${type}.edgelist

  echo "Analyzing $type network..."
  docker run --rm -v $DATA_DIR:/data py3plex:latest \
    stats /data/network_${type}.edgelist --measure all \
    --output /data/stats_${type}.json
done

echo "Comparison complete!"
```

### CI/CD Integration

#### GitHub Actions

Example workflow for network analysis:

```
# .github/workflows/network-analysis.yml
name: Network Analysis with Docker

on: [push, pull_request]

jobs:
  analyze:
```

```
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Build Docker image
        run: docker build -t py3plex:latest .

      - name: Create test network
        run: |
          mkdir -p data
          docker run --rm -v $PWD/data:/data py3plex:latest \
            create --nodes 50 --layers 2 --output /data/test.edgelist

      - name: Run analysis
        run: |
          docker run --rm -v $PWD/data:/data py3plex:latest \
            stats /data/test.edgelist --measure all --output /data/stats.json

      - name: Upload results
        uses: actions/upload-artifact@v3
        with:
          name: analysis-results
          path: data/
```

### GitLab CI

Example GitLab CI configuration:

```
# .gitlab-ci.yml
image: docker:latest

services:
  - docker:dind

variables:
  DOCKER_DRIVER: overlay2

stages:
  - build
  - analyze

build:
  stage: build
  script:
    - docker build -t py3plex:latest .
    - docker save py3plex:latest > py3plex-image.tar
  artifacts:
    paths:
      - py3plex-image.tar
```

```
analyze:
  stage: analyze
  script:
    - docker load < py3plex-image.tar
    - mkdir -p data
    - docker run --rm -v $PWD/data:/data py3plex:latest create --nodes 100 --layers 3 --outp
    - docker run --rm -v $PWD/data:/data py3plex:latest stats /data/network.edgelist --measu
  artifacts:
    paths:
      - data/
```

#### Additional Resources

- Py3plex Documentation[25]
- CLI Tutorial
- Docker Documentation[26]
- Docker Compose Documentation[27]

#### Support

For issues related to:

- **Py3plex functionality**: Open an issue at https://github.com/SkBlaz/py3plex/issues
- **Docker setup**: Check this guide first, then open an issue if problems persist
- **General questions**: See the main README.md and documentation

### 3.8.3 Performance and Scalability Best Practices

This guide shows how to tune py3plex for large multilayer networks: memory usage, runtime, visualization, and tool choices. The advice targets typical sparse, research-scale graphs; adjust down if your network is unusually dense.

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[25] https://skblaz.github.io/py3plex/
[26] https://docs.docker.com/
[27] https://docs.docker.com/compose/

  - *Stratified Sampling (Preserve Layer Distribution)*
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### Network Scale Guidelines

py3plex is optimized for research-scale networks. These ranges are rough heuristics to pick appropriate tactics; dense graphs behave like larger networks than their node count alone suggests because memory and runtime scale with edge count.

Table 11: Network Scale Performance

| Network Size | Performance | Visualization | Recommendations |
|---|---|---|---|
| Small (<100 nodes) | Excellent | Fast, detailed | Dense visualization is fine |
| Medium (100-1k nodes) | Good | Fast, balanced | Default settings work well |
| Large (1k-10k nodes) | Good | Slower, minimal | Use sparse matrices, sampling |
| Very Large (>10k nodes) | Variable | Very slow | Sampling required, consider igraph/graph-tool |

### Sparse Matrix Backend

#### Why Sparse Matrices?

Most real-world networks are **sparse** (few edges compared to possible edges). Using sparse matrices keeps storage and computation proportional to the number of non-zero entries rather than the square of node count:

- **Cuts memory** by an order of magnitude for typical networks
- **Speeds up** matrix operations (multiplication, inversion)
- **Enables** analysis of larger networks before hitting RAM limits

**Example:**

```
import numpy as np
from scipy.sparse import csr_matrix

# Dense representation (10k x 10k network)
# Memory: 10_000^2 x 8 bytes = ~800 MB for float64

# Sparse representation (with 1% density)
# Memory: tens of MB (format-dependent), not hundreds
```

#### Automatic Sparse Matrix Usage

py3plex **automatically uses sparse matrices** for:

- Supra-adjacency matrix operations
- Large network storage (>1000 nodes)
- Matrix-based algorithms (PageRank, spectral methods)

If SciPy's sparse support is unavailable, py3plex falls back to dense matrices—install `scipy` to keep large workloads memory-efficient and avoid silent slowdowns.

**Verify sparse usage:**

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
network.load_network("large_network.csv", input_type="multiedgelist")
```

```
# Get sparse adjacency matrix
adj_sparse = network.get_sparse_adjacency_matrix()

print(f"Matrix size: {adj_sparse.shape}")
print(f"Non-zero entries: {adj_sparse.nnz}")
print(f"Sparsity: {1 - adj_sparse.nnz / (adj_sparse.shape[0]**2):.2%}")
```

### Force Sparse Operations

For custom algorithms, explicitly use sparse operations and avoid converting back to dense arrays mid-computation:

```
# Assumes ``network`` is already loaded; map nodes to indices once
from scipy.sparse import lil_matrix
from scipy.sparse import linalg as sparse_linalg
import numpy as np

# Map nodes to indices once
node_to_idx = {node: idx for idx, node in enumerate(network.core_network.nodes())}
n_nodes = len(node_to_idx)

# Create sparse adjacency matrix
adj = lil_matrix((n_nodes, n_nodes))

for u, v in network.core_network.edges():
    i, j = node_to_idx[u], node_to_idx[v]
    adj[i, j] = 1
    adj[j, i] = 1  # Undirected

# Convert to efficient format for operations
adj_csr = adj.tocsr()

# Sparse matrix operations scale better than dense for large, sparse graphs
result = adj_csr @ adj_csr  # Matrix multiplication
eigvals = sparse_linalg.eigs(adj_csr, k=10)  # Top eigenvalues
```

## Network Sampling

### When to Sample

Sampling helps when:

- The network is too large to visualize (>5k nodes)
- Algorithms stall or exceed your runtime budget (>10 minutes)
- Memory usage is excessive (>8 GB RAM)
- You need quick exploratory analysis before full runs

Skip sampling when you must preserve exact counts or connectivity-sensitive metrics (diameter, exact shortest paths); run the full network once resources allow.

### Random Node Sampling

Use when you want an unbiased glimpse of the network at smaller scale.

```
# Assumes ``network`` is loaded as in the random sampling example
import random
from py3plex.core import multinet

# Load full network
network = multinet.multi_layer_network()
network.load_network("large_network.csv", input_type="multiedgelist")

# Sample up to 1000 random nodes
all_nodes = list(network.get_nodes())
sample_nodes = random.sample(all_nodes, min(1000, len(all_nodes)))

# Create subnetwork
subnetwork = network.get_subnetwork(sample_nodes)

print(f"Original: {len(all_nodes)} nodes")
print(f"Sample: {len(sample_nodes)} nodes")
print(f"Sampling ratio: {len(sample_nodes)/len(all_nodes):.1%}")
```

### Stratified Sampling (Preserve Layer Distribution)

Use when layer proportions matter (multiplex data).

```
import random

# Sample proportionally from each layer
layers = network.get_layer_names()
sample_nodes_per_layer = {}

for layer in layers:
    layer_nodes = [n for n in network.get_nodes() if n[1] == layer]
    if not layer_nodes:
        continue
    sample_size = max(1, len(layer_nodes) // 10)  # 10% sample, at least 1
    sample_nodes_per_layer[layer] = random.sample(layer_nodes, sample_size)

# Combine samples
all_samples = [node for nodes in sample_nodes_per_layer.values() for node in nodes]
subnetwork = network.get_subnetwork(all_samples)
```

### Hub-Based Sampling (Keep Important Nodes)

Use when high-degree nodes dominate the phenomenon you study (traffic, influence).

```
# Uses the loaded ``network``; keeps high-degree nodes (hubs)
import networkx as nx
```

```
# Sample high-degree nodes (hubs)
degrees = dict(network.core_network.degree())

# Sort by degree and take up to top 1000
sorted_nodes = sorted(degrees.items(), key=lambda x: x[1], reverse=True)
hub_nodes = [node for node, deg in sorted_nodes[:min(1000, len(sorted_nodes))]]

subnetwork = network.get_subnetwork(hub_nodes)
print(f"Sampled {len(hub_nodes)} hubs")
```

### Algorithm Optimization

#### Choose Efficient Algorithms

Choose algorithms with complexity close to O(n + m) where possible. The table uses `n` for nodes and `m` for edges:

Table 12: Algorithm Complexity

| Algorithm | Complexity | Recommendations |
|---|---|---|
| Degree centrality | O(n + m) | Fast, use freely |
| Betweenness centrality | O(nm) | Slow for large networks, sample first or approximate |
| PageRank | O(iterations x m) | Fast if sparse, limit iterations |
| Community detection (Louvain) | O(m log n) | Fast, recommended |
| Shortest paths (all pairs) | O(n^2 m) | Very slow, use sampling or approximate |
| Force-directed layout | O(n^2) | Slow for >5k nodes, use alternatives |

**Example - Fast centrality:**

```
import networkx as nx

G = network.core_network  # assumes network is already loaded

# FAST: Degree centrality
degree_cent = nx.degree_centrality(G)  # O(n+m) - instant

# SLOW: Betweenness centrality
# For large networks, sample first or use approximate algorithm
if G.number_of_nodes() < 1000:
    between_cent = nx.betweenness_centrality(G)
else:
    # Use approximate algorithm
    between_cent = nx.betweenness_centrality(G, k=100)  # Sample 100 nodes
```

#### Limit Algorithm Iterations

Iteration limits prevent runaway runtimes on large, sparse graphs while keeping results stable enough for exploration. Lower `tol` increases accuracy but may require more iterations; pair it with a sensible `max_iter` cap.

```
import networkx as nx

# PageRank with iteration limit
pagerank = nx.pagerank(
    network.core_network,
    max_iter=50,        # Limit iterations
    tol=1e-4            # Tolerance for convergence
)

# Community detection with resolution limit
from py3plex.algorithms.community_detection import community_louvain
communities = community_louvain.best_partition(
    network.core_network,
    resolution=1.0      # Adjust resolution parameter
)
```

### Parallel Processing

#### Multi-Core Processing

Use joblib for embarrassingly parallel node/edge operations; cap `n_jobs` to avoid oversubscribing shared machines or container limits:

```
from joblib import Parallel, delayed
import networkx as nx

def compute_node_centrality(node, graph):
    """Compute centrality for a single node."""
    # Custom centrality computation
    neighbors = list(graph.neighbors(node))
    return node, len(neighbors)

# Parallel processing
nodes = list(network.core_network.nodes())
results = Parallel(n_jobs=-1)(  # Use all cores
    delayed(compute_node_centrality)(node, network.core_network)
    for node in nodes
)

centralities = dict(results)
```

### GPU Acceleration (Advanced)

For very large networks with a CUDA-capable GPU, try GPU acceleration. Keep data sparse to avoid blowing GPU memory and move only the operations that benefit from GPU throughput:

```
# Install CuPy for GPU NumPy operations
pip install cupy-cuda11x  # Replace 11x with your CUDA version
```

```
# Requires NVIDIA GPU with CUDA
try:
    import cupy as cp
    import numpy as np
    from cupyx.scipy.sparse import csr_matrix as gpu_csr

    # Keep sparse; convert SciPy CSR to CuPy CSR
    adj_cpu = network.get_sparse_adjacency_matrix().astype(np.float32)
    gpu_adj = gpu_csr(adj_cpu)

    # GPU-accelerated sparse matrix operations
    gpu_result = gpu_adj @ gpu_adj

    # Transfer back to CPU if needed
    result = gpu_result.get()

except ImportError:
    print("CuPy not available, using CPU")
```

Keep GPU arrays in `float32` and sparse formats to fit device memory; fall back to CPU if allocations fail.

### Visualization Optimization

#### Reduce Visual Complexity

Favor minimal styling for large graphs to avoid overplotting:

```
from py3plex.visualization.multilayer import draw_multilayer_default

# For large networks (>1000 nodes)
draw_multilayer_default(
    network.get_layers(),
    node_size=3,              # Tiny nodes
    labels=False,             # No labels
    edge_size=0.3,            # Thin edges
    alphalevel=0.2,           # Very transparent
    remove_isolated_nodes=True  # Remove disconnected
)
```

#### Save to File Instead of Display

```
import matplotlib.pyplot as plt

# Don't show interactively (faster)
fig, ax = plt.subplots(1, 1, figsize=(10, 8))
draw_multilayer_default(
    network.get_layers(),
    display=False,  # Don't show
    axis=ax
)

# Save directly to file
plt.savefig('network.png', dpi=150, bbox_inches='tight')
plt.close()  # Free memory
```

#### Use Lower Resolution

```
import matplotlib.pyplot as plt

# For quick exploration, use low DPI
plt.figure(figsize=(8, 6), dpi=72)  # Low resolution

# For publications, use high DPI
plt.figure(figsize=(10, 8), dpi=300)  # High resolution
```

### Memory Management

#### Monitor Memory Usage

```
import psutil
import os
from py3plex.core import multinet

# Requires ``psutil``; install it if missing

def get_memory_usage():
    """Get current memory usage in MB."""
    process = psutil.Process(os.getpid())
    return process.memory_info().rss / 1024 / 1024

print(f"Memory before loading: {get_memory_usage():.1f} MB")

network = multinet.multi_layer_network()
network.load_network("large_network.csv", input_type="multiedgelist")

print(f"Memory after loading: {get_memory_usage():.1f} MB")
```

#### Free Memory When Done

```
# Delete network when no longer needed
del network

# Force garbage collection
import gc
gc.collect()
```

#### Use Generators Instead of Lists

```
# BAD: Loads all nodes into memory
all_nodes = list(network.get_nodes())
for node in all_nodes:
    process(node)

# GOOD: Processes nodes one at a time
for node in network.get_nodes():
    process(node)
```

### Batch Processing

#### Process Networks in Batches

For multiple networks:

```
import os
import gc

network_files = ["net1.csv", "net2.csv", "net3.csv"]  # Extend with your files

results = []
for i, file in enumerate(network_files):
    print(f"Processing {i+1}/{len(network_files)}: {file}")

    # Load network
    network = multinet.multi_layer_network()
    network.load_network(file, input_type="multiedgelist")

    # Compute statistics
    result = analyze_network(network)
    results.append(result)

    # Free memory
    del network
    gc.collect()

    # Save intermediate results every 10 networks
    if (i + 1) % 10 == 0:
        save_results(results, f"results_batch_{i+1}.json")
```

Persist partial outputs regularly so an interrupted job does not lose prior work.

### Benchmark Results

#### Illustrative Performance Benchmarks

Tested on: Intel i7-10700K, 32 GB RAM, Python 3.10. These figures illustrate order-of-magnitude behavior; disk speed, graph density, and algorithm parameters will change your results.

Table 13: Operation Performance

| Operation | 100 nodes | 1k nodes | 10k nodes | 100k nodes |
|---|---|---|---|---|
| Load from CSV | <1s | <1s | 2s | 20s |
| Basic statistics | <1s | <1s | <1s | 3s |
| Degree centrality | <1s | <1s | 1s | 10s |
| PageRank | <1s | <1s | 2s | 25s |
| Louvain communities | <1s | 1s | 5s | 60s |
| Visualization (sparse) | <1s | 2s | 15s | N/A* |

* Visualization not recommended for >10k nodes without sampling

### Scaling Recommendations

Based on network size:

**<1k nodes:**

- Use any algorithms
- Full visualization
- No sampling needed

**1k-10k nodes:**

- Use sparse matrices
- Minimal visualization
- Sample for some algorithms

**10k-100k nodes:**

- Sparse matrices required
- Sample for visualization
- Use approximate algorithms
- Consider igraph for speed

**>100k nodes:**

- Use specialized tools (igraph, graph-tool, NetworKit)
- Sample heavily for py3plex operations
- Focus on specific analyses

### Alternative Tools for Scale

#### When py3plex Isn't Enough

For networks >100k nodes, consider exporting to a faster backend for heavy algorithms and re-importing results:

**igraph** (C-based, very fast):

```
import igraph as ig

# Often 10-100x faster for large networks
g = ig.Graph.Read_GraphML("large_network.graphml")
communities = g.community_multilevel()

# Export back to py3plex if needed
# (via GraphML or edge list)
```

**graph-tool** (C++, fastest):

```
import graph_tool.all as gt

# Fastest for >1M edges
g = gt.load_graph("large_network.graphml")
state = gt.minimize_blockmodel_dl(g)
communities = state.get_blocks()
```

**NetworKit** (C++, parallel):

```
import networkit as nk

# Excellent for parallel algorithms
G = nk.readGraph("large_network.edgelist", nk.Format.EdgeList)
communities = nk.community.detectCommunities(G)
```

### Quick Performance Checklist

#### Before Running on Large Network

```
[ ] Enable sparse matrices (automatic for most ops)
[ ] Sample network if >10k nodes
[ ] Choose efficient algorithms (degree > betweenness)
[ ] Limit visualization detail
[ ] Monitor memory usage
[ ] Use batch processing for multiple networks
[ ] Consider alternative tools if >100k nodes
```

### Optimization Order

1. **Use sparse matrices** (biggest impact, usually automatic)
2. **Sample network** (if >10k nodes)
3. **Choose efficient algorithms** (avoid O(n^3) operations)

4. **Parallelize** (if multi-core available)
5. **GPU acceleration** (only if CUDA GPU available)

#### Next Steps

- ../user_guide/io_and_formats - Efficient data loading
- ../user_guide/visualization - Optimize visualizations
- *Docker Usage Guide* - Docker deployment for production

For performance issues, open an issue on GitHub Issues[28].

### 3.8.4 GUI: Web Interface for Network Exploration

**Warning**

**Development Mode Only:** The GUI is intended for local development and exploration. Do not expose it to the public internet without proper authentication and security hardening. See gui_deployment for production configuration.

The py3plex GUI provides a web-based interface for interactive multilayer network exploration.

Use this page to see how the GUI fits into the documentation set and pick a setup path.

**Features:**

- Load networks from various file formats without code
- Interactive, zoomable visualizations
- Compute statistics via point-and-click menus
- Detect communities and visualize them
- Export results for further analysis

**This section covers:**

- *Py3plex GUI* — Loading data, exploring networks, running analyses
- gui_deployment — Running locally, Docker, production setup
- gui_api_reference — Backend API documentation
- gui_architecture — How the GUI is built
- gui_testing — Testing infrastructure

#### Running the GUI Locally

Prerequisites:

- Docker and Docker Compose installed locally
- Ports 8080 (GUI), 5555 (Flower), 8000 (API), and 6379 (Redis) are free

Choose one of the following:

**Option 1 — Docker quick start (recommended):**

[28] https://github.com/SkBlaz/py3plex/issues

```
# Clone the repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex/gui

# Copy environment configuration
cp .env.example .env

# Start all services (wait for containers to report "healthy")
make up

# Open browser to http://localhost:8080
```

**Option 2 — Install from source (contributors):**

```
# Develop against source
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex/gui
cp .env.example .env
make up
```

The GUI will be available at `http://localhost:8080`. See gui_deployment for detailed setup instructions and troubleshooting.

### GUI Actions vs py3plex APIs

The GUI provides a point-and-click interface for common py3plex operations. The examples below show representative Python calls once you have a `network` instance loaded:

| GUI Action | Equivalent Python API |
|---|---|
| **Load Network** (upload file) | `network = multinet.multi_layer_network(). load_network(file_path, input_type="...")` |
| **Compute Statistics** → Basic Stats | `network.basic_stats()` |
| **Compute Statistics** → Layer Density | `from py3plex.algorithms.statistics import multilayer_statistics as mls mls. layer_density(network, layer_name)` |
| **Detect Communities** → Louvain | `from py3plex.algorithms.community_detection import community_louvain partition = community_louvain. best_partition(network.core_network)` |
| **Compute Centrality** → Degree | `from py3plex.algorithms.multilayer_algorithms. centrality import MultilayerCentrality calc = MultilayerCentrality(network) degree = calc. overlapping_degree_centrality()` |
| **Visualize Network** | `from py3plex.visualization.multilayer import draw_multilayer_default draw_multilayer_default([network], display=True)` |
| **Export** → JSON | `network.save_network("output.json", output_type="json")` |

This mapping helps you transition from GUI exploration to programmatic analysis.

### When to Use the GUI

**Use GUI for:**

- Domain experts who prefer not to write Python
- Quick exploratory analysis of new datasets
- Demonstrations and teaching
- Collaborative work with mixed technical backgrounds

**Use Python library for:**

- Production pipelines and automation
- Reproducible analysis workflows
- Advanced customization
- Large-scale processing

Start with gui_deployment to run the GUI, then *Py3plex GUI* for usage.

## 3.8.5 Py3plex GUI

A production-ready web-based GUI for **py3plex** multilayer network analysis, running locally via Docker Compose. Upload multilayer graphs, explore them visually, and run common analytics from your browser.

### Visual Overview

The Py3plex GUI provides an intuitive web interface for multilayer network analysis. Below are screenshots of the main interface pages:

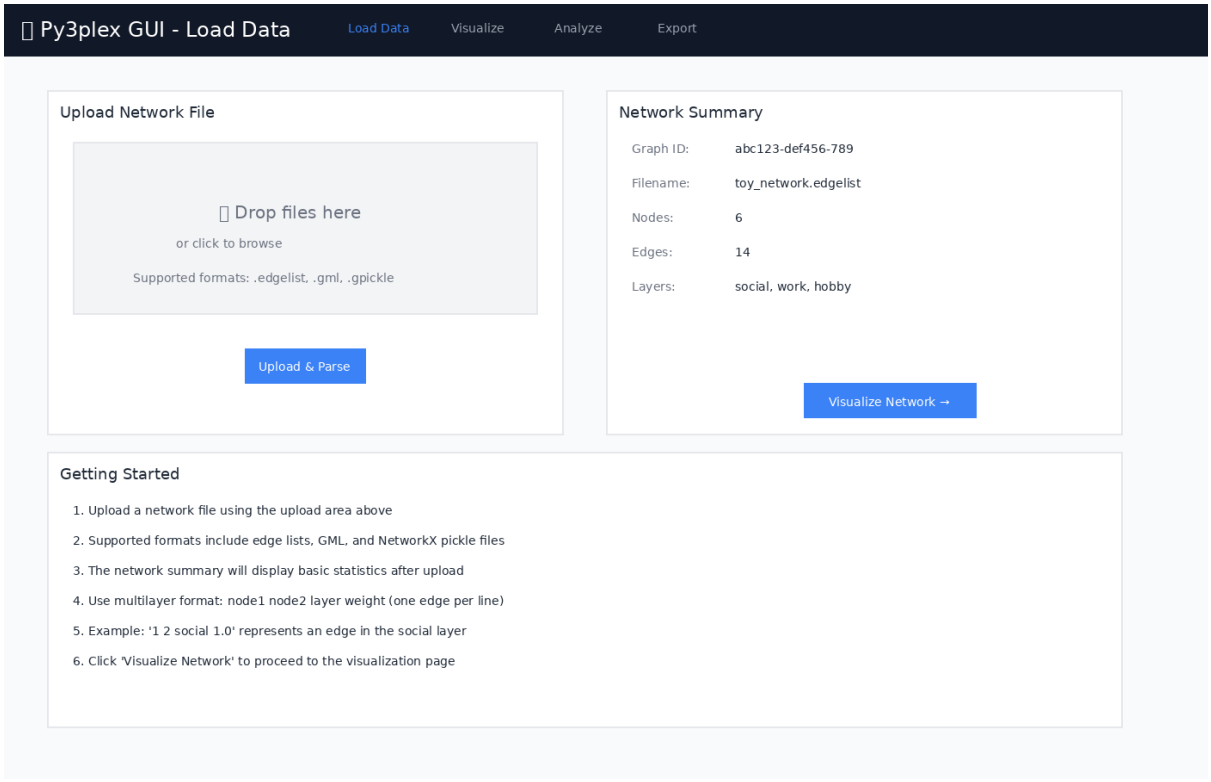



*Figure 1: Load Data page for uploading network files in various formats*

### Prerequisites

- Docker (>= 20.10)
- Docker Compose (>= 2.0)
- 4GB RAM minimum
- Ports available: 8080 (GUI), 5555 (Flower), 8000 (API), 6379 (Redis)

### Quickstart

Use this sequence after the prerequisites above are installed and ports are free:

```
# Navigate to the gui directory
cd gui

# Copy environment configuration
cp .env.example .env

# Start all services (builds automatically)
make up

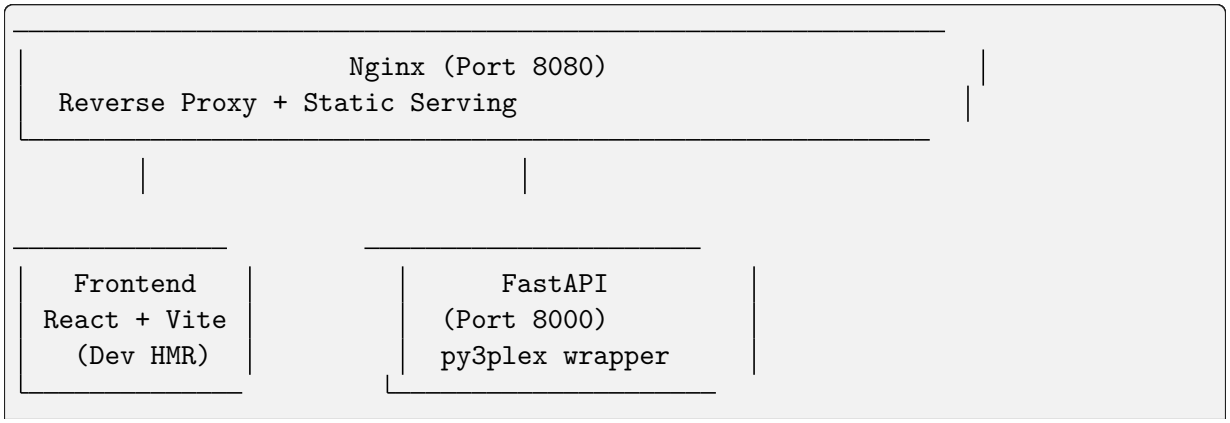
# Open in browser
#  Application: http://localhost:8080
# Data Job Monitor (Flower): http://localhost:5555
```

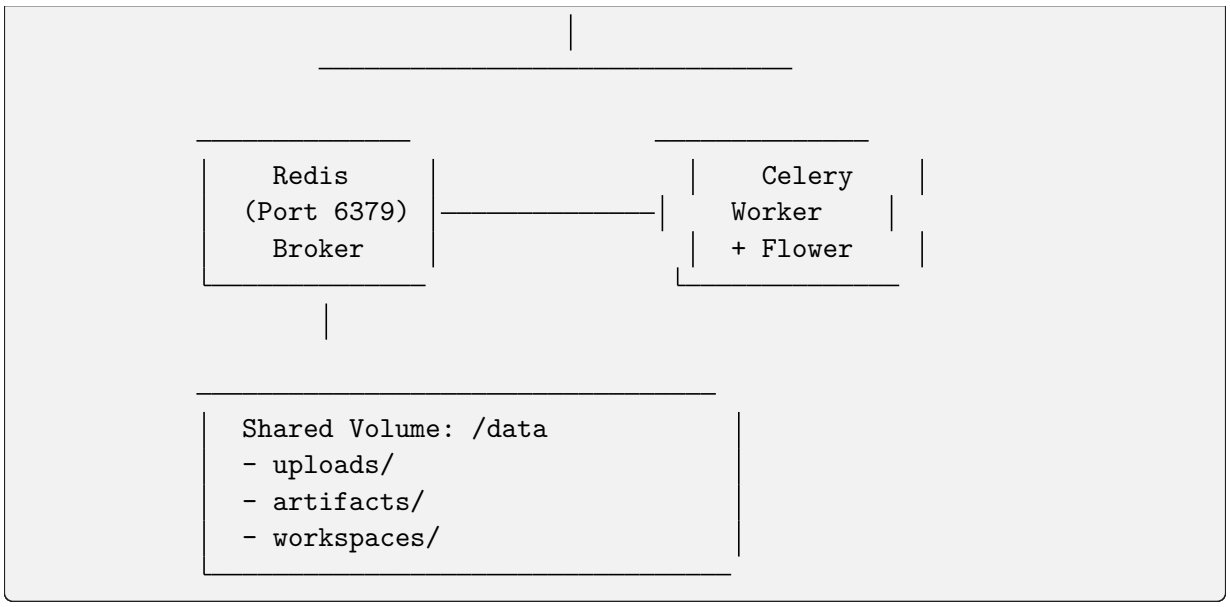
**First time?** The build takes ~3-5 minutes. Subsequent starts are instant. Wait for containers to report `healthy` before using the UI.

That's it! The application will be running with:

- React + TypeScript frontend with hot reload
- FastAPI backend with py3plex integration
- Celery workers for async jobs
- Redis broker
- Nginx reverse proxy

### Architecture






**Key Components:**

- **Frontend**: React 18 + Vite + TypeScript + Tailwind CSS
- **API**: FastAPI (Python 3.12) with py3plex integration
- **Worker**: Celery for long-running analysis jobs
- **Broker**: Redis for job queue
- **Proxy**: Nginx for unified access
- **Monitoring**: Flower dashboard for job inspection

### Features

### Data Loading

- Upload network files (.edgelist, .txt, .gml, .gpickle)
- Automatic format detection
- Multilayer network support
- Real-time file preview

### Visualization

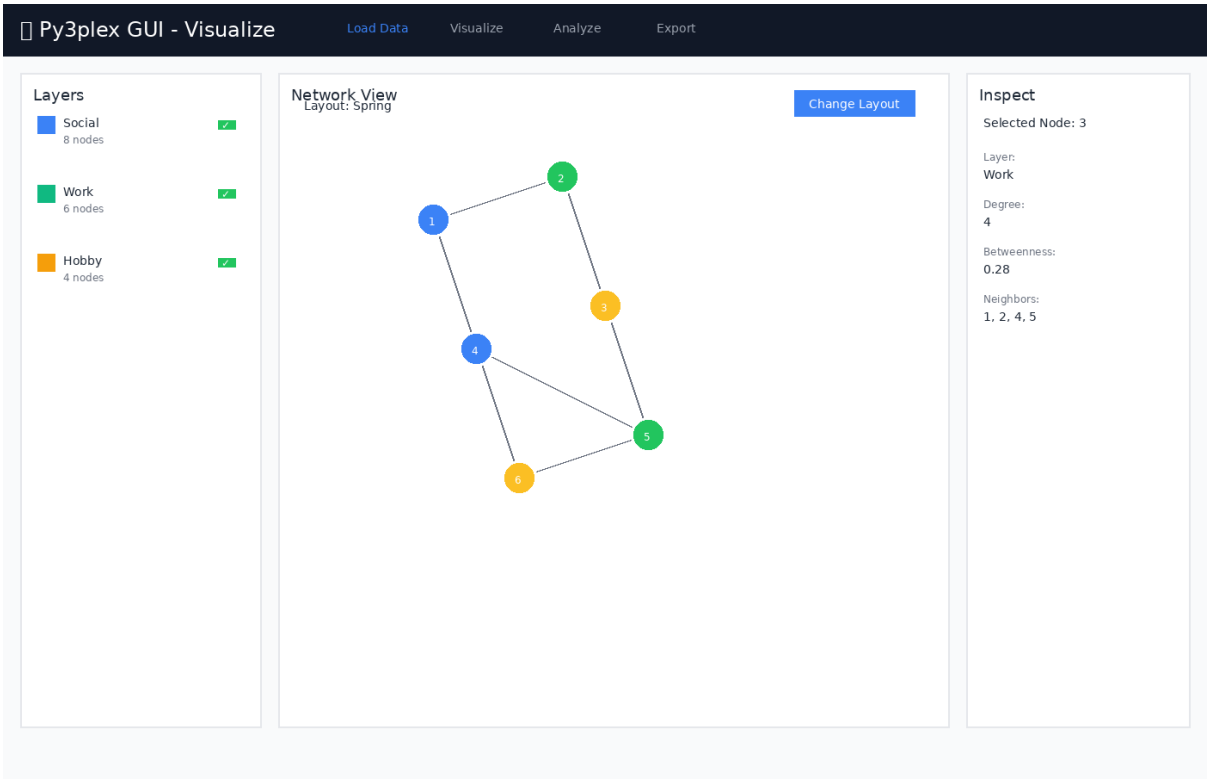



*Figure 2: Network visualization with layer controls and node inspection panel*

**Features:**

- Layer-centric view
- Interactive node/edge inspection
- Configurable layouts
- Position caching

### Analysis (Async via Celery)

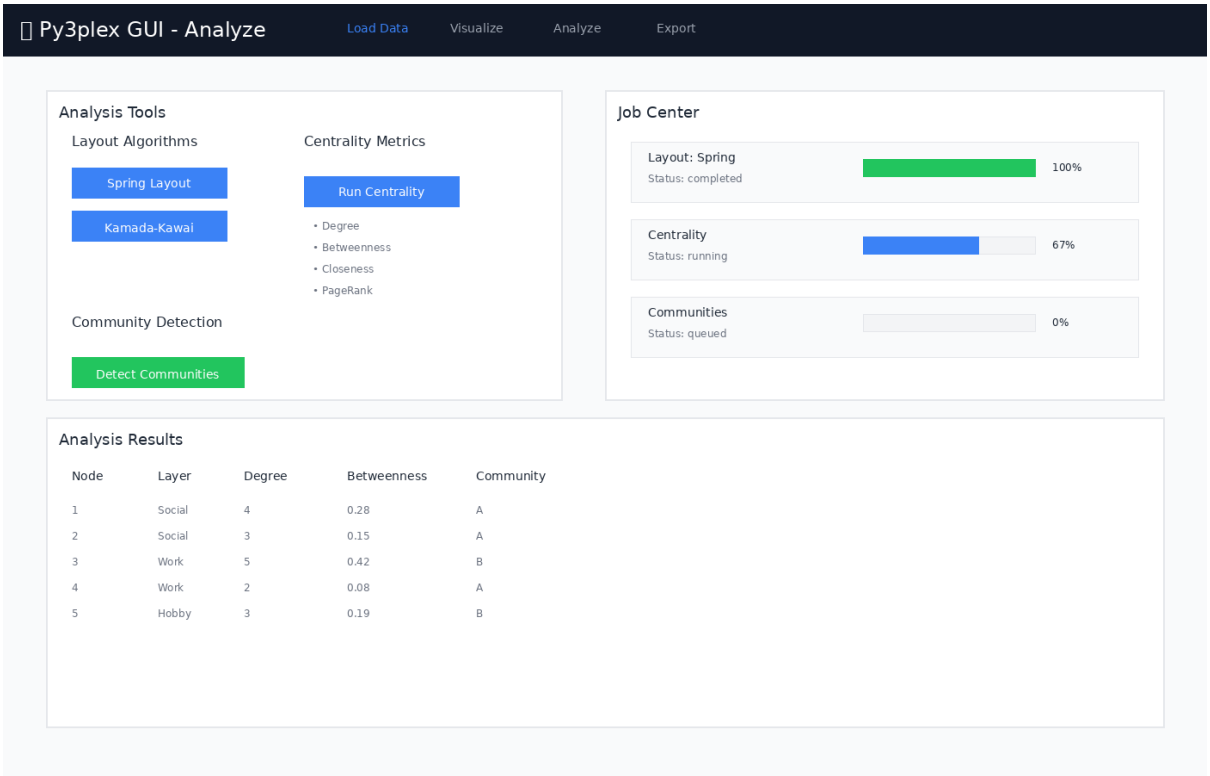



*Figure 3: Analysis page with job controls and real-time progress tracking*

**Available Analysis Tools:**

- **Layouts**: Spring, Kamada-Kawai, Circular, Random
- **Centrality**: Degree, Betweenness, Closeness, Eigenvector, PageRank
- **Community Detection**: Louvain, Label Propagation, Greedy Modularity
- Real-time progress tracking
- Result caching

### Export

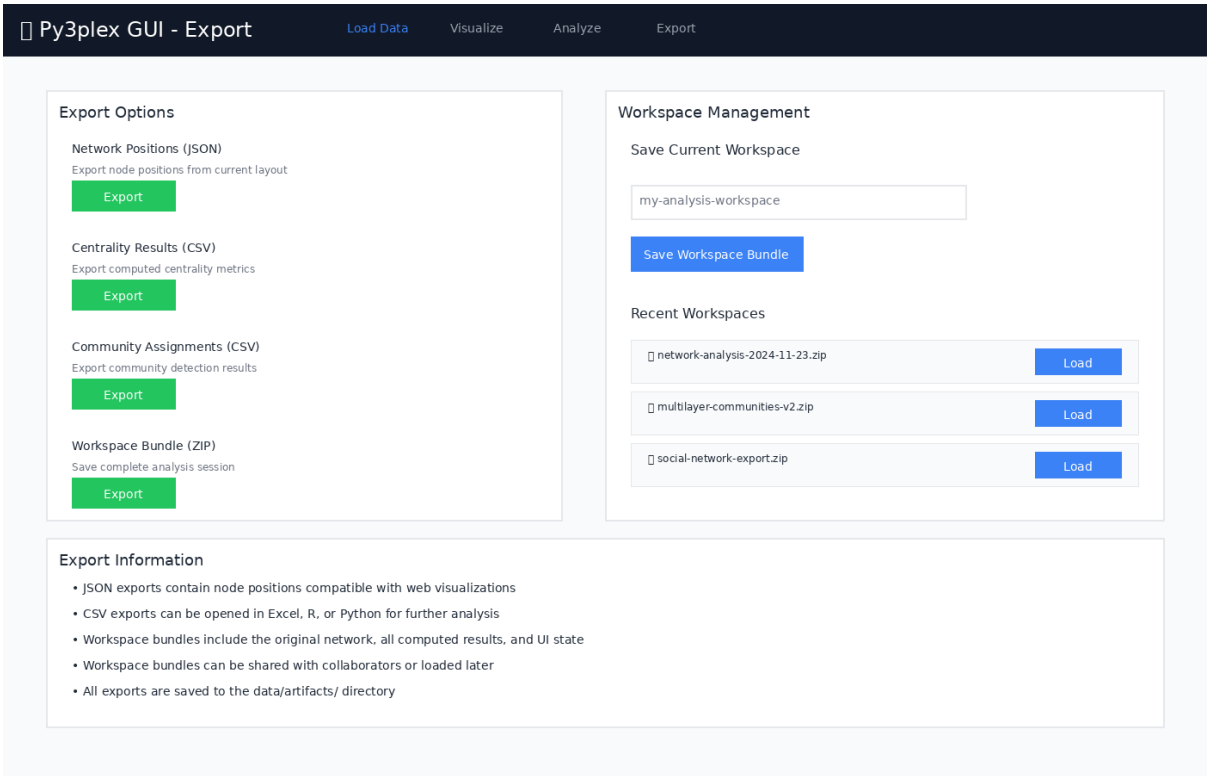



*Figure 4: Export page for saving analysis results and workspace bundles*

**Export Options:**

- CSV summaries (centrality, communities)
- JSON position data
- PNG snapshots (planned)
- Workspace bundles (data + state + results)

### Workspace Bundles

Save and restore complete analysis sessions. A bundle includes:

- Original network file
- Computed layouts
- Centrality results
- Community assignments
- UI view state

Export a bundle from the **Export** page and re-import it later to restore the exact session state.

### Job Monitoring

The GUI includes Flower dashboard for real-time monitoring of background jobs:

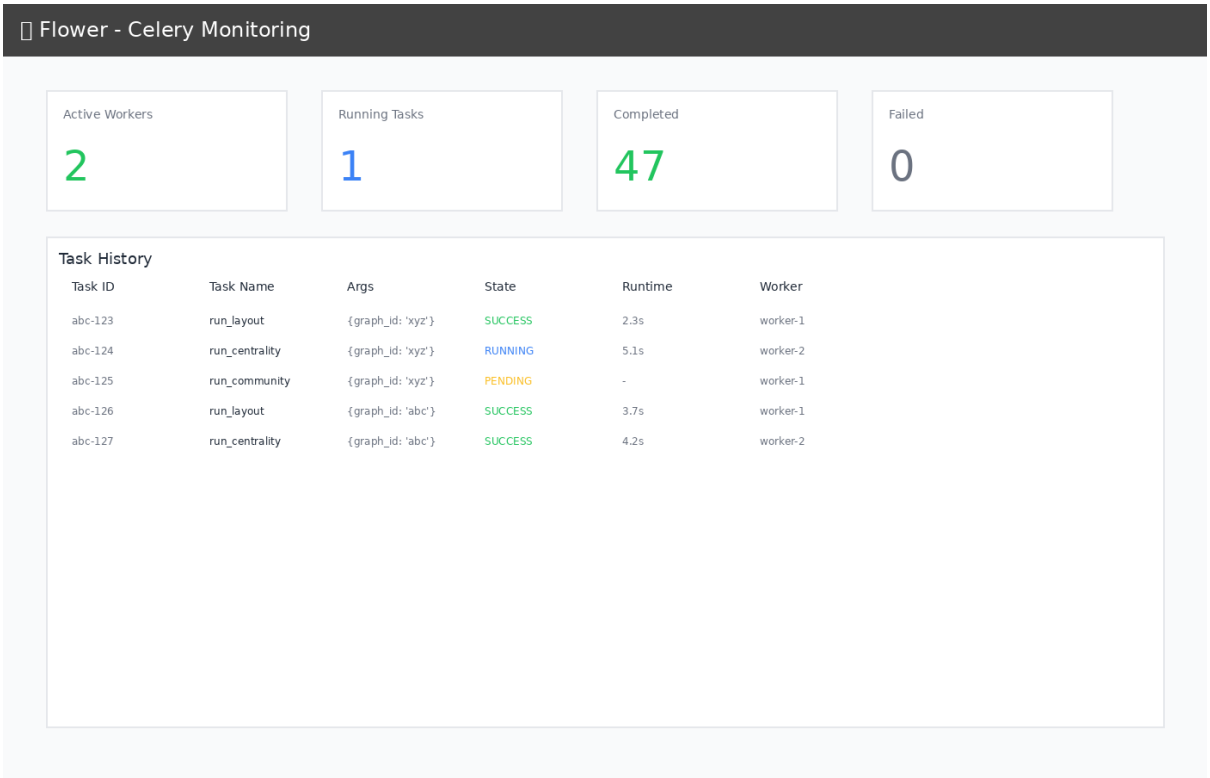



*Figure 5: Flower dashboard showing task history and worker status*

**Monitoring Features:**

- Track task execution in real-time
- View worker status and performance
- Inspect task history and results
- Monitor task queue and completion rates
- Access at http://localhost:5555 when running

### Development Guide

### Project Structure

```
gui/
├── docker-compose.yml          # Main orchestration
├── compose.gpu.yml             # GPU override (optional)
├── Makefile                    # Convenience commands
├── .env.example                # Configuration template
├── nginx/
│   └── nginx.conf              # Reverse proxy config
```

```
├── api/
│   ├── Dockerfile.api          # API container
│   ├── pyproject.toml          # Python dependencies
│   └── app/
│       ├── main.py             # FastAPI app
│       ├── schemas.py          # Pydantic models
│       ├── deps.py             # DI & config
│       ├── routes/             # API endpoints
│       ├── services/           # Business logic (py3plex wrappers)
│       ├── workers/            # Celery tasks
│       └── utils/              # Helpers
├── worker/
│   └── Dockerfile              # Worker container
├── frontend/
│   ├── Dockerfile.frontend     # Frontend container
│   ├── package.json            # Node dependencies
│   ├── vite.config.ts          # Vite config
│   └── src/
│       ├── App.tsx             # Root component
│       ├── lib/api.ts          # API client
│       ├── pages/              # Page components
│       └── components/         # Reusable components
└── data/                       # Shared volume (gitignored)
    ├── uploads/
    ├── artifacts/
    └── workspaces/
```

### Makefile Commands

```
make setup          # Copy .env.example to .env
make up             # Start all services (build if needed)
make down           # Stop and remove containers + volumes
make restart        # Restart all services
make build          # Rebuild Docker images
make logs           # Tail logs from all services

# Development helpers
make bash-api       # Shell into API container
make bash-worker    # Shell into worker container
make bash-frontend  # Shell into frontend container

# Testing
make test-api       # Run API tests
make e2e            # Run end-to-end tests (WIP)

# Cleanup
make clean          # Remove containers, volumes, and data
```

### Local Development Against py3plex

The py3plex repository root is bind-mounted read-only into containers at `/workspace`:

```
# In Dockerfile.api
ARG PY3PLEX_PATH=/workspace
RUN pip install -e ${PY3PLEX_PATH}
```

**Benefits:**

- Changes to py3plex core reflect immediately (no rebuild)
- Editable install for development
- Isolated GUI code under `gui/`

**Note:** The mount is read-only to prevent accidental writes from containers.

### Configuration

#### Environment Variables (.env)

```
# API
API_WORKERS=2               # Uvicorn workers
MAX_UPLOAD_MB=512           # Max file size

# Celery
CELERY_CONCURRENCY=2        # Worker threads
REDIS_URL=redis://redis:6379/0

# Frontend
VITE_API_URL=http://localhost:8080/api
```

### GPU Support (Optional)

Enable NVIDIA GPU acceleration:

```
docker compose -f docker-compose.yml -f compose.gpu.yml up
```

**Requirements:**

- NVIDIA Docker runtime installed
- CUDA-compatible GPU

### Data Formats

#### Accepted Input Formats

**1. Edge List (.edgelist, .txt)**

```
# Multilayer format: node1 node2 layer weight
1 2 social 1.0
1 3 work 1.0
2 3 hobby 0.5
```

```
# Simple format: node1 node2
A B
B C
C D
```

Use whitespace-separated columns. Include the layer column for multilayer graphs; omit it for single-layer files. Weights are optional and default to 1.0 if absent.

2. **GML (.gml)**
   - Graph Modeling Language
   - Preserves attributes and metadata
3. **NetworkX Pickle (.gpickle)**
   - Native NetworkX serialization
   - Fastest for large graphs

### Example Dataset

Try the included toy network (6 nodes, 14 edges, 3 layers):

```
# From host
curl -F "file=@gui/toy_network.edgelist" http://localhost:8080/api/upload

# Or use the Web UI at http://localhost:8080
```

### Testing

### Continuous Integration

GUI tests run automatically on GitHub Actions for any changes to the `gui/` directory. Pushes and pull requests that touch GUI files trigger the workflow:

```
# Workflow: .github/workflows/gui-tests.yml
- API unit tests (pytest)
- Integration tests (Docker Compose)
- Frontend build verification
```

**Status**:

[29] Tests include:

- Health endpoint validation
- File upload with toy network
- Layout job execution and completion
- Centrality computation
- Service health checks

[29] https://github.com/SkBlaz/py3plex/actions/workflows/gui-tests.yml

### API Tests

```
# Run inside API container
make bash-api
pytest ci/api-tests/ -v

# Or directly
docker compose exec api pytest ci/api-tests/ -v
```

### Integration Tests

The CI workflow runs full integration tests (Docker Compose brings up the full stack and exercises uploads + analysis):

```
# With the stack running (`make up`):
cd gui
# Upload the sample network
curl -F "file=@toy_network.edgelist" http://localhost:8080/api/upload
# Kick off layout, centrality, and community detection jobs
# Verify each job reaches status=completed and returns results
```

### Frontend Tests (WIP)

```
# Playwright smoke tests
make e2e
```

### Manual Testing Workflow

For a step-by-step manual checklist, see `docfiles/gui/gui_testing.rst`. Quick smoke flow:

1. **Upload** `toy_network.edgelist` via Web UI
2. **Visualize** the network (expect 6 nodes, 14 edges, 3 layers)
3. **Analyze**:
   - Run Spring Layout
   - Compute Centrality (degree + betweenness)
   - Detect Communities (Louvain)
4. **Monitor** jobs at http://localhost:5555 (Flower) and confirm they transition to `completed` without errors
5. **Export** workspace bundle

### Production Deployment

### Static Build (Recommended)

For production, serve pre-built frontend assets instead of the Vite dev server:

```
# Modify Dockerfile.frontend to use multi-stage build
FROM node:20-alpine AS builder
```

```
WORKDIR /app
COPY frontend/ .
RUN npm install && npm run build

FROM nginx:alpine
COPY --from=builder /app/dist /usr/share/nginx/html
COPY nginx/nginx.conf /etc/nginx/nginx.conf
```

Update `docker-compose.yml`:

```
frontend:
  build:
    context: .
    dockerfile: Dockerfile.frontend.prod
  # Remove dev server volume mounts
```

#### Security Considerations

- Set `CORS allow_origins` to specific domains (not `*`)
- Add authentication/authorization middleware
- Use HTTPS (add TLS termination in Nginx)
- Validate file uploads strictly
- Set resource limits per user/job
- Enable Celery task rate limiting

**Current Status**: Development mode - suitable for local use only. Do not expose to the public internet without proper security hardening.

#### Troubleshooting

##### Issue: Port already in use

```
# Find process using port 8080
lsof -ti:8080 | xargs -r kill

# Or change port in docker-compose.yml
ports:
  - "8081:80"  # Use 8081 instead
```

##### Issue: Permission denied on /data

```
# Fix volume permissions
sudo chown -R $(whoami):$(whoami) gui/data/
```

#### Issue: py3plex import error in containers

```
# Verify mount
docker compose exec api ls -la /workspace

# Reinstall if needed
docker compose exec api pip install -e /workspace
```

#### Issue: Frontend can't reach API

Check Nginx proxy config:

```
docker compose exec nginx cat /etc/nginx/nginx.conf

# Test API directly
curl http://localhost:8000/api/health
```

### API Documentation

Interactive docs available at:

- **Swagger UI**: http://localhost:8080/api/docs
- **ReDoc**: http://localhost:8080/api/redoc

### Key Endpoints

```
POST   /api/upload                          # Upload network file
GET    /api/graphs/{id}/summary             # Graph statistics
POST   /api/graphs/{id}/layout              # Compute layout (async)
POST   /api/graphs/{id}/analysis/centrality # Compute centrality (async)
POST   /api/graphs/{id}/analysis/community  # Detect communities (async)
GET    /api/jobs/{id}                       # Poll job status
POST   /api/workspaces/save                 # Save workspace bundle
```

### Example: Run Analysis

```
# 1. Upload
GRAPH_ID=$(curl -F "file=@toy_network.edgelist" \
  http://localhost:8080/api/upload | jq -r .graph_id)

# 2. Start layout job
JOB_ID=$(curl -X POST \
  -H "Content-Type: application/json" \
  -d '{"algorithm":"spring","seed":42,"dimensions":2}' \
  http://localhost:8080/api/graphs/$GRAPH_ID/layout | jq -r .job_id)

# 3. Poll job
curl http://localhost:8080/api/jobs/$JOB_ID
```

```
# 4. Get positions
curl http://localhost:8080/api/graphs/$GRAPH_ID/positions
```

### Contributing

This GUI is part of the py3plex[30] project.

**Guidelines:**

- Keep all GUI code under `gui/`
- Do not modify py3plex core
- Follow existing code style (Black, Ruff for Python; ESLint for TypeScript)
- Add tests for new features
- Update documentation with new features

### License

This GUI follows the py3plex repository license:

- **GUI code** (under `gui/`): BSD-3-Clause (same as py3plex)
- **Dependencies**: See individual package licenses

### Acknowledgments

Built with:

- py3plex[31] - Multilayer network analysis
- FastAPI[32] - Modern Python web framework
- Celery[33] - Distributed task queue
- React[34] - UI library
- Vite[35] - Frontend build tool
- Tailwind CSS[36] - Utility-first CSS

### Support

- **Issues**: https://github.com/SkBlaz/py3plex/issues
- **Docs**: https://skblaz.github.io/py3plex/
- **py3plex Paper**: Applied Network Science 2019[37]

[30] https://github.com/SkBlaz/py3plex
[31] https://github.com/SkBlaz/py3plex
[32] https://fastapi.tiangolo.com/
[33] https://docs.celeryproject.org/
[34] https://react.dev/
[35] https://vitejs.dev/
[36] https://tailwindcss.com/
[37] https://doi.org/10.1007/s41109-019-0203-7

### 3.8.6 Developer & Contributor Docs

Use this page as your landing point for development setup, code architecture, and contribution guidelines. Follow the linked pages in order so you can move from setup to contributing without guessing the next step.

**This section covers:**

- *Development Guide* — Environment setup, running tests, submitting changes
- *Architecture and Design* — How py3plex is organized internally (modules, data flow, extension points)
- repo_layout — Directory structure and where key files live
- contributing — Bug reports, code style, review process

#### Types of Contributions

- **Bug reports** — Include steps to reproduce, expected vs. actual behavior, and environment details (OS, Python, py3plex version)
- **Documentation** — Improve explanations, runnable examples, and doc tests
- **Code fixes** — Address bugs, safety gaps, or performance bottlenecks
- **New features** — Add algorithms, visualizations, I/O formats, or plugins
- **Examples** — Write tutorials and end-to-end workflows others can run locally
- **Community support** — Help others in issues and discussions with concrete guidance

#### Getting Started

1. Read *Development Guide* to set up your environment and workflow.
2. Pick a small, scoped task (docs, tests, or a single bug) to get familiar with the process.
3. Browse GitHub issues labeled "good first issue" or propose a doc improvement you can finish quickly.
4. Comment on the issue you plan to tackle so others know it is in progress, and keep the thread updated with findings.
5. Ask questions in the relevant issue or discussion thread when blocked; link any experiments or logs that help others assist you.

See contributing for the complete process.

### 3.8.7 Development Guide

This guide covers **development workflows**, **Makefile commands**, **testing**, and **contributing** to py3plex. Keep it handy as your single reference while coding, testing, and writing docs.

#### Getting Started

Clone the repository and install in development mode. Requirements: Python 3.8+ and `pip`; `make` is recommended for the shortcuts below. Keep your virtual environment active so `make` uses the same interpreter every time.

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
```

```
# Setup development environment (creates .venv)
make setup

# Install package in editable mode with dev dependencies
make dev-install
```

If you prefer manual setup, create and activate `.venv` yourself, then run `pip install -e .[dev]` from the repository root. Whichever path you choose, activate the environment before running any `make` target so the same interpreter is used consistently.

### Makefile Commands

The Makefile provides a streamlined development workflow. Run targets from the repository root with your virtual environment active:

```
# View all available commands
make help

# Environment setup
make setup          # Create virtual environment
make dev-install    # Install in editable mode

# Code quality
make format         # Auto-format code
make lint           # Run linters

# Testing
make test           # Run tests with coverage
make coverage       # Open coverage report

# Documentation
make docs           # Build Sphinx documentation

# Build & publish
make clean          # Remove build artifacts
make build          # Build distributions
make publish        # Publish to PyPI

# CI
make ci             # Run full CI checks (format + lint + tests)
```

Makefile highlights:

- Uses `.venv/bin/` tools when available, otherwise falls back to global tools
- Colorized output with clear **success/warning/error** messages
- Works on **Linux** and **macOS**
- `make ci` is the superset; use it before pushing. Use `make format` / `make lint` / `make test` for quicker local loops during development.

### Testing

Start with the Makefile target, then drill down with pytest as needed. Favor deterministic seeds for generators so results are reproducible. Use narrow targets while iterating, then run the full suite.

```
# Recommended: run all tests with coverage
make test

# Equivalent pytest invocations
python -m pytest tests/ -v
python -m pytest tests/ --cov=py3plex --cov-report=html

# Run a specific test module or node
python -m pytest tests/test_core.py -k "degree"
python -m pytest tests/test_core.py::test_degree  # single test
```

The coverage HTML report is written to `htmlcov/index.html`.

### Code Quality

Tools configured in `pyproject.toml`:

- **Black**: Code formatting
- **Ruff**: Fast linting
- **isort**: Import sorting
- **Mypy**: Type checking
- **Pytest**: Testing with coverage

Before committing, run the quality loop from an active virtual environment. Reinstall dev dependencies if a tool is missing.

```
make format  # Auto-format code
make lint    # Check code quality
make test    # Run tests
make ci      # Run all checks (use before pushing)
```

### Contributing

We welcome contributions! Recommended workflow:

1. Sync your local `main` with `upstream/main`
2. Create a feature branch
3. Make focused changes with tests and docs
4. Run `make format`, `make lint`, `make test` (or `make ci`) from your active environment
5. Commit with clear messages that explain what changed and why
6. Push to your fork
7. Open a Pull Request

**Code Guidelines**:

- Follow existing code style (enforced by Black and Ruff)

- Add tests for new features
- Update documentation as needed
- Keep changes focused and atomic
- Write clear commit messages

### Documentation

Build Sphinx documentation:

```
# Using Makefile (recommended from repository root)
make docs

# Manual build from docfiles/
cd docfiles
sphinx-build -b html . _build/html
```

If the build fails because Sphinx is missing, (re)install the development extras in your active virtual environment, then rerun the command.

Open `docfiles/_build/html/index.html` in your browser to view the output.

### Documentation Style Guide

When contributing to documentation, follow these conventions for consistency:

**Naming Conventions:**

- Use "Quickstart" (one word) instead of "Quick Start" or "Quickstart Guide"
- Use "multilayer" (no hyphen) consistently, not "multi-layer" or "multi layer"
- Use short, direct section titles without boilerplate

**Tone and Structure:**

- Write first paragraphs that directly explain why the reader is on this page
- Avoid boilerplate like "This chapter will introduce…" or "In this section, you will learn…"
- Start with concrete examples rather than abstract explanations
- When relevant, point readers to clear paths for different audiences (new users, experienced users, advanced users) so they can skip to what they need

**Code Examples:**

All code examples must include:

1. **Imports** — Always show required imports
2. **Network creation** — Build a tiny example network from scratch or via helper
3. **Core operation** — Run one focused operation
4. **Expected output** — Show what the user should see

**Example structure:**

```
# 1. Imports
from py3plex.core import multinet
```

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# 2. Create network
network = multinet.multi_layer_network(directed=False)
network.add_nodes([{'source': 'A', 'type': 'layer1'}])
network.add_edges([
    {'source': 'A', 'target': 'B', 'source_type': 'layer1', 'target_type': 'layer1'}
])

# 3. Run operation
density = mls.layer_density(network, 'layer1')
print(f"Density: {density:.3f}")
```

**Expected output:**

```
Density: 1.000
```

**Marking Experimental APIs:**

Use warning directives for experimental or unstable features:

```
.. warning::

   **Experimental Feature:** This API is under active development and may change in future v
```

**Cross-references:**

- Link to related sections using `:doc:` for documentation pages
- Link to algorithm details using `:ref:` for labeled sections
- When documenting algorithms, link to the Algorithm Roadmap as the conceptual entry point

**Output accuracy:**

- Use real outputs from the current version of py3plex (no placeholders)
- Include the smallest network or dataset needed to reproduce the output

**Version Labels:**

- Use consistent version strings throughout (e.g., "py3plex 1.0.0 documentation")
- Update version in `conf.py` and `pyproject.toml` together

### Resources

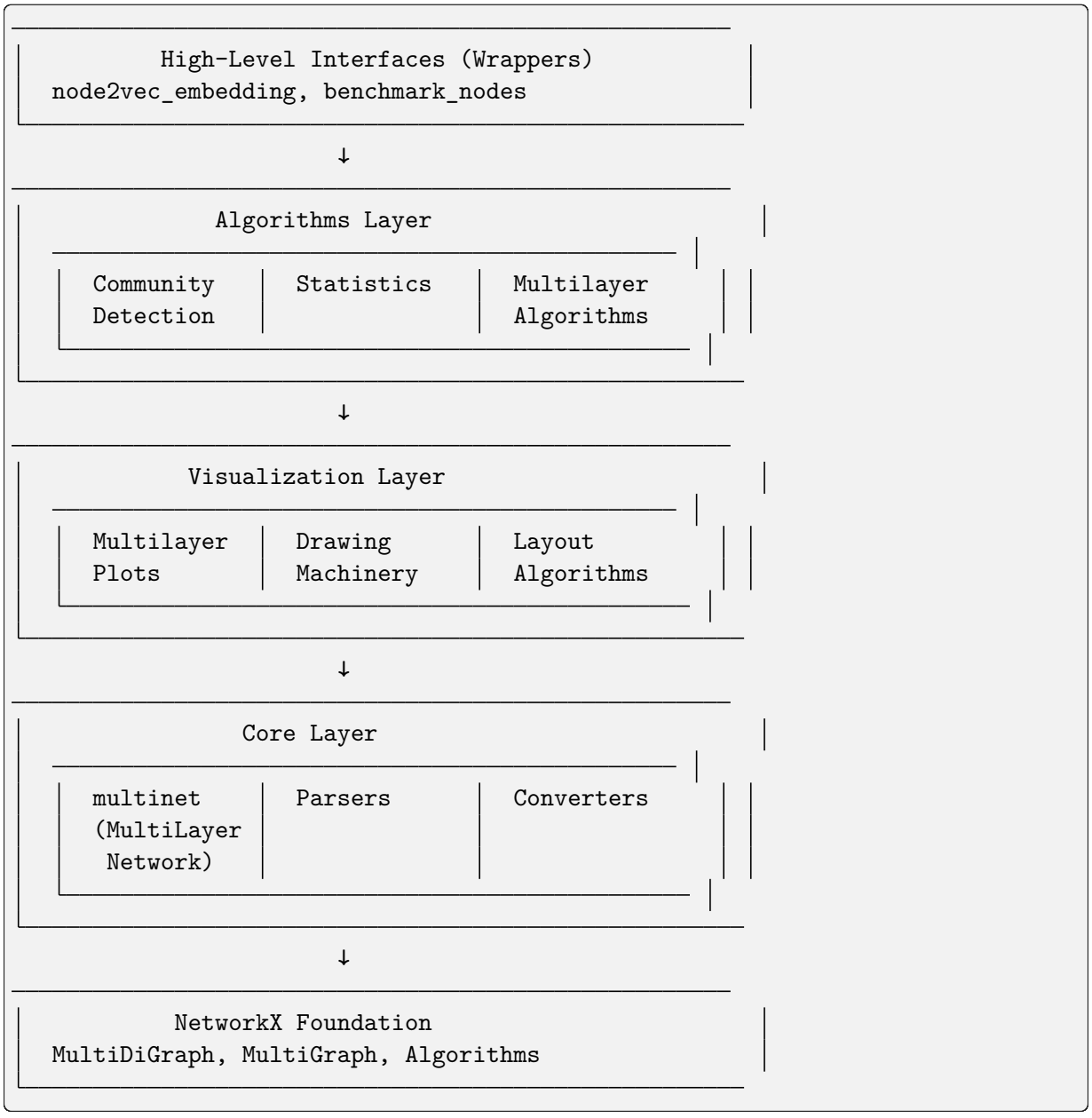
- **Repository**: https://github.com/SkBlaz/py3plex
- **Documentation**: https://skblaz.github.io/py3plex/
- **Issues**: https://github.com/SkBlaz/py3plex/issues
- **Examples**: https://github.com/SkBlaz/Py3Plex/tree/master/examples

### 3.8.8 Architecture and Design

This document describes the **system architecture**, **design patterns**, and **extension points** of py3plex. It explains how the layers fit together, what each layer owns, and where to plug in new capabilities without leaking responsibilities across layers.

#### System Overview

py3plex uses a **modular, layered architecture** with explicit boundaries. Higher layers depend on lower ones, never the reverse; data flows downward for computation and back upward for presentation:




**Reading the diagram:** NetworkX primitives sit at the base. The core layer wraps them in a multilayer-aware data model. Algorithms, visualization, and wrappers build on that model and never mutate it implicitly; wrappers orchestrate complete tasks while delegating computation to the lower layers.

### Architectural Layers

Each layer has a narrow contract: the core encodes data, algorithms consume that encoding, visualization renders results, and wrappers orchestrate complete tasks. When extending py3plex, start from the lowest layer you need and only depend upward.

### Core Layer

**Purpose:** Fundamental data structures and I/O operations. Everything else depends on this layer, so invariants and encoding live here.

**Key Components:**

- `multinet.py` - The `multi_layer_network` class (core facade)
- `parsers.py` - Input/output for various formats
- `converters.py` - Format conversion utilities
- `random_generators.py` - Random network generators
- `HINMINE/` - Heterogeneous network decomposition helpers

**Responsibilities:**

- Network construction and mutation through a single API
- File I/O (GraphML, GML, GEXF, edge lists, etc.)
- Layer management (string  integer IDs, delimiter handling)
- Matrix representations (adjacency, supra-adjacency)
- NetworkX integration and type selection (directed vs. undirected)
- Cache ownership (e.g., supra adjacency, embeddings) and invalidation hooks triggered by mutations

**Design Pattern: Facade Pattern** — `multi_layer_network` exposes one consistent interface while hiding NetworkX wiring and encoding details

### Algorithms Layer

**Purpose:** Network analysis algorithms optimized for multilayer networks. Algorithms assume core-layer encoding and never adjust it themselves.

**Key Components:**

- `community_detection/` - Community detection algorithms
- `statistics/` - Network statistics and metrics
- `multilayer_algorithms/` - Multilayer-specific methods
- `node_ranking/` - Centrality and ranking measures
- `general/` - General-purpose algorithms (random walks, etc.)

**Responsibilities:**

- Community detection (Louvain, Infomap, Label Propagation)
- Statistical analysis (multilayer density, inter-layer correlation, etc.)
- Centrality computation (degree, betweenness, PageRank, and variants)
- Random walks and embeddings

- Network decomposition primitives
- Result caching where appropriate (never mutating the underlying graph)

**Design Pattern: Strategy Pattern** — interchangeable algorithms share a common interface

### Visualization Layer

**Purpose:** Network plotting and rendering for multilayer structures. Visualization consumes algorithm outputs but does not compute new graph state.

**Key Components:**

- `multilayer.py` - High-level multilayer plotting
- `drawing_machinery.py` - Core drawing primitives
- `layout_algorithms.py` - Layout computation
- `colors.py` - Color scheme generators
- `fa2/` - ForceAtlas2 layout

**Responsibilities:**

- Diagonal projection plots and multilayer-specific layouts
- Force-directed and ForceAtlas2 layouts
- Matrix visualizations
- Color mapping and legend helpers
- Interactive plots (via Plotly)

**Design Pattern: Template Method Pattern** — layout algorithms follow a shared skeleton with overridable steps

### Wrappers Layer

**Purpose:** High-level interfaces for common workflows so users can run end-to-end tasks without touching internals. Wrappers compose algorithms and visualization, but defer state changes to the core.

**Key Components:**

- `node2vec_embedding.py` - Node2Vec embedding generation
- `benchmark_nodes.py` - Node classification benchmarking

**Responsibilities:**

- Simplified interfaces for multi-step workflows
- Integration with external tools
- Benchmarking and evaluation shortcuts

**Design Pattern: Facade Pattern** — hides orchestration and sensible defaults behind a small API

### Core Data Structure

#### The multi_layer_network Class

Central to py3plex, this class wraps the underlying NetworkX graph and enforces multilayer encoding invariants:

```
class multi_layer_network:
    def __init__(self, directed=True, label_delimiter="---", coupling_weight=1.0):
        self.core_network = nx.MultiDiGraph() if directed else nx.MultiGraph()
        self.layer_name_map = {}  # Bidirectional mapping
        self.label_delimiter = label_delimiter
        self.coupling_weight = coupling_weight
        self.embedding = None
        self.labels = None
```

**Key Attributes:**

- `core_network` - Underlying NetworkX graph
- `layer_name_map` - Maps layer names to integer IDs
- `label_delimiter` - Separator for node-layer encoding (default: `"---"`)
- `coupling_weight` - Default weight for inter-layer edges
- `embedding` - Cached node embedding matrix
- `labels` - Node classification labels

**Encoding Scheme and Invariants:**

- Nodes are represented as `(node_id, layer)` tuples in Python.
- When serialized to flat text (files, labels), tuples are joined with the delimiter: `"{node_id}{delimiter}{layer}"`. Avoid using the delimiter inside raw IDs.
- Layer names are mapped to integers in `layer_name_map` for stable ordering.
- `core_network` stays a NetworkX `MultiGraph`/`MultiDiGraph`; avoid injecting derived attributes that are not part of the graph definition.
- Inter-layer edges use `coupling_weight` unless explicitly weighted; modifying coupling should clear related caches.
- Any mutation (adding/removing nodes, relabeling layers) should invalidate cached matrices or embeddings.

### Design Patterns

#### Facade Pattern

**Used in:** `multi_layer_network`, wrappers

**Purpose:** Provide a simplified interface to complex subsystems and enforce a single entry point for mutations.

```
# Complex underlying operations hidden behind simple interface
network = multinet.multi_layer_network()
network.add_edges(edges, input_type='list')  # Handles parsing, encoding, validation
stats = network.basic_stats()  # Aggregates multiple NetworkX calls behind one call
```

### Strategy Pattern

**Used in:** Algorithms, layout computation

**Purpose:** Interchangeable algorithms following a common interface; callers pick by name without changing call sites.

```
# Different community detection strategies
def detect_communities(network, method='louvain'):
    strategies = {
        'louvain': community_louvain.best_partition,
        'infomap': community_wrapper.infomap_communities,
        'label_prop': label_propagation.propagate
    }
    return strategies[method](network.core_network)  # Adding a new strategy means adding o
```

### Template Method Pattern

**Used in:** Visualization, layout algorithms

**Purpose:** Define algorithm skeleton, allow customization in subclasses; shared steps live in the base class.

```
class LayoutAlgorithm:
    def compute(self, graph):
        self.initialize(graph)
        self.iterate()
        return self.finalize()

    def initialize(self, graph):
        raise NotImplementedError

    def iterate(self):
        raise NotImplementedError

    def finalize(self):
        raise NotImplementedError
```

### Dependency Injection

**Used in:** Configuration, algorithm parameters

**Purpose:** Inject dependencies rather than hard-coding so testing and theming stay configurable.

```
# Configuration injected rather than hard-coded
from py3plex.config import DEFAULT_COLORS, LAYOUT_PARAMS

def draw_network(network, colors=None, layout_params=None):
    colors = colors or DEFAULT_COLORS
    layout_params = layout_params or LAYOUT_PARAMS
    # Use injected configuration without touching global state
```

### Data Flow

#### Typical Workflow

py3plex workflows follow a predictable sequence: load → compute → analyze → visualize → export. Each step uses the layer beneath it and should avoid mutating lower layers unless explicitly intended.

1. **Input:** Load or create network, keeping encoding consistent

```
network = multinet.multi_layer_network()
network.load_network("data.graphml", input_type="graphml")
```

2. **Processing:** Apply algorithms against the core graph without re-encoding

```
communities = community_louvain.best_partition(network.core_network)
centrality = calc.multilayer_degree_centrality(network)
```

3. **Analysis:** Compute statistics on the derived results

```
density = mls.layer_density(network, 'layer1')
correlation = mls.inter_layer_degree_correlation(network, 'layer1', 'layer2')
```

4. **Visualization:** Render results; visualization functions expect immutable inputs

```
draw_multilayer_default([network], display=True)
```

5. **Output:** Export results using the same delimiter and layer naming

```
network.save_network("output.graphml", output_type="graphml")
```

#### State Management

**Immutable Operations:** Most algorithms don't modify the network or its caches

```
# These don't modify the network
centrality = calc.multilayer_degree_centrality(network)
communities = community_louvain.best_partition(network.core_network)
```

**Mutable Operations:** Some operations modify network state (nodes, edges, layer mappings, caches)

```
# These modify the network
network.add_edges(new_edges, input_type='list')
network.aggregate_layers(['L1', 'L2'], 'combined')
```

After running mutable operations, clear or recompute cached matrices/embeddings before downstream analysis. If you need isolation, copy the network or work on a subgraph before running destructive operations.

### Extension Points

#### Custom Algorithms

Add new algorithms by following existing patterns (pure functions returning dictionaries or NetworkX objects). Keep inputs typed as `multi_layer_network` to reuse encoding and caching, and avoid mutating the passed network unless the function is explicitly transformative. Document input assumptions (directed vs. undirected, weighted vs. unweighted) and expose new callables in the relevant package `__init__` when you want them importable by name.

```python
# py3plex/algorithms/my_module/my_algorithm.py
from py3plex.core.multinet import multi_layer_network

def my_centrality(network: multi_layer_network) -> dict:
    """
    Custom centrality measure.

    Parameters
    ----------
    network : multi_layer_network
        Input network

    Returns
    -------
    dict
        Node centrality scores
    """
    G = network.core_network
    centrality = {}

    for node in G.nodes():
        # Implement custom logic
        centrality[node] = compute_score(node, G)

    return centrality
```

#### Custom Visualizations

Create custom plots using drawing machinery. Treat the network as read-only and reuse shared layout helpers to keep visuals consistent with built-in plots:

```python
from py3plex.visualization import drawing_machinery as dm
import matplotlib.pyplot as plt

def my_custom_plot(network):
    """Custom visualization."""
    fig, ax = plt.subplots(figsize=(10, 8))

    # Compute layout
    pos = dm.compute_layout(network.core_network, 'force')

    # Draw elements
```

```
    dm.draw_nodes(ax, network.core_network, pos, node_size=50)
    dm.draw_edges(ax, network.core_network, pos, edge_width=1)
    dm.draw_labels(ax, pos, labels=network.get_node_labels())

    plt.show()
```

### Custom Parsers

Add support for new file formats. Normalize layer names, respect `label_delimiter`, and raise the appropriate domain exceptions so callers can distinguish parsing failures from missing data:

```
# py3plex/core/parsers.py
def parse_my_format(input_file, **kwargs):
    """
    Parse custom file format.

    Parameters
    ----------
    input_file : str
        Path to input file

    Returns
    -------
    multi_layer_network
        Parsed network
    """
    network = multi_layer_network()

    with open(input_file, 'r') as f:
        for line in f:
            # Parse line and add to network
            pass

    return network
```

### Configuration System

#### Centralized Configuration

`py3plex/config.py` provides centralized configuration:

```
# Default color palettes (8 options including colorblind-safe)
DEFAULT_COLORS = 'Set1'
COLORBLIND_SAFE = 'colorblind'

# Visualization defaults
DEFAULT_NODE_SIZE = 20
DEFAULT_EDGE_WIDTH = 1.0
DEFAULT_ALPHA = 0.7

# Layout parameters
```

```
LAYOUT_PARAMS = {
    'force': {'iterations': 500, 'optimal_distance': 1.0},
    'fa2': {'iterations': 1000, 'gravity': 1.0}
}

# Performance settings
SPARSE_THRESHOLD = 1000  # Use sparse matrices above this node count
MEMORY_WARNING_THRESHOLD = 10000  # Warn for large dense matrices
```

Usage:

```
from py3plex.config import DEFAULT_COLORS, LAYOUT_PARAMS

colors = DEFAULT_COLORS
iterations = LAYOUT_PARAMS['force']['iterations']
```

Prefer importing needed values rather than mutating module globals. For per-run overrides, pass parameters explicitly into drawing or layout helpers.

### Testing Architecture

#### Test Organization

Tests mirror the architecture: core primitives first, then algorithms and workflows.

```
tests/
├── test_core_functionality.py      # Core data structure tests
├── test_multilayer_*.py            # Multilayer algorithm tests
├── test_random_walks.py            # Random walk tests
├── test_io_*.py                    # I/O and parsing tests
├── test_config_api.py              # Configuration tests
└── test_utils.py                   # Utility function tests
```

#### Test Patterns

**Unit Tests:** Test individual functions in isolation

```
def test_layer_density():
    network = create_test_network()
    density = mls.layer_density(network, 'layer1')
    assert 0 <= density <= 1
```

**Integration Tests:** Test workflows across modules

```
def test_community_detection_workflow():
    network = load_network("test_data.graphml")
    communities = community_louvain.best_partition(network.core_network)
    assert len(communities) > 0
```

**Property-Based Tests:** Test invariants

```
def test_centrality_normalization():
    network = create_random_network()
    centrality = calc.multilayer_degree_centrality(network)
    # Centrality values should not be negative; normalized variants stay within [0, 1]
    assert all(v >= 0 for v in centrality.values())
```

### Performance Considerations

#### Lazy Evaluation

Expensive operations are computed on-demand and cached on the instance:

```
class multi_layer_network:
    @property
    def supra_adjacency(self):
        if self._supra_adj_cache is None:
            self._supra_adj_cache = self._compute_supra_adjacency()
        return self._supra_adj_cache
```

#### Sparse Matrices

Use sparse representations for large networks; the threshold is configurable in `config.py`:

```
def get_supra_adjacency_matrix(self, sparse=True):
    if sparse or len(self.get_nodes()) > SPARSE_THRESHOLD:
        return scipy.sparse.csr_matrix(adj)
    return np.array(adj)
```

Keep the chosen representation consistent downstream to avoid repeated densesparse conversions.

#### Vectorization

Prefer NumPy vectorized operations to avoid Python loops:

```
# Bad: Python loop
degrees = [sum(1 for _ in G.neighbors(node)) for node in nodes]

# Good: Vectorized
degrees = np.array(list(dict(G.degree()).values()))
```

### Logging Infrastructure

#### Centralized Logging

`py3plex/logging_config.py` provides structured logging:

```
import logging
from py3plex.logging_config import get_logger

logger = get_logger(__name__)
```

```
logger.info("Processing network with %d nodes", num_nodes)
logger.warning("Large network detected, using sparse matrices")
logger.error("Invalid layer: %s", layer_name)
```

Call `get_logger` once per module; configure logging once in the application entrypoint to avoid duplicate handlers.

### Log Levels

- **DEBUG:** Detailed diagnostic information
- **INFO:** General informational messages
- **WARNING:** Warning messages (e.g., performance concerns)
- **ERROR:** Error messages
- **CRITICAL:** Critical errors

### Error Handling

### Custom Exceptions

`py3plex/exceptions.py` defines domain-specific exceptions:

```
class NetworkError(Exception):
    """Base exception for network errors."""
    pass

class LayerNotFoundError(NetworkError):
    """Raised when layer doesn't exist."""
    pass

class InvalidFormatError(NetworkError):
    """Raised when file format is invalid."""
    pass
```

Usage:

```
from py3plex.exceptions import LayerNotFoundError

def get_layer(self, layer_name):
    if layer_name not in self.layer_name_map:
        raise LayerNotFoundError(f"Layer '{layer_name}' not found")
    return self.layer_name_map[layer_name]
```

Prefer raising domain-specific exceptions for recoverable errors so callers can distinguish parsing failures, missing layers, or invalid formats.

### Future Architecture

#### Planned Improvements

1. **Backend Registry:** Support for igraph, cugraph backends
2. **Streaming API:** Process networks larger than memory
3. **Distributed Computing:** Dask/Ray integration for large-scale analysis
4. **Plugin System:** Easy addition of third-party algorithms
5. **Type System:** Full type hints coverage (currently 65%)

These roadmap items are exploratory; expect details and APIs to evolve.

### See Also

- contributing - Contributing guidelines
- ../development - Development workflow
- ../core - Core API documentation

CHAPTER

FOUR

# CITATION

If you use py3plex in your research, please cite:

```bibtex
@Article{Skrlj2019,
  author={Skrlj, Blaz and Kralj, Jan and Lavrac, Nada},
  title={Py3plex toolkit for visualization and analysis of multilayer networks},
  journal={Applied Network Science},
  year={2019},
  volume={4},
  number={1},
  pages={94},
  doi={10.1007/s41109-019-0203-7}
}
```

CHAPTER

FIVE

# INDICES AND TABLES

- genindex
- modindex
- search